# Dark Matter Detection in Solids

## Volume III

## Computations

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# Plan of the three volumes

The book is in three volumes. Volume I, *Foundations* (Chaps. 1 to 8), teaches the quantum mechanics, solid-state physics and response theory that the rates rest on, and ends with the master rate formula and the response framework built on it (Chap. 8). Volume II, *Signals and Detection* (Chaps. 9 to 22), derives the rate of each excitation channel and follows the deposited energy to the measured signal and to an exclusion limit. Volume III, *Computations* (Chaps. 23 to 33), gives the first-principles methods and the specification of each calculation behind the numbers of the other two.

Chapter numbers and appendix letters run through the three volumes, so a reference names its target without the volume: Sect. 4.4 is in Volume I and Sect. 10.2 in Volume II. Appendices A and B (conventions and notation) are printed in every volume, Appendices C and D (mathematical tools and long derivations) in Volume I, and Appendices E and F (long derivations and an index of literature formulas) in Volume II.

# Conventions and units

We use natural units, $\hbar = c = k_{\mathrm{B}} = 1$, unless stated otherwise, and quote physical values in conventional units with SI prefixes.

A widely used formula from the literature appears in a gray box, a Limit Check, that plots it against the book's more complete result. A Limit Check has three parts: the formula, a plot, and one sentence. The formula is written in the book's notation, and a one-line citation names the equation of the source where it was first given. The plot compares the formula with a fuller result for the same quantity, computed in the book from a crystal calculation or from a model that keeps the physics the formula drops, or measured. Most boxes plot the ratio of the two, so that the formula is the horizontal line at one; some plot both curves, when their shapes matter more than their ratio. The horizontal axis carries the *controlling parameter*, the variable along which the relative difference of the two runs from negligible at one end of the axis to order one at the other. The *leading correction*, the first term of that difference in the controlling parameter, is drawn as a third curve, usually dotted or dashed. The gray band marks the values of the controlling parameter at which the formula agrees with the fuller result, to a tolerance stated in the caption, usually 10 %, and the caption prints the edge of the band. In Limit Check 1.1 the free-recoil formula $\omega = q^2/2M$ is compared with a nucleus bound in an Einstein potential, against the recoil energy in units of the phonon energy, and the leading correction is the rms width $(E_R\omega_0)^{1/2}$ of the deposit. The sentence under the plot says what breaks down outside the band, and the steps from the book's general result to the formula are derived in the running text next to the box.

Inside the gray band the formula is established to the stated tolerance, and that statement is as strong as the fuller result. Where the fuller result is a model, the edge of the band is an estimate of where the formula fails, not a bound. Where nothing is shaded, the formula fails over the whole plotted range, and the box gives the size of the failure instead. Some boxes compare two published forms of one result, or test a sum rule; they are read in the same way, with the exact relation as the fuller result. Appendix F lists the most used of these formulas with their source equations and the sizes of their leading corrections.

# Part I

# Methods

# Chapter 23

# Density-functional theory

A dark-matter rate in silicon needs a few dozen numbers about the crystal: a gap good to tens of meV, a loss function out to 100 eV, density vertices at momentum transfers of tens of keV, electron–phonon couplings on grids of $10^9$ points, and the energy a struck atom leaves behind. Almost none of them is measured everywhere it is needed. Each is computed by a method with its own approximation, numerical parameters and typical failure. Chaps. 23 to 29 follow each method through one calculation, almost always of silicon, so that a number in the text can be traced to what it neglects and to the check that would expose it. Chaps. 30 to 33 go the other way and specify, quantity by quantity, the calculation behind each number a rate uses, with its inputs, its method and its convergence targets.

The calculation at clamped nuclei, with independent particles and no phonons, the base point of Sect. 8.5, is the density-functional theory of Sect. 3.4. The bands and wavefunctions behind computed dark-matter rates come from it, and the gap is its best-known failure: a density-functional calculation of silicon in the local-density approximation puts the gap at 0.52 eV, less than half the measured 1.17 eV. The computed gap would put the reach of a silicon detector at 0.15 MeV instead of 0.33 MeV (Worked example 3.7). The Hartree–Fock equations and the theorems of Hohenberg and Kohn behind Sect. 3.4 are proved first; then a Kohn–Sham calculation of silicon is followed through its numerical parameters to a corrected gap.

## 23.1 Hartree–Fock equations and Koopmans' theorem

The Hartree–Fock equations Eq. (3.60) are the minimum condition for the energy of a Slater determinant with respect to its orbitals [1], and Koopmans' theorem [2] follows from deleting one orbital from the determinant. For the electron gas both take closed form. The derivation follows Ref. [3, Chap. 2].

### 23.1.1 The energy of a determinant

Take $N_e$ electrons with clamped nuclei. The Hamiltonian is a one-body part, $h = -\nabla^2/2m_e + U_1(\boldsymbol{r})$, with $U_1$ the attraction of the nuclei, summed over the electrons, plus the Coulomb pair potential $U_2(\boldsymbol{r}-\boldsymbol{r}') = e^2/(4\pi|\boldsymbol{r}-\boldsymbol{r}'|)$ summed over pairs. Let $|\Phi_0\rangle$ be the determinant Eq. (2.29) of $N_e$ orthonormal orbitals $\psi_\gamma(x)$, with $x = (\boldsymbol{r}, s)$ and $\int \mathrm{d}x = \sum_s \int \mathrm{d}^3r$. Its energy was worked out in Sect. 2.3: the one-body operator contributes $\sum_\gamma \langle\gamma|h|\gamma\rangle$, and the two-body operator contributes the direct term minus the exchange term, summed over pairs of occupied

orbitals. In position space,

$$E[\Phi_0] = \sum_\gamma \int \mathrm{d}x\, \psi_\gamma^*(x)\, h\, \psi_\gamma(x) \\ + \frac{1}{2}\sum_{\gamma\gamma'} \int \mathrm{d}x\, \mathrm{d}x'\, U_2(\boldsymbol{r}-\boldsymbol{r}')\Big[|\psi_\gamma(x)|^2|\psi_{\gamma'}(x')|^2 - \psi_\gamma^*(x)\psi_{\gamma'}^*(x')\psi_{\gamma'}(x)\psi_\gamma(x')\Big], \quad (23.1)$$

where both sums run over the occupied orbitals. The terms with $\gamma = \gamma'$ cancel inside the bracket, so no electron interacts with itself.

The density matrix $\varrho(x,x') = \sum_\gamma \psi_\gamma(x)\psi_\gamma^*(x')$ has the electron density on its diagonal, $\boldsymbol{n}(\boldsymbol{r}) = \sum_s \varrho(x,x)$, and the direct term is the classical electrostatic energy of that density. The exchange term is $-\frac{1}{2}\int \mathrm{d}x\, \mathrm{d}x'\, U_2\, |\varrho(x,x')|^2$, so $\varrho$ carries all the dependence of the energy on the occupied orbitals. It is unchanged when the occupied orbitals are mixed among themselves by a unitary matrix $\mathcal{M}$, $\psi_\gamma \to \sum_{\gamma'} \psi_{\gamma'}\mathcal{M}_{\gamma'\gamma}$, because $\mathcal{M}\mathcal{M}^\dagger = 1$ leaves $\varrho$ unchanged. The determinant itself changes only by the phase $\det \mathcal{M}$. The energy is therefore a function of the occupied subspace, not of the individual orbitals.

### 23.1.2 The variational equations

By the variational principle, $E[\Phi_0] \geq E_0$ for every determinant, where $E_0$ is the exact ground-state energy. The Hartree–Fock determinant is the one that makes $E[\Phi_0]$ stationary, and its lowest stationary point is the Hartree–Fock ground state. The orbitals must stay orthonormal, and we impose this with a Hermitian matrix of Lagrange multipliers $\lambda_{\gamma'\gamma}$, making stationary

$$E[\Phi_0] - \sum_{\gamma\gamma'} \lambda_{\gamma'\gamma}\Big(\int \mathrm{d}x\, \psi_\gamma^*(x)\psi_{\gamma'}(x) - \delta_{\gamma\gamma'}\Big).$$

A complex function and its conjugate can be varied independently. The derivative with respect to $\psi_\gamma^*(x)$ of the one-body term is $h\psi_\gamma(x)$. In the double sum, $\psi_\gamma^*$ appears once in the slot $\gamma$ and once in the slot $\gamma'$, and since $U_2$ is symmetric the two contributions are equal and cancel the factor $\frac{1}{2}$. The direct term gives $\int \mathrm{d}x'\, U_2(\boldsymbol{r}-\boldsymbol{r}')\, \boldsymbol{n}(\boldsymbol{r}')\, \psi_\gamma(x)$, and the exchange term gives $-\int \mathrm{d}x'\, U_2(\boldsymbol{r}-\boldsymbol{r}')\, \varrho(x,x')\, \psi_\gamma(x')$. Setting the total derivative to zero,

$$h_{\mathrm{HF}}\, \psi_\gamma = \sum_{\gamma'} \lambda_{\gamma'\gamma}\, \psi_{\gamma'}, \qquad h_{\mathrm{HF}}\, \psi(x) = \Big[-\frac{\nabla^2}{2m_e} + U_1(\boldsymbol{r}) + v_{\mathrm{H}}(\boldsymbol{r})\Big]\psi(x) + \int \mathrm{d}x'\, \Sigma_{\mathrm{x}}(x,x')\, \psi(x'),$$

with the Hartree potential and the exchange (Fock) term

$$v_{\mathrm{H}}(\boldsymbol{r}) = \int \mathrm{d}^3r'\, \frac{e^2\, \boldsymbol{n}(\boldsymbol{r}')}{4\pi|\boldsymbol{r}-\boldsymbol{r}'|}, \qquad \Sigma_{\mathrm{x}}(x,x') = -\frac{e^2\, \varrho(x,x')}{4\pi|\boldsymbol{r}-\boldsymbol{r}'|}.$$

The operator $h_{\mathrm{HF}}$ is Hermitian, since $\Sigma_{\mathrm{x}}(x',x)^* = \Sigma_{\mathrm{x}}(x,x')$. Projecting the equation onto $\psi_{\gamma''}$ gives $\lambda_{\gamma''\gamma} = \langle\gamma''|h_{\mathrm{HF}}|\gamma\rangle$, a Hermitian matrix. A unitary rotation of the occupied orbitals leaves $h_{\mathrm{HF}}$ unchanged, because it depends on them only through $\boldsymbol{n}$ and $\varrho$, and we can choose the rotation that diagonalizes $\lambda$. In that basis the result takes its canonical form,

$$h_{\mathrm{HF}}\, \psi_\gamma(x) = E_\gamma\, \psi_\gamma(x), \qquad \boldsymbol{n}(\boldsymbol{r}) = \sum_s \sum_{\gamma\,\mathrm{occ}} |\psi_\gamma(\boldsymbol{r},s)|^2, \quad \varrho(x,x') = \sum_{\gamma\,\mathrm{occ}} \psi_\gamma(x)\psi_\gamma^*(x'). \quad (23.2)$$

The equations are nonlinear, since $h_{\mathrm{HF}}$ is built from its own occupied eigenfunctions, and they are solved by iteration until the orbitals that go in are the orbitals that come out. The occupied set is the $N_e$ eigenfunctions of lowest $E_\gamma$; the other eigenfunctions of the same operator are the empty, or virtual, orbitals.

Acting on an occupied orbital $\psi_\gamma$, the term $\gamma' = \gamma$ of $\Sigma_\mathrm{x}$ is $-\int \mathrm{d}x'\, U_2 |\psi_\gamma(x')|^2 \psi_\gamma(x)$, which cancels the self-repulsion contained in $v_\mathrm{H}$. More generally, $-|\varrho(x,x')|^2/\varrho(x,x)$ is a hole in the density of the other electrons of the same spin around an electron at $x$, the exchange hole. Since $\int \mathrm{d}x''\, \varrho(x,x'')\varrho(x'',x') = \varrho(x,x')$ for orthonormal orbitals, it integrates over $x'$ to exactly one missing electron. The Fock term $\Sigma_\mathrm{x}$ is the first-order self-energy of Sect. 4.7 in the Coulomb interaction; Appendix D.9 obtains it from a diagram.

### 23.1.3 Orbital energies and Koopmans' theorem

Taking the expectation value of Eq. (23.2) in $\psi_\gamma$ leads to the orbital energy,

$$E_\gamma = \langle\gamma|h|\gamma\rangle + \sum_{\gamma'\,\mathrm{occ}} \Big[\langle\gamma\gamma'|U_2|\gamma\gamma'\rangle - \langle\gamma\gamma'|U_2|\gamma'\gamma\rangle\Big],$$

the kinetic and nuclear energy of the orbital plus its direct and exchange interaction with every occupied orbital. Summing $E_\gamma$ over the occupied orbitals counts every pair twice, and comparison with Eq. (23.1) gives

$$E_\mathrm{HF} = \sum_{\gamma\,\mathrm{occ}} E_\gamma - \frac{1}{2} \sum_{\gamma\gamma'\,\mathrm{occ}} \Big[\langle\gamma\gamma'|U_2|\gamma\gamma'\rangle - \langle\gamma\gamma'|U_2|\gamma'\gamma\rangle\Big]. \tag{23.3}$$

The total energy is not the sum of the orbital energies, and the difference is the double counting of the interaction.

Now remove the electron in an occupied orbital $a$ and keep all the other orbitals as they are. The new state is a determinant of $N_e - 1$ orbitals, and its energy is Eq. (23.1) without every term that contains $a$. These are the one-body term $\langle a|h|a\rangle$ and, in the double sum, the terms with $\gamma = a$ or $\gamma' = a$, which are equal in pairs and together give the full sum over $\gamma'$ of the interaction of $a$ with the others. That is $E_a$. Adding an electron in an empty orbital $r$ adds the same kind of terms, with the sum over the $N_e$ occupied orbitals. With the orbitals frozen, therefore,

$$E^{(a)}_{N_e-1} - E_{N_e} = -E_a, \qquad E^{(r)}_{N_e+1} - E_{N_e} = E_r. \tag{23.4}$$

The occupied eigenvalues are minus the removal energies and the empty ones are the addition energies. This is Koopmans' theorem. Its hypothesis is the frozen orbitals. In a real removal the other orbitals relax around the hole, and the new determinant of lowest energy is not the one obtained by deleting a column.

In a crystal the relaxation drops out. The orbitals are Bloch states $\psi_{n\boldsymbol{k}}$, each spread over the $N$ cells of the crystal, and removing one electron changes $n$ and $\varrho$ in each cell by an amount of relative order $1/N$. The orbitals respond at first order in that change. The energy, however, is stationary with respect to the orbitals, so the energy gained by relaxing them is of second order, $1/N^2$ per orbital, and summed over the orbitals of one band it vanishes as $1/N$. Hartree–Fock band energies are therefore the removal and addition energies of the Hartree–Fock description of the crystal, and their difference across the gap is its fundamental gap.

### 23.1.4 The electron gas

In jellium, Eq. (2.47), plane waves solve Eq. (23.2) for any occupied set of plane waves, since $\varrho$ is then invariant under translations, and the filled Fermi sphere is the natural set. The density is then uniform, its Hartree potential is cancelled by the background (Worked example 2.5), and $\varrho(x,x') = \delta_{ss'}\mathcal{V}^{-1}\sum_{p<k_\mathrm{F}} \mathrm{e}^{\mathrm{i}\boldsymbol{p}\cdot(\boldsymbol{r}-\boldsymbol{r}')}$ depends only on $\boldsymbol{r}-\boldsymbol{r}'$. Acting on a

plane wave $\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$, the exchange term multiplies it by a number. With $U_2$ written through its transform, $\mathscr{V}^{-1}\sum_{\boldsymbol{k}'} v_{\mathrm{C}}(k')\,\mathrm{e}^{-\mathrm{i}\boldsymbol{k}'\cdot\boldsymbol{r}}$, and the sum over $\boldsymbol{p}$ turned into an integral,

$$\Sigma_{\mathrm{x}}(k) = -\frac{1}{\mathscr{V}}\sum_{p<k_{\mathrm{F}}} v_{\mathrm{C}}(|\boldsymbol{k}-\boldsymbol{p}|) = -\int_{p<k_{\mathrm{F}}} \frac{\mathrm{d}^3p}{(2\pi)^3}\,\frac{e^2}{k^2+p^2-2kpu},$$

with $u$ the cosine of the angle between $\boldsymbol{k}$ and $\boldsymbol{p}$. The sum runs over one spin, since $\varrho$ is diagonal in spin: only electrons of the same spin exchange. The angular integral is $\int_{-1}^{1}\mathrm{d}u\,(k^2+p^2-2kpu)^{-1} = (kp)^{-1}\ln|(k+p)/(k-p)|$, which leaves

$$\Sigma_{\mathrm{x}}(k) = -\frac{e^2}{4\pi^2 k}\int_0^{k_{\mathrm{F}}}\mathrm{d}p\;p\,\ln\Big|\frac{k+p}{k-p}\Big| = -\frac{e^2k_{\mathrm{F}}^2}{4\pi^2k}\int_0^1\mathrm{d}t\;t\,\ln\Big|\frac{x+t}{x-t}\Big|, \qquad x=\frac{k}{k_{\mathrm{F}}}.$$

The integrand has the antiderivative $\frac{1}{2}(t^2-x^2)\ln|(x+t)/(x-t)| + xt$, as differentiation shows: the derivative of the logarithm, $2x/(x^2-t^2)$, times $\frac{1}{2}(t^2-x^2)$ is $-x$, which cancels the derivative of $xt$. Between $t=0$ and 1 it gives $x+\frac{1}{2}(1-x^2)\ln|(1+x)/(1-x)| = 2xF_{\mathrm{L}}(x)$, with the function $F_{\mathrm{L}}$ of Eq. (C.72) evaluated at $x$. Hence

$$\Sigma_{\mathrm{x}}(k) = -\frac{e^2k_{\mathrm{F}}}{2\pi^2}\,F_{\mathrm{L}}\!\left(\frac{k}{k_{\mathrm{F}}}\right), \qquad E_{\boldsymbol{k}}^{\mathrm{HF}} = \frac{k^2}{2m_e} + \Sigma_{\mathrm{x}}(k). \tag{23.5}$$

The exchange energy is most negative at the bottom of the band, $\Sigma_{\mathrm{x}}(0) = -e^2k_{\mathrm{F}}/2\pi^2$, since $F_{\mathrm{L}}(0)=1$, and it falls to half that value at the Fermi surface, since $F_{\mathrm{L}}(1)=\frac{1}{2}$.

The occupied band is widened, from $E_{\mathrm{F}}$ to $E_{\mathrm{F}}+e^2k_{\mathrm{F}}/4\pi^2 = E_{\mathrm{F}}+\alpha k_{\mathrm{F}}/\pi$. At the valence density of silicon, with $k_{\mathrm{F}} = 3.57\,\mathrm{keV}$ and $E_{\mathrm{F}} = 12.5\,\mathrm{eV}$ (Sect. 2.5), the widening is 8.29 eV and the band is 20.8 eV wide. And the slope of $F_{\mathrm{L}}$ diverges logarithmically at $x=1$ (Appendix C.8), so the group velocity $\partial E_{\boldsymbol{k}}^{\mathrm{HF}}/\partial k$ is infinite at the Fermi surface and the density of states there vanishes. A sharp Fermi surface turns the long range of the bare exchange, $v_{\mathrm{C}}\propto 1/k^2$, into a singular $\Sigma_{\mathrm{x}}(k)$. In the real metal the other electrons screen the exchange, the logarithm disappears, and the velocity at the Fermi surface stays close to $k_{\mathrm{F}}/m_e$. The screened exchange is the GW self-energy of Sect. 4.8.

The exchange energy of the gas follows from Eq. (23.3). The kinetic part averages to $\frac{3}{5}E_{\mathrm{F}}$ per electron. Since $\Sigma_{\mathrm{x}}$ is the interaction of one electron with all the others, the interaction energy is half its average over the occupied states: $E_{\mathrm{x}}/N_e = \frac{1}{2}\langle\Sigma_{\mathrm{x}}\rangle$. The average of $F_{\mathrm{L}}(x)$ over the Fermi sphere is $\int_0^1\mathrm{d}x\,3x^2F_{\mathrm{L}}(x)$. Its first part is $\frac{1}{2}$. The second is $\frac{3}{4}\int_0^1\mathrm{d}x\,(x-x^3)\ln[(1+x)/(1-x)]$; integrating by parts with $-(1-x^2)^2/4$ as the antiderivative of $x-x^3$, which vanishes at $x=1$, turns it into $\frac{3}{4}\int_0^1\mathrm{d}x\,\frac{1}{2}(1-x^2) = \frac{1}{4}$. The average is $\frac{3}{4}$, and

$$\frac{E_{\mathrm{x}}}{N_e} = -\frac{3e^2k_{\mathrm{F}}}{16\pi^2} = -\frac{3\alpha k_{\mathrm{F}}}{4\pi} = -\frac{0.916}{r_s}\,\mathrm{Ry}, \tag{23.6}$$

with $\mathrm{Ry} = \alpha^2m_e/2 = 13.61\,\mathrm{eV}$ and $k_{\mathrm{F}} = (9\pi/4)^{1/3}/(r_sa_0)$, which follows from $n_e = 3/(4\pi r_s^3a_0^3)$ and Eq. (2.71). At the silicon valence density, $r_s = 2.0$ and $E_{\mathrm{x}}/N_e = -6.22\,\mathrm{eV}$, comparable in size to the kinetic energy per electron, $\frac{3}{5}E_{\mathrm{F}} = 7.5\,\mathrm{eV}$. The local-density approximation Eq. (3.63) is built on Eq. (23.6).

## 23.2 Hohenberg–Kohn theorems and the Kohn–Sham construction

The two theorems of Hohenberg and Kohn [4] and the Kohn–Sham equations Eq. (3.62) [5] are proved here in the constrained-search formulation of Levy [6], which needs fewer assumptions than the original proof. The derivation follows Ref. [7, Chaps. 6 and 7].

### 23.2.1 The density determines the potential

Take $N_e$ electrons in an external potential $U_1(\boldsymbol{r})$, the attraction of the clamped nuclei, with

$$H = H_{\text{kin}} + H_{ee} + \sum_{i=1}^{N_e} U_1(\boldsymbol{r}_i), \qquad H_{\text{kin}} = \sum_i \frac{\boldsymbol{p}_i^2}{2m_e},$$

and $H_{ee}$ the Coulomb repulsion of Eq. (2.44). Only the last term depends on the material, and it enters every expectation value through the density alone: $\langle\Psi|\sum_i U_1(\boldsymbol{r}_i)|\Psi\rangle = \int \mathrm{d}^3r\, U_1(\boldsymbol{r})\, n(\boldsymbol{r})$, with $n(\boldsymbol{r})$ the electron density of $\Psi$.

Suppose two potentials $U_1$ and $U_1'$, which differ by more than a constant, had nondegenerate ground states $\Psi$ and $\Psi'$ with the same density $n$. The two states are different. If they were equal, subtracting the two Schrödinger equations would give $\sum_i[U_1(\boldsymbol{r}_i) - U_1'(\boldsymbol{r}_i)]\Psi = (E_0 - E_0')\Psi$, so that the sum would be constant wherever $\Psi \neq 0$, which for a ground state is almost everywhere, and the potentials would differ by a constant. Since the ground state of $H$ is nondegenerate, $\Psi'$ has a strictly higher energy in $H$:

$$E_0 < \langle\Psi'|H|\Psi'\rangle = \langle\Psi'|H'|\Psi'\rangle + \int \mathrm{d}^3r\,[U_1(\boldsymbol{r}) - U_1'(\boldsymbol{r})]n(\boldsymbol{r}) = E_0' + \int \mathrm{d}^3r\,[U_1 - U_1']n.$$

The same argument with the primed and unprimed quantities exchanged gives $E_0' < E_0 + \int \mathrm{d}^3r\,[U_1' - U_1]n$. Adding the two inequalities, the integrals cancel and $E_0 + E_0' < E_0' + E_0$, a contradiction. Hence the ground-state density determines $U_1$ up to a constant [4]. It also fixes $N_e = \int \mathrm{d}^3r\, n$, so it fixes $H$, and through $H$ every property of the system.

### 23.2.2 The variational principle

The universal functional of Sect. 3.4 is

$$\mathcal{F}[n] = \min_{\Psi\to n} \langle\Psi|H_{\text{kin}} + H_{ee}|\Psi\rangle, \tag{23.7}$$

where the minimum is over all normalized antisymmetric states of $N_e$ electrons that have the density $n$ [6]. Every reasonable density, nonnegative, integrating to $N_e$ and with a finite kinetic energy, comes from some determinant, so $\mathcal{F}$ is defined on all of them. For a given potential let

$$E_U[n] = \mathcal{F}[n] + \int \mathrm{d}^3r\, U_1(\boldsymbol{r})\, n(\boldsymbol{r}).$$

For any $n$, the state $\Psi_n$ that realizes the minimum in Eq. (23.7) has $\langle\Psi_n|H|\Psi_n\rangle = E_U[n]$, and by the ordinary variational principle this is at least $E_0$. The minimum over all states, on the other hand, can be done in two steps, first over the states with a given density and then over the densities:

$$E_0 = \min_{\Psi}\langle\Psi|H|\Psi\rangle = \min_n\Big[\min_{\Psi\to n}\langle\Psi|H_{\text{kin}} + H_{ee}|\Psi\rangle + \int \mathrm{d}^3r\, U_1 n\Big] = \min_n E_U[n].$$

The ground-state energy is the minimum of $E_U$ over densities, and the minimum is reached at the ground-state density. This is the second theorem of Hohenberg and Kohn, here without the assumption of a nondegenerate ground state, and without the requirement that every trial density be the ground-state density of some potential. Where $\mathcal{F}$ is differentiable, the minimum under the constraint $\int n = N_e$ obeys $\delta\mathcal{F}/\delta n(\boldsymbol{r}) + U_1(\boldsymbol{r}) = \mu_{\text{ch}}$, with the chemical potential as the Lagrange multiplier.

### 23.2.3 The Kohn–Sham equations

The independent-electron kinetic energy is the same constrained search over determinants $\Phi$ alone,

$$K_\mathrm{s}[\boldsymbol{n}] = \min_{\Phi\to\boldsymbol{n}} \langle\Phi|H_\mathrm{kin}|\Phi\rangle,$$

and the Hartree energy is $E_\mathrm{H}[\boldsymbol{n}] = \frac{1}{2}\int \mathrm{d}^3r\,\mathrm{d}^3r'\,e^2\boldsymbol{n}(\boldsymbol{r})\boldsymbol{n}(\boldsymbol{r}')/(4\pi|\boldsymbol{r}-\boldsymbol{r}'|)$. The exchange–correlation energy is $E_\mathrm{xc}[\boldsymbol{n}] = \mathcal{F}[\boldsymbol{n}] - K_\mathrm{s}[\boldsymbol{n}] - E_\mathrm{H}[\boldsymbol{n}]$.

Write the minimizing determinant through its orbitals $\psi_\gamma$, with occupations $n_\gamma$ equal to one for the $N_e$ occupied orbitals and zero otherwise, and the density $\boldsymbol{n}(\boldsymbol{r}) = \sum_\gamma n_\gamma \sum_s |\psi_\gamma(\boldsymbol{r},s)|^2$. Then

$$E_U = \sum_\gamma n_\gamma \int \mathrm{d}x\,\psi_\gamma^*(x)\Big(-\frac{\nabla^2}{2m_e}\Big)\psi_\gamma(x) + \int \mathrm{d}^3r\,U_1\boldsymbol{n} + E_\mathrm{H}[\boldsymbol{n}] + E_\mathrm{xc}[\boldsymbol{n}],$$

which depends on the orbitals explicitly through the kinetic term and implicitly through $\boldsymbol{n}$. We vary $\psi_\gamma^*(x)$, with Lagrange multipliers for orthonormality chosen diagonal, as in Sect. 23.1. The chain rule gives $\delta E_\mathrm{H}/\delta\psi_\gamma^*(x) = n_\gamma v_\mathrm{H}(\boldsymbol{r})\psi_\gamma(x)$, with the Hartree potential $v_\mathrm{H} = \delta E_\mathrm{H}/\delta\boldsymbol{n}$ of Sect. 23.1, and similarly for $E_\mathrm{xc}$. Dividing by $n_\gamma$ gives the Kohn–Sham equations Eq. (3.62), with the density built from the $N_e$ orbitals of lowest $E_\gamma$. Filling the lowest orbitals assumes that the minimizing density is the ground-state density of independent electrons in some local potential, the noninteracting $v$-representability of the density; for the crystals of this book it is taken for granted. In a crystal, $\gamma$ becomes the band and crystal momentum $n\boldsymbol{k}$, $E_\gamma$ the band energy $\varepsilon_{n\boldsymbol{k}}$ and $n_\gamma$ the occupation $f_{n\boldsymbol{k}}$, and the density is Eq. (23.10). The equations are nonlinear, since $V_\mathrm{KS}$ depends on the density, and they are iterated to self-consistency.

### 23.2.4 Total energy and the eigenvalues

Multiplying Eq. (3.62) by $n_\gamma\psi_\gamma^*$, integrating and summing gives $\sum_\gamma n_\gamma E_\gamma = K_\mathrm{s} + \int \mathrm{d}^3r\,V_\mathrm{KS}\boldsymbol{n}$, and $\int v_\mathrm{H}\boldsymbol{n} = 2E_\mathrm{H}$. The ground-state energy is therefore

$$E_0 = \sum_\gamma n_\gamma E_\gamma - E_\mathrm{H}[\boldsymbol{n}] - \int \mathrm{d}^3r\,v_\mathrm{xc}(\boldsymbol{r})\boldsymbol{n}(\boldsymbol{r}) + E_\mathrm{xc}[\boldsymbol{n}]. \tag{23.8}$$

As in Hartree–Fock, Eq. (23.3), the sum of the eigenvalues counts the interaction twice and the other terms correct for it.

The eigenvalues are Lagrange multipliers, and Eq. (3.62) alone does not make them removal or addition energies. One exact statement connects them to energies. Let the occupations $n_\gamma$ take fractional values and keep the orbitals self-consistent at each set of $n_\gamma$. Then $E_U$ is a function of the occupations, and

$$\frac{\partial E_U}{\partial n_\gamma} = E_\gamma, \tag{23.9}$$

which is Janak's theorem [8]. The explicit dependence on $n_\gamma$, through the kinetic term and through $\boldsymbol{n}$, gives the expectation value of the Kohn–Sham Hamiltonian in $\psi_\gamma$, which is $E_\gamma$. The implicit dependence through the orbitals gives, by Eq. (3.62),

$$\sum_{\gamma'} n_{\gamma'}E_{\gamma'}\,\frac{\partial\langle\psi_{\gamma'}|\psi_{\gamma'}\rangle}{\partial n_\gamma} = 0,$$

because every orbital stays normalized. For the exact functional the ground-state energy of a fractional number of electrons is a straight line between integers [9]. Integrating Eq. (23.9) from $N_e-1$ to $N_e$ in the highest occupied level, whose eigenvalue is then constant, gives $E_{N_e} - E_{N_e-1} = E_\mathrm{HOMO}$, with $E_\mathrm{HOMO}$ the eigenvalue of the highest occupied molecular orbital

(HOMO): the highest occupied Kohn–Sham eigenvalue is minus the ionization energy $I$. In the same way, the eigenvalue of the level that fills between $N_e$ and $N_e+1$ is minus the electron affinity $A$. No other eigenvalue has an exact meaning, and the Kohn–Sham gap differs from the fundamental gap by the derivative discontinuity of Eq. (3.65).

### 23.2.5 The local exchange potential

The potential of the local-density approximation Eq. (3.63) is

$$v_{\mathrm{xc}}^{\mathrm{LDA}}(\boldsymbol{r}) = \frac{\mathrm{d}\big[n\,\varepsilon_{\mathrm{xc}}(n)\big]}{\mathrm{d}n}\bigg|_{n=n(\boldsymbol{r})}.$$

The exchange part of $\varepsilon_{\mathrm{xc}}$ is Eq. (23.6), $\varepsilon_{\mathrm{x}}(n) = -3e^2k_{\mathrm{F}}/16\pi^2$ with the local $k_{\mathrm{F}} = (3\pi^2 n)^{1/3}$, which follows from Eq. (2.71). Since $n\,\varepsilon_{\mathrm{x}} \propto n^{4/3}$, the exchange potential is

$$v_{\mathrm{x}}^{\mathrm{LDA}} = \frac{4}{3}\,\varepsilon_{\mathrm{x}} = -\frac{e^2k_{\mathrm{F}}}{4\pi^2} = \Sigma_{\mathrm{x}}(k_{\mathrm{F}}),$$

the Hartree–Fock exchange of an electron at the Fermi surface of the gas, Eq. (23.5) at $F_{\mathrm{L}}(1) = \frac{1}{2}$. At the valence density of silicon, $v_{\mathrm{x}}^{\mathrm{LDA}} = -8.29\,\mathrm{eV}$. Unlike the Fock term, the local exchange potential does not cancel the self-repulsion of an electron in $v_{\mathrm{H}}$ exactly, and the residual self-interaction is one source of the error of the local-density approximation in localized states [10].

## 23.3 Kohn–Sham calculations in practice

A density-functional calculation of silicon at its measured lattice constant returns, in seconds on a few processor cores, a total energy, an electron density and a set of Kohn–Sham bands with an indirect gap near 0.6 eV, against the 1.17 eV measured at low temperature [11, Eq. (11)]. The work of a calculation lies in knowing which outputs to trust, how far, and what each numerical parameter does to them. The calculation of silicon used here has the PBE functional, the norm-conserving silicon pseudopotential of the SG15 library [12] with four valence electrons per atom, and the diamond structure with two atoms per cell.

### 23.3.1 Computed quantities

The Kohn–Sham equations Eq. (3.62) are solved for the periodic parts $u_{n\boldsymbol{k}}$ of the Bloch states on a grid of crystal momenta. The states define the density

$$n(\boldsymbol{r}) = \frac{2}{n_k}\sum_{\boldsymbol{k}}\sum_{n} f_{n\boldsymbol{k}}\,|u_{n\boldsymbol{k}}(\boldsymbol{r})|^2, \tag{23.10}$$

in which the factor 2 counts spin, $n_k$ is the number of grid points and $f_{n\boldsymbol{k}}$ are the occupations. Since $u_{n\boldsymbol{k}}$ is normalized over the cell (Appendix A.5), $n$ integrates over $\Omega$ to the number of valence electrons per cell. The density fixes the Kohn–Sham potential $V_{\mathrm{KS}}$, the potential fixes the states, and the loop is iterated until input and output agree. The outputs deserve different levels of trust: ground-state quantities, eigenvalues and states, and the responses built from both. The total energy, the density, and the forces and stresses derived from the energy are ground-state quantities, exact in principle for the exact functional (Sect. 3.4); their error is that of the approximate functional and of the numerics. The eigenvalues $\varepsilon_{n\boldsymbol{k}}$ and states $\psi_{n\boldsymbol{k}}$ are auxiliary. Apart from the highest occupied level, they have no exact physical meaning, and the band gap they give is too small even for the exact functional (Eq. (3.65)). The response quantities built from the states, such as $\varepsilon_\infty$, the phonons or the density vertex $\rho_{mn}(\boldsymbol{k},\boldsymbol{q})$, inherit the errors of both.

Table 23.1: Lattice constants and band gaps of silicon, germanium and gallium arsenide for three functionals, without spin–orbit coupling. Lattice constants from LDA and PBE are all-electron values from Ref. [17, Table II], those from HSE from Ref. [18, Table I]. Gaps are evaluated at each functional's own relaxed lattice constant, from Ref. [18, Table I], whose LDA lattice constant of silicon is 5.410 Å; a zero gap means that the functional makes the crystal a metal. The static-lattice constants are the measured ones extrapolated to zero temperature with the zero-point anharmonic expansion removed [17, Table II]; they are not measured at any temperature, and they lie below the lattice constants measured at low temperature. They are used only inside calculations, as the cell of a static lattice. The room-temperature lattice constants are measured: silicon [19, lattice parameter of silicon], germanium [20, Sect. II A] and GaAs [21]; they are the values for observables and the cell of the shared lattice setting of Sect. 30.1. The measured gaps are low-temperature values [18, Table I].

| | lattice constant (Å) | | | | | gap (eV) | | |
|---|---|---|---|---|---|---|---|---|
| | LDA | PBE | HSE | static lattice | room $T$ (expt.) | LDA | HSE | measured |
| Si | 5.403 | 5.466 | 5.451 | 5.416 | 5.431 | 0.51 | 1.22 | 1.17 |
| Ge | 5.623 | 5.759 | 5.706 | 5.640 | 5.658 | 0.00 | 0.54 | 0.74 |
| GaAs | 5.605 | 5.746 | 5.711 | 5.638 | 5.653 | 0.43 | 1.18 | 1.52 |

### 23.3.2 Functionals

A calculation chooses one of the approximations to the exchange–correlation energy $E_{\mathrm{xc}}$ of Sect. 3.4, and the choice sets the accuracy of everything downstream. The gradient dependence of PBE [13] is on $|\nabla n|/n^{4/3}$, built to satisfy exact constraints of the electron gas. In the screened hybrid HSE [14] the Coulomb kernel $e^2/4\pi r$ is split, with an error function, into a part that decays on a length of a few Å and a long-range remainder, and the exact exchange acts with the short-range part only. The screening reflects what happens in a semiconductor, where the exchange interaction between distant electrons is screened by the others (Sect. 4.8).

The trends are systematic. LDA binds too strongly and gives lattice constants that are slightly too small, by 0.2 % to 0.6 % for the materials of Table 23.1; PBE binds too weakly and gives them too large, by 0.9 % to 2.1 %. Over the elemental crystals, PBE equilibrium volumes are too large by 3.6 % on average [15, Table IV], about 1.2 % in the lattice constant. The semilocal functionals underestimate gaps, LDA and PBE alike, and close the small gap of germanium altogether; HSE recovers most of the gap. Against the measured gaps of 41 semiconductors and insulators, HSE has a mean absolute error of 0.38 eV, falling to 0.18 eV for the typical semiconductors of the set, whose gaps are measured at low temperature [16, Table I and text]. Table 23.1 collects the numbers for silicon, germanium and GaAs. In silicon the hybrid lattice constant is 0.6 % too large and its gap 0.05 eV too large. In germanium, where LDA gives a metal, the hybrid reopens the gap but leaves it 0.2 eV too small.

The price of a hybrid is the exact exchange. A semilocal potential acts on a state $\psi_{n\boldsymbol{k}}$ by a multiplication in real space. The Fock operator instead needs, for every pair of the state with an occupied state $\psi_{m\boldsymbol{k}'}$, the potential of the pair density $\psi^*_{m\boldsymbol{k}'}\psi_{n\boldsymbol{k}}$. Each such potential is one Poisson solve, a fast Fourier transform (FFT) of the pair density, a division by $|\boldsymbol{k}-\boldsymbol{k}'+\boldsymbol{G}|^2$ and an FFT back. With $N_{\mathrm{occ}}$ occupied bands, applying the Fock operator to all $N_{\mathrm{b}}$ bands at one $\boldsymbol{k}$ therefore costs about $n_k N_{\mathrm{occ}} N_{\mathrm{b}}$ FFTs, against $N_{\mathrm{b}}$ for a semilocal potential, a factor $n_k N_{\mathrm{occ}}$ more. For silicon on an $8^3$ grid this factor is $512 \times 4 \approx 2 \times 10^3$. In practice the inner sum over $\boldsymbol{k}'$ runs over a coarser grid than the outer one, which reduces the factor by one or two orders of magnitude. A hybrid calculation of a two-atom cell therefore costs 10 to $10^3$ times a semilocal one, minutes to hours where PBE takes seconds.

In this book the choice is usually PBE (or LDA) for structures, phonons and electron–phonon couplings, where semilocal functionals are accurate to a few percent. Wherever an excitation energy sets a threshold, a gap correction, a hybrid functional or a quasiparticle calculation is used instead. The two choices do not mix freely: a phonon computed with PBE and a threshold computed with HSE describe two slightly different crystals, and the lattice constant at which both are evaluated must be stated.

### 23.3.3 Plane waves and the cutoff

The periodic part of each Bloch state is expanded in plane waves,

$$u_{n\boldsymbol{k}}(\boldsymbol{r}) = \Omega^{-1/2} \sum_{\boldsymbol{G}} c_{n\boldsymbol{k}}(\boldsymbol{G})\, \mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}, \qquad \frac{|\boldsymbol{k}+\boldsymbol{G}|^2}{2m_e} \le E_{\mathrm{cut}},$$

truncated at the kinetic-energy cutoff $E_{\mathrm{cut}}$, the one parameter of the basis. Codes quote $E_{\mathrm{cut}}$ in rydberg ($1\,\mathrm{Ry} = 13.606\,\mathrm{eV}$) or hartree (two rydberg). The number of plane waves follows from counting reciprocal-lattice points in a sphere of radius $k_{\mathrm{cut}} = (2m_e E_{\mathrm{cut}})^{1/2}$. Each point occupies a volume $(2\pi)^3/\Omega$ of reciprocal space, so

$$N_{\mathrm{PW}} \simeq \frac{4\pi k_{\mathrm{cut}}^3/3}{(2\pi)^3/\Omega} = \frac{\Omega\, k_{\mathrm{cut}}^3}{6\pi^2}. \tag{23.11}$$

For silicon, $\Omega = 40.05\,\text{Å}^3$ (Sect. 3.1) is 270.3 cubic bohr. At $E_{\mathrm{cut}} = 60\,\mathrm{Ry}$, with $\hbar^2/2m_e = 1\,\mathrm{Ry\,bohr}^2$, $k_{\mathrm{cut}} = \sqrt{60}$ inverse bohr and Eq. (23.11) gives 2121 plane waves per state. The density is a product of two states and has Fourier components up to $2k_{\mathrm{cut}}$, eight times as many, about $1.7 \times 10^4$. These set the size of the FFT grid on which the potential is applied. A component $\boldsymbol{G}$ has integer coordinates $m_i = \boldsymbol{G}\cdot\boldsymbol{a}_i/2\pi$ with $|m_i| \le 2k_{\mathrm{cut}}|\boldsymbol{a}_i|/2\pi$, which is 17.9 for $|\boldsymbol{a}_i| = a/\sqrt{2}$. The indices then run from $-17$ to 17, so the grid needs at least 35 points along each primitive vector, in practice 36, the next size that the FFT handles efficiently, and $36^3 \approx 4.7 \times 10^4$ points in all.

The cutoff is a variational parameter. For a fixed pseudopotential, enlarging the basis can only lower the total energy, so the energy converges from above and monotonically. The change between successive cutoffs estimates the error only once the scan has reached its asymptotic regime: in germanium the change from 50 to 60 Ry, 1.3 meV, is less than half the error that remains at 60 Ry, 3.1 meV. Table 23.2 shows the scan for silicon and germanium. Silicon converges to 1 meV per atom at 50 Ry, and to 0.1 meV at about 60 Ry. Germanium, whose pseudopotential carries the ten 3d electrons in the valence, needs 80 Ry for 0.2 meV, and at 30 Ry it is wrong by almost 3 eV per atom. The pressure behaves differently in the two cases: in silicon it is stable to 0.3 kbar from 30 Ry, while in germanium it is not even of the right sign at 30 Ry. The positive pressures at the measured lattice constant, 24 kbar and 38 kbar, are the PBE tendency to expand the lattice seen from the other side.

Two things go wrong with the cutoff: different quantities converge at different rates, and the stress picks up a Pulay term. Energy differences converge faster than energies, since most of the cutoff error sits near the nuclei and is the same in both structures, whereas absolute quantities, and quantities that probe the region near the nuclei, converge more slowly. The density vertex at large momentum transfer is the extreme case, with the cutoff condition Eq. (24.1). At a fixed cutoff the number of plane waves changes with the cell volume, so the basis is not the same at two lattice constants, and the computed stress carries a spurious negative contribution that vanishes only at convergence. A structural relaxation must therefore be done at a cutoff where the stress, and not only the energy, is converged.

Table 23.2: Convergence of the total energy and pressure with the plane-wave cutoff in the representative PBE calculations of silicon and germanium at their measured room-temperature lattice constants, 5.431 Å and 5.658 Å (Table 23.1). The norm-conserving pseudopotentials carry four (Si) and fourteen (Ge, with 3d) valence electrons, and the grid has $8^3$ points $\boldsymbol{k}$. $\Delta E$ is the total energy per atom minus its value at 120 Ry.

| | Si | | Ge | |
|---|---|---|---|---|
| $E_{\text{cut}}$ (Ry) | $\Delta E$ (meV) | $P$ (kbar) | $\Delta E$ (meV) | $P$ (kbar) |
| 30 | 8.36 | 23.6 | 2880 | −1615 |
| 40 | 2.52 | 23.4 | 11.7 | 27.5 |
| 50 | 0.71 | 23.6 | 4.42 | 37.6 |
| 60 | 0.14 | 23.7 | 3.10 | 37.2 |
| 70 | 0.02 | 23.7 | 0.49 | 37.7 |
| 80 | 0.01 | 23.7 | 0.22 | 38.0 |
| 100 | 0.00 | 23.7 | 0.04 | 38.0 |

### 23.3.4 Brillouin-zone sampling

Sums over the zone become averages over a uniform grid of $n_k = n_1 n_2 n_3$ points, possibly shifted from the origin as proposed by Monkhorst and Pack [22]. Appendix C.2 derived the error of such a grid, Eq. (C.15): it is the sum of the real-space Fourier components of the integrand at the superlattice vectors $\boldsymbol{R}$ of the $n_1 \times n_2 \times n_3$ block of cells that the grid implies. For an insulator these decay exponentially, so a modest grid suffices. In the representative silicon calculation at 60 Ry, the total energy changes by 1.6 meV per atom from $8^3$ to $10^3$, by 0.3 meV from $10^3$ to $12^3$ and by 0.06 meV from $12^3$ to $14^3$. Symmetry reduces the work: on a Γ-centered $12^3$ grid, the 48 operations of the diamond structure leave 72 inequivalent points.

The grid needed depends on the quantity, however, and a converged total energy says little about a converged response. The electronic dielectric constant $\varepsilon_\infty$ (Sect. 4.4), computed with density-functional perturbation theory at the PBE lattice constant, shows this. It is 14.36 on an $8^3$ grid, 13.57 on $10^3$, 13.29 on $12^3$ and 13.20 on $14^3$; at the measured lattice constant it reaches 13.04 on $18^3$. The total energy has converged to 1 meV per atom long before $\varepsilon_\infty$ converges to one percent. The reason is that a response involves transitions between occupied and empty states, energy denominators $1/(\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}})$ that are largest near the minimum direct gap, and a grid must resolve the region of the zone where they are largest. The same holds, more severely, for any quantity that probes a small region of the zone, such as the conduction-band minimum of silicon at 0.84 of the way to X or the valley edges of a phonon-assisted window.

### 23.3.5 Metals and smearing

In a metal the occupations jump from one to zero at the Fermi surface. The integrand of Eq. (C.15) then has a discontinuity, its Fourier components decay only as a power of $|\boldsymbol{R}|$, and a grid converges slowly and erratically. The remedy is to replace the step by a smooth function of width $\sigma_{\text{sm}}$, $f_{n\boldsymbol{k}} = \tilde{f}((\mu_{\text{ch}} - \varepsilon_{n\boldsymbol{k}})/\sigma_{\text{sm}})$, which makes the integrand smooth on the scale of the grid. The smearing changes the answer, and the change must be controlled. The cleanest case is Fermi–Dirac smearing, $\tilde{f}(x) = 1/(\mathrm{e}^{-x} + 1)$, where $\sigma_{\text{sm}}$ is a fictitious temperature. The quantity that a self-consistent calculation minimizes is then the free energy $F_{\text{sm}} = E - \sigma_{\text{sm}} S_{\text{sm}}$, with the entropy $S_{\text{sm}}$ of the occupations. Let the density of states be smooth on the scale of $\sigma_{\text{sm}}$, with $\mathcal{N}_{\text{F}}$ per spin and volume at the Fermi level (Sect. 2.5), so $2\Omega\mathcal{N}_{\text{F}}$ per cell. The

Sommerfeld expansion then gives

$$E(\sigma_{\rm sm}) = E(\sigma_{\rm sm}=0) + \frac{\pi^2}{3}\,\Omega\mathcal{N}_{\rm F}\,\sigma_{\rm sm}^2, \qquad F_{\rm sm}(\sigma_{\rm sm}) = E(\sigma_{\rm sm}=0) - \frac{\pi^2}{3}\,\Omega\mathcal{N}_{\rm F}\,\sigma_{\rm sm}^2, \quad (23.12)$$

up to terms of order $\sigma_{\rm sm}^4$, with $E(\sigma_{\rm sm}=0)$ the energy per cell of the unsmeared metal. The first relation is the electronic heat of a metal, with $\sigma_{\rm sm}$ in the role of the temperature; the second follows from the electronic entropy at the same order, $S_{\rm sm} = (2\pi^2/3)\Omega\mathcal{N}_{\rm F}\sigma_{\rm sm}$. The errors of $E$ and $F_{\rm sm}$ are equal and opposite, and the average $\frac{1}{2}(E + F_{\rm sm})$ is correct to order $\sigma_{\rm sm}^4$. The forces, however, are derivatives of $F_{\rm sm}$, not of $E$, so they carry the full $\sigma_{\rm sm}^2$ error.

Other smearing functions remove the $\sigma_{\rm sm}^2$ term from the start. The functions of Methfessel and Paxton [23] are Gaussians multiplied by Hermite polynomials, whose low moments vanish, at the price of occupations that are not confined to $[0,1]$. The "cold smearing" of Marzari and Vanderbilt [24, Eq. (1)],

$$\frac{\mathrm{d}\tilde{f}}{\mathrm{d}x} = \frac{1}{\sqrt{\pi}}\,\exp\Big[-\Big(x - \frac{1}{\sqrt{2}}\Big)^2\Big]\,(2 - \sqrt{2}\,x), \quad (23.13)$$

is the broadening that replaces the delta function. The occupations it integrates to are never negative, although they can exceed one, and it is chosen so that the free energy has no quadratic dependence on $\sigma_{\rm sm}$, and neither do its derivatives, forces and stresses included [24]. A width of a few tenths of an eV is common for a structural calculation of a simple metal. Such a smearing must not be confused with a physical temperature. For a quantity that depends on the occupations near the Fermi surface, such as the density of states at $\mu_{\rm ch}$ of aluminum (Sect. 7.2) or a Fermi-surface average of the electron–phonon coupling, the smearing and the grid must be converged together. The tetrahedron method of Appendix C.2 [25] avoids smearing altogether. For an insulator such as silicon, all of this is unnecessary: occupations are fixed at one and zero, and smearing an insulator is an error.

### 23.3.6 Self-consistency

The density that goes into $V_{\rm KS}$ and the density that comes out of the states must agree. A calculation starts from a guess, usually a superposition of atomic densities, and iterates. Feeding the output straight back in diverges in all but the smallest systems, and the reason is screening. Let the input density differ from the self-consistent one by $\delta n_{\rm in}$. Its Hartree potential $v_{\rm C}\,\delta n_{\rm in}$ changes the states, and the output density differs by $\delta n_{\rm out} = R^0 v_{\rm C}\,\delta n_{\rm in}$, with $R^0$ the static independent-particle response of Sect. 4.6 (we drop the exchange–correlation kernel, which does not change the argument). The dielectric matrix of the random-phase approximation is $\varepsilon = 1 - v_{\rm C}R^0$ (Sect. 4.4); the same product in the other order, $\varepsilon^{\rm T} = 1 - R^0 v_{\rm C}$, acts on densities and has the same eigenvalues. The output error is $(1 - \varepsilon^{\rm T})\,\delta n_{\rm in}$. Linear mixing takes the new input as $n_{\rm in} + \beta_{\rm mix}(n_{\rm out} - n_{\rm in})$, and the error then evolves as

$$\delta n_{\rm in}^{(i+1)} = (1 - \beta_{\rm mix}\,\varepsilon^{\rm T})\,\delta n_{\rm in}^{(i)}. \quad (23.14)$$

The iteration converges if $|1 - \beta_{\rm mix}\varepsilon_\lambda| < 1$ for every eigenvalue $\varepsilon_\lambda$ of the static dielectric matrix, that is for $\beta_{\rm mix} < 2/\max_\lambda \varepsilon_\lambda$. In an insulator the largest eigenvalue stays finite as the cell grows, of the order of $\varepsilon_\infty$, and a fixed mixing parameter works. In a metal, the Thomas–Fermi form $\varepsilon(\boldsymbol{G}) = 1 + k_{\rm TF}^2/G^2$ (Sect. 4.4) grows without limit at small $G$, and the smallest $G$ of a cell of length $L$ is $2\pi/L$. A large metallic cell therefore needs $\beta_{\rm mix} \lesssim 2(2\pi/k_{\rm TF}L)^2$, and long-wavelength charge oscillates back and forth between iterations, a failure known as charge sloshing. The cure of Kerker [26] is to mix each Fourier component with $\beta_{\rm mix}G^2/(G^2 + k_{\rm TF}^2)$, with a wavevector of the order of $k_{\rm TF}$ in the denominator, which cancels the $1/G^2$ of the

Thomas–Fermi dielectric function, so that $\beta_{\mathrm{mix}}\varepsilon(\boldsymbol{G}) \approx \beta_{\mathrm{mix}}$ for all $\boldsymbol{G}$. Production codes combine such a preconditioner with a quasi-Newton update that uses the history of previous iterations to build an approximate inverse of $\varepsilon$, the modified Broyden method of Johnson [27] or the closely related Pulay mixing.

The convergence threshold of the loop must match the use of the result. A total energy to 0.1 meV needs a loose threshold, since the energy is variational and its error is quadratic in the density error. A response calculation started from the converged ground state needs a tight one, since the first-order change of the states is computed against the ground-state Hamiltonian, and a residual error in the density appears linearly in the response.

### 23.3.7 Cost

The work of an iteration is dominated by applying the Hamiltonian, local and nonlocal parts, to the $N_{\mathrm{b}}$ bands at each of the $n_{\boldsymbol{k}}^{\mathrm{irr}}$ inequivalent $\boldsymbol{k}$ points, and by keeping those bands orthonormal. Applying the local part of the Hamiltonian to one band takes two FFTs, of order $N_{\mathrm{PW}} \log N_{\mathrm{PW}}$ each. The nonlocal projectors of the pseudopotential take of order $N_{\mathrm{PW}} N_{\mathrm{proj}}$, with $N_{\mathrm{proj}}$ the number of projectors in the cell. Keeping the bands orthonormal, with the diagonalization in the subspace they span, takes of order $N_{\mathrm{b}}^2 N_{\mathrm{PW}} + N_{\mathrm{b}}^3$ [28]. The cost per iteration is therefore

$$t \propto n_{\boldsymbol{k}}^{\mathrm{irr}} \left( N_{\mathrm{b}} N_{\mathrm{PW}} \log N_{\mathrm{PW}} + N_{\mathrm{b}} N_{\mathrm{PW}} N_{\mathrm{proj}} + N_{\mathrm{b}}^2 N_{\mathrm{PW}} \right). \tag{23.15}$$

Each term grows with the size of the cell. The number of bands and of projectors is proportional to the number of atoms $N_{\mathrm{at}}$, and so is $N_{\mathrm{PW}}$ at fixed cutoff, by Eq. (23.11). The last two terms therefore grow as $N_{\mathrm{at}}^3$ and dominate in large cells, while $n_{\boldsymbol{k}}^{\mathrm{irr}}$ falls as $1/N_{\mathrm{at}}$ for a fixed sampling density. Raising the cutoff increases $N_{\mathrm{PW}}$ as $E_{\mathrm{cut}}^{3/2}$, which is why the 3d electrons of germanium, at 80 Ry against 60 Ry, cost a factor $(4/3)^{3/2} \approx 1.5$ in plane waves on top of the larger number of bands. Spin–orbit coupling doubles both the length of each state, now a two-component spinor, and the number of bands, and multiplies the $N_{\mathrm{b}}^2 N_{\mathrm{PW}}$ term by eight. For a two-atom cell all this is small: the self-consistent calculation of silicon at 60 Ry on a $12^3$ grid converges in eight iterations and about four seconds on four cores. Cost becomes a constraint only for supercells, hybrids, the many empty states of a quasiparticle calculation, and the dense grids of response and electron–phonon calculations.

### 23.3.8 Bands and effective masses

Once the density is converged, it is held fixed and the Kohn–Sham Hamiltonian is diagonalized at any set of $\boldsymbol{k}$ points, on lines through the zone for a band structure, or on a fine grid around an extremum. This non-self-consistent step is cheap, because no iteration is needed, and it adds no approximation: the Kohn–Sham Hamiltonian is fixed by the converged density, and once the density is known it can be evaluated at any $\boldsymbol{k}$.

An effective mass is the inverse curvature of a band at an extremum, $(m^{*-1})_{\alpha\beta} = \partial^2 \varepsilon_{n\boldsymbol{k}} / \partial k_\alpha \partial k_\beta$ (Sect. 3.3). It is computed by finite differences, by $\boldsymbol{k}\cdot\boldsymbol{p}$ theory or by Wannier interpolation. The three-point finite-difference formula along a direction,

$$\frac{1}{m^*} \simeq \frac{\varepsilon(k_0 + h) - 2\varepsilon(k_0) + \varepsilon(k_0 - h)}{h^2}, \tag{23.16}$$

has a truncation error $h^2 \varepsilon''''/12$ from the next term of the Taylor series, and an error of about $4\delta\varepsilon/h^2$ if the eigenvalues carry an error $\delta\varepsilon$. A small step amplifies the noise of the eigenvalues, a large step samples the departure from a parabola, and the step must be chosen between the two. A least-squares parabola through several points around the extremum is

more robust than Eq. (23.16), and varying the fitting range tests both errors at once. The $\boldsymbol{k}\cdot\boldsymbol{p}$ route (Sect. 3.3) uses perturbation theory with the computed velocity matrix elements, which gives the full tensor in one calculation at the extremum, provided enough bands are included and the velocities are corrected for the nonlocal pseudopotential (Appendix A.5). Wannier interpolation (Chap. 27) yields the band at any $\boldsymbol{k}$ and its derivatives analytically.

The masses inherit the errors of the functional. In $\boldsymbol{k}\cdot\boldsymbol{p}$ theory the curvature at an extremum comes from the coupling to the other bands at the same $\boldsymbol{k}$, each term weighted by the inverse energy separation. A mass set by a small direct gap, such as the conduction-band mass of GaAs at Γ, therefore follows the gap. A semilocal functional that puts the gap too low gives a mass that is too light, and a method that overestimates the gap gives one that is too heavy. Masses are checked against cyclotron resonance, and gaps against optical absorption or photoemission, before they are used in a rate.

**Worked example 23.1: Silicon from input to bands**

The representative calculation uses the PBE functional, the SG15 norm-conserving pseudopotential with four valence electrons, a cutoff of 60 Ry (816 eV) for the states and 240 Ry for the density, fixed occupations of the four valence bands, a Γ-centered $12^3$ grid (72 points by symmetry), scalar-relativistic potentials without spin–orbit coupling, and a stopping criterion of $10^{-12}$ Ry on the change of the total energy. A relaxation of the lattice at this cutoff gives $a = 5.477$ Å, 0.85 % above the measured room-temperature 5.431 Å (Sect. 3.1; the zero-temperature value of Table 23.1, with the zero-point expansion removed, is 5.416 Å), and 0.2 % above the all-electron PBE value of that table, 5.466 Å. At the relaxed lattice constant the valence-band maximum is the threefold Γ state, and the conduction-band minimum lies on the line from Γ to X. A scan of that line in steps of $0.005 \times 2\pi/a$ puts the minimum at $k_c = (0.840 \pm 0.003) \times 2\pi/a$, against the measured position near 0.85 (Sect. 3.3), and the indirect gap at 0.602 eV, about half of the measured 1.17 eV. The direct gap at Γ is 2.55 eV. At the measured lattice constant, with the same pseudopotential at 40 Ry, where band energies are converged to about 2 meV, the indirect gap is 0.557 eV: the 0.85 % expansion moves it by about 45 meV.

The longitudinal mass of the conduction valley follows from the same scan. A parabola fitted to the nine eigenvalues within $\pm 0.02 \times 2\pi/a$ of the minimum gives $m_{\parallel}^{*} = 0.955\, m_e$; widening the window to 13 and 17 points gives 0.958 and 0.951. The spread, below one percent, is the combined truncation and rounding error of the fit, and $m_{\parallel}^{*} = 0.95\, m_e$ is the PBE value at this lattice constant. The measured value is $0.92\, m_e$ [29]. The PBE mass is 4 % too heavy, while the gap is 49 % too small: the longitudinal mass of the silicon valleys is insensitive to the error of the fundamental gap, and the functional error that matters for this material is the gap itself.

### 23.3.9 Pitfalls

A semilocal functional can get the order of the bands wrong, and no later correction of the gap alone repairs that. In germanium, PBE without spin–orbit coupling at the measured lattice constant places the Γ conduction state 0.047 eV above the valence-band maximum and the L state at 0.191 eV, while in experiment the L valley is lowest, at 0.74 eV [18, Table I], and Γ lies above it. At the larger PBE lattice constant the gap closes and the crystal becomes a metal. A rigid upward shift of the conduction bands keeps the wrong order, so any germanium band input needs bands that pass the check of Sect. 31.1 (valley order, both gaps, masses at L and Γ), with spin–orbit coupling. A hybrid at the standard exchange fraction inverts Γ

and L even with spin–orbit coupling, and a shift of the 4s-like orbitals fitted to both gaps gets the masses wrong (Sect. 31.1); a quasiparticle calculation is the next route under that check. The spin–orbit splitting of the valence-band maximum is 0.296 eV in germanium against 0.044 eV in silicon (Sect. 3.3). The scalar-relativistic potentials of the worked example omit the splitting altogether; a calculation that includes it needs fully relativistic pseudopotentials and two-component spinors.

The gap error does not stay in the band structure. In germanium the overscreening becomes a failure: the near-metallic PBE state gives a dielectric constant of order $10^4$ that does not converge with the grid. Either a corrected gap or the measured screening is used in such cases.

Smearing hides a metal. A semiconductor computed with smeared occupations, as some codes do by default, converges smoothly even when the functional has closed its gap, and the result looks normal unless the band structure is inspected. With fixed occupations, the same calculation struggles to converge, which is the useful symptom.

## 23.4 Quasiparticle corrections

For an electronic threshold the correction of the Kohn–Sham gap, more than half an eV in silicon (Fig. 3.9), is the largest a calculation makes. The $GW$ self-energy $\Sigma = \mathrm{i}\mathsf{G}W^{\mathrm{scr}}$, with $W^{\mathrm{scr}} = \varepsilon^{-1}v_{\mathrm{C}}$, and the electron–electron lifetime from $\mathrm{Im}\,\Sigma$ are defined in Sect. 4.8. The corrections run from the scissor shift to a one-shot $G_0W_0$ calculation and its self-consistent successors, and each changes the energies, the lifetimes and the matrix elements of a dark-matter rate.

### 23.4.1 Computed quantities

The quantities are the quasiparticle energies $E_{\mathrm{qp}}(n\boldsymbol{k})$, the positions of the poles of the one-particle Green's function $\mathsf{G}$ (Sect. 4.7), with their residues $z_{\mathrm{qp}}$ and their widths. In practice they are computed as corrections to the Kohn–Sham energies, and the Kohn–Sham states are kept. A quasiparticle calculation changes the inputs of every later step (the threshold $E_{\mathrm{min}}(\boldsymbol{q})$, the energy denominators of a response, the screening) but not its approximation level.

### 23.4.2 The scissor shift in practice

The accuracy of the scissor shift Eq. (3.66) in silicon, about 0.1 eV across the bands, is partly fortuitous: the energy dependence of $\Sigma$ and its nonlocality would each make the correction strongly state-dependent, and in silicon the two almost cancel [30, Sect. V B]. The excess of the $f$-sum by $\Delta_{\mathrm{scis}}\sum_c |\rho_{cv}|^2$ is the extra term Eq. (24.13) generated by a nonlocal operator. A scissor-corrected response therefore violates the $f$-sum rule unless the transition strengths are rescaled with the energies, and a response used in an $f$-sum check must be built consistently (Sect. 26.1).

The shift also changes the screening. In the independent-particle response each transition contributes $|\rho_{cv}|^2$ divided by its energy, so with $\rho_{cv}$ fixed and the energies raised, $\varepsilon_\infty - 1$ decreases. The PBE value of silicon, 13.04 against the measured 11.7, moves in the direction of the measurement. Whether it lands above or below 11.7 depends on the rest of the calculation, local fields included, so a scissored screening is not guaranteed to be closer to the measured one.

Shifting a static-lattice calculation to the measured gap and then adding the electron–phonon renormalization of Sects. 13.7 and 25.3 counts it twice; in silicon the zero-point part is about 60 meV (Worked example 25.3), small next to the gap but not against the window

widths of Chap. 13. An optical absorption edge sits below the fundamental gap by the exciton binding energy (Sect. 4.9), and a scissor to the optical edge puts the free-pair continuum too low.

### 23.4.3 One-shot $G_0W_0$

The workhorse for semiconductors is the one-shot $G_0W_0$ method of Hybertsen and Louie [31]: $\mathsf{G}$ and $W^{\mathrm{scr}}$ are built once from the Kohn–Sham states, and the energies are corrected to first order.

The input is a Kohn–Sham calculation, extended to many empty bands, of the order of a hundred per atom (the count is made below). The screening is built from the static independent-particle response $R^0_{\boldsymbol{GG'}}(\boldsymbol{q},\omega=0)$ of Sect. 4.6, a sum over pairs of occupied and empty states of products of the plane-wave matrix elements

$$\rho_{mn}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G}) = \langle \psi_{m,\boldsymbol{k}+\boldsymbol{q}} | \mathrm{e}^{\mathrm{i}(\boldsymbol{q}+\boldsymbol{G})\cdot\boldsymbol{r}} | \psi_{n\boldsymbol{k}} \rangle, \tag{23.17}$$

divided by the energy differences [32, Eqs. (7) and (8)]. These are the density vertices of Sect. 3.2 at the unreduced momentum $\boldsymbol{q}+\boldsymbol{G}$ (in the periodic gauge, $\psi_{m,\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{G}} = \psi_{m,\boldsymbol{k}+\boldsymbol{q}}$). The same objects that enter a dark-matter rate enter the screening of the electron's own interaction, and the large-$q$ caveats of Sect. 24.2 apply to both. From $R^0$ follow the dielectric matrix of the random-phase approximation, $\varepsilon_{\boldsymbol{GG'}} = \delta_{\boldsymbol{GG'}} - v_{\mathrm{C}}(\boldsymbol{q}+\boldsymbol{G}) R^0_{\boldsymbol{GG'}}$ (Sect. 4.6), its inverse, and the screened interaction $W^{\mathrm{scr}}_{\boldsymbol{GG'}}(\boldsymbol{q}) = (\varepsilon^{-1})_{\boldsymbol{GG'}}(\boldsymbol{q})\, v_{\mathrm{C}}(\boldsymbol{q}+\boldsymbol{G'})$ [32, Eq. (12)].

The frequency dependence of $W^{\mathrm{scr}}$ comes from a model or from a frequency grid. In the generalized plasmon-pole model of Hybertsen and Louie, each element of $\mathrm{Im}(\varepsilon^{-1})_{\boldsymbol{GG'}}(\boldsymbol{q},\omega)$ at $\omega>0$ is a single delta function at a frequency $\tilde{\omega}_{\boldsymbol{GG'}}(\boldsymbol{q})$, and the real part follows by the Kramers–Kronig relation. The two parameters of each element, the weight and the frequency of its pole, are fixed by two exact constraints: the static value $(\varepsilon^{-1})_{\boldsymbol{GG'}}(\boldsymbol{q},0)$ just computed, and the $f$-sum rule for $\varepsilon^{-1}$, $\int_0^\infty \mathrm{d}\omega\, \omega\, \mathrm{Im}(\varepsilon^{-1})_{\boldsymbol{GG'}} = -(\pi/2)\,\omega_p^2\,[(\boldsymbol{q}+\boldsymbol{G})\cdot(\boldsymbol{q}+\boldsymbol{G'})/|\boldsymbol{q}+\boldsymbol{G}|^2]\, n(\boldsymbol{G}-\boldsymbol{G'})/n(0)$, with $n(\boldsymbol{G})$ the Fourier components of the density [30, Eqs. (83)–(86)]. No parameter is adjusted. The model works because in sp semiconductors most of the spectral weight of $\mathrm{Im}\,\varepsilon^{-1}$ sits in the plasmon, near the free-electron plasma frequency of the valence electrons, which for the eight valence electrons in the $40.05\,\text{Å}^3$ cell of silicon is $\omega_p = 16.6\,\mathrm{eV}$. The alternative is to compute $\varepsilon^{-1}$ at many frequencies, on the real axis or on the imaginary axis followed by an analytic continuation. The continuation is usually a rational (Padé) fit, and Appendix C.5 explains why such fits are unstable and must be checked against sum rules.

The self-energy is a sum over bands, $\boldsymbol{q}$ and $\boldsymbol{G}$ of products of the matrix elements Eq. (23.17), weighted by $v_{\mathrm{C}}$ and by the frequency structure of $\varepsilon^{-1}$. It is organized as a screened-exchange part, a sum over occupied states only, and a Coulomb-hole part, a sum over all bands, occupied and empty [32, Eqs. (19)–(22)]. Codes evaluate separately the bare exchange $\Sigma_{\mathrm{x}}$, the Fock exchange of Sect. 3.4 built from the occupied Kohn–Sham states, and the correlation part $\Sigma_{\mathrm{c}}$ that $W^{\mathrm{scr}} - v_{\mathrm{C}}$ adds to it.

The quasiparticle equation, which corrects the energies, replaces $v_{\mathrm{xc}}$ in the Kohn–Sham equation by $\Sigma(E)$, evaluated at the quasiparticle energy itself. Keeping only the diagonal matrix elements in the Kohn–Sham basis, which assumes that the Kohn–Sham states are close to the quasiparticle states, and expanding $\Sigma(E)$ to first order about $\varepsilon_{n\boldsymbol{k}}$ gives [30, Eqs. (57) and (58)]

$$E_{\mathrm{qp}}(n\boldsymbol{k}) = \varepsilon_{n\boldsymbol{k}} + z_{\mathrm{qp}}\ \mathrm{Re}\langle \psi_{n\boldsymbol{k}} | \Sigma(\varepsilon_{n\boldsymbol{k}}) - v_{\mathrm{xc}} | \psi_{n\boldsymbol{k}} \rangle, \qquad z_{\mathrm{qp}} = \left[ 1 - \frac{\partial\, \mathrm{Re}\langle \psi_{n\boldsymbol{k}} | \Sigma(\omega) | \psi_{n\boldsymbol{k}} \rangle}{\partial \omega} \right]^{-1}_{\omega=\varepsilon_{n\boldsymbol{k}}}. \tag{23.18}$$

Table 23.3: Minimum band gaps of silicon, germanium and GaAs in LDA, in one-shot *GW* calculations started from LDA, and from experiment, in eV, as collected in Ref. [30, Table I]. The *GW* column gives the range of the independent calculations of Refs. [31, 33–35]. The LDA gap of germanium is negative (the crystal is a metal); for GaAs the LDA gap is the one of Table 23.1, 0.43 eV. The last two columns are the *GW* calculations of Shirley, Zhu and Louie [36], in which the core–valence exchange and correlation are treated with the local-density approximation ("core in LDA") or with a core-polarization potential, as collected in Ref. [30, Table II]: the silicon gap from Γ to the conduction minimum at 0.85 X, and the direct (Γ) and indirect (L) gaps of germanium, with spin–orbit coupling. The measured direct gap of germanium is 0.89 eV [30, Table II].

| | LDA | *GW* | measured | *GW*, core in LDA | *GW*, core polarization |
|---|---|---|---|---|---|
| Si | 0.52 | 1.24–1.29 | 1.17 | 0.85 X 1.29 | 0.85 X 1.16 |
| Ge | $< 0$ | 0.65–0.75 | 0.744 | Γ 0.53, L 0.58 | Γ 0.85, L 0.73 |
| GaAs | – | 1.22–1.58 | 1.52 | – | – |

The residue $z_{\mathrm{qp}}$ is 0.7 to 0.8 in semiconductors [30, Sect. V B]. The slope of $\mathrm{Re}\,\Sigma$ is negative, so $z_{\mathrm{qp}} < 1$ reduces the correction: neglecting the energy dependence of $\Sigma$, as the static Coulomb-hole plus screened-exchange (COHSEX) approximation does, overestimates the gap correction [30, Sect. V B]. The off-diagonal elements, when kept, turn Eq. (23.18) into a small matrix diagonalization in the Kohn–Sham basis [32, Eqs. (17) and (18)], which changes the states as well as the energies.

Table 23.3 shows what these $G_0W_0$ calculations achieve for silicon, germanium and GaAs. In silicon they correct most of the LDA error and overshoot the measured gap by 0.07 to 0.12 eV. In germanium and GaAs the independent calculations scatter by 0.1 and 0.36 eV, a measure of the sensitivity of the method to its numerical and physical choices rather than of its intrinsic accuracy. One such choice is the treatment of the core. The calculation of Shirley, Zhu and Louie [36] includes spin–orbit coupling and resolves both gaps. In it, approximating the exchange and correlation of the valence electrons with the 3d and deeper shells of germanium within LDA puts the direct gap below the indirect one, the wrong order. Treating the core polarization explicitly restores the order and brings both gaps within 0.04 eV of experiment, and it lowers the silicon gap from 1.29 to 1.16 eV (Table 23.3, last two columns). The core electrons must therefore be treated beyond LDA in the self-energy, either in the valence or through a core-polarization model, and not only in the density.

### 23.4.4 Self-consistency and the starting point

A one-shot calculation inherits its starting point. $G_0W_0$ on LDA states underestimates gaps systematically once core states are treated adequately, and it can fail qualitatively in transition-metal compounds [37]. Updating the energies in $\mathsf{G}$ and $W^{\mathrm{scr}}$ until they are self-consistent (eigenvalue self-consistency) removes part of the dependence. The quasiparticle self-consistent *GW* method of van Schilfgaarde, Kotani and Faleev goes further and determines the best static, Hermitian one-particle Hamiltonian from the self-energy itself, so that the result no longer depends on the starting functional [37]. Its errors are small and systematic. It overestimates the gaps of sp semiconductors by about 0.2 eV for transitions between Γ points and 0.1 eV for the others, and it slightly underestimates $\varepsilon_\infty$. The reason given is that the random-phase approximation of $W^{\mathrm{scr}}$ omits the electron–hole attraction, which makes the screening too weak and the gap too large [37]. A starting point that is metallic, LDA or PBE germanium, is the case where the choice matters most: the first $R^0$ then contains

intraband transitions that an insulator does not have, and either a hybrid-functional start or a self-consistent scheme is used.

Current practice differs from the calculations of Table 23.3 in three ways [38, Sects. IV C–F, V and XI D]. The quasiparticle energies converge only as the inverse of the number of plane waves in the basis for the empty states and the response, so that the few hundred empty bands per atom of the early calculations can leave errors of tenths of an eV. Modern calculations extrapolate in $1/N_{\mathrm{PW}}$, or avoid the sum over empty states with a Sternheimer equation or a static remainder; converged in this way, $G_0W_0$ started from LDA gives 1.10 eV for silicon [39, Sect. IV C], below the measured gap rather than above it. The frequency integral is done by contour deformation or on the imaginary axis rather than with a plasmon pole. And the missing electron–hole attraction in $W^{\mathrm{scr}}$ is put back as a vertex correction: adding the ladder diagrams of the Bethe–Salpeter equation (Sect. 4.9) to the polarizability of quasiparticle self-consistent *GW* removes most of its overestimate of the gaps and its underestimate of $\varepsilon_\infty$, and gives 1.08 eV for the silicon gap near X against the measured 1.17 eV [40, Table 1].

### 23.4.5 Lifetimes

The imaginary part of the self-energy gives the width of the quasiparticle. Near the pole, the linearized quasiparticle energy of Eq. (23.18) acquires an imaginary part $z_{\mathrm{qp}} \operatorname{Im}\Sigma$, and the probability that the electron is still in the state decays as $\exp(-2z_{\mathrm{qp}}|\operatorname{Im}\Sigma|\,t)$. The decay rate of the electron, its inverse lifetime, is therefore $2z_{\mathrm{qp}}|\operatorname{Im}\Sigma(E_{\mathrm{qp}}(n\boldsymbol{k}))|$. By the cutting rule of Sect. 4.8, $\operatorname{Im}\Sigma$ in the *GW* approximation is the golden-rule rate of the decay of the electron into a lower state with the emission of an excitation of the screened interaction: an electron–hole pair or a plasmon. An electron in the conduction band of an insulator can create an electron–hole pair only if it has the energy to cross the gap. Its energy above the conduction-band minimum must exceed $E_{\mathrm{g}}$, and the conservation of crystal momentum raises the true threshold above this bound. Below it the electron–electron width vanishes, and only phonon emission (Sect. 6.3) relaxes the electron. Above it, impact ionization opens, and its rate grows with the phase space of the final pairs. These two rates, as functions of energy, are the input of the hot-carrier cascades that set the mean energy per electron–hole pair $\varepsilon_{eh}$ (Sect. 29.3).

The plasmon-pole model, with all the weight of $\operatorname{Im}\varepsilon^{-1}$ in delta functions at the plasmon energies, gives $\operatorname{Im}\Sigma = 0$ except at those energies, and it cannot produce lifetimes [30, Sect. III C]. A lifetime needs the full frequency dependence of $\varepsilon^{-1}$ near the energies of the pairs that are created, which for silicon starts at the gap. A full-frequency calculation costs more than a plasmon-pole one by about the number of frequencies at which $\varepsilon^{-1}$ is computed.

### 23.4.6 Matrix elements

The diagonal approximation of Eq. (23.18) keeps the Kohn–Sham states and changes the energies only, and its effect on matrix elements follows from that alone. The density vertex $\rho_{mn}(\boldsymbol{k},\boldsymbol{q})$ is unchanged. The velocity changes, and its change can be computed exactly within the approximation. For eigenstates of a Hamiltonian $H$, the interband velocity is $\boldsymbol{v}_{mn} = \mathrm{i}\langle m|[H,\boldsymbol{r}]|n\rangle = \mathrm{i}(\varepsilon_m - \varepsilon_n)\langle m|\boldsymbol{r}|n\rangle$ for $m \neq n$. Between Bloch states at the same $\boldsymbol{k}$ the interband position matrix element is well defined, and this relation, applied to the Kohn–Sham Hamiltonian, gives it as $\langle m|\boldsymbol{r}|n\rangle = -\mathrm{i}\boldsymbol{v}^{\mathrm{KS}}_{mn}/(\varepsilon_m - \varepsilon_n)$. A quasiparticle Hamiltonian that is diagonal in the Kohn–Sham basis has the same $\langle m|\boldsymbol{r}|n\rangle$ and the energies $E_{\mathrm{qp}}$, so

$$\boldsymbol{v}^{\mathrm{qp}}_{mn}(\boldsymbol{k}) = \boldsymbol{v}^{\mathrm{KS}}_{mn}(\boldsymbol{k})\,\frac{E_{\mathrm{qp}}(m\boldsymbol{k}) - E_{\mathrm{qp}}(n\boldsymbol{k})}{\varepsilon_{m\boldsymbol{k}} - \varepsilon_{n\boldsymbol{k}}}, \qquad m \neq n. \tag{23.19}$$

For a rigid shift this is the scissor rescaling Eq. (3.67). For a state-dependent correction it rescales each transition by its own factor. The velocity vertex enters the absorption of dark matter at small momentum transfer and the optical limit of $\rho_{mn}$ (Sect. 3.2). Eq. (23.19) keeps the optical limit of the density vertex, $\boldsymbol{q} \cdot \boldsymbol{v}_{mn}/(E_m - E_n)$, equal to that of the uncorrected calculation, as it must be when the states are unchanged. The intraband velocities, the group velocities, follow from the dispersion of the corrected bands. Since the corrections are computed on a coarse grid, they are interpolated to fine grids with Wannier functions (Sect. 27.2), with the same gauge as every other interpolated quantity (Sect. 27.1).

Effective masses change with the dispersion. The $\boldsymbol{k} \cdot \boldsymbol{p}$ argument of Sect. 23.3 connects a mass at a direct gap to the gap, and the quasiparticle self-consistent calculation of GaAs shows the connection: it overestimates the gap, 1.77 eV against 1.52 eV, and the conduction-band mass at $\Gamma$, $0.077\,m_e$ [37, Fig. 2 and text] against the measured $0.067\,m_e$.

### 23.4.7 Convergence and cost

A $G_0W_0$ calculation is converged in the number of empty bands, the dielectric cutoff, the grid of crystal momenta and the frequency treatment, and these parameters are coupled. The number of empty bands $N_\mathrm{b}$ enters the sums for $R^0$ and for $\Sigma_\mathrm{c}$. The dielectric cutoff is the kinetic energy $E^{\varepsilon}_\mathrm{cut}$ below which the vectors $\boldsymbol{q} + \boldsymbol{G}$ of the matrix $\varepsilon_{\boldsymbol{G}\boldsymbol{G}'}$ are kept. The two are not independent. The highest empty band included should have an energy of about the dielectric cutoff, since a plane wave $\boldsymbol{q} + \boldsymbol{G}$ resolves transitions into states up to that energy [32, Sect. 4]. A free-electron count makes the link concrete: each plane wave with kinetic energy below $E$ contributes one band below $E$, so the number of bands below $E$ is about $N_\mathrm{PW}(E)$ of Eq. (23.11). For silicon, a dielectric cutoff of 10 Ry keeps 144 vectors $\boldsymbol{G}$ and asks for about 150 bands; 15 Ry keeps 265 and asks for about 270. The sum for the Coulomb-hole part of $\Sigma_\mathrm{c}$ converges much more slowly with $N_\mathrm{b}$ than the sum for $R^0$. A test that converges $R^0$ alone, or one parameter with the other held fixed, can therefore stop at a value far from the converged one [32, Sect. 4 and Fig. 8]. The grid of $\boldsymbol{k}$ and $\boldsymbol{q}$ points is the same for both. The divergence of $v_\mathrm{C}(\boldsymbol{q})$ at $\boldsymbol{q} \to 0$ is handled by averaging $v_\mathrm{C}$ and $W^\mathrm{scr}$ over the small cell around $\boldsymbol{q} = 0$ and by computing the head and wings of $\varepsilon_{\boldsymbol{G}\boldsymbol{G}'}$ (its $\boldsymbol{G} = \boldsymbol{G}' = 0$ element and the rest of the $\boldsymbol{G} = 0$ row and column) at $\boldsymbol{q} \to 0$ from the optical limit of the vertex. The frequency treatment is either a plasmon pole or a frequency grid. The bare exchange $\Sigma_\mathrm{x}$ is a separate case: it involves the unscreened interaction, whose $1/|\boldsymbol{q} + \boldsymbol{G}|^2$ falls off slowly, and its sum over $\boldsymbol{G}$ runs to the cutoff of the pair densities rather than to the dielectric cutoff.

The cost follows from the structure of the sums [32, Sect. 4]. The number of occupied and of empty bands, and the number of plane waves, each grow in proportion to the number of atoms $N_\mathrm{at}$. Computing the matrix elements Eq. (23.17) for all pairs of bands is $N_\mathrm{at}^2$ pairs times an FFT of size $N_\mathrm{at} \log N_\mathrm{at}$, so $N_\mathrm{at}^3 \log N_\mathrm{at}$. Summing them into $R^0_{\boldsymbol{G}\boldsymbol{G}'}$ is $N_\mathrm{at}^2$ pairs for each of $N_\mathrm{at}^2$ pairs of $\boldsymbol{G}$ vectors, so $N_\mathrm{at}^4$, done as one dense matrix product; the inversion of $\varepsilon$ is $N_\mathrm{at}^3$ per $\boldsymbol{q}$ and per frequency. The self-energy of one state is $N_\mathrm{at}^3$, and all states in an energy window add a factor $N_\mathrm{at}$. A $G_0W_0$ calculation thus scales formally as the fourth power of the system size, against the third power of Eq. (23.15), with a prefactor set by the hundreds of empty bands. For the two-atom cell of silicon all of this remains modest. For the supercells of defects and for the dense grids of a finite-momentum exciton calculation, it is the step that limits what can be computed.

**Worked example 23.2: The silicon gap from three corrections**

A scissor, a hybrid functional and a $G_0W_0$ calculation each lead from the PBE calculation of Worked example 23.1 to a gap for a silicon threshold. The scissor to the measured

gap at zero temperature uses $\Delta_{\text{scis}} = 0.568\,\text{eV}$ at the PBE lattice constant, or 0.613 eV at the measured one. It reproduces the measured gap by construction, includes the zero-point renormalization through the measurement, and is accurate to about 0.1 eV for other bands [33]. The hybrid functional HSE gives 1.22 eV at its own lattice constant (Table 23.1). A one-shot $G_0W_0$ calculation from LDA gives 1.10 eV converged, from a static lattice. Two corrections separate it from the measurement: spin–orbit coupling, which these calculations omit, and the zero-point renormalization. Spin–orbit coupling splits the sixfold valence-band maximum (with spin) into a fourfold level at $+\Delta_{\text{SO}}/3$ and a twofold level at $-2\Delta_{\text{SO}}/3$; the weighted mean, $4 \times \Delta_{\text{SO}}/3 - 2 \times 2\Delta_{\text{SO}}/3 = 0$, is the level without spin–orbit coupling. The gap therefore closes by $\Delta_{\text{SO}}/3$, 15 meV in silicon and, with $\Delta_{\text{SO}} = 0.296\,\text{eV}$, 99 meV in germanium (Sect. 3.3). The zero-point renormalization lowers the gap of a real crystal relative to the static lattice by about 60 meV. For a dark-matter threshold in silicon, the three agree to about 0.1 eV. That spread is the theory uncertainty of the threshold if the scissor is not anchored to a measurement, and Worked example 3.7 gives what it does to the minimum mass.

What a quasiparticle calculation does not include, the interaction of the excited electron with its hole, is the subject of the Bethe–Salpeter equation of Sect. 4.9 and of Sect. 26.1.

## Summary

The Hartree–Fock equations are the minimum condition for the energy of a Slater determinant, and in a crystal Koopmans' theorem makes their bands the removal and addition energies of the Hartree–Fock state; in the electron gas they take closed form. The theorems of Hohenberg and Kohn and the Kohn–Sham equations of Sect. 3.4 follow from the constrained search of Levy. A measured gap already contains the zero-point renormalization, and a scissor to it must not add that renormalization a second time.

Every numerical parameter of the calculation is converged on the quantity that is used, not on the total energy: the $8^3$ grid that converges the energy of silicon to 2 meV per atom leaves $\varepsilon_\infty$ nine percent too high. A scissor, the hybrid functional HSE and a one-shot $G_0W_0$ calculation give the silicon gap to about 0.1 eV of one another. That spread, and the screening, $\varepsilon_\infty$ of 13.0 computed against 11.7 measured, are the theory uncertainties that this chapter leaves in an electronic rate of silicon.

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# Chapter 24

# Pseudopotentials and core states

A momentum transfer of 30 keV resolves the electron density on a length scale $1/q = 0.066$ Å. That is smaller than the $2p$ shell of silicon, whose radial density peaks 0.21 Å from the nucleus, and it is the scale on which the valence states of silicon change sign twice to stay orthogonal to the core. Dark matter of a few MeV reaches such momentum transfers, and a heavy mediator, unlike a light one, does not suppress them against the few-keV momentum transfers of Sect. 1.3. The density vertex $\rho_{mn}(\boldsymbol{k}, \boldsymbol{q})$ of Eq. (3.34) at large $q$ is therefore a statement about the wavefunctions within a fraction of an angstrom of the nuclei. The calculations of Chap. 23, on the other hand, are built to describe the bonds between the atoms, and they do so by throwing that region away.

## 24.1 Plane waves and the core region

A plane-wave code stores each periodic part as the Fourier series Eq. (3.22), truncated at the kinetic-energy cutoff of Sect. 23.3, $|\boldsymbol{k} + \boldsymbol{G}| \leq k_{\mathrm{cut}} = (2m_e E_{\mathrm{cut}})^{1/2}$. The basis treats all regions of the cell alike and improves systematically as the cutoff is raised. In Eq. (3.23) the kinetic energy is diagonal in $\boldsymbol{G}$, while the potential is diagonal in $\boldsymbol{r}$, and a fast Fourier transform between a grid of $\boldsymbol{G}$ and a grid of points in the cell applies each where it is diagonal. This is what makes plane waves the standard basis for crystals.

### 24.1.1 The cutoff and the vertex

The price is paid near the nuclei. A state confined to a region of size $r_0$ has momentum components up to a few times $1/r_0$, and the innermost orbitals of an atom are small. A hydrogen-like $1s$ orbital of nuclear charge $Z$, $(p_0^3/\pi)^{1/2}\mathrm{e}^{-p_0 r}$ with $p_0 = Z\alpha m_e$, has the radius $1/p_0$. Its Fourier transform follows from $\int \mathrm{d}^3 r\, \mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{r}}\mathrm{e}^{-p_0 r} = 8\pi p_0/(p^2+p_0^2)^2$, and its momentum density, normalized to $\int \mathrm{d}^3 p = 1$, is $(8/\pi^2)\, p_0^5/(p^2+p_0^2)^4$. For silicon, $p_0 = 14 \times 3.73\,\mathrm{keV} = 52\,\mathrm{keV}$. Worse, the density falls as slowly as $p^{-8}$, a power law that comes from the cusp of the orbital at the nucleus, and so do the squared coefficients $|c_{n\boldsymbol{k}}(\boldsymbol{G})|^2$ of any state that has weight there. The valence states inherit a version of this tail. A $3s$ electron of silicon must be orthogonal to the $1s$ and $2s$ orbitals, and it does this by changing sign at 0.08 and 0.38 Å from the nucleus; a $3p$ electron changes sign at 0.38 Å to stay orthogonal to $2p$. These oscillations, and the cusp, are what a momentum transfer of tens of keV sees. To resolve the silicon $1s$ orbital alone, a plane-wave basis needs wavevectors up to a few times $Z/a_0$. With $3Z/a_0 = 42$ inverse bohr, Eq. (23.11) asks for a cutoff near 1800 Ry and $3 \times 10^5$ plane waves per state, 160 times the 2121 of the converged calculation of silicon, and the cost of each band grows at least in proportion.

The vertex in plane waves is the sum Eq. (3.35). A term contributes only if both coefficients exist: the component $\boldsymbol{k}+\boldsymbol{G}$ of the initial state and the component $\boldsymbol{k}+\boldsymbol{G}+\boldsymbol{q}$ of the final state must both lie inside the cutoff sphere. Two points of a sphere of radius $k_{\mathrm{cut}}$ are at most $2k_{\mathrm{cut}}$ apart, and the computed vertex vanishes identically for $q > 2k_{\mathrm{cut}}$, which at 60 Ry is 58 keV. This bound would be reached if the two states had their weight at opposite ends of the sphere, and the kinematics of dark matter keeps them from it. By Eq. (1.22), a momentum transfer $q$ deposits at most $\omega = qv_{\max}$, 80 eV at $q = 30$ keV, far below the free-electron energy $q^2/2m_e = 0.88$ keV at that momentum. The final state is a low-lying conduction state. Its plane-wave components are concentrated at momenta below a scale $p_f$ set by its energy above the bottom of the valence band, $p_f \simeq [2m_e(\omega + E_{\mathrm{F}})]^{1/2}$ with $E_{\mathrm{F}} = 12.5$ eV the free-electron value at the valence density (Sect. 2.5). In Eq. (3.35) the final-state coefficient at $\boldsymbol{k}+\boldsymbol{G}+\boldsymbol{q}$ is then appreciable for $|\boldsymbol{k}+\boldsymbol{G}+\boldsymbol{q}| \lesssim p_f$, and the initial state must supply the momentum, through its components at $|\boldsymbol{k}+\boldsymbol{G}| \simeq q$. The same holds with the two states exchanged. The vertex at momentum transfers up to $q_{\max}$ is converged once the basis contains these components,

$$k_{\mathrm{cut}} \gtrsim q_{\max} + p_f, \qquad E_{\mathrm{cut}} \gtrsim \frac{(q_{\max}+p_f)^2}{2m_e}. \tag{24.1}$$

The large-$q$ vertex is the overlap of the high-momentum tail of one state with the low-momentum body of the other, as Sect. 3.2 anticipated. A cutoff converged for total energies says nothing about whether that tail is in the basis, and Eq. (24.1) has to be checked separately for every rate calculation. Whether the tail is physical is another matter, which depends on the potential the states were computed in.

### 24.1.2 Pseudopotentials

The core electrons of silicon, $1s^2 2s^2 2p^6$, are bound by 99 eV and more and take no part in bonding; their orbitals hardly change between the atom and the crystal. The valence electrons, however, feel them through orthogonality. A $3s$ state must be orthogonal to $1s$ and $2s$, and its two nodes inside 0.4 Å carry a large kinetic energy there. That kinetic energy acts like a repulsion that nearly cancels the attraction of the nucleus inside the core, and outside the core the valence electron sees a nucleus screened by ten electrons. Phillips and Kleinman made this precise: subtracting the core states from a valence state leaves a smooth function that obeys a Schrödinger equation with a weak, repulsive correction added to the potential inside the core [1]. The cancellation is why the nearly-free-electron picture of Sect. 3.2 works for silicon at all, and why the empirical pseudopotential of Sect. 3.3 needs only three Fourier components.

A *pseudopotential* turns the cancellation into a construction. It replaces the nucleus and the core electrons by an effective potential whose valence states are smooth and nodeless inside a *core radius* $r_c$ and identical to the true, all-electron states outside it. The core states are dropped from the calculation, and with them the oscillations of the valence states that kept them orthogonal. The construction starts from an all-electron calculation of the atom and is carried out for each angular momentum $l$ separately. Hamann, Schlüter and Chiang formulated the conditions that make the result useful [2]. Two of them do the work. For a reference configuration of the atom, the pseudo and all-electron valence states must have the same eigenvalue and agree beyond $r_c$, and they must hold the same charge inside it. Their logarithmic derivatives at $r_c$ then agree, and so do the first energy derivatives of these. The charge condition, *norm conservation*, is the one with content, and it implies the last.

To see this, write the radial Schrödinger equation for $u_l(r;\varepsilon) = rR_l(r)$ at energy $\varepsilon$ in a

potential $V_l(r)$, which may differ from one $l$ to the next,

$$-\frac{1}{2m_e}\,u_l'' + \Big[V_l(r) + \frac{l(l+1)}{2m_e r^2} - \varepsilon\Big]u_l = 0, \tag{24.2}$$

with primes for $\partial/\partial r$, and differentiate it with respect to $\varepsilon$. The derivative $\dot{u}_l = \partial u_l/\partial\varepsilon$ obeys the same equation with the source $u_l$ on the right-hand side. Multiplying the equation for $u_l$ by $\dot{u}_l$, the equation for $\dot{u}_l$ by $u_l$, and subtracting, the potential drops out and leaves $(\dot{u}_l u_l' - u_l \dot{u}_l')' = 2m_e u_l^2$. Both functions vanish at the origin, and integrating to $r_c$,

$$\frac{\partial}{\partial\varepsilon}\,\frac{u_l'(r_c)}{u_l(r_c)} = \frac{\dot{u}_l' u_l - u_l' \dot{u}_l}{u_l^2}\bigg|_{r_c} = -\frac{2m_e}{u_l(r_c)^2}\int_0^{r_c} \mathrm{d}r\, u_l^2. \tag{24.3}$$

The logarithmic derivative at $r_c$ fixes how a wave arriving from outside is scattered by everything inside, and Eq. (24.3) ties its energy dependence to the charge inside the sphere. A pseudo-state that matches the all-electron state in value and slope at $r_c$ at the reference energy, and has the same charge inside, therefore scatters like the true atom to first order in the energy away from the reference. A potential fitted to the free atom then works in the crystal, where the valence energies are shifted by the bonding. Two or more reference energies per channel extend the range of energies over which it works. Vanderbilt introduced them in his ultrasoft scheme, which relaxes the norm-conservation condition of Hamann, Schlüter and Chiang and restores the missing charge with augmentation functions [3]. Hamann's optimized potentials keep norm conservation in a generalized form: the overlaps $\int_0^{r_c} \mathrm{d}r\, u_i u_j$ of the pseudo and all-electron reference states agree for every pair $i, j$ [4, Eq. (23)].

The potential this produces is *nonlocal*. Orthogonality acts only within one angular momentum: a $3s$ electron is pushed out by the $1s$ and $2s$ cores, a $3p$ electron by $2p$, and a $3d$ electron by nothing, since silicon has no $d$ core. The effective potential is then different in each channel, $\sum_l V_l(r) P_l$ with $P_l$ the projector on angular momentum $l$ about the nucleus, and in practice it is cast into a local part plus a few separable projections, the form Eq. (24.9) of Kleinman and Bylander [5]. A nonlocal potential does not commute with $\boldsymbol{r}$, and we have already met two consequences: the velocity is no longer $\boldsymbol{p}/m_e$ (Sect. 3.2), and the $f$-sum rule acquires a correction (Sect. 4.3). The density vertex needs none, since $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ couples to the pseudo-states by multiplication, as in any Hamiltonian; its only error is the missing core region.

The core radius sets the trade-off. A large $r_c$ makes the pseudo-states smooth and the cutoff low, but asks one potential to describe a large part of the atom, and it transfers less well to other environments. The cutoff is decided by the Fourier tail of the pseudo-states, and the optimized construction of Hamann chooses the shape inside $r_c$ to minimize the kinetic energy above a wavevector $q_{\mathrm{opt}}$ (Eq. (24.10)). In practice, the potentials of the plane-wave calculations of this book come from the SG15 library [6]. The silicon potential keeps the four $3s$ and $3p$ electrons, with $r_c = 1.27\,\text{Å}$ for $s$ and $1.89\,\text{Å}$ for $p$, and $q_{\mathrm{opt}} = 19$ and $13\,\mathrm{keV}$. With it, the total energy converges to $0.1\,\mathrm{meV}$ per atom at about $60\,\mathrm{Ry}$ (Table 23.2).

Fig. 24.1(a) shows what is lost, for a norm-conserving silicon potential built with the SG15 core radii. The pseudo-orbitals agree with the all-electron ones beyond $r_c$ and are smooth and nodeless inside, where the true orbitals oscillate; the charge inside $r_c$ is the same, 0.662 of the $3s$ electron and 0.767 of the $3p$ electron. In momentum space, panel (b), the two agree to 10 % below $2\,\mathrm{keV}$ for $3s$ and $5\,\mathrm{keV}$ for $3p$, where most of the weight is. Above, the pseudo-orbitals fall off much faster. Above $10\,\mathrm{keV}$ the all-electron $3s$ orbital keeps 7 % of its weight and the pseudo-orbital 0.14 %; at $30\,\mathrm{keV}$ the all-electron momentum densities exceed the pseudo ones by more than four orders of magnitude. A pseudopotential calculation converged in $E_{\mathrm{cut}}$ converges to the pseudo answer, and at large $q$ the pseudo answer is wrong.

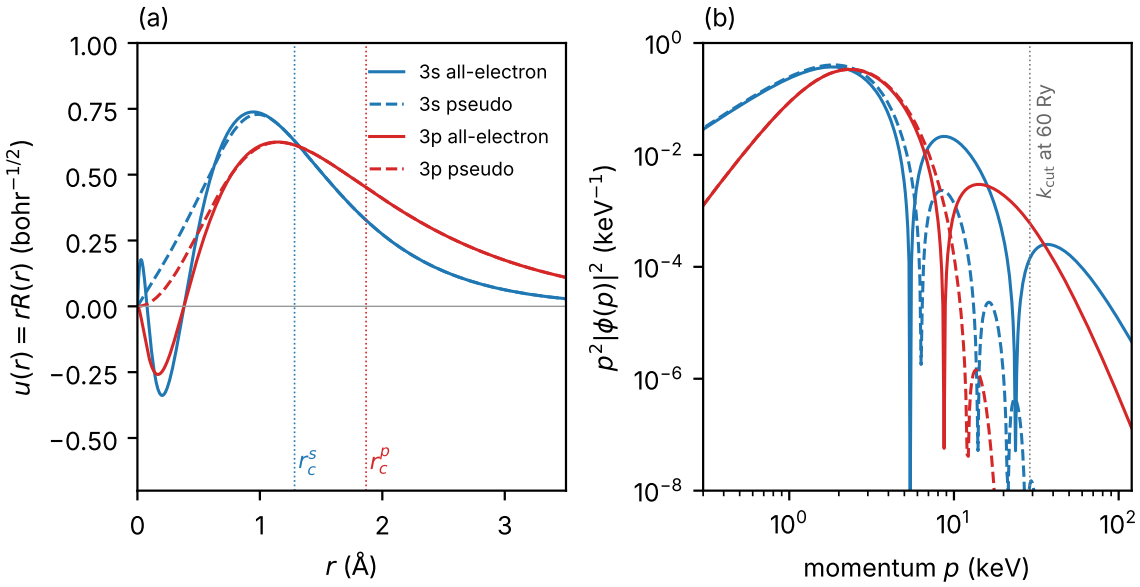



Figure 24.1: The valence orbitals of the silicon atom near the nucleus, all-electron (solid) and pseudo (dashed), for $3s$ (blue) and $3p$ (red). (a) Radial functions $u = rR$. The all-electron orbitals change sign at 0.08 and 0.38 Å ($3s$) and at 0.38 Å ($3p$) to stay orthogonal to the core; the pseudo-orbitals are nodeless inside the core radii $r_c$ (dotted). Outside them they equal the all-electron ones, and they carry the same charge inside. (b) Momentum distributions $p^2|\phi(p)|^2$, normalized to $\int \mathrm{d}p\, p^2|\phi|^2 = 1$; dotted, the plane-wave cutoff $k_{\mathrm{cut}} = 28.9\,\mathrm{keV}$ at which the total energy of silicon converges (60 Ry). Free atom, all-electron calculation with the PBE functional, scalar-relativistic (mass–velocity and Darwin terms kept, spin–orbit coupling dropped), configuration [Ne] $3s^2 3p^2$. The pseudopotential is a single-projector norm-conserving potential of the Troullier–Martins form [7], generated for this figure at the core radii of the SG15 potential of the plane-wave calculations, 1.27 and 1.89 Å for $s$ and $p$, to within 0.02 Å; it reproduces the all-electron eigenvalues. The SG15 potential has the same radii but a different shape inside them.

### 24.1.3 Projector augmented waves

The core region that the pseudopotential discards can be restored. Outside the core spheres the pseudo-state is the true state, and inside each sphere the true state is fixed, up to a few coefficients, by the atom from which the pseudopotential was built. Blöchl's *projector-augmented-wave* (PAW) method makes the map from one to the other explicit and linear [8]. For each atom it uses three sets of functions, labeled by an index $j$ that collects the site, the angular momentum $lm$ and a reference energy. The *all-electron partial waves* $\phi_j^{\mathrm{AE}}$ are solutions of the radial equation of the atom at the reference energies. The *pseudo partial waves* $\phi_j^{\mathrm{PS}}$ are smooth functions that coincide with them outside the sphere. The *projectors* $\tilde{p}_j$ are localized inside the sphere and dual to the pseudo partial waves, $\langle \tilde{p}_i | \phi_j^{\mathrm{PS}} \rangle = \delta_{ij}$. The pseudo partial waves are assumed complete for smooth functions inside the sphere, $\sum_j |\phi_j^{\mathrm{PS}}\rangle\langle \tilde{p}_j| = 1$ there. The all-electron state is then

$$|\psi_{n\boldsymbol{k}}\rangle = |\tilde{\psi}_{n\boldsymbol{k}}\rangle + \sum_j \big(|\phi_j^{\mathrm{AE}}\rangle - |\phi_j^{\mathrm{PS}}\rangle\big)\langle \tilde{p}_j | \tilde{\psi}_{n\boldsymbol{k}}\rangle, \tag{24.4}$$

where $\tilde{\psi}_{n\boldsymbol{k}}$ is the smooth pseudo-state of the calculation. Outside the spheres the bracket vanishes and $\psi = \tilde{\psi}$. Inside a sphere, completeness lets us write the pseudo-state as its one-center expansion $\tilde{\psi}^1 = \sum_j \phi_j^{\mathrm{PS}} \langle \tilde{p}_j | \tilde{\psi}\rangle$, so that Eq. (24.4) subtracts it and puts in its place $\psi^1 = \sum_j \phi_j^{\mathrm{AE}} \langle \tilde{p}_j | \tilde{\psi}\rangle$, the same combination of all-electron partial waves. The nodes and the cusp come back with the $\phi_j^{\mathrm{AE}}$.

We can compute matrix elements without ever building $\psi$ on a fine grid. Write $\psi = \tilde{\psi} + \psi^1 - \tilde{\psi}^1$ for both states, and take an operator $A$ that is local, so that it does not connect the inside of a sphere with the outside. In a cross term such as $\langle \tilde{\psi}_m | A | \psi_n^1 - \tilde{\psi}_n^1 \rangle$ the difference lives inside the sphere, where $\tilde{\psi}_m = \tilde{\psi}_m^1$, and the cross term equals $\langle \tilde{\psi}_m^1 | A | \psi_n^1 - \tilde{\psi}_n^1 \rangle$. Doing the same in the other cross term, the three terms that contain a difference combine into $\langle \psi_m^1 | A | \psi_n^1 \rangle - \langle \tilde{\psi}_m^1 | A | \tilde{\psi}_n^1 \rangle$, and

$$\langle \psi_m | A | \psi_n \rangle = \langle \tilde{\psi}_m | A | \tilde{\psi}_n \rangle + \sum_{ij} \langle \tilde{\psi}_m | \tilde{p}_i \rangle \big( \langle \phi_i^{\mathrm{AE}} | A | \phi_j^{\mathrm{AE}} \rangle - \langle \phi_i^{\mathrm{PS}} | A | \phi_j^{\mathrm{PS}} \rangle \big) \langle \tilde{p}_j | \tilde{\psi}_n \rangle, \tag{24.5}$$

with the partial-wave matrix elements taken inside the spheres. The smooth part is computed in plane waves at a modest cutoff. The rest is a sum of one-center terms, computed once for each species on a radial grid fine enough for the cusp.

**All-electron density vertex**

With $A = \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ and $\boldsymbol{r} = \boldsymbol{\tau}_\kappa + \boldsymbol{x}$ in the sphere of atom $\kappa$ of the cell, Eq. (24.5) gives the vertex of Eq. (3.34) between all-electron states as

$$\rho_{mn}(\boldsymbol{k},\boldsymbol{q}) = \langle \tilde{\psi}_{m,\boldsymbol{k}+\boldsymbol{q}} | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} | \tilde{\psi}_{n\boldsymbol{k}} \rangle + \sum_\kappa \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa} \sum_{ij} \langle \tilde{\psi}_{m,\boldsymbol{k}+\boldsymbol{q}} | \tilde{p}_{\kappa i} \rangle \, \Delta_{\kappa,ij}(\boldsymbol{q}) \, \langle \tilde{p}_{\kappa j} | \tilde{\psi}_{n\boldsymbol{k}} \rangle, \tag{24.6}$$

with the one-center matrix

$$\Delta_{\kappa,ij}(\boldsymbol{q}) = \langle \phi_{\kappa i}^{\mathrm{AE}} | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}} | \phi_{\kappa j}^{\mathrm{AE}} \rangle - \langle \phi_{\kappa i}^{\mathrm{PS}} | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}} | \phi_{\kappa j}^{\mathrm{PS}} \rangle. \tag{24.7}$$

Here we have written the atom index explicitly, $\tilde{p}_{\kappa i}$ for the projectors and $\phi_{\kappa i}$ for the partial waves of atom $\kappa$. The projections $\langle \tilde{p}_{\kappa j} | \tilde{\psi}_{n\boldsymbol{k}} \rangle$ are taken with the states normalized over one cell (Appendix A.5). Every one of the $N$ cells contributes the same one-center terms by Bloch's theorem, each with weight $1/N$ for states normalized over the crystal, and the cell normalization adds them up; the phase is that of the atom in the reference cell. The

plane wave about the nucleus expands into spherical Bessel functions $j_l(qx)$ times spherical harmonics (Eq. (C.22)), so each element of $\Delta_{\kappa,ij}$ is a sum of radial integrals of $j_l(qx)$ against products of partial waves, times Gaunt coefficients (integrals of three spherical harmonics). These are computed once per species and per $|q|$.

How large is the correction? Expand $\Delta_{\kappa,ij}(\boldsymbol{q})$ in $\boldsymbol{q}\cdot\boldsymbol{x}$. The zeroth-order term is the difference of overlaps inside the sphere. For partial waves that obey the generalized norm-conservation condition it vanishes; a PAW data set does not require that condition, and the term is then the charge that the smooth part lacks inside the sphere. The first-order term is a difference of dipole moments of partial-wave products, of relative size $qr_c$, weighted by the fraction of the state inside the sphere; the next is a quadrupole difference, of order $(qr_c)^2$. The pseudo vertex is accurate for $qr_c \ll 1$, below about 1 keV for the silicon potential above, and the correction grows from there. Well above $q_{\text{opt}}$ it is the whole answer, since the pseudo vertex has decayed. In between lies the range where most electron-recoil rates are decided. Griffin and coworkers computed the all-electron vertices of silicon and germanium in this way, from PAW calculations, and the valence-to-conduction rates for dark matter of 1 GeV. For a heavy mediator, which weights large $q$, the reconstruction makes the differential rate much higher than the pseudo result at energy deposits above about 15 eV, so that the rate integrated above a threshold is enhanced even for thresholds as low as about 10 eV [9, Sects. 1 and 3.1]. Together with the core states it raises it by more than an order of magnitude at higher thresholds, especially in germanium, while the light-mediator rate, dominated by small $q$, changes less [9, Sects. 3.1 and 4]. A norm-conserving potential can be reconstructed with the same formula, using the partial waves of the atom from which it was built. The reconstruction is as good as its ingredients. It assumes a frozen core. It restores the core region only within the span of a few partial waves per channel, which is accurate near the reference energies and less so far above them. And it does not add the core states themselves, which are absent from the pseudopotential calculation. Sect. 26.2 gives the practice and its convergence.

**Worked example 24.1: The cutoff for momentum transfers up to 30 keV**

A momentum transfer $q = 30\,\text{keV}$ needs $2m_\chi v_{\max} \geq q$, that is $m_\chi \geq q/2v_{\max} = 5.7\,\text{MeV}$ for $v_{\max} = 2.650 \times 10^{-3}$, and it deposits at most $\omega = qv_{\max} = 79.5\,\text{eV}$. The final state lies at most 79.5 eV above the valence-band maximum, or about 92 eV above the bottom of the valence band. Its plane-wave components are concentrated below

$$p_f = [2m_e(\omega + E_\text{F})]^{1/2} = (2 \times 0.511\,\text{MeV} \times 92\,\text{eV})^{1/2} = 9.7\,\text{keV}.$$

By Eq. (24.1), the basis must hold the initial-state components up to $q + p_f = 39.7\,\text{keV}$:

$$E_\text{cut} \gtrsim \frac{(39.7\,\text{keV})^2}{2 \times 511\,\text{keV}} = 1.54\,\text{keV} = 113\,\text{Ry}.$$

With $k_\text{cut} = 10.6/a_0$ and the silicon cell of $270.3\,a_0^3$, in units of the Bohr radius $a_0 = 0.529\,\text{Å}$, Eq. (23.11) gives about 5500 plane waves per state, 2.6 times the 2121 at which the total energy has converged to 0.1 meV per atom. The same estimate counts the bands below $\omega$: the free-electron states inside a sphere of radius $p_f$ number $\Omega p_f^3/6\pi^2 \approx 80$ per spin.
At 60 Ry, $k_\text{cut} = 28.9\,\text{keV}$. The vertex at 30 keV is not zero there, since it vanishes identically only beyond $2k_\text{cut} = 58\,\text{keV}$, but it lacks the initial-state components between 28.9 and 39.7 keV. Raising the cutoff to 113 Ry converges it, to the pseudo answer. In the atom of Fig. 24.1, the pseudo-orbitals carry less than $10^{-7}$ of their weight above 30 keV, the all-electron orbitals 0.7 ($3s$) and 0.4 % ($3p$): the converged pseudo vertex at 30 keV

misses almost all of the true one. Only the reconstruction Eq. (24.6) restores the tail, and the cutoff estimate applies to the reconstructed states. The silicon and germanium calculation of Griffin and coworkers meets these requirements: pseudo-states at a cutoff of 1 keV, reconstructed all-electron states up to 2 keV, which hold momentum components up to $(2m_e \times 2\,\mathrm{keV})^{1/2} = 45\,\mathrm{keV}$ [9, Sect. 2.1].

### 24.1.4 Core and semicore states

Below the valence bands of silicon lie the atomic shells $2p$, $2s$ and $1s$. Their binding energies relative to the valence-band maximum, measured by photoemission, are 99.4 eV for $2p$ (the $2p_{3/2}$ level; $2p_{1/2}$ is at 99.8 eV), 150 eV for $2s$ and 1839 eV for $1s$ [10, Table 1-1]. Their radial densities peak at 0.21, 0.24 and 0.038 Å from the nucleus, against about 1 Å for the $3s$ and $3p$ valence orbitals. Half of the $2p$ electrons have momenta above 14 keV, and half of the $1s$ electrons above 37 keV; for the valence electrons the median is 2 to 3 keV. These are free-atom numbers, from the all-electron calculation of Fig. 24.2, but the core shells change little in the crystal. A pseudopotential calculation contains none of these states.

Whether they matter is decided by kinematics. An electron in a shell at depth $E_{\mathrm{core}}$ below the valence-band maximum needs at least $\omega = E_{\mathrm{g}} + E_{\mathrm{core}}$ to reach the conduction band. By Eq. (1.22), a particle of mass $m_\chi$ deposits $\omega$ at momentum transfer $q$ only if $\omega/q + q/2m_\chi \leq v_{\mathrm{max}}$. Since the second term is positive, the momentum transfer must exceed $\omega/v_{\mathrm{max}}$; minimizing the left-hand side over $q$, at $q = (2m_\chi\omega)^{1/2}$, gives the threshold mass $2\omega/v_{\mathrm{max}}^2$ of Eq. (1.24). With $v_{\mathrm{max}} = 2.650 \times 10^{-3}$, the silicon $2p$ shell needs $q \geq 38\,\mathrm{keV}$ for any mass and opens at $m_\chi = 29\,\mathrm{MeV}$, where the only usable momentum transfer is $(2m_\chi\omega)^{1/2} = 2\omega/v_{\mathrm{max}} = 76\,\mathrm{keV}$. The $2s$ shell needs $q \geq 57\,\mathrm{keV}$ and opens at 43 MeV, with 114 keV there, and the $1s$ shell opens above 0.5 GeV. Below about 29 MeV, dark matter scatters only on the four valence electrons of each silicon atom. Above, the $2p$ shell is reached through the tail of its momentum distribution: 3 % of its weight lies above 38 keV.

Germanium is different. Its $3d$ shell lies only 29 eV below the valence-band maximum [10, Table 1-1], with a radial density peaked at 0.27 Å. It needs $q \geq 11\,\mathrm{keV}$ for any mass, 22 keV at the threshold mass $m_\chi = 8.5\,\mathrm{MeV}$, and three quarters of its momentum distribution lies above 11 keV. Ten electrons per atom, kinematically accessible to light dark matter and probed near the peak of their momentum distribution, are what the field calls a *semicore* shell: too deep to bond, too shallow and too extended to be frozen without loss. A germanium pseudopotential can freeze it, with four valence electrons, or keep it in the valence, with fourteen and a higher cutoff. Only the second choice puts the $3d$ states into the calculated response. For a heavy mediator, and once their large-momentum components are restored, the $3d$ electrons dominate the total rate of germanium for $m_\chi \gtrsim 30\,\mathrm{MeV}$ at a one-pair threshold [9, Sects. 1 and 3.3], a point to which Sect. 10.6 returns. The deeper shells, $3p$ and below in germanium and all core shells of silicon, are added as atomic orbitals arranged in Bloch sums (Eq. (26.11)).

Fig. 24.2 also explains why the valence states need the core region. Above about 5 keV the core electrons outnumber the valence electrons in both atoms, and at 30 keV the silicon $2p$ shell holds forty times more weight per keV than the valence shell. What weight the valence shell has there comes from the inner lobes that keep it orthogonal to the core. These are the lobes that a pseudopotential removes and that Eq. (24.6) restores.

### 24.1.5 Relativistic corrections

A $1s$ electron of germanium moves at about $Z\alpha = 0.23$ of the speed of light, and the core electrons that a large momentum transfer probes are the fastest electrons of the atom. How

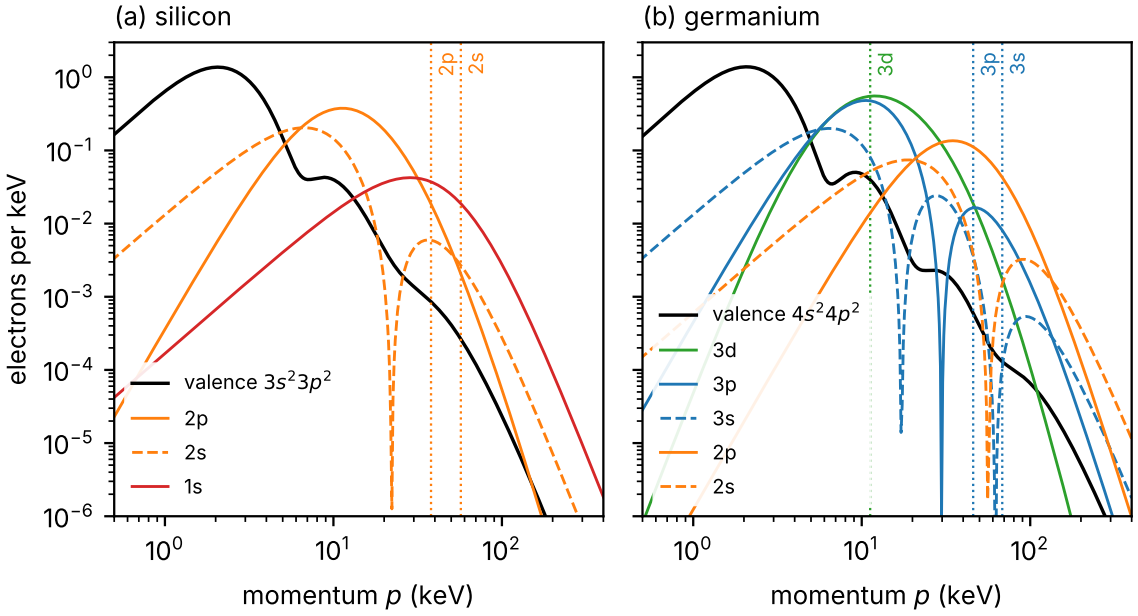



Figure 24.2: How the electrons of each atomic shell are distributed in momentum, for (a) silicon and (b) germanium: electrons per keV of momentum magnitude, $N_{nl}\, p^2 |\phi_{nl}(p)|^2$, with $N_{nl}$ the occupation of the shell and $\phi_{nl}$ the radial momentum-space orbital normalized to $\int \mathrm{d}p\, p^2 |\phi_{nl}|^2 = 1$. The orbitals are those of the free atoms, from an all-electron, scalar-relativistic calculation with the PBE functional (configurations [Ne] $3s^2 3p^2$ and [Ar] $3d^{10} 4s^2 4p^2$; germanium $1s$, which opens above $q = 4\,\mathrm{MeV}$, is not shown). Black: the valence shell. Dotted vertical lines: the smallest momentum transfer at which dark matter of any mass can lift an electron of the shell of the same color into the conduction band, $(E_\mathrm{g} + E_\mathrm{core})/v_\mathrm{max}$ (twice that at the threshold mass), with $v_\mathrm{max} = 2.650 \times 10^{-3}$, gaps of 1.17 and 0.744 eV and measured binding energies relative to the valence-band maximum [10, Table 1-1]. Above about 5 keV the core electrons outnumber the valence electrons in both atoms. The silicon $2p$ shell opens at 38 keV, in the tail of its distribution; the germanium $3d$ shell opens at 11 keV, near its peak.

large are the relativistic corrections to their states? For a single electron in the potential energy $U_1(\boldsymbol{r})$ of the nuclei and the other electrons, the Dirac equation $(\boldsymbol{\alpha}_\mathrm{D}\cdot\boldsymbol{p}+\beta_\mathrm{D}m_e+U_1)\psi = E\psi$ (Appendix C.15) couples a large upper component $\psi_\mathrm{A}$ to a small lower one $\psi_\mathrm{B}$,

$$\boldsymbol{\sigma}\cdot\boldsymbol{p}\,\psi_\mathrm{B} = (\epsilon - U_1)\,\psi_\mathrm{A}, \qquad \boldsymbol{\sigma}\cdot\boldsymbol{p}\,\psi_\mathrm{A} = (2m_e + \epsilon - U_1)\,\psi_\mathrm{B},$$

with $\epsilon = E - m_e$ the energy without the rest mass. The second equation gives $\psi_\mathrm{B} = (2m_e+\epsilon-U_1)^{-1}\boldsymbol{\sigma}\cdot\boldsymbol{p}\,\psi_\mathrm{A}$. Wherever $|\epsilon - U_1| \ll 2m_e$, which fails within $Z\alpha/2m_e$ of a nucleus, we expand to first order, $\psi_\mathrm{B} \simeq (2m_e)^{-1}[1-(\epsilon-U_1)/2m_e]\boldsymbol{\sigma}\cdot\boldsymbol{p}\,\psi_\mathrm{A}$, and insert it into the first equation. With $(\boldsymbol{\sigma}\cdot\boldsymbol{p})^2 = p^2$,

$$\Big[\frac{p^2}{2m_e} + U_1 - \frac{\boldsymbol{\sigma}\cdot\boldsymbol{p}\,(\epsilon-U_1)\,\boldsymbol{\sigma}\cdot\boldsymbol{p}}{4m_e^2}\Big]\psi_\mathrm{A} = \epsilon\psi_\mathrm{A}.$$

This is not yet a Schrödinger equation, since the energy appears on both sides and $\psi_\mathrm{A}$ is not normalized: the probability is $\int(|\psi_\mathrm{A}|^2+|\psi_\mathrm{B}|^2) = \int\psi_\mathrm{A}^\dagger(1+p^2/4m_e^2)\psi_\mathrm{A}$ to this order. The normalized two-component state is $\Phi = (1+p^2/8m_e^2)\psi_\mathrm{A}$. We substitute $\psi_\mathrm{A} = (1-p^2/8m_e^2)\Phi$ and multiply the equation from the left by $1-p^2/8m_e^2$, keeping terms to order $1/m_e^2$ relative to $p^2/2m_e$. The kinetic term becomes $p^2/2m_e - p^4/8m_e^3$; the potential becomes $U_1 - (p^2U_1 + U_1p^2)/8m_e^2$; the term $-\epsilon p^2/4m_e^2$ on the left cancels the same term that the right-hand side, $\epsilon(1-p^2/4m_e^2)\Phi$, acquires. What remains is

$$H\Phi = \Big[\frac{p^2}{2m_e} + U_1 - \frac{p^4}{8m_e^3} - \frac{p^2U_1+U_1p^2}{8m_e^2} + \frac{\boldsymbol{\sigma}\cdot\boldsymbol{p}\,U_1\,\boldsymbol{\sigma}\cdot\boldsymbol{p}}{4m_e^2}\Big]\Phi = \epsilon\Phi.$$

The last two terms simplify with $[\boldsymbol{p}, U_1] = -\mathrm{i}\nabla U_1$. Moving $\boldsymbol{p}$ to the right, $p^2U_1 = U_1p^2 - 2\mathrm{i}\nabla U_1\cdot\boldsymbol{p} - \nabla^2U_1$, and with $(\boldsymbol{\sigma}\cdot\boldsymbol{a})(\boldsymbol{\sigma}\cdot\boldsymbol{b}) = \boldsymbol{a}\cdot\boldsymbol{b} + \mathrm{i}\boldsymbol{\sigma}\cdot(\boldsymbol{a}\times\boldsymbol{b})$, $\boldsymbol{\sigma}\cdot\boldsymbol{p}\,U_1\,\boldsymbol{\sigma}\cdot\boldsymbol{p} = U_1p^2 - \mathrm{i}\nabla U_1\cdot\boldsymbol{p} + \boldsymbol{\sigma}\cdot(\nabla U_1\times\boldsymbol{p})$. The terms in $U_1p^2$ and in $\nabla U_1\cdot\boldsymbol{p}$ cancel between the two, and

$$H = \frac{p^2}{2m_e} + U_1 \underbrace{-\frac{p^4}{8m_e^3}}_{H_\mathrm{mv}} \underbrace{+\frac{\nabla^2U_1}{8m_e^2}}_{H_\mathrm{Dar}} \underbrace{+\frac{\boldsymbol{\sigma}\cdot(\nabla U_1\times\boldsymbol{p})}{4m_e^2}}_{H_\mathrm{so}}. \tag{24.8}$$

The *mass–velocity* term $H_\mathrm{mv}$ is the next term of the expansion of $(p^2+m_e^2)^{1/2}$: a fast electron is heavier. The *Darwin* term $H_\mathrm{Dar}$ smears the potential over the Compton wavelength $1/m_e$. For the nuclear attraction $-Z\alpha/r$ it is the contact term $(\pi Z\alpha/2m_e^2)\,\delta^3(\boldsymbol{r})$, felt only by $s$ electrons. In a central potential $\nabla U_1\times\boldsymbol{p} = r^{-1}(\mathrm{d}U_1/\mathrm{d}r)\boldsymbol{L}$, and with $\boldsymbol{\sigma} = 2\boldsymbol{S}$ the last term is the spin–orbit coupling Eq. (3.53), now with its coefficient, including the Thomas factor, derived rather than quoted. The Foldy–Wouthuysen transformation reaches the same three operators by a unitary transformation, systematically to any order, and also gives the corresponding correction to $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ (Appendix C.15).

The size of each term follows from the velocity. For a hydrogen-like orbital, the virial theorem makes the mean kinetic energy equal to the binding energy, $\langle p^2\rangle/2m_e = (Z\alpha)^2m_e/2n^2$, so that $\langle v^2\rangle = (Z\alpha/n)^2$. The mass–velocity and Darwin terms are of relative order $\langle v^2\rangle$, and so is the change of the wavefunction near the nucleus. For $1s$ both can be evaluated. Since $p^2/2m_e = \epsilon - U_1$ for the eigenstate, $\langle p^4\rangle = 4m_e^2\langle(\epsilon-U_1)^2\rangle$. With $\epsilon = -(Z\alpha)^2m_e/2$, $\langle U_1\rangle = 2\epsilon$ and $\langle U_1^2\rangle = (Z\alpha)^2\langle r^{-2}\rangle = 2(Z\alpha)^4m_e^2$, this is $5(Z\alpha)^4m_e^4$, and $\langle H_\mathrm{mv}\rangle = -\frac{5}{8}(Z\alpha)^4m_e$. The Darwin term is $(\pi Z\alpha/2m_e^2)|\psi_{1s}(0)|^2 = +\frac{1}{2}(Z\alpha)^4m_e$, with $|\psi_{1s}(0)|^2 = (Z\alpha m_e)^3/\pi$. Each is a fraction of order $(Z\alpha)^2$ of the binding energy $(Z\alpha)^2m_e/2$, and they nearly cancel: their sum deepens the level by a quarter of $(Z\alpha)^2$ of its binding energy, the $1s$ term of the Dirac fine-structure formula. Relativistic effects are therefore of order $(Z\alpha)^2$ in the state and of order $(Z\alpha)^2/4$ in the energy of a $1s$ shell.

In practice, pseudopotentials and PAW data sets are generated from *scalar-relativistic* atoms [11]: the mass–velocity and Darwin terms are kept in the radial equation, which stays independent of spin, and spin–orbit coupling is dropped. The core-region shape of the valence states, and the partial waves of Eq. (24.4), then contain the leading relativistic contraction, and a reconstructed vertex inherits it. Spin–orbit coupling is added, when needed, with fully relativistic potentials.

For the core shells, spin–orbit coupling is the visible effect. A shell with $l \geq 1$ splits into $j = l \pm \frac{1}{2}$, by $(l + \frac{1}{2})\xi_{\mathrm{SO}}$, the $p$ splitting $\frac{3}{2}\xi_{\mathrm{SO}}$ of Sect. 3.3 generalized. Near the nucleus $r^{-1}\mathrm{d}U_1/\mathrm{d}r \simeq Z\alpha/r^3$, and for an inner shell, which lives there, $\langle r^{-3}\rangle$ grows as $Z^3$, so that $\xi_{\mathrm{SO}}$ grows as $Z^4$. The hydrogen-like $2p$ splitting $(Z\alpha)^4 m_e/32$ of Sect. 3.3 is an upper bound for a real atom, in which the outer part of the orbital sees a screened charge: it would be 1.7 eV for silicon and 47 eV for germanium. A Dirac–Kohn–Sham calculation of the atoms yields 0.65 and 31.9 eV, and the measured germanium $2p$ doublet, at 1217.0 and 1248.1 eV, is split by 31.1 eV [10, Table 1-1]. A core edge is therefore a doublet. In the $2p$ edge of silicon the two components lie a few tenths of an eV apart, while those of the germanium $3d$ edge, the one light dark matter reaches, lie 0.6 eV apart, with the statistical weights $4 : 6$ of $3d_{3/2}$ and $3d_{5/2}$.

**Worked example 24.2: Relativistic corrections for silicon and germanium core states**

With $\alpha = 1/137.036$, the squared velocity of a $1s$ electron is

$$(Z\alpha)^2 = \Big(\frac{14}{137.036}\Big)^2 = 0.0104 \quad \text{(silicon)},$$
$$(Z\alpha)^2 = \Big(\frac{32}{137.036}\Big)^2 = 0.0545 \quad \text{(germanium)}.$$

These are the relative sizes of the mass–velocity and Darwin terms, and of the relativistic change of the $1s$ orbitals: 1 % in silicon, 5 % in germanium. The binding energy shifts by $(Z\alpha)^2/4$ of itself, 0.26 and 1.36 %. All-electron calculations of the free atoms with the PBE functional, with and without the mass–velocity and Darwin terms, check this. The $1s$ level of silicon moves from $-1781.2$ to $-1785.9$ eV, by 0.26 %, and that of germanium from $-10\,803$ to $-10\,961$ eV, by 1.46 %. The screening of the nucleus by the other electrons changes the binding energies, but hardly the ratio.

The germanium $3d$ shell is not hydrogen-like, and we estimate an effective charge from its size. A hydrogen-like orbital has $\langle r\rangle = [3n^2 - l(l+1)]/2Z_{\mathrm{eff}}$ in bohr, $10.5/Z_{\mathrm{eff}}$ for $3d$. The atomic calculation gives $\langle r\rangle = 0.737$ bohr, so $Z_{\mathrm{eff}} = 14.2$, and

$$\langle v^2\rangle = \Big(\frac{Z_{\mathrm{eff}}\alpha}{n}\Big)^2 = \Big(\frac{14.2}{3 \times 137.036}\Big)^2 = 1.2 \times 10^{-3}.$$

The direct relativistic correction to the $3d$ states is about 0.1 %. The atomic calculation shows a larger, indirect effect of the opposite sign: the scalar-relativistic Kohn–Sham $3d$ level lies at $-29.31$ eV instead of $-30.15$ eV, 2.8 % less bound. The $s$ and $p$ shells inside it contract, screen the nucleus more strongly, and push the $3d$ level up. Spin–orbit coupling splits the level by 0.59 eV in the Dirac–Kohn–Sham atom, against the measured 0.6 eV, or 2 % of the binding energy [10, Table 1-1].

Dark matter lighter than 0.3 GeV reaches every shell except silicon $1s$ and germanium $2p$ and below. Each shell it reaches needs relativistic corrections of at most a few percent in energy, and less in its states. A scalar-relativistic pseudopotential or PAW data set includes the leading part, and the spin–orbit doublet of the $3d$ edge is the one correction that a spectrum resolves.

## 24.2 Construction and testing of pseudopotentials

What a pseudopotential gives up is the true valence wavefunction near the nucleus, exactly where the density vertex $\rho_{mn}(\boldsymbol{k},\boldsymbol{q})$ of a heavy-mediator rate is evaluated, at momentum transfers of tens of keV. There a pseudopotential that is excellent for total energies, bands and phonons can fail without any sign in those quantities.

### 24.2.1 The frozen core

A pseudopotential acts on the valence states through a local radial potential $V_{\mathrm{loc}}(r)$ and a nonlocal part in the separable form of Kleinman and Bylander [5],

$$V_{\mathrm{NL}} = \sum_{\kappa} \sum_{ij} |\tilde{p}_{\kappa i}\rangle D_{\kappa,ij} \langle \tilde{p}_{\kappa j}|. \tag{24.9}$$

The projector functions $\tilde{p}_{\kappa i}(\boldsymbol{r})$, the nonlocal counterparts of the PAW projectors of Sect. 24.1, are localized within a core radius $r_c$ of nucleus $\kappa$ and carry an angular momentum $l$ each. The coefficients $D_{\kappa,ij}$ form a small matrix. $V_{\mathrm{NL}}$ is the nonlocal pseudopotential of Appendix A.5.

The approximation behind all of this is the frozen core. The core electrons are assumed to keep the orbitals they have in the isolated atom, unaffected by the crystal, the valence electrons, or a displacement of the nucleus. Its errors are of three kinds: transferability, the missing core region of the valence states, and the missing core states. Transferability means that a potential built for the atom must describe the atom in the crystal, at other bond lengths and charge states. Tests control this error. The missing core region of the valence states leaves total energies and bands almost untouched but changes matrix elements of operators that probe short distances, such as $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ at large $q$. The all-electron reconstruction Eq. (24.6) repairs it. The core states themselves are absent from the spectrum, so a pseudopotential calculation of a response has no core edges. They are added back as atomic orbitals (Sect. 26.2). Core polarization by a displaced nucleus, which enters the large-$q$ lattice response, is identically zero in a frozen-core calculation and has to be computed separately (Sect. 25.2).

### 24.2.2 Construction

A pseudopotential is built from an all-electron calculation of the isolated atom, followed by the construction of pseudo-wavefunctions, the removal of the valence screening and the separable form. The atomic calculation is scalar-relativistic (that is, with the mass–velocity and Darwin terms of Sect. 24.1 but without spin–orbit coupling) and uses a reference configuration such as $[\mathrm{Ne}]\,3s^2 3p^2$ for silicon. The choice of valence comes with it. Silicon has an obvious split, four valence electrons above a core bound by about 100 eV and more. Germanium does not: its 3d shell (Sect. 24.1) overlaps the 4s and 4p orbitals in space and is polarized by the crystal.

The pseudo-wavefunction of each angular momentum $l$ equals the all-electron one beyond $r_c$, joins it smoothly at $r_c$, and is nodeless inside (Sect. 24.1). Within these constraints, the shape inside $r_c$ is chosen to make the plane-wave expansion converge fast. In the optimized norm-conserving construction of Hamann [4], the pseudo-wavefunction is expanded in a few spherical Bessel functions inside $r_c$. The coefficients minimize the residual kinetic energy above a wavevector $q_{\mathrm{opt}}$ [4, Eq. (1)],

$$E^{\mathrm{res}}(q_{\mathrm{opt}}) = \int_{q_{\mathrm{opt}}}^{\infty} \mathrm{d}q\, q^4\, |\phi(q)|^2, \qquad \phi(q) = 4\pi \int_0^{\infty} \mathrm{d}r\, r^2 j_l(qr)\, \phi(r), \tag{24.10}$$

subject to norm conservation and continuity at $r_c$. Here $\phi(r)$ is the radial pseudo-wavefunction and $j_l$ the spherical Bessel function of Appendix C.3. Up to a constant, $q^4|\phi(q)|^2$ is the

kinetic-energy density in momentum space, $q^2/2m_e$ times the radial measure $q^2|\phi(q)|^2$, so $E^{\mathrm{res}}$ is the kinetic energy missed by a basis truncated at $q_{\mathrm{opt}}$. The cutoff of the solid is then set by $q_{\mathrm{opt}}$, and the tail above it decays fast. The same construction uses two projectors per angular momentum, which reproduce the scattering of the all-electron atom at two energies, typically a bound state and a scattering state above it, rather than one.

Next, the valence part of the potential is removed. The potential that reproduces the pseudo-wavefunctions is a screened one: it contains the Hartree and exchange–correlation potentials of the valence density of the reference atom, which must be subtracted, since the crystal supplies its own. The Hartree potential is linear in the density and subtracts cleanly. The exchange–correlation potential is not, and where the core and valence densities overlap, subtracting the valence part alone leaves an error that changes with the valence configuration. The nonlinear core correction of Louie, Froyen and Cohen [12] keeps a smoothed partial core density in the exchange–correlation term for this reason. Finally, the semilocal potential is cast into the separable form Eq. (24.9), which reduces the cost of applying it from a double sum over plane waves to a few projections (Eq. (23.15)).

Ultrasoft pseudopotentials and the PAW method relax these conditions to lower the cutoff. The ultrasoft scheme (Sect. 24.1) pays for it with a generalized eigenvalue problem, and Kresse and Joubert showed how ultrasoft potentials follow from the PAW method as an approximation [13]. Norm-conserving potentials remain the simplest to use in density-functional perturbation theory, in electron–phonon calculations and in many-body perturbation theory, where every extra term of the ultrasoft and PAW formalisms has to be carried through each derivation [4].

The potentials of the representative calculations come from the SG15 library of Schlipf and Gygi [6], generated with Hamann's construction, with parameters optimized automatically against all-electron results. The silicon potential has four valence electrons, s and p channels with $r_c = 2.405$ and 3.565 bohr (1.27 Å and 1.89 Å), two projectors per channel, $q_{\mathrm{opt}} = 5.11$ and 3.50 inverse bohr, and no nonlinear core correction. The germanium potential has fourteen valence electrons, s, p and d channels with $r_c$ between 2.54 and 2.67 bohr, two projectors each, and $q_{\mathrm{opt}}$ up to 6.36 inverse bohr for the 3d channel. The squares of these wavevectors, 26 Ry and 12 Ry for silicon and 40 Ry for the germanium 3d, are the scale at which the residual kinetic energy has been made small. They are below the cutoffs at which the total energy converges in Table 23.2, 60 Ry and 80 Ry, because the residual Eq. (24.10) decays above $q_{\mathrm{opt}}$ but does not vanish.

### 24.2.3 Testing

A pseudopotential is tested on the atom, for transferability, and in the solid before it is used. The logarithmic derivative $\mathrm{d}\ln R_l(r)/\mathrm{d}r$ of the radial solution at a radius just beyond $r_c$, plotted against energy for the all-electron and the pseudo atom, measures how well the potential scatters at energies other than its reference energies. The two curves must agree over the whole window of energies the calculation will use, which for a response calculation includes empty states tens of eV above the gap. A separable potential can also bind a spurious state, a ghost, below or among the physical ones, because the eigenstates of Eq. (24.9) are not ordered in energy by their number of nodes [4]. A ghost appears as a spurious step in the logarithmic derivative, or in the crystal as a flat band. The PseudoDojo tables were scanned for ghosts up to about 200 eV above the Fermi level, a range chosen for optical and *GW* calculations, and a few heavy elements still show one in that range, between 20 and 80 eV [14, Table 8]. Ghosts at still higher energies are unavoidable [14].

For transferability, the potential, built in one configuration, is applied to excited and ionized configurations of the atom, and the excitation energies are compared with all-electron

values. In the solid, the most systematic test is the $\Delta$ gauge of Lejaeghere and coworkers. It is the root-mean-square difference between two equations of state $E(\mathcal{V})$ of the same crystal [15, Eq. (15)],

$$\Delta_{\text{gauge}} = \left[\frac{1}{\Delta\mathcal{V}} \int_{\mathcal{V}_0-\Delta\mathcal{V}/2}^{\mathcal{V}_0+\Delta\mathcal{V}/2} \mathrm{d}\mathcal{V}\, \left(E_1(\mathcal{V}) - E_2(\mathcal{V})\right)^2\right]^{1/2}, \qquad \Delta\mathcal{V} = 0.12\,\mathcal{V}_0, \tag{24.11}$$

with both curves set to zero at their minima. The window spans $\pm 6\,\%$ of the reference volume $\mathcal{V}_0$. It folds the errors of the volume, the bulk modulus and its pressure derivative into one number in meV per atom. Different all-electron codes agree with each other to 0.3–0.5 meV per atom on average over the periodic table. The norm-conserving PseudoDojo tables reach a mean of about 0.7 meV per atom against an all-electron reference (excluding the magnetic Cr, Mn and Fe) [14]. A potential must also be used with the functional with which it was generated; Hamann found that using a PBE-generated potential in an LDA calculation increased the differences from the all-electron results [4].

The standard tests are necessary but not sufficient for a dark-matter calculation. They probe the total energy and its derivatives with respect to volume, and the bands near the gap. Such a calculation also needs phonons, which probe the potential under small displacements of the nucleus. It needs empty states far above the gap, which probe the logarithmic derivatives at high energy. And it needs the density vertex at momentum transfers up to tens of keV, which probes the core region. The last of these is not part of any standard test.

### 24.2.4 Semicore states

When a shell sits between the deep core and the valence, its treatment changes both the accuracy and the cost. Keeping the germanium 3d shell in the valence improves transferability, since the shell responds to its environment; it puts the 3d states into the band structure, where they contribute to the response above an energy deposit of about 30 eV (Sect. 10.6); and it costs. The fourteen electrons need seven occupied bands per atom instead of two, and the localized d states need a higher cutoff, 80 Ry where silicon needs 60 Ry (Table 23.2). Gallium arsenide, with the Ga 3d in the valence, is harder still: in the same kind of calculation the energy still changes by 25 meV per atom between 100 and 120 Ry. The alternative, a frozen 3d shell with a nonlinear core correction, gives a cheaper calculation of the valence and conduction bands and a response without the 3d channel, which must then be added as core states.

### 24.2.5 The density vertex at large momentum

The pseudo vertex is accurate for $qr_c \ll 1$ (Sect. 24.1) and must be checked beyond. In the opposite limit, $q \gg q_{\text{opt}}$, the pseudo-wavefunctions have almost no Fourier components by construction, Eq. (24.10). The pseudo vertex then falls off much faster than the true one, whose large-$q$ tail comes from the nodes and the cusp of the valence states near the nucleus. For the SG15 silicon potential $1/r_c$ is 1.6 and 1.0 keV, and $q_{\text{opt}}$ is 19 and 13 keV. A dark-matter particle of 10 MeV at the maximum speed in the Earth frame, $v_{\text{max}} = 2.65 \times 10^{-3}$ (Sect. 1.3), transfers at most $2m_\chi v_{\text{max}} \approx 53$ keV, and the typical momentum transfer of electron recoil is of the order of $\alpha m_e = 3.7$ keV (Sect. 1.3). Much of the rate therefore comes from momentum transfers between $1/r_c$ and $q_{\text{opt}}$, where the pseudo vertex of this potential is neither guaranteed accurate nor negligible, and a heavy-mediator rate weights the upper end of that range. A potential with a smaller $r_c$ pushes the problem to larger $q$ at the price of a higher cutoff.

In the reconstruction of Griffin and coworkers for silicon and germanium (Worked example 24.1), the deeper core states, 2p and below in silicon and 3d and below in germanium

in their treatment, were modeled as atomic orbitals in Bloch sums. The atomic description agreed with the reconstructed crystal states up to the reconstruction cutoff [9, Sect. 2.2].

### 24.2.6 Nonlocal sum rules

A nonlocal potential changes the commutators on which sum rules rest, and a test of a pseudopotential calculation against a sum rule must account for it. For a local potential, the double commutator of the Hamiltonian with $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ involves only the kinetic energy and gives the $f$-sum rule of Sect. 4.3. The plane wave $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ is unitary. Inserting a complete set of states $|m\rangle$ of the one-electron Hamiltonian $H$ gives $\sum_m (\varepsilon_m - \varepsilon_n)|\langle m|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|n\rangle|^2 = \langle n|\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}[H, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}]|n\rangle = \langle n|\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}H\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} - H|n\rangle$, and the same with $-\boldsymbol{q}$ in place of $\boldsymbol{q}$. Adding the two,

$$\tfrac{1}{2}\sum_m (\varepsilon_m - \varepsilon_n)\big(|\langle m|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|n\rangle|^2 + |\langle m|\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|n\rangle|^2\big) = \tfrac{1}{2}\langle n|\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}H\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} + \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}H\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} - 2H|n\rangle. \tag{24.12}$$

The kinetic energy is shifted by the plane wave, $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\boldsymbol{p}\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} = \boldsymbol{p} + \boldsymbol{q}$, so its contribution is $\frac{1}{2}\langle(\boldsymbol{p}+\boldsymbol{q})^2 + (\boldsymbol{p}-\boldsymbol{q})^2 - 2\boldsymbol{p}^2\rangle/2m_e = q^2/2m_e$. A local potential commutes with the plane wave and drops out. A nonlocal one does not, and leaves

$$\frac{q^2}{2m_e} + \delta_{\mathrm{NL}}(\boldsymbol{q}), \qquad \delta_{\mathrm{NL}}(\boldsymbol{q}) = \tfrac{1}{2}\langle n|\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}V_{\mathrm{NL}}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} + \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}V_{\mathrm{NL}}\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} - 2V_{\mathrm{NL}}|n\rangle, \tag{24.13}$$

on the right-hand side of Eq. (24.12). In the separable form Eq. (24.9), the conjugated potential $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}V_{\mathrm{NL}}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ has the projectors $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\tilde{p}_{\kappa i}$, so $\delta_{\mathrm{NL}}$ is a sum of projections and costs almost nothing to evaluate. At small $q$ it is of order $q^2$, like the kinetic term, and it does not vanish when all bands are summed: the pseudo-Hamiltonian obeys a modified $f$-sum rule. The all-electron Hamiltonian, whose potential is local, obeys the unmodified one. For a single Bloch state the same sum rule, without the symmetrization in $\pm\boldsymbol{q}$, keeps a term linear in $q$ set by the group velocity; Eq. (26.13) gives that form, and summed over a filled band it reduces to Eq. (24.12).

The representative silicon calculation shows the size of the effect. With the SG15 potential and the states at $\boldsymbol{k} = 0$, the left-hand side of Eq. (24.12), divided by $q^2/2m_e$, is 0.81 with 16 bands and 0.88 with 100 at $|\boldsymbol{q}| = 0.1\,\text{Å}^{-1}$ along [100], and nearly the same at $0.5\,\text{Å}^{-1}$. The sum still rises slowly at 100 bands, so part of the deficit is band truncation, and part is $\delta_{\mathrm{NL}}$. The two must be separated, by computing Eq. (24.13) directly, before the sum rule can be used as a convergence test of the number of bands.

The limit $q \to 0$ of the same algebra gives the velocity, which carries the nonlocal commutator $\mathrm{i}[V_{\mathrm{NL}}, \boldsymbol{r}]$ of Appendix A.5; Eq. (26.9) gives its plane-wave form. The density vertex needs no such correction (Sect. 24.1): a scalar potential couples to the density by multiplication, as in a local Hamiltonian [16, Eq. (5)].

## Summary

The states of a crystal are computed in plane waves with pseudopotentials. A vertex at momentum transfer $q$ needs plane waves up to about $q$ plus the momentum of the final state, far beyond the cutoff that converges a total energy. A pseudopotential replaces the core region by a smooth potential whose valence states agree with the true ones outside a core radius, and norm conservation fixes their charge there. Above about a keV, the density vertex depends on the oscillations of the valence states near the nuclei, which the all-electron reconstruction of the projector-augmented-wave method restores, and core and semicore shells are added as atomic orbitals where kinematics opens them. Relativistic corrections grow as $(Z\alpha)^2$ and stay

at a few percent for every shell that dark matter lighter than about 0.3 GeV can reach. A sum rule tested with a nonlocal Hamiltonian carries its nonlocal term. For silicon the vertex beyond $q \sim 1/r_c$ is one of the dominant theory uncertainties of an electronic rate.

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# Chapter 25

# Phonons from first principles

Every number that Chap. 5 attaches to the lattice of silicon, from the 15.5 THz optical phonon at the zone center [1] to the Debye–Waller exponent at a momentum transfer of tens of keV, comes out of one kind of calculation: the first-order change of the Kohn–Sham potential when the nuclei of one species are displaced in a wave of crystal momentum $\boldsymbol{Q}$. For dark matter the result matters at both ends of the spectrum: the sound speeds that set the reach of acoustic-phonon searches (Sect. 9.2) and the optical branches of polar crystals (Sect. 5.4) all come out of density-functional perturbation theory (DFPT). The same linear-response equations that give a phonon also give, with an electric field or a strain in place of the displacement, the Born charges, $\varepsilon_\infty$ and the elastic constants, and, extended slightly, the changes of the empty states that a phonon-assisted amplitude needs (Sect. 13.2). The calculation is for a harmonic crystal on a static lattice; the anharmonic and finite-temperature methods of Sect. 25.3 all start from its harmonic phonons, Born charges and electron–phonon potentials.

## 25.1 Frozen phonons and linear response

The dispersion of silicon in Fig. 5.3 came from sixteen calculations, each in the two-atom primitive cell, one for each inequivalent wavevector of a $6 \times 6 \times 6$ grid. The same force constants could have been obtained by moving atoms in a supercell of $6^3 = 216$ primitive cells, 432 atoms, and recording the forces. The cost of a density-functional calculation grows as the cube of the number of atoms $N_{\mathrm{at}}$ in the cell times the number of $\boldsymbol{k}$ points. A cell $n$ times larger needs $n$ times fewer $\boldsymbol{k}$ points for the same sampling, as long as more than one is left. Each self-consistent calculation of that supercell would therefore cost $216^3/216 = 216^2 \approx 5 \times 10^4$ times one of the primitive cell. The two routes measure the same thing, the second derivatives of the energy surface $E^{\mathrm{ad}}(u)$ of Sect. 5.1, from finite differences of forces or from the linear response of the electrons to a displacement wave. The Sternheimer equation makes the linear-response route, DFPT, cheap, and Fourier interpolation turns a few dynamical matrices into the whole dispersion. Our account follows Baroni and coworkers [2] and Giustino [3].

### 25.1.1 Frozen phonons

The most direct way to get a force constant is to displace an atom and measure the forces on all the others. The force on the nucleus $(\boldsymbol{R}, \kappa')$ along $\alpha$ when only the nucleus $(\boldsymbol{0}, \kappa)$ is displaced by $h$ along $\beta$ is, by Eq. (5.13) carried one order further,

$$F^{\alpha}_{\boldsymbol{R}\kappa'}(h) = -\Phi^{(2),\alpha\beta}_{\kappa'\kappa}(\boldsymbol{R},\boldsymbol{0})\,h - \tfrac{1}{2}\Phi^{(3),\alpha\beta\beta}_{\kappa'\kappa\kappa}(\boldsymbol{R},\boldsymbol{0},\boldsymbol{0})\,h^2 - \tfrac{1}{6}\Phi^{(4),\alpha\beta\beta\beta}_{\kappa'\kappa\kappa\kappa}(\boldsymbol{R},\boldsymbol{0},\boldsymbol{0},\boldsymbol{0})\,h^3 + O(h^4),$$

where $\Phi^{(3)}$ and $\Phi^{(4)}$ are the third and fourth derivatives of $E^{\mathrm{ad}}$ (Sect. 5.6). The forces are the Hellmann–Feynman forces of Eq. (5.10), which a code returns at the cost of one

self-consistent calculation. The quotient $-F/h$ has an error of first order in $h$, from the cubic term. Displacing by $+h$ and by $-h$ and taking the difference cancels all even powers of $h$:

$$\Phi^{(2),\alpha\beta}_{\kappa'\kappa}(\boldsymbol{R},\boldsymbol{0}) = -\frac{F^{\alpha}_{\boldsymbol{R}\kappa'}(h) - F^{\alpha}_{\boldsymbol{R}\kappa'}(-h)}{2h} + O(h^2). \tag{25.1}$$

This is the *frozen-phonon* method in its finite-displacement form [2, Sect. IV.B]. Because each calculation returns the force on every nucleus, one displacement gives a whole column of the force-constant matrix, $3N_{\mathrm{at}}$ numbers for a cell of $N_{\mathrm{at}}$ atoms; an energy alone would give one number. Symmetry reduces the work further. In silicon both atoms of the cell are equivalent under the space group, and the site symmetry $T_d$ contains the threefold rotation about [111] that turns $x$ into $y$ and $z$. A single displacement of one atom along $x$, made with both signs, therefore fixes every force constant the supercell can hold. The step $h$ is a compromise: large enough that the force exceeds the numerical noise of the calculation, typically 0.01 Å to 0.03 Å, and small enough that the $O(h^2)$ term of Eq. (25.1), set by $\Phi^{(4)}$, is negligible.

A displaced atom in a periodic calculation is not alone. A supercell built from $N_Q \times N_Q \times N_Q$ primitive cells repeats with the lattice of supercell translations $\boldsymbol{T}$, and moving the nucleus $(\boldsymbol{0},\kappa)$ moves all its images $(\boldsymbol{T},\kappa)$ with it. The force recorded on $(\boldsymbol{R},\kappa')$ is then the sum over images. With $\Phi^{(2)}(\boldsymbol{R},\boldsymbol{T}) = \Phi^{(2)}(\boldsymbol{R}-\boldsymbol{T},\boldsymbol{0})$ by translation invariance (Sect. 5.2), the supercell measures

$$\Phi^{\mathrm{SC},\alpha\beta}_{\kappa'\kappa}(\boldsymbol{R}) = \sum_{\boldsymbol{T}} \Phi^{(2),\alpha\beta}_{\kappa'\kappa}(\boldsymbol{R}+\boldsymbol{T},\boldsymbol{0}), \tag{25.2}$$

a sum of the true constant and its copies displaced by every supercell translation. The true $\Phi^{(2)}$ cannot be recovered from $\Phi^{\mathrm{SC}}$ in general, but the dynamical matrix can, at special wavevectors. Write each cell vector of Eq. (5.16) as $\boldsymbol{R} = \boldsymbol{R}_s + \boldsymbol{T}$, with $\boldsymbol{R}_s$ one of the $N_Q^3$ cells of the supercell. If $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{T}} = 1$ for every $\boldsymbol{T}$, the phase depends only on $\boldsymbol{R}_s$, the sum over $\boldsymbol{T}$ acts on $\Phi^{(2)}$ alone, and

$$D^{\mathrm{dyn},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{Q}) = \frac{1}{\sqrt{M_\kappa M_{\kappa'}}} \sum_{\boldsymbol{R}_s} \Phi^{\mathrm{SC},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{R}_s)\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}_s}$$

holds exactly, with $\Phi^{\mathrm{SC}}_{\kappa\kappa'}(\boldsymbol{R}_s)$ read off the supercell with the roles of the two nuclei exchanged by pair symmetry. The wavevectors with $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{T}} = 1$ are the $N_Q^3$ points of a uniform grid in the zone, the zone center among them. A frozen-phonon calculation in an $N_Q^3$ supercell therefore gives the exact phonons on an $N_Q^3$ grid of wavevectors and nothing more. At the zone center the primitive cell is the supercell. The zone-boundary point X of silicon, $\boldsymbol{Q} = (0,0,1)\,2\pi/a$, has $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}} = -1$ for half of the cell vectors and needs a cell twice as large, as the doubled period of the folded chain in Worked example 5.2 already showed. Long wavelengths need large cells. Force constants of range $L$ need a supercell of side $\sim L$, with $N_{\mathrm{at}} \propto L^3$ atoms. By the rule of the opening paragraph each calculation then costs $\propto N_{\mathrm{at}}^3$ times the number of $\boldsymbol{k}$ points: $\propto L^6$ while the $\boldsymbol{k}$ grid can still be thinned, and $\propto L^9$ once a single $\boldsymbol{k}$ point suffices, the regime of the ninth-power scaling quoted by Baroni and coworkers [2, Sect. IV.B] and worked out in Sect. 25.2. A dark-matter calculation needs the acoustic branches precisely near the zone center, where this route is most expensive.

The energy itself, rather than the force, yields a frequency in one stroke when the pattern of the displacement is fixed by symmetry. In this energy form of the frozen-phonon method [2, Sect. IV.B] the atoms are displaced along a known normal mode, and the energy is fitted as a function of its amplitude. The zone-center optical mode of silicon is such a mode.

**Worked example 25.1: The optical frequency of silicon from a frozen phonon**

At the zone center the two atoms of the cell move against each other, and by symmetry the polarization is fixed up to its direction. For a displacement along [111], atom 1 moves by $-(u/2)\hat{\boldsymbol{n}}$ and atom 2 by $+(u/2)\hat{\boldsymbol{n}}$, with $\hat{\boldsymbol{n}} = (1,1,1)/\sqrt{3}$, so that the bond along [111] changes its length by $u$. Every cell moves in the same way, and the primitive cell is the whole calculation. The energy per cell is $E^{\mathrm{ad}}_{\mathrm{cell}}(u) = \frac{1}{2}\Phi_{\mathrm{opt}}u^2 + \frac{1}{6}\Phi^{(3)}_{\mathrm{opt}}u^3 + \frac{1}{24}\Phi^{(4)}_{\mathrm{opt}}u^4 + \dots$, measured from the minimum. In the PBE calculation of Fig. 5.2, $E^{\mathrm{ad}}_{\mathrm{cell}} = 5.943\,\mathrm{meV}$ at $u = +0.03\,\text{Å}$ and $6.316\,\mathrm{meV}$ at $u = -0.03\,\text{Å}$. The bond resists compression more than stretching.

From a single energy, $2E^{\mathrm{ad}}_{\mathrm{cell}}(u)/u^2 = \Phi_{\mathrm{opt}} + \Phi^{(3)}_{\mathrm{opt}}u/3 + \dots$, we find $\Phi_{\mathrm{opt}} = 13.21\,\mathrm{eV\,Å^{-2}}$ from the stretched bond and $14.04\,\mathrm{eV\,Å^{-2}}$ from the compressed one. The spread is the cubic term: half the difference, $0.415\,\mathrm{eV\,Å^{-2}} = |\Phi^{(3)}_{\mathrm{opt}}|u/3$, puts $\Phi^{(3)}_{\mathrm{opt}}$ at $-41.5\,\mathrm{eV\,Å^{-3}}$, the value of Sect. 5.1. The symmetric combination removes it,

$$\frac{E^{\mathrm{ad}}_{\mathrm{cell}}(u) + E^{\mathrm{ad}}_{\mathrm{cell}}(-u) - 2E^{\mathrm{ad}}_{\mathrm{cell}}(0)}{u^2} = \Phi_{\mathrm{opt}} + \frac{\Phi^{(4)}_{\mathrm{opt}}u^2}{12} + \dots = 13.62\,\mathrm{eV\,Å^{-2}},$$

and the same combination at $u = 0.06\,\text{Å}$ is $13.64\,\mathrm{eV\,Å^{-2}}$. The two symmetric estimates differ by $0.02\,\mathrm{eV\,Å^{-2}}$, within the precision of the computed energies, so the quartic term has to come from a fit to the whole curve. Polynomial fits to all the computed energies and forces, over ranges of $|u|$ from 0.1 Å to 0.3 Å, put the quartic constant $\Phi^{(4)}_{\mathrm{opt}}$ at $35\,\mathrm{eV\,Å^{-4}}$ to $42\,\mathrm{eV\,Å^{-4}}$, about $40\,\mathrm{eV\,Å^{-4}}$, so the quartic correction $\Phi^{(4)}_{\mathrm{opt}}u^2/12$ is $0.003\,\mathrm{eV\,Å^{-2}}$ at 0.03 Å, 0.02 %. The coordinate $u$ carries the reduced mass $M_\kappa/2$, since each atom moves by $u/2$ and the kinetic energy is $\frac{1}{2}(M_\kappa/2)\dot{u}^2$. With $M_\kappa = 28.0855\,\mathrm{u}$,

$$\omega_{\mathrm{opt}} = \left(\frac{2\Phi_{\mathrm{opt}}}{M_\kappa}\right)^{1/2} = 63.68\,\mathrm{meV}.$$

The central difference of the Hellmann–Feynman forces, Eq. (25.1) applied to the coordinate $u$, yields $13.63\,\mathrm{eV\,Å^{-2}}$ and 63.71 meV from the same two calculations. DFPT at the same settings, the $4^3$ calculation of Fig. 5.2(b), finds 63.7 meV, and the calculation of Fig. 5.3, with a higher cutoff and a finer grid, 63.69 meV. Neutron scattering finds 64.1 meV [1], 0.6 % higher, a difference set by the functional and the static lattice, not by the method. The one-sided estimates, by contrast, miss by 1.5 % at the smallest step and by 3 % at 0.06 Å. A finite-difference force constant is accurate only when it is symmetric.

### 25.1.2 Force constants as a density response

A second derivative of the energy needs only the first-order change of the electron density. Differentiate the Hellmann–Feynman force Eq. (5.10) once more. Its electrostatic form contains the nuclear positions in two places, the electron–nucleus potential and the nucleus–nucleus repulsion, and the electron density, which also changes when a nucleus moves. In a Kohn–Sham calculation the electron–nucleus term is the external potential $V_{\mathrm{ext}}(\boldsymbol{r})$ of the electrons, the potential $U_1$ of Eq. (3.62) and the operator $V_{eN}$ of Eq. (5.2). Writing $E^{\mathrm{ad}} = E_{NN} + E_{\mathrm{el}}$, with $E_{NN}$ the repulsion of the nuclei and $\partial E_{\mathrm{el}}/\partial u_i = \int \mathrm{d}^3r\, n(\boldsymbol{r})\, \partial V_{\mathrm{ext}}(\boldsymbol{r})/\partial u_i$ by the theorem, for any two displacement coordinates $u_i, u_j$ [2, Eq. (10)],

$$\frac{\partial^2 E^{\mathrm{ad}}}{\partial u_i\, \partial u_j} = \int \mathrm{d}^3r\, \frac{\partial n(\boldsymbol{r})}{\partial u_j}\, \frac{\partial V_{\mathrm{ext}}(\boldsymbol{r})}{\partial u_i} + \int \mathrm{d}^3r\, n(\boldsymbol{r})\, \frac{\partial^2 V_{\mathrm{ext}}(\boldsymbol{r})}{\partial u_i\, \partial u_j} + \frac{\partial^2 E_{NN}}{\partial u_i\, \partial u_j}. \tag{25.3}$$

The terms with $n$ and with $E_{NN}$ need only the ground state: the density at equilibrium and the positions of the nuclei, whose repulsion is an Ewald sum. The term with $\partial n/\partial u_j$ needs the change of the density to first order in a displacement, and nothing else. With a nonlocal pseudopotential (Sect. 24.1) the integrals of $n$ against derivatives of $V_{\text{ext}}$ become sums $2\sum_v \langle\psi_v|\partial V_{\text{ext}}|\psi_v\rangle$ over occupied states, and the structure is unchanged.

A crystal is perturbed one wavevector at a time. Let the nuclei of species $\kappa$ move along $\alpha$ in a wave, $u^{\beta}_{\boldsymbol{R}\kappa'} = \delta_{\kappa\kappa'}\delta_{\alpha\beta}\,\lambda\,\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$. The derivative with respect to its amplitude is

$$\partial_{\kappa\alpha,\boldsymbol{Q}} = \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\,\frac{\partial}{\partial u^{\alpha}_{\boldsymbol{R}\kappa}}.$$

Applied to the bare potential, with $v_\kappa$ the potential of one nucleus (or pseudo-ion) of species $\kappa$, it yields $\partial_{\kappa\alpha,\boldsymbol{Q}} V_{\text{ext}}(\boldsymbol{r}) = -\sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\,\partial_\alpha v_\kappa(\boldsymbol{r}-\boldsymbol{R}-\boldsymbol{\tau}_\kappa)$, which is a lattice-periodic function times $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{r}}$. The double sum $\sum_{\boldsymbol{R}\boldsymbol{R}'} \mathrm{e}^{-\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}'}\Phi^{(2)}(\boldsymbol{R},\boldsymbol{R}')$ collapses by translation invariance to $N\sum_{\boldsymbol{R}}\Phi^{(2)}(\boldsymbol{0},\boldsymbol{R})\,\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$, so the dynamical matrix Eq. (5.16) is a second derivative with respect to two such waves [2, Eq. (82)],

$$D^{\mathrm{dyn},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{Q}) = \frac{1}{N\sqrt{M_\kappa M_{\kappa'}}}\,\partial_{\kappa\alpha,-\boldsymbol{Q}}\,\partial_{\kappa'\beta,\boldsymbol{Q}}\,E^{\mathrm{ad}}. \tag{25.4}$$

Inserted in Eq. (25.3), its electronic part contains the integral of $[\partial_{\kappa\alpha,\boldsymbol{Q}}V_{\text{ext}}]^*\,\partial_{\kappa'\beta,\boldsymbol{Q}}n$ over the crystal [2, Eq. (86)], with $[\partial_{\boldsymbol{Q}}V]^* = \partial_{-\boldsymbol{Q}}V$ since the potential is real. The density response to the wave at $\boldsymbol{Q}$ carries the same crystal momentum: the perturbed crystal is invariant under a lattice translation combined with the phase $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$, and so is its first-order density. Different $\boldsymbol{Q}$ decouple, and each is a problem in the primitive cell.

The density response is the static bubble of Sect. 4.6. Kohn–Sham electrons move in $V_{\text{KS}}$, and a change of it changes their density by $\delta n = R^0\,\delta V_{\text{KS}}$, with $R^0$ the independent-particle response Eq. (4.63) at $\omega = 0$. The potential itself responds to the density through the Hartree and exchange–correlation terms,

$$\partial_{\kappa\alpha,\boldsymbol{Q}}V_{\text{KS}}(\boldsymbol{r}) = \partial_{\kappa\alpha,\boldsymbol{Q}}V_{\text{ext}}(\boldsymbol{r}) + \int \mathrm{d}^3r'\,\left[v_{\text{C}}(\boldsymbol{r}-\boldsymbol{r}') + f_{\text{xc}}(\boldsymbol{r},\boldsymbol{r}')\right]\partial_{\kappa\alpha,\boldsymbol{Q}}n(\boldsymbol{r}'), \tag{25.5}$$

where $f_{\text{xc}}$ is the kernel Eq. (4.69) in its static form, the second functional derivative $\delta^2E_{\text{xc}}/\delta n(\boldsymbol{r})\delta n(\boldsymbol{r}')$ of the ground-state functional [3, Eqs. (42)–(46)]. With $f_{\text{Hxc}} = v_{\text{C}} + f_{\text{xc}}$, solving $\delta n = R^0(\delta V_{\text{ext}} + f_{\text{Hxc}}\delta n)$ for $\delta n$ yields $\delta n = R\,\delta V_{\text{ext}}$, with $R = R^0 + R^0 f_{\text{Hxc}} R$ the Dyson equation Eq. (4.70) at zero frequency. Inserting $\partial n = R\,\partial V_{\text{ext}}$ into the first term of Eq. (25.3), the electronic force constant is

$$\begin{aligned} N\sqrt{M_\kappa M_{\kappa'}}\,D^{\mathrm{dyn,el},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{Q}) = &\iint \mathrm{d}^3r\,\mathrm{d}^3r'\,\left[\partial_{\kappa\alpha,\boldsymbol{Q}}V_{\text{ext}}(\boldsymbol{r})\right]^* R(\boldsymbol{r},\boldsymbol{r}';\omega=0)\,\partial_{\kappa'\beta,\boldsymbol{Q}}V_{\text{ext}}(\boldsymbol{r}') \\ &+ \int \mathrm{d}^3r\,n(\boldsymbol{r})\,\partial_{\kappa\alpha,-\boldsymbol{Q}}\partial_{\kappa'\beta,\boldsymbol{Q}}V_{\text{ext}}(\boldsymbol{r}), \end{aligned}$$

the microscopic theory of force constants of Pick, Cohen and Martin [4]. The bare nuclei and the rigid ground-state density give a stiff lattice; the static response $R$ is negative, since electrons move away from where their potential energy rises, and the electrons that flow to screen the displaced nuclei soften it. In a metal the screening is strong enough at long wavelength to turn the bare ionic plasma frequency into a sound wave. Within density-functional theory the static kernel is exactly $\delta^2E_{\text{xc}}/\delta n\delta n$, so a DFPT phonon contains the full static electronic response of the functional, and its error is the error of the functional. And since the electrons are relaxed at every frozen configuration, the adiabatic phonon frequency already includes the static part of the electron–phonon self-energy; Sect. 6.4 uses this to avoid counting it twice.

### 25.1.3 The Sternheimer equation

The bubble Eq. (4.63) is a sum over pairs of occupied and empty states, and a converged sum over empty states is expensive. DFPT avoids it. Start from the change of one occupied Kohn–Sham state. Linearizing $(H_{\mathrm{KS}} - \varepsilon_v)\psi_v = 0$ leads to [2, Eq. (25)]

$$(H_{\mathrm{KS}} - \varepsilon_v)\,|\delta\psi_v\rangle = -(\delta V_{\mathrm{KS}} - \delta\varepsilon_v)\,|\psi_v\rangle, \qquad \delta\varepsilon_v = \langle\psi_v|\delta V_{\mathrm{KS}}|\psi_v\rangle. \tag{25.6}$$

Solved by expanding in eigenstates it is ordinary first-order perturbation theory, $\delta\psi_v = \sum_{m\neq v}\psi_m\langle\psi_m|\delta V_{\mathrm{KS}}|\psi_v\rangle/(\varepsilon_v - \varepsilon_m)$. In the density, $\delta\boldsymbol{n} = 2\sum_v(\psi_v^*\delta\psi_v + \psi_v\delta\psi_v^*)$ with the factor two for spin, the terms in which $m$ is another occupied state $v'$ cancel. The pair $(v, v')$ contributes $\psi_v^*\psi_{v'}\langle\psi_{v'}|\delta V_{\mathrm{KS}}|\psi_v\rangle/(\varepsilon_v - \varepsilon_{v'})$; the complex conjugate of the pair $(v', v)$ has the same numerator, since $\delta V_{\mathrm{KS}}$ is Hermitian, and the opposite denominator. A rotation among occupied states does not change the density, and only the component of $\delta\psi_v$ in the empty manifold matters [2, Eq. (29)]. Call it $\hat{P}^{\mathrm{c}}\delta\psi_v = \sum_c \psi_c\langle\psi_c|\delta V_{\mathrm{KS}}|\psi_v\rangle/(\varepsilon_v - \varepsilon_c)$, with $\hat{P}^{\mathrm{c}} = 1 - \hat{P}^{\mathrm{occ}}$ the projector on the empty states. Acting on it with $H_{\mathrm{KS}} - \varepsilon_v$ turns each denominator into $-1$:

$$(H_{\mathrm{KS}} - \varepsilon_v)\,\hat{P}^{\mathrm{c}}|\delta\psi_v\rangle = -\sum_c |\psi_c\rangle\langle\psi_c|\delta V_{\mathrm{KS}}|\psi_v\rangle = -\hat{P}^{\mathrm{c}}\,\delta V_{\mathrm{KS}}\,|\psi_v\rangle.$$

This is a linear equation for $\hat{P}^{\mathrm{c}}\delta\psi_v$ whose right-hand side contains only the occupied state $\psi_v$. The empty states have disappeared, and so has $\delta\varepsilon_v$, which multiplies $\psi_v$ and is removed by the projector. The operator $H_{\mathrm{KS}} - \varepsilon_v$ is positive on the empty manifold of an insulator, where every eigenvalue exceeds $\varepsilon_v$, but singular or negative on the occupied one. Adding $a_{\mathrm{sh}}\hat{P}^{\mathrm{occ}}$ changes nothing for a solution that lies in the empty manifold, and with $a_{\mathrm{sh}}$ larger than the valence bandwidth every eigenvalue $\varepsilon_{v'} + a_{\mathrm{sh}} - \varepsilon_v$ on the occupied manifold becomes positive too. The operator is then positive definite, and the equation can be solved by conjugate gradients with the cost of a few applications of $H_{\mathrm{KS}}$ [2, Eq. (30)].

For a displacement wave the bookkeeping is in crystal momentum. The perturbation $\partial_{\kappa\alpha,\boldsymbol{Q}}V_{\mathrm{KS}}$ carries $\boldsymbol{Q}$, so its matrix element between $\psi_{v\boldsymbol{k}}$ and a state of crystal momentum $\boldsymbol{k}'$ vanishes unless $\boldsymbol{k}' = \boldsymbol{k} + \boldsymbol{Q}$ (Sect. 3.2). The change of $\psi_{v\boldsymbol{k}}$ lives at $\boldsymbol{k} + \boldsymbol{Q}$, the projector is the one on the occupied states at $\boldsymbol{k} + \boldsymbol{Q}$, and $\delta\varepsilon_{v\boldsymbol{k}}$ vanishes unless $\boldsymbol{Q} = 0$, where the projector removes it.

**Sternheimer equation of density-functional perturbation theory**

For each occupied state $\psi_{v\boldsymbol{k}}$ and each perturbation $(\kappa, \alpha, \boldsymbol{Q})$, the change $|\partial\psi_{v\boldsymbol{k}}\rangle$, of crystal momentum $\boldsymbol{k} + \boldsymbol{Q}$ and orthogonal to the occupied states, solves

$$(H_{\mathrm{KS}} + a_{\mathrm{sh}}\hat{P}^{\mathrm{occ}}_{\boldsymbol{k}+\boldsymbol{Q}} - \varepsilon_{v\boldsymbol{k}})\,|\partial\psi_{v\boldsymbol{k}}\rangle = -(1 - \hat{P}^{\mathrm{occ}}_{\boldsymbol{k}+\boldsymbol{Q}})\,\partial_{\kappa\alpha,\boldsymbol{Q}}V_{\mathrm{KS}}\,|\psi_{v\boldsymbol{k}}\rangle, \tag{25.7}$$

with $\hat{P}^{\mathrm{occ}}_{\boldsymbol{k}+\boldsymbol{Q}} = \sum_{v'}|\psi_{v',\boldsymbol{k}+\boldsymbol{Q}}\rangle\langle\psi_{v',\boldsymbol{k}+\boldsymbol{Q}}|$ and $a_{\mathrm{sh}}$ larger than the valence bandwidth. The first-order density is

$$\partial_{\kappa\alpha,\boldsymbol{Q}}\boldsymbol{n}(\boldsymbol{r}) = 4\sum_{v\boldsymbol{k}}\psi^*_{v\boldsymbol{k}}(\boldsymbol{r})\,\partial\psi_{v\boldsymbol{k}}(\boldsymbol{r}), \tag{25.8}$$

and Eqs. (25.5), (25.7) and (25.8) are solved to self-consistency.

Eq. (25.7) is the equation of Baroni and coworkers [2, Eq. (31)] and of Giustino [3, Eq. (48)]. The factor four in Eq. (25.8) is two for spin and two from time reversal: the term $\psi_{v\boldsymbol{k}}\,(\partial_{-\boldsymbol{Q}}\psi_{v\boldsymbol{k}})^*$ of the density equals $\psi^*_{v,-\boldsymbol{k}}\,\partial_{\boldsymbol{Q}}\psi_{v,-\boldsymbol{k}}$ and doubles the sum over $\boldsymbol{k}$ [2, Eq. (35)]. Writing $\partial\psi_{v\boldsymbol{k}} = \mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{Q})\cdot\boldsymbol{r}}\,\partial u_{v\boldsymbol{k}}$ turns every quantity into a lattice-periodic function,

and the calculation runs in the primitive cell at any $\boldsymbol{Q}$, commensurate or not, with a cost per perturbation comparable to that of the ground state.

The self-consistency loop has the structure of a ground-state calculation: Eq. (25.7) yields $\partial\psi$ for a trial potential, Eq. (25.8) the density, Eq. (25.5) a new potential. The loop starts from the bare $\partial V_{\text{ext}}$ and converges to the screened $\partial V_{\text{KS}}$, which is also the electron–phonon vertex of Sect. 6.2. Sect. 25.2 describes what a run returns, its convergence parameters and its cost.

### 25.1.4 The $2n+1$ theorem

The Hellmann–Feynman theorem gets a first derivative of the energy from the unperturbed state alone, and Eq. (25.3) gets a second derivative from the first-order density. Both are the lowest cases of one statement. Let a perturbation of strength $\lambda$ change the energy functional to $E[\psi;\lambda]$, where $\psi$ stands for all the occupied orbitals, and let $\psi(\lambda)$ be its minimum, so that the ground-state energy is $E_0(\lambda) = E[\psi(\lambda);\lambda]$. Expand $\psi(\lambda) = \psi^{(0)} + \lambda\psi^{(1)} + \lambda^2\psi^{(2)} + \dots$ and $E_0(\lambda) = \sum_l \lambda^l E^{(l)}$.

**$2n+1$ theorem**

The derivatives of the wavefunctions up to order $n$ determine the derivatives of the energy up to order $2n+1$ [5, 6], [2, Sect. II.F.1].

The proof uses only stationarity. Truncate the expansion of the orbitals at order $n$, $\tilde{\psi} = \sum_{l\le n} \lambda^l \psi^{(l)}$, so that $\tilde{\psi} - \psi(\lambda) = O(\lambda^{n+1})$. Since $E[\,\cdot\,;\lambda]$ is stationary at $\psi(\lambda)$, its first variation there vanishes and a change of the orbitals by $\delta\psi$ changes it only at second order, $E[\psi(\lambda)+\delta\psi;\lambda] - E_0(\lambda) = O(\delta\psi^2)$. With $\delta\psi = O(\lambda^{n+1})$ the error is $O(\lambda^{2n+2})$, and $E[\tilde{\psi};\lambda]$, expanded in $\lambda$, agrees with $E_0(\lambda)$ through order $\lambda^{2n+1}$. The orbitals must stay orthonormal, which the truncated $\tilde{\psi}$ does only to order $n$. The argument goes through for a functional written for non-orthonormal orbitals, or with Lagrange multipliers expanded in the same way; this is the part Gonze worked out for density-functional theory at arbitrary order [5].

For $n = 0$ the theorem is Eq. (5.10): the unperturbed orbitals fix the energy to first order. For $n = 1$ the first-order orbitals of the Sternheimer equation fix the energy to third order. The second derivatives are the force constants, and the third are the cubic constants $\Phi^{(3)}$ that make phonons decay (Sect. 5.6); one DFPT run per wavevector thus contains, in principle, the anharmonic couplings as well, without a second-order Sternheimer equation [2, Eq. (123)]. For two different perturbations the same reasoning shows that a mixed second derivative needs the first-order response to only one of them, as Eq. (25.3) needed $\partial\boldsymbol{n}/\partial u_j$ but not $\partial\boldsymbol{n}/\partial u_i$. The force on a nucleus per unit electric field, the Born charge of Sect. 5.4, can therefore be computed from the response to the displacement or from the response to the field, whichever is cheaper.

Stationarity also controls the numerical error. The second-order energy can be written as a functional of the trial $\delta\psi$ that is stationary at the solution of Eq. (25.7), and its error is then quadratic in the error of $\delta\psi$ [2, Sect. II.C.3]. The expression Eq. (25.3), which codes usually evaluate, is not of this form: it is linear in $\partial\boldsymbol{n}$, and an error in the first-order density enters the force constant linearly. That is why the self-consistency of a phonon calculation is converged far more tightly than that of a ground state.

### 25.1.5 Interpolation and the sum rule

A dispersion curve, a density of states or a Debye–Waller sum needs $D^{\text{dyn}}(\boldsymbol{Q})$ at thousands of wavevectors, and a Sternheimer calculation at each is out of the question. The way out is

Eq. (25.2) read backward. Compute $D^{\mathrm{dyn}}$ by DFPT on the uniform $N_Q^3$ grid. The inverse of Eq. (5.16) transforms it to real-space force constants [2, Eq. (99)],

$$\Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\mathbf{0},\boldsymbol{R}) = \frac{\sqrt{M_\kappa M_{\kappa'}}}{N_Q^3} \sum_{\boldsymbol{Q}\in\mathrm{grid}} D^{\mathrm{dyn},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{Q})\, \mathrm{e}^{-\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}. \tag{25.9}$$

Inserting Eq. (5.16) and using $N_Q^{-3}\sum_{\boldsymbol{Q}\in\mathrm{grid}} \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot(\boldsymbol{R}'-\boldsymbol{R})} = \sum_{\boldsymbol{T}} \delta_{\boldsymbol{R}',\boldsymbol{R}+\boldsymbol{T}}$, where $\boldsymbol{T}$ runs over the translations of the $N_Q^3$ supercell, the transform returns

$$\frac{\sqrt{M_\kappa M_{\kappa'}}}{N_Q^3} \sum_{\boldsymbol{Q}\in\mathrm{grid}} D^{\mathrm{dyn},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{Q})\, \mathrm{e}^{-\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}} = \sum_{\boldsymbol{T}} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\mathbf{0},\boldsymbol{R}+\boldsymbol{T}),$$

the aliased force constants Eq. (25.2) of a frozen-phonon calculation in the same supercell. DFPT on an $N_Q^3$ grid and finite displacements in an $N_Q^3$ supercell contain exactly the same information. They differ in cost, and DFPT can add any other $\boldsymbol{Q}$ at will. To interpolate, each aliased constant is assigned to the image $\boldsymbol{R}$ that lies in the Wigner–Seitz cell of the supercell, with equal weights on images at equal distance, and Eq. (5.16) is summed with these constants at any $\boldsymbol{Q}$ [2, Sect. II.D.3]. The result is exact on the grid. Off the grid its error is the part of $\Phi^{(2)}$ that reaches beyond the Wigner–Seitz cell, which the aliasing has folded back onto shorter vectors. The interpolation therefore works as well as the force constants are short ranged, and their range is the smoothness of $D^{\mathrm{dyn}}(\boldsymbol{Q})$ seen from the other side of the Fourier transform. In a nonpolar insulator, where $D^{\mathrm{dyn}}$ is a smooth function of $\boldsymbol{Q}$, the constants fall off rapidly with distance. In a metal the Kohn anomalies, kinks of $D^{\mathrm{dyn}}$ at wavevectors that connect two parallel pieces of the Fermi surface, make them oscillate and decay slowly. In a polar crystal the dipoles of the displaced charges make them fall as the inverse cube of the distance, and $D^{\mathrm{dyn}}$ is nonanalytic at $\boldsymbol{Q}=0$. The remedy there is to subtract the dipole part, whose form is known exactly, interpolate the short-ranged remainder, and add the dipole part back (Sect. 5.4, Eq. (25.13)).

Fig. 25.1 tests this on silicon, with grids cut out of the single $6^3$ calculation of Fig. 5.3. The $2^3$ grid has three inequivalent points, Γ, X and L. At those points its frequencies agree with the $6^3$ ones to the last digit, 17.06 meV for the transverse acoustic mode at X, as exactness on the grid requires. Between them it fails: at the three test points off all grids, its frequencies differ from direct DFPT by up to 4.0 meV, a quarter of the transverse acoustic frequency. The $3^3$ grid, with four inequivalent points, misses X and puts the transverse acoustic mode there 14 % low, at 14.75 meV; at the test points, however, it is within 1.2 meV. The $6^3$ grid reproduces them to 0.3 meV, 1.5 % at worst, which is below the 2.6 % mean difference between this functional and experiment (Sect. 5.2). Panel (c) shows why. The on-site block is 23.3 eV Å$^{-2}$ and the nearest-neighbor block 8.0 eV Å$^{-2}$. Beyond the second neighbors most blocks fall below 0.1 eV Å$^{-2}$, with one family of exceptions: the neighbors along the ⟨110⟩ zigzag chains of bonds, 0.59 eV Å$^{-2}$ at $a/\sqrt{2} = 3.84$ Å, 0.21 eV Å$^{-2}$ at $\sqrt{19}\,a/4 = 5.92$ Å and still 0.12 eV Å$^{-2}$ at $\sqrt{2}a = 7.68$ Å, 1.5 % of the nearest-neighbor value at more than three bond lengths. These long-range constants along ⟨110⟩ are a known feature of the diamond and zincblende structures [2, Sect. V.A.1], and they are what the $2^3$ and $3^3$ supercells cannot hold. They control the flat transverse acoustic branch, which is where the coarse grids fail most.

The acoustic sum rule Eq. (5.18) is exact for $E^{\mathrm{ad}}$ but not for its computed approximation. The violation comes chiefly from the exchange–correlation energy, which is integrated on a real-space grid that is fixed in space. A rigid translation of the crystal relative to the grid changes the energy by a small periodic amount, and the computed $\sum_{\boldsymbol{R}\kappa'} \Phi^{(2)}_{\kappa\kappa'}(\mathbf{0},\boldsymbol{R})$ is a small matrix $\Delta^{\mathrm{ASR}}_\kappa$ instead of zero. At $\boldsymbol{Q}=0$ the acoustic modes then have $\omega^2 \sim \Delta^{\mathrm{ASR}}_\kappa/M_\kappa$ instead

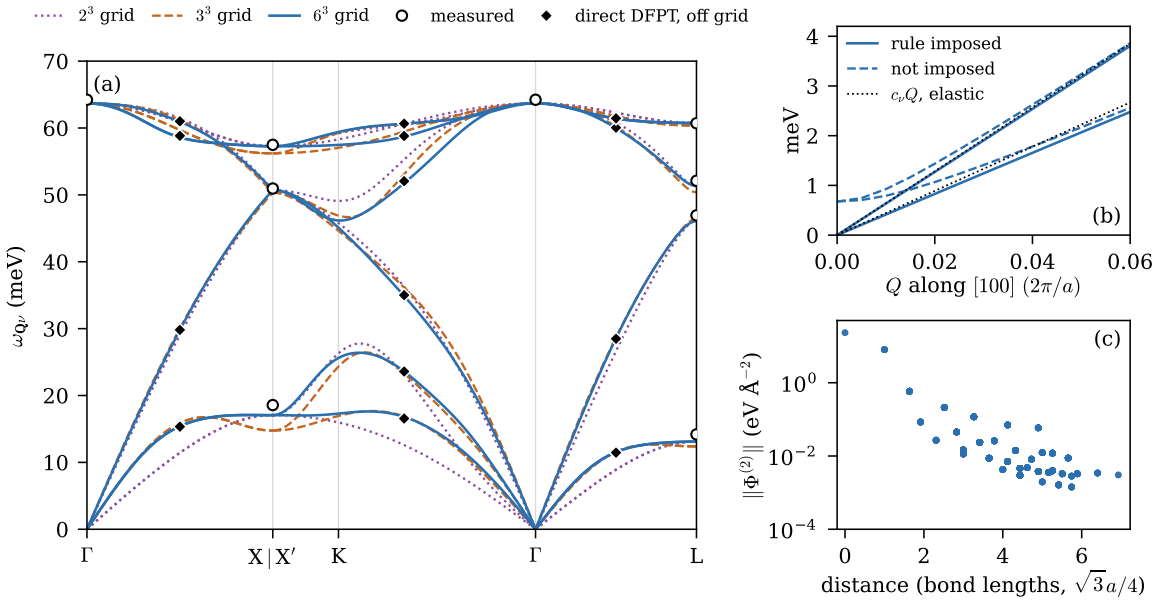



Figure 25.1: Fourier interpolation of the force constants of silicon. All curves come from one set of dynamical matrices: density-functional perturbation theory with the PBE functional, an SG15 norm-conserving pseudopotential, a 60 Ry cutoff, a $12^3$ grid of crystal momenta, $a = 5.431$ Å and the natural mass of silicon, computed on a $6^3$ grid of wavevectors (the calculation of Fig. 5.3). (a) Dispersion interpolated from the full $6^3$ grid (solid), from its $3^3$ subgrid (dashed) and from its $2^3$ subgrid (dotted), which contains only Γ, X and L; acoustic sum rule imposed in each. Diamonds: frequencies from direct calculations, with the same settings, at three wavevectors that lie on none of the grids, $(0, 0, \frac{1}{2})$, $(\frac{1}{2}, \frac{1}{2}, 0)$ and $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ in units of $2\pi/a$. Circles: frequencies measured by inelastic neutron scattering at room temperature [1, 7], as tabulated in Ref. [8, Table I]. (b) Acoustic branches along [100] near Γ from the $6^3$ grid, with the sum rule imposed (solid) and not imposed (dashed); dotted, $\omega = c_\nu Q$ with the measured sound speeds 8.43 km s$^{-1}$ (longitudinal) and 5.84 km s$^{-1}$ (transverse). (c) Norm of the $3 \times 3$ blocks $\Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\mathbf{0}, \boldsymbol{R})$ of the $6^3$ grid against the distance $|\boldsymbol{R} + \boldsymbol{\tau}_{\kappa'} - \boldsymbol{\tau}_\kappa|$. The distance is that of the shortest image in the supercell, in units of the bond length $\sqrt{3}a/4 = 2.35$ Å.

of zero, 0.675 meV (0.163 THz) in Fig. 25.1(b). The simplest repair subtracts the violation from the on-site constants,

$$\Phi^{(2),\alpha\beta}_{\kappa\kappa}(\mathbf{0},\mathbf{0}) \;\to\; \Phi^{(2),\alpha\beta}_{\kappa\kappa}(\mathbf{0},\mathbf{0}) - \sum_{\boldsymbol{R}\kappa'} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\mathbf{0},\boldsymbol{R}), \tag{25.10}$$

after which Eq. (5.18) holds exactly. The repair is not unique. When $\Delta^{\mathrm{ASR}}_{\kappa}$ is not symmetric in $\alpha\beta$, Eq. (25.10) breaks the pair symmetry of Sect. 5.2. Codes therefore also offer the smallest change of all the constants that restores both rules; in silicon $\Delta^{\mathrm{ASR}}_{\kappa}$ is a multiple of the identity by the cubic site symmetry, and the on-site repair keeps it. Neither repair imposes the rotational invariance of Sect. 5.2.

The repair matters most where dark matter looks. Near $\Gamma$ the unrepaired branches follow $\omega^2 \approx \omega^2_{\mathrm{ASR}} + c^2_\nu Q^2$, with $\omega_{\mathrm{ASR}} = 0.675\,\mathrm{meV}$, and they meet the straight lines only for $\omega \gg \omega_{\mathrm{ASR}}$. At $Q = 0.02\,(2\pi/a)$ the transverse acoustic frequency is 1.07 meV instead of 0.83 meV, 29 % high; at $Q = 0.1\,(2\pi/a)$ the difference has dropped to 1 %. With the rule imposed, the slopes along [100] are 5.44 and $8.34\,\mathrm{km\,s^{-1}}$, the PBE sound speeds of Sect. 5.2. An acoustic-phonon search with a meV threshold samples exactly the region that the unrepaired constants get wrong (Sect. 9.2). The size of $\omega_{\mathrm{ASR}}$ before the repair is also a cheap diagnostic of the real-space grid.

## 25.2 Density-functional perturbation theory in practice

A converged run of the self-consistent Sternheimer equation of Sect. 25.1 returns the screened potential, from which the vertex follows; the dynamical matrix; and, at $\boldsymbol{Q} = 0$, the response to a uniform field.

### 25.2.1 Outputs of a linear-response run

One perturbation is labeled by a phonon wavevector $\boldsymbol{Q}$, a nucleus $\kappa$ and a Cartesian direction $\alpha$. For each one, Eq. (25.7), the first-order density Eq. (25.8) [3, Eq. (49)] and the screened potential Eq. (25.5) are iterated to self-consistency, exactly as a ground-state calculation is, and the cost of one perturbation is of the same order as that of the ground-state density [2, Sect. II.C].

The screened potential $\partial_{\kappa\alpha,\boldsymbol{Q}} V_{\mathrm{KS}}$ comes for all $3N_{\mathrm{at}}$ perturbations at this $\boldsymbol{Q}$, with $N_{\mathrm{at}}$ the number of atoms per cell, and its matrix elements between Bloch states, weighted by the polarization vectors, are the electron–phonon vertex $g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})$ of Eq. (A.25). The dynamical matrix $D^{\mathrm{dyn}}(\boldsymbol{Q})$ of Eq. (A.22) follows from the first-order density by Eq. (25.3). Diagonalizing it yields $\omega_{\boldsymbol{Q}\nu}$ and $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})$. The response to a uniform electric field gives the electronic dielectric tensor $\varepsilon_\infty$ and the Born effective charges $Z^*_\kappa$. DFPT frequencies inherit the strengths and the failures of the ground state (Sect. 23.3).

### 25.2.2 Born charges and dielectric tensors

A uniform electric field $\boldsymbol{\mathcal{E}}$ cannot be written as a periodic potential, since $-e\boldsymbol{\mathcal{E}}\cdot\boldsymbol{r}$ grows without bound. DFPT avoids the problem because Eq. (25.7) needs only matrix elements of the perturbation between different eigenstates, and those of $\boldsymbol{r}$ are well defined through the commutator,

$$\langle \psi_{c\boldsymbol{k}} | \boldsymbol{r} | \psi_{v\boldsymbol{k}} \rangle = \frac{\langle \psi_{c\boldsymbol{k}} | [H_{\mathrm{KS}}, \boldsymbol{r}] | \psi_{v\boldsymbol{k}} \rangle}{\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}}}, \tag{25.11}$$

which is the relation between the position and velocity matrix elements of Sect. 3.2 [2, Eq. (37)]. For a local potential $[H_{\mathrm{KS}}, \boldsymbol{r}] = -\mathrm{i}\boldsymbol{p}/m_e$; a nonlocal pseudopotential (Sects. 24.1 and 24.2)

adds its own commutator, and omitting it biases every quantity built from the optical limit, the dielectric constant first among them. The projected state $\hat{P}^{\mathrm{c}}\boldsymbol{r}|\psi_{v\boldsymbol{k}}\rangle$ is itself the solution of a Sternheimer equation with $[H_{\mathrm{KS}},\boldsymbol{r}]|\psi_{v\boldsymbol{k}}\rangle$ on the right-hand side [2, Eq. (44)]. With it, the field enters Eq. (25.7) like any other perturbation, and its self-consistent solution is the induced macroscopic polarization $\boldsymbol{P}$.

The macroscopic quantities then follow by differentiation. In the Heaviside–Lorentz units of the book (Appendix A.2), with $\boldsymbol{D}=\boldsymbol{\mathcal{E}}+\boldsymbol{P}$,

$$\varepsilon_{\infty}^{\alpha\beta}=\delta_{\alpha\beta}+\left.\frac{\partial P^{\alpha}}{\partial \mathcal{E}^{\beta}}\right|_{\boldsymbol{u}=0}, \qquad e\,Z_{\kappa}^{*,\alpha\beta}=\Omega\left.\frac{\partial P^{\beta}}{\partial u_{\kappa}^{\alpha}}\right|_{\boldsymbol{\mathcal{E}}=0}=\left.\frac{\partial F_{\kappa}^{\alpha}}{\partial \mathcal{E}^{\beta}}\right|_{\boldsymbol{u}=0}, \tag{25.12}$$

where $\Omega$ is the cell volume and $\boldsymbol{u}_{\kappa}$ a uniform displacement of sublattice $\kappa$; Ref. [2, Eqs. (94)–(96)] writes the same relations in Gaussian units, with $4\pi$ in front of $\partial P/\partial\mathcal{E}$. The equality of the two forms of $Z_{\kappa}^{*}$ in Eq. (25.12), as the polarization per displacement and as the force per field, holds because both are the mixed second derivative of one energy, so a code may compute whichever is cheaper. The quantities $\varepsilon_{\infty}$ and $Z_{\kappa}^{*}$ are all that the nonanalytic part of the dynamical matrix needs (Sect. 5.4): in Heaviside–Lorentz units

$$D_{\kappa\kappa'}^{\mathrm{dyn,na},\alpha\beta}(\boldsymbol{Q}\to 0)=\frac{e^{2}}{\Omega\sqrt{M_{\kappa}M_{\kappa'}}}\,\frac{(Z_{\kappa}^{*}\cdot\boldsymbol{Q})_{\alpha}\,(Z_{\kappa'}^{*}\cdot\boldsymbol{Q})_{\beta}}{\boldsymbol{Q}\cdot\varepsilon_{\infty}\cdot\boldsymbol{Q}}, \tag{25.13}$$

which is Ref. [2, Eq. (98)] with $4\pi e_{\mathrm{G}}^{2}\to e^{2}$ and the mass factors of Eq. (A.22). The same two tensors build the long-range vertex $g^{\mathrm{L}}$ of Sect. 6.2 and of Sect. 27.2.

Measured frequencies give an independent check on $Z^{*}$. For a cubic crystal with two atoms per cell, $Z_{1}^{*}=-Z_{2}^{*}=Z^{*}$ times the identity, and Eq. (25.13) gives the LO–TO splitting of Sect. 5.4,

$$\omega_{\mathrm{L}}^{2}-\omega_{\mathrm{T}}^{2}=\frac{e^{2}Z^{*2}}{\Omega\,\varepsilon_{\infty}\,\mu_{\mathrm{red}}}, \tag{25.14}$$

with $\mu_{\mathrm{red}}=M_{1}M_{2}/M_{\mathrm{cell}}$ the reduced mass of the pair. With the measured $\omega_{\mathrm{L}}$, $\omega_{\mathrm{T}}$ and $\varepsilon_{\infty}$ (or $\varepsilon_{0}$, through the Lyddane–Sachs–Teller relation of Sect. 5.4), Eq. (25.14) fixes $|Z^{*}|$. Where a rate depends on a near-cancellation, as the electron-coupled scalar in GaAs does through the vertex $Z_{\kappa}-Z_{\kappa}^{*}$ (Sect. 11.8), this check matters more than any convergence parameter. A rate $\propto(Z_{\kappa}-Z_{\kappa}^{*})^{2}$ has relative error $2\,\delta Z_{\kappa}^{*}/|Z_{\kappa}-Z_{\kappa}^{*}|$, so a one-percent error in $Z_{\kappa}^{*}$ becomes a large error in the rate when $|Z_{\kappa}-Z_{\kappa}^{*}|$ is a small fraction of $Z_{\kappa}^{*}$.

The Berry-phase theory of polarization (Sect. 5.4) is the other route to $Z_{\kappa}^{*}$. The electronic polarization of an insulator is

$$\boldsymbol{P}_{\mathrm{el}}=-\frac{e\,g_{\mathrm{deg}}}{(2\pi)^{3}}\sum_{v}\int_{\mathrm{BZ}}\mathrm{d}^{3}k\,\langle u_{v\boldsymbol{k}}|\mathrm{i}\nabla_{\boldsymbol{k}}|u_{v\boldsymbol{k}}\rangle, \tag{25.15}$$

with $g_{\mathrm{deg}}=2$ for spin, and the integrand a Berry connection of the periodic parts $u_{v\boldsymbol{k}}$ [9, Eqs. (89)–(90)]. On a grid of $\boldsymbol{k}$ the derivative is replaced by the phase of a product of overlaps $\langle u_{v\boldsymbol{k}}|u_{v'\boldsymbol{k}+\Delta\boldsymbol{k}}\rangle$ along strings of points, which is gauge invariant. A Born charge is a difference of two polarizations at small displacements of opposite sign, $\pm\delta u$, and the two must sit on the same branch of $\boldsymbol{P}_{\mathrm{el}}$, which is defined only modulo the quantum $eg_{\mathrm{deg}}\boldsymbol{R}/\Omega$ for a lattice vector $\boldsymbol{R}$ [9, Sect. V.A.1]; with $\delta u$ a small fraction of a bohr this is automatic. Linear response and finite differences of Eq. (25.15) agree within numerical accuracy [2, Sect. II.D.2], and the Berry-phase route is the one that also works at finite fields and for ferroelectrics.

The acoustic sum rule for charges and the measured $\varepsilon_{\infty}$ cost nothing to check. The sum rule, $\sum_{\kappa}Z_{\kappa}^{*}=0$, holds exactly only at convergence; its violation on a finite $\boldsymbol{k}$ grid is the cleanest measure of how well $\varepsilon_{\infty}$ and $Z^{*}$ are converged. In the comparison of $\varepsilon_{\infty}$ with

experiment the functional dominates. Local and semilocal functionals put the gap too low (Sect. 3.4), the energy denominators of Eq. (25.11) come out too small, and the material is overscreened. The direction follows from the static limit of the Kramers–Kronig relation of Sect. 4.3,

$$\varepsilon_1(\omega = 0) - 1 = \frac{2}{\pi} \int_0^\infty \frac{\varepsilon_2(\omega')}{\omega'} \, \mathrm{d}\omega' : \tag{25.16}$$

shifting the absorption spectrum $\varepsilon_2$ down in energy by a fraction of an eV, at roughly fixed weight, increases the integral. For silicon, a DFT calculation finds $\varepsilon_\infty = 12.89$ [10], against the measured room-temperature value of 11.7 [11] (Sect. 4.4). When $\varepsilon_\infty$ enters a rate through Eq. (25.13) or the Fröhlich vertex, the measured value is usually the better input.

### 25.2.3 Force-constant interpolation

The Fourier interpolation of Sect. 25.1 is exact only if the true force constants vanish beyond the $N_Q^3$ supercell, so the grid is chosen large enough that they have decayed at its edge. A rule of thumb is to include neighbors out to two or three bond lengths and to check the interpolated frequencies against direct DFPT runs at a few $\boldsymbol{Q}$ off the grid [2, Sect. II.D.3]. The long-range constants along the $\langle 110 \rangle$ chains of bonds (Fig. 25.1) make silicon need larger grids than its short bonds suggest.

In a polar crystal the nonanalytic term Eq. (25.13) is subtracted from the computed $D^{\mathrm{dyn}}(\boldsymbol{Q})$ in its Ewald-summed form before the transform and added back analytically at the target $\boldsymbol{Q}$; this restores the LO–TO splitting and its dependence on the direction of $\boldsymbol{Q} \to 0$.

Codes impose the acoustic sum rule after the fact, by Eq. (25.10) or by a minimal projection. The size of the correction is a diagnostic: the acoustic frequencies at $\Gamma$ before it should be small compared with the lowest frequency the application needs, and a value that grows instead of shrinking with the cutoff signals an unconverged real-space grid. The correction is applied to $D^{\mathrm{dyn}}$ and to $Z^*$, never as a zeroing of $g$ at $\boldsymbol{Q} \to 0$: the acoustic vertex vanishes there by its own sum rule (Sect. 6.2), and its leading term, linear in $\boldsymbol{Q}$, is physical.

The cost argument for DFPT over frozen phonons is a scaling law. A frozen-phonon calculation of the force constants out to a range $R_{\mathrm{IFC}}$ (in units of the cell size) needs supercells of $\sim R_{\mathrm{IFC}}^3$ cells, $N_{\mathrm{at}} R_{\mathrm{IFC}}^3$ atoms; with ground-state cost cubic in the number of atoms and $3N_{\mathrm{at}}$ independent displacements, the total scales as $3N_{\mathrm{at}}^4 R_{\mathrm{IFC}}^9$. DFPT works in the primitive cell on a grid of $\sim R_{\mathrm{IFC}}^3$ wavevectors, each costing $\sim N_{\mathrm{at}}^3$ per perturbation, which totals $\sim 3N_{\mathrm{at}}^4 R_{\mathrm{IFC}}^3$ [2, Sect. IV.B], a factor $R_{\mathrm{IFC}}^6$ less.

### 25.2.4 Displacement correlations from DFPT phonons

The Debye–Waller exponent $2W_\kappa(\boldsymbol{q}) = \boldsymbol{q} \cdot B_\kappa \cdot \boldsymbol{q}$ and the displacement correlation $\mathcal{C}_{\kappa\kappa'}(t)$ of Sect. 5.5 are sums over all phonons of the zone. Inserting the mode expansion Eq. (A.20) into $B_\kappa = \langle \boldsymbol{u}_\kappa \boldsymbol{u}_\kappa \rangle$ and using $\langle B_{\boldsymbol{Q}\nu} B_{-\boldsymbol{Q}\nu} \rangle = 2n_{\boldsymbol{Q}\nu} + 1$ in the harmonic thermal state,

$$B_\kappa^{\alpha\beta} = \frac{1}{N} \sum_{\boldsymbol{Q}\nu} \frac{2n_{\boldsymbol{Q}\nu} + 1}{2M_\kappa \, \omega_{\boldsymbol{Q}\nu}} \, e_{\kappa\nu}^{\alpha}(\boldsymbol{Q}) \, e_{\kappa\nu}^{\beta}(\boldsymbol{Q})^*, \tag{25.17}$$

and $\mathcal{C}_{\kappa\kappa'}(t)$ is the same sum with the phases $\mathrm{e}^{\mp \mathrm{i}\omega_{\boldsymbol{Q}\nu} t}$ on the emission and absorption parts and the cross products $e_{\kappa\nu} e_{\kappa'\nu}^*$. These are post-processing sums: the frequencies and polarization vectors come from the interpolated force constants on a fine grid, and the cost is negligible. The approximation is $n_A = 0$, with the thermal occupation exact for the harmonic crystal; at high temperature or for light, soft lattices the anharmonic methods of Sect. 25.3 replace the harmonic phonons in Eq. (25.17).

The catch is the zone center, where the acoustic frequencies vanish as $\omega = c_\nu Q$ and the summand of Eq. (25.17) diverges. The divergence is integrable in three dimensions, but a uniform grid samples it badly. How badly follows from replacing the sum over the grid cell around $\Gamma$ by an integral over a sphere of the same volume, of radius $Q_c$ with $(4\pi/3)Q_c^3 = (2\pi)^3/(N_Q^3\Omega)$. With the acoustic polarization $\boldsymbol{e}_{\kappa\nu} = (M_\kappa/M_{\text{cell}})^{1/2}\hat{\boldsymbol{e}}_\nu$ of Appendix A.6, an isotropic average over directions, and $\sum_\nu$ over the three acoustic branches, the contribution of that sphere to one diagonal component is

$$\Delta B^{\alpha\alpha} = \frac{\Omega}{2\pi^2}\int_0^{Q_c} Q^2\,\mathrm{d}Q\;\frac{1}{3}\sum_\nu \frac{2n_{\boldsymbol{Q}\nu}+1}{2M_{\text{cell}}c_\nu Q} \simeq \begin{cases} \dfrac{\Omega\,Q_c^2}{24\pi^2 M_{\text{cell}}}\displaystyle\sum_\nu \frac{1}{c_\nu}, & T = 0, \\ \dfrac{\Omega\,Q_c\,T}{6\pi^2 M_{\text{cell}}}\displaystyle\sum_\nu \frac{1}{c_\nu^2}, & T \gg c_\nu Q_c, \end{cases} \tag{25.18}$$

where the high-temperature form uses $2n+1 \simeq 2T/\omega$. Since $Q_c \propto 1/N_Q$, the error of a calculation that simply omits the $\Gamma$ cell falls as $N_Q^{-2}$ at zero temperature but only as $N_Q^{-1}$ at room temperature. For silicon, with $\Omega = a^3/4 = 40.05\,\text{Å}^3$, $M_{\text{cell}}$ two silicon masses, a representative sound speed of $6\,\text{km}\,\text{s}^{-1}$ for all three branches (the range is 3 to $10\,\text{km}\,\text{s}^{-1}$ across materials, Sect. 1.3) and an $8^3$ grid, $Q_c = 0.142\,\text{Å}^{-1}$ and the room-temperature term is $\Delta B^{\alpha\alpha} \approx 3.6\times10^{-4}\,\text{Å}^2$, while the zero-temperature term is twenty times smaller, $1.9\times10^{-5}\,\text{Å}^2$. (The high-temperature form applies since $c\,Q_c$ corresponds to 65 K.) The remedy is to add Eq. (25.18), evaluated with the computed sound speeds, as an analytic correction, or to use a grid fine enough that the residual is below the target accuracy. The same care applies to $\mathcal{C}_{\kappa\kappa'}(t)$ at long times, which are controlled by the smallest frequencies.

### 25.2.5 Wavefunction changes beyond the density

The vertex of Eq. (A.25) needs only $\partial_{\kappa,\boldsymbol{Q}} V_{\text{KS}}$ and two Bloch states, so every DFPT run already contains it [3, Sect. III.B.3]. Codes write the potentials to disk for each irreducible $\boldsymbol{Q}$ of the coarse grid, one complex function on the real-space grid per perturbation, and a later step forms the matrix elements and interpolates them (Sect. 27.2). The storage is modest by modern standards: for a two-atom cell, six perturbations per $\boldsymbol{Q}$, each a complex function on a real-space grid of $\sim 10^5$ points, are about 10 MB per wavevector, so a few hundred irreducible wavevectors fit in a few gigabytes.

A transition amplitude needs more than the density does. The density vertex of Sect. 3.2, $\rho_{cv}(\boldsymbol{k},\boldsymbol{q}) = \langle\psi_{c,\boldsymbol{k}+\boldsymbol{q}}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\psi_{v\boldsymbol{k}}\rangle$, changes under a frozen displacement of crystal momentum $\boldsymbol{Q}$ by

$$\partial\rho_{cv} = \langle\partial\psi_c|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\psi_v\rangle + \langle\psi_c|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\partial\psi_v\rangle, \tag{25.19}$$

which is the adiabatic (Herzberg–Teller) form of the phonon-assisted amplitude of Sect. 13.2, valid to relative order $\omega_{\boldsymbol{Q}\nu}/\delta_{\text{off}}$, with $\delta_{\text{off}}$ the energy by which the intermediate states are off shell. Both terms need the complete first-order change of an individual state, while Eq. (25.7) returns only part of one of them. The full change of an occupied state is

$$|\partial\psi_{v\boldsymbol{k}}\rangle = \sum_{m\notin\deg(v\boldsymbol{k})} |\psi_{m,\boldsymbol{k}+\boldsymbol{Q}}\rangle\,\frac{\langle\psi_{m,\boldsymbol{k}+\boldsymbol{Q}}|\partial_{\kappa,\boldsymbol{Q}}V_{\text{KS}}|\psi_{v\boldsymbol{k}}\rangle}{\varepsilon_{v\boldsymbol{k}} - \varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}}}, \tag{25.20}$$

with the states degenerate with $\psi_{v\boldsymbol{k}}$ excluded. DFPT computes the part with $m$ empty, by Eq. (25.7), without ever forming the sum. The part with $m$ occupied, $\hat{P}^{\text{occ}}_{\boldsymbol{k}+\boldsymbol{Q}}|\partial\psi_{v\boldsymbol{k}}\rangle$, is left out because it cancels in the density (Sect. 25.1). In Eq. (25.19) these terms do not cancel, because $\psi_c$ singles out one state. They are, however, cheap. The sum runs over the few occupied bands at $\boldsymbol{k}+\boldsymbol{Q}$, and the numerators are the occupied–occupied matrix elements of the potential

that DFPT has already stored. Two cautions apply. The denominators $\varepsilon_{v\boldsymbol{k}} - \varepsilon_{v',\boldsymbol{k}+\boldsymbol{Q}}$ vanish wherever two valence bands at $\boldsymbol{k}$ and $\boldsymbol{k}+\boldsymbol{Q}$ cross, and near such crossings the adiabatic form fails: there the dynamical denominators $\varepsilon_{v\boldsymbol{k}} - \varepsilon_{v',\boldsymbol{k}+\boldsymbol{Q}} \pm \omega_{\boldsymbol{Q}\nu}$ of the full amplitude (Sects. 13.2 and 13.6) must be used instead. And at $\boldsymbol{Q}=0$ the degenerate subspace has to be treated in the same gauge as the states that enter $\rho_{cv}$ (Appendix A.5).

The change of an empty state is not produced by standard DFPT at all, since empty states do not contribute to the density. Its equation is a Sternheimer equation with the converged potential on the right,

$$(H_{\mathrm{KS}} - \varepsilon_{c\boldsymbol{k}})|\partial\psi_{c\boldsymbol{k}}\rangle = -(1 - \hat{P}_c^{\mathrm{deg}})\,\partial_{\kappa,\boldsymbol{Q}} V_{\mathrm{KS}}\,|\psi_{c\boldsymbol{k}}\rangle, \tag{25.21}$$

with $\hat{P}_c^{\mathrm{deg}}$ the projector on the states at $\boldsymbol{k}+\boldsymbol{Q}$ degenerate with $\psi_{c\boldsymbol{k}}$. No self-consistency is needed, because $\partial V_{\mathrm{KS}}$ is already the screened potential of the occupied manifold, and that is the main saving. The operator on the left, however, is indefinite: $\varepsilon_{c\boldsymbol{k}}$ lies inside the spectrum of $H_{\mathrm{KS}}$, with all the valence states and part of the conduction bands below it, and the conjugate-gradient solvers used for Eq. (25.7) do not converge on an indefinite operator. One way out splits the space: a window $\mathcal{W}$ of bands at $\boldsymbol{k}+\boldsymbol{Q}$, containing every state below $\varepsilon_{c\boldsymbol{k}}$ and the conduction bands up to some energy above it, is summed explicitly with Eq. (25.20), where the Wannier-interpolated vertices of Sect. 27.2 make the terms cheap. The remainder obeys

$$\hat{P}_{\mathcal{W}}^{\perp}(H_{\mathrm{KS}} - \varepsilon_{c\boldsymbol{k}})\hat{P}_{\mathcal{W}}^{\perp}\,|\partial\psi_{c\boldsymbol{k}}^{\perp}\rangle = -\hat{P}_{\mathcal{W}}^{\perp}\,\partial_{\kappa,\boldsymbol{Q}} V_{\mathrm{KS}}\,|\psi_{c\boldsymbol{k}}\rangle, \tag{25.22}$$

with $\hat{P}_{\mathcal{W}}^{\perp}$ the projector on the complement of the window. Every state left in the complement lies above $\varepsilon_{c\boldsymbol{k}}$, so the projected operator is positive definite and the standard solvers apply. The remainder carries the high-energy intermediate states, which matter for large momentum transfers, where $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ couples to bands far above the gap. The window is a convergence parameter: the sum of the explicit and the Sternheimer parts must not depend on it.

The same equations at finite frequency give dynamical response without empty states. For a perturbation oscillating as $\mathrm{e}^{-\mathrm{i}\omega t}$, first-order time-dependent perturbation theory (Sect. 2.4) leads to two components of the change of each occupied state,

$$(H_{\mathrm{KS}} - \varepsilon_{v\boldsymbol{k}} \mp \omega)|\partial\psi_{v\boldsymbol{k}}^{\pm}\rangle = -(1 - \hat{P}^{\mathrm{occ}})\,\partial V_{\mathrm{KS}}(\pm\omega)\,|\psi_{v\boldsymbol{k}}\rangle, \tag{25.23}$$

with the density change built from both and the potential updated as in Eq. (25.5) with an adiabatic kernel. Below the gap the operators remain positive definite on the empty manifold, because $\varepsilon_c - \varepsilon_v - \omega > 0$ for every pair, and the solution is as cheap as a static one. Above the gap a small broadening $\eta_{\mathrm{b}}$, $\omega \to \omega + \mathrm{i}\eta_{\mathrm{b}}$, is added and the linear systems become complex and non-Hermitian. Frequency-dependent Sternheimer calculations of this kind give the inverse frequency moments and the drag mass of Sect. 12.5, and the screened interaction in Sternheimer formulations of the methods of Sect. 23.4.

### 25.2.6 Elastic constants

The sound speeds of Sect. 11.1 follow from the elastic tensor $C_{ijkl}$ through the Christoffel equation of Appendix C.18; for a cubic crystal along [100] it reduces to $\rho_T c_{\mathrm{LA}}^2 = C_{11}$ and $\rho_T c_{\mathrm{TA}}^2 = C_{44}$ in Voigt notation. Three routes give $C_{ijkl}$. The slopes of the interpolated acoustic branches are the cheapest and the least accurate, since the small-$Q$ part of the dispersion is precisely where the Fourier interpolation of Eq. (25.9) is weakest, and the slopes converge only slowly with $N_Q$. Finite strains are the most reliable: strains of a few tenths of a percent of both signs are applied, the internal coordinates are relaxed at each strain, and the stress is fitted. The internal relaxation matters for any structure with internal degrees of freedom, and in the diamond structure it affects $C_{44}$ because the two sublattices shift against

each other under shear. Since relaxation can only lower the energy at fixed strain, the relaxed constants are never larger than the clamped ones. DFPT with strain as the perturbation returns the clamped-ion tensor and the internal-strain couplings in one run, from which the relaxed tensor follows by eliminating the internal displacements [2, Sect. II.E.1].

A cheap check against the Voigt-order error of Sect. 11.1 is to compute $\rho_T c^2$ from the tensor along one direction and compare it with the slope of the computed dispersion along the same direction.

### 25.2.7 Convergence parameters and cost

Five numerical parameters control a DFPT calculation. Each must be converged on the quantity of interest, not on the total energy, and with a tolerance fixed before the scan. The plane-wave cutoff of Sect. 23.3 sets the basis; the test is the frequency at $\Gamma$ and at one zone-boundary point at two cutoffs. The $\boldsymbol{k}$ grid sets the Brillouin-zone sum in Eq. (25.8); for an insulator the density and the frequencies converge on modest grids, but $\varepsilon_\infty$ and $Z^*$ need noticeably finer ones, because the response to a field involves the velocity matrix elements and energy denominators of Eq. (25.11) at every $\boldsymbol{k}$, and these converge slowly where the direct gap is small [2, Sect. II.C.2]. Metals need smearing of the occupations (Sect. 23.3) and a correspondingly finer grid. The $\boldsymbol{Q}$ grid sets the range of the interpolated force constants and is tested by direct runs off the grid. The self-consistency threshold of Eq. (25.5) must be much tighter than in a ground-state run, because an error in the first-order density enters the force constants linearly (Sect. 25.1). Where a frequency is small, as for the acoustic modes near $\Gamma$ or a soft mode, a small error in the force constant is a large relative error in the frequency, or makes it imaginary.

The lattice constant is a parameter in its own right. Phonons are usually computed at the theoretical equilibrium of the functional. In diamond and zincblende crystals the forces vanish by symmetry at any lattice constant, but only at the equilibrium does the stress vanish, so that the second derivatives are taken at a minimum of the same energy surface. Each frequency moves with the lattice constant through its Grüneisen parameter (Sect. 25.3); the generalized-gradient approximation, which expands lattice constants, lowers the branches with positive Grüneisen parameter, and for silicon the local-density approximation already reproduces the measured dispersion closely while the generalized-gradient approximation slightly worsens it [2, Sect. V.A].

The cost of a complete phonon calculation is the number of irreducible wavevectors on the $\boldsymbol{Q}$ grid, times $3N_{\mathrm{at}}$ perturbations (fewer, by symmetry, at high-symmetry wavevectors), times the cost of a ground-state calculation with the reduced symmetry of the perturbed crystal. That reduced symmetry is the hidden factor: a perturbation at a general $\boldsymbol{Q}$ leaves few symmetry operations, so the $\boldsymbol{k}$ sum runs over many more points than in the ground state. An $N_Q^3$ grid has about $N_Q^3/48$ irreducible points for a cubic crystal when $N_Q$ is large; an $8^3$ grid for silicon has 29, a count that follows from applying the 48 operations of the cubic group to the grid points and counting the distinct orbits. With six perturbations per wavevector, fewer at the high-symmetry points, and the three field perturbations at $\Gamma$, a complete silicon calculation on that grid costs at most about 180 self-consistent linear-response solutions, each of the order of one ground-state calculation at reduced symmetry. The cost grows with the fourth power of the number of atoms per cell [2, Sect. IV.B].

**Worked example 25.2: A DFPT calculation of silicon**

We assemble the settings of two published silicon calculations and the checks they pass. The first used the local-density approximation with plane-wave pseudopotentials, a

35 Ry cutoff and a $14^3$ grid for the ground-state density, and computed the dynamical matrices and the potential changes on a $6^3$ grid of wavevectors [12]. The second used the generalized-gradient approximation at 35 Ry (45 Ry for the local-density approximation), again a $6^3$ wavevector grid, and found that an $18^3$ grid of $\boldsymbol{k}$ points was needed for the Born charges to vanish [10]. A 35 Ry cutoff corresponds to plane waves up to $|\boldsymbol{G}| = (2m_e \times 476\,\mathrm{eV})^{1/2} = 22\,\mathrm{keV}$, between five and six times the shortest reciprocal-lattice vector $|\boldsymbol{G}_{111}| = \sqrt{3}\,2\pi/a = 3.95\,\mathrm{keV}$ (Sect. 3.1), which is ample for phonons but, as Sects. 26.1 and 26.2 explain, not automatically for a density vertex at large momentum transfer.

The vanishing of the Born charges is a symmetry statement that the numerics must reproduce. The two atoms of the diamond structure are exchanged by inversion through the bond center. $Z^*_\kappa$ is the derivative of a polar vector with respect to a polar vector, which is even under inversion, so $Z^*_1 = Z^*_2$. Each atom sits at a site of tetrahedral symmetry, so each tensor is a multiple of the identity. Charge neutrality, $Z^*_1 + Z^*_2 = 0$, then forces both to vanish. On a coarse $\boldsymbol{k}$ grid they do not, and since $Z^*$ enters Eq. (25.13), a residual Born charge produces a spurious LO–TO splitting in a crystal that has none. The $\boldsymbol{k}$ grid is therefore fixed by the Born charge, not by the frequencies.

The dielectric constant is the second check, and here the functional limits the result. The PBE calculation of Worked example 23.1 reaches 13.04 at the measured lattice constant on an $18^3$ grid, within about one percent of the 12.89 of Ref. [10]. The 11 % excess over the measured 11.7 is the overscreening of Eq. (25.16) and does not shrink with any numerical parameter. The frequencies are the third check. The measured optical frequency at $\Gamma$ is 15.5 THz [1], and the local-density dispersion agrees with neutron scattering across the zone [2, Sect. V.A].

## 25.3 Anharmonic and finite-temperature methods

The band gap of silicon near zero temperature is about 60 meV smaller than the gap of the same crystal with its nuclei clamped at their equilibrium positions: the zero-point motion of the lattice closes it, and the measured values are 62 and 64 meV [13, Table VI]. Between zero and room temperature the gap closes by a further 50 meV (Sect. 1.3), and the phonons themselves shift and broaden. A detector at 10 mK sits deep in the zero-point regime, where none of these effects are thermal and all of them are quantum. The anharmonic small parameter $\eta_{\mathrm{anh}}$ of Sect. 5.6 is small for silicon, and the lattice-side corrections are correspondingly small; the electron-side ones are not.

### 25.3.1 Quasi-harmonic approximation

In the quasi-harmonic approximation the free energy per cell, $\mathcal{F}$, at cell volume $\Omega$ is the harmonic one, with frequencies computed at that volume [14, Eq. (4)],

$$\mathcal{F}(\Omega, T) = E_0^{\mathrm{ad}}(\Omega) + \frac{1}{N}\sum_{\boldsymbol{Q}\nu} \frac{\omega_{\boldsymbol{Q}\nu}(\Omega)}{2} + \frac{T}{N}\sum_{\boldsymbol{Q}\nu} \ln\bigl[1 - \mathrm{e}^{-\omega_{\boldsymbol{Q}\nu}(\Omega)/T}\bigr], \tag{25.24}$$

where $E_0^{\mathrm{ad}}(\Omega)$ is the clamped-nucleus energy per cell, the minimum of the energy surface of Sect. 5.1 at that volume, and the sums run over the $N$ points of the phonon grid. The equilibrium volume minimizes $\mathcal{F}$ at each $T$. Differentiating, the volume thermal expansion

coefficient is

$$\alpha_{\mathrm{th}} = \frac{1}{NB\Omega} \sum_{\boldsymbol{Q}\nu} \omega_{\boldsymbol{Q}\nu}\, \gamma^{\mathrm{G}}_{\boldsymbol{Q}\nu}\, \frac{\partial n_{\boldsymbol{Q}\nu}}{\partial T}, \qquad \gamma^{\mathrm{G}}_{\boldsymbol{Q}\nu} = -\frac{\Omega}{\omega_{\boldsymbol{Q}\nu}} \frac{\partial \omega_{\boldsymbol{Q}\nu}}{\partial \Omega}, \tag{25.25}$$

with $B$ the isothermal bulk modulus and $\gamma^{\mathrm{G}}_{\boldsymbol{Q}\nu}$ the mode Grüneisen parameters [14, Eqs. (6) and (9)]. A perfectly harmonic crystal has $\gamma^{\mathrm{G}} = 0$ and no thermal expansion. The calculation is a set of harmonic calculations: DFPT phonons at five to seven volumes around the minimum, a fit of $E_0^{\mathrm{ad}}(\Omega)$ and of each $\omega_{\boldsymbol{Q}\nu}(\Omega)$, and a minimization of Eq. (25.24). The cost is that number of DFPT runs. In silicon the transverse acoustic branch has negative Grüneisen parameters, and its large weight in the density of states at low frequency makes the thermal expansion negative at low temperature, a behavior the quasi-harmonic approximation reproduces [14, Sect. 5.1].

The approximation holds as long as the frequencies depend on the volume but not on the amplitude of vibration, that is, as long as the phonons are well-defined harmonic excitations at each volume. It fails for modes that are unstable in the harmonic approximation, near structural transitions and near melting, which is where the next two methods take over. The zero-point term of Eq. (25.24) expands the lattice at $T = 0$, and this expansion changes the gap: for silicon the zero-point lattice expansion alone shifts the gap by $+9\,\mathrm{meV}$ [15, Table S2], opposite to the electron–phonon renormalization. And the same Grüneisen parameters set how sensitive a phonon calculation is to the lattice constant it uses: by Eq. (25.25), a relative error $\delta a/a$ in the lattice constant shifts each frequency by $\delta\omega_{\boldsymbol{Q}\nu}/\omega_{\boldsymbol{Q}\nu} = -3\gamma^{\mathrm{G}}_{\boldsymbol{Q}\nu}\,\delta a/a$. In silicon the transverse acoustic modes, with $\gamma^{\mathrm{G}} < 0$, and the optical modes, with $\gamma^{\mathrm{G}} > 0$, move in opposite directions, so that no single choice between the theoretical and the measured lattice constant improves both. A calculation that needs the transverse acoustic branch accurately, as the phonon-assisted thresholds of Chap. 13 do, has to test both lattice constants and quote the spread.

### 25.3.2 Self-consistent harmonic approximation

When the anharmonicity is not small, the harmonic expansion about the minimum is the wrong starting point, and the self-consistent harmonic approximation (SCHA) replaces it by the best harmonic description of the actual distribution of nuclei. It minimizes the Gibbs–Bogoliubov bound on the free energy (Sect. 5.6) [16, Eq. (2)] over the thermal states of harmonic Hamiltonians,

$$H_{\mathrm{trial}} = (\text{nuclear kinetic energy}) + \frac{1}{2} \sum_{ab} (X_a - \bar{X}_a)\, \Phi^{(2)}_{\mathrm{aux},ab}\, (X_b - \bar{X}_b), \tag{25.26}$$

with $X_a$ the $3N_{\mathrm{at}}$ Cartesian coordinates of the nuclei of a supercell of $N_{\mathrm{at}}$ atoms, and the centroids $\bar{X}_a$ and the auxiliary force constants $\Phi^{(2)}_{\mathrm{aux}}$ as variational parameters [16, Eqs. (4) and (7)]. Minimizing the bound over $\bar{X}$ and $\Phi^{(2)}_{\mathrm{aux}}$ gives the SCHA free energy, the equilibrium positions at temperature $T$, and an effective harmonic problem whose quantum zero-point motion is included by construction. At the minimum $\Phi^{(2)}_{\mathrm{aux}}$ satisfies the self-consistency condition of Sect. 5.6: it equals the average of the true second derivatives of the energy surface over the Gaussian distribution it generates.

The averages are computed stochastically. A population of $N_{\mathrm{conf}}$ configurations is drawn in a supercell from the Gaussian distribution of the current trial Hamiltonian, each together with its mirror image about the centroids to reduce the statistical error; DFT supplies the energy, forces and stress of each; and the gradients of $\mathcal{F}$ with respect to $\bar{X}$ and $\Phi^{(2)}_{\mathrm{aux}}$ follow from these forces. As the parameters move, the same configurations are reweighted by the ratio of the new to the old Gaussian density, until the effective number of samples, $N_{\mathrm{eff}} = (\sum_j w_j)^2 / \sum_j w_j^2$ for

weights $w_j$, drops below about half of $N_{\mathrm{conf}}$; then a new population is drawn [16, Eqs. (21), (38) and (39)]. The cost is entirely in the DFT forces: a population holds several hundred to several thousand configurations, and most materials converge within three populations [16, Sect. III B]. The convergence parameters are the supercell size, $N_{\mathrm{conf}}$, and the threshold on $N_{\mathrm{eff}}/N_{\mathrm{conf}}$.

The auxiliary frequencies of $\Phi^{(2)}_{\mathrm{aux}}$ are not the physical phonon energies. They describe the Gaussian distribution of the nuclei; the frequencies measured by a probe at small energy follow from the curvature of the free energy with respect to the centroids, which adds a term with third-order force constants [16, Sect. IV]. Mistaking one for the other is the most common error in reading SCHA output. For silicon the SCHA changes the phonons little; it is needed for light atoms, soft modes and quantum crystals, the targets of Sect. 11.5.

### 25.3.3 Temperature-dependent effective potential

The temperature-dependent effective potential (TDEP) method fits effective force constants to forces sampled at temperature $T$. Molecular dynamics in a supercell of $N_{\mathrm{at}}$ atoms supplies, at each of $N_t$ time steps, the displacements $\boldsymbol{u}_i$ and the DFT forces $\boldsymbol{F}_i$ on every atom $i$ of the supercell. The effective second-order force constants $\tilde{\Phi}^{(2)}_{ij}$ are those that minimize the mismatch

$$\Delta_{\mathrm{fit}} = \frac{1}{N_t}\sum_{t,i}\Big|\boldsymbol{F}_i(t) + \sum_j \tilde{\Phi}^{(2)}_{ij}\,\boldsymbol{u}_j(t)\Big|^2, \tag{25.27}$$

since a harmonic energy surface gives $\boldsymbol{F}_i = -\sum_j \tilde{\Phi}^{(2)}_{ij}\boldsymbol{u}_j$. This is a linear least-squares problem, overdetermined once $N_t > 3N_{\mathrm{at}}$ [17, Eq. (3), with the opposite sign convention for the force constants]. Symmetry reduces the unknowns to a few per coordination shell, and the same fit extends to third and fourth order. The fitted constants absorb the anharmonicity sampled at that temperature into effective harmonic ones; the phonons and free energy follow as in the harmonic case [18]. The convergence parameters are the number of time steps, which Ref. [17, Fig. 1] tracks through the change in the fitted constants, the supercell size, and the range of the force constants kept in the fit.

The method inherits the statistics of its sampling. Classical molecular dynamics samples a classical distribution, with mean squared displacement $T/(M_\kappa\omega^2_{\boldsymbol{Q}\nu})$ per mode, while the quantum harmonic value is $(2n_{\boldsymbol{Q}\nu}+1)/(2M_\kappa\omega_{\boldsymbol{Q}\nu})$. Their ratio is $2T/\omega_{\boldsymbol{Q}\nu}$ for $T \ll \omega_{\boldsymbol{Q}\nu}$. For silicon phonons of 10 to 60 meV at 10 mK, the ratio is below $2\times10^{-4}$, so a classical sample sees an almost static lattice and misses the zero-point motion that sets every renormalization at detector temperatures. At low temperature the configurations must be drawn from the quantum distribution instead, as the SCHA does, or from path-integral molecular dynamics (Sect. 29.1). The cost of TDEP is that of the molecular dynamics, $N_t$ force calculations of which only those separated by more than a correlation time are independent.

Third-order force constants also give the three-phonon linewidths $\gamma^{\mathrm{anh}}_{\boldsymbol{Q}\nu}$ of Sect. 5.6. They are fitted in the same way, or computed by finite differences in supercells, one supercell calculation per pair of displaced atoms allowed by symmetry, with a cutoff radius on the interacting triplets. The linewidth then needs a sum over pairs of final phonons on a dense $\boldsymbol{Q}$ grid with an energy delta function, and the same rules for broadening and grid spacing apply as for electronic spectra (Appendix C.2).

### 25.3.4 Allen–Heine–Cardona calculations

The AHC theory of Sect. 6.3 gives the shift of a band energy at second order in the electron–phonon coupling, summed over all phonons with the thermal weight $2n_{\boldsymbol{Q}\nu}+1$. In the adiabatic

form, which neglects the phonon energy next to electronic energy differences, the Fan–Migdal part is

$$\Delta\varepsilon^{\mathrm{FM}}_{n\boldsymbol{k}}(T) = \frac{1}{N}\sum_{\boldsymbol{Q}\nu}(2n_{\boldsymbol{Q}\nu}+1)\sum_{m}\frac{|g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})|^2}{\varepsilon_{n\boldsymbol{k}}-\varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}}}, \tag{25.28}$$

with the vertex of Eq. (A.25), and the Allen–Heine part adds the diagonal element of the second-order vertex $g^{\mathrm{AH}}$ with the same weight. In the rigid-ion approximation the second-order vertex is rewritten through first-order matrix elements at $\boldsymbol{Q}=0$, using the translation invariance of the crystal, so that no second derivative of the potential is computed [13, Eqs. (11) and (15)] (Sect. 6.3). The AHC literature calls this term the Debye–Waller term; in this book Debye–Waller refers only to $\mathrm{e}^{-2W_\kappa}$ (Appendix A.6). The nonadiabatic form keeps $\pm\omega_{\boldsymbol{Q}\nu}$ in the denominators of Eq. (25.28), separately for phonon emission and absorption, and gives the renormalization a lifetime as well [13, Eq. (16)].

The sum over $m$ runs over all bands. In practice it is split: the bands within an active window are summed explicitly, and the contribution of all higher bands is obtained from a Sternheimer equation for the first-order change of the wavefunction projected on the space above the window, as DFPT does for phonons (Sect. 25.2). The split is exact in the adiabatic limit; in the nonadiabatic form the Sternheimer part drops the phonon energy, and the active window must be large enough for that to be harmless [13, below Eq. (16)].

The sum over $\boldsymbol{Q}$ is the hard part. The Fan–Migdal denominators vanish when $\varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}}=\varepsilon_{n\boldsymbol{k}}$, which happens for $m=n$ at $\boldsymbol{Q}\to 0$ and wherever another band is degenerate with the state. Near these points the integrand is singular, and the result converges slowly with the number $N_Q^3$ of grid points, at a rate that depends on the band and on the form of the theory; a small imaginary part $\mathrm{i}\delta$ in the denominators smooths the integrand, at the price of a bias that must be extrapolated to $\delta\to 0$ together with $1/N_Q\to 0$ [13, Sect. III and Table II]; the broadening is a numerical device, never quoted as a lifetime. A nonzero Born effective charge makes the vertex diverge as $1/Q$ at small $Q$. In a nonpolar crystal the charges vanish by symmetry, but a DFPT calculation always leaves small residual charges that violate the charge-neutrality sum rule, and these must be removed before the $\boldsymbol{Q}$ sum, or the integral diverges at band extrema [13, Sect. III]. In a polar crystal the divergence is physical: the adiabatic integral does not converge at all, and only the nonadiabatic form gives a finite result [13]. For polar materials the nonadiabatic, Fröhlich-like part often dominates the zero-point renormalization [15].

The cost is that of DFPT at every $\boldsymbol{Q}$ of a dense grid: for silicon, Ref. [13] goes up to a $100\times100\times100$ grid, 22 776 irreducible points, to converge the values of its Table VI. Wannier interpolation of the vertices (Sect. 27.2) reduces this to DFPT on a coarse grid. The results for silicon are in Worked example 25.3.

### 25.3.5 Special displacements

The perturbative route needs vertices and a converged sum over phonon wavevectors. The nonperturbative alternative computes the electronic property directly in a supercell whose nuclei are displaced, and averages over displacements. In the adiabatic and harmonic approximations, the Williams–Lax expression for a temperature-dependent property, here the imaginary part of the dielectric function, is a Gaussian average over the normal coordinates $x_\nu$ of the supercell,

$$\varepsilon_2(\omega;T) = \prod_{\boldsymbol{Q}\nu}\int \mathrm{d}x_{\boldsymbol{Q}\nu}\,\frac{\mathrm{e}^{-x^2_{\boldsymbol{Q}\nu}/2\sigma^2_{\boldsymbol{Q}\nu}}}{\sqrt{2\pi}\,\sigma_{\boldsymbol{Q}\nu}}\,\varepsilon_2(\omega;\{x\}), \qquad \sigma^2_{\boldsymbol{Q}\nu}=\frac{2n_{\boldsymbol{Q}\nu}+1}{2\omega_{\boldsymbol{Q}\nu}}, \tag{25.29}$$

where the $x_{\boldsymbol{Q}\nu}$ are the mass-weighted normal coordinates of the supercell, labeled by the wavevectors $\boldsymbol{Q}$ of the primitive-cell phonons it contains, and $\varepsilon_2(\omega;\{x\})$ is computed with the

nuclei frozen at the displaced positions [19, Eqs. (3) and (4), with mass-weighted coordinates]. The same formula holds for a band energy or any other property computed at fixed nuclei.

The average needs only one deterministic configuration. In a large supercell, the average of a smooth function over the Gaussian distribution of Eq. (25.29) is dominated by configurations in which every normal coordinate has the magnitude $\sigma_{\boldsymbol{Q}\nu}$, and the cross terms between pairs of modes cancel if the signs alternate in a suitable order [19, Sect. III]. The resulting special, or ZG, displacement of nucleus $\kappa$ in cell $\boldsymbol{R}$ of a supercell of $N$ primitive cells is

$$\boldsymbol{u}_{\boldsymbol{R}\kappa}^{\mathrm{ZG}} = \sum_{\boldsymbol{Q}\in\mathcal{B},\nu} \varsigma_{\boldsymbol{Q}\nu} \left[\frac{2n_{\boldsymbol{Q}\nu}+1}{2NM_\kappa\omega_{\boldsymbol{Q}\nu}}\right]^{1/2} 2\,\mathrm{Re}\big[\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\,\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\big], \tag{25.30}$$

in the conventions of Eq. (A.20) [20, Eq. (2)]. The set $\mathcal{B}$ contains one wavevector from each pair $\pm\boldsymbol{Q}$ of the grid commensurate with the supercell, and the signs $\varsigma_{\boldsymbol{Q}\nu} = \pm 1$ alternate from one mode to the next in a prescribed order. Wavevectors equal to their own time-reversal partners, such as $\boldsymbol{Q} = 0$, are left out: they describe standing waves that break the crystal symmetry, and their weight vanishes as the supercell grows [20, Sects. II and VII B]. The mean squared displacement of each atom, averaged over the supercell, then equals its thermal value from Eq. (A.20). The method needs the phonons of the primitive cell on the commensurate grid, from DFPT, and one DFT calculation of the displaced supercell with an electronic $\boldsymbol{k}$ sampling appropriate to its size.

The convergence parameter is the supercell size. Including or excluding the wavevectors with $\boldsymbol{Q} = -\boldsymbol{Q}$ modulo $\boldsymbol{G}$ brackets the converged value from above and below, and the width of the bracket is a direct error estimate [20, Sect. VII B]. With the $\boldsymbol{Q} = -\boldsymbol{Q}$ modes included, degenerate bands such as the threefold valence maximum of silicon split, and the splitting is an artifact [20, Sect. VII B]. The phases of the eigenvectors must be fixed consistently from mode to mode before the signs are assigned; Ref. [19, Sect. II] chooses the first nonzero component of each eigenvector positive. Since the displaced supercell is a static calculation, it needs no sum over phonon wavevectors with vanishing denominators, and the adiabatic divergence of polar materials is absent: for GaAs the zero-point renormalization converges smoothly with supercell size, though a $6 \times 6 \times 6$ supercell, 432 atoms, is needed to come within ten percent [20, Sect. VII]. The cost is one supercell calculation, which scales as the cube of the number of atoms for the diagonalization; an $8 \times 8 \times 8$ supercell of silicon holds 1024 atoms and costs about $8^3 = 512$ times more than a $4 \times 4 \times 4$ one at fixed accuracy per atom.

**Phonon-assisted absorption.** With Eq. (25.29) applied to $\varepsilon_2$, a single ZG supercell gives the temperature-dependent absorption spectrum including indirect transitions. The displaced supercell breaks translational symmetry, so that a photon can connect states that differ in crystal momentum by a phonon wavevector of the supercell grid. For silicon at room temperature, an $8 \times 8 \times 8$ supercell with 40 random $\boldsymbol{k}$ points, a scissor of 0.74 eV and a Gaussian broadening of 30 meV reproduces the measured absorption coefficient over seven orders of magnitude [19, Sect. II and VII]. Being adiabatic, the method does not distinguish the absorption of a phonon from its emission; it cannot reproduce the two thresholds $E_\mathrm{g} \pm \omega_{\boldsymbol{Q}\nu}$ of the Hall–Bardeen–Blatt theory near the onset [19, Sect. II], which is precisely the region that sets the phonon-assisted thresholds of Chap. 13. At 10 mK the phonon-absorption branch is absent, and the onset is at $E_\mathrm{g} + \omega_{\boldsymbol{Q}\nu}$ (Sect. 6.5). Special displacements therefore give the spectrum and its renormalization away from the edge, while the edge itself needs the nonadiabatic treatment with explicit phonon energies of Sect. 27.2 (Sect. 13.7).

**Zero-point renormalization from special displacements.** The band energies of the displaced supercell, unfolded onto the primitive zone as a spectral function, give the renormalized

bands $\varepsilon^{\mathrm{ren}}_{n\boldsymbol{k}}(T)$ directly. Unlike the perturbative sum, this includes all orders in the displacements, within the adiabatic approximation, and needs no electron–phonon vertices. It misses the nonadiabatic correction; in silicon the adiabatic and nonadiabatic AHC renormalizations differ by 8 meV [13, Table VI].

**Worked example 25.3: The zero-point renormalization of the silicon gap**

For the indirect gap of silicon at $T = 0$, perturbative AHC theory with the local-density approximation, Sternheimer equations for the high bands and grids up to $100 \times 100 \times 100$ phonon wavevectors, extrapolated to $\delta \to 0$, gives $-64.3$ meV in the adiabatic form and $-56.2$ meV in the nonadiabatic one [13, Table VI]. The ZG displacement in an $8 \times 8 \times 8$ supercell, 1024 atoms, gives $-57$ meV, made of a rise of the valence maximum by 35 meV and a drop of the conduction minimum, near 0.87 of the way from Γ to X, by 22 meV; a $10 \times 10 \times 10$ supercell brackets the converged value between 55 and 65 meV [20, Sect. VII B]. A $4 \times 4 \times 4$ supercell, 128 atoms, is already within ten percent of the converged value [20, Sect. VII]. The measured renormalization is $-62$ or $-64$ meV, depending on how the data are extrapolated [13, Table VI].

The measured value contains both the electron–phonon renormalization and the lattice expansion, which contributes +9 meV (Eq. (25.24)). The lattice term is therefore added to the computed fixed-volume value before the two are compared [15]: $-56 + 9 = -47$ meV for the nonadiabatic AHC result and $-57 + 9 = -48$ meV for the ZG displacement, against $-62$ to $-64$ meV. Equivalently, the electron–phonon part implied by the measurement is $-71$ to $-73$ meV, and the calculations underestimate it by about 20 %, a shortfall that Sect. 6.3 traces to the overscreening of the vertex by the small Kohn–Sham gap. The cost differs by orders of magnitude: tens of thousands of DFPT calculations in the primitive cell for the perturbative route, against DFPT phonons on an $8 \times 8 \times 8$ grid and one 1024-atom supercell for the special displacement. A measured low-temperature gap, such as the 1.17 eV of silicon (Sect. 1.3), already contains the zero-point renormalization (Sect. 3.4), and a computed renormalization is added only to a clamped-nucleus gap, from *GW* or from a hybrid functional. Between 10 mK and 1 K all occupations $n_{\boldsymbol{Q}\nu}$ of silicon are of order $\mathrm{e}^{-\omega_{\boldsymbol{Q}\nu}/T}$ with $\omega_{\boldsymbol{Q}\nu}/T \gtrsim 10^2$ for every phonon above 10 meV, and the thermal part of the renormalization is negligible; only the acoustic modes at small $Q$ are thermally occupied, and their contribution to Eq. (25.28) vanishes with their phase space. What remains temperature dependent at those temperatures is the population of the final states, not their energies.

## Summary

Force constants are second derivatives of the Born–Oppenheimer energy, obtained from finite differences of forces in a supercell or from the linear response of the electrons to a displacement wave. A frozen-phonon supercell of $N_Q^3$ cells gives the exact phonons on an $N_Q^3$ grid of wavevectors and nothing more, at a cost that grows as $L^6$ to $L^9$ with the range $L$ of the force constants. DFPT solves the Sternheimer equation for the change of the occupied states, with no sum over empty states, and one perturbation costs about as much as the ground-state density; by the $2n + 1$ theorem the first-order orbitals also fix the cubic constants. Fourier interpolation with the acoustic sum rule turns dynamical matrices on a coarse grid into the whole dispersion. The same runs give the Born charges and $\varepsilon_\infty$, through the commutator with its nonlocal term, the potential changes that the electron–phonon vertex needs, and, by finite strain, the elastic constants. In silicon the $\boldsymbol{k}$ grid is fixed by the requirement that the Born

charges vanish, as symmetry demands, and not by the frequencies. The quasi-harmonic and self-consistent harmonic approximations, TDEP, Allen–Heine–Cardona theory and special displacements all start from these harmonic phonons. For the zero-point renormalization of the silicon gap the last two give −56 and −57 meV at fixed volume, about 20 % short of the electron–phonon part, −71 to −73 meV, that the measurement implies once the lattice expansion is removed.

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# Chapter 26

# Dielectric response and form factors

The valence electrons of silicon, eight per primitive cell of volume $\Omega = a^3/4 = 40.05\,\text{Å}^3$, have a plasma frequency $\omega_p = (4\pi\alpha\, n_e/m_e)^{1/2} = 16.6\,\text{eV}$, and the loss function $L(\boldsymbol{q},\omega)$ of Sect. 4.4 has its main peak near that energy at small momentum. Where exactly the peak sits, how sharp it is, and how much spectral weight lies near the gap, where the dark-matter rate is largest for light masses, depend on how the response is computed. The bands and wavefunctions of Chaps. 23 and 24 reach an electron-recoil rate through the density vertices. The independent-particle form factor built from them is where every electronic rate of Chap. 10 starts, and screening and local fields use the same vertices as input.

## 26.1 Dielectric response from first principles

The hierarchy of many-body approximations of Sects. 4.6 and 4.9 runs from the independent-particle bubble through the random-phase approximation with and without local fields and time-dependent density-functional kernels to the Bethe–Salpeter equation. Each becomes a calculation with its own inputs and cost.

### 26.1.1 Computed quantities

Every approximation starts from the crystal bubble of Sect. 4.6, the Adler–Wiser form of the independent-particle response [1, 2],

$$R^0_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q},\omega) = \frac{g_{\text{deg}}}{N\Omega}\sum_{\boldsymbol{k}}\sum_{n,m}(f_{n\boldsymbol{k}} - f_{m,\boldsymbol{k}+\boldsymbol{q}})\,\frac{\rho_{mn}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G})^*\,\rho_{mn}(\boldsymbol{k},\boldsymbol{q}+\boldsymbol{G}')}{\omega + \varepsilon_{n\boldsymbol{k}} - \varepsilon_{m,\boldsymbol{k}+\boldsymbol{q}} + \mathrm{i}\eta_{\text{b}}}, \tag{26.1}$$

with $\boldsymbol{q}$ in the first zone, $N$ the number of $\boldsymbol{k}$ points, $g_{\text{deg}} = 2$ for spin, the density vertices $\rho_{mn}$ of Sect. 3.2, and a broadening $\eta_{\text{b}}$ that tends to $0^+$ in the exact expression and is finite on a grid. The random-phase dielectric matrix is $\varepsilon_{\boldsymbol{G}\boldsymbol{G}'} = \delta_{\boldsymbol{G}\boldsymbol{G}'} - v_{\text{C}}(\boldsymbol{q}+\boldsymbol{G})\,R^0_{\boldsymbol{G}\boldsymbol{G}'}$, the proper response being approximated by the bubble (Sect. 4.4). Time-dependent density-functional theory replaces it by the solution of the matrix Dyson equation

$$R = R^0 + R^0 f_{\text{Hxc}} R, \qquad (f_{\text{Hxc}})_{\boldsymbol{G}\boldsymbol{G}'} = v_{\text{C}}(\boldsymbol{q}+\boldsymbol{G})\,\delta_{\boldsymbol{G}\boldsymbol{G}'} + (f_{\text{xc}})_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q}), \tag{26.2}$$

from which $\varepsilon^{-1} = 1 + v_{\text{C}}R$ in matrix form; with $f_{\text{xc}} = 0$ it reduces to the random-phase result. From the inverse matrix come the macroscopic dielectric function $\varepsilon_{\text{mac}} = 1/(\varepsilon^{-1})_{\boldsymbol{0}\boldsymbol{0}}$ and the loss function. For a momentum transfer outside the first zone, $\boldsymbol{q} = \boldsymbol{q}_r + \boldsymbol{G}_q$ in the notation of Sect. 3.1, the loss function that enters the rate is the diagonal element of the inverse at $\boldsymbol{G}_q$,

$$L(\boldsymbol{q},\omega) = \mathrm{Im}\big[-(\varepsilon^{-1})_{\boldsymbol{G}_q\boldsymbol{G}_q}(\boldsymbol{q}_r,\omega)\big], \tag{26.3}$$

so a calculation at dark-matter momentum transfers of several keV needs the matrix at reciprocal vectors far from the origin, not just its head (Sect. 10.2). The Bethe–Salpeter equation leaves the two-point language altogether. The many-body approximation is not an expansion in a small number (Sect. 8.5). Its error is estimated from the spread between approximations, and for that estimate to mean anything, every approximation must be converged with the same numerical parameters.

### 26.1.2 Convergence parameters

The wavefunction cutoff, the local-field cutoff, the band count, the $\boldsymbol{k}$ grid and the gap control Eqs. (26.1) and (26.2), and they interact.

*The wavefunction cutoff* limits the largest momentum transfer at which $\rho_{mn}$ can be trusted. A density vertex at $\boldsymbol{q}+\boldsymbol{G}$ is an overlap of two sets of plane-wave coefficients displaced by $\boldsymbol{q}+\boldsymbol{G}$, and when the displacement approaches the cutoff momentum, $k_{\mathrm{cut}} = (2m_e E_{\mathrm{cut}})^{1/2}$, few coefficients overlap. Within that range the vertex is still only as good as the pseudo-wavefunction is near the nucleus, which is the subject of Sects. 24.2 and 26.2.

*The local-field cutoff* sets the size $N_G$ of the matrix $\varepsilon_{\boldsymbol{G}\boldsymbol{G}'}$, through a separate, smaller energy cutoff on $|\boldsymbol{q}+\boldsymbol{G}|$. Building the matrix costs $\propto N_G^2$ per transition and inverting it $\propto N_G^3$ per $(\boldsymbol{q},\omega)$. The test is the change of the quantity of interest, $\varepsilon_{\mathrm{mac}}$ near the gap or $L$ at the target $\boldsymbol{q}$, as $N_G$ grows. At large $|\boldsymbol{q}|$ the matrix must contain reciprocal vectors around $\boldsymbol{G}_q$, not only around the origin.

*The number of empty bands* truncates the sum over $m$ in Eq. (26.1). Its test is a sum rule: $\int_0^\infty \mathrm{d}\omega\, \omega\, \varepsilon_2 = (\pi/2)\omega_p^2$ and the same holds for $\omega L$ (Sect. 4.4), and a truncated band sum misses the weight at high energy. A free-electron count gives the scale. Above the bottom of the valence band, $N_{\mathrm{b}}$ bands per spin fill a sphere of radius $k$ with $\Omega k^3/(6\pi^2) = N_{\mathrm{b}}$, so the highest included band lies at $k^2/2m_e$ above the bottom.

*The $\boldsymbol{k}$ grid* is usually the bottleneck in a semiconductor [3]. The bands disperse strongly, and Eq. (26.1) is a sum of sharp poles that must merge into a continuum. The broadening $\eta_{\mathrm{b}}$ must exceed the spacing of the transition energies on the grid, of order $|\nabla_{\boldsymbol{k}}(\varepsilon_m - \varepsilon_n)|\,\Delta k$; a spectrum computed with too small an $\eta_{\mathrm{b}}$ for its grid shows spurious peaks. For finite $\boldsymbol{q}$, the first-zone part $\boldsymbol{q}_r$ must be a difference of grid points, so the $\boldsymbol{k}$ grid also fixes the momentum resolution. A coarse grid with a moderate $\eta_{\mathrm{b}}$ suffices where the spectrum is smooth on the scale of $\eta_{\mathrm{b}}$; the region near the gap is not, and it needs much finer sampling, often on a shifted or locally refined grid.

*The gap* enters every denominator. Kohn–Sham gaps are too small (Sect. 3.4), and the usual remedies are a rigid scissor shift $\Delta_{\mathrm{scis}}$ of the empty bands, quasiparticle energies from Sect. 23.4, or a functional designed for gaps. A scissor keeps the states and raises the transition energies, and its effect on the $f$-sum check follows from that. The density vertices are unchanged, so in the optical limit the spectrum moves rigidly, $\varepsilon_2(\omega) = \varepsilon_2^{\mathrm{KS}}(\omega - \Delta_{\mathrm{scis}})$, and its first moment becomes $\int \mathrm{d}\omega\, \omega\, \varepsilon_2^{\mathrm{KS}}(\omega - \Delta_{\mathrm{scis}}) = (\pi/2)\omega_p^2 + \Delta_{\mathrm{scis}} \int \mathrm{d}\omega\, \varepsilon_2^{\mathrm{KS}}$, larger by $\Delta_{\mathrm{scis}} \int \mathrm{d}\omega\, \varepsilon_2^{\mathrm{KS}}$. The excess is the $f$-sum rule of a nonlocal operator, the scissor, and a band-convergence test on a scissored spectrum subtracts it. Codes that build the optical limit from velocity matrix elements must rescale them as in Eq. (23.19); otherwise the spectrum acquires a spurious factor $[(\omega - \Delta_{\mathrm{scis}})/\omega]^2$ and is no longer the rigid shift that unchanged states imply. A density vertex computed directly from plane-wave coefficients at finite $\boldsymbol{q}$ needs no such care, since it depends on the states and not on their energies.

*Semicore and core states* matter at large $\omega$ and large $\boldsymbol{q}$. The silicon $2p$ and germanium $3d$ shells are frozen into the core by most pseudopotentials. Their contribution to $L$ starts at their binding energies, and their contribution to the $f$-sum adds to the valence one. A loss function for $\omega$ above a few tens of eV, or at $|\boldsymbol{q}|$ where core orbitals overlap the probe, needs

them in the valence or an all-electron treatment (Sect. 10.6).

### 26.1.3 Local fields and exchange–correlation kernels

Local fields are the difference between the inverse of the averaged matrix and the average of the inverse (Sect. 4.6). Computationally they cost one matrix inversion per $(\boldsymbol{q}, \omega)$, and they change the spectrum where the response is inhomogeneous on the scale of the cell. In the optical spectrum of silicon they lower and shift the two main peaks to higher energy, which moves the random-phase result further from experiment [3]. In the dark-matter rate of silicon and germanium the difference between the loss function with and without local fields is small on the scale of the projected reach [4]. In the silicon tables of that calculation the local fields lower the screened rate by a factor of about 1.1 at 1 MeV and 1.46 at 10 MeV for a heavy mediator, against 9 and 2.3 for the ratio of the unscreened to the screened rate (Sect. 10.2.2); at 10 MeV the step comes from momentum transfers of two to three reciprocal-lattice vectors, which carry the heavy-mediator rate.

The adiabatic local-density kernel adds little to the random-phase result for the absorption of semiconductors: its spectra are extremely close to those of the random-phase approximation with local fields [5]. For energy-loss spectra of solids the same kernel does well, but largely because the random-phase approximation with local fields already agrees with experiment there [5], which is good news for the loss function of a dark-matter rate. The reason is that the kernel is short-ranged, while the electron–hole attraction that produces excitons is long-ranged. A kernel of the form $f_{\mathrm{xc}}(\boldsymbol{r}, \boldsymbol{r}') = -\alpha_{\mathrm{xc}}/(4\pi|\boldsymbol{r} - \boldsymbol{r}'|)$, which behaves as $-\alpha_{\mathrm{xc}}/|\boldsymbol{q} + \boldsymbol{G}|^2$ in reciprocal space, fits the measured optical spectrum of silicon almost as well as the Bethe–Salpeter equation, with quasiparticle energies in the bubble and the empirical value $\alpha_{\mathrm{xc}} = 0.2$ [5], a fit for silicon rather than a universal constant. The cost of any two-point kernel is that of the random-phase calculation: the matrices have the same size, and only the Dyson equation changes. This is the attraction of TDDFT for dark-matter loss functions on fine $(\boldsymbol{q}, \omega)$ grids, and the reason its kernels are worth testing against the Bethe–Salpeter equation on a few points.

### 26.1.4 Optical conductivity and large momentum

Bosonic absorption (Sect. 10.7) and the crystal Migdal response (Sect. 12.3) take opposite limits of one calculation. For absorption of a bosonic dark-matter particle the relevant quantity is the optical conductivity $\sigma_1 = \omega\varepsilon_2$ at $\boldsymbol{q} \to 0$, where the transverse and longitudinal dielectric functions coincide (Sect. 4.4). Computing it needs the head of Eq. (26.1) in the optical limit, with the velocity matrix elements and their nonlocal commutator, a fine $\boldsymbol{k}$ grid near the absorption edge, and the approximation that the energy range demands: near the gap the Bethe–Salpeter equation, well above it the random-phase approximation with local fields. Measured optical data are available for most candidate targets, and where they are, they fix $\sigma_1$ better than any calculation. The calculation is then needed where the data stop: at low temperature, below the measured range, or for anisotropic crystals where only some tensor components were measured.

The opposite limit is the loss function at momenta far beyond the zone, up to tens of keV, and energies up to about 100 eV, which the crystal Migdal response of Sect. 12.3 convolves with the nuclear motion. Every convergence parameter above is pushed at once. The cutoff must hold momentum components well above the largest $|\boldsymbol{q}|$, the band sum must reach 100 eV above the gap and beyond, and the semicore shells must be in the valence. In addition, the local-field matrix must be centered on $\boldsymbol{G}_q$, which at $|\boldsymbol{q}| = 30$ keV lies more than seven shortest reciprocal vectors from the origin. The pseudo-wavefunctions are no longer adequate there,

and the all-electron reconstruction becomes part of the response calculation. The cost grows with all of these at once, and in practice such a calculation is done on a coarse grid of $|\boldsymbol{q}|$ and a few directions, with the convergence tested on the $f$-sum rule at each $\boldsymbol{q}$.

### 26.1.5 The Bethe–Salpeter equation

The Bethe–Salpeter equation of Sect. 4.9 is an eigenvalue problem in the space of transitions $(v, c, \boldsymbol{k})$. Its Hamiltonian, in the resonant (Tamm–Dancoff) form, is

$$H^{\mathrm{exc}}_{vc\boldsymbol{k},v'c'\boldsymbol{k}'} = (\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}})\,\delta_{vv'}\delta_{cc'}\delta_{\boldsymbol{k}\boldsymbol{k}'} + g_{\mathrm{deg}}\,K^{\mathrm{x}}_{vc\boldsymbol{k},v'c'\boldsymbol{k}'} - K^{\mathrm{d}}_{vc\boldsymbol{k},v'c'\boldsymbol{k}'}, \tag{26.4}$$

with the energies from quasiparticle bands (Sect. 23.4), an exchange kernel $K^{\mathrm{x}}$ built from the Coulomb kernel without its $\boldsymbol{G} = 0$ component, and a direct kernel $K^{\mathrm{d}}$ built from the statically screened interaction $W^{\mathrm{scr}}$ at the momentum $\boldsymbol{k} - \boldsymbol{k}'$ of the scattered pair (Sect. 4.9); the factor $g_{\mathrm{deg}}$ holds for singlets. Its eigenvalues are the exciton energies $E_\lambda$ and its eigenvectors the envelopes $A^\lambda_{cv\boldsymbol{k}}$, and the macroscopic dielectric function is a sum over $\lambda$ of oscillator strengths $|\sum_{cv\boldsymbol{k}} A^\lambda_{cv\boldsymbol{k}}\rho_{cv}(\boldsymbol{k},\boldsymbol{q})^*|^2$ over $\omega - E_\lambda$. At finite $\boldsymbol{q}$ the transitions are $(v\boldsymbol{k}) \to (c, \boldsymbol{k}+\boldsymbol{q})$ and the same structure gives the excitonic loss function (Sect. 10.5).

The cost has two parts. Building the kernel requires the screened interaction $\varepsilon^{-1}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{k} - \boldsymbol{k}')$, which is a random-phase calculation of its own, and the matrix elements of $K^{\mathrm{d}}$ between all pairs of transitions, $\propto (N_v N_c N_k)^2$ entries, each a sum over $N_G^2$ reciprocal vectors when the full inverse is used. Solving it costs $\propto (N_v N_c N_k)^3$ for a full diagonalization, or $\propto (N_v N_c N_k)^2$ per step of an iterative (Lanczos–Haydock) method that delivers the spectrum without the eigenvectors. The transition-basis dimension $N_v N_c N_k$ is the number to watch. For the optical spectrum of silicon below 4.5 eV, four valence and two conduction bands are sufficient, neglecting the off-diagonal elements of $\varepsilon^{-1}$ in $K^{\mathrm{d}}$ changes nothing visible, and quasiparticle corrections can be interpolated onto the fine grid of 2,048 $\boldsymbol{k}$ points in the full zone [3]. The dimension is then $2{,}048 \times 4 \times 2 \approx 1.6 \times 10^4$, and a dense diagonalization of that size takes $\sim 10^{13}$ operations; the $\boldsymbol{k}$ grid, not the cutoff, is the bottleneck [3].

Symmetry can be used to reduce the zone, but the spectrum is extremely sensitive to inconsistent phases when wavefunctions are rotated, especially for degenerate bands, and only a subgroup of the operations can be used safely [3]. This is the gauge rule of Appendix A.5. A finite $\boldsymbol{k}$ grid overestimates individual structures [3].

Bound excitons near the gap are harder still. The envelope of the silicon exciton, of radius 4.2 nm, has a width in crystal momentum of about one percent of the zone (Sect. 4.9). Resolving it needs close to a hundred points per direction near the band edges, and so of order $10^6$ points if the grid were uniform. Calculations of such excitons therefore use grids refined only around the band extrema, or the effective-mass limit, with the Bethe–Salpeter equation reserved for the continuum and the finite-momentum structure (Sect. 10.5).

### 26.1.6 Molecular $GW$ and Bethe–Salpeter calculations

Molecular crystals (Sect. 16.4) are often treated molecule by molecule, with the environment added afterwards. The same $GW$ and Bethe–Salpeter equations are then solved in a basis of atom-centered Gaussian orbitals instead of plane waves. The $GW$ quasiparticle energies of molecules improve the Kohn–Sham gaps to within a few tenths of an eV at a cost that scales as the fourth power of the system size, and the Bethe–Salpeter singlet excitation energies that follow have typical errors of 0.1 eV to 0.3 eV, with a mean error of 0.2 eV on a standard benchmark set of about 200 excitations [6]. An iterative search for the lowest eigenstates costs the same as in TDDFT, $O(N_{\mathrm{orb}}^4)$ operations with $N_{\mathrm{orb}}$ the number of orbitals, when resolution-of-the-identity techniques are used to build the kernel [6]. The convergence

parameter specific to this setting is the Gaussian basis: diffuse functions are needed for the empty states and for the screening, and results are tested along a family of basis sets of increasing size. The crystal environment enters through its dielectric screening, as a reaction field added to the bare Coulomb interaction [6], or through a periodic calculation of the full crystal. Sect. 16.4 compares the periodic result with measurement for anthracene: 3.3 eV against the measured 3.1 eV for the lowest singlet of the crystal.

### 26.1.7 Silicon

**Worked example 26.1: The loss function of silicon from Kohn–Sham states**

A published dark-matter calculation of the silicon loss function used a plane-wave basis with a 500 eV cutoff, a Γ-centered $8 \times 8 \times 8$ grid of $\boldsymbol{k}$ points, 70 bands per spin, a functional designed for band gaps with a scissor shift to the measured gap, and the random-phase dielectric matrix with local fields at all $\boldsymbol{q}$ of the grid [4]. For the rate, the matrix was reduced to an angular average of its diagonal elements over all $\boldsymbol{q}_r + \boldsymbol{G}$ of equal length [4, Eq. (24)].

The cutoff corresponds to $k_{\mathrm{cut}} = (2m_e \times 500\,\mathrm{eV})^{1/2} = 22.6\,\mathrm{keV}$, the range quoted by the authors [4]. Kinematics allows momentum transfers up to $2m_\chi v_{\max}$, 5.3 keV per MeV of dark-matter mass (Sect. 1.3), but the valence density vertices fall off beyond a few times $\alpha m_e = 3.7\,\mathrm{keV}$ (Sects. 10.1 and 24.1), and the cutoff covers that range, provided the pseudo-wavefunctions are accurate there. By the free-electron count above, the 70 bands reach $k = 4.7\,\text{Å}^{-1}$, which is 84 eV above the bottom of the valence band, about 70 eV above the valence maximum since the occupied bands are about 12 eV wide (Sect. 3.3). The authors traced the shortfall of the computed loss function at large energy to this truncation [4]. An $8^3$ grid has spacing $|\boldsymbol{b}_i|/8 = 0.25\,\text{Å}^{-1}$, or 0.49 keV, which is the resolution in $\boldsymbol{q}_r$; it is adequate for a loss function dominated by energies of several eV, and coarse for the structure within a few tenths of an eV of the gap, where excitonic and phonon-assisted effects live (Sect. 13.1).

The $f$-sum rule is the cleanest single check. With $\omega_p = 16.6\,\mathrm{eV}$ from the valence density, the integral $\int_0^\infty \mathrm{d}\omega\, \omega L(\boldsymbol{q}, \omega) = (\pi/2)\omega_p^2$ at small $\boldsymbol{q}$ is $433\,\mathrm{eV}^2$. A calculation that recovers, say, 0.9 of it is missing a tenth of the weight, all at high energy, and that missing weight is a bound on what the band truncation can do to a rate at large $\omega$.

The optical spectrum gives the complementary test near the gap: only the electron–hole attraction of the Bethe–Salpeter equation restores the measured line shape and absolute intensity [3]. The same hierarchy matters for a dark-matter rate in proportion to the weight it gives to energies within an eV or two of the gap, which is why the many-body spread is quoted as an uncertainty for light masses and becomes negligible for heavier ones.

### 26.1.8 Pitfalls

The head element $\varepsilon_{\mathbf{00}}$ used in place of $1/(\varepsilon^{-1})_{\mathbf{00}}$ drops the local fields, and the element at $\boldsymbol{G} = 0$ used in place of $\boldsymbol{G}_q$ for a momentum transfer beyond the first zone computes the response at the wrong momentum. A broadening too large for the grid smears the onset, which is the part of the spectrum that light dark matter sees. The response computed here screens the density vertex, the electron–phonon vertex of Sect. 27.2 and, through $W^{\mathrm{scr}}$, the quasiparticle energies of Sect. 23.4, so every one of these errors propagates into the rates built on them.

## 26.2 Crystal form factors

An electron-recoil rate of silicon at independent-particle level is a sum of $|\rho_{cv}(\boldsymbol{k},\boldsymbol{q})|^2$ over valence states $v\boldsymbol{k}$, conduction states $c\boldsymbol{k}'$ and momentum transfers $\boldsymbol{q}$, each term weighted by an energy delta function. With four valence bands, 52 conduction bands, a $10\times10\times10$ grid of crystal momenta and every reciprocal-lattice vector up to $q = 30\,\mathrm{keV}$, the sum has about $5\times10^{11}$ terms (Worked example 26.2). Each term is an overlap of two sets of plane-wave coefficients. The work is in restoring the core region that the pseudopotential removed, adding the core shells, and knowing when the sum has converged; the bands come from the calculation of Sect. 23.3, and the screening that turns the bare vertex into a measured loss function from Sect. 26.1.

### 26.2.1 The density vertex

The object is the density vertex of Eq. (A.16),

$$\rho_{mn}(\boldsymbol{k},\boldsymbol{q}) = \langle\psi_{m,\boldsymbol{k}+\boldsymbol{q}}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\psi_{n\boldsymbol{k}}\rangle,$$

with $\boldsymbol{q}$ never reduced to the first zone (Appendices A.3 and A.4). From it the independent-particle structure factor follows as a sum over pairs (Sect. 10.1); the crystal form factor $f_{\mathrm{crys}}$ of the literature is the same object in another normalization, Eq. (A.44); and the vertices of the spin and current couplings of Sect. 10.4 are built the same way, with $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ multiplied by a Pauli matrix or by the velocity. Screening divides the squared potential by $|\varepsilon(\boldsymbol{q},\omega)|^2$, which is $\varepsilon_\infty^2$ as $q\to0$ and approaches one only at large $q$ (Sect. 4.4), so an unscreened form factor is an input to that correction, not a substitute for it.

### 26.2.2 Plane-wave evaluation

A plane-wave code returns, for each band $n$ and each $\boldsymbol{k}$ of its grid, the coefficients $c_{n\boldsymbol{k}}(\boldsymbol{G})$ of

$$u_{n\boldsymbol{k}}(\boldsymbol{r}) = \frac{1}{\sqrt{\Omega}}\sum_{|\boldsymbol{k}+\boldsymbol{G}|^2\le 2m_eE_{\mathrm{cut}}} c_{n\boldsymbol{k}}(\boldsymbol{G})\,\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}, \qquad \sum_{\boldsymbol{G}}|c_{n\boldsymbol{k}}(\boldsymbol{G})|^2 = 1, \tag{26.5}$$

which is the normalization of Eq. (A.14) and the code convention of Appendix A.3 (coefficients of $\mathrm{e}^{+\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}$, the opposite of the book's lattice Fourier components). The cutoff $E_{\mathrm{cut}}$ is the one of Sect. 24.1. With $\boldsymbol{k}+\boldsymbol{q}$ folded back into the zone, $\boldsymbol{k}+\boldsymbol{q} = \boldsymbol{k}'+\boldsymbol{G}_{\mathrm{U}}$ with $\boldsymbol{k}'$ a point of the grid and $\boldsymbol{G}_{\mathrm{U}}$ a reciprocal-lattice vector, and by Eq. (A.15) the vertex is $\int_\Omega u^*_{m\boldsymbol{k}'}\mathrm{e}^{\mathrm{i}\boldsymbol{G}_{\mathrm{U}}\cdot\boldsymbol{r}}u_{n\boldsymbol{k}}$, and inserting Eq. (26.5) the cell integral of $\mathrm{e}^{\mathrm{i}(\boldsymbol{G}_{\mathrm{U}}+\boldsymbol{G}-\boldsymbol{G}')\cdot\boldsymbol{r}}$ is $\Omega\,\delta_{\boldsymbol{G}',\boldsymbol{G}+\boldsymbol{G}_{\mathrm{U}}}$:

$$\rho_{mn}(\boldsymbol{k},\boldsymbol{q}) = \sum_{\boldsymbol{G}} c^*_{m\boldsymbol{k}'}(\boldsymbol{G}+\boldsymbol{G}_{\mathrm{U}})\,c_{n\boldsymbol{k}}(\boldsymbol{G}), \qquad \boldsymbol{q} = \boldsymbol{k}'-\boldsymbol{k}+\boldsymbol{G}_{\mathrm{U}}. \tag{26.6}$$

The sum runs over the vectors present in both basis sets. Coefficients are matched by their Miller indices, the integer coordinates of $\boldsymbol{G}$ in the basis $\boldsymbol{b}_i$, since the two states are stored on different lists of $\boldsymbol{G}$. A term with $\boldsymbol{G}+\boldsymbol{G}_{\mathrm{U}}$ outside the basis of $\boldsymbol{k}'$ is zero in the representation, so Eq. (26.6) is exact for the functions the code represents; its error is the basis truncation itself, which we discuss below. This is the expression of Ref. [7, Eq. (3.16)], in the notation of Appendix A.5.

Each value of Eq. (26.6) costs one dot product of length $N_{\mathrm{PW}}$, the number of plane waves per state. All the $\boldsymbol{G}_{\mathrm{U}}$ for one pair $(m\boldsymbol{k}', n\boldsymbol{k})$ come at once from a fast Fourier transform: the product $u^*_{m\boldsymbol{k}'}(\boldsymbol{r})\,u_{n\boldsymbol{k}}(\boldsymbol{r})$ is formed on the real-space grid and transformed, and the component at $\boldsymbol{G}_{\mathrm{U}}$ is $\rho_{mn}$ for $\boldsymbol{q} = \boldsymbol{k}'-\boldsymbol{k}+\boldsymbol{G}_{\mathrm{U}}$. The cost per pair drops from $N_{\mathrm{PW}}N_{\boldsymbol{G}}$

to of order $N_{\mathrm{FFT}} \log N_{\mathrm{FFT}}$, where $N_{\boldsymbol{G}}$ is the number of $\boldsymbol{G}_{\mathrm{U}}$ needed and $N_{\mathrm{FFT}}$ the number of grid points [8, below Eq. (3.15)]. The real-space grid has to resolve the product, whose components reach twice the wavefunction cutoff. For a cubic cell it covers a cube of side $4\sqrt{2m_eE_{\mathrm{cut}}}$ in reciprocal space, which holds $8\Omega(2m_eE_{\mathrm{cut}})^{3/2}/\pi^3$ points, while the sphere of Eq. (26.5) holds $N_{\mathrm{PW}} = \Omega(2m_eE_{\mathrm{cut}})^{3/2}/(6\pi^2)$, so $N_{\mathrm{FFT}} \approx (48/\pi)N_{\mathrm{PW}}$. The fcc grid runs along the primitive vectors, whose lengths multiply to $\sqrt{2}\,\Omega$, and holds $\sqrt{2}$ times more, $N_{\mathrm{FFT}} \approx (48\sqrt{2}/\pi)N_{\mathrm{PW}} \approx 22\,N_{\mathrm{PW}}$: about $36^3$ points at 60 Ry, as in Sect. 23.3.

**Which $\boldsymbol{q}$ a grid can reach.** Eq. (26.6) needs the state at $\boldsymbol{k}' = \boldsymbol{k} + \boldsymbol{q} - \boldsymbol{G}_{\mathrm{U}}$. If both $\boldsymbol{k}$ and $\boldsymbol{k}'$ are taken from one uniform grid of $n_k = n_1n_2n_3$ points (Appendix C.2), the first-zone part of $\boldsymbol{q}$ is a difference of grid vectors, and $\boldsymbol{q}$ is resolved in steps of $|\boldsymbol{b}_i|/n_i$. A form factor tabulated on a finer or arbitrary set of $\boldsymbol{q}$ needs a second, shifted set of states: one non-self-consistent calculation at the points $\boldsymbol{k} + \boldsymbol{q}$, as density-functional perturbation theory does for phonon wavevectors (Sect. 25.2). The states at $\boldsymbol{k}'$ must be computed, or unfolded from the irreducible wedge by a symmetry operation, with the phases that the operation carries. In the diamond structure some operations include a fractional translation $\boldsymbol{t}_{\mathrm{f}}$, and each coefficient of a rotated Bloch state picks up a phase $\mathrm{e}^{-\mathrm{i}\boldsymbol{k}_{\mathrm{rot}}\cdot\boldsymbol{t}_{\mathrm{f}}}$, with $\boldsymbol{k}_{\mathrm{rot}}$ the rotated wavevector $\boldsymbol{k} + \boldsymbol{G}$. Computing the full grid without symmetry avoids the bookkeeping at a cost of the order of the symmetry factor, 48 for silicon.

**Gauge and degeneracy.** Each Bloch state carries an arbitrary phase, and $\rho_{mn}$ picks up the phase difference of its two states, while $|\rho_{mn}|^2$ does not. A rate that sums $|\rho_{mn}|^2$ is therefore gauge independent, provided that the sum over $m$ and $n$ covers complete degenerate sets: a degenerate subspace may be rotated by any unitary matrix, and only the trace over it is fixed. At the valence maximum of silicon the threefold degeneracy at $\Gamma$ makes this concrete; a sum restricted to one of the three states there is meaningless. Amplitudes that interfere, such as the second-order amplitudes of phonon-assisted processes in which $\rho_{mn}$ multiplies an electron–phonon vertex, need $\rho$ and $g$ in one gauge, computed from the same stored states (Appendix A.5 and Sect. 27.2).

### 26.2.3 Velocity matrix elements and nonlocal potentials

The momentum matrix element follows from Eq. (26.5) in the same way,

$$\boldsymbol{p}_{mn}(\boldsymbol{k}) = \sum_{\boldsymbol{G}} c^*_{m\boldsymbol{k}}(\boldsymbol{G})\,(\boldsymbol{k} + \boldsymbol{G})\,c_{n\boldsymbol{k}}(\boldsymbol{G}), \tag{26.7}$$

but the velocity is not $\boldsymbol{p}_{mn}/m_e$ in a pseudopotential calculation. The velocity operator is $\boldsymbol{v} = \mathrm{i}[H, \boldsymbol{r}]$. For $H = \boldsymbol{p}^2/2m_e + V_{\mathrm{loc}}(\boldsymbol{r}) + V_{\mathrm{NL}}$, with $V_{\mathrm{NL}}$ the nonlocal part of the pseudopotential (Appendix A.5), the local potential commutes with $\boldsymbol{r}$ and

$$\boldsymbol{v} = \frac{\boldsymbol{p}}{m_e} + \mathrm{i}[V_{\mathrm{NL}}, \boldsymbol{r}]. \tag{26.8}$$

The commutator is evaluated in the $\boldsymbol{k}$ representation. Define $V_{\mathrm{NL}}(\boldsymbol{k}) \equiv \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} V_{\mathrm{NL}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$, the operator that acts on $u_{n\boldsymbol{k}}$. Differentiating, $\nabla_{\boldsymbol{k}} V_{\mathrm{NL}}(\boldsymbol{k}) = \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\,\mathrm{i}[V_{\mathrm{NL}}, \boldsymbol{r}]\,\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$, so that

$$\boldsymbol{v}_{mn}(\boldsymbol{k}) = \frac{\boldsymbol{p}_{mn}(\boldsymbol{k})}{m_e} + \langle u_{m\boldsymbol{k}} | \nabla_{\boldsymbol{k}} V_{\mathrm{NL}}(\boldsymbol{k}) | u_{n\boldsymbol{k}} \rangle. \tag{26.9}$$

For the separable form Eq. (24.9) that plane-wave codes use, with projectors $\tilde{p}_{\kappa i}$ centered on the atoms and coefficients $D_{\kappa,ij}$, the gradient acts only on the Fourier transforms of the

projectors, $\tilde{p}_{\kappa i}(\boldsymbol{k}+\boldsymbol{G})$, and costs about as much as applying $V_{\mathrm{NL}}$ once. The correction is not small. In diamond, the absorption spectrum computed with $\boldsymbol{p}/m_e$ instead of $\boldsymbol{v}$ differs by 15 to 20 percent [9, Fig. 1], and Ref. [9, Eq. (20)] shows that a long-wavelength field couples to $\boldsymbol{v}$, not to $\boldsymbol{p}/m_e$.

For the dark-matter vertices the correction has opposite consequences for the density and for the current. The density vertex needs none. The operator $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ is local, and Eq. (26.6) evaluated at small $q$ reproduces $\boldsymbol{q}\cdot\boldsymbol{v}_{mn}/(\varepsilon_{m\boldsymbol{k}}-\varepsilon_{n\boldsymbol{k}})$ with the full velocity of Eq. (26.9), since the derivation of Eq. (A.17) used only $\mathrm{i}[H,\boldsymbol{r}]=\boldsymbol{v}$. The comparison of the two sides at $q\sim 10^{-3}\,|\boldsymbol{b}_i|$ is a test that catches a missing nonlocal term in a velocity routine. A coupling to the electron current, on the other hand, is ambiguous with a nonlocal Hamiltonian. The paramagnetic current $j_e(\boldsymbol{q})$ of Sect. 4.3 is built from $\boldsymbol{p}/m_e$; gauge invariance requires an additional nonlocal term, which Ref. [9, Eqs. (18) and (19)] derives as a phase $\exp(\mathrm{i}\int \boldsymbol{A}\cdot\mathrm{d}\boldsymbol{x})$ along the straight line between the two arguments of $V_{\mathrm{NL}}(\boldsymbol{r},\boldsymbol{r}')$. Its long-wavelength limit is the commutator of Eq. (26.8). At finite $q$ the term depends on $\boldsymbol{q}$ through the extent of the projectors, of order $q\,r_c$ with $r_c$ the core radius of the pseudopotential (Sect. 24.1), and codes that compute current responses at finite $q$ either include it or restrict themselves to $q\,r_c\ll 1$.

**The scissor shift.** The scissor operator $\Delta_{\mathrm{scis}}\sum_c|\psi_c\rangle\langle\psi_c|$ is nonlocal too. It rescales each interband velocity by $(\varepsilon_{c\boldsymbol{k}}-\varepsilon_{v\boldsymbol{k}}+\Delta_{\mathrm{scis}})/(\varepsilon_{c\boldsymbol{k}}-\varepsilon_{v\boldsymbol{k}})$, the rigid-shift case of Eq. (23.19), and leaves the density vertex and its optical limit unchanged. An optical spectrum built from unscaled velocities acquires the spurious factor of Sect. 26.1. For silicon in the local-density approximation, with $\Delta_{\mathrm{scis}}=0.74\,\mathrm{eV}$ [10, Sect. VII] and a direct gap at $\Gamma$ of 2.57 eV [11, Table V], the factor is 1.29 at $\Gamma$, and $|\boldsymbol{v}_{cv}|^2$ changes by 66 percent.

### 26.2.4 All-electron reconstruction

The pseudo vertex is reliable only for $q\,r_c\ll 1$ (Sect. 24.1). The PAW map Eq. (24.4) restores the all-electron state inside the core spheres, in one of two ways. The first route expands the all-electron state itself in plane waves up to a second, larger cutoff and then applies Eq. (26.6) unchanged. Ref. [8, Sect. 2.1.1] does this for silicon and germanium with a second cutoff of 2 keV, which by $q_{\max}=(2m_eE)^{1/2}$ reaches $q\approx 45\,\mathrm{keV}$. The second route works with the matrix elements, Eq. (24.6). With $\boldsymbol{r}=\boldsymbol{\tau}_\kappa+\boldsymbol{x}$ in each sphere, the Rayleigh expansion Eq. (C.22) of $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}}$ reduces each element of the one-center matrix Eq. (24.7) to Gaunt coefficients times the radial integrals

$$\int_0^{r_c}\mathrm{d}x\,x^2\,j_l(qx)\,\left[\phi^{\mathrm{AE}}_{\kappa i}(x)\phi^{\mathrm{AE}}_{\kappa j}(x)-\phi^{\mathrm{PS}}_{\kappa i}(x)\phi^{\mathrm{PS}}_{\kappa j}(x)\right], \tag{26.10}$$

with $\phi(x)$ the radial parts of the partial waves of Eq. (24.4) and $j_l$ the spherical Bessel functions of Appendix C.3. The integrals live on the radial grid of the atomic calculation, which resolves the nodes of the all-electron partial waves, so the correction has no plane-wave cutoff at all; it is limited by the completeness of the partial waves, usually two per angular momentum. They are computed once per species and per $q$, and the per-state work is in the projections $\langle\tilde{p}_{\kappa j}|\tilde{\psi}_{n\boldsymbol{k}}\rangle$, which the code already stores.

How much the reconstruction matters depends on which $q$ the rate weights. For a heavy mediator, $F_{\mathrm{DM}}=1$, and the factor $q$ of the measure in Eq. (A.43) gives large momentum transfers more weight than any lighter mediator does, and the reconstruction raises the silicon rate even above low thresholds (Sect. 24.2). For a light mediator, $F_{\mathrm{DM}}^2\propto q^{-4}$ weights the smallest momentum transfers, and the reconstruction changes the rate little except at high thresholds, where only large $q$ is kinematically allowed. Norm-conserving pseudopotentials can be reconstructed in the same way when the generator writes out its all-electron and pseudo partial waves.

### 26.2.5 Core and semicore states

Kinematics decides when the core shells matter (Sect. 24.1): the silicon $2p$ shell opens at $m_\chi = 29\,\mathrm{MeV}$ and needs $q \geq 38\,\mathrm{keV}$; the germanium $3d$ shell, only 29 eV below the valence maximum, opens at 8.5 MeV and needs $q \geq 11\,\mathrm{keV}$. Silicon $2p$ transitions therefore probe the wavefunctions at momenta beyond the reach of a typical plane-wave basis (Sect. 24.1), and a pseudopotential calculation does not contain the shell at all.

Such shells are either kept in the pseudopotential or taken from the atom. A *semicore* shell, such as germanium $3d$, can be kept in the valence of the pseudopotential, at the price of a higher cutoff, and is then computed with the other bands and reconstructed as above. A deeper shell is taken from the isolated atom. Its Bloch states are the lattice sums

$$\psi_{\kappa nlm,\boldsymbol{k}}(\boldsymbol{r}) = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot(\boldsymbol{R}+\boldsymbol{\tau}_\kappa)}\, \psi^{\mathrm{at}}_{\kappa nlm}(\boldsymbol{r} - \boldsymbol{R} - \boldsymbol{\tau}_\kappa), \tag{26.11}$$

of the atomic orbitals $\psi^{\mathrm{at}}_{\kappa nlm}$ [8, Eq. (2.8)], labeled by the atomic quantum numbers instead of a band index; the orbitals barely overlap, so the band is flat and its energy is the atomic level. Where both treatments exist, the atomic orbitals reproduce the reconstructed crystal wavefunctions up to the reconstruction cutoff, about 45 keV (Sect. 24.2). The vertex between a core state and a conduction state is then $\int \mathrm{d}^3 r\, \psi^*_{c,\boldsymbol{k}+\boldsymbol{q}} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \psi_{\kappa nlm,\boldsymbol{k}}$, with the conduction state from the crystal calculation; above the highest computed band the final states are approximated by plane waves, in the spirit of Eq. (2.24), with or without a Coulomb factor [8, Sects. 2.3 and 3.4]. Relativistic corrections to the core orbitals are of relative size $(Z\alpha)^2$, about 1 % for the silicon $1s$ shell and 5 % for germanium $1s$ (Worked example 24.2). The approximation leaves out the relaxation of the other electrons around the core hole, which shifts and reshapes the edge; that final-state effect lies beyond independent particles.

### 26.2.6 Atomic ionization form factors

For a target made of isolated atoms, such as a noble liquid, the rate of Sect. 2.2 needs the ionization form factor $|f_{\mathrm{ion}}(p_{\mathrm{ej}}, q)|^2$ of Eq. (2.22) for each occupied shell. The initial orbitals are Hartree–Fock orbitals (Sect. 3.4), almost always the Roothaan–Hartree–Fock solutions of Bunge and collaborators [12], which expand each radial function in Slater-type orbitals,

$$R^{\mathrm{at}}_{nl}(r) = a_0^{-3/2} \sum_j C_{jln}\, \frac{(2Z_{jl})^{\tilde{n}_{jl}+1/2}}{\sqrt{(2\tilde{n}_{jl})!}} \Big(\frac{r}{a_0}\Big)^{\tilde{n}_{jl}-1} \mathrm{e}^{-Z_{jl} r/a_0}, \tag{26.12}$$

with tabulated coefficients $C_{jln}$, exponents $Z_{jl}$ and integers $\tilde{n}_{jl}$ [13, Eq. (B33)]. The final state is a continuum state of energy $E_{\mathrm{ej}} = p^2_{\mathrm{ej}}/2m_e$, expanded in partial waves $l'$. In the simplest choice it is the exact Coulomb wave of a hydrogenic potential $-Z_{\mathrm{eff}}\alpha/r$, with $Z_{\mathrm{eff}} = n\,(E_B/13.6\,\mathrm{eV})^{1/2}$ fixed by the binding energy $E_B$ of the shell [13, Eqs. (B34) and (B35)]; a better choice solves the radial equation in the self-consistent potential of the ion. The radial integrals of $j_L(qr)$ between the two radial functions are then evaluated for every $(l, l', L)$ allowed by the Gaunt coefficients.

The convergence parameters are the partial-wave cutoff $l'_{\mathrm{max}}$, the grids in $q$ and $E_{\mathrm{ej}}$, and the accuracy of the radial integrals. Ref. [13, footnote 11] sums to $l'_{\mathrm{max}} = 7$; the number needed grows with $q$, since a plane wave of momentum $q$ reaches angular momenta up to about $q$ times the orbital radius, and has to be checked at the largest $q$ of the table. The $q$ grid must extend to the kinematic maximum $2m_\chi v_{\mathrm{max}}$ of the heaviest mass considered, 5.3 MeV at 1 GeV, and is best logarithmic. The radial integrands oscillate rapidly at large $q$, and an integration split into intervals can misread an accidental cancellation over one interval

as convergence; Ref. [13, Appendix B 5] therefore requires two consecutive intervals to agree and switches to a series representation at large $q$. The cost is negligible next to any crystal calculation: a few shells, a few partial waves, and one-dimensional integrals.

The approximation behind the table is that of frozen orbitals. The binding energies are single-particle (Koopmans) energies, the other electrons do not relax when one is removed, and the final state does not feel the polarization of the ion. It is also nonrelativistic, which matters for the inner shells of xenon, where $(Z\alpha)^2 \approx 0.16$. A subtler problem is orthogonality. A continuum state of the model potential $-Z_{\mathrm{eff}}\alpha/r$ is not an eigenstate of the Hartree–Fock Hamiltonian and is not orthogonal to the bound orbital, so that $f_{\mathrm{ion}}$ does not vanish as $q \to 0$, as it must for $\langle f|i\rangle = 0$. Orthogonalizing the continuum state to the occupied orbitals removes the spurious small-$q$ weight, which a light mediator would otherwise amplify. For silicon these atomic tables play a second role: the deep shells of the crystal in Eq. (26.11) are built from the same orbitals.

### 26.2.7 Convergence parameters and checks

A crystal form factor is controlled by the cutoff, the number of conduction bands, the $\boldsymbol{k}$ grid and the treatment of the energy delta function. The cutoff $E_{\mathrm{cut}}$ sets the largest momentum the vertex contains, $(2m_e E_{\mathrm{cut}})^{1/2}$, which is 31 keV at 70 Ry; Ref. [7, Appendix C] varies it from 30 to 110 Ry and finds the silicon rate at $m_\chi = 1\,\mathrm{GeV}$ changed by less than one percent above 70 Ry. The number of conduction bands $N_c$ sets the largest energy deposition the spectrum contains; with 52 conduction bands the lowest band left out lies about 57 eV above the gap, and the computed spectrum ends there [7, Sect. 4.2]. A spectrum above the highest band is zero, not small, and a rate integrated over $\omega$ is missing everything beyond it. The $\boldsymbol{k}$ grid sets both the resolution in $q$ and the sampling of the onset near $E_{\mathrm{min}}(q)$; Ref. [7, Appendix C] finds the grid matters only at low masses and low thresholds, where few transitions are kinematically allowed. It tests the grid by shaking every grid point randomly by up to half the grid spacing and comparing ten such grids. Finally, the delta function of energy needs a broadening or a tetrahedron integration, with the width tied to the grid spacing as described in Appendix C.2, or a binning: Ref. [7, Sect. 4.2] uses bins of 0.1 eV in $\omega$ and $0.02\,\alpha m_e = 75\,\mathrm{eV}$ in $q$.

Most errors are caught by the small-$q$ limit and by a sum rule. The small-$q$ check is the normalization $\rho_{nn}(\boldsymbol{k}, 0) = 1$ with $\rho_{mn}(\boldsymbol{k}, \boldsymbol{q})$ for $m \neq n$ vanishing linearly in $q$, with the slope given by the velocity, as discussed above. The sum rule holds for each occupied state and measures the effect of truncating the bands. Since $\rho_{mn}(\boldsymbol{k}, \boldsymbol{q}) = \langle u_{m,\boldsymbol{k}+\boldsymbol{q}}|u_{n\boldsymbol{k}}\rangle$ (Eq. (A.16)), and the $u_{m,\boldsymbol{k}+\boldsymbol{q}}$ form a complete set of eigenstates of $H(\boldsymbol{k}+\boldsymbol{q}) = \mathrm{e}^{-\mathrm{i}(\boldsymbol{k}+\boldsymbol{q})\cdot\boldsymbol{r}} H \mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{q})\cdot\boldsymbol{r}}$,

$$\sum_m (\varepsilon_{m,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{n\boldsymbol{k}})\,|\rho_{mn}(\boldsymbol{k},\boldsymbol{q})|^2 = \langle u_{n\boldsymbol{k}}|H(\boldsymbol{k}+\boldsymbol{q}) - H(\boldsymbol{k})|u_{n\boldsymbol{k}}\rangle = \boldsymbol{q}\cdot\boldsymbol{v}_n^{\mathrm{g}}(\boldsymbol{k}) + \frac{q^2}{2m_e} + \delta_{\mathrm{NL}}, \quad (26.13)$$

where the sum over $m$ runs over all bands and $\delta_{\mathrm{NL}}$ is the part of $\langle u_{n\boldsymbol{k}}|V_{\mathrm{NL}}(\boldsymbol{k}+\boldsymbol{q}) - V_{\mathrm{NL}}(\boldsymbol{k})|u_{n\boldsymbol{k}}\rangle$ beyond first order in $q$. The kinetic part of $H(\boldsymbol{k}) = (\boldsymbol{p}+\boldsymbol{k})^2/2m_e + V_{\mathrm{loc}} + V_{\mathrm{NL}}(\boldsymbol{k})$ gives $\boldsymbol{q}\cdot(\boldsymbol{p}+\boldsymbol{k})/m_e + q^2/2m_e$, and the first-order part of the nonlocal term combines with it into the group velocity, by the Hellmann–Feynman relation $\nabla_{\boldsymbol{k}}\varepsilon_{n\boldsymbol{k}} = \langle u_{n\boldsymbol{k}}|\nabla_{\boldsymbol{k}} H(\boldsymbol{k})|u_{n\boldsymbol{k}}\rangle$. Summed over a filled band the linear term cancels. Transitions between two occupied states cancel in pairs, since time reversal gives $|\rho_{nm}(-\boldsymbol{k}-\boldsymbol{q}, \boldsymbol{q})| = |\rho_{mn}(\boldsymbol{k}, \boldsymbol{q})|$ with the opposite energy difference. What remains is the valence $f$-sum rule of a pseudo-Hamiltonian, the band-summed form of Eq. (24.12): $q^2/2m_e$ per valence electron, plus the nonlocal term, which is absent for the all-electron Hamiltonian of Eq. (A.38). The right side of Eq. (26.13) is a ground-state expectation value and costs nothing; the left side with $N_c$ conduction bands falls short of it by the weight of the bands left out. The shortfall falls slowly with $N_c$, because at momentum transfer $q$ much of the missing weight sits near the free-electron energy $q^2/2m_e$.

### 26.2.8 Cost and pitfalls

The count of vertices sets the cost. With both $\boldsymbol{k}$ and $\boldsymbol{k}'$ on a grid of $n_k$ points, $N_v$ valence and $N_c$ conduction bands, and $N_{\boldsymbol{G}}$ reciprocal-lattice vectors within the largest $q$, there are $n_k^2 N_v N_c N_{\boldsymbol{G}}$ vertices, computed with $n_k^2 N_v N_c$ fast Fourier transforms of $N_{\mathrm{FFT}}$ points each. Directional averages can be taken over the irreducible wedge of $\boldsymbol{k}$, with the full set of $\boldsymbol{k}'$. The vertices themselves are far too many to store, and they are accumulated into bins of $(q, \omega)$, or of $(\boldsymbol{q}, \omega)$ for a directional response, as they are computed; a tabulated crystal form factor is that binned sum. The all-electron correction adds, per state, the projections onto the atomic projectors, and per $q$ the radial integrals of Eq. (26.10).

Three errors are not caught by the checks above. A $\boldsymbol{q}$ reduced to the first zone assigns an Umklapp vertex to the wrong momentum transfer (Appendix A.3). A spectrum cut by the highest band, or a $q$ range cut by $E_{\mathrm{cut}}$, looks converged in its own window and silently loses everything outside it. And normalizations differ between codes and papers: plane-wave codes often fold the spin factor 2 into the $\boldsymbol{k}$ weights [7, Eq. (4.2)], and the conversions between $f_{\mathrm{crys}}$ and $\bar{S}$ are in Eq. (A.44). For germanium, which semilocal functionals leave without a gap [8, Sect. 2.1], the band order must be fixed before any rigid shift (Sect. 23.3).

**Worked example 26.2: The silicon form factor**

We set up a valence-to-conduction calculation for silicon and count its cost. The cell volume is $\Omega = a^3/4 = 40.05\,\text{Å}^3$ at $a = 5.431\,\text{Å}$. A cutoff of 70 Ry, 952 eV, reaches $(2m_e E_{\mathrm{cut}})^{1/2} = 31.2\,\mathrm{keV} = 15.8\,\text{Å}^{-1}$ and holds $N_{\mathrm{PW}} = \Omega(2m_e E_{\mathrm{cut}})^{3/2}/6\pi^2 \approx 2700$ plane waves per state. The non-self-consistent run computes 4 valence and 52 conduction bands on a $10 \times 10 \times 10$ grid without symmetry, $n_k = 1000$; the coefficients take $1000 \times 56 \times 2700$ complex numbers, 2.4 GB in double precision, which fits in memory. The Kohn–Sham gap is about 0.6 eV in the representative PBE calculation (Worked example 23.1), and a scissor $\Delta_{\mathrm{scis}} = E_{\mathrm{g}} - E_{\mathrm{g}}^{\mathrm{KS}}$ moves the conduction bands up to the measured low-temperature gap, 1.17 eV (Sect. 1.3).

The reciprocal-lattice vectors within $q_{\max} = 30\,\mathrm{keV}$ number $\frac{4\pi}{3} q_{\max}^3 \Omega/(2\pi)^3 \approx 2400$, since the zone of silicon has volume $(2\pi)^3/\Omega = 47.6\,\mathrm{keV}^3$. The vertex count is $n_k^2 N_v N_c N_{\boldsymbol{G}} = 10^6 \times 4 \times 52 \times 2400 \approx 5 \times 10^{11}$. They come from $n_k^2 N_v N_c = 2.1 \times 10^8$ transforms on grids of $N_{\mathrm{FFT}} \approx 22 N_{\mathrm{PW}} \approx 6 \times 10^4$ points. At about $5 N_{\mathrm{FFT}} \log_2 N_{\mathrm{FFT}} \approx 5 \times 10^6$ floating-point operations per transform, the total is $1 \times 10^{15}$ operations, about three hours on a node that sustains $10^{11}$ operations per second, and less with the irreducible wedge. The calculation is cheap; the binned output, at the bin widths quoted above, has about $500 \times 400$ entries up to 30 keV.

The calculation is then tested at small $q$, against the sum rule and against the cutoff. At $q = 10^{-3}\,|\boldsymbol{b}_i|$ the ratio $|\rho_{cv}|/q$ must equal $|\hat{\boldsymbol{q}} \cdot \boldsymbol{v}_{cv}|/(\varepsilon_c - \varepsilon_v)$ with the velocity of Eq. (26.9). For each valence state the truncated left side of Eq. (26.13), with its nonlocal term on the right, measures the band truncation. And the rate at $m_\chi = 1\,\mathrm{GeV}$ for a heavy mediator, recomputed at a higher cutoff and with all-electron reconstruction, fixes where the pseudo vertex stops being adequate. Kinematics tells us where to look: at 10 MeV, with $\omega$ just above the gap, momentum transfers between 0.45 keV and 53 keV are allowed, but most of the valence weight lies at a few keV, and the cutoff test decides how much of the rate comes from beyond 30 keV; the $2p$ shell, which would need $q \geq 38\,\mathrm{keV}$, is still closed below 29 MeV.

## Summary

Every response calculation starts from the Adler–Wiser bubble of Kohn–Sham states. The random-phase dielectric matrix follows from it, local fields cost one matrix inversion per $(\boldsymbol{q}, \omega)$, and time-dependent density-functional kernels and the Bethe–Salpeter equation on quasiparticle bands add exchange and the attraction of electron and hole. The wavefunction cutoff, the local-field cutoff, the band count, the $\boldsymbol{k}$ grid and the gap control the result and interact. In a semiconductor the $\boldsymbol{k}$ grid is usually the bottleneck. The band count is tested by the $f$-sum rule, and a scissor raises the first moment of $\varepsilon_2$ by $\Delta_{\text{scis}} \int \mathrm{d}\omega\, \varepsilon_2^{\text{KS}}$. The independent-particle form factor of silicon up to 30 keV is a sum of about $5 \times 10^{11}$ overlaps of plane-wave coefficients, some $10^{15}$ operations. The density vertex needs no correction for the nonlocal pseudopotential, while the velocity carries its commutator, which changes the absorption spectrum of diamond by 15 to 20 percent. The all-electron reconstruction is computed either with a second plane-wave cutoff or with radial integrals that have no cutoff at all. A form factor is tested at small $q$ against the velocity, against the sum rule of each occupied state, and against a higher cutoff with reconstruction.

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# Chapter 27

# Wannier interpolation

A conduction electron of silicon 10 meV above the band minimum has a crystal momentum within $(2m_\perp^* \times 10\,\mathrm{meV})^{1/2} = 44\,\mathrm{eV}$ of $\boldsymbol{k}_\mathrm{c}$ in the transverse direction (Table 3.1), 2 % of $2\pi/a$. A uniform grid that places a point in every such pocket needs about $|\boldsymbol{b}_i|/44\,\mathrm{eV} \approx 90$ points along each reciprocal primitive vector, close to $10^6$ in the zone, and a phonon-assisted rate pairs each of them with as many phonon wavevectors (Sects. 6.5 and 13.3). Every point would be a plane-wave diagonalization, and every pair a perturbation calculation. Such grids are affordable only because the Bloch states of a group of bands can be recombined into orbitals localized within a cell or two, the Wannier functions. In their basis the Hamiltonian couples nearby cells and little else, and its Fourier series returns the bands at any $\boldsymbol{k}$ for the price of diagonalizing an $8 \times 8$ matrix. They are the computed tight-binding orbitals that Sect. 3.2 anticipated, orthonormal, with hopping integrals taken from a first-principles calculation instead of fitted.

## 27.1 Wannier functions

Wannier functions are not unique, and asking for the most localized set fixes the freedom; the interpolation they allow is exact when the Hamiltonian has finite range in their basis, and every quantity interpolated with them must be built in one gauge. We follow Refs. [1–3].

### 27.1.1 From Bloch states to localized orbitals

Take first a single band $n$ that is separated from all others by gaps throughout the zone. Its $N$ Bloch states each spread over the whole crystal with the same weight in every cell. A superposition of them with phases that add up in one cell and cancel elsewhere is a wave packet localized near that cell:

$$w_n(\boldsymbol{r} - \boldsymbol{R}) = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{k}} \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\, \psi_{n\boldsymbol{k}}(\boldsymbol{r}), \tag{27.1}$$

with the sum over the $N$ crystal momenta of the Born–von Karman grid, Eq. (3.13). The left side is written as a function of $\boldsymbol{r} - \boldsymbol{R}$ because it is one: Bloch's theorem in the form $\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\psi_{n\boldsymbol{k}}(\boldsymbol{r} - \boldsymbol{R})$ turns the right side into $N^{-1/2}\sum_{\boldsymbol{k}} \psi_{n\boldsymbol{k}}(\boldsymbol{r} - \boldsymbol{R})$, the function for $\boldsymbol{R} = 0$ shifted by $\boldsymbol{R}$. All the Wannier functions of a band are copies of one function, one per cell. The transform inverts with the lattice sum. On the grid of Eq. (3.13) the same geometric sums that proved Eq. (3.14) give $\sum_{\boldsymbol{k}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} = N\delta_{\boldsymbol{R},0}$ for lattice vectors of one Born–von Karman crystal. Multiplying Eq. (27.1) by $N^{-1/2}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}$, summing over $\boldsymbol{R}$ and using this identity to

collapse the sum over crystal momenta,

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\, w_n(\boldsymbol{r}-\boldsymbol{R}). \tag{27.2}$$

This is the Bloch sum of Eq. (3.28), with the Wannier function in place of the atomic orbital. Unlike atomic orbitals, Wannier functions are orthonormal. Writing $w_{n\boldsymbol{R}}$ for $w_n(\boldsymbol{r}-\boldsymbol{R})$,

$$\langle w_{n\boldsymbol{R}} | w_{n'\boldsymbol{R}'} \rangle = \frac{1}{N} \sum_{\boldsymbol{k}\boldsymbol{k}'} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R} - \mathrm{i}\boldsymbol{k}'\cdot\boldsymbol{R}'} \langle \psi_{n\boldsymbol{k}} | \psi_{n'\boldsymbol{k}'} \rangle = \frac{1}{N} \sum_{\boldsymbol{k}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot(\boldsymbol{R}-\boldsymbol{R}')}\, \delta_{nn'} = \delta_{\boldsymbol{R}\boldsymbol{R}'}\delta_{nn'}.$$

The overlap matrix of the tight-binding problem, Eq. (3.29), is then the identity. The normalization of Eq. (3.20) gives $w_n$ the dimension $\Omega^{-1/2}$, the same as $u_{n\boldsymbol{k}}$, and a function localized within a few cells has a norm that does not depend on the size of the crystal.

The construction has a loose end. A Bloch state is defined only up to a phase, and $\psi_{n\boldsymbol{k}} \to \mathrm{e}^{\mathrm{i}\theta_n(\boldsymbol{k})}\psi_{n\boldsymbol{k}}$ describes the same physical state at every $\boldsymbol{k}$. It changes the Wannier function, however, since Eq. (27.1) adds the states with new relative phases. In the periodic gauge of Eq. (A.15), with $\theta_n(\boldsymbol{k}+\boldsymbol{G}) = \theta_n(\boldsymbol{k})$, the summand $\psi_{n\boldsymbol{k}}(\boldsymbol{r})$ at fixed $\boldsymbol{r}$ is a periodic function of $\boldsymbol{k}$. The value of $w_n$ in cell $\boldsymbol{R}$ is then a Fourier coefficient of that function, and its decay with $|\boldsymbol{R}|$ is set by how smooth the function is. If $\theta_n$ jumps at random from one $\boldsymbol{k}$ to the next, as the phases of a numerical diagonalization do, the sum is a random walk and $w_n$ spreads over the whole crystal. If the periodic parts $u_{n\boldsymbol{k}}$ are smooth, the Wannier function decays faster than any power of $|\boldsymbol{R}|$, and if they are analytic in a strip of half-width $1/\xi_W$ around the real $\boldsymbol{k}$ axis it decays as $\mathrm{e}^{-|\boldsymbol{R}|/\xi_W}$. In two and three dimensions such an analytic choice exists if and only if the Chern invariants of the bands vanish [4]. A Chern invariant is a topological integer of a group of bands, a count of how often the phases of its Bloch states must wind across a plane of the zone, and it vanishes in every crystal with time-reversal symmetry, silicon among them. A Wannier function is not a property of the band alone: it depends on a choice of gauge, and a good choice makes it exponentially localized.

A single isolated band is the exception. The four valence bands of silicon touch at $\Gamma$, where three of them are degenerate (Sect. 3.3), and cross elsewhere; no one of them is smooth across the zone. Their sum, the projector $P(\boldsymbol{k}) = \sum_{n=1}^{4} |u_{n\boldsymbol{k}}\rangle\langle u_{n\boldsymbol{k}}|$ on the four-dimensional space they span, is smooth, since the group is separated from the conduction bands by the gap at every $\boldsymbol{k}$. The way out is to give up Wannier functions per band and build $N_W$ functions for the whole group of $N_W$ bands, mixing the bands at each $\boldsymbol{k}$ with a unitary $N_W \times N_W$ matrix:

$$w_n(\boldsymbol{r}-\boldsymbol{R}) = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{k}} \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\, \psi^W_{n\boldsymbol{k}}(\boldsymbol{r}), \qquad \psi^W_{n\boldsymbol{k}} = \sum_m U_{mn}(\boldsymbol{k})\, \psi_{m\boldsymbol{k}}, \tag{27.3}$$

which is the convention of Eq. (A.19). The gauge matrix $U(\boldsymbol{k})$ is periodic, $U(\boldsymbol{k}+\boldsymbol{G}) = U(\boldsymbol{k})$; the phases $\mathrm{e}^{\mathrm{i}\theta_n(\boldsymbol{k})}$ are its diagonal special case. The rotated states $\psi^W_{n\boldsymbol{k}}$ are still Bloch functions at $\boldsymbol{k}$, but they are no longer eigenstates of $H_\mathrm{e}$, and $n$ is no longer a band index: it labels a Wannier function. Translations, the inverse transform and orthonormality carry over from the single band, with $\psi^W$ in place of $\psi$. Since a unitary mixing within the group leaves $P(\boldsymbol{k})$ unchanged, the $N_W$ Wannier functions of a cell, together with their copies in all other cells, span exactly the states of the group [3, Eqs. (8)–(10)].

A first smooth gauge comes from projection. Choose $N_W$ trial orbitals $\varphi^\mathrm{tr}_n(\boldsymbol{r})$ located where the Wannier functions are expected, such as $sp^3$ hybrids on the atoms of silicon or Gaussians at the bond centers, and compute the $N_W \times N_W$ matrix $A^\mathrm{pr}_{mn}(\boldsymbol{k}) = \langle \psi_{m\boldsymbol{k}} | \varphi^\mathrm{tr}_n \rangle$. Its Löwdin orthonormalization

$$U(\boldsymbol{k}) = A^\mathrm{pr}(\boldsymbol{k}) \left[ A^{\mathrm{pr}\dagger}(\boldsymbol{k}) A^\mathrm{pr}(\boldsymbol{k}) \right]^{-1/2}, \tag{27.4}$$

is unitary, and it is smooth in $\boldsymbol{k}$ because the trial orbitals are localized [3, Eqs. (16) and (17)]. The arbitrary phases of the $\psi_{m\boldsymbol{k}}$ cancel in Eq. (27.4): a phase on $\psi_{m\boldsymbol{k}}$ multiplies row $m$ of $A^{\mathrm{pr}}$ and is undone in $\psi^W$. The construction fails where $A^{\mathrm{pr}\dagger}A^{\mathrm{pr}}$ becomes nearly singular, that is, where the group has almost no weight on some trial orbital. For the four valence bands of silicon, four Gaussians at the bond centers are enough, and the resulting Wannier functions are centered on the four bonds of a cell, each a bonding combination of two $sp^3$ hybrids [3, Fig. 2].

### 27.1.2 Maximal localization

Projection leaves the result tied to the trial orbitals. Marzari and Vanderbilt removed this arbitrariness with a criterion. They chose the gauge that minimizes the total quadratic spread of the Wannier functions of one cell around their centers [1, Eq. (11)],

$$\Omega_W = \sum_{n=1}^{N_W} \big[\langle r^2\rangle_n - \bar{\boldsymbol{r}}_n^2\big], \qquad \bar{\boldsymbol{r}}_n = \langle w_{n\boldsymbol{0}}|\boldsymbol{r}|w_{n\boldsymbol{0}}\rangle, \qquad \langle r^2\rangle_n = \langle w_{n\boldsymbol{0}}|r^2|w_{n\boldsymbol{0}}\rangle. \tag{27.5}$$

The functions that minimize $\Omega_W$ are the *maximally localized Wannier functions*, the solid-state version of the Foster–Boys orbitals of quantum chemistry.

The position operator needs care in a crystal. On the Born–von Karman torus $\boldsymbol{r}$ is not a periodic function. It does not map Bloch states into Bloch states, and $\langle\psi_{n\boldsymbol{k}}|\boldsymbol{r}|\psi_{n\boldsymbol{k}}\rangle$ depends on where the box is cut. For a localized function the matrix elements of $\boldsymbol{r}$ and $r^2$ are well defined, as long as the function vanishes near the edges of the box, and expressed through Bloch states the position operator becomes a derivative with respect to $\boldsymbol{k}$. To see how, let the crystal grow, $N \to \infty$, so that $N^{-1}\sum_{\boldsymbol{k}} \to \Omega(2\pi)^{-3}\int_{\mathrm{BZ}} \mathrm{d}^3k$, and write $u^W_{n\boldsymbol{k}} = \sum_m U_{mn}(\boldsymbol{k}) u_{m\boldsymbol{k}}$ for the periodic parts of the rotated states. Eq. (27.3) with Eq. (3.20) reads $w_n(\boldsymbol{r}) = \Omega(2\pi)^{-3}\int \mathrm{d}^3k\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} u^W_{n\boldsymbol{k}}(\boldsymbol{r})$. Writing $\boldsymbol{r}\,\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} = -\mathrm{i}\nabla_{\boldsymbol{k}}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$ and integrating by parts in $\boldsymbol{k}$,

$$\boldsymbol{r}\, w_n(\boldsymbol{r}) = \frac{\Omega}{(2\pi)^3}\int \mathrm{d}^3k\ \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\ \mathrm{i}\nabla_{\boldsymbol{k}} u^W_{n\boldsymbol{k}}(\boldsymbol{r}).$$

The boundary terms cancel because $\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}u^W_{n\boldsymbol{k}}$ is periodic in $\boldsymbol{k}$ in the periodic gauge. Multiplying by $w^*_m(\boldsymbol{r}-\boldsymbol{R})$, written in the same way with a second momentum $\boldsymbol{k}'$, and integrating over all space leaves $\int \mathrm{d}^3r\, \mathrm{e}^{\mathrm{i}(\boldsymbol{k}-\boldsymbol{k}')\cdot\boldsymbol{r}} f(\boldsymbol{r})$ with $f = u^{W*}_{m\boldsymbol{k}'}\nabla_{\boldsymbol{k}} u^W_{n\boldsymbol{k}}$ periodic. Only the $\boldsymbol{G}=0$ component of $f$ survives for $\boldsymbol{k}$ and $\boldsymbol{k}'$ in the zone, and the integral is $(2\pi)^3\Omega^{-1}\delta^3(\boldsymbol{k}-\boldsymbol{k}')\int_\Omega f$. Hence [1, Eqs. (4) and (8)]

$$\langle w_{m\boldsymbol{R}}|\boldsymbol{r}|w_{n\boldsymbol{0}}\rangle = \mathrm{i}\,\frac{\Omega}{(2\pi)^3}\int \mathrm{d}^3k\ \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\,\langle u^W_{m\boldsymbol{k}}|\nabla_{\boldsymbol{k}} u^W_{n\boldsymbol{k}}\rangle_\Omega, \qquad \langle r^2\rangle_n = \frac{\Omega}{(2\pi)^3}\int \mathrm{d}^3k\ \|\nabla_{\boldsymbol{k}} u^W_{n\boldsymbol{k}}\|^2_\Omega, \tag{27.6}$$

where the second form is the norm of $\boldsymbol{r}w_n$ obtained in the same way, and $\langle\cdot|\cdot\rangle_\Omega$ is an integral over one cell. The center $\bar{\boldsymbol{r}}_n$ is the zone average of the Berry connection $\mathrm{i}\langle u^W_{n\boldsymbol{k}}|\nabla_{\boldsymbol{k}} u^W_{n\boldsymbol{k}}\rangle$. A gauge change $\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}'}$ on $u^W_{n\boldsymbol{k}}$, which keeps $\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}u^W_{n\boldsymbol{k}}$ periodic in $\boldsymbol{k}$ only for a lattice vector $\boldsymbol{R}'$, moves the center by $\boldsymbol{R}'$. The center is defined only modulo a lattice vector. This is the ambiguity of the Berry-phase polarization (Eq. (25.15)): the sum of the centers of the valence group, times $-2e/\Omega$ for spin-degenerate bands, is the electronic polarization.

A calculation knows the states on a grid, and the derivatives in Eq. (27.6) become finite differences between neighboring points $\boldsymbol{k}$ and $\boldsymbol{k}+\Delta\boldsymbol{k}$. All the information is in the overlaps $M_{mn}(\boldsymbol{k},\Delta\boldsymbol{k}) = \langle u^W_{m\boldsymbol{k}}|u^W_{n,\boldsymbol{k}+\Delta\boldsymbol{k}}\rangle_\Omega$, and with weights $\lambda_{\Delta\boldsymbol{k}}$ fixed by the geometry of the grid,

$\sum_{\Delta\boldsymbol{k}} \lambda_{\Delta\boldsymbol{k}} \Delta k_\alpha \Delta k_\beta = \delta_{\alpha\beta}$, the forms used in practice are [1, Eqs. (31) and (32)]

$$\begin{aligned} \bar{\boldsymbol{r}}_n &= -\frac{1}{n_k} \sum_{\boldsymbol{k},\Delta\boldsymbol{k}} \lambda_{\Delta\boldsymbol{k}}\, \Delta\boldsymbol{k}\, \mathrm{Im}\ln M_{nn}(\boldsymbol{k},\Delta\boldsymbol{k}), \\ \langle r^2\rangle_n &= \frac{1}{n_k} \sum_{\boldsymbol{k},\Delta\boldsymbol{k}} \lambda_{\Delta\boldsymbol{k}} \Big\{ 1 - |M_{nn}|^2 + \big[\mathrm{Im}\ln M_{nn}\big]^2 \Big\}, \end{aligned} \tag{27.7}$$

for a grid of $n_k$ points, which is a crystal of $n_k$ cells. They agree with Eq. (27.6) to first and second order in $\Delta k$, and they are chosen, among many such expressions, because the logarithm makes the center shift by exactly $\boldsymbol{R}'$ when the Wannier function is translated by $\boldsymbol{R}'$. The branch of the logarithm, defined modulo $2\pi$, is the discrete form of the ambiguity of the center.

The spread splits into a part that no gauge can change and a part that the gauge controls. Inserting $1 = P + Q$ between the two factors of $r^2 = \sum_\alpha r_\alpha r_\alpha$, with $P = \sum_{m\boldsymbol{R}} |w_{m\boldsymbol{R}}\rangle\langle w_{m\boldsymbol{R}}|$ the projector on the whole group, and $Q = 1 - P$, gives $\langle r^2\rangle_n = \sum_\alpha \langle w_{n\boldsymbol{0}}|r_\alpha Q r_\alpha|w_{n\boldsymbol{0}}\rangle + \sum_{m\boldsymbol{R}} |\langle w_{m\boldsymbol{R}}|\boldsymbol{r}|w_{n\boldsymbol{0}}\rangle|^2$. The term with $(m,\boldsymbol{R}) = (n,\boldsymbol{0})$ is $\bar{\boldsymbol{r}}_n^2$, and [1, Eqs. (13)–(16)]

$$\Omega_W = \Omega_W^{\mathrm{I}} + \tilde{\Omega}_W, \qquad \Omega_W^{\mathrm{I}} = \sum_\alpha \mathrm{Tr_c}\big[P r_\alpha Q r_\alpha\big], \qquad \tilde{\Omega}_W = \sum_n \sum_{(m,\boldsymbol{R})\neq(n,\boldsymbol{0})} \big|\langle w_{m\boldsymbol{R}}|\boldsymbol{r}|w_{n\boldsymbol{0}}\rangle\big|^2, \tag{27.8}$$

with $\mathrm{Tr_c}$ a trace per cell. Both parts are positive. The first depends on the projector $P$ alone, which no unitary mixing within the group changes, and it is gauge invariant; it measures how fast the space spanned by the group turns as $\boldsymbol{k}$ moves through the zone. Minimizing $\Omega_W$ therefore means minimizing $\tilde{\Omega}_W$, the sum of the squared matrix elements of $\boldsymbol{r}$ between different Wannier functions. In one dimension $\tilde{\Omega}_W$ can be made to vanish, by choosing the Wannier functions as eigenstates of the projected position operator $PxP$. In three dimensions $PxP$, $PyP$ and $PzP$ do not commute, and the maximally localized functions are the best compromise between diagonalizing all three [1, Sect. III]. The condition for that compromise follows from an infinitesimal gauge change $U(\boldsymbol{k}) \to U(\boldsymbol{k})[1 + \mathrm{d}W(\boldsymbol{k})]$, with $\mathrm{d}W$ anti-Hermitian. Maximal localization requires $\mathrm{d}\Omega_W = 0$ for every $\mathrm{d}W(\boldsymbol{k})$, that is [1, Eq. (52)]

$$\sum_{\Delta\boldsymbol{k}} \lambda_{\Delta\boldsymbol{k}} \Big( \mathcal{A}\big[\mathsf{R}(\boldsymbol{k},\Delta\boldsymbol{k})\big] - \mathcal{S}\big[\mathsf{T}(\boldsymbol{k},\Delta\boldsymbol{k})\big] \Big) = 0 \quad \text{at every } \boldsymbol{k}, \tag{27.9}$$

with $\mathsf{R}_{mn} = M_{mn} M_{nn}^*$, $\mathsf{T}_{mn} = (M_{mn}/M_{nn})\,[\mathrm{Im}\ln M_{nn} + \Delta\boldsymbol{k}\cdot\bar{\boldsymbol{r}}_n]$, and the superoperators $\mathcal{A}[B] = (B - B^\dagger)/2$ and $\mathcal{S}[B] = (B + B^\dagger)/2\mathrm{i}$. The left side, multiplied by four, is the gradient of $\Omega_W$ with respect to $\mathrm{d}W(\boldsymbol{k})$; codes follow it downhill from the projected gauge of Eq. (27.4), which also steers them away from the false minima that a random starting gauge can fall into.

For the eight $sp^3$ functions of silicon a further step is needed. The four valence bands are an isolated group, but the four lowest conduction bands are not: above the conduction minimum they cross and mix with higher bands of other character, and no set of eight bands is separated from the rest everywhere in the zone. The eight-dimensional space to be Wannierized must first be chosen at each $\boldsymbol{k}$ from a larger set of bands, which makes $U(\boldsymbol{k})$ a rectangular matrix, with more rows than columns. This is *disentanglement*. Souza, Marzari and Vanderbilt chose the space by the same smoothness criterion. Among all $N_W$-dimensional subspaces of the bands in an outer energy window, they take the one that minimizes the gauge-invariant spread [2, Eqs. (8) and (9)],

$$\Omega_W^{\mathrm{I}} = \frac{1}{n_k} \sum_{\boldsymbol{k},\Delta\boldsymbol{k}} \lambda_{\Delta\boldsymbol{k}}\, \mathrm{Tr}\big[P(\boldsymbol{k})\, Q(\boldsymbol{k}+\Delta\boldsymbol{k})\big], \tag{27.10}$$

where $P(\boldsymbol{k})$ projects on the chosen subspace at $\boldsymbol{k}$ and $Q = 1 - P$. Each trace is the spillage of the subspace at one point out of the subspace at its neighbor, and the minimum is the subspace that turns least from point to point. States inside an inner, or frozen, window are kept in the subspace unchanged, so that the bands there are reproduced exactly; $\tilde{\Omega}_W$ is then minimized within the chosen subspace as before.

Fig. 27.1 shows one of the resulting functions for silicon. It is an $sp^3$-like hybrid on one atom, with its center 0.80 Å from the nucleus. Its main lobe points away from the bonds, opposite to one of them and toward the empty tetrahedral site of the diamond structure, and small lobes of opposite sign sit on the bonds. Its amplitude falls by two orders of magnitude within 6 Å. The hybrids that chemistry would draw, pointing along the bonds, are also a stationary point of $\Omega_W$, but a less localized one: on a $6^3$ grid their total spread is 19.8 Å$^2$, against 16.2 Å$^2$ for the functions of Fig. 27.1, with the same $\Omega_W^{\mathrm{I}} = 14.0$ Å$^2$. Which stationary point a minimization finds depends on the trial orbitals it starts from, an example of the false minima mentioned above. Started from symmetric $sp^3$ hybrids, or from randomly placed trial orbitals, the eight functions end pointing away from the bonds. Started from hybrids with the same orientation on both atoms, which point toward the bonds on one atom and away from them on the other, the minimization stops at a mixed stationary point. The four functions on one atom point away from the bonds, and the four on the other point along them. Both sets are real. Maximally localized functions in a crystal with time-reversal symmetry are found to be real, but no general proof exists [1]; reality, a common check that a minimization has converged, therefore does not tell the two sets apart. Where the windows go, which trial orbitals start the minimization and how the result is checked are questions of practice, taken up in Sect. 27.2.

### 27.1.3 Interpolation of the bands

The Hamiltonian between Wannier functions follows from Eq. (27.3). Since $H_{\mathrm{e}}$ does not connect different $\boldsymbol{k}$, and $\langle \psi^W_{m\boldsymbol{k}} | H_{\mathrm{e}} | \psi^W_{n\boldsymbol{k}} \rangle = \sum_{m'} U^*_{m'm}(\boldsymbol{k}) \varepsilon_{m'\boldsymbol{k}} U_{m'n}(\boldsymbol{k})$, the double sum over $\boldsymbol{k}$ collapses to one:

$$H^W_{mn}(\boldsymbol{R}) \equiv \langle w_{m\boldsymbol{0}} | H_{\mathrm{e}} | w_{n\boldsymbol{R}} \rangle = \frac{1}{N} \sum_{\boldsymbol{k}} \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} \left[ U^\dagger(\boldsymbol{k})\, E(\boldsymbol{k})\, U(\boldsymbol{k}) \right]_{mn}, \qquad E(\boldsymbol{k}) = \mathrm{diag}(\varepsilon_{n\boldsymbol{k}}). \tag{27.11}$$

This real-space Hamiltonian is an $N_W \times N_W$ matrix for each lattice vector. Its diagonal elements at $\boldsymbol{R} = 0$ are the on-site energies of the Wannier functions, the others are hopping integrals, and since the Wannier functions are localized it decays with $|\boldsymbol{R}|$. Its Fourier series is, by the lattice sum,

$$H^W(\boldsymbol{k}) \equiv \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\, H^W(\boldsymbol{R}) = U^\dagger(\boldsymbol{k})\, E(\boldsymbol{k})\, U(\boldsymbol{k}), \tag{27.12}$$

a Hermitian $N_W \times N_W$ matrix whose eigenvalues are the band energies $\varepsilon_{n\boldsymbol{k}}$ and whose eigenvectors, the columns of $U^\dagger(\boldsymbol{k})$, convert the Wannier gauge back into band states [3, Eqs. (98) and (99)]. Eq. (27.12) is the tight-binding eigenvalue problem of Eq. (3.30), with orthonormal orbitals, so that $S^{\mathrm{ov}} = 1$, and with every hopping computed. In practice the Bloch states come from a plane-wave Kohn–Sham calculation (Sect. 3.4) on a uniform grid; the code supplies the overlaps $M_{mn}$ and the projections $A^{\mathrm{pr}}_{mn}$, a separate program minimizes the spread, and $H^W(\boldsymbol{R})$ follows from Eq. (27.11) on the same grid.

The point of Eq. (27.12) is that its left side can be evaluated at any $\boldsymbol{k}$, not only at the points of the grid on which $H^W(\boldsymbol{R})$ was computed. On a coarse grid of $n_k = n_1 n_2 n_3$ points,

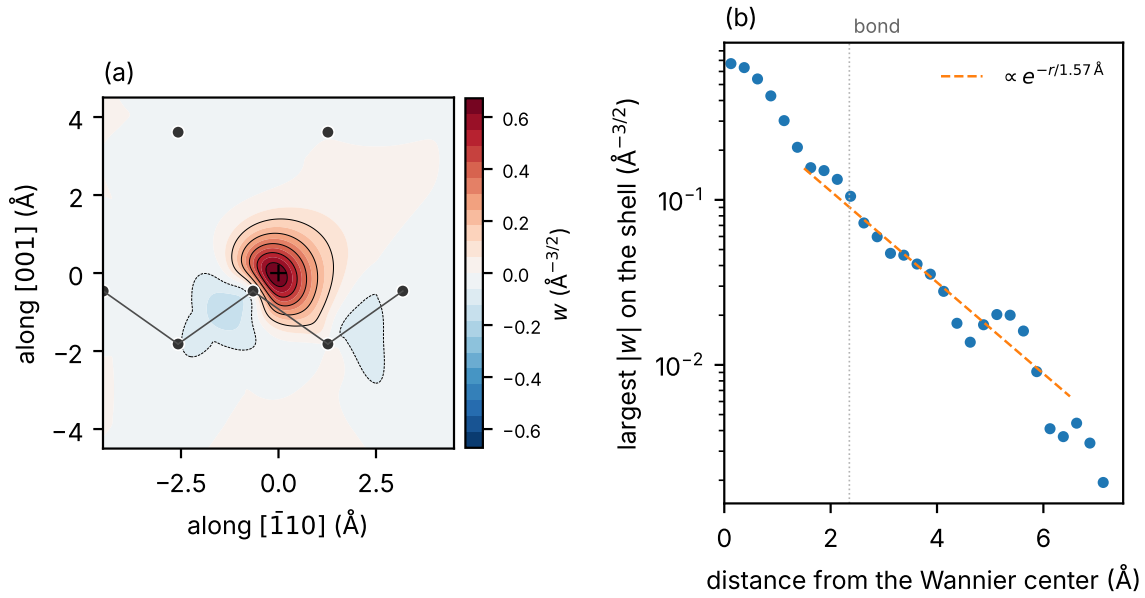



Figure 27.1: A maximally localized $sp^3$ Wannier function of silicon, one of eight disentangled from the 16 lowest bands. (a) The function on the (110) plane through the atom on which it sits (cross: Wannier center; dots: atoms in the plane; lines: bonds in the plane). The plane contains the $[\bar{1}11]$ axis of the function and the bond opposite to it. The function is real and normalized to one; red and blue are opposite signs. (b) Largest $|w|$ on spherical shells 0.25 Å wide around the center, against distance, with an exponential fit between 2 and 5.5 Å (dashed), whose decay length is 1.57 Å. Calculation: PBE, norm-conserving SG15 pseudopotential, 40 Ry cutoff, $a = 5.431$ Å, $6^3$ coarse grid, outer window up to 12 eV and frozen window up to 2.5 eV above the valence-band maximum, minimization started from $sp^3$ hybrids on both atoms pointing away from the bonds [5–8]. The spread of each function is 2.02 Å$^2$; of the total $\Omega_W = 16.2$ Å$^2$, the gauge-invariant part $\Omega_W^{\mathrm{I}}$ is 14.0 Å$^2$.

the sum in Eq. (27.11) defines the coarse-grid matrix

$$\tilde{H}^W(\boldsymbol{R}) = \frac{1}{n_k} \sum_{j=1}^{n_k} \mathrm{e}^{-\mathrm{i}\boldsymbol{k}_j\cdot\boldsymbol{R}}\, H^W(\boldsymbol{k}_j).$$

Inserting Eq. (27.12) for $H^W(\boldsymbol{k}_j)$ and using the lattice sum on the coarse grid, $n_k^{-1}\sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{k}_j\cdot\boldsymbol{R}} = \sum_{\boldsymbol{L}_{\mathrm{sc}}} \delta_{\boldsymbol{R},\boldsymbol{L}_{\mathrm{sc}}}$, where $\boldsymbol{L}_{\mathrm{sc}} = \sum_i l_i n_i \boldsymbol{a}_i$ ($l_i$ integers) are the lattice vectors of the supercell that the coarse grid makes periodic, we find

$$\tilde{H}^W(\boldsymbol{R}) = \sum_{\boldsymbol{L}_{\mathrm{sc}}} H^W(\boldsymbol{R} + \boldsymbol{L}_{\mathrm{sc}}). \tag{27.13}$$

The coarse grid does not lose the hoppings; it folds them onto one supercell, so that a hopping to $\boldsymbol{R} + \boldsymbol{L}_{\mathrm{sc}}$ is added to the hopping to $\boldsymbol{R}$. This aliasing is the only error of the interpolation. Suppose $H^W(\boldsymbol{R})$ vanishes beyond some range $R_{\max}$, and the supercell is large enough that every $\boldsymbol{R}$ with $|\boldsymbol{R}| \le R_{\max}$ lies inside its Wigner–Seitz cell. Then at most one term of Eq. (27.13) is nonzero for each $\boldsymbol{R}$ of that cell, and $\tilde{H}^W(\boldsymbol{R}) = H^W(\boldsymbol{R})$ there. A lattice vector on the boundary of the Wigner–Seitz cell belongs to two or more cells; its term is shared among its images with weights that add up to one, which restores the same result.

**Exactness of Wannier interpolation**

If the real-space Hamiltonian $H^W(\boldsymbol{R})$ has finite range $R_{\max}$, and the supercell of the coarse grid contains the sphere of radius $R_{\max}$ in its Wigner–Seitz cell, then

$$\sum_{\boldsymbol{R}\in\mathrm{WS}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\, \tilde{H}^W(\boldsymbol{R}) = H^W(\boldsymbol{k}) \quad \text{for every } \boldsymbol{k}, \tag{27.14}$$

and its eigenvalues are the exact bands $\varepsilon_{n\boldsymbol{k}}$, on the coarse grid and between its points. The sum runs over the lattice vectors of the Wigner–Seitz supercell, with fractional weights on its boundary.

Real Wannier functions are not of finite range. They decay exponentially, and so does $H^W(\boldsymbol{R})$. The error of the interpolation is then the aliased tail of Eq. (27.13), of the size of $H^W$ at a distance of half a supercell, and it falls exponentially as the coarse grid is refined; Worked example 27.1 shows both behaviors in a model. The gauge enters at this point. The bands at the grid points do not depend on $U(\boldsymbol{k})$, but the range of $H^W(\boldsymbol{R})$ does, and the most localized Wannier functions make the shortest-ranged Hamiltonian and the fastest convergence. With disentanglement, the interpolated bands are exact only inside the frozen window; outside it they are the bands of the selected subspace.

Fig. 27.2 applies this to silicon, where the convergence turns out slower than the size of the Wannier functions suggests. The four bond-centered functions of the valence bands have a total spread of $7.58\,\text{Å}^2$ on a $6^3$ grid. Interpolated from $4^3$, $6^3$ and $8^3$ grids, their bands deviate from a direct calculation by at most 350, 120 and 51 meV, about a factor 2.5 for every two added points per direction. The reason is visible in $H^W(\boldsymbol{R})$. Its elements fall to 0.02 eV at 5.4 Å in most directions, but along the twelve $\langle 110\rangle$ directions, in which the bonds of the diamond structure form zigzag chains, they are 0.12 eV two lattice vectors away (7.7 Å) and 0.03 eV three away (11.5 Å). These are the hoppings that the coarse grids fold back. For the eight $sp^3$ functions on the $8^3$ grid, the valence bands are within 40 meV and the conduction bands below 1.5 eV within 84 meV. The largest error sits near the conduction minimum, which lies between the grid points, and calculations of rates use denser coarse grids for the conduction bands for this reason.

Derivatives come for free. Differentiating Eq. (27.12) analytically, $\nabla_{\boldsymbol{k}} H^W(\boldsymbol{k}) = \sum_{\boldsymbol{R}} \mathrm{i}\boldsymbol{R}\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} H^W(\boldsymbol{R})$, and by the Hellmann–Feynman argument of Eq. (3.41) its diagonal elements in the band basis are the group velocities. The interband velocities need more. By Eq. (3.42), $\boldsymbol{v}_{mn} = (\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m\boldsymbol{k}})\langle u_{m\boldsymbol{k}} | \nabla_{\boldsymbol{k}} u_{n\boldsymbol{k}} \rangle$ for $m \neq n$, and the overlap on the right is interpolated from the matrix elements of $\boldsymbol{r}$ between Wannier functions, Eq. (27.6), rotated with the same $U(\boldsymbol{k})$ [3, Eqs. (106)–(109)]; the Hamiltonian alone would treat the Wannier functions as point-like orbitals.

### 27.1.4 One gauge for all interpolated quantities

Any operator that conserves crystal momentum interpolates the same way. If $O_{mn}(\boldsymbol{k}) = \langle \psi_{m\boldsymbol{k}} | O | \psi_{n\boldsymbol{k}} \rangle$ is known on the grid, its Wannier-gauge form $U^\dagger(\boldsymbol{k}) O(\boldsymbol{k}) U(\boldsymbol{k})$ is transformed to real space as in Eq. (27.11), summed back at the new $\boldsymbol{k}$ as in Eq. (27.12), and rotated into the band basis with the eigenvectors of $H^W(\boldsymbol{k})$ at that point. Vertices that change the crystal momentum, the density vertex $\rho_{mn}(\boldsymbol{k}, \boldsymbol{q})$ and the electron–phonon vertex, carry one gauge matrix at each end, $U^\dagger(\boldsymbol{k}+\boldsymbol{q})\, \rho(\boldsymbol{k}, \boldsymbol{q})\, U(\boldsymbol{k})$, and are interpolated in the same way (Eq. (27.15)). The step back into the band basis uses the eigenvectors of the interpolated Hamiltonian. As a result, the band index of an interpolated vertex refers to the interpolated bands only if the vertex and the Hamiltonian were rotated into the Wannier gauge with the same $U(\boldsymbol{k})$. That is the rule of Appendix A.5, and the reason is short. Suppose $O(\boldsymbol{k})$ was computed from states that differ from those behind $U(\boldsymbol{k})$ by phases, $\psi_{m\boldsymbol{k}} \to \mathrm{e}^{\mathrm{i}\theta_m(\boldsymbol{k})} \psi_{m\boldsymbol{k}}$, as two independent diagonalizations produce. With $D(\boldsymbol{k}) = \operatorname{diag} \mathrm{e}^{\mathrm{i}\theta_m(\boldsymbol{k})}$, the matrix that is transformed is $U^\dagger D^\dagger O D\, U$, the operator in the gauge $DU$. That gauge jumps at random from point to point, its real-space form does not decay, and Eq. (27.13) folds an unbounded tail onto the supercell. The interpolated matrix elements between the grid points are then wrong by amounts as large as the elements themselves. At the grid points the moduli $|O_{mn}|$ of nondegenerate bands survive, because there the error is only a phase. A check made on the grid does not reveal it. For degenerate bands, such as the valence-band maximum of silicon at $\Gamma$, $D$ is a unitary mixing and even the moduli are lost. The interpolated energies carry no trace of the mismatch at all, since $H^W$ was rotated with its own states.

A first-order rate contains only $|\rho_{mn}|^2$ between fixed bands, so a gauge mismatch spoils it between the grid points but not on them. A second-order amplitude is not protected even on the grid. It sums products such as $g_{cm\nu} \rho_{m v}$ over the intermediate band $m$ (Sect. 6.5), and the phase of band $m$ cancels between the two factors only if both were built from the same state. Every quantity that enters one amplitude is therefore computed from one set of wavefunctions and rotated with one set of gauge matrices. The density vertex depends on the momentum transfer $\boldsymbol{q}$, which is a parameter, not a crystal momentum of the grid, and at $q \sim q_0$ the factor $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ varies over a single Wannier function. Its real-space form must then be built separately for each $\boldsymbol{q}$.

**Worked example 27.1: A two-band chain interpolated from a coarse grid**

A chain of lattice constant $a$ has two orthonormal orbitals per cell, on sites a and b, with on-site energies $E_\mathrm{a} = -E_\mathrm{b} = 0.5\,\mathrm{eV}$, a hopping matrix element $-t_\mathrm{hop}$ between the two sites of a cell and $-t'_\mathrm{hop}$ between site a of one cell and site b of the cell to its left, with $t_\mathrm{hop} = 2.0\,\mathrm{eV}$ and $t'_\mathrm{hop} = 1.0\,\mathrm{eV}$. The numbers are illustrative, not fitted to a material. In

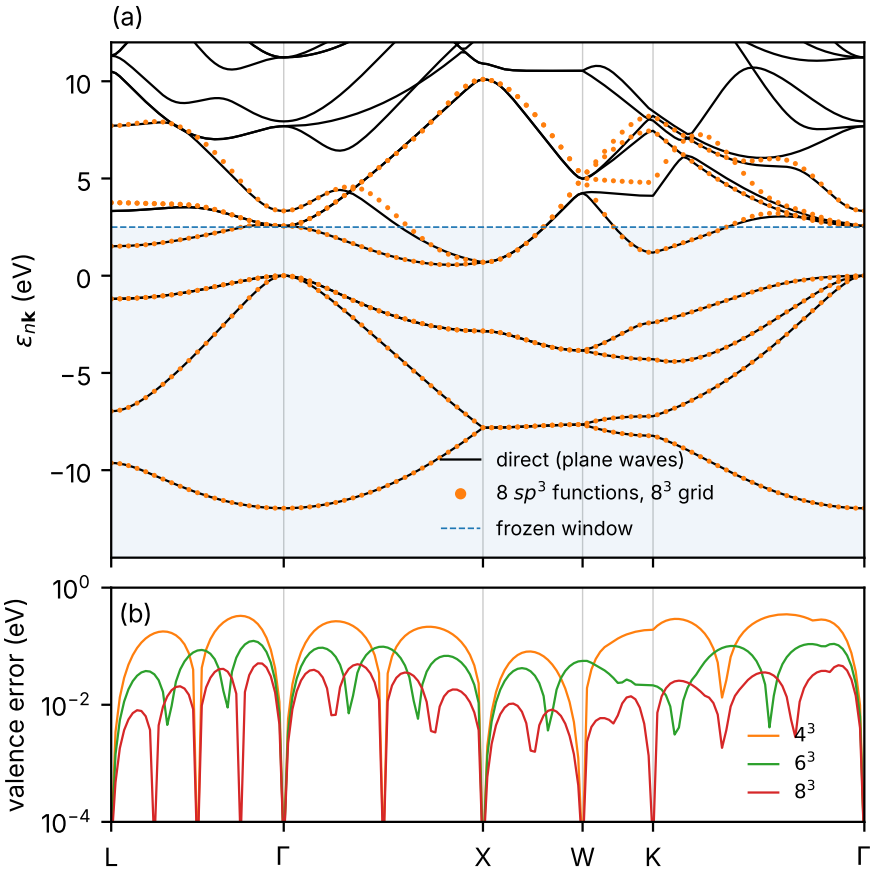



Figure 27.2: Wannier interpolation of the bands of silicon. (a) Bands computed directly in plane waves at each point of the path (lines) and interpolated from the eight $sp^3$ Wannier functions of Fig. 27.1, built on an $8^3$ coarse grid (dots), relative to the valence-band maximum. Inside the frozen window (shaded, up to 2.5 eV) the Wannier bands follow the four valence and the lowest conduction bands; above it they follow the selected subspace, not the true bands, and the higher bands (lines without dots) are outside the Wannier manifold. (b) Largest error of the four valence bands interpolated from four bond-centered Wannier functions, an isolated group, for coarse grids of $4^3$, $6^3$ and $8^3$ points. It vanishes at Γ, X and L, which lie on all three grids. PBE calculation as in Fig. 27.1; its gap, 0.557 eV at $0.843\,(2\pi/a)$ along ΓX, is the Kohn–Sham gap of Sect. 3.4, not the measured 1.17 eV.

the basis of Bloch sums of the two orbitals, Eq. (3.29) gives

$$H(k) = \begin{pmatrix} E_\mathrm{a} & -t_\mathrm{hop} - t'_\mathrm{hop}\mathrm{e}^{-\mathrm{i}ka} \\ -t_\mathrm{hop} - t'_\mathrm{hop}\mathrm{e}^{\mathrm{i}ka} & -E_\mathrm{a} \end{pmatrix},$$
$$\varepsilon_{\pm,k} = \pm\big(E_\mathrm{a}^2 + t_\mathrm{hop}^2 + t_\mathrm{hop}'^2 + 2t_\mathrm{hop}t'_\mathrm{hop}\cos ka\big)^{1/2}.$$

The two bands span $\pm 1.118\,\mathrm{eV}$ to $\pm 3.041\,\mathrm{eV}$, with a gap of $2.236\,\mathrm{eV}$ at $k = \pi/a$. We treat the model as a code would. On a grid of $n_k$ points $k_j = 2\pi j/(n_k a)$ we diagonalize $H(k_j)$ and give each eigenvector a random phase. We then build the gauge by projection, Eq. (27.4), on the two orbitals, and compute $\tilde{H}^W(R)$ on the Wigner–Seitz supercell, $R = -\lfloor n_k/2\rfloor a, \ldots, \lfloor n_k/2\rfloor a$, with weight $\frac{1}{2}$ on the two end points when $n_k$ is even. Finally we diagonalize the interpolated $H^W(k)$ at 2001 points between $-\pi/a$ and $\pi/a$ and compare with $\varepsilon_{\pm,k}$.

*Two Wannier functions.* For the pair of bands the projection returns $U(k_j) = C^\dagger(k_j)$, with $C$ the eigenvector matrix, and the random phases cancel. The Wannier functions are the two orbitals themselves, and $H^W(R)$ has range $a$: on-site energies and $-t_\mathrm{hop}$ at $R = 0$, $-t'_\mathrm{hop}$ at $R = \pm a$, nothing beyond. The exactness condition of Eq. (27.14) requires the supercell to hold $R = -a, 0, a$ inside its Wigner–Seitz cell, that is $n_k \geq 3$. With $n_k = 3$, 4 and 6 the largest band error is $1 \times 10^{-15}\,\mathrm{eV}$ to $3 \times 10^{-15}\,\mathrm{eV}$, rounding error. With $n_k = 2$ the hoppings to $R = +a$ and $R = -a$ fall on the same boundary point, they are not equal (one couples a to b, the other b to a), and the error is $0.24\,\mathrm{eV}$.

*One Wannier function.* The lower band alone is an isolated band, and its Wannier function is a combination of both orbitals in many cells. Its $H^W(R)$ is the set of Fourier coefficients of $\varepsilon_{-,k}$, which do not depend on the gauge. It decays as $|R|^{-3/2}\mathrm{e}^{-|R|/\xi_W}$, where $1/\xi_W$ is the distance of the nearest branch point of $\varepsilon_{-,k}$ from the real axis, at $\cos ka = -(E_\mathrm{a}^2 + t_\mathrm{hop}^2 + t_\mathrm{hop}'^2)/(2t_\mathrm{hop}t'_\mathrm{hop}) = -1.3125$, so that $a/\xi_W = \operatorname{arccosh} 1.3125 = 0.771$. In numbers, $|H^W(R)| = 2.19$, 0.468, 0.0525 and $0.0120\,\mathrm{eV}$ for $R/a = 0$ to 3, and the ratio of successive values approaches $(R/(R+1))^{3/2}\mathrm{e}^{-a/\xi_W}$, with $\mathrm{e}^{-a/\xi_W} = 0.462$. The interpolation error is now never zero. It is $51\,\mathrm{meV}$ for $n_k = 4$, $4.7\,\mathrm{meV}$ for 8, $0.64\,\mathrm{meV}$ for 12 and $2.4\,\mu\mathrm{eV}$ for 24, approaching a factor $\mathrm{e}^{-a/2\xi_W} = 0.68$ per added grid point, as the aliased tail at half a supercell predicts; at small $n_k$ the power-law factor makes the fall somewhat faster. A tolerance of $1\,\mathrm{meV}$ needs 12 points.

*One gauge.* For the gauge test we take the matrix $\partial H/\partial k$ of the model, whose band-diagonal elements are the group velocities, and interpolate its interband element $|(\partial H/\partial k)_{+-}|$, which ranges between 0.27 and $1.00\,\mathrm{eV}\,a$ along the zone. With $n_k = 6$ we compute it in the band basis of one set of random phases. Rotated into the Wannier gauge with the eigenvectors of that set, it interpolates to within $10^{-15}\,\mathrm{eV}\,a$ everywhere. Rotated instead with the eigenvectors of a second, independent set, which describe the same states with other phases, it is correct in modulus at the six grid points and off between them by amounts that depend on the random phases, up to 0.3 to $0.8\,\mathrm{eV}\,a$ in different draws, as large as the matrix element itself. The interpolated bands are exact in both cases.

The interpolated energies, velocities and vertices on fine grids are the raw material of the phonon-assisted rates of Sects. 6.5 and 13.3. Wannier functions make the electron–phonon vertex as short-ranged as the Hamiltonian, which is what lets those grids be filled (Sect. 6.2), except for the long-range dipole part of polar crystals, which is split off analytically (Sect. 27.2).

## 27.2 Wannier functions in practice

The first-principles phonon-assisted absorption spectrum of silicon was converged to an energy resolution of 14 meV with $40 \times 40 \times 40$ grids for both the electron wavevector $\boldsymbol{k}$ and the phonon wavevector $\boldsymbol{Q}$ [9]. That is $64{,}000^2 \approx 4 \times 10^9$ pairs $(\boldsymbol{k}, \boldsymbol{Q})$, each with a vertex $g_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q})$ for every pair of bands and every branch. A direct DFPT calculation costs about one ground-state calculation per perturbation and wavevector, so 64,000 wavevectors are already out of reach, and the matrix elements between all those states even more so. The same study computed the vertices directly on a $6^3$ grid and produced the rest by Wannier interpolation. In practice the work is in choosing and testing the Wannier functions and in keeping every interpolated quantity in one gauge.

### 27.2.1 Computed quantities

A Wannier calculation turns quantities known on a coarse uniform grid into real-space matrices, and those back into the same quantities anywhere in the zone. The bands are interpolated through Eqs. (27.11) and (27.12), and the eigenvectors of $H^W(\boldsymbol{k})$ define $U(\boldsymbol{k})$ on the fine grid. For the electron–phonon vertex, the coarse-grid vertices are rotated with the coarse-grid $U$ matrices and the phonon displacement patterns are undone, which leaves a matrix in two lattice vectors, one for the electron hop and one for the displaced atom [10, Eqs. (177)–(179)]. Back on a fine grid,

$$g_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q}) = \sum_{\boldsymbol{R}_e, \boldsymbol{R}_p} \mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{R}_e + \boldsymbol{Q}\cdot\boldsymbol{R}_p)} \sum_{m'n'\kappa\alpha} U_{mm'}(\boldsymbol{k}+\boldsymbol{Q})\, g^W_{m'n'\kappa\alpha}(\boldsymbol{R}_e, \boldsymbol{R}_p)\, U^\dagger_{n'n}(\boldsymbol{k})\, \frac{e^\alpha_{\kappa\nu}(\boldsymbol{Q})}{\sqrt{2M_\kappa \omega_{\boldsymbol{Q}\nu}}}, \tag{27.15}$$

in the normalization of Eq. (A.25). The phonon frequencies and polarization vectors on the fine grid come from the interpolated force constants of Sect. 25.2. The same construction interpolates the velocity matrix elements of Sect. 3.2, which the optical limit needs, and, with the restriction of Sect. 27.1, the density vertex $\rho_{mn}(\boldsymbol{k}, \boldsymbol{q})$.

### 27.2.2 Projections and disentanglement

The four bond-centered Wannier functions of the valence bands of silicon need no disentanglement, and they are the natural choice for the valence part of a polarization calculation (Eq. (25.15)). A dark-matter rate needs the conduction bands as well, and with them the eight disentangled $sp^3$ functions.

Projection onto atomic $sp^3$ orbitals alone reproduces the bands of silicon well except where higher bands with some $sp^3$ character mix in; forcing the whole valence manifold into the frozen window removes those deviations [3, Fig. 5]. For an electron-recoil calculation the inner (frozen) window should contain every state that can be a final state at the energies of interest, plus a margin, since a conduction state outside the frozen window is reproduced only approximately.

The number of Wannier functions grows with the energy range that must be described. For phonon-assisted absorption up to the direct gap and beyond, Ref. [9] extracted 26 Wannier functions from 34 bands, reproducing the band structure from 10 eV below to 30 eV above the Fermi level. The range is set by the intermediate states. A second-order amplitude sums over all bands $m$, and the Wannier manifold truncates that sum; the remainder above the manifold can be added by the projected Sternheimer equation Eq. (25.22).

Every set of Wannier functions passes four checks before use. The spread functional $\Omega_W$ must be converged and the individual spreads small compared with the cell. The centers must sit where the symmetry says, on the bonds or the atoms. For a time-reversal-symmetric

crystal the maximally localized functions can be chosen real, and a large ratio of imaginary to real parts signals a poor minimum. Above all, the interpolated bands must agree with a direct calculation at points off the coarse grid, inside the frozen window, to the tolerance the application needs. In silicon the last check has a twist: the conduction minimum lies on the $\Delta$ line at about 0.85 of the way from $\Gamma$ to X, not at a high-symmetry point, so a coarse grid that does not resolve it misplaces the valley. Ref. [11] needed a $12^3$ coarse $\boldsymbol{k}$ grid for the conduction bands for this reason, while $6^3$ was enough for the phonon wavevectors.

### 27.2.3 Interpolating vertices, and the long-range parts

The decay of $g^W(\boldsymbol{R}_e, \boldsymbol{R}_p)$ decides whether Eq. (27.15) converges. In the electron vector $\boldsymbol{R}_e$ it decays at least as fast as the Wannier functions themselves. In the atom vector $\boldsymbol{R}_p$ it decays as the screened potential of a displaced atom. In a nonpolar semiconductor or insulator that potential falls off at least as fast as a quadrupole, $R_p^{-3}$, so the vertices in reciprocal space stay finite as $\boldsymbol{Q} \to 0$ and the interpolation converges [10, Sect. VI.A.2]. For diamond, coarse grids of $4^3$, $6^3$ and $8^3$ points interpolate the vertex of the highest optical branch along high-symmetry lines onto the direct DFPT values, with the $4^3$ grid already close [10, Fig. 6]. For a new material, the same comparison, of $g$ from direct DFPT at a few $(\boldsymbol{k}, \boldsymbol{Q})$ off the coarse grid with the interpolated values, is the only evidence that the interpolation converges.

In a polar crystal the displaced atom carries a dipole, its potential decays as $R_p^{-2}$, and the vertex of the longitudinal optical branch diverges as $1/Q$: the Fröhlich coupling. No finite real-space sum reproduces a divergence, so the long-range part is split off analytically. With the book's normalization,

$$g^{\mathrm{L}}_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q}) = \mathrm{i}\,\frac{e^2}{\Omega} \sum_{\kappa} \frac{1}{\sqrt{2M_\kappa \omega_{\boldsymbol{Q}\nu}}} \sum_{\boldsymbol{G} \neq -\boldsymbol{Q}} \frac{\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q}) \cdot Z^*_\kappa \cdot (\boldsymbol{Q}+\boldsymbol{G})}{(\boldsymbol{Q}+\boldsymbol{G}) \cdot \varepsilon_\infty \cdot (\boldsymbol{Q}+\boldsymbol{G})} \, \langle \psi_{m,\boldsymbol{k}+\boldsymbol{Q}} | \mathrm{e}^{\mathrm{i}(\boldsymbol{Q}+\boldsymbol{G})\cdot(\boldsymbol{r}-\boldsymbol{\tau}_\kappa)} | \psi_{n\boldsymbol{k}} \rangle, \tag{27.16}$$

with $\boldsymbol{\tau}_\kappa$ the position of atom $\kappa$ in the cell. It is the potential of the bound charge $-\nabla \cdot \boldsymbol{P}$ of the displacement wave, with $\boldsymbol{P} = e\,\boldsymbol{u}_\kappa \cdot Z^*_\kappa / \Omega$ per cell, screened by $\varepsilon_\infty$ and acting on an electron of charge $-e$, for the displacement per unit $B_{\boldsymbol{Q}\nu}$ of Eq. (A.20); it is Ref. [10, Eq. (180)] in Heaviside–Lorentz units and the normalization of Appendix A.6 [12]. The procedure subtracts $g^{\mathrm{L}}$ from the coarse-grid vertices, interpolates the smooth short-range remainder $g^{\mathrm{S}}$, and adds $g^{\mathrm{L}}$ back on the fine grid. The overlap in Eq. (27.16) is needed at the fine-grid points; for small $\boldsymbol{Q}+\boldsymbol{G}$ it is $[U(\boldsymbol{k}+\boldsymbol{Q})U^\dagger(\boldsymbol{k})]_{mn}$ in the Wannier gauge [10, Sect. VI.A.3], and the sum over $\boldsymbol{G}$ is cut off by a Gaussian factor in practice. The inputs $Z^*_\kappa$ and $\varepsilon_\infty$ come from Sect. 25.2, with the caveat about the functional's $\varepsilon_\infty$ stated there.

Besides its dipole, a displaced atom carries a dynamical quadrupole, which is nonzero even in silicon, where every Born charge vanishes, and its potential gives vertices that are finite as $\boldsymbol{Q} \to 0$ but depend on the direction of $\boldsymbol{Q}$. Such a nonanalytic angular dependence is long-ranged in real space and is not removed by a finer coarse grid: densifying the grid confines the spurious oscillations of the interpolated potential to a smaller region around $\Gamma$ without reducing their amplitude [13]. In silicon, omitting the quadrupole term changes the computed electron mobility by about 10 %, and in GaAs by about 30 % [13]. Codes that support it subtract the quadrupolar term along with the dipole; where the code does not, the error must be estimated from the small-$Q$ behavior of directly computed vertices.

Quasiparticle corrections are interpolated on the same Wannier basis and with the same $U(\boldsymbol{k})$ as every other quantity of the amplitude (Sect. 27.1). Since they change the energy denominators but not the wavefunctions, velocity matrix elements must be rescaled by the ratio of quasiparticle to Kohn–Sham transition energies, Eq. (23.19) [9].

### 27.2.4 Density vertices at large momentum

In the Wannier gauge the density vertex $\rho_{mn}(\boldsymbol{k},\boldsymbol{q})$ is the matrix $\langle w_{m'\boldsymbol{0}}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|w_{n'\boldsymbol{R}}\rangle$, Fourier transformed in $\boldsymbol{R}$. For small $\boldsymbol{q}$ one may expand $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \approx 1 + \mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}$, and the vertex follows from the overlaps and the position matrix elements already computed for the Wannier functions; this is the small-$\boldsymbol{q}$ form used for $g^{\mathrm{L}}$ above. The expansion is controlled by $|\boldsymbol{q}|$ times the spatial extent of a Wannier function, which is of the order of a bond length, 2.35 Å in silicon. At the momentum transfers of electron recoil, a keV and above, $|\boldsymbol{q}| \times 2.35\,\text{Å} \approx 1.2$ at 1 keV and grows linearly beyond, so the expansion fails, and the real-space matrix must be built anew for each $\boldsymbol{q}$ (Sect. 27.1).

The practical route therefore computes $\rho_{mn}(\boldsymbol{k},\boldsymbol{q})$ directly on the coarse $\boldsymbol{k}$ grid for each target $\boldsymbol{q}$, from the plane-wave coefficients of the states at $\boldsymbol{k}$ and at $\boldsymbol{k}+\boldsymbol{q}$, rotates it with the same $U$ matrices into the Wannier gauge, transforms it to real space in $\boldsymbol{R}$, and interpolates in $\boldsymbol{k}$ alone. The cost is one non-self-consistent calculation at the shifted grid $\boldsymbol{k}+\boldsymbol{q}$ per target $\boldsymbol{q}$, and the number of targets is set by the resolution needed in $|\boldsymbol{q}|$ and in direction. For a cubic crystal the directions can be restricted to the irreducible wedge.

### 27.2.5 Counting matrix elements and memory

**Worked example 27.2: Matrix elements of a converged silicon rate**

The setup of Ref. [9] has $N_W = 26$ Wannier functions, coarse $6^3$ grids for $\boldsymbol{k}$ and $\boldsymbol{Q}$, and fine $40^3$ grids for both, with two atoms and six branches. We count what a phonon-assisted rate needs and where it must live.

*Fine-grid vertices.* There are $N_k N_Q = 64{,}000^2 = 4.1 \times 10^9$ pairs $(\boldsymbol{k},\boldsymbol{Q})$, and at each pair $N_W^2 \times 6 = 4{,}056$ complex vertices. At 16 bytes per complex number this is $1.7 \times 10^{13}$ numbers, or 270 TB. Nobody stores it. Codes loop over $\boldsymbol{Q}$ (or over $\boldsymbol{k}$), interpolate the vertices for all partners at once, accumulate the rate, and discard them. Even one $\boldsymbol{Q}$ with all 64,000 partners and all band pairs is 4 GB, which is why the band indices are restricted at once to the bands that can be initial, final or intermediate states at the energies of interest.

*Real-space vertices.* The matrix $g^W(\boldsymbol{R}_e,\boldsymbol{R}_p)$ of Eq. (27.15) has one entry for each pair of lattice vectors in the Wigner–Seitz supercells of the two coarse grids, about $6^3 = 216$ vectors each, times $N_W^2$ band pairs and six displacement directions: $216^2 \times 676 \times 6 \approx 1.9 \times 10^8$ complex numbers, 3 GB. This is the object that is kept, and it must fit in the memory of one node, or be distributed over several.

*Interpolation work.* Eq. (27.15) is done in two stages. For each $\boldsymbol{Q}$ the sum over $\boldsymbol{R}_p$ costs $216 \times 216 \times 676 \times 6 \approx 1.9 \times 10^8$ complex multiply–adds; for each $\boldsymbol{k}$ at that $\boldsymbol{Q}$ the sum over $\boldsymbol{R}_e$ costs $216 \times 676 \times 6 \approx 8.8 \times 10^5$, plus the $U$ rotations, which are smaller. Over all pairs the second stage dominates: $4.1 \times 10^9 \times 8.8 \times 10^5 \approx 3.6 \times 10^{15}$ multiply–adds, about $3 \times 10^{16}$ floating-point operations. Restricting $\boldsymbol{Q}$ to the irreducible wedge divides this by the 48 operations of the cubic group, to $6 \times 10^{14}$, and a pair of energy windows does far more.

*Energy windows.* Near the absorption edge only valence states within a few tenths of an eV of the valence maximum and conduction states within the same range of the six conduction minima contribute. For a parabolic band of mass $m^*$, the states within $\delta$ of the edge fill a sphere of radius $k_\delta = (2m^*\delta)^{1/2}$, a fraction $(4\pi/3)k_\delta^3/V_{\mathrm{BZ}}$ of the zone, with $V_{\mathrm{BZ}} = (2\pi)^3/\Omega = 6.19\,\text{Å}^{-3}$. With $\delta = 0.3$ eV and masses of a few tenths of $m_e$, say $0.5\,m_e$ for the holes and, per conduction valley, the density-of-states mass $(m_\parallel^* m_\perp^{*2})^{1/3} = (0.92 \times 0.19^2)^{1/3}\,m_e = 0.32\,m_e$, the fractions are $5.3 \times 10^{-3}$ for one valence

band and $6 \times 2.7 \times 10^{-3} = 1.6 \times 10^{-2}$ for the six valleys. On the $40^3$ grid this leaves about 340 valence and 1,040 conduction points per band, so $3.5 \times 10^5$ pairs instead of $4.1 \times 10^9$, four orders of magnitude fewer. The window, not the interpolation, is what makes fine grids affordable near an edge.

### 27.2.6 Phonon linewidths and the Eliashberg function

The phonon linewidth $\gamma^{\mathrm{ep}}_{\boldsymbol{Q}\nu}$ and the Eliashberg function $\alpha^2F(\omega)$ of Sect. 6.4 are Fermi-surface averages of $|g|^2$: double sums over $\boldsymbol{k}$ with two delta functions that place both the initial and the final electron on the Fermi surface. They are the hardest integrals in this section, because a delta function in three dimensions samples a two-dimensional surface, and a product of two samples the intersection of two surfaces. In practice each delta function is replaced by a Gaussian of width $\sigma$. The width must be larger than the spacing of the fine-grid energies near the Fermi surface, of order $|\nabla_{\boldsymbol{k}}\varepsilon|\,\Delta k$ with $\Delta k = |\boldsymbol{b}_i|/N_{\mathrm{fine}}$ the fine-grid step and $N_{\mathrm{fine}}$ the number of fine-grid points per direction, or the sum is noise. It must also be smaller than the energy scale on which $|g|^2$ and the density of states vary, or the result is biased. Convergence means decreasing $\sigma$ while refining the grid until the result is independent of both.

For metals with complicated Fermi surfaces the requirement is severe. For lead, the DFPT phonons needed $14^3$ $\boldsymbol{k}$ points and a $10^3$ wavevector grid with a Gaussian broadening of 0.025 Ry, the electron–phonon vertices were interpolated from $6^3$ coarse grids, and the coupling strength $\lambda_{\mathrm{ep}}$ converged only with 400,000 quasi-random $\boldsymbol{k}$ points and 27,000 random wavevectors [14, Fig. 12]. Quasi-random (Sobol) sampling converged faster than uniform grids in that study [14], since it avoids the systematic aliasing of a regular grid with the Fermi surface. For an undoped insulator at $T = 0$ the electron–phonon linewidth below the gap vanishes (Sect. 6.4), and the calculation is needed only for metals, superconductors and doped semiconductors.

### 27.2.7 Convergence of the phonon-assisted dielectric function

The phonon-assisted $\varepsilon_2^{\mathrm{PA}}(\omega)$ of Sect. 6.5 is a double sum over $\boldsymbol{k}$ and $\boldsymbol{Q}$ of the squared two-path amplitude, with one energy delta function [9, Eqs. (1)–(3)]. Its convergence parameters are the two fine grids, the broadening of the delta function, the imaginary part $\Gamma_{\mathrm{w}}$ given to the intermediate states, and the number of intermediate bands. The energy resolution is set by the grid: in silicon, $40^3$ grids for both sums resolve features of 14 meV, which is what the onset needs, while the smoother spectrum between the indirect and direct gaps converges at $24^3$ with a resolution of 30 meV [9]. In that range the result is insensitive to the intermediate-state width, set to a constant 100 meV [9], because the intermediate states are far from resonance; near the direct gap they are not, and the width matters (Sect. 13.6).

A coarse grid that looks converged for bands and phonons can still be too coarse for vertices, and a fine grid that resolves a 30 meV spectrum can be too coarse for a 14 meV onset. Each convergence statement belongs to the quantity, the energy range and the resolution for which it was made.

The comparison with experiment is where the physics outside the calculation shows. The computed spectra of Ref. [9] had to be shifted rigidly by 0.15 eV to 0.23 eV to match the measured onsets. The shift absorbs the remaining error of the quasiparticle gap, of order 0.1 eV, and the electron–phonon renormalization of the gap, 0.05 eV to 0.1 eV between 0 and 400 K, which the calculation did not include [9]. A computed absorption edge without such a shift is accurate only to the accuracy of the gap.

### 27.2.8 A phonon-assisted electron-recoil calculation

A phonon-assisted electron recoil (Sect. 13.2) replaces the photon of Sect. 6.5 by the dark-matter density vertex at finite $\boldsymbol{q}$, and the four paths of its amplitude contain $\rho$, $g$ and the energies at three crystal momenta. The calculation chains the steps above, and each link has its own check. It starts from the ground state and DFPT on a coarse wavevector grid (Sect. 25.2), with the frequencies checked against neutron data and the Born charges against zero. Bands on a coarse $\boldsymbol{k}$ grid fine enough to place the conduction minimum fix the Wannier manifold, whose frozen window covers the final and intermediate states and whose bands are tested off the grid. The coarse-grid vertices $g$ are rotated into the Wannier gauge with their dipole and quadrupole parts removed, and tested against direct DFPT off the grid. For each target $\boldsymbol{q}$, $\rho_{mn}(\boldsymbol{k},\boldsymbol{q})$ is computed from the plane-wave coefficients and rotated with the same $U(\boldsymbol{k})$, and it too is tested against direct values off the grid. For each $\boldsymbol{q}$ the window is the range $\omega < E_{\min}(\boldsymbol{q})$ with no zero-phonon final state (Sect. 13.1), cut to the fine-grid pockets of states that can contribute; its edges are in Sect. 13.9. In that window the four-path amplitude is summed over the Wannier manifold, with the remainder above it from the projected Sternheimer equation Eq. (25.22), into the phonon-assisted loss function on a grid in $(|\boldsymbol{q}|,\omega)$.

Three controls close the chain. As $\boldsymbol{q}\to 0$ the density vertex reduces to its optical limit (Sect. 3.2), and the computed loss function must reproduce the phonon-assisted $\varepsilon_2^{\mathrm{PA}}$ of the same code and, after the gap shift above, the measured indirect absorption (Sect. 13.5). With $g$ set to zero the phonon-assisted rate must vanish identically, which catches any zero-phonon contamination of the window. The third is the $f$-sum rule for the density vertices of one occupied state in its nonlocal form, Eq. (26.13), whose shortfall measures the truncation of the intermediate-state sum once the nonlocal term $\delta_{\mathrm{NL}}$ is included.

The cost of a silicon run follows from the counts above. The DFPT stage is the one of Worked example 25.2, a few hundred linear-response solutions. The density vertices need one non-self-consistent calculation per target $\boldsymbol{q}$; a grid of $|\boldsymbol{q}|$ in steps of $\sim 0.1\,\mathrm{keV}$ up to a few keV and a few tens of directions in the irreducible wedge gives several hundred targets, and this stage dominates the plane-wave part of the work. The amplitude stage is a sum over pocket pairs. Near the edge the relevant states lie within $\sim 0.1\,\mathrm{eV}$ of the band extrema, and the window fractions of Worked example 27.2, which scale as $\delta^{3/2}$, drop by a factor $3^{3/2}\approx 5$ from their values at 0.3 eV. On a $120^3$ fine grid this leaves about $1.8\times10^3$ valence points per band and $5.4\times10^3$ conduction points. Three valence bands lie within 0.1 eV of the maximum, the heavy-hole, light-hole and split-off bands (44 meV below the others), so there are $\sim 3\times10^7$ pairs per $\boldsymbol{q}$. Momentum conservation fixes the phonon wavevector of each pair, $\boldsymbol{Q}=\boldsymbol{k}'-\boldsymbol{k}-\boldsymbol{q}$ up to a reciprocal-lattice vector, so each pair needs its own interpolated vertices. Restricted to one external band and $N_W$ intermediate bands, the four paths need about $4\times216\times26\times6\approx1.3\times10^5$ complex multiply–adds of interpolation per pair, plus a smaller amount for $\rho$, which is $\sim 1\times10^6$ floating-point operations. That count assumes the $\boldsymbol{R}_p$ stage of Eq. (27.15) is already done for the pair's $\boldsymbol{Q}$; it runs once per distinct $\boldsymbol{Q}$ of the pocket-difference set, at $216^2\times26\times6\approx7\times10^6$ multiply–adds per external band, and stays below the $\boldsymbol{R}_e$ stage for up to $\sim10^5$ distinct $\boldsymbol{Q}$. Per $\boldsymbol{q}$ the total is $\sim 3\times10^{13}$ operations, and for several hundred targets $\sim 1\times10^{16}$. At a sustained $10^{13}$ operations per second, typical of one modern accelerator, that is $10^3\,\mathrm{s}$, a quarter of an hour; at $5\times10^{10}$, typical of a few processor cores, it is two to three days. The estimate scales as $\delta^3$ with the energy range above the edge and as $N_{\mathrm{fine}}^6$ with the fine grid, so these two choices, not the plane-wave stages, decide the budget.

## Summary

Wannier functions recombine the Bloch states of a group of bands into orthonormal orbitals localized within a cell or two. They depend on a gauge $U(\boldsymbol{k})$, which maximal localization of the spread $\Omega_W$ fixes, and disentanglement extends them to bands that cross others, as the conduction bands of silicon do. The Hamiltonian between them, $H^W(\boldsymbol{R})$, has short range, and its Fourier series interpolates the bands exactly when that range fits inside the supercell of the coarse grid, and with an exponentially small error otherwise. Every quantity interpolated together must be rotated with one gauge. Vertices are interpolated in the same way once their long-range dipole and quadrupole parts are split off, and the density vertex is rebuilt for each momentum transfer. Near an absorption edge an energy window around the band extrema cuts the number of pairs $(\boldsymbol{k}, \boldsymbol{Q})$ on $40^3$ grids by four orders of magnitude.

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# Chapter 28

# Magnets, superconductors and defects

Silicon, the running example of the book, has neither magnetic order nor a Fermi surface; a spin-polarized calculation returns zero moment, and $\alpha^2F$ vanishes identically for an undoped insulator at $T = 0$ (Sect. 6.4). The magnetic and superconducting targets of Sects. 7.1 and 7.2 and of Chaps. 14 and 15 are worked here on NiO and aluminum. Defect properties come from supercells whose defect must be isolated from its own images, electrostatically for charged states and dynamically for a recoil.

## 28.1 Magnetic and superconducting parameters

The magnon band of NiO is set almost entirely by one number, the antiferromagnetic exchange between next-nearest-neighbor nickel ions, and the density-functional values of it collected in one comparison span a factor of two and a half, from the same crystal structure, depending on the functional and on the Hubbard correction used [1, Table III]. The superconducting gap of aluminum, by contrast, is computed from first principles to within a few percent of the measured 0.18 meV [2, Table III], but only after one parameter, the Coulomb pseudopotential, is fixed by the measured critical temperature.

### 28.1.1 DFT $+U$ and the Hubbard parameter

Semilocal functionals spread the $d$ electrons of transition-metal oxides over too many orbitals. In NiO they put nickel $d$ states at the top of the valence band and give a small gap of the wrong character, while DFT $+U$ puts oxygen $p$ states there and gives the charge-transfer gap that experiments see [3, Sect. V]. The correction acts on the occupation matrices $n^{I\sigma}_{mm'} = \sum_{v\boldsymbol{k}} f_{v\boldsymbol{k}} \langle \psi^{\sigma}_{v\boldsymbol{k}} | P^{I}_{mm'} | \psi^{\sigma}_{v\boldsymbol{k}} \rangle$ of the localized orbitals of site $I$ with spin $\sigma$, where $P^{I}_{mm'}$ projects on the atomic orbitals $m$ and $m'$ of the shell [3, Eqs. (5) and (8)]. In its simplest rotationally invariant form it adds

$$E_U = \frac{U_{\mathrm{H}}}{2} \sum_{I\sigma} \mathrm{Tr}\big[ n^{I\sigma} (1 - n^{I\sigma}) \big] \tag{28.1}$$

to the energy functional [3, Eq. (9)]. In the basis that diagonalizes $n^{I\sigma}$ the trace is $\sum_i \lambda_i (1 - \lambda_i)$ over its eigenvalues $0 \le \lambda_i \le 1$, so the correction penalizes fractional occupations and drives each orbital toward full or empty. The physical content is a restoration of the piecewise linearity of the energy as a function of the occupation of the shell, which semilocal functionals violate.

The parameter $U_{\mathrm{H}}$ can be computed rather than fitted. In the linear-response method of Cococcioni and de Gironcoli, a small potential shift $\delta V_I^U$ is applied to the localized orbitals of one site in a supercell, and the response of the occupations $n_J$ is computed twice: once self-consistently, giving the response matrix $R_{IJ}^U = \partial n_I / \partial(\delta V_J^U)$, and once at fixed Kohn–Sham potential, giving the bare response $R_{IJ}^{U,0}$. The Hubbard parameter is the difference of their inverses,

$$U_{\mathrm{H}} = \left[ (R^{U,0})^{-1} - (R^U)^{-1} \right]_{II}, \tag{28.2}$$

which removes the curvature of the energy that comes from rehybridization of the noninteracting bands [3, Eqs. (15), (18) and (19)]. The convergence parameter is the supercell size, since the perturbation of one site must not interact with its images. Density-functional perturbation theory replaces the supercells by a sum over wavevectors in the primitive cell and reaches the same $U_{\mathrm{H}}$ with better numerical control: for nickel in NiO, the two routes agree to $5 \times 10^{-3}$ eV, and the value converges from 7.90 to 8.17 eV as the supercell grows from $2 \times 2 \times 2$ to $4 \times 4 \times 4$ or the equivalent $\boldsymbol{q}$ grid is refined [4, Table I].

The value is not unique. It depends on the projectors that define $n^{I\sigma}$, on the functional and on the pseudopotential [4, Sect. V], and on whether $U_{\mathrm{H}}$ is recomputed self-consistently from the DFT $+U$ ground state. For the same nickel shell in NiO, published linear-response values include 4.6 eV with nonorthogonal atomic projectors [3, Sect. V], 8.17 eV with Löwdin-orthogonalized projectors from a semilocal ground state [4, Table I], and 6.26 eV computed self-consistently [5, Sect. II]. A $U_{\mathrm{H}}$ is meaningful only together with its projectors, and a calculation that takes $U_{\mathrm{H}}$ from one code and uses it in another changes the physics.

### 28.1.2 Exchange constants

The constants below are in the convention of the Heisenberg model Eq. (7.6). The literature uses every combination of sign, of single or double counting of pairs, and of spin vectors of length $|\boldsymbol{S}|$ or of unit length. The conversions between them are factors of 2 and of $|\boldsymbol{S}|^2$ [6, Table 1], and a value of $J^{\mathrm{ex}}$ quoted without its Hamiltonian is not a number.

**Energy mapping.** The most direct route computes the total energies of several collinear magnetic configurations in DFT $+U$ and equates their differences to those of Eq. (7.6) with classical spins. In the fcc lattice of nickel ions in NiO, with $J_1$ between the 12 nearest neighbors and $J_2$ between the 6 next-nearest neighbors, three configurations fix both. In the ferromagnet each ion has all its neighbors parallel. In the antiferromagnet of type I, with ferromagnetic (001) planes alternating in sign, 4 of the 12 nearest neighbors are parallel and 8 antiparallel, and all 6 next-nearest neighbors, at $\pm a$ along the cube axes, lie in planes of the same sign. In the type-II structure of the ground state, with ferromagnetic (111) planes alternating, half of the nearest neighbors are parallel and all next-nearest neighbors antiparallel. Counting each pair once, the energies per ion are

$$E_{\mathrm{FM}} = (6J_1 + 3J_2)|\boldsymbol{S}|^2, \qquad E_{\mathrm{AFI}} = (-2J_1 + 3J_2)|\boldsymbol{S}|^2, \qquad E_{\mathrm{AFII}} = -3J_2|\boldsymbol{S}|^2, \tag{28.3}$$

relative to a common reference, and therefore

$$J_1 = \frac{E_{\mathrm{FM}} - E_{\mathrm{AFI}}}{8|\boldsymbol{S}|^2}, \qquad J_2 = \frac{E_{\mathrm{FM}} + 3E_{\mathrm{AFI}} - 4E_{\mathrm{AFII}}}{24|\boldsymbol{S}|^2}. \tag{28.4}$$

These are Ref. [1, Eqs. (6) and (7)] converted to Eq. (7.6): that reference uses unit vectors, counts each pair twice and puts a minus sign in front, so that its $J_i$ equal $-J_i/2$ of ours for $|\boldsymbol{S}| = 1$. The two measured values for NiO that it collects, from Shanker and Singh and from Hutchings and Samuelsen [7], are $J_2 = 17.3$ and 19.0 meV in this convention.

More configurations resolve more neighbors. A cleaner variant flips one chosen pair of spins in a large supercell, in the four combinations $\uparrow\uparrow$, $\uparrow\downarrow$, $\downarrow\uparrow$ and $\downarrow\downarrow$, and isolates the coupling of that pair from all others; Ref. [5, Sect. II] uses it for NiO in $4\times4\times4$ supercells. The mapping has three weaknesses. The magnetic moment computed in DFT is not $2|\boldsymbol{S}|$ Bohr magnetons, and the classical $|\boldsymbol{S}|^2$ in Eq. (28.3) must be replaced by a choice, $|\boldsymbol{S}|^2$ or $|\boldsymbol{S}|(|\boldsymbol{S}|+1)$, that changes the constants by up to a factor of two for $|\boldsymbol{S}|=1$. The moments may change size from one configuration to the next, which a Heisenberg model cannot describe. And the result inherits the full dependence on $U_{\mathrm{H}}$: in NiO the superexchange scales roughly as $t_{\mathrm{hop}}^2/U_{\mathrm{H}}$ (Sect. 7.1), so the spread of $U_{\mathrm{H}}$ above becomes a spread of $J_2$, and the density-functional values of Ref. [1, Table III] range from 11 to 27 meV in our convention.

**Spin spirals.** A spin spiral of wavevector $\boldsymbol{Q}$, with every moment tilted by a cone angle $\theta$ from a fixed axis and rotated in phase by $\boldsymbol{Q}\cdot\boldsymbol{R}$ from cell to cell, can be computed in the primitive cell when spin–orbit coupling is neglected, by a generalized Bloch theorem. For a lattice with one magnetic ion per cell, Eq. (7.6) has $\boldsymbol{S}_{\boldsymbol{R}}\cdot\boldsymbol{S}_0=|\boldsymbol{S}|^2[\sin^2\theta\cos(\boldsymbol{Q}\cdot\boldsymbol{R})+\cos^2\theta]$ and therefore

$$E(\boldsymbol{Q},\theta)-E(0,\theta)=\frac{|\boldsymbol{S}|^2\sin^2\theta}{2}\big[J^{\mathrm{ex}}(\boldsymbol{Q})-J^{\mathrm{ex}}(0)\big],\qquad J^{\mathrm{ex}}(\boldsymbol{Q})=\sum_{\boldsymbol{R}\neq0}J^{\mathrm{ex}}_{0\boldsymbol{R}}\,\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}},\tag{28.5}$$

per ion, the single-sublattice case of Ref. [8, Eq. (7)]. A set of spirals on a grid of $\boldsymbol{Q}$ maps out $J^{\mathrm{ex}}(\boldsymbol{Q})$, and a Fourier transform gives the real-space constants out to a range set by the grid. Small cone angles keep the perturbation of the ground state small, which reduces the change of the moments that plagues collinear mapping [8, Sect. II]. For a ferromagnet, linear spin-wave theory in the convention of Eq. (7.6) has $\omega(\boldsymbol{Q})=|\boldsymbol{S}|[J^{\mathrm{ex}}(\boldsymbol{Q})-J^{\mathrm{ex}}(0)]$, so the spiral energy is the magnon energy directly: $\omega(\boldsymbol{Q})=2[E(\boldsymbol{Q},\theta)-E(0,\theta)]/(|\boldsymbol{S}|\sin^2\theta)$, the frozen-magnon relation. For a collinear antiferromagnet with two sublattices, the same theory has

$$\omega(\boldsymbol{Q})=|\boldsymbol{S}|\Big\{\big[J^{\mathrm{ex}}_{\mathrm{AB}}(0)-J^{\mathrm{ex}}_{\mathrm{AA}}(0)+J^{\mathrm{ex}}_{\mathrm{AA}}(\boldsymbol{Q})\big]^2-|J^{\mathrm{ex}}_{\mathrm{AB}}(\boldsymbol{Q})|^2\Big\}^{1/2},\tag{28.6}$$

with $J^{\mathrm{ex}}_{\mathrm{AA}}(\boldsymbol{Q})$ and $J^{\mathrm{ex}}_{\mathrm{AB}}(\boldsymbol{Q})$ the sums of $J^{\mathrm{ex}}_{0\boldsymbol{R}}\,\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$ over the sites $\boldsymbol{R}\neq0$ of the same and of the other sublattice; with $J^{\mathrm{ex}}_{\mathrm{AB}}=0$ it reduces to the ferromagnetic form. The spirals then need one cone angle per sublattice [8, Eq. (8)], and the two sums follow from their energies.

**Green's functions.** The third route needs only the ground state. Rotating two moments by small angles changes the energy, by the magnetic force theorem, by the change of the single-particle energies at fixed potential, which second-order perturbation theory expresses through the Green's function of the ground state. The result is the formula of Liechtenstein, Katsnelson, Antropov and Gubanov, which in a Wannier basis reads

$$J_{ij}^{\mathrm{LKAG}}=-\frac{1}{4\pi}\int_{-\infty}^{\mu_{\mathrm{ch}}}\mathrm{d}\varepsilon\;\mathrm{Im}\,\mathrm{Tr}\big[\Delta_i\,\mathsf{G}^{\uparrow}_{ij}(\varepsilon)\,\Delta_j\,\mathsf{G}^{\downarrow}_{ji}(\varepsilon)\big]\tag{28.7}$$

in the convention of its source [6, Eq. (19) and Table 1], $H=-\sum_{i\neq j}J_{ij}^{\mathrm{LKAG}}\,\hat{\boldsymbol{e}}_i\cdot\hat{\boldsymbol{e}}_j$ with unit vectors and each pair counted twice, so that $J_{ij}^{\mathrm{ex}}=-2J_{ij}^{\mathrm{LKAG}}/|\boldsymbol{S}|^2$ in the convention of Eq. (7.6). Here $\Delta_i$ is the exchange splitting of the on-site Hamiltonian of ion $i$, $\mathsf{G}^{\sigma}_{ij}$ the block of the spin-$\sigma$ Green's function between the Wannier orbitals of ions $i$ and $j$, the trace runs over orbitals, and the integral runs up to the chemical potential $\mu_{\mathrm{ch}}$ (Sect. 2.5). One ground-state calculation yields every pair at once, ferromagnetic or antiferromagnetic, and decomposes each constant by orbital. The formula describes infinitesimal rotations about the ground state, while energy mapping measures large ones, and the two differ where the

moments are not rigid. For NiO, linear spin-wave theory with constants from energy mapping overestimates the measured magnon energies by about 14 percent, and with constants from the Green's-function formula by up to 40 percent [5, Sect. III]. The convergence parameters of Eq. (28.7) are the Wannier basis, which must contain the magnetic $d$ orbitals and the ligand $p$ orbitals that mediate superexchange, and the $\boldsymbol{k}$ grid of the Green's function, which sets the range of $|\boldsymbol{R}_i - \boldsymbol{R}_j|$ that is free of images.

**Spin response.** The most complete route skips the Heisenberg model. Time-dependent density-functional perturbation theory computes the transverse spin susceptibility of the DFT $+U$ ground state as a function of $(\boldsymbol{Q}, \omega)$, and its poles are the magnons; constants are then fitted to the dispersion only for comparison [5, Sect. II]. This is the spin analogue of the dielectric response of Sect. 26.1, and it costs correspondingly more.

### 28.1.3 Eliashberg theory

The gap and critical temperature of a phonon-mediated superconductor follow from the Eliashberg equations, the Nambu–Gor'kov form of the electron self-energy with the phonon propagator of Sect. 6.4 and the Coulomb repulsion. On the imaginary axis, at the fermionic Matsubara frequencies $\omega_j^{\mathrm{F}}$ of Sect. 4.7, and for an isotropic gap restricted to the Fermi surface, they are two coupled equations for a mass-renormalization function $Z^{\mathrm{El}}(\mathrm{i}\omega_j^{\mathrm{F}})$ and a gap function $\Delta^{\mathrm{El}}(\mathrm{i}\omega_j^{\mathrm{F}})$,

$$Z^{\mathrm{El}}(\mathrm{i}\omega_j^{\mathrm{F}}) = 1 + \frac{\pi T}{\omega_j^{\mathrm{F}}} \sum_{j'} \frac{\omega_{j'}^{\mathrm{F}}}{\sqrt{(\omega_{j'}^{\mathrm{F}})^2 + \Delta^{\mathrm{El}}(\mathrm{i}\omega_{j'}^{\mathrm{F}})^2}}\, \lambda_{\mathrm{ep}}(\omega_j^{\mathrm{F}} - \omega_{j'}^{\mathrm{F}}), \tag{28.8}$$

$$Z^{\mathrm{El}}(\mathrm{i}\omega_j^{\mathrm{F}})\, \Delta^{\mathrm{El}}(\mathrm{i}\omega_j^{\mathrm{F}}) = \pi T \sum_{j'} \frac{\Delta^{\mathrm{El}}(\mathrm{i}\omega_{j'}^{\mathrm{F}})}{\sqrt{(\omega_{j'}^{\mathrm{F}})^2 + \Delta^{\mathrm{El}}(\mathrm{i}\omega_{j'}^{\mathrm{F}})^2}} \Big[\lambda_{\mathrm{ep}}(\omega_j^{\mathrm{F}} - \omega_{j'}^{\mathrm{F}}) - \mu_{\mathrm{C}}^*\Big], \tag{28.9}$$

with the frequency-dependent coupling

$$\lambda_{\mathrm{ep}}(\omega^{\mathrm{B}}) = \int_0^\infty \mathrm{d}\omega\, \frac{2\omega\, \alpha^2 F(\omega)}{\omega^2 + (\omega^{\mathrm{B}})^2}, \tag{28.10}$$

evaluated at the bosonic differences $\omega^{\mathrm{B}} = \omega_j^{\mathrm{F}} - \omega_{j'}^{\mathrm{F}}$, so that $\lambda_{\mathrm{ep}}(0)$ is the coupling constant $\lambda_{\mathrm{ep}}$ of Sect. 6.4. These are the isotropic forms of the Fermi-surface-restricted equations of Ref. [9, Eqs. (41), (42) and (48)]; the literature writes $Z$, $\Delta$, $\lambda$ and $\mu^*$, symbols that this book uses for other things. The highest temperature at which Eq. (28.9) has a nonzero solution is $T_c$, and the gap on the real axis follows by analytic continuation of $\Delta^{\mathrm{El}}(\mathrm{i}\omega_j^{\mathrm{F}})$, usually with Padé approximants (Appendix C.5).

The Coulomb pseudopotential $\mu_{\mathrm{C}}^*$ is the weak point. It replaces the screened Coulomb repulsion between paired electrons, renormalized down from its bare Fermi-surface value by the retardation of the phonon interaction, and in most calculations it is not computed but chosen, with values from 0.1 to 0.2 giving reasonable agreement with experiment [9, Sect. 4]. Since $T_c$ depends on it exponentially, a computed $T_c$ with a chosen $\mu_{\mathrm{C}}^*$ is a consistency check on $\alpha^2 F$ rather than a prediction. Two alternatives remove the parameter: computing $\mu_{\mathrm{C}}^*$ as a Fermi-surface average of the screened interaction, and density-functional theory for superconductors [9, Sect. 4].

A quick estimate replaces the equations by the McMillan formula in the form of Allen and

Dynes,

$$T_c = \frac{\omega_{\log}}{1.2} \exp\left[-\frac{1.04(1+\lambda_{\mathrm{ep}})}{\lambda_{\mathrm{ep}} - \mu_{\mathrm{C}}^*(1+0.62\lambda_{\mathrm{ep}})}\right], \qquad \omega_{\log} = \exp\left[\frac{2}{\lambda_{\mathrm{ep}}}\int_0^\infty \mathrm{d}\omega\, \frac{\alpha^2F(\omega)}{\omega} \ln\omega\right], \tag{28.11}$$

which is Ref. [9, Eq. (30)], a fit to numerical solutions of the Eliashberg equations useful for preliminary estimates.

**Convergence and cost.** The input $\alpha^2F$ is a double Fermi-surface average of $|g|^2$, with two energy delta functions, and it converges only on fine grids of $\boldsymbol{k}$ and $\boldsymbol{Q}$, which in practice means Wannier interpolation of the vertices (Sect. 27.2). The anisotropic equations, with a gap that varies over the Fermi surface, need the vertices on commensurate fine grids, restricted to an energy window of a few times the largest phonon energy around the Fermi level [9, Sect. 4]. The Matsubara sum needs frequencies up to a cutoff, commonly ten times the largest phonon frequency, with convergence tests recommended [9, Sect. 4]. Since the spacing is $2\pi T$, the number of frequencies grows as $1/T$, and low-$T_c$ superconductors are expensive: for aluminum, with phonons up to 40.5 meV ($\omega_{\max} \approx 470\,\mathrm{K}$ [2, Sect. III]), a cutoff of ten times that holds about 750 positive frequencies at 1 K and 7500 at 0.1 K. Sparse sampling of the Matsubara axis reduces the count [9, Sect. 4]. The analytic continuation to the real axis is the last numerical hazard: Padé approximants can produce spurious structure near the gap edge, which is where the pair-breaking absorption of Chap. 15 begins, and a real-axis solution or a check against the imaginary-axis data is needed there.

**Worked example 28.1: Aluminum from its Eliashberg function**

Linear-response calculations of aluminum give $\lambda_{\mathrm{ep}} = 0.44$, against 0.42 from tunneling, and a logarithmic phonon frequency $\omega_{\log} = 270\,\mathrm{K}$ [2, Tables II and III]. Solving the Eliashberg equations with this $\alpha^2F$, the measured $T_c = 1.18\,\mathrm{K}$ is reproduced with $\mu_{\mathrm{C}}^* = 0.12$, and the zero-temperature gap is then 0.18 meV, equal to the measured value [2, Table III]. The weak-coupling ratio of Sect. 7.2 agrees, $\Delta_{\mathrm{BCS}} = 1.764\,T_c = 0.179\,\mathrm{meV}$, as it should for $\lambda_{\mathrm{ep}} = 0.44$.

Eq. (28.11) with the same three numbers gives $T_c = (270/1.2)\,\mathrm{e}^{-5.213}\,\mathrm{K} = 1.22\,\mathrm{K}$, within four percent of the full solution. Its sensitivity to $\mu_{\mathrm{C}}^*$ is the logarithmic derivative of the exponent, $\partial \ln T_c/\partial\mu_{\mathrm{C}}^* = -1.04(1+\lambda_{\mathrm{ep}})(1+0.62\lambda_{\mathrm{ep}})/[\lambda_{\mathrm{ep}} - \mu_{\mathrm{C}}^*(1+0.62\lambda_{\mathrm{ep}})]^2 = -23$. A change of $\mu_{\mathrm{C}}^*$ by $\pm 0.02$, well inside the range used in the literature, moves $T_c$ by a factor of 0.60 or 1.53, from 0.74 K at $\mu_{\mathrm{C}}^* = 0.14$ to 1.87 K at 0.10. The gap and $T_c$ of aluminum are measured, and the first-principles $\alpha^2F$ enters where a measured quantity is not available, in the frequency dependence of the phonon-assisted pair-breaking absorption of Chap. 15.

Wherever a magnetic or superconducting target is ranked by a computed parameter, the spread over $U_{\mathrm{H}}$ and mapping, or over $\mu_{\mathrm{C}}^*$, is the theory uncertainty to carry (Sect. 8.8).

## 28.2 Defects

A carbon pair with a bound hydrogen, the T center of silicon, emits its zero-phonon line at 935 meV; a hybrid-functional calculation in a 512-atom supercell puts it at 985 meV [10]. In the Stillinger–Weber potential, displacing a silicon atom from its site takes between 17.5 and 63.5 eV, depending on the direction in which it is struck [11]. Both numbers are defect properties that enter the book: the first in the color centers and dopants that serve as targets and readouts (Sects. 7.4 and 16.6), the second in the energy that a nuclear recoil stores in the

lattice (Chap. 19). The effective-mass theory of shallow donors, which no supercell reaches, is in Sect. 7.4 and Appendix D.17.

### 28.2.1 Formation energies and charge states

The formation energy of a defect $D$ in charge state $z_\mathrm{D}$, in units of $e$, compares two supercells, with and without the defect,

$$E_\mathrm{D}^\mathrm{f}(z_\mathrm{D}) = E_\mathrm{D}(z_\mathrm{D}) - E_\mathrm{host} - \sum_s n_s\,\mu_s^\mathrm{at} + z_\mathrm{D}\big(\varepsilon_\mathrm{VBM} + \mu_\mathrm{ch}\big) + E_\mathrm{corr}, \tag{28.12}$$

where $E_\mathrm{D}$ and $E_\mathrm{host}$ are the total energies of the defective and the host supercell, $n_s$ atoms of species $s$ are added ($n_s > 0$) or removed ($n_s < 0$) and exchanged with a reservoir at chemical potential $\mu_s^\mathrm{at}$, the electron chemical potential $\mu_\mathrm{ch}$ is measured from the valence maximum $\varepsilon_\mathrm{VBM}$, and $E_\mathrm{corr}$ is the finite-size correction of the next subsection [12, Eq. (3)]. The atomic chemical potentials are bounded by the stability of the host against its elements and against competing phases, and $\mu_\mathrm{ch}$ ranges over the gap [12, Sect. 3]. Where two charge states have equal formation energy, $E_\mathrm{D}^\mathrm{f}(z_\mathrm{D}) = E_\mathrm{D}^\mathrm{f}(z'_\mathrm{D})$, Eq. (28.12) gives the thermodynamic transition level

$$\mu_\mathrm{ch}(z_\mathrm{D}/z'_\mathrm{D}) = \frac{E_\mathrm{D}^\mathrm{f}(z_\mathrm{D})\big|_{\mu_\mathrm{ch}=0} - E_\mathrm{D}^\mathrm{f}(z'_\mathrm{D})\big|_{\mu_\mathrm{ch}=0}}{z'_\mathrm{D} - z_\mathrm{D}}, \tag{28.13}$$

the electron chemical potential at which the defect changes charge with full relaxation of the lattice. It is the quantity that photoluminescence quenching and capture experiments measure, and it differs from a Kohn–Sham eigenvalue, which carries no relaxation and no correction.

The defect is relaxed in a supercell as close to cubic as possible, built from the conventional cell, at the volume of the host [12, Sect. 3]. The work lies in the number of charge and spin states, which must all be tried, since a defect in a nonmagnetic host may have a spin-polarized ground state. The gap problem of semilocal functionals is more serious for defects than for bulk bands, because defect levels sit inside the gap and their position relative to both band edges matters; hybrid functionals, which give silicon a gap of 1.114 eV [10], are the standard for levels and optical energies.

### 28.2.2 Charged defects in a periodic cell

A charged defect in a supercell is a periodic array of charges, neutralized by a uniform background so that the electrostatic energy is finite. Both the interaction with the images and the background are artifacts. The leading term is the energy of a simple-cubic lattice of point charges $z_\mathrm{D}e$ in a compensating background and a dielectric medium. In Heaviside–Lorentz units, with $e^2 = 4\pi\alpha$, the Coulomb energy of two charges $e$ at distance $r$ in vacuum is $\alpha/r$, and the Madelung energy per charge of the array is $-z_\mathrm{D}^2\alpha\,\alpha_\mathrm{M}/(2\varepsilon L_\mathrm{sc})$, with $\alpha_\mathrm{M} = 2.8373$ the Madelung constant of the simple-cubic lattice [13, Sect. II]. The correction removes it:

$$E_\mathrm{corr} \simeq \frac{z_\mathrm{D}^2\,\alpha\,\alpha_\mathrm{M}}{2\varepsilon\,L_\mathrm{sc}} + O(L_\mathrm{sc}^{-3}), \tag{28.14}$$

where the next term, of order $z_\mathrm{D}L_\mathrm{sc}^{-3}$ times the quadrupole moment of the defect charge, comes from the shape of that charge; this is the Makov–Payne form, quoted in Gaussian units in Ref. [12, Eq. (8)]. The dielectric constant is $\varepsilon_0$ for a relaxed defect, whose screening includes the ions, and $\varepsilon_\infty$ for a vertical transition at frozen nuclei; in silicon the two are equal, 11.7. The bare point-charge term often overcorrects: for the doubly charged carbon vacancy in diamond it does so clearly [13, Sect. III B], because the defect charge is not a point. The quadrupole term is hard to determine, and a second artifact remains: the average

electrostatic potential of a charged cell is set by convention, so that band edges and defect levels of different cells must be aligned. The schemes in current use, by Freysoldt, Neugebauer and Van de Walle and by Kumagai and Oba, model the defect charge as a Gaussian, compute its periodic energy analytically, and align the potential far from the defect, read from the plane-averaged or the atomic-site potentials, with the model potential [14, Sect. II and Eq. (1)]. The Kumagai–Oba version also handles an anisotropic dielectric tensor, and shows that the conventional potential-alignment term is already contained in the image-charge correction [14, Sect. IV]. Without a correction, formation energies of charged defects converge as $1/L_{\mathrm{sc}}$, and their spread between correction schemes translates into defect densities that differ by orders of magnitude [12, Sect. 3].

### 28.2.3 Color centers

The optical transition of a color center connects two states of a few localized electrons. Its zero-phonon energy $E_{\mathrm{ZPL}}$ is the difference of the total energies of the relaxed excited and ground states. When both states are well described by single Slater determinants, the excited state is computed by constraining the occupations, promoting an electron from one Kohn–Sham orbital to another and relaxing the structure in that configuration; this is the ΔSCF method. For the negatively charged nitrogen–vacancy center of diamond, with its triplet transition between single-determinant states, a 512-site supercell gives $E_{\mathrm{ZPL}} = 1.757\,\mathrm{eV}$ with a semilocal functional and 2.035 eV with the HSE hybrid functional, against the measured 1.945 eV [15, Sect. 4.1]. For the T center of silicon, the same approach in a 512-atom supercell with HSE and Γ-point sampling gives 985 meV against 935 meV, and the $(0/-1)$ transition level at 1.07 eV above the valence maximum [10]. The radiative lifetime, from the transition dipole of the two Kohn–Sham orbitals and the Wigner–Weisskopf rate, comes out as 1.3 to 5 µs, against a measured 0.9 µs [10].

The same two relaxed geometries give the vibronic coupling. The Huang–Rhys factor $S_{\mathrm{HR}}$ of Sect. 5.5 is the number of phonons emitted on average when the lattice relaxes after the transition, $S_{\mathrm{HR}} = \sum_k \omega_k \Delta Q_k^2/2$, with $\Delta Q_k$ the projection of the mass-weighted displacement between the two minima on the vibrational mode $k$ of frequency $\omega_k$; the zero-phonon fraction of the emission is $\mathrm{e}^{-S_{\mathrm{HR}}}$. Its convergence with supercell size is slow, because the relaxation reaches into long-wavelength acoustic modes that a small cell does not contain. For the nitrogen–vacancy center, a 64-site supercell gives $S_{\mathrm{HR}} = 3.02$, 20 percent below the converged 3.67, and the converged value needs an embedding scheme: the forces of the transition from a 512-site hybrid-functional calculation, and the vibrational modes of supercells of up to 10 648 sites from force constants computed with a semilocal functional [15, Sects. 2.3 and 4.2]. The measured value is $3.85 \pm 0.05$ [15, Sect. 3].

ΔSCF has a clear limit. States that are not single determinants, such as the singlet states of the nitrogen–vacancy center, need a correlated treatment of an active space of defect orbitals, embedded in the screened host, and the density transition matrix elements that a dark-matter excitation needs, $|\langle F|n_e(\boldsymbol{q})|I\rangle|^2$ over keV momenta, need the many-body states themselves rather than their energies (Sect. 16.6). Shallow donors are the opposite limit: their effective-mass envelope extends over several nanometers, far beyond any supercell, and a supercell calculation can only supply the central-cell correction to the effective-mass theory of Appendix D.17.

### 28.2.4 Displacement thresholds

The displacement threshold $E_{\mathrm{d}}$ of Sect. 7.4 is the smallest kinetic energy that, given to one atom in a given direction, leaves a stable defect. It is computed by molecular dynamics over

many directions. Of three common silicon potentials, Stillinger–Weber reproduces the density-functional thresholds best [11], and it still lies 4 to 6 eV above the density-functional threshold along $\langle 111 \rangle$ (Sect. 19.1.3). Sect. 29.1 gives the protocol, its cost and its approximations, among them the electronic excitation that Born–Oppenheimer dynamics leaves out (Chap. 19).

### 28.2.5 Tunneling defects in metals

A defect that tunnels between two configurations $L$ and $R$, such as a light interstitial or an atom in a disordered film, couples to the conduction electrons of a metal through the change of their scattering potential. That coupling damps the tunneling and is described by the ohmic constant $\alpha_{\mathrm{ohm}}$ of Sect. 16.8. In the Born limit, $\alpha_{\mathrm{ohm}} = 4K_{\mathrm{b}}$ with $K_{\mathrm{b}} = (\mathcal{N}_{\mathrm{F}} V_{\mathrm{d}} \Omega)^2$, where $\mathcal{N}_{\mathrm{F}}$ is the density of states per spin and volume at the Fermi level and $V_{\mathrm{d}}$ half the difference of the contact potentials of the two configurations, which sit at $\pm V_{\mathrm{d}}$ [16, Eq. (4)]. Beyond the Born limit the change of potential is described by the scattering matrices $S_L$ and $S_R$ of the conduction electrons at the Fermi energy in the two configurations, with $S = \mathrm{e}^{2\mathrm{i}\delta}$ for a channel of phase shift $\delta$, and by the eigenvalues $\mathrm{e}^{2\mathrm{i}\tilde{\delta}_\nu}$ of $S_R S_L^\dagger$, so that $\tilde{\delta}_\nu$ is the difference of the phase shifts in channel $\nu$. A channel whose phase shift changes by $\tilde{\delta}$ reduces the long-time overlap of the two Fermi seas as $t^{-(\tilde{\delta}/\pi)^2}$, the orthogonality exponent of Anderson [17], and an ohmic bath of coupling $\alpha_{\mathrm{ohm}}$ produces $t^{-2\alpha_{\mathrm{ohm}}}$. Equating the two over all channels and both spins,

$$\alpha_{\mathrm{ohm}} = \frac{1}{2\pi^2} \sum_{\sigma} \sum_{\nu} \tilde{\delta}_\nu^2 , \tag{28.15}$$

which in the Born limit, with one channel and phase shifts $\mp\pi\mathcal{N}_{\mathrm{F}} V_{\mathrm{d}} \Omega$ in the two configurations, so that $\tilde{\delta} = 2\pi\mathcal{N}_{\mathrm{F}} V_{\mathrm{d}} \Omega$, returns $\alpha_{\mathrm{ohm}} = 4(\mathcal{N}_{\mathrm{F}} V_{\mathrm{d}} \Omega)^2 = 4K_{\mathrm{b}}$. A first-principles calculation needs the two relaxed configurations, from a nudged-elastic-band search, identical relaxation of the host around both, and the scattering phases at the Fermi energy, either from a Green's-function method or from the overlap of the two many-electron ground states of a supercell as a function of its size. The orthogonality exponent appears only with fixed occupations: $\boldsymbol{k}$ sampling and smearing of the Fermi surface blur the logarithm on which it rests and must be removed or extrapolated.

**Worked example 28.2: A charged defect in a silicon supercell**

We estimate the image-charge error for a singly charged defect, $z_{\mathrm{D}} = \pm 1$, in cubic supercells of silicon built from the conventional cell, $L_{\mathrm{sc}} = na$ with $a = 5.431\,\text{Å}$, and with $\alpha\hbar c = 14.40\,\mathrm{eV\,Å}$ and $\varepsilon_0 = 11.7$. Eq. (28.14) gives

$$E_{\mathrm{corr}} = \frac{2.8373 \times 14.40\,\mathrm{eV\,Å}}{2 \times 11.7 \times L_{\mathrm{sc}}} = 0.161\,\mathrm{eV},\ 0.107\,\mathrm{eV},\ 0.080\,\mathrm{eV}$$

for 64, 216 and 512 atoms ($n = 2$, 3, 4). Even the 512-atom cell, the size used for the T center, carries a point-charge correction of 80 meV, larger than the 50 meV by which its computed zero-phonon line misses the measurement. Since the point-charge term can overcorrect (above), production calculations use the Gaussian-charge schemes; what the estimate shows is that the error does not become negligible at any affordable size. It falls only as $1/L_{\mathrm{sc}}$, that is, as $N_{\mathrm{at}}^{-1/3}$, so that halving it means eight times more atoms. The cost of a hybrid-functional calculation grows faster than the cube of the number of atoms, and a 4096-atom cell is out of reach where a 512-atom cell is routine; the correction, not the cell, is what converges the result. The neutral T center needs none of this, and its error budget is the functional; its $(0/-1)$ level at 1.07 eV does, since the charged state enters it [10].

## Summary

In NiO semilocal functionals give a small gap of the wrong character, and DFT $+U$ restores the charge-transfer gap. The Hubbard parameter is computed by linear response, 8.17 eV for nickel with Löwdin-orthogonalized projectors, but it depends on the projectors, the functional and the pseudopotential, and published linear-response values for the same shell run from 4.6 to 8.17 eV. Exchange constants follow from energy mapping of collinear configurations, from spin spirals, from the Green's-function formula of Liechtenstein and coworkers or from the poles of the spin susceptibility, and each inherits the dependence on $U_{\mathrm{H}}$. In aluminum the linear-response $\alpha^2F$ gives $\lambda_{\mathrm{ep}} = 0.44$, and the Eliashberg equations reproduce the measured $T_c$ of 1.18 K and gap of 0.18 meV once the Coulomb pseudopotential is set to 0.12. A change of $\mu^*_{\mathrm{C}}$ by $\pm 0.02$ moves $T_c$ by a factor of 0.60 or 1.53, so a computed $T_c$ is a consistency check on $\alpha^2F$ rather than a prediction. The spread over $U_{\mathrm{H}}$ and mapping, or over $\mu^*_{\mathrm{C}}$, is the theory uncertainty of a target ranked by a computed parameter. Defect energies come from supercells. A charged defect needs an image-charge correction that falls only as $1/L_{\mathrm{sc}}$ and is still 80 meV for a singly charged defect in 512 atoms of silicon, and the optical transitions of color centers whose states are single determinants come from ΔSCF. Shallow donors extend beyond any supercell and keep the effective-mass theory.

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# Chapter 29

# Recoils, transport and cascades

What a detector records after an energy deposit is decided in three stages. A struck atom leaves the region where the force constants of Sect. 5.2 apply and may end as a vacancy and an interstitial that store several eV. A hot electron multiplies into electron–hole pairs by impact ionization while it loses energy to phonons. The phonons and charges that result then cross the crystal, over microseconds, before a sensor absorbs them. Chaps. 18 to 20 derive the physics of these stages. The numbers they need come from simulations that follow many trajectories: the displacement threshold and the stored energy from molecular dynamics, the collection kernels from a transport Monte Carlo, the pair energy and the Fano factor from a cascade Monte Carlo. Each simulation is followed through silicon.

## 29.1 Recoil molecular dynamics

A silicon atom struck with 100 eV leaves its site at 26 km/s, comes within about 1 Å of a neighbor, and within a few picoseconds has either fallen back into the lattice or left behind a vacancy and an interstitial that store several eV between them. Nothing in this sequence is a small perturbation of the harmonic crystal. Bonds break and re-form, and the outcome is decided by the shape of the Born–Oppenheimer energy surface far from equilibrium. Sect. 19.1 hands the recoil over to this dynamics, and Sects. 19.4.4 and 19.5 need four of its products: the displacement threshold $E_\mathrm{d}$ as a function of recoil direction (Sect. 7.4), the distribution of the stored energy $E_\mathrm{st}$ at a given recoil energy $E_R$, the configurations of the defects that survive, and the barriers along the paths by which they recombine or move. Each comes from many trajectories on an energy surface fitted to density-functional theory, with the close collisions patched in by a screened Coulomb repulsion, and from the statistics of those trajectories over recoil directions.

### 29.1.1 Computed quantities

Molecular dynamics integrates Newton's equations for the nuclei on a single energy surface,

$$M_\kappa \, \ddot{\boldsymbol{R}}_I = -\frac{\partial E^\mathrm{ad}(\{\boldsymbol{R}\})}{\partial \boldsymbol{R}_I}, \tag{29.1}$$

where $\boldsymbol{R}_I$ is the position of nucleus $I$ of species $\kappa$ and $E^\mathrm{ad}(\{\boldsymbol{R}\})$ is the electronic ground-state energy at fixed nuclear positions, the Born–Oppenheimer surface of Sect. 5.1. A recoil simulation starts from a crystal equilibrated at the base temperature, gives one atom (the primary knock-on atom) the kinetic energy $E_R$ in a chosen direction $\hat{\boldsymbol{r}}_R$, integrates Eq. (29.1) until the excess energy has spread through the cell, and records what changed. The quantities

read off a single trajectory are whether a defect survived, which one, and the potential energy it stores. Repeated over directions and energies, the trajectories give the probability $p_{\mathrm{FP}}(E_R, \hat{\boldsymbol{r}}_R)$ that a Frenkel pair survives, the threshold surface $E_{\mathrm{d}}(\hat{\boldsymbol{r}}_R)$, and the distribution of $E_{\mathrm{st}}$ that Sect. 19.4.4 bounds with calibration data.

Density-functional theory (Sect. 23.3) gives $E^{\mathrm{ad}}$ and its gradients at a cost that grows as the cube of the number of atoms. A plane-wave calculation with $N_{\mathrm{at}}$ atoms holds a number of occupied bands and a number of plane waves that are both proportional to $N_{\mathrm{at}}$, and orthogonalizing the bands costs their number squared times the basis size. A recoil cascade needs thousands of atoms, to keep the recoil away from its periodic images, and thousands of trajectories, to sample directions. Direct density-functional molecular dynamics can afford a few directions in a cell of a few hundred atoms, and serves as the reference; the production runs use an interatomic potential that reproduces $E^{\mathrm{ad}}$ at a cost linear in $N_{\mathrm{at}}$.

### 29.1.2 Place on the hierarchy

In the language of the truncation label $\mathfrak{T}$ of Sect. 8.5, Eq. (29.1) is the adiabatic limit, $n_B = 0$: the electrons stay in their ground state for every nuclear configuration and never take energy from the nuclei. It keeps the anharmonicity of $E^{\mathrm{ad}}$ to all orders, which the expansion of Sect. 5.6 cannot, and it sits beyond the impulse regime of the lattice label $\mathcal{K}$, since the struck atom does not merely recoil from its site but collides with its neighbors. Two approximations come with it. The nuclei are classical, so zero-point motion is absent and the lattice starts at rest apart from thermal motion. For recoils above 10 eV this does not matter for the mean threshold, since the kinetic energies exceed the vibrational energies of a few tens of meV by orders of magnitude [1, Sect. II A]; it does set the width of the threshold step, 0.7 eV at 12.5 eV (Sect. 19.1.1). It does matter for the slow processes at millikelvin, where barriers are crossed by tunneling rather than by thermal activation (the defect paths below). The second approximation is that energy given to the electrons is lost from the description. At 1 keV a silicon recoil already transfers part of its energy to electronic excitations, a channel that needs the nonadiabatic treatment of Sect. 19.2; molecular dynamics then supplies the nuclear trajectory on which the electronic losses are evaluated.

### 29.1.3 Interatomic potentials

An interatomic potential replaces $E^{\mathrm{ad}}$ by a sum of local terms,

$$E^{\mathrm{ad}}(\{\boldsymbol{R}\}) \approx \sum_I E^{\mathrm{at}}_\kappa(\mathcal{N}_I), \tag{29.2}$$

where $\mathcal{N}_I$ is the set of positions of the neighbors of atom $I$ within a cutoff radius $r_{\mathrm{c}}$, measured from $\boldsymbol{R}_I$. Locality is what makes the cost linear in $N_{\mathrm{at}}$. It holds for covalent and metallic solids because the density matrix decays with distance, and it fails for long-range electrostatics in ionic materials, which are then added as an explicit Coulomb term.

The classical forms write $E^{\mathrm{at}}_\kappa$ as a handful of terms with physically motivated shapes. The Stillinger–Weber potential for silicon is a pair term plus a three-body term that penalizes deviations from the tetrahedral angle [2]; Tersoff-type potentials let the strength of a bond depend on its local coordination. With some ten parameters fitted to a few properties, they reproduce the crystal well and the configurations far from it less well. For the defect energies and planar faults of silicon, the analytical potentials differ from density-functional values by tens of percent for at least some of the defects [3, Sect. III E].

A machine-learned potential keeps Eq. (29.2) but lets $E^{\mathrm{at}}_\kappa$ be a flexible function of the neighborhood, fitted to density-functional data. The neighborhood is first encoded in a

descriptor that is invariant under rotations, reflections and permutations of like atoms. Examples are symmetry functions of the neighbor distances and angles feeding a neural network [4], the smooth overlap of atomic positions feeding a kernel regression [5], or a systematic expansion in body-ordered invariants, the atomic cluster expansion [6]. Equivariant message-passing networks build the descriptor from features that rotate with the structure and are passed between neighbors over several layers [7, 8]. The families differ in cost and in how much data they need; they share Eq. (29.2) and therefore share its failure modes.

The Gaussian approximation potential for silicon [3] is a well-documented example and the one we use as the silicon case. Its energy is a fixed pair term plus a kernel sum,

$$E^{\mathrm{ad}} = \sum_{I<J} V^{(2)}(|\boldsymbol{R}_I - \boldsymbol{R}_J|) + \sum_{I} \sum_{s=1}^{M_{\mathrm{rep}}} \alpha_s \, K_{\mathrm{GAP}}(\mathcal{N}_I, \mathcal{N}_s), \tag{29.3}$$

with $K_{\mathrm{GAP}}$ a similarity kernel between the neighborhood of atom $I$ and $M_{\mathrm{rep}}$ representative neighborhoods $\mathcal{N}_s$ drawn from the training data, and coefficients $\alpha_s$ fitted by regularized least squares [3, Eq. (1)]. The pair term $V^{(2)}$ is purely repulsive and is fitted to the density-functional energy of an isolated pair of silicon atoms; it carries the steep exchange repulsion at short distance and leaves the bonding to the kernel term [3, Sect. II A]. The cutoff is 5 Å.

### 29.1.4 Training sets

Each density-functional calculation in the training set yields a total energy, six components of the stress and three force components per atom, and all of them enter the fit with tolerances that set their relative weight. For silicon the database holds 2475 periodic cells with 171 815 atomic environments [3, Table I]. They cover the diamond crystal at strains and temperatures up to melting, other crystal phases ($\beta$-tin, simple hexagonal, bc8, fcc, hcp and others), liquid and amorphous cells, low-index surfaces with their reconstructions and decohesion paths, the vacancy, the divacancy and the interstitials in cells of 63 to 217 atoms, crack tips and a screw dislocation. The reference is the PW91 functional, with a plane-wave cutoff of 250 eV [3, Sect. II B].

What a recoil needs from this list is the defect region and the short-distance region. The defects are there by design. The short distances are not, and thermal sampling cannot put them there. At 1.0 Å two silicon atoms repel each other with about 51 eV (Eq. (29.4) below), while the hottest training configurations, liquid silicon near 2000 K, have $T \approx 0.17$ eV. A thermal ensemble visits such a pair separation with a Boltzmann weight of order $\mathrm{e}^{-51/0.17}$, which is zero for every practical purpose. A training set assembled from equilibrium and thermal configurations therefore leaves the potential unconstrained exactly where a recoil cascade begins.

### 29.1.5 Close collisions and the Ziegler–Biersack–Littmark splice

At separations well below a bond length the energy of two atoms is dominated by the Coulomb repulsion of the nuclei, screened by the electrons that remain between them. The standard description is the universal screened potential of Ziegler, Biersack and Littmark (ZBL) [9, Chap. 2],

$$V_{\mathrm{ZBL}}(r) = \frac{Z_1 Z_2 \, \alpha}{r} \, \phi\!\left(\frac{r}{a_{\mathrm{U}}}\right), \qquad a_{\mathrm{U}} = \frac{0.8854 \, a_0}{Z_1^{0.23} + Z_2^{0.23}}, \tag{29.4}$$

with $\phi(x) = 0.1818\,\mathrm{e}^{-3.2x} + 0.5099\,\mathrm{e}^{-0.9423x} + 0.2802\,\mathrm{e}^{-0.4029x} + 0.02817\,\mathrm{e}^{-0.2016x}$, $a_0$ the Bohr radius and $Z_{1,2}$ the nuclear charges. It is a fit to the repulsion of many atom pairs computed from their free-atom charge densities, which makes it universal in the sense that one screening

function serves all pairs once lengths are measured in $a_{\mathrm{U}}$. For silicon on silicon, $a_{\mathrm{U}} = 0.128\,\text{Å}$, and the unscreened Coulomb energy $Z^2\alpha/r = 2822\,\text{eV Å}/r$ is reduced by $\phi$ to 51 eV at 1.0 Å and 9.6 eV at 1.5 Å.

In a head-on collision of the recoiling atom with a neighbor at rest, the two equal masses share the kinetic energy $E_R/2$ of the center-of-mass frame, and the turning point $r_{\min}$ solves $V_{\mathrm{ZBL}}(r_{\min}) = E_R/2$. Solved numerically with Eq. (29.4) for silicon, it is 1.49, 1.20, 1.00, 0.83, 0.49 and 0.18 Å at $E_R = 20$, 50, 100, 200, 1000 and 10 000 eV (Table 29.1), against the bond length of 2.35 Å. Head-on collisions are the extreme case, and glancing ones stay farther apart. Near the threshold, $E_R \approx 20$ eV, the atoms come to about 1.5 Å, inside the bonding region that a good training set covers, while above 100 eV they reach the core region that no thermal training set covers.

The two descriptions are joined by a smooth switch,

$$E^{\mathrm{ad}} = \sum_{I<J} [1 - f_{\mathrm{sw}}(r_{IJ})]\, V_{\mathrm{ZBL}}(r_{IJ}) + f_{\mathrm{sw}}(r_{IJ})\, E^{\mathrm{ad}}_{\mathrm{ML}}, \qquad f_{\mathrm{sw}}(r) = \frac{1}{1 + \mathrm{e}^{-b_{\mathrm{sw}}(r - r_{\mathrm{sw}})}}, \quad (29.5)$$

written here schematically: in practice the switch multiplies the pair part of the machine-learned energy and the many-body part is left to vanish at short distance by its training. The center $r_{\mathrm{sw}}$ and steepness $b_{\mathrm{sw}}$ are chosen so that the potential is pure $V_{\mathrm{ZBL}}$ below about 1 Å and pure machine-learned energy above about 1.5 Å. The design rule follows from the turning points. Threshold dynamics must not feel the splice, so the machine-learned energy must be fully restored above the $r_{\min}$ of a 20 eV collision, 1.49 Å. It also must not leave a minimum, a shoulder or a jump in the force anywhere in between, which would deflect every collision that passes through it. The splice is checked against density-functional energies of a silicon pair, and of a crystal with one atom pushed toward a neighbor, at separations between 0.8 and 1.8 Å. A potential whose own pair term was already fitted to the density-functional dimer, as in Eq. (29.3), is usually close to $V_{\mathrm{ZBL}}$ near 1 Å, and the check then confirms rather than repairs.

### 29.1.6 Integrating a recoil

The equations of motion are integrated with the velocity Verlet scheme, which is time-reversible and conserves a shadow energy close to the true one, so that the total energy of an isolated cell drifts only through round-off and through forces that are not smooth. The time step is set by the fastest atom. A silicon atom with 100 eV moves at $v = (2E_R/M_\kappa)^{1/2} = 26.2\,\text{km/s}$, which is 0.26 Å per femtosecond; at 1 keV it moves 0.83 Å/fs. Resolving a collision whose turning point lies within a fraction of an Å of the steep part of $V_{\mathrm{ZBL}}$ requires steps in which no atom moves more than a few hundredths of an Å: 0.05 Å per step means 0.19 fs at 100 eV and 0.06 fs at 1 keV. After a few hundred femtoseconds the fastest atom carries a few eV, and the step can grow to the 1 fs that thermal silicon allows, about a sixty-fifth of the shortest phonon period (65 fs, from the 64.1 meV optical phonon). Recoil codes therefore adapt the step to the current maximum velocity and force, a practice the silicon and germanium threshold studies follow [1, Table I]. The check is energy conservation: with the thermostat switched off, the total energy of the cell must stay constant to well below the stored energies being measured, which for this application means a small fraction of an eV over the whole run.

The cell has to absorb the recoil. Its excess energy, $E_R$ minus whatever is stored, is spread over the cell as heat, and it must leave through a boundary layer held at the base temperature by a thermostat. The silicon and germanium studies use cells of $8 \times 8 \times 8$ conventional cubes (4096 atoms), a thermostatted border 6 Å thick, and runs of 20 ps [1, Sect. II A and Table I]. Three things set these numbers. The recoil must stop well inside the cell, so that it neither crosses the periodic boundary nor reaches the thermostat while still energetic; at 100 eV the

primary travels a few Å to about a nanometer, and $8a = 4.3\,\mathrm{nm}$ is ample. The heat must be removed without reflecting back from the thermostat as a coherent pulse, which a too thin or too stiff border does. And the run must last until the potential energy of the cell has settled; the same study fixed its run lengths from test runs longer than 5 ps, by the time the potential energy took to come within 1 eV of its final value [1, Sect. II A].

### 29.1.7 Cascade statistics

A single trajectory answers a yes-or-no question for one direction and one energy. The physical quantities are averages over the direction of the recoil, which a dark-matter or neutron scattering event draws from its own angular distribution, and over the position of the struck atom within the unit cell. The standard protocol draws $N_{\mathrm{dir}}$ directions uniformly on the sphere and, for each, increases $E_R$ in steps of about 1 eV through the threshold region and more coarsely above it. The published silicon and germanium scans use more than 1000 directions [1, Sect. II A].

Whether a defect survived is decided by an occupation count: each atom of the final configuration is assigned to the Wigner–Seitz cell of the nearest perfect-lattice site, and any site with occupation other than one marks a vacancy or an interstitial. The method has no free parameter and does not count the bond-switching defects that relax away [1, Sect. II A]. From the counts we obtain $p_{\mathrm{FP}}(E_R)$, the fraction of directions with a surviving defect, with the binomial error $[p_{\mathrm{FP}}(1 - p_{\mathrm{FP}})/N_{\mathrm{dir}}]^{1/2}$, at most 0.016 for 1000 directions. Along many directions the defect appears at one energy, disappears at a slightly higher energy because the interstitial now lands close enough to its vacancy to recombine, and reappears above that. Silicon and germanium show some of this recombination; sapphire shows so much that a direction-resolved threshold is ill-defined [1, Sect. II B]. The quantity the rate formulas of Sect. 19.4.4 need is therefore $p_{\mathrm{FP}}(E_R, \hat{\boldsymbol{r}}_R)$ itself, and $E_{\mathrm{d}}(\hat{\boldsymbol{r}}_R)$ is quoted as its lowest crossing. With the Stillinger–Weber potential the minimum threshold over all directions is 18 eV in silicon and 13 eV in germanium (with a modified Stillinger–Weber potential); above it the defect probability rises steeply, though more slowly than in diamond, because of recombination [1, Table II, Fig. 1 and Sect. II B]. Both numbers are properties of the potentials, and the germanium value is model-conditional. Kadribasic and coworkers give 17.5 and 12.5 eV for the same two potentials, the minima of threshold surfaces averaged over eight runs [10].

### 29.1.8 Stored energy and the quench protocol

The stored energy of a trajectory is the potential energy the cell keeps once all motion has died out. Taking the difference of potential energies at the end and at the start of a finite run, as the published scans do [1, Sect. II A], includes whatever heat has not yet left the cell. The size of that error follows from equipartition: in the harmonic crystal the potential energy is $\frac{3}{2}N_{\mathrm{at}}T$, so a residual temperature rise of 1 K in 4096 atoms adds $\frac{3}{2} \times 4096 \times 8.6 \times 10^{-5}\,\mathrm{eV} = 0.53\,\mathrm{eV}$, comparable to the uncertainty of the stored energy itself. A quench removes it. The final configuration is relaxed to its local minimum by energy minimization: conjugate gradients, or damped dynamics with the velocities set to zero whenever the power $-\nabla E^{\mathrm{ad}} \cdot \dot{\boldsymbol{R}}$ turns negative. The perfect crystal is relaxed the same way, and

$$E_{\mathrm{st}} = E^{\mathrm{ad}}_{\mathrm{min}}(\mathrm{final}) - E^{\mathrm{ad}}_{\mathrm{min}}(\mathrm{perfect}). \tag{29.6}$$

The minimum reached is the inherent structure of the final configuration: thermal motion is removed and the defects are kept, as long as the barriers between them exceed the small

forces of the minimizer. For the experiments of Sect. 19.4.4, which measure heat at millikelvin long after the event, Eq. (29.6) is the relevant quantity.

A second protocol choice concerns the reference energies. A cascade census produces vacancies, interstitials of several types, close Frenkel pairs and small clusters, each with a formation energy that the potential predicts with some error. The stored energy of a trajectory is roughly the sum of its defects' energies plus their interactions. To make results from different potentials, or from potential and density functional, comparable, one table of configurational energies of the recurring defects is computed once with density-functional theory in large relaxed cells, and every census is checked against it. For the silicon potential of Eq. (29.3), the formation energies of the vacancy and of the hexagonal, tetrahedral and dumbbell interstitials lie within 7 % of the density-functional reference [3, Sect. III E]. A defect storing several eV then carries an error of a few tenths of an eV from the potential alone, which is the floor on how precisely a single-pair stored energy can be quoted.

### 29.1.9 Defect paths and barriers

Once defects survive, Sect. 19.5 needs to know how they move and recombine. The tool is the nudged elastic band: a chain of $N_{\mathrm{im}}$ replicas of the cell, between two relaxed end states, is relaxed with spring forces along the chain and the true force perpendicular to it, until the chain lies on the minimum-energy path. In the climbing-image variant the highest replica is driven uphill along the path and converges onto the saddle point, whose energy relative to the initial minimum is the barrier $V_{\mathrm{b}}$ [11]. End states come from the cascade census, which at $E_R = 100\,\mathrm{eV}$ already supplies close pairs at many separations and orientations. Paths are first mapped with the machine-learned potential, at a cost of a few thousand force calls each, and the saddle points that matter are recomputed with density-functional theory in cells of 64 to 216 atoms. The silicon potential reproduces the density-functional ordering of the hexagonal and tetrahedral interstitials, which are related by a short displacement along a diffusion path, though it underestimates their energy difference; the analytical potentials make larger errors, and many reverse the order [3, Sect. III E].

A barrier becomes a rate through harmonic transition-state theory, $\Gamma = \Gamma_0\,\mathrm{e}^{-V_{\mathrm{b}}/T}$, with an attempt rate $\Gamma_0$ from the vibrational frequencies at the minimum and at the saddle. At the operating temperature of a cryogenic detector this is useless for any barrier of chemical size: at 10 mK, $T = 0.86\,\mu\mathrm{eV}$, and a barrier of 0.1 eV gives a Boltzmann factor of $\mathrm{e}^{-1.2\times10^{5}}$. What moves defects at millikelvin is tunneling through low barriers, or activation by the athermal phonons of a later event. The first needs the barrier shape and the tunneling mass along the path, which the nudged elastic band provides and Sect. 19.5 turns into a rate; the second needs the energy that reaches the defect, which the transport of Sect. 29.2 provides.

### 29.1.10 Convergence parameters and cost

For the potential: the size and coverage of the training set, the cutoff $r_{\mathrm{c}}$, the descriptor resolution, and the splice $(r_{\mathrm{sw}}, b_{\mathrm{sw}})$. For the dynamics: the time-step criterion, the cell size, the border thickness and the run length. For the statistics: $N_{\mathrm{dir}}$, the energy grid, and the sampling of the struck atom over the inequivalent sites of the cell. For the stored energy: the quench tolerance. Each is converged by varying it and checking that $p_{\mathrm{FP}}$ and the mean $E_{\mathrm{st}}$ move by less than their statistical errors.

The cost is set by the number of force evaluations. The silicon Gaussian approximation potential costs about 100 ms per atom per force call on one core [3, Sect. V]. A 4096-atom cell then needs 410 s per step, and a trajectory of $2 \times 10^4$ steps ($\sim 20\,\mathrm{ps}$ at a typical 1 fs step after the collision phase) needs $8 \times 10^6$ core-seconds, about $2 \times 10^3$ core-hours. A scan of

Table 29.1: Recoil molecular dynamics in silicon: the inputs and the quantities a production study converges.

| Quantity | Silicon |
|---|---|
| Bond length; shortest phonon period | 2.35 Å; 65 fs |
| Head-on turning point $r_{\min}$ at $E_R =$ 20, 50, 100, 200, 1000, 10 000 eV | 1.49, 1.20, 1.00, 0.83, 0.49, 0.18 Å (Eq. (29.4)) |
| Speed of the struck atom at 100 and 1000 eV | 0.26 and 0.83 Å/fs |
| Training set of the machine-learned potential | 2475 cells, 171 815 environments, PW91 [3] |
| Point-defect formation energies vs density-functional theory | within 7 % [3] |
| Cost per atom per force call | 100 ms on one core [3] |
| Cell, border, run length | 4096 atoms, 6 Å, 20 ps [1] |
| Directions sampled | $> 1000$ [1] |
| Minimum threshold (Stillinger–Weber) | 18 eV [1] |
| Residual heat per K in 4096 atoms | 0.53 eV (equipartition) |

1000 directions at 20 energies is $2 \times 10^4$ trajectories, or $5 \times 10^7$ core-hours. A production study with a machine-learned potential therefore has to use one of the faster descriptor families, on graphics processors, and has to spend its trajectories where $p_{\mathrm{FP}}$ changes fastest. A density-functional reference trajectory in a 216-atom cell costs far more per atom, but only a few directions are needed to validate the potential.

### 29.1.11 Pitfalls

The most common failure is a potential that is accurate at equilibrium and wrong in the region a recoil explores (Sect. 19.1.3). The remedy is validation on the quantity itself: thresholds along a few directions from density-functional molecular dynamics, the approach of the density-functional reference study for silicon [12].

### 29.1.12 Silicon

Table 29.1 collects the silicon numbers of this section.

## 29.2 Phonon and charge transport Monte Carlo

A 100 eV deposit in a silicon crystal ends, after the cascades of Chaps. 18 and 19, as some $10^4$ acoustic phonons of a few meV and, in a biased detector, a few tens of electrons and holes drifting toward the electrodes. The sensors on the surface see neither the deposit nor the cascade. They see phonons that arrive over microseconds after scattering off isotopes, decaying, bouncing off surfaces and crossing into metal films, and charges that arrive after drifting, emitting phonons of their own, and sometimes getting trapped. Chap. 20 derives the physics of each step; combined in one simulation of a whole detector, they turn an event into the arrival distributions called collection kernels. The method is Monte Carlo solution of the Boltzmann equation for phonons and carriers, and the silicon case is worked with the rate constants and code conventions of the public detector-physics toolkit G4CMP [13, 14].

### 29.2.1 Computed quantities

The state of a phonon is its wavevector $\boldsymbol{Q}$, its branch $\nu$ and its position; the state of a carrier is its crystal momentum, its valley or band, and its position. Between events each particle moves on a straight line with its group velocity (a carrier also accelerates in the applied field). Events are drawn at random with the rates of the scattering processes, and each event replaces the particle by its products. With many particles, the procedure samples the solution of the linear Boltzmann equation

$$\partial_t f_i + \boldsymbol{v}_i \cdot \nabla_{\boldsymbol{r}} f_i + \boldsymbol{F}_i \cdot \nabla_{\boldsymbol{k}} f_i = \Big(\frac{\partial f_i}{\partial t}\Big)_{\text{coll}}, \tag{29.7}$$

for the occupation $f_i(\boldsymbol{r}, \boldsymbol{k}, t)$ of each species $i$, with $\boldsymbol{v}_i$ its group velocity, $\boldsymbol{F}_i$ the force on it ($\pm e\boldsymbol{\mathcal{E}}$ for carriers in a field $\boldsymbol{\mathcal{E}}$, zero for phonons) and a collision term built from golden-rule rates [15]. The equation is linear because the athermal populations are dilute: phonons scatter off the static disorder and decay spontaneously, carriers emit into an empty phonon bath, and no process depends on the occupation of the other particles. The outputs are tallies at the boundaries: for every sensor, the energy and time of each phonon it absorbs, and for every electrode, the charge induced on it.

### 29.2.2 Place on the hierarchy

Every rate in the collision term is a lowest-order golden-rule rate. Isotope scattering is the second-order Born rate of a phonon off the random mass defects of Sect. 11.6. Anharmonic decay is the rate of the cubic term of the lattice potential (Sect. 5.6), at the lowest order at which it does not vanish, the cubic vertex squared. Neganov–Trofimov–Luke emission and intervalley scattering are one-phonon emission by a carrier through a deformation potential, $n_g = 2$ in the notation of the truncation label $\mathfrak{T}$ of Sect. 8.5, with $n_B = 0$. The transport itself is incoherent: each particle is a wavepacket with a well-defined momentum, and interference between successive events is dropped. This holds when the scattering rate is small compared with the particle's frequency. For a silicon phonon at 1 THz the isotope rate is $2.4 \times 10^6\,\mathrm{s}^{-1}$ (Eq. (29.8) below) against $2\pi \times 10^{12}\,\mathrm{s}^{-1}$; even at the top of the acoustic band, 15 THz, the ratio is $\sim 10^{-3}$. It also needs mean free paths long compared with the wavelength, which they are by a factor $4 \times 10^5$ at 1 THz (a wavelength of 5.4 nm against a mean free path of 2.2 mm). The Boltzmann description is therefore safe for the bulk; its approximations are concentrated in the rates and in the surface models, which is where the pitfalls below lie.

### 29.2.3 Phonon rates

Isotope scattering and anharmonic decay remove a phonon from its ballistic path. Isotope scattering redirects it, and may change its branch, at the rate [16], [14, Eq. (6)]

$$\Gamma_{\text{iso}} = B_{\text{iso}} \left(\frac{E_{\text{ath}}}{h}\right)^4, \qquad B_{\text{iso}} = 4\pi^3\, g_{\text{md}}\, \Omega_{\text{at}}\, \langle c^{-3} \rangle, \tag{29.8}$$

where $E_{\text{ath}}/h$ is the phonon frequency, $\Omega_{\text{at}}$ the volume per atom, $\langle c^{-3} \rangle = \frac{1}{3} \sum_\nu c_\nu^{-3}$ the average of the inverse cubed sound speed over the three acoustic branches $\nu$ (the source writes $1/\langle c^3 \rangle$ with the polarization-averaged cubed speed; the two agree when the speeds are equal, and $\langle c^{-3} \rangle$ is the average the Debye density of states contains, Sect. 20.2), and $g_{\text{md}} = \langle (M - \bar{M})^2 \rangle / \bar{M}^2$ the mass variance of the isotope mixture over the squared mean mass [17, Eqs. (3)–(5)]. It is the Rayleigh law of a point scatterer, valid as long as the wavelength is long compared with the interatomic spacing. For natural silicon, $g_{\text{md}} = 2.02 \times 10^{-4}$, $\Omega_{\text{at}} = \Omega/2 = 20.0\,\text{Å}^3$ and,

Table 29.2: Bulk phonon rates in natural silicon from Eqs. (29.8) and (29.9) with $B_{\mathrm{iso}} = 2.43 \times 10^{-42}\,\mathrm{s}^3$ and $A_{\mathrm{anh}} = 7.41 \times 10^{-56}\,\mathrm{s}^4$ [14]. The anharmonic rate applies to longitudinal phonons. The isotope mean free path uses the transverse speed 5.4 km/s [14].

| $E_{\mathrm{ath}}/h$ (THz) | $E_{\mathrm{ath}}$ (meV) | $\Gamma_{\mathrm{iso}}$ ($\mathrm{s}^{-1}$) | isotope mean free path | $\Gamma_{\mathrm{anh}}$ ($\mathrm{s}^{-1}$) |
|---|---|---|---|---|
| 0.5 | 2.07 | $1.5 \times 10^5$ | 3.6 cm | $2.3 \times 10^3$ |
| 1 | 4.14 | $2.4 \times 10^6$ | 2.2 mm | $7.4 \times 10^4$ |
| 3 | 12.4 | $2.0 \times 10^8$ | 27 µm | $1.8 \times 10^7$ |
| 10 | 41.4 | $2.4 \times 10^{10}$ | 0.22 µm | $7.4 \times 10^9$ |

with $c_{\mathrm{LA}} = 9.0$ km/s and $c_{\mathrm{T}} = 5.4$ km/s, $\langle c^{-3} \rangle = 1/(2.13 \times 10^{11}\,\mathrm{m^3\,s^{-3}}) = 4.69 \times 10^{-12}\,\mathrm{s^3\,m^{-3}}$, and Eq. (29.8) gives $B_{\mathrm{iso}} = 124.0 \times 2.02 \times 10^{-4} \times 2.00 \times 10^{-29}\,\mathrm{m^3} \times 4.69 \times 10^{-12}\,\mathrm{s^3\,m^{-3}} = 2.35 \times 10^{-42}\,\mathrm{s}^3$, in line with the tabulated $2.33 \times 10^{-42}\,\mathrm{s}^3$ and with the value $2.43 \times 10^{-42}\,\mathrm{s}^3$ that most transport codes use [17, Table 4], [14, Table I]. The direction and branch of the scattered phonon are then drawn with the weight $|\hat{\boldsymbol{e}} \cdot \hat{\boldsymbol{e}}'|^2$ of the polarization vectors of the initial and final modes, divided by the cube of the final frequency [14, Eq. (7)]. Repeated isotope scattering drives the branch populations of silicon to slow transverse 55 %, fast transverse 35 % and longitudinal 10 % [14, Sect. III E].

Anharmonic decay splits a longitudinal phonon into two of lower energy, at the rate [18], [14, Eq. (8)]

$$\Gamma_{\mathrm{anh}} = A_{\mathrm{anh}} \left( \frac{E_{\mathrm{ath}}}{h} \right)^5 , \tag{29.9}$$

with $A_{\mathrm{anh}}$ fixed by the second- and third-order elastic constants in the isotropic continuum [14, Eq. (8)]. Transverse phonons hardly decay at low frequency. The two channels, $\mathrm{L} \to \mathrm{L} + \mathrm{T}$ and $\mathrm{L} \to \mathrm{T} + \mathrm{T}$, have energy distributions of the daughters that the isotropic approximation fixes in closed form. For $\mathrm{L} \to \mathrm{L}' + \mathrm{T}'$, with $x = E_{\mathrm{L}'}/E_{\mathrm{L}}$ and $\delta = c_{\mathrm{LA}}/c_{\mathrm{T}}$, the rate is distributed as $x^{-2}(1-x^2)^2[(1+x)^2 - \delta^2(1-x)^2][1 + x^2 - \delta^2(1-x)^2]^2$ over $(\delta-1)/(\delta+1) < x < 1$, and the $\mathrm{T} + \mathrm{T}$ channel has a polynomial distribution of the same kind with coefficients set by the elastic constants [14, Eqs. (9) and (12)]. The daughter directions then follow from energy and momentum conservation; for the longitudinal daughter, $\cos\theta_{\mathrm{L}'} = [1 + x^2 - \delta^2(1-x)^2]/(2x)$ [14, Eq. (10)]. For silicon, Tamura's isotropic calculation gives $A_{\mathrm{anh}} = 7.41 \times 10^{-56}\,\mathrm{s}^4$, as tabulated by Leman [14, Table I]; a recent recalculation from cryogenic elastic constants gives $1.15 \times 10^{-55}\,\mathrm{s}^4$ [17, Table 3]. No direct measurement exists for silicon, so $A_{\mathrm{anh}}$ is uncertain by at least the factor 1.5 between the two.

Table 29.2 evaluates Eqs. (29.8) and (29.9) for silicon. The ratio of the two rates, $\Gamma_{\mathrm{anh}}/\Gamma_{\mathrm{iso}} = (A_{\mathrm{anh}}/B_{\mathrm{iso}})(E_{\mathrm{ath}}/h)$, equals $(E_{\mathrm{ath}}/h)/33\,\mathrm{THz}$ and is below one everywhere in the acoustic band. Decay is nevertheless what lowers the frequency, since isotope scattering conserves energy. A 10 THz longitudinal phonon decays within 0.13 ns. Its daughters near 3 THz live some 55 ns before decaying again, scattering off isotopes about ten times in between, and below 1 THz the decay time exceeds 10 µs, longer than the time a phonon needs to reach a surface of a centimeter-sized crystal.

### 29.2.4 Free flight, focusing and surfaces

Between events a phonon moves with the group velocity of its branch, obtained from the Christoffel equation of continuum elasticity, $\rho_T \omega^2 \hat{e}_i = C_{ijkl} Q_j Q_k \hat{e}_l$, for the unit polarization $\hat{\boldsymbol{e}}$ of the elastic wave and the elastic tensor $C_{ijkl}$ (Appendix C.18). In a cubic crystal the group velocity is not parallel to $\boldsymbol{Q}$, and an isotropic distribution of wavevectors becomes a strongly peaked distribution of energy flux, the phonon focusing that concentrates ballistic

phonons along certain crystal directions [17, Sect. 2.4]. Using the continuum dispersion is an approximation for the phonons near the top of the acoustic band, whose dispersion bends over; it is good for the few-THz phonons that dominate after the first decays.

A phonon that reaches a surface is reflected specularly, reflected diffusely, down-converted into lower-energy phonons, or transmitted into whatever is attached. The bare crystal surface is described by the probabilities of these outcomes, which are inputs of the simulation, fitted to measured pulse shapes for a given surface preparation. An interface to a sensor film is described by a transmission probability, taken from the acoustic mismatch model or the diffuse mismatch model [19]. Sect. 20.3 derives both and the reciprocity they obey. Measured silicon-to-aluminum values, $\bar{t}_{12} P_{\mathrm{abs}} = 0.3$–$0.55$, are absorption probabilities and not transmissions [20], and a simulation that uses them must use them in that role.

### 29.2.5 Films

Inside a superconducting film the phonon breaks Cooper pairs, the resulting quasiparticles relax by emitting phonons, and phonons with energy above $2\Delta_{\mathrm{BCS}}$ break further pairs; this is the Kaplan cascade of Sect. 20.4 [21]. A full simulation tracks quasiparticles and phonons in the film. The lumped model used in detector codes treats the film as a single object with a thickness $d_{\mathrm{f}}$, a gap $\Delta_{\mathrm{BCS}}$ and an energy-dependent phonon mean free path $\ell_{\mathrm{f}}(E_{\mathrm{ath}})$. An incident phonon is absorbed with the probability $P_{\mathrm{abs}}$ of Eq. (20.31) and otherwise escapes back into the substrate. An absorbed phonon's energy goes to two quasiparticles, which share it with the BCS density of states weight; quasiparticles below $3\Delta_{\mathrm{BCS}}$ are counted as collected, and those above emit a phonon and continue [13, Sect. 2.4]. Phonons below $2\Delta_{\mathrm{BCS}}$ cannot break pairs and are either absorbed directly with a set probability or returned to the substrate. The output is the quasiparticle yield $Y_{\mathrm{qp}}(E_{\mathrm{ath}})$, sampled event by event, and the phonons the film sends back into the crystal.

### 29.2.6 Carrier transport

In a biased detector the electrons and holes of the cascade drift in the field $\boldsymbol{\mathcal{E}}$, computed beforehand for the electrode geometry by a finite-element solution of the Laplace equation. A carrier accelerates until it moves faster than the longitudinal sound speed, and then emits acoustic phonons, the acoustic analogue of Cherenkov radiation (Sect. 20.1). These are the Neganov–Trofimov–Luke phonons, and they carry the work $eV_{\mathrm{bias}}$ that the bias does on each pair, the Luke gain of Sect. 19.4 [22]. For a carrier with an isotropic mass $m$ and momentum $k$, and the acoustic deformation potential $D^n$ of its band (Sect. 6.2) taken isotropic, the golden-rule emission rate is $\Gamma_{\mathrm{L}}(k)$ of Eq. (20.10), with $k_{\mathrm{s}} = m\, c_{\mathrm{LA}}$ the momentum of a carrier moving at the sound speed. The polar angle $\theta$ of the emitted phonon, measured from $\boldsymbol{k}$, is drawn from $P(\theta) \propto (\cos\theta - k_{\mathrm{s}}/k)^2 \sin\theta$ for $0 \le \theta \le \arccos(k_{\mathrm{s}}/k)$ [13, Eq. (9)]. The rate vanishes below $k_{\mathrm{s}}$ and grows as $k^2$ at large $k$. It counts longitudinal phonons only; a carrier faster than the transverse sound speed, which is lower, can also emit transverse ones, and a model that omits them shifts the Luke phonons toward the longitudinal branch.

Holes in silicon and germanium are close to isotropic, and Eq. (20.10) applies directly. Electrons sit in anisotropic valleys, six along $\langle 100 \rangle$ in silicon and four along $\langle 111 \rangle$ in germanium, each with the mass tensor of components $m^*_{\parallel}$ along the valley axis and $m^*_{\perp}$ across it (Sect. 3.3). A Herring–Vogt transformation rescales the momentum along each principal axis by $(m_{\mathrm{c}}/m^*_i)^{1/2}$, with the conductivity mass $3/m_{\mathrm{c}} = 1/m^*_{\parallel} + 2/m^*_{\perp}$, and turns the valley into an isotropic one. The field and the emission are handled there as for holes, and the results are transformed back [23], [13, Eqs. (11)–(12)]. The transformation is not unitary, and the emitted phonon's momentum is transformed back by keeping its magnitude, which conserves

energy, and transforming only its direction [14, Sect. V C]. Electrons also scatter between valleys by emitting or absorbing intervalley phonons, which turns their trajectories toward the field direction. The intervalley rates are parametrized, from measurements in silicon and from a model in germanium, rather than computed from first principles [13, Sect. 2.3]; the drift speed against field is then a check of the whole carrier model, which reproduces measurements in germanium [13, Fig. 7].

Impurities enter as two processes, each with a mean free path that is fitted to the device. One is trapping, in which a carrier is captured by a neutral impurity and stops, and the other is impact ionization of an impurity, in which a carrier frees a second charge from a charged one [13, Eqs. (13)–(16)]. Trapping reduces the collected charge and the Luke phonons, impact ionization increases them, and both fill the regions between the single-pair peaks of a charge-resolving silicon detector, from whose shape the four or six mean free paths are fitted [13, Fig. 6]. In a uniform field with trapping alone, the simulation must reproduce the Hecht relation of Sect. 20.1; that is the first check of a trapping model.

The step size for carriers is limited by the requirement that the rate stays nearly constant during a step. Taking a step of half the current mean free time conserves energy, in the sense that the energy of the Luke phonons emitted equals the work done by the field, and longer steps produce too few Luke phonons [14, Sect. V D]. The number of Luke phonons is large: each pair crossing a bias $V_{\mathrm{bias}}$ releases $eV_{\mathrm{bias}}$, and a code that assumes 2 meV per Luke phonon in estimating its event size [13, Sect. 3.4] expects $100\,\mathrm{eV}/2\,\mathrm{meV} = 5 \times 10^4$ phonons per pair at 100 V.

### 29.2.7 Collection kernels

The collection kernel of Sect. 20.5 is the probability that energy deposited at $\boldsymbol{r}_{\mathrm{ev}}$, in phonons of energy $E_{\mathrm{ath}}$, is absorbed in sensor $s$ within a time window. It is tallied by launching phonons of each energy from a grid of event positions and counting where they end. For charge, the analogous kernel is the induced charge on each electrode, computed from the trajectories with Ramo's theorem and the weighting potentials $\varphi_s$ of Sect. 20.1. Kernels separate the transport from the event: once tabulated, they are convolved with the deposit and the phonon spectrum of any event, from the created-phonon spectra of Chap. 19 to the down-converted spectra of Sect. 20.2, without rerunning the transport.

The statistical error of a kernel bin with $n$ absorbed out of $N_{\mathrm{l}}$ launched phonons is binomial, $[n(1 - n/N_{\mathrm{l}})]^{1/2}/N_{\mathrm{l}}$. For the collection contrasts of Sect. 20.5, which are differences between kernels of similar size, the useful technique is to run the two cases with common random numbers, so that their statistical fluctuations are correlated and cancel in the difference.

### 29.2.8 Validation and the closure checks

Four checks cost nothing: energy, charge, rate and step closure. For energy closure, the deposit, plus the work of the bias on the carriers, must equal the energy absorbed in the sensors, plus the energy lost at bare surfaces, plus the energy in trapped carriers and in particles still in flight at the end, to the round-off of the tallies. A missing term in this balance signals a process that loses energy without booking it, which is the most common silent bug in such codes. For charge closure, the charge induced on all electrodes, summed with the trapped charges weighted by their weighting potentials, must equal the charge created. For rate closure, in a large block with surfaces removed and one process switched on at a time, the measured mean free paths must follow Eqs. (29.8) and (29.9) with the input constants, with the slopes $E_{\mathrm{ath}}^{-4}$ and $E_{\mathrm{ath}}^{-5}$; this has been done for the materials of the recent rate tabulation [17, Sect. 3]. For step closure, halving every time step must leave the kernels unchanged within

their statistical errors.

Beyond these internal checks, a run must report its agreement with external benchmarks that test the physics rather than the bookkeeping. They are carrier drift velocities against field, the Hecht curve for a known trap density, the ballistic focusing pattern of a thin crystal, the quasiparticle yield $Y_{\mathrm{qp}}(E_{\mathrm{ath}})$ against the Kaplan cascade of Sect. 20.4, and, where it exists, a measured pulse shape.

### 29.2.9 Cost

The cost is set by the number of particle steps. A 10 keV deposit in silicon that has down-converted to 1 THz phonons (4.1 meV) holds $2.4 \times 10^6$ phonons. In a crystal a centimeter across, each phonon scatters off isotopes about every 2 mm (Table 29.2) and bounces off the surfaces many times before a sensor covering a few percent of the surface absorbs it. At a hundred steps per phonon the event needs $\sim 10^8$–$10^9$ steps, and a biased detector adds the $5 \times 10^4$ Luke phonons per pair per 100 V. With full tracking, a few-keV event takes a day or more of central processing unit (CPU) time, and with the down-sampling described next about an hour per event [13, Sect. 3.4 and Fig. 8]. Collection kernels are much cheaper than events, since each launched phonon is an independent sample and no event needs to be complete.

Down-sampling generates only a fraction $f$ of the phonons and carriers and gives each the weight $1/f$ [13, Sect. 3.4]. It conserves the mean of every tally. It does not conserve the fluctuations: a weighted event has a variance $1/f$ times too large for the collected energy, and a variance computed from weighted samples is not the physical variance. Kernels and means may be computed with down-sampling; the measured-charge Fano factor $F_{\mathrm{meas}}$ of Sect. 20.1, and any statement about event-by-event resolution, may not.

### 29.2.10 Pitfalls

No closure check sees the errors in three inputs: the sound speed used to convert a decay calculation to the fifth-power law, the starting phonon spectrum, and the surface, intervalley and trapping parameters. The fifth-power law of Eq. (29.9) amplifies any inconsistency in that sound speed, as the recalculation found for sapphire [17, Sect. 3.1]. The transport is usually started from phonons at the Debye frequency, on the grounds that the first decays erase the details [14, Sect. III B], which is safe for kernels at long times but not for the prompt signal near the event, the subject of the memory-loss arguments of Sect. 20.2. The surface parameters are fits to the device in which they were measured, so that transferring them to a crystal from a different boule is an assumption to be stated.

### 29.2.11 Silicon

Table 29.3 lists the silicon inputs of a phonon and charge transport run, with the checks that a run must report alongside its kernels.

Table 29.3: Phonon and charge transport in silicon: inputs, the quantities they control, and the checks a run must report. Rates are from Eqs. (29.8) and (29.9) with the cited constants.

| Input or check | Silicon |
|---|---|
| $B_{\mathrm{iso}}$; $A_{\mathrm{anh}}$ | $2.43 \times 10^{-42}\,\mathrm{s}^3$; $7.41 \times 10^{-56}\,\mathrm{s}^4$ to $1.15 \times 10^{-55}\,\mathrm{s}^4$ [14, 17] |
| Sound speeds L, T | 9.0, 5.4 km/s [14] |
| Isotope mean free path at 1 THz | 2.2 mm |
| Equilibrium branch populations slow transverse (ST) : fast transverse (FT) : longitudinal | 55 : 35 : 10 [14] |
| Si $\rightarrow$ Al absorption probability $\bar{t}_{12}P_{\mathrm{abs}}$ | 0.3–0.55 [20] |
| Luke phonons per pair at 100 V | $5 \times 10^4$ (at 2 meV each) |
| Carrier model | six $\langle 100 \rangle$ electron valleys, Herring–Vogt; isotropic holes; fitted intervalley, trapping and impact-ionization mean free paths |
| Closure checks | energy, induced charge, rate slopes $E_{\mathrm{ath}}^{-4}$ and $E_{\mathrm{ath}}^{-5}$, time step |
| Benchmarks | drift velocity, Hecht curve, focusing pattern, $Y_{\mathrm{qp}}(E_{\mathrm{ath}})$, pulse shape |

## 29.3 Hot-carrier cascades from first principles

An electron given 50 eV in silicon ends, a picosecond later, as about 13 electron–hole pairs at the band edges and some 35 eV of phonons. Which share goes where, and how much the pair number fluctuates from one event to the next, is decided by a competition between two processes at every energy along the way. One is impact ionization, in which the hot carrier gives part of its energy to a valence electron and lifts it across the gap; the other is phonon emission, in which it gives a few tens of meV to the lattice. Sect. 18.1 writes the cascade as a branching process whose kernels are master-formula rates with the carrier as the projectile, and Sect. 18.2 defines the mean energy per pair $\varepsilon_{\mathrm{eh}}$ and the Fano factor $F_{\mathrm{eh}}$ as its first two moments. The kernels are computed from the band structure, the screened interaction and the electron–phonon vertex, and silicon is the case where both measurements and a calibrated phenomenological model exist to compare against.

### 29.3.1 Computed quantities

For a seed class $\Lambda$, the set of initial electron and hole states that the primary excitation produces (Sect. 18.3), the calculation produces the distribution $P_{\mathrm{casc}}(N_{\mathrm{eh}}|\Lambda)$ of the final number of pairs. Its moments are the measured quantities,

$$\varepsilon_{\mathrm{eh}} = \frac{E_0}{\langle N_{\mathrm{eh}} \rangle}, \qquad F_{\mathrm{eh}} = \frac{\langle N_{\mathrm{eh}}^2 \rangle - \langle N_{\mathrm{eh}} \rangle^2}{\langle N_{\mathrm{eh}} \rangle}, \tag{29.10}$$

for a seed of total energy $E_0$, in the limit of large $E_0$ where both become independent of it. The cascade needs the quasiparticle bands $\varepsilon_{n\boldsymbol{k}}$ from the band edges up to the highest energy a carrier reaches, tens of eV for the cascades of Chap. 18; the impact-ionization rate $\Gamma_{\mathrm{ii}}(n\boldsymbol{k})$ of a carrier in state $n\boldsymbol{k}$, with the distribution of its three products over final states; and the phonon emission and absorption rate $\Gamma_{\mathrm{ph}}(n\boldsymbol{k})$, with its final states and phonon branches.

The cascade Monte Carlo then follows each carrier from state to state until every carrier is below the impact-ionization threshold $\theta_{\mathrm{ii}} E_{\mathrm{g}}$ (Sect. 18.1), after which it can only lose energy to phonons and cannot add a pair.

### 29.3.2 Place on the hierarchy

Each kernel is a golden-rule rate at a definite place on the truncation label $\mathfrak{T}$ of Sect. 8.5. The impact-ionization rate is the decay of a quasiparticle into a lower-energy quasiparticle and an electron–hole pair through the dynamically screened interaction $W^{\mathrm{scr}}$ of Sect. 4.8; with $W^{\mathrm{scr}}$ in the random-phase approximation, its many-body approximation is $\mathcal{X} = \mathrm{RPA}$, applied to quasiparticle rather than Kohn–Sham energies. The phonon rate is one-phonon emission or absorption, $n_g = 2$, with $n_B = 0$. The cascade itself is a sequence of independent golden-rule events: the carrier is on shell between events, and interference between successive events is dropped. This is the same incoherent limit as the transport of Sect. 29.2, and it holds when the total scattering rate of a carrier is small compared with its energy above the band edge, the condition that the off-shell parameter be small. A rate of $10^{14}\,\mathrm{s}^{-1}$ corresponds to a width $\Gamma = 0.066\,\mathrm{eV}$ in natural units, so a carrier 5 eV above the conduction-band minimum meets the condition even at rates ten times larger. A carrier a few tens of meV above the impact-ionization threshold, whose energy relative to that threshold is comparable to the width, does not, and the threshold region is where the incoherent picture is least reliable.

### 29.3.3 Quasiparticle bands

The impact-ionization threshold is set by the gap, and the rate just above it rises steeply with the energy above threshold, so the gap must be right before anything else is. Semilocal density-functional theory places the conduction bands of silicon too low, with an indirect gap of about half the measured 1.17 eV (Sect. 23.3). Quasiparticle corrections in the $G_0W_0$ approximation (Sect. 23.4) repair the gap and the ordering [24]. A self-consistent variant gives 1.23 eV for silicon without spin–orbit coupling, and a single empirical scaling of the self-energy correction brings the gap to experiment and the other tabulated conduction-band energies of silicon to within 0.2 eV of measurement [25, Table I]. A rigid scissor shift fixes the gap by assuming that the correction is the same for every conduction state, which has to be checked over the tens of eV that the cascade spans.

The band window is large. The number of bands below an energy $E$ above the band bottom is about the number of plane waves in a sphere of radius $k_E = (2m_eE)^{1/2}$ in the unit cell, $\Omega k_E^3/(6\pi^2)$. For silicon, with $\Omega = a^3/4 = 40.05\,\text{Å}^3$, this is about 8 bands up to 20 eV, 32 up to 50 eV and 90 up to 100 eV. All of these bands, on a dense $\boldsymbol{k}$ grid, enter both kernels.

### 29.3.4 The impact-ionization kernel

The rate at which a quasiparticle in an empty state $n\boldsymbol{k}$ decays is set by the imaginary part of its self-energy $\Sigma$ (Sect. 4.7). In the $GW$ approximation, $\Sigma = \mathrm{i}GW^{\mathrm{scr}}$, the imaginary part for an unoccupied state is [25, Eqs. (1)–(2)]

$$\begin{aligned} \Gamma_{\mathrm{ii}}(n\boldsymbol{k}) &= 2 z_{\mathrm{qp}} \,|\operatorname{Im}\Sigma_{n\boldsymbol{k}}|, \\ \operatorname{Im}\Sigma_{n\boldsymbol{k}} &= -\pi \sum_{n'\boldsymbol{k}'} \iint \mathrm{d}^3r\,\mathrm{d}^3r'\,\psi^*_{n\boldsymbol{k}}(\boldsymbol{r})\psi_{n'\boldsymbol{k}'}(\boldsymbol{r}) \\ &\times \operatorname{Im} W^{\mathrm{scr}}(\boldsymbol{r},\boldsymbol{r}',\varepsilon_{n\boldsymbol{k}} - \varepsilon_{n'\boldsymbol{k}'})\,\psi^*_{n'\boldsymbol{k}'}(\boldsymbol{r}')\psi_{n\boldsymbol{k}}(\boldsymbol{r}'), \end{aligned} \tag{29.11}$$

with the sum over empty final states $n'\boldsymbol{k}'$ between the Fermi level and $\varepsilon_{n\boldsymbol{k}}$, and $z_{\mathrm{qp}}$ the quasiparticle residue of Eq. (23.18), typically about 0.8 [25, Sect. II]. The physical content

is in $\operatorname{Im} W^{\mathrm{scr}}$: at energy transfer $\omega = \varepsilon_{n\boldsymbol{k}} - \varepsilon_{n'\boldsymbol{k}'}$ it is the spectral weight of the excitations the carrier can create, which below the plasmon energy are electron–hole pairs and above it are mostly plasmons. In silicon the plasmon loss peak lies at 16.6 eV in electron energy-loss spectra [26, Table I] and at 16.4 eV in the loss function from optical constants (Sect. 4.4). For a carrier more than a plasmon energy above the band edge Eq. (29.11) includes plasmon emission automatically; the plasmon then decays into a pair.

The golden-rule route writes the rate directly as a golden-rule sum over the two electrons and one hole of the final state. It uses the matrix element of the screened interaction between the initial and final pairs, direct and exchange terms combined as $|M|^2 = \frac{1}{2}(|M_{\mathrm{D}}|^2 + |M_{\mathrm{E}}|^2 + |M_{\mathrm{D}} - M_{\mathrm{E}}|^2)$, and momentum conservation $\boldsymbol{k} = \boldsymbol{k}' + \boldsymbol{k}_1 - \boldsymbol{k}_2$ [27], [25, Eq. (5)]. For weak $\operatorname{Im} W^{\mathrm{scr}}$ the two expressions differ by the factor $z_{\mathrm{qp}}$ and by the exchange term, which Eq. (29.11) omits; together these make the two rates differ by a factor between 0.5 and 1 [25, Sect. II]. Omitting exchange alone multiplies the rate by no less than 0.75 [25, Sect. II]. On the logarithmic scale on which impact-ionization rates are usually compared this is a small difference. For $F_{\mathrm{eh}}$, which depends on how the rate rises above threshold relative to the phonon rate, it is not negligible, and the choice belongs among the stated approximations of a calculation.

The cascade needs more than the total rate. When an ionization event is selected, the energies and momenta of the three products are drawn from the integrand of the rate, resolved by final state. The kernel is then a table of probabilities. For each initial $(n, \boldsymbol{k})$ it holds the probabilities of the final $(n', \boldsymbol{k}')$ of the scattered carrier and of the transfer $(\boldsymbol{q}, \omega)$ to the pair; the pair's electron and hole are then drawn with the weight of the joint density of states at that $(\boldsymbol{q}, \omega)$. The screened interaction enters at all transfers up to the carrier energy, so $W^{\mathrm{scr}}$ is needed with full frequency dependence from 0 to tens of eV and with local fields, which is the dielectric calculation of Sect. 26.1 carried to high frequency.

### 29.3.5 The electron–phonon kernel

The phonon rate of a carrier in state $n\boldsymbol{k}$ is the golden-rule rate of Sect. 6.2,

$$\Gamma_{\mathrm{ph}}(n\boldsymbol{k}) = 2\pi \sum_{m\nu} \int \frac{\mathrm{d}^3 Q}{V_{\mathrm{BZ}}} \, |g_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q})|^2 \Big[ (n_{\boldsymbol{Q}\nu} + 1)\, \delta(\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}} - \omega_{\boldsymbol{Q}\nu}) + n_{\boldsymbol{Q}\nu}\, \delta(\varepsilon_{n\boldsymbol{k}} - \varepsilon_{m,\boldsymbol{k}+\boldsymbol{Q}} + \omega_{\boldsymbol{Q}\nu}) \Big], \tag{29.12}$$

with $V_{\mathrm{BZ}}$ the zone volume, the emission term carrying $n_{\boldsymbol{Q}\nu}+1$ and the absorption term $n_{\boldsymbol{Q}\nu}$. At the millikelvin temperatures of a detector, $n_{\boldsymbol{Q}\nu} = 0$ for the optical and zone-boundary phonons that dominate, and only emission survives. The vertex $g_{mn\nu}$ comes from density-functional perturbation theory (Sect. 25.2) and is interpolated to fine grids with Wannier functions (Sect. 27.2) [28, 29].

The catch for a cascade is the energy range. The Wannier set that interpolates the band edges, eight $sp^3$-like orbitals per silicon cell spanning the valence bands and the lowest four conduction bands (Sect. 27.2), does not represent states tens of eV up, and its interpolated $g$ is meaningless there. Either the Wannier manifold is enlarged, with disentanglement over a window that contains tens of entangled bands, or the vertex is computed directly on a coarser grid at high energy, where the rates are smoother functions of $\boldsymbol{k}$ because the density of final states averages over many bands. Above a few eV the phonon rate is dominated by the emission of optical and intervalley phonons of about 60 meV [26, Sect. II B], so the energy lost per phonon event is nearly fixed, and what the calculation must get right is the rate, not the detailed spectrum.

### 29.3.6 Full-band cascade Monte Carlo

With the two kernels tabulated, the cascade is sampled as in full-band device simulation [30], without a field. Each carrier is a state $(n, \boldsymbol{k})$; its total rate is $\Gamma_{\mathrm{tot}} = \Gamma_{\mathrm{ii}} + \Gamma_{\mathrm{ph}}$. The time to its next event is drawn by self-scattering: a constant ceiling $\Gamma_{\max} \geq \Gamma_{\mathrm{tot}}$ is chosen, a time is drawn from $\Gamma_{\max} \mathrm{e}^{-\Gamma_{\max} t}$, and at that time a real event is accepted with probability $\Gamma_{\mathrm{tot}}/\Gamma_{\max}$, otherwise nothing happens [15]. Without a field the times only order the events and do not change the outcome, but they are kept because the same code with a field gives the drift of Sect. 29.2. An accepted event is impact ionization with probability $\Gamma_{\mathrm{ii}}/\Gamma_{\mathrm{tot}}$, which replaces the carrier by three, or phonon emission, which lowers its energy by $\omega_{\boldsymbol{Q}\nu}$ and moves it to $m, \boldsymbol{k} + \boldsymbol{Q}$. Holes are followed in the valence bands with their own kernels, and the hole band width, about 12 eV in silicon, limits the energy a hole can take [26, Sect. II A]. A carrier that falls below $\theta_{\mathrm{ii}} E_{\mathrm{g}}$ is removed from the list with its pair counted. The cascade ends when the list is empty.

The seed matters at low energy. Sect. 18.1 explains why the cascade must start from the final states of the primary process, with their momentum and band structure, rather than from a carrier of given energy. The energy sharing between the first electron and hole sets how much energy is above threshold, and below about 20 eV in silicon different sharing assumptions give different pair yields [26, Sect. IV B]. The seed distribution of Sect. 18.3 is drawn into the initial carrier list, and $P_{\mathrm{casc}}(N_{\mathrm{eh}}|\Lambda)$ inherits the seed class $\Lambda$.

### 29.3.7 The single-parameter model

Silicon has a cascade model that reproduces the measured pair energy, and it is the benchmark. The model of Alig and Bloom, as implemented by Ramanathan and Kurinsky [26], replaces both kernels by free-particle forms with a single optical phonon of energy $\omega_{\mathrm{AB}} = 63$ meV; it is the free-particle limit of the random-$k$ cascade of Sect. 18.1. For a carrier at energy $E$ above its band edge, the ratio of the phonon and impact-ionization rates is then [26, Eq. (8)]

$$\frac{\Gamma_{\mathrm{ph}}(E)}{\Gamma_{\mathrm{ii}}(E)} = A_{\mathrm{AB}} \, \frac{105}{2\pi} \, \frac{(E - \omega_{\mathrm{AB}})^{1/2}}{(E - E_{\mathrm{g}})^{7/2}}, \tag{29.13}$$

with energies in eV and one constant $A_{\mathrm{AB}}$, the ratio of the squared phonon and Coulomb matrix elements times phase-space factors, and the probability that the next event is an ionization is $[1 + \Gamma_{\mathrm{ph}}/\Gamma_{\mathrm{ii}}]^{-1}$ [26, Eq. (10)]. Alig and coworkers fitted $A_{\mathrm{AB}} = 5.2$ to the measured pair energy of silicon [31, Sect. II], and Ramanathan and Kurinsky find that this value reproduces the pair energies measured at room temperature [26, Table I and Sect. IV A]. They quote it in $\mathrm{eV}^2$, with which the ratio in Eq. (29.13) carries a leftover $\mathrm{eV}^{-1}$. Hot-electron injection data confirm the model up to 5 eV [26, Sect. III and Table I]. The cascade of Sect. 18.4 fixes it instead by the measured pair energy at 123 K in its own solution of the model, $A_{\mathrm{AB}} = 5.703\,\mathrm{eV}^3$ (Eq. (18.20)). The original value 5.2 makes the ionization probability per event 0.05 at 3 eV, 0.36 at 5 eV, 0.77 at 8 eV and 0.88 at 10 eV, from Eq. (29.13) with the millikelvin gap $E_{\mathrm{g}} = 1.17$ eV. Phonon emission wins below about 6 eV, where the two probabilities cross, and the many 63 meV losses a carrier suffers on its way down from there are what puts $\varepsilon_{\mathrm{eh}}$ at three times the gap.

A charge-resolving silicon sensor at 123 K measures $\varepsilon_{\mathrm{eh}} = (3.749 \pm 0.001)$ eV and $F_{\mathrm{eh}} = 0.119 \pm 0.002$ [32]. With $A_{\mathrm{AB}} = 5.703\,\mathrm{eV}^3$ tuned to the first at $E_{\mathrm{g}} = 1.160$ eV, the model predicts $F_{\mathrm{eh}} = 0.114$, $2.4\sigma$ low (Sect. 18.4). This is the comparison the first-principles cascade must improve on. Since its kernels contain no fitted constant, matching $\varepsilon_{\mathrm{eh}}$ is a test of the kernels, and $F_{\mathrm{eh}}$ is a second, independent one.

### 29.3.8 Convergence parameters

For the bands: the quasiparticle correction, the number of bands (Eq. (29.10) needs every band a carrier can reach) and the $\boldsymbol{k}$ grid. For $W^{\mathrm{scr}}$: the plane-wave cutoff of the dielectric matrix, which must resolve momentum transfers up to about twice the largest carrier momentum, the number of bands in the polarizability, the frequency grid from 0 to the largest carrier energy, and the $\boldsymbol{q}$ grid. For the kernels: the $\boldsymbol{k}$ grid on which $\Gamma_{\mathrm{ii}}$ and $\Gamma_{\mathrm{ph}}$ are tabulated, and the broadening of the energy-conserving delta functions. Near threshold the impact-ionization rate is a sum over few final states that conserve both energy and momentum. On a finite grid it scatters strongly from one $\boldsymbol{k}$ to the next; with 1728 $\boldsymbol{k}$ points in the zone the numerical error is about a factor two at rates below $10^{10}\,\mathrm{s}^{-1}$ [25, Fig. 2]. The scatter is a physical feature of the band structure smeared by the grid, and it averages out at higher energies, where many final states contribute [25, Sect. III].

The statistical parameter is the number of simulated cascades $N_{\mathrm{casc}}$. The mean $\langle N_{\mathrm{eh}}\rangle$ converges quickly, since its relative error is $[F_{\mathrm{eh}}/(\langle N_{\mathrm{eh}}\rangle N_{\mathrm{casc}})]^{1/2}$. The variance converges slowly: for a nearly Gaussian pair number, the relative error of a sample variance is $(2/N_{\mathrm{casc}})^{1/2}$. Resolving $F_{\mathrm{eh}}$ to $\pm 0.002$ at 0.119, the precision of the silicon measurement, needs a relative error of 1.7 % and therefore $N_{\mathrm{casc}} \approx 2/0.017^2 \approx 7 \times 10^3$ cascades at each seed energy, and more if the comparison is to be made at the two-standard-deviation level.

### 29.3.9 Cost

The dominant cost is the full-frequency screened interaction. For carriers up to 50 eV in silicon, the momentum transfers reach $2k_E = 7.2\,\text{Å}^{-1}$, and a dielectric matrix that resolves them holds $N_G \approx \Omega(2k_E)^3/(6\pi^2) \approx 260$ plane waves. Stored at 200 frequencies on the 29 irreducible points of an $8^3$ $\boldsymbol{q}$ grid, in double-precision complex numbers, the matrices $W^{\mathrm{scr}}_{\boldsymbol{GG'}}(\boldsymbol{q},\omega)$ occupy $29 \times 260^2 \times 200 \times 16\,\mathrm{byte} \approx 6\,\mathrm{GB}$, and computing them requires the polarizability summed over about 30 empty bands on a $\boldsymbol{k}$ grid of similar density. For carriers up to 100 eV the band count triples, $N_G$ grows to about 730 and $N_G^2$ eightfold. The kernel tables add a sum over final states for every tabulated initial state: with $N_k$ points and $N_b$ bands it scales as $N_k^2 N_b^2$ before symmetry, which for the 1728-point grid above and 30 bands is about $3 \times 10^9$ terms, each a contraction with $W^{\mathrm{scr}}$. The phonon kernel is a standard Wannier-interpolated calculation (Sect. 27.2), cheaper than the electronic one once the high-energy vertex problem is solved. The cascade Monte Carlo is cheap: a 5 keV seed makes some 1300 pairs and a few times $10^4$ phonon events, so that $10^4$ cascades amount to some $6 \times 10^8$ events, negligible next to the tabulation of the kernels.

### 29.3.10 Pitfalls

The gap must match the detector temperature, 1.17 eV at millikelvin against 1.12 eV at room temperature [26, Table I], since every rate near threshold depends on the energy above $\theta_{\mathrm{ii}} E_{\mathrm{g}}$. Two double countings are easy to make: the $GW$ rate already contains plasmon emission, so no plasmon channel is added; and the first-principles phonon rate of Eq. (29.12) replaces, and is not added to, a phenomenological one. Two models that leave out the same physics do not validate each other's $F_{\mathrm{eh}}$. The full-band cascade of Sect. 18.3.3, with no fitted constant and a pair energy of 4.15 eV, gives the same $F_{\mathrm{eh}} = 0.114$ as the single-parameter model, and both lack the plasmon channel, excitonic final states and momentum-dependent phonon couplings (Sect. 18.4).

Table 29.4: The hot-carrier cascade in silicon. Band and matrix sizes are free-electron estimates from the text. The ionization probabilities follow from Eq. (29.13) with the room-temperature value $A_{\mathrm{AB}} = 5.2$ [26] and $E_{\mathrm{g}} = 1.17\,\mathrm{eV}$. The model Fano factor follows from the 123 K calibration $A_{\mathrm{AB}} = 5.703\,\mathrm{eV}^3$ of Sect. 18.4; the measurement is at 123 K and 5.9 keV [32].

| Quantity | Silicon |
|---|---|
| Gap at low temperature; quasiparticle gap without spin–orbit | 1.17 eV; 1.23 eV [25] |
| Plasmon energy by electron energy loss; hole band width | 16.6 eV; 12 eV [26] |
| Bands up to 20, 50, 100 eV | 8, 32, 90 |
| $N_G$ and storage of $W^{\mathrm{scr}}$ for 50 eV carriers | 260; 6 GB (29 $\boldsymbol{q}$, 200 frequencies) |
| $\boldsymbol{k}$ grid; threshold-region error | 1728 points; factor 2 below $10^{10}\,\mathrm{s}^{-1}$ [25] |
| Ionization probability per event at 3, 5, 8 eV | 0.05, 0.36, 0.77 (Eq. (29.13)) |
| $\varepsilon_{\mathrm{eh}}$, $F_{\mathrm{eh}}$: model | 3.749 eV (fitted), 0.114 |
| $\varepsilon_{\mathrm{eh}}$, $F_{\mathrm{eh}}$: measured | $(3.749 \pm 0.001)$ eV, $0.119 \pm 0.002$ |
| Cascades for $\delta F_{\mathrm{eh}} = 0.002$ | $7 \times 10^3$ per seed energy |

### 29.3.11 Silicon

Table 29.4 summarizes the silicon inputs, the reference values and the sizes derived in this section.

## Summary

Recoil molecular dynamics integrates Newton's equations on the Born–Oppenheimer surface, with classical nuclei and electrons that never take energy from them, using a machine-learned potential trained on density-functional cells and spliced to the Ziegler–Biersack–Littmark repulsion at short range. The displacement threshold is the lowest crossing of the defect probability, scanned over more than 1000 directions, and the stored energy is read after a quench, since 1 K of residual heat in 4096 atoms adds 0.53 eV. The transport Monte Carlo samples the linear Boltzmann equation for phonons and carriers with lowest-order rates: isotope scattering and anharmonic decay for phonons, one-phonon emission through a deformation potential for carriers. It returns collection kernels that are convolved with any event, and the closure of energy, charge, rates and time step costs nothing and catches the most common errors. The cascade Monte Carlo takes impact ionization from the imaginary part of the *GW* self-energy on quasiparticle bands, and phonon emission from the electron–phonon vertex of DFPT. Its benchmark in silicon is the single-parameter model, which, tuned to the measured pair energy of 3.749 eV, predicts a Fano factor of 0.114 against the measured $0.119 \pm 0.002$. A first-principles cascade has no fitted constant, so matching $\varepsilon_{\mathrm{eh}}$ tests its kernels and $F_{\mathrm{eh}}$ is a second, independent test.

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# Part II

# Specifications

# Chapter 30

# Lattice and special targets

One harmonic calculation per crystal by the DFPT of Chap. 25, about 100 core-hours for silicon, supplies every lattice channel of Chap. 11 for silicon, germanium and gallium arsenide (Sect. 30.1). The electronic specifications of Chap. 31 start from the same ground state and phonons. The anharmonic and disordered solids of Sect. 11.5 and the targets of Chaps. 14 to 16 have no such common base. Their specifications cost from CPU minutes, for the conserving response of clean aluminum and the ejection factors of graphene, to $10^5$ to $10^6$ hours on graphics processors for the two-level systems of a glass. At millikelvin the response of a quantum crystal is bounded only by the band of its exact moments (Sect. 30.2). The specifications of Chaps. 30 to 32 share one rule: each calculation is accepted against an exact limit or sum rule and, wherever one exists, a measurement made at the same kinematics, never against the estimate it replaces.

## 30.1 Lattice libraries

Every lattice channel of Chap. 11 uses the same few objects for a given crystal: force constants, the frequencies and polarization vectors they give, the per-atom phonon density of states, the displacement tensors $B_\kappa$, and, in a polar crystal, the Born charges and $\varepsilon_\infty$. The multiphonon tables of Sect. 11.3 follow from them in minutes. For the candidate targets named in Sect. 17.5 most of the same numbers are not tabulated, and some of the tables need more than harmonic DFPT: all-electron densities at large momentum, path-integral momentum distributions, isotope-resolved Hessians and frequency-dependent Born charges. The DFPT method is that of Sect. 25.2, the anharmonic and path-integral methods those of Sect. 25.3.

### 30.1.1 The shared harmonic base

One harmonic lattice per material serves every lattice channel, and a mismatch between channels computed from different lattices is an error that no single channel reveals. The base consists of the force constants $\Phi_{\kappa\kappa'}(\Delta\boldsymbol{R})$, the eigenpairs $(\omega_{\boldsymbol{Q}\nu}, \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q}))$, the per-atom density of states $D_\kappa(\omega)$, the tensors $B_\kappa$ and their off-site generalization $B_{\kappa\kappa'}(\Delta\boldsymbol{R})$ (Eq. (11.24)), and for GaAs $Z^*_\kappa$ and $\varepsilon_\infty$. On the truncation label $\mathfrak{T}$ (Sect. 8.5) it sits at $n_B = 0$ and $n_A = 0$, static DFPT on the harmonic Born–Oppenheimer surface. The setting is the one of the representative calculations of Sect. 23.3 and Sect. 5.2, shared by every specification of Chaps. 30 to 32: PBE, SG15 norm-conserving pseudopotentials at 60 Ry, and the measured room-temperature lattice constants of Table 23.1, 5.431 Å (Si), 5.658 Å (Ge) and 5.653 Å (GaAs), with the GaAs frequencies also reported at the static-lattice constant 5.638 Å, at which Sect. 5.4 quotes them. The germanium and gallium pseudopotentials carry the 3d shell, which converges the total energy more slowly (Sect. 23.3), so for those two the frequencies are

checked once at 80 Ry. Silicon uses a $12^3$ $\boldsymbol{k}$ grid and an $8^3$ wavevector grid (29 irreducible points), which the phonon-assisted window needs (Sect. 31.1); germanium and GaAs use $8^3$ and $6^3$. DFPT runs to a self-consistency threshold of $10^{-16}$, with the electric-field perturbation for GaAs, the potential changes kept on disk for the vertices, and Fourier interpolation with the acoustic sum rule imposed on the force constants. GaAs needs the nonanalytic term of Eq. (25.13), without which its longitudinal-optical branch is wrong at small $\boldsymbol{Q}$.

One low-symmetry wavevector of silicon at 40 Ry with an $8^3$ $\boldsymbol{k}$ grid takes about eight minutes on four cores; scaled by $(60/40)^{3/2}$ for the cutoff and $(12/8)^3$ for the grid, the 29 wavevectors take about 24 h on four cores, roughly 100 core-hours, in less than 2 GB. Germanium takes 13 to 16 minutes per wavevector on four cores at its smaller grids, about 15 core-hours, and GaAs with the electric field about 20.

The frequencies change by less than 0.1 meV between the two coarsest wavevector grids, and they are compared with neutron data at Γ, X and L (silicon [1], germanium [2]) with a tolerance per branch, since the error of PBE at the measured lattice constant differs from branch to branch. The optical and longitudinal-acoustic branches pass at 3 %. The transverse-acoustic branches of silicon are 8 % soft at X and L in this setting (Sect. 5.2), and pass at 10 %; those of germanium are 11.6 to 13.5 % soft on the $4^3$ grid and pass at 15 % until the test of Sect. 31.1 has separated the grid error from the functional. Where a transverse-acoustic frequency sets an edge, the neutron-scaled branch is the reference and the unscaled one is reported beside it. The phase convention matters for every coherent quantity downstream: the eigenvector carries the cell phase $\mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$, and the intracell phase $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}$ is taken at the unreduced $\boldsymbol{q}$ (Appendix A.3).

The deformation-potential tensors $D^n$ of Eq. (6.13) belong with the base, since they set the soft acoustic limit of the phonon-assisted amplitude (Sect. 13.6 and Appendix E.8). They are computed by frozen strain: each of the six strain components is applied at ±0.5 % to the two-atom cell with the internal coordinate relaxed, and the band energies at the conduction minimum and at Γ are referenced to the average electrostatic potential. Only differences between the conduction and valence tensors enter at the band extrema, so the absolute alignment from a slab calculation is needed only for single-band checks against published deformation potentials [3]. At the extrema the group velocity vanishes, and the comoving intraband vertex of Eq. (E.28) reduces to the symmetric tensor that frozen strain computes; away from them it carries the Galilean term $\boldsymbol{v}_n^{\mathrm{g}} \otimes \boldsymbol{p}_{nn}$, which frozen strain does not produce and which is added from the band velocities and momentum matrix elements. Valley degeneracy is lifted under shear strain, and the bands must be followed by character, not by index. The run takes minutes, and its size decides which terms dominate the soft limit: with deformation potentials of order 9 eV and a longitudinal sound speed of 8.43 km s$^{-1}$, the ratio $D/c_\nu$ is 300 to 350 keV (Sect. 13.6), far above the momentum transfers of a few keV, so the deformation terms dominate the soft limit by two orders of magnitude.

### 30.1.2 Multiphonon tables and coherence

At momentum transfers far above the zone boundary a nuclear recoil excites many phonons, and in the incoherent approximation of Sect. 11.3 the response of each atom follows from its own density of states. On a cubic site the directional density of states is a third of $D_\kappa(\omega)$, so the per-atom density of states is the complete input. The library holds $S(\boldsymbol{q}, \hat{\boldsymbol{q}}, \omega)$ per cell, in eV$^{-1}$; the per-volume $S$ of Sect. 4.2, in eV$^2$ in natural units, is this divided by the cell volume Ω. The library also holds the probability $P_{\mathrm{ph}}(n|\boldsymbol{q}, \omega)$ that $n$ phonons were created, the branch content per atom, the coherent fraction and the coherence correction below, on a grid of 400 logarithmic points in $q$ from 0.1 to 300 keV, $\omega$ from 0 to 1 eV in steps of 0.1 meV, and temperatures of 10 mK, 1 K and 77 K. Deep in the impulse regime, for $x_R > 16$, the table

switches to the impulse Gaussian with the third- and fourth-cumulant corrections whose inputs come from the path-integral calculation below. The level is $\mathcal{K}$ = multiphonon, incoherent, with the coherent one-phonon term added and $n_A = 0$.

At zero temperature the multiphonon expansion is a forward recursion in $n$; at finite temperature the time-domain form $\mathrm{e}^{-2W_\kappa}[\mathrm{e}^{\mathcal{C}_{\kappa\kappa}(t)} - 1]$ of Sect. 5.5 is Fourier transformed with a Gaussian regulator in time. The densities of states come from $48^3$ grids with tetrahedra. The code is about 300 lines on top of the base and runs in minutes. Its validation is an Einstein site, whose $n$-phonon weights are known in closed form at any temperature and are reproduced to $10^{-5}$. The convergence tests are the $f$-sum rule of the lattice, $\int \omega S \,\mathrm{d}\omega$ fixed by the recoil energy, to $10^{-4}$ at every $q$, and a time window $t_{\max} \geq 20\pi/\delta\omega$ for the bin width $\delta\omega$. The exponent $2W_\kappa$ uses $B_\kappa$ with the thermal factor $2n_{\boldsymbol{Q}\nu} + 1$; the elastic Bragg peak is removed analytically; and GaAs needs a separate $W_\kappa$ for each species. When the Migdal channel is computed separately, the lattice response is multiplied by $1 - P_{\mathrm{Mig}}$ and uses the dressed mass.

Below the zone boundary the incoherent approximation fails pointwise for two-phonon emission, because the two phonons from neighboring atoms interfere. The coherent two-phonon response is a Brillouin-zone sum over one phonon wavevector, the second fixed by momentum conservation, and its $\omega$ integral has an exact real-space form. With the static correlation tensor $B_{\kappa\kappa'}(\Delta\boldsymbol{R})$ of Eq. (11.24), the $n$-phonon weight is the lattice sum Eq. (11.25), per cell, and its ratio $\delta_n^{\mathrm{coh}}(q)$ to the incoherent weight is the exact coherence correction for bins that cover the whole $n$-phonon band. At zero temperature in three dimensions $B_{\kappa\kappa'}$ falls as $\Delta R^{-2}$, so the sum converges absolutely for $n \geq 2$. The lattice sum Eq. (11.25) is not used at finite temperature, where the thermal factor spoils the convergence, nor at $n = 1$; at 1 K and 77 K the coherence correction comes from the pointwise zone sum below, which converges at any temperature. The pointwise response $S^{(2)}_{\mathrm{coh}}(\boldsymbol{q}, \omega)$ is computed for silicon, then germanium and GaAs, on 40 logarithmic values of $|\boldsymbol{q}|$ from 0.2 to 10 keV, the 20 or so irreducible directions of a 590-point Lebedev grid, and bins of 0.5 meV, with $B_{\kappa\kappa'}$ to 40 Å by FFT from $48^3$ eigenpairs and the zone sum on $48^3$ with a Gaussian of 0.25 meV. The tests are a change below 2 % per bin from $32^3$ through $48^3$ to $64^3$ and below $10^{-3}$ in $\delta_n^{\mathrm{coh}}$ under the real-space radius. The acoustic point $\boldsymbol{Q} = 0$ has an integrable singularity that a mesh samples badly: it is excluded and a small sphere around it is added analytically. The whole run costs a few core-hours after the base.

The same tables, projected on cubic harmonics, give the anisotropy of the lattice response for Sect. 11.7. In the harmonic incoherent part a cubic site is isotropic at every phonon order, so the components with $\ell \geq 4$ vanish there to numerical noise, below $10^{-6}$ of the isotropic part; a larger value is a symmetry-breaking artifact of the mesh. The anisotropy comes only from the coherent terms, chiefly the one-phonon term. At long wavelength its $\ell = 4$ component in silicon is $S_4/S_0 = 0.028$ with the measured elastic constants and 0.025 with the PBE force constants at $q \leq 0.3\, q_{\mathrm{BZ}}$, and $|S_6/S_0|$ is below $10^{-3}$ (Sect. 11.7); the components are tabulated as $S_\ell/S_0$ on the cubic harmonics $Y^{\mathrm{c}}_\ell$. The projection uses the 590-point Lebedev grid, exact to degree 41, with $\ell \leq 12$ for cubic crystals and $\ell \leq 8$ for sapphire and SiC, and $\omega$ bins wider than the change of the dispersion between neighboring directions.

### 30.1.3 The charge vertex at large momentum

Two corrections to the lattice vertex need all-electron densities. The first is the valence charge deformation of Sect. 11.3: whether the electron cloud follows a displaced nucleus at $q$ far above the zone boundary. Split into a rigid neutral atom and a deformation, the charge vertex of atom $\kappa$ is $q_\alpha[Z_\kappa - f^{\mathrm{at}}_\kappa(q)] - \Delta_{\kappa\alpha}(\boldsymbol{q})$, with $f^{\mathrm{at}}_\kappa$ the all-electron atomic form factor, and the deformations obey an exact sum rule: summed over the atoms of the cell

at $\boldsymbol{Q} = 0$, the density changes give $G_\alpha$ times the Fourier component of the electron density at $\boldsymbol{G}$, the response to a uniform translation. The deformation $\Delta_{\kappa\alpha}(\boldsymbol{Q}+\boldsymbol{G})$ is wanted for Si, Ge and GaAs at $|\boldsymbol{Q}+\boldsymbol{G}| \le 20\,\mathrm{keV}$, about 725 reciprocal-lattice vectors per $\boldsymbol{Q}$ in silicon and 821 in germanium and GaAs. Up to about 10 keV it is the DFPT density change of the base, Fourier transformed per $\boldsymbol{Q}$, minus $q f_\kappa^{\mathrm{at}}$. At all $q$ it is the symmetric difference of all-electron densities with one atom displaced by $\pm 0.01$ Å in a 216-atom cell, from PAW with all-electron reconstruction or from a full-potential code. The PAW density on the FFT grid aliases its core-region components above about 10 keV, so those components are taken as the smooth-part FFT plus one-center radial transforms per atom. The density cutoff must reach 1.5 times the largest $q$, at least 65 Ry, or what the PAW dataset needs if that is more. The result must satisfy the sum rule to $10^{-3}$ and be linear in $\pm u$. A frozen-core calculation gives no core polarization at all; that part, the polarization of the core by the residual field at the moving nucleus, comes from a Sternheimer (Dalgarno–Lewis) calculation on the all-electron core orbitals, checked by the sum rule that its dipole moment equals the core polarizability. The DFPT part costs nothing extra, the supercells about $10^2$ core-hours, the cores minutes.

The second correction is the finite-$q$ factor of the electron density at the nucleus (Sect. 12.5), which enters the cross term between nuclear and electronic recoil. Its spherical atomic part is known from all-electron atoms; the crystal anisotropy and the relativistic enhancement are not. The crystal valence adds an anisotropy of tetrahedral symmetry, estimated at $1.3 \times 10^{-3}$ of the factor in silicon; it is computed from the valence all-electron density (PAW minus the atomic core, or a full-potential code) to $|\boldsymbol{G}| \le 40/a_0$ on a Lebedev grid, converged to $10^{-5}$ of the density at the nucleus under the cutoff and the augmentation radius. The relativistic density averaged over the finite nucleus raises the contact density by the factor $\mathcal{R}_{\mathrm{rel}}$ of Eq. (12.44), 8 % in silicon and 40 to 45 % in Ge, Ga and As. It comes from a finite-nucleus Dirac calculation of the atom with a Fermi or uniform nuclear charge of the measured radius, with the innermost radial point below a hundredth of the nuclear radius; the choice of nuclear model changes it by less than 1 %. A relativistic point-nucleus density diverges at the origin, and its value at the first grid point is an artifact of the grid. Norm-conserving pseudo-densities are unusable here. The crystal part costs about 10 core-hours, each atom minutes.

### 30.1.4 Nuclear momentum distributions

The width and skewness of the impulse peak of Sect. 11.4 are set by the quantum motion of the nuclei, which harmonic theory gives only to leading order. Path-integral molecular dynamics with i-PI [4], driving a machine-learned potential (Sect. 29.1) trained on PBE or SCAN (strongly constrained and appropriately normed) forces in 216-atom cells, gives the momentum distribution of each species to all orders in the anharmonicity: the centroid-virial estimator yields the kinetic energy along $\hat{\boldsymbol{q}}$, the open-path estimator the momentum distribution and the fourth moment, and the analytic Hessian averaged over beads the mean curvature of the Born–Oppenheimer surface, which is the third cumulant of the impulse peak. At 10 K silicon would need about 250 beads, from $\omega_{\max}/(k_B T P) \lesssim 0.3$, so we run at 50 to 100 K with a colored-noise thermostat, which brings the bead number to 16 to 32, and correct to zero temperature with the harmonic difference from the base. The kinetic energy anchors the anharmonic template, and the anharmonic change of the kinetic energy of silicon is at most $\pm 0.2$ % (Sect. 11.5); its error, statistical plus that of the harmonic-difference correction, must therefore be at most 0.2 %. The potential is validated against DFT on displaced configurations out to three standard deviations of the sampled distribution, since extrapolation beyond the training set biases the fourth cumulant. The open-path estimator of the fourth moment is noisy and needs at least $10^6$ steps, and the whole calculation costs about $10^3$ core-hours per material.

### 30.1.5 Isotope-resolved Hessians

A vector coupled to baryon minus lepton number sees the isotope disorder of the lattice (Sect. 11.6), and its absorption rate $\Gamma_{\mathrm{iso}}(\omega)$ is computed exactly by diagonalizing the Hessian of a disordered supercell. For natural silicon the supercell has 4096 atoms ($8 \times 8 \times 8$ conventional cells, 12,288 degrees of freedom, about ten minutes per dense diagonalization), with the force constants of the base and ten random isotope configurations; the vertex is the mass-subtracted one, with the mean taken mass-weighted, otherwise a spurious acoustic term appears. The check is the exact single-site (Lifshitz) Green function, which gives the leading correlation correction between vertex and eigenvector; with the PBE density of states it is $+8.0\,\%$ at 50 meV (Sect. 11.6). The rate must change by less than $2\,\%$ from 10 to 20 configurations and from 4096 to 13,824 atoms, the larger cell by the kernel polynomial method; the whole calculation costs about 10 core-hours.

The competing electromagnetic absorption at the same frequency is two-phonon absorption, and the bound on the ratio in Sect. 11.6 uses room-temperature data where the millikelvin spectrum is needed. The zero-temperature two-phonon $\varepsilon_2(\omega)$ of silicon follows from the second-order dipole moments $\partial Z^*/\partial u$, by the first-principles route of Deinzer and Strauch [5]. The derivatives come from finite differences of DFPT Born charges in displaced 64- and 216-atom cells, with displacements of $\pm 0.005$, $\pm 0.01$ and $\pm 0.02$ Å and linearity to $2\,\%$ (in silicon $Z^* = 0$ at equilibrium, so the first order is linear in $u$), at 80 Ry, with $4^3$ and $2^3$ $\boldsymbol{k}$ grids checked against $6^3$ and $3^3$. The two-phonon sum uses only the sum processes, which survive at zero temperature, on $32^3$ and $48^3$ grids with Gaussians of 0.25 to 1 meV extrapolated to zero. The test comes first: the same calculation with room-temperature occupations, sum and difference processes, and quasi-harmonic frequencies reproduces the measured room-temperature absorption to $20\,\%$ between 40 and 60 meV, and only then is the zero-temperature curve used. The two cells cost about $10^2$ and $2 \times 10^3$ core-hours.

### 30.1.6 Dynamical Born charges

A recoiling ion in a polar crystal radiates through its Born charge at the photon frequency, not at zero frequency (Sect. 12.6). What is needed is $Z^*_\kappa(\omega)$ at zero macroscopic field for $\omega$ from 0.05 eV to the gap, with the refractive index. At small $\omega$, $Z^*_\kappa(\omega) = Z^*_\kappa(0) - \omega^2 C_\kappa + O(\omega^4)$, and a Cauchy–Schwarz bound on the same spectral sums gives for GaAs $|Z^*(\omega) - Z^*(0)| \le 0.9\,(\omega/1\,\mathrm{eV})^2$ against $Z^*(0) \approx 2.1$: at photon energies near 1 eV the dispersion is not negligible a priori and must be computed. Frequency-dependent DFPT computes it as a mixed response to an electric field and an atomic displacement, solved by shifted Sternheimer equations (Eq. (25.23)), at the RPA and then with the adiabatic local-density kernel. The PBE gap would place the resonance of $Z^*(\omega)$ far below the true edge, so the bands carry a scissor to the measured gap, and the response is taken at zero macroscopic field, since with the field the photon line is counted twice. Imaginary frequencies give $C_\kappa$ by fits in $\omega^2$; real frequencies near the gap use a damping of 10 to 20 meV, extrapolated to zero. $Z^*(0)$ and $\varepsilon_\infty$ must agree with static DFPT to $1\,\%$, and $\sum_\kappa Z^*_\kappa(\omega)$ must tend to 0 as $\omega \to 0$ and to $\sum_\kappa Z_\kappa$ as $\omega \to \infty$. Each frequency batch costs about 50 core-hours for GaAs.

For a scalar coupled to electrons, the single-phonon rate of GaAs is a near-cancellation (Sect. 11.8). The vertex enters through $d = v_{\mathrm{Ga}}/M_{\mathrm{Ga}} - v_{\mathrm{As}}/M_{\mathrm{As}}$ with $v_\kappa = Z_\kappa - Z^*_\kappa/\varepsilon_\infty$, which is $-1.21 \times 10^{-3}$ per u for $Z^* = 2.11$ and $\varepsilon_\infty = 10.88$, because the nuclear charges per unit mass of gallium and arsenic nearly coincide (Sect. 11.8). A $10\,\%$ change in $Z^*/\varepsilon_\infty$ changes $d$ by $44\,\%$, so a rate good to $20\,\%$ needs $Z^*/\varepsilon_\infty$ to $2.3\,\%$. PBE-DFPT cannot deliver this, since its $\varepsilon_\infty$ for GaAs is tens of percent too large. The combination is taken instead from

Table 30.1: Cutoffs and costs of the zone-center and phonon-grid DFPT runs for the candidates beyond the shared base, with SG15 norm-conserving pseudopotentials. Costs are in core-hours; on one accelerator each small-cell run takes minutes to an hour. The last four rows are batches.

| material | cutoff (Ry) | cell | cost (core-hours) |
|---|---|---|---|
| diamond, 3C-SiC, LiF, NaF | 90–100 | 2 atoms | 2–20 |
| 4H-SiC | 90 | 8 atoms | 40–80 |
| $Al_2O_3$ | 100 | 10 atoms | 120–240 |
| $CaWO_4$, $Li_2MoO_4$ | W, Mo semicore | primitive cells | $1.2 \times 10^3$–$5.4 \times 10^3$ |
| magnetic, DFT $+U$ (YIG dominant) | per material | 4–80 atoms | $1.1 \times 10^4$–$10^5$ |
| 23 further candidates | per material | small cells | $10^3$–$5 \times 10^3$ |
| thermal contraction, five volumes | as above | all candidates | $5 \times 10^3$–$2.5 \times 10^4$ |

measured quantities through the Lyddane–Sachs–Teller relation [6],

$$\frac{Z^*}{\varepsilon_\infty} = \Big[\frac{\Omega\,\mu_{\mathrm{red}}(\omega_{\mathrm{L}}^2 - \omega_{\mathrm{T}}^2)}{e^2\varepsilon_\infty}\Big]^{1/2}, \tag{30.1}$$

in Heaviside–Lorentz units ($e^2 = 4\pi\alpha$), with $\mu_{\mathrm{red}}$ the reduced mass of the pair and $\omega_{\mathrm{L}}$, $\omega_{\mathrm{T}}$ the zone-center longitudinal and transverse optical frequencies, all at low temperature, since $\varepsilon_\infty$ changes by 2 to 3 % between room temperature and 4 K. With $\varepsilon_\infty$ to 1 % and the Raman frequencies to 0.1 % the requirement is met; DFPT supplies the sign of $Z^*_{\mathrm{Ga}}$ and a consistency check. The phonon widths come from three-phonon calculations, to 10 %. Rigid-ion charges change $d$ by a factor of three and must not be used, and the gallium isotopes add a broadband channel of the isotope type above.

### 30.1.7 Data for the candidate targets

The comparison of Sect. 17.5 and the target table of Sect. 21.6 need, for each candidate, $\varepsilon_\infty$ and $\varepsilon_0$, the Born-charge tensors, the zone-center transverse and longitudinal optical frequencies, the sound speeds along the principal directions, $\omega_{\max}$, the phonon density of states, the elastic tensor, phonon lifetimes and the thermal contraction to millikelvin. Where a measured value exists, it is used; DFPT fills the rest. Each candidate gets a relaxation, a zone-center DFPT run with the electric field ($\varepsilon_\infty$, $Z^*$, the optical modes and, through the mode sum, $\varepsilon_0$), and $4^3$ and $6^3$ wavevector grids for the slopes, sound speeds, $\omega_{\max}$ and density of states. Table 30.1 lists the cutoffs, set where the total energy converges to 1 meV per atom and the pressure to 1 kbar.

PBE overestimates $\varepsilon_\infty$ systematically, by 11 % in silicon (Sect. 25.2), so measured dielectric constants are used for ranking and the computed ones are diagnostics. The Lyddane–Sachs–Teller relation, $\varepsilon_0/\varepsilon_\infty = \prod_j \omega_{\mathrm{L},j}^2/\omega_{\mathrm{T},j}^2$, holds per principal axis and checks each run; a non-cubic crystal has a pair of dielectric constants per axis. In soft-mode crystals ($SrTiO_3$, the lead chalcogenides) the DFPT $\varepsilon_0$ is unusable, and the measured low-temperature value replaces it; an infrared mode below 20 $\mathrm{cm}^{-1}$ flags such a case. Heavy elements (Pb, Bi, Te, W) need spin–orbit coupling at least as a check. The scintillating crystals $CaWO_4$ and $Li_2MoO_4$ are treated like the oxides, with the tungsten or molybdenum semicore in the valence and a measured elastic tensor where one exists. For the magnetic candidates (YIG, NiO, MnO, $Cr_2O_3$, $NiPS_3$) DFPT runs with a Hubbard $U$ from linear response [7] on the magnetic ground state, with antiferromagnetic cell doubling for NiO and MnO; the dependence of $\varepsilon$ and $Z^*$ on $U$ is reported as the uncertainty, and measured moments and infrared modes are the checks.

Elastic tensors come from strain–stress calculations or DFPT. Each is positive definite and in the Voigt order; at least one public database lists the shear components in a different

order, a trap when tables are merged. For a cubic crystal the longitudinal sound speed in any direction is at least $\sqrt{B/\rho_{\mathrm{T}}}$ (Sect. 11.1), which tests both computed and quoted values. Where a ranking turns on a computed tensor, as for $TeO_2$ against the measured $CaWO_4$ in Sect. 21.6, the measured tensor is the requirement and the offset of computed tensors, 4.9 % on average for silicon, germanium and diamond (Sect. 17.6.4), is the uncertainty until it exists. CsI, whose sound speeds set its nuclear-recoil edge, enters with its measured elastic constants at 4.2 K, $C_{11} = 27.37$, $C_{12} = 7.93$ and $C_{44} = 8.25$ GPa; what it lacks is the density of the cold cell, which the thermal contraction below supplies.

Phonon lifetimes come from three-phonon perturbation theory, on-shell, which is $n_A = 2$ on the truncation label: third-order force constants from phono3py [8] on $3 \times 3 \times 3$ supercells, or from a machine-learned potential, converged in supercell size and interaction cutoff. The anharmonic decay constant of silicon [9] is the check; for germanium the published values spread by a factor of ten. The cost is $10^3$ to $10^4$ core-hours per material, and the lifetimes enter Sect. 11.8. The thermal contraction integrated from 300 K to millikelvin, which stores elastic energy between a crystal and its holder, comes from the quasi-harmonic approximation (Eq. (25.24)) with DFPT at about five volumes, with the PBE and PBEsol spread as the uncertainty and measured expansion coefficients (Si, Ge, sapphire) as checks. The quasi-harmonic approximation fails near soft modes, and silicon expands on cooling below about 120 K, so the sign of the integrand must be checked.

## 30.2 Anharmonic, disordered and quantum solids

Solid helium near melting has a Lindemann ratio of about a quarter: the zero-point amplitude of an atom is about a quarter of the distance to its neighbor (Sect. 11.5), and the harmonic lattice is not a valid starting point. Strontium titanate and potassium tantalate have soft modes whose frequencies depend on temperature at order one; lithium hydride has a light atom whose zero-point motion is large. For these targets the multiphonon response of Sect. 11.5 has no small parameter. Glasses lack the lattice that the response tables assume, and a random medium can scatter a strongly interacting particle coherently over its whole volume.

### 30.2.1 Strongly anharmonic and quantum crystals

Every integer frequency moment of the self response is an equal-time correlation function. The moments $\mathscr{M}^k$, $k = 0, \ldots, 4$, of the self response of Sect. 8.3 are therefore exact from a static path-integral simulation, for any anharmonicity, any temperature and any Lindemann ratio. This holds in the Born–Oppenheimer approximation, where the nuclei move on a smooth energy surface and every moment is finite; with point nuclei and the full Coulomb interaction the fourth moment diverges (Sect. 8.3). The Markov–Krein band of Sect. 11.5 then brackets every response with these five moments; it bounds bin-integrated rates in the bulk of the response, and threshold rates only from above. The calculation therefore has two parts, exact moments and an approximate real-time response tested against them.

For $KTaO_3$, $SrTiO_3$ and LiH we want $\mathscr{M}^0$–$\mathscr{M}^4$ of the self response at $|\boldsymbol{q}| = 0.5$ to 20 keV, the band per $\boldsymbol{q}$ at the millikelvin operating temperature, and the self response $S_{\mathrm{self}}(\boldsymbol{q}, \omega)$ (per volume, in $\mathrm{eV}^2$, the units of Sect. 4.2) at 50 and 100 K, where the real-time methods below hold. On the truncation label $\mathfrak{T}$ (Sect. 8.5) the moments are exact in $n_A$; the dynamics is approximate at all orders of $n_A$, and the moment test measures its error. The static part is path-integral molecular dynamics in i-PI [4] with a machine-learned potential (Sect. 29.1) trained on PBE and validated on displaced configurations to three standard deviations of the path-integral distribution. The cells are cubic $n \times n \times n$ supercells with $n = 4$, 6 and 8 (320, 1080 and 2560 atoms) for the perovskites, even $n$ so that the soft mode at the R point is

commensurate, and $4^3$ to $6^3$ conventional cells (512 to 1728 atoms) for LiH. The number of beads $P$ satisfies $\beta\omega_{\max}/P \lesssim 0.3$; for $SrTiO_3$ at 50 K with $\omega_{\max} \approx 100$ meV that is $P \approx 80$, doubled until $\mathscr{M}^2$ changes by less than 1 %. Colored-noise thermostats may accelerate the static moments but must not be used in the dynamics, which they alter.

The dynamic part compares three approximate methods. Thermostatted ring-polymer molecular dynamics damps only the internal ring-polymer modes, with friction $\gamma_k = \lambda\omega_k$ at $\lambda = 1/2$ and checks at 1/4 and 1, which preserves the exact limits of the method [10]; the centroid is not thermostatted during the dynamics. Centroid molecular dynamics, with the time step reduced threefold, is the second method, and their difference is the method uncertainty. The third is the stochastic self-consistent harmonic approximation [11], first static, then with the dynamical bubble, then the full time-dependent form [12]; the bubble-to-full difference measures the order. The time step is at most $0.1/\omega_P$ of the stiffest free ring-polymer mode, typically 0.1 to 0.25 fs, with centroid energy drift below $10^{-4}$ per ps. Resolving $\delta\omega = 0.5$ meV needs trajectories of $20\pi/\delta\omega \approx 80$ ps at $q \lesssim q_{\mathrm{BZ}}$, shorter at large $q$ where the correlation decays; 100 independent initial conditions give 1 % statistical error on bin integrals. Every approximate response must lie inside the band of its exact moments, and the response must change by less than 2 % from one cell size to the next while the finite-size gap $2\pi c_\nu/L$ stays below the lowest frequency bin. A response outside the band is flagged, not used.

Solid $^4$He is computed the same way with the Aziz pair potential, which needs no fitting. The simulation is path-integral molecular dynamics at a molar volume of 21 $\mathrm{cm^3\,mol^{-1}}$ and 1 K, 180 to 432 atoms and $P = 64 \to 128$ beads (the Debye energy is 2 meV, so $\beta\omega_{\mathrm{D}} \approx 25$), with an open-path estimator for the momentum distribution. The band is computed for the self and the coherent response; the coherent moments are the static structure factor, the $f$-sum, the current correlation and the third-moment correlator, all equal-time. Bose exchange is negligible for the statics at this temperature but is checked once by permutation sampling. At 1 to 40 GPa, where the Lindemann ratio drops to about 0.1, the self-consistent harmonic force constants from a potential trained on dispersion-corrected PBE replace the static Hessian in the phonon response of Sect. 11.5, converged to 2 % in the frequencies and 3 % in the displacement correlations.

Ring-polymer and centroid methods give Kubo-transformed correlations; converting to $S$ multiplies by $\beta\omega/(1 - \mathrm{e}^{-\beta\omega})$, which amplifies their error by $\beta\omega$, so they are usable only for $T \gtrsim \omega/10$ and not at a 10 mK operating temperature. Their short-time error for the nonlinear operator $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{u}}$ is already $O(t^3)$. The time-dependent self-consistent method excites at most two phonons at a time, so at $q \gg q_{\mathrm{BZ}}$ it can supply only the dressed one-phonon propagator inside the Gaussian form. Tunneling splittings in double wells are missed by centroid-like methods. At millikelvin, therefore, the response of these targets is bounded by the band alone. The ring-polymer runs cost $10^3$ to $10^4$ core-hours per material and temperature, and centroid dynamics three times more. The self-consistent harmonic runs cost about $10^2$ core-hours per material, path-integral helium at melting about $10^3$, and high-pressure helium about $10^4$ per pressure. The memory is a few gigabytes. The moments and bands enter Sect. 11.5.

### 30.2.2 Weak anharmonicity

In silicon the anharmonicity is small, and the question is the size of its first correction to the multiphonon response above the one-phonon band, where the low-mass reach is set. Time reversal fixes the structure. For a target invariant under time reversal $S_{\kappa\kappa}(\boldsymbol{q},\omega) = S_{\kappa\kappa}(-\boldsymbol{q},\omega)$, and a term of first order in the cubic force constants $\Phi^{(3)}$ is of third degree in $\boldsymbol{q}$, so it cancels. The leading correction is therefore of second order in $\Phi^{(3)}$ or first order in $\Phi^{(4)}$, the level $n_A = 2$ of Sect. 8.5: the one-phonon propagator dressed by the bubble and loop self-energies,

the connected four-point function, and the square of the connected three-point function. We want the relative correction $\delta_{\mathrm{anh}}(\boldsymbol{q},\omega) = S/S^{\mathrm{harm}} - 1$ of Sect. 11.5, for silicon first, then germanium and gallium arsenide.

The cubic force constants come from finite displacements in a $4 \times 4 \times 4$ primitive supercell (128 atoms) with a pair cutoff of 5 Å, about 300 supercell DFT runs, through phono3py [8]; the quartic ones from hiPhive [13] or TDEP. The bubble self-energy gives the full spectral function on a $24^3$ grid, not on-shell linewidths, and the quartic loop gives the shift. The three- and four-point functions are mode sums over four branch labels with contour kernels built from the thermal propagator, about $10^3$ core-hours. The dense cubic constants of the 128-atom cell have $384^3$ entries, 0.45 GB in double precision; the quartic set within the cutoff is smaller. The spread under the supercell from $3^3$ to $4^3$, the cutoff from 4 to 5 Å and the grid from $24^3$ to $32^3$ is the uncertainty of the correction. The first-order cubic term must vanish numerically, which checks the placement of the time arguments on the contour, and the correction must satisfy the exact linear constraints of Sect. 11.5, anchored by the kinetic energy of the struck atom from the path-integral calculation of Sect. 30.1. The total cost is $10^3$ to $10^4$ core-hours.

### 30.2.3 Glass models

Amorphous $SiO_2$, silicon and silicon nitride need structures before they need responses. The models are made by melt-quench molecular dynamics with a machine-learned potential (Sect. 29.1), 648 to 5184 atoms and at least five independent quenches, validated on the measured density and pair distribution. From each model come the full Hessian and the Born charges $Z^*_I$ of every atom, either by DFPT on 192- to 648-atom subcells or from a charge-equilibration model fitted to them. The Hessian of 5184 atoms has $15552^2$ entries, 1.9 GB in double precision, and its dense diagonalization costs about $4 \times 10^{12}$ floating-point operations.

The infrared coefficient $C^{\mathrm{IR}}(\omega) = A^{\mathrm{IR}} + B^{\mathrm{IR}}\omega^2$ of glass absorption below the Ioffe–Regel frequency, Eqs. (16.64) and (16.65), is the first quantity that enters Sect. 16.7. $A^{\mathrm{IR}}$ and $B^{\mathrm{IR}}$ follow from the Born-charge structure factor $S^Z_\nu(k)$ of static configurations and a charge model, without the Hessian, on wavevectors $k \le 0.3\,\text{Å}^{-1}$ of the periodic cell; the Hessian gives the full $C^{\mathrm{IR}}(\omega)$ as a cross-check up to the boson peak. The second quantity is the distinct part of the response at two and three phonons. It needs the configurational averages, at fixed pair distance $r$, of the longitudinal and transverse parts of the pair displacement tensor, its longitudinal–transverse vector and its traceless transverse part, binned in $\Delta r = 0.02$ Å up to 15 Å; the third is the site-resolved phonon density of states. The ensemble is validated against the measured terahertz absorption of the same glass [14].

The pair distribution, the moments, the density of states and $C^{\mathrm{IR}}$ change by less than 5 % between 1728 and 5184 atoms; the box is at least 4 nm, so that $2\pi/L$ lies below the correlation length of the charges; and $\sum_I Z^*_I = 0$ in every configuration. The 5184-atom box of $SiO_2$ is 4.3 nm across; the 1728-atom box, 3.0 nm, serves only as the smaller member of the size test. A locally neutral charge model forces $A^{\mathrm{IR}} = 0$ by construction and must not be used; the model has to reproduce the site-charge fluctuations of the subcell DFPT. The finite-size gap $2\pi c_\nu/L$, 3 to 4 meV at 4 nm, removes the lowest modes, and the dependence on quench rate is a model systematic. The cost is $10^4$ core-hours for the molecular dynamics, $10^3$ for the subcell Born charges and $10^2$ per model afterward.

### 30.2.4 Coherent force from a random medium

A dark-matter particle with a large cross-section on nuclei sees a solid as the optical potential $V_{\rm opt}$ of Eq. (16.68), and the Born treatment of scattering off density fluctuations holds only for $\sigma_{\chi N} \ll \sigma_{\rm crit}$ (Eq. (16.69)). Above it the force on a test mass comes from reflection at its faces. A plane wave with normal velocity $v_n$ meets the step with reflection probability $R(v_n) = |(k_n - k')/(k_n + k')|^2$, $k'^2 = k_n^2 - 2\mu_{\chi N} V_{\rm opt}$; a slab many de Broglie wavelengths thick reflects, averaged over its thickness, $R_{\rm slab} = 2R/(1+R)$. Each reflected particle returns momentum $2m_\chi v_n$, and the flux on a face of outward normal $\hat{\boldsymbol{n}}$ is $\rho_\chi/m_\chi$ times the inward normal velocity, so the pressure is

$$P_{\rm wall}(\hat{\boldsymbol{n}}) = 2\rho_\chi \int \mathrm{d}^3 v\, f_\chi(\boldsymbol{v})\, (\boldsymbol{v}\cdot\hat{\boldsymbol{n}})^2 R_{\rm slab}(\boldsymbol{v}\cdot\hat{\boldsymbol{n}})\, \Theta(-\boldsymbol{v}\cdot\hat{\boldsymbol{n}}), \tag{30.2}$$

while transmitted particles leave with their initial velocity and transfer nothing on average. What the equivalence-principle test measures is the differential acceleration $\Delta a(\sigma_{\chi N}, m_\chi)$, in $\mathrm{m\,s^{-2}}$, of the two cylinders of the MICROSCOPE equivalence-principle test (a platinum alloy and Ti–6Al–4V), summed over faces and averaged over the standard halo with the Earth's velocity. The scale is set by $\rho_\chi v^2 \approx 5 \times 10^{-11}\,\mathrm{Pa}$.

The calculation evaluates Eq. (30.2) per face of the axisymmetric geometry, with one-dimensional transfer matrices along the local normal (the curvature radius is far larger than the de Broglie wavelength, 60 µm at 20 eV), and a four-dimensional Monte Carlo over velocity and face. The code is about 300 lines and runs in CPU minutes. Below $\sigma_{\rm crit}$ the same code evaluates the distorted-wave Born form, with the density correlation $\langle \delta n\, \delta n \rangle$ taken from small-angle X-ray scattering on the actual alloy. As $V_{\rm opt}/E_{\rm kin} \to 0$ the result must reproduce the Born random-medium rate of Sect. 16.7. Roughness on the de Broglie scale makes the reflection diffuse, and the nested cylinders reflect particles back and forth; both are included in the Monte Carlo.

### 30.2.5 Chiral phonons in molecular frameworks

A chiral phonon carries angular momentum $L^{\rm ph}_{\boldsymbol{Q}\nu} = -\mathrm{i}\sum_\kappa \boldsymbol{e}^*_{\kappa\nu}(\boldsymbol{Q}) \times \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})$. The proposal of Sect. 14.6 reads out the phonons that end, after anharmonic decay, in a long-lived chiral band with $|L^{\rm ph}| > L_{\rm th}$ [15]. Its yield is the mean number of such phonons per phonon created in mode $\lambda = (\boldsymbol{Q}\nu)$, averaged over the created spectrum. Although the decay branches, the expectations obey a linear system. Let $\mathsf{M}_{\lambda\lambda'}$ be the mean number of daughters in $\lambda'$ per decay of $\lambda$: the three-phonon rates into unordered pairs, counted twice for identical daughters, plus the elastic isotope rate, all divided by the total rate of $\lambda$, which includes boundary loss and sensor capture. Let $b_\lambda = 1$ for modes in the terminal chiral set, those with $|L^{\rm ph}| > L_{\rm th}$ and a decay time longer than the observation time $t_{\rm obs}$, and $b_\lambda = 0$ otherwise, with the rows of $\mathsf{M}$ zero for terminal modes. Conditioning on the first decay gives $\mathcal{N}^{\rm ch}_\lambda = b_\lambda + \sum_{\lambda'} \mathsf{M}_{\lambda\lambda'} \mathcal{N}^{\rm ch}_{\lambda'}$, so

$$\boldsymbol{\mathcal{N}}^{\rm ch} = (\mathbb{1} - \mathsf{M})^{-1} \boldsymbol{b}. \tag{30.3}$$

Without isotope scattering $\mathsf{M}$ is nilpotent in energy order. With it, $\mathbb{1} - \mathsf{M}$ is still invertible as long as every closed set of elastically connected modes has some decay or loss.

The first framework is $\mathrm{InF_3}$(4,4′-bipyridine), whose chiral phonon bands Romao et al. computed [15]. The rates come from phono3py on the framework cell, at $n_A = 2$ (Fermi's golden rule with two cubic vertices) with harmonic $L^{\rm ph}$, on a $10 \times 10 \times 20$ grid; Eq. (30.3) is a sparse solve. Cubic force constants from DFT cost about $10^5$ core-hours for a metal–organic framework; a machine-learned potential validated on 50 DFT supercells brings this to $10^3$. The grid, the cutoff of the cubic constants and the smearing of energy conservation are converged,

and the sensitivity to $t_{\mathrm{obs}}$ is reported. A direct Monte Carlo of the branching process with the same rates must reproduce $\boldsymbol{\mathcal{N}}^{\mathrm{ch}}$ within its statistical error, and the energy in the terminal set plus the energy lost must equal the energy created. $L^{\mathrm{ph}}$ is not unique inside a degenerate subspace, so it is diagonalized there. It is not conserved in decay, so the label is recomputed for each daughter. And the code must state whether its rates are per ordered or per unordered pair, since the factor two moves the answer at order one.

Frameworks with lanthanide ions, whose host and ion Sect. 14.6 selects, add one-phonon amplitudes through the crystal field. Let $P_{\mathrm{K}}(\boldsymbol{u})$ be the projector on the ground Kramers doublet, of energy $E_{\mathrm{K}}$, of the crystal-field Hamiltonian $H_{\mathrm{CF}}(\boldsymbol{u})$ of the $4f$ shell at displacement $\boldsymbol{u}$. Let $O$ be the spin $\boldsymbol{s}_e$ or the orbital moment $\boldsymbol{L}$, and $R = \sum_{c \notin \mathrm{K}} |c\rangle\langle c| / (E_{\mathrm{K}} - E_c)$. First-order perturbation theory gives $\partial P_{\mathrm{K}} = R\, \partial H_{\mathrm{CF}}\, P_{\mathrm{K}} + P_{\mathrm{K}}\, \partial H_{\mathrm{CF}}\, R$, and the amplitude follows from $\partial(P_{\mathrm{K}} O P_{\mathrm{K}})$ contracted with the phonon eigenvector, $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q}) / \sqrt{2 N M_\kappa \omega_{\boldsymbol{Q}\nu}}$. The projector form avoids derivatives of individual doublet states, which depend on an arbitrary phase. The inputs are the derivatives of the 27 Stevens parameters ($k = 2, 4, 6$) and of the $g$ tensor with respect to symmetry-adapted displacements of an embedded cluster (the ion, its first coordination shell and whole ligands, in Ewald point charges with model potentials on boundary cations), computed by central differences of $\pm 0.01$ Å. The electronic structure is a state-averaged complete-active-space self-consistent field (CASSCF) calculation over the ground term with spin–orbit mixing, a perturbative correction of the crystal-field splittings, a scalar-relativistic Hamiltonian and ANO-type basis sets. The derivatives must be linear over 0.005 to 0.02 Å to 2 % and stable to 10 % under enlargement of the active space, with the cluster extended to its second shell and the basis enlarged as further checks. The roots must be tracked by overlap, because they can exchange order under displacement, and $\boldsymbol{s}_e$ and $\boldsymbol{L}$ are kept separate since dark matter couples to them differently. Each geometry costs 0.5 to 2 core-hours, $10^2$ to $10^3$ per inequivalent site. Both results enter Sect. 14.6.

## 30.3 Magnetic, superconducting and Dirac targets

Clean aluminum needs three numbers for its loss function near the pair-breaking edge, the density of states $\mathcal{N}_{\mathrm{F}}$ (free-electron value), the Fermi velocity $v_{\mathrm{F}} = 2.0 \times 10^6\,\mathrm{m\,s^{-1}}$ and the gap $\Delta_{\mathrm{BCS}} = 0.18\,\mathrm{meV}$, and no first-principles calculation at all; its conserving response is a two-dimensional integral that runs in minutes. Amorphous WSi, at the other extreme, needs molecular dynamics, DFT, the RPA and a Bogoliubov–de Gennes solution on each quenched cell. The magnets and Dirac materials lie in between. Their rates are written in the book through a few model parameters (exchange constants, anisotropies, form factors, Dirac velocities and background screening), and the calculations below supply those parameters and the responses that go beyond the model. Table 30.3 lists them with their costs.

### 30.3.1 The conserving response of aluminum

The conserving density response of Sect. 15.1 adds a phase-mode term to the bare quasiparticle bubble, and the two cancel at $\boldsymbol{q} = 0$. We want its vertex fraction $\mathfrak{v}$ on the grid of that section: $y = \omega / \Delta_{\mathrm{BCS}}$ from 2 to 50 and $q v_{\mathrm{F}} / \Delta_{\mathrm{BCS}}$ from its kinematic minimum $y\, v_{\mathrm{F}} / v_{\max}$ to $10^3$, together with the full $\mathrm{Im}\, \varepsilon^{\mathrm{cons}}$, its Kramers–Kronig transform and the loss function $L^{\mathrm{cons}}$ in its resummed form at the edge. The electrons of a typical film are diffusive on the scale of the probe (Sect. 15.3), so the calculation includes impurity scattering. On the truncation label $\mathfrak{T}$ (Sect. 8.5) it is $\mathcal{X} =$ RPA in the Coulomb channel with the pairing ladder, on the BCS reference, quasiclassical (constant $\mathcal{N}_{\mathrm{F}}$, spherical Fermi surface, $T = 0$), clean or with Born impurity scattering.

Each clean bubble is an integral over the angle $\mu$ between $\boldsymbol{k}$ and $\boldsymbol{q}$ and over the band energy $\xi$, with $\xi' = \xi + v_{\mathrm{F}} q\mu + q^2/2m_e$, $E = (\xi^2 + \Delta_{\mathrm{BCS}}^2)^{1/2}$, and the pair-creation coherence factors $\frac{1}{2}[1 - (\xi\xi' - \Delta_{\mathrm{BCS}}^2)/(EE')]$ for the density vertex and $\frac{1}{2}[1 + (\xi\xi' + \Delta_{\mathrm{BCS}}^2)/(EE')]$ for the phase vertex. Gauss–Legendre quadrature in $\mu$ and adaptive quadrature in $\xi$ evaluate it. The imaginary parts are taken at the roots of $E + E' = \omega$, solved analytically, with no broadening, and the real parts are principal values with the pole subtracted. The phase–phase bubble diverges logarithmically in the $\xi$ cutoff, but only its difference from the value at $(0, 0)$ enters, so the cutoff is taken at $10^3 \Delta_{\mathrm{BCS}}$ and the logarithm is cancelled analytically; the value at $(0, 0)$ itself must be computed with the same cutoff as the gap equation, where it equals the inverse pairing coupling, or the Goldstone cancellation fails and $\mathfrak{v}$ acquires a spurious constant. The Kramers–Kronig integral runs on a logarithmic grid to $10^3 q v_{\mathrm{F}}$, with the Lindhard normal state above $q v_{\mathrm{F}}$. The sign of i0 on the antiresonant pole is that of the retarded function, with which the time-ordered form agrees only for $\omega > 0$ at $T = 0$, and the bare and conserving bubbles are never mixed between $\mathrm{Im}\,\varepsilon$ and $\mathrm{Re}\,\varepsilon$, since that breaks positivity.

Impurities enter through the Born self-energy and the impurity ladder of Sect. 15.3, whose normal state is the Mermin function Eq. (4.62) at the relaxation rate $\gamma_{\mathrm{rel}} = v_{\mathrm{F}}/\ell_e$. The table is computed in the clean limit and for $\ell_e = 1\,\mu\mathrm{m}$, 300 nm and 30 nm, with the edge values divided by $\mathfrak{g}_{\mathrm{I}}$ times that normal state, as in Table 15.1.

The conserving response must vanish at $q \to 0$ for every $\omega$, and with the ladder switched off it must return the bare bubble with the density coherence factor [16, Eq. (7)]. $\mathfrak{v}$ must be stable to $10^{-3}$ under doubling of both grids, the Kramers–Kronig tail must stay below $10^{-3}$ of $\mathrm{Re}\,\varepsilon$, and the loss-function table must be converged to 1 %. With impurities the normalized edge ratio has to reproduce the conserving values of Sect. 15.3 at the same mean free paths, and $L$ must still vanish at $2\Delta_{\mathrm{BCS}}$. The table replaces the exclusion of the cells nearest the edge in Sect. 15.2, for the mean free path of the film at hand.

### 30.3.2 Amorphous WSi

A $\mathrm{W}_x\mathrm{Si}_{1-x}$ film has $k_{\mathrm{F}}\ell_e$ of a few (1.8–5.5 from its measured transport), and at $q = 1$ to 5 keV, where $q$ is comparable to both $k_{\mathrm{F}}$ and the screening wavenumber, no relaxation-time form of $\varepsilon$ is valid (Sect. 15.3). The rate needs the configurational average $\langle L \rangle(\boldsymbol{q}, \omega)$ of the loss function, not the loss function of the averaged $\varepsilon$, for $q = 0.5$ to 5 keV and $\omega$ from $2\Delta_{\mathrm{BCS}}$ to 50 eV, together with the distribution $P_\Delta$ of the local gap. On $\mathfrak{T}$ the normal state is $n_B = 0$, $\mathcal{X} = \mathrm{RPA}$ with local fields, averaged over amorphous configurations, and the gap is a Bogoliubov–de Gennes mean field with an attractive interaction. The superconducting factor is applied locally with $P_\Delta$, which Anderson's theorem justifies where the mean free path is far shorter than the coherence length.

A machine-learned potential (Sect. 29.1) is trained on PBE molecular dynamics of 216-atom cells at the film composition, $x \approx 0.2$–$0.25$, and 512- to 1000-atom cells are melt-quenched at no more than $10^{13}\,\mathrm{K\,s^{-1}}$, at least five times, and validated on the measured density and the W–Si and W–W pair distributions. Each quench gets a DFT ground state (SG15, with the tungsten 5s5p semicore in the valence) at $\Gamma$ plus a $2^3$ grid, and a non-self-consistent run with at least three times as many bands as are occupied, so that the semicore states, which contribute above about 35 eV, are in the response. Its Kubo–Greenwood conductivity $\sigma_1(\omega)$ is the optical control, compared with film ellipsometry and sheet resistance. The RPA dielectric matrix (Yambo or BerkeleyGW, Sect. 26.1) is computed directly at the reciprocal vectors of the supercell, which are $2\pi/L \approx 0.5$ keV apart for $L = 25$ Å, so no folding is needed above that, with the $\boldsymbol{G}$ cutoff converged to $q + 3$ keV and the bands to $\omega_{\max} + 30$ eV. Local fields are large in a disordered cell, and the loss function is $\mathrm{Im}[-(\varepsilon^{-1})_{\mathbf{00}}]$, never $\mathrm{Im}[-1/\varepsilon_{\mathbf{00}}]$. A Wannier tight-binding model of each cell, solved in the Bogoliubov–de Gennes approximation with the

attractive interaction fitted to the measured film $T_c$, gives $\Delta(\boldsymbol{r})$ and $P_\Delta$. The $f$-sum per $q$ must hold to 2 % and the measured $\sigma_1$ must be reproduced; the spread of $\langle L \rangle$ across quenches and from 512 to 1000 atoms, required below 5 %, is the error bar. The RPA matrices, at tens of gigabytes each for 1000 atoms, dominate the cost (Table 30.3), and the result enters Sect. 15.3.

### 30.3.3 Holstein absorption with Eliashberg functions

Above $2\Delta_{\mathrm{BCS}}$ a superconductor absorbs a dark photon by breaking a pair and emitting a phonon, with the conductivity $\sigma_1^{\mathrm{S}}(\omega)$ of Eq. (15.24). The rate within a few meV of the edge depends on the shape of $\alpha^2 F_{\mathrm{tr}}(\omega')$ below a few meV, where the free-electron model holds $1.25 \times 10^4$ times the weight of the $\omega'^4$ law (Sect. 15.4). We therefore want $\sigma_1^{\mathrm{S}}(\omega)/\sigma_n$ for aluminum and niobium with $\alpha^2 F_{\mathrm{tr}}$ from first principles, also in its strong-coupling form, where the pair weight becomes $\int \mathrm{d}E\,[N^{\mathrm{El}}(E)N^{\mathrm{El}}(E') + P^{\mathrm{El}}(E)P^{\mathrm{El}}(E')]$ with the Eliashberg density of states and pair amplitude of Sect. 15.4.4. On $\mathfrak{T}$ it is $n_g = 2$, one electron–phonon vertex and one current vertex, with isotropic Eliashberg dressing.

EPW [17] computes the vertices by DFPT on an $8^3$ grid and interpolates them to fine grids of $60^3$ $\boldsymbol{k}$ and $30^3$ $\boldsymbol{Q}$ for aluminum and $40^3$ and $20^3$ for niobium, for both $\alpha^2 F$ and the transport function with its weight $1 - \cos\theta$. The $30^3$ grid does not resolve the $\omega'^4$ regime, so the transport function below a few meV is recomputed on a refined grid of small $\boldsymbol{Q}$ around $\Gamma$, and it must follow the $\omega'^4$ law there. That law holds only if the vertex obeys the acoustic sum rule Eq. (6.12), which makes it vanish for $\boldsymbol{Q} \to 0$ between states on the Fermi surface, so the sum rule is checked on the interpolated vertex. The isotropic Eliashberg equations [18] are solved on the real axis, since Padé continuation produces artifacts near $2\Delta_{\mathrm{BCS}}$, and Eq. (15.24) is then integrated with a phonon step no larger than $\Delta_{\mathrm{BCS}}/20$. The transport coupling $\lambda_{\mathrm{tr}}$ must match the value 0.39 from the resistivity of aluminum, against which the free-electron model gives 0.32 (Sect. 15.4); $T_c$ must come within 10 % of experiment with a single Coulomb pseudopotential $\mu^*$, $\sigma_1$ must be stable to 5 % under doubling of the fine grids, and the normal-state $\sigma_1$ must lie within 20 % of the measured optical conductivity of aluminum between 0.2 and 3 eV. Which renormalization factors survive in the strong-coupling form is open in Sect. 15.4.4; in the normal metal the factor is $(1 + \lambda_{\mathrm{tr}})^{-2} = 0.52$ in the rate, and the calculation reports the superconducting result with and without the factor $1/(1 + \lambda_{\mathrm{tr}})$ in the current, so that their difference measures it.

### 30.3.4 Magnet inputs from DFT $+U$

The magnon rates of Sects. 14.1 and 14.2 need exchange constants $J_{ij}^{\mathrm{ex}}$, single-ion anisotropies and $g_{\mathrm{ion}}$ tensors. For YIG, NiO, MnO, $Cr_2O_3$ and $NiPS_3$ these come from inelastic neutron or resonance fits where they exist, and otherwise from DFT $+U$ (Sect. 28.1), in the Heisenberg convention of the book, $J^{\mathrm{ex}} > 0$ antiferromagnetic with each pair counted once. The Hubbard $U$ is computed by linear response [7], and the exchange constants by four-state energy mapping or by the Green-function method of TB2J [19] on Wannier functions, in the magnetic unit cell at the measured lattice constant, shell by shell until a further shell no longer moves the computed magnon band by more than its spread over $U$. That spread is the main systematic, and the measured ordering temperature and, where available, the measured magnon band are the checks. The orbital moments of $Cr^{3+}$ and $Ni^{2+}$ enter $g_{\mathrm{ion}}$ and are computed with spin–orbit coupling. Spin–orbit coupling also gives the Dzyaloshinskii–Moriya constants of $Cr_2O_3$, by four-state mapping with the two spins of each bond placed in the plane normal to the component sought, which the spin model of Sect. 14.6 needs together with the exchange and anisotropy constants for $T \ll T_{\mathrm{CM}}$. For a target without ultrasonic data, the magnetoelastic

constants $B_1$ and $B_2$ of Sect. 14.2 follow from the strain derivative of the magnetic anisotropy energy, computed with spin–orbit coupling at a few strained cells, with strains of $\pm 0.5\,\%$ as for the deformation potentials; where ultrasonic data exist, they check it.

The same linear spin-wave code that serves the two-magnon response below gives the uniform modes that absorption into magnons needs (Sect. 14.3). For a sample of finite size these carry the demagnetizing field of its shape, and in a body that is not an ellipsoid the dipolar lattice sums over the real sample spread the weight over several magnetostatic modes, so the code takes the sample shape and the dipolar sums as inputs. In the ferromagnetic limit the uniform mode must sit exactly at the Larmor frequency, and in the two-sublattice antiferromagnet the uniform transverse weight per antiparallel pair must equal $|\boldsymbol{S}|\omega_{\mathrm{AFMR}}/(2\omega_E + \omega_A)$. YIG, with 80 atoms in its magnetic cell, dominates the cost.

### 30.3.5 Two-magnon responses

At linear spin-wave order a ferromagnet at $T = 0$ has no two-magnon density response, while an antiferromagnet has a two-magnon continuum whose weight relative to the one-magnon line is 0.146 to 0.029 for $|\boldsymbol{S}| = 1/2$ to $5/2$ in a simple-cubic lattice (Sect. 14.1). The quartic terms of the Holstein–Primakoff expansion add a two-magnon interaction of relative order $1/|\boldsymbol{S}|$, which Sect. 14.2 resums as a ladder in the pair channel of total momentum $\boldsymbol{q}$; its poles below the continuum are two-magnon bound states. Whether NiO has such a state, and the interacting response $S^{(2m)}(\boldsymbol{q}, \omega)$ in $\mathrm{eV}^2$, are open; we want them, with the bound-state energies in meV and their weights, for YIG, NiO, $MnF_2$ and, as an uncontrolled benchmark, $\alpha$-$RuCl_3$. On $\mathfrak{T}$ the reference is linear spin-wave theory with the pair ladder, and for noncollinear order the cubic vertex is added.

Linear spin-wave theory with Colpa's diagonalization [20] (SpinW [21] or Sunny), with the inputs of Sect. 30.3.4, gives the reference. The quartic vertex is derived symbolically from the spin Hamiltonian, and the Holstein–Primakoff and Dyson–Maleev vertices agree only on shell, so one of them is used throughout. The ladder is one linear solve per $(\boldsymbol{q}, \omega)$ on $30^3$ to $40^3$ grids, about $3 \times 10^7$ operations per $\boldsymbol{q}$, and exact diagonalization of the two-magnon sector in small cells checks it. The zero-point reduction $\Delta S$, a sum over Bogoliubov coefficients, is converged to $10^{-3}$ on a $40^3$ grid. The sum rule on $\langle \boldsymbol{S}^2 \rangle$ must hold to $1\,\%$, and the bound-state energies must be stable to $1\,\%$ when the grid is doubled. YIG, with 20 iron ions per primitive cell, needs the full multi-sublattice transformation, and a matrix that is not positive definite signals a wrong classical ground state. The results enter Sect. 14.2.

### 30.3.6 Orbital form factors

Ions with unquenched orbital moments, the $4f$ ions and some $3d$ ions with strong spin–orbit coupling, scatter with form factors beyond the dipole form $\langle j_0 \rangle + c_2 \langle j_2 \rangle$ at $q \gtrsim 1/\langle r \rangle_{4f} \approx 3\,\mathrm{keV}$. Expanding $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ in spherical Bessel functions and applying the Wigner–Eckart theorem inside the ground $J$ multiplet reduces the spin and orbital matrix elements to the radial integrals $\langle j_K(q) \rangle = \int r^2 \mathrm{d}r\, R_{n\ell}(r)^2 j_K(qr)$ with $K$ even up to $2\ell = 6$, times angular coefficients. The integrals for $K = 0, \ldots, 6$ (dimensionless) and the multiplet-projected matrix elements are wanted for the ions of the magnetic targets of Sect. 14.1, with the crystal field treated as a perturbation within $J$; crystal-field mixing of $J$ multiplets matters for the light rare earths. On $\mathfrak{T}$ this is a single-ion calculation in the ground multiplet at clamped nuclei, and the magnetic response that uses it is at linear spin-wave order. Dirac–Fock calculations with the General-purpose Relativistic Atomic Structure Package (GRASP) or fully relativistic DFT $+U$ Wannier projections give the radial functions. Dark matter that couples to the orbital moment couples to the orbital current, not to $\boldsymbol{\ell}$ times the density, and the operator is built

that way. The integrals must satisfy $\langle j_0(0)\rangle = 1$ and the orbital-moment sum rules to $10^{-3}$, and agree with the tabulated dipole form factors to 2 % for $q \leq 1\,\mathrm{keV}$.

### 30.3.7 Finite-momentum optical activity

In a chiral crystal the current response has an antisymmetric part odd in $\boldsymbol{q}$, $\mathrm{i}\epsilon_{\alpha\beta\gamma}\mathcal{G}^{\gamma}_{\mathrm{OA}}(\boldsymbol{q},\omega)$ in $S_{j^\alpha j^\beta}$ (Eq. (14.45)), which changes sign between the two enantiomers and gives the difference of their rates in Sect. 14.6. DFPT gives only its slope at $q = 0$, the natural optical activity [22]. We want $\mathcal{G}_{\mathrm{OA}}(\boldsymbol{q},\omega)$, with the units of $S_{j^\alpha j^\beta}$, and the full current tensor at $\pm\boldsymbol{q}$ for tellurium or $\alpha$-quartz and for the magnetoelectric crystal $Cr_2O_3$, at $q \leq 10\,\mathrm{keV}$ and $\omega \leq 50\,\mathrm{eV}$. On $\mathfrak{T}$ the longitudinal part is at $\mathcal{X}$ = RPA with local fields and the transverse part at independent particles with local fields, both with spin–orbit coupling. Noncollinear DFT with fully relativistic SG15 pseudopotentials gives the states. The current vertex $\langle \boldsymbol{k}+\boldsymbol{q}|\{\boldsymbol{p}+\mathrm{i}[V_{\mathrm{NL}},\boldsymbol{r}],\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\}/2|\boldsymbol{k}\rangle$ needs the nonlocal commutator (Sect. 26.2) and is evaluated on Wannier-interpolated states in the same gauge at $+\boldsymbol{q}$ and $-\boldsymbol{q}$, and Yambo adds the local fields of the longitudinal channel. The sign of the odd part follows the sign convention for $\boldsymbol{q}$ of Appendix A. The odd part must vanish to numerical noise in silicon, which is centrosymmetric, and its linear term must reproduce the DFPT optical activity; the grid goes from $16^3$ to $24^3$ until the odd part changes by less than 5 %.

### 30.3.8 The $ZrTe_5$ edge

The excitonic binding of a gapped Dirac material is set by $\mathrm{Ry_D} = \Delta_{\mathrm{D}}\tilde{\alpha}^2/4$, the Rydberg of a pair with reduced mass $\Delta_{\mathrm{D}}/(2v_{\mathrm{D}}^2)$ and coupling $\alpha/\varepsilon_{\mathrm{bg}}$, with $\tilde{\alpha} = \alpha/(\varepsilon_{\mathrm{bg}}v_{\mathrm{D}})$ the effective coupling of Eq. (7.47). For $ZrTe_5$ the central parameters are those of the measured crystal structure (Sect. 16.1.3, the first $ZrTe_5$ row of Table 30.2): the half-gap $\Delta_{\mathrm{D}} = 11.8\,\mathrm{meV}$, the principal velocities $1.1\times10^{-3}$, $4.5\times10^{-4}$ and $1.0\times10^{-3}$ with geometric mean $v_{\mathrm{D}} = 7.9\times10^{-4}$, the energy cutoff $\tilde{\Lambda} = 0.191\,\mathrm{eV}$, $g_{\mathrm{deg}} = 2$, and $\varepsilon_{\mathrm{bg}} = 50.5$, the harmonic mean of the dielectric tensor. Then $\tilde{\alpha} \approx 0.18$, $\mathrm{Ry_D} \approx 0.10\,\mathrm{meV}$, and the exciton radius is $a_{\mathrm{D}} = 2v_{\mathrm{D}}/(\tilde{\alpha}\Delta_{\mathrm{D}}) \approx 1.4\times10^3\,\text{Å}$. A $24^3$ grid over the Brillouin zone has a spacing of $1.9\times10^{-2}\,\text{Å}^{-1}$, 27 times $1/a_{\mathrm{D}}$, so an ab initio BSE cannot resolve the threshold. What is wanted instead is $L(\boldsymbol{q},\omega)$ of the anisotropic massive-Dirac model for $|\boldsymbol{q}| \leq \Lambda_{\mathrm{D}}$ and $2\Delta_{\mathrm{D}} \leq \omega \leq 2\Delta_{\mathrm{D}} + 50\,\mathrm{meV}$ (at $\boldsymbol{q}\to 0$; at finite $\boldsymbol{q}$ the invariant pair energy replaces $\omega$), with the bound-state energies and weights. The box contains the region up to 1.33 times the edge where the Sommerfeld factor exceeds 1.5 (Sect. 16.1).

$G_0W_0$ with spin–orbit coupling on the 12-atom cell ($12\times12\times4$ grid, refined near $\Gamma$) fits the direction-resolved velocities $v_{\mathrm{D},i}$, the half-gap, the cutoffs $\Lambda_{\mathrm{D},i}$ and the background tensor $\varepsilon_{\mathrm{bg},ii}$. Since $\mathrm{Ry_D}$ lies far below the optical phonons, the right background is the static tensor with the ionic part, and the $G_0W_0$ stage computes it that way as its own check on the tabulated one; remote bands belong in $\varepsilon_{\mathrm{bg}}$, not also in the $GW$ screening. The threshold region, within about 10 $\mathrm{Ry_D}$ of $2\Delta_{\mathrm{D}}$, is the Elliott–Sommerfeld problem of an anisotropic hydrogen atom with masses $\Delta_{\mathrm{D}}/(2v_{\mathrm{D},i}^2)$ and dielectric tensor $\varepsilon_{\mathrm{bg},ii}$, solved on a radial grid. Away from threshold the BSE of the model, with the Coulomb interaction screened by $\varepsilon_{\mathrm{bg}}$ plus the RPA Dirac polarization, runs on a logarithmically graded radial grid from $k_{\min} \leq 0.1/a_{\mathrm{D}}$ to $\Lambda_{\mathrm{D}}$, about 200 radial points times 590 Lebedev directions, matched to the analytic solution over 5 to 10 $\mathrm{Ry_D}$; a uniform grid would give a spurious absence of bound states. The spread of $v_{\mathrm{D},i}$ and $\Delta_{\mathrm{D}}$ between one-shot and eigenvalue-self-consistent $GW$, and between the measured-structure and relaxed rows, is propagated through both stages. On $\mathfrak{T}$ this is $\mathcal{X}$ = BSE with static screening, the controlled first correction in $\tilde{\alpha}$; beyond it no controlled method exists at $\tilde{\alpha}$ of order one, so that stage gives only an estimate of the theory uncertainty. $\mathrm{Ry_D}$ and the first

Table 30.2: Dirac parameters of $ZrTe_5$ [23, Tables 1 and 2]: half-gap, principal velocities (units of $c$), diagonal background dielectric tensor and degeneracy. The first row is the measured crystal structure, the central row of Sect. 16.1.3; the second is the relaxed cell.

| material | $\Delta_{\mathrm{D}}$ (meV) | $v_{\mathrm{D},x}, v_{\mathrm{D},y}, v_{\mathrm{D},z}$ $(10^{-4})$ | $\varepsilon_{\mathrm{bg},xx}, \varepsilon_{\mathrm{bg},yy}, \varepsilon_{\mathrm{bg},zz}$ | $g_{\mathrm{deg}}$ |
|---|---|---|---|---|
| $ZrTe_5$, measured structure | 11.8 | 11, 4.5, 10 | 308.4, 20.75, 126.1 | 2 |
| $ZrTe_5$, relaxed cell | 15.6 | 11, 4.4, 9.1 | 308.4, 20.75, 126.1 | 2 |

Table 30.3: Magnetic, superconducting and Dirac calculations: targets, main grids, acceptance tests and cost in core-hours. KK, Kramers–Kronig.

| quantity | targets | grids and parameters | acceptance test | cost |
|---|---|---|---|---|
| vertex fraction $\mathfrak{v}$, $L^{\mathrm{cons}}$ | Al, clean and $\ell_e$ = 1 µm, 300 nm, 30 nm | $y \in [2, 50]$; $qv_{\mathrm{F}}/\Delta_{\mathrm{BCS}} \leq 10^3$; KK to $10^3 qv_{\mathrm{F}}$ | $q \to 0$ cancellation; bare limit; edge values of Sect. 15.3 | minutes |
| $\langle L \rangle$, $P_\Delta$ | a-$W_xSi_{1-x}$ | 512–1000 atoms, ≥5 quenches | $f$-sum 2 %; film $\sigma_1$ | $1.6\times10^4$–$1.6\times10^5$ |
| $\sigma_1^{\mathrm{S}}(\omega)$, $\alpha^2F_{\mathrm{tr}}$ | Al, Nb | $\boldsymbol{k}$ $60^3$, $\boldsymbol{Q}$ $30^3$ (Al), refined small $\boldsymbol{Q}$ | $\lambda_{\mathrm{tr}}$ 0.39; acoustic sum rule; $T_c$ 10 %; optical $\sigma_1$ | $10^3$ |
| $J^{\mathrm{ex}}$, anisotropy, Dzyaloshinskii–Moriya, $B_1$, $B_2$, $g_{\mathrm{ion}}$ | YIG, NiO, MnO, $Cr_2O_3$, $NiPS_3$ | magnetic cell; strained cells | $T_{\mathrm{N}}$, $T_{\mathrm{C}}$; magnon bands; Larmor; two sublattices | $10^3$–$5\times10^3$ |
| $S^{(2m)}$, bound states | YIG, NiO, $MnF_2$ | $30^3$–$40^3$ | $\langle \boldsymbol{S}^2 \rangle$ 1 %; grid doubling | $10^2$ |
| $\langle j_K(q) \rangle$ | $4f$ and spin–orbit ions | $K \leq 6$ | $\langle j_0(0) \rangle = 1$; dipole tables | <10 per ion |
| $\mathcal{G}_{\mathrm{OA}}$ | Te or $\alpha$-$SiO_2$; $Cr_2O_3$ | $16^3$–$24^3$ | zero in Si | $10^3$–$10^4$ |
| $L$ near $2\Delta_{\mathrm{D}}$ | $ZrTe_5$ | log radial, 590 directions | $\mathrm{Ry}_{\mathrm{D}}$ 5 %; $f$-sum | $10^4$ |

three bound states must be stable to 5 % when $k_{\min}$ is halved and the radial points doubled, the Sommerfeld factor must be continuous to 2 % across the matching window, and the model $f$-sum must hold to $10^{-3}$. The model stages take minutes and the $G_0W_0$ about $10^4$ core-hours; the same run gives $\varepsilon_{\mathrm{bg}}$ of $Bi_2Se_3$.

## 30.4 Dopants, color centers and small targets

A final state that the band picture does not contain occurs in donor clusters near the metal–insulator transition, molecular excitons with charge-transfer admixture, the ejected electron of a graphene sheet, the self-trapped exciton that becomes a Frenkel pair, and a molecule that dissociates in its cage (Chap. 16). Table 30.4 summarizes the calculations.

### 30.4.1 Disorder-averaged donor responses

Phosphorus donors in silicon at $n_D = 10^{18}$ and $1.8\times10^{18}\,\mathrm{cm}^{-3}$ sit at $n_D/n_c \approx 0.3$–$0.6$, where the dilute response of Sect. 16.5 fails at order one near the edge: donor pairs and clusters broaden the lines into an impurity band. What replaces it is the configurational average $\langle S^D(\boldsymbol{q},\omega) \rangle$ per donor, in $\mathrm{eV}^{-1}$, the per-cell convention of Sect. 30.1 with the donor in place

of the cell, at $n_B = 0$ and $n_g = 0$ on $\mathfrak{T}$, with a Hartree mean field on the donor electrons and host screening. Each donor carries the six-valley orbitals of Sect. 16.5, split into $A_1$, $E$ and $T_2$, at Poisson-random positions. The intervalley terms of the transition density sit at $\boldsymbol{q} = \boldsymbol{k}_\iota - \boldsymbol{k}_{\iota'} + \boldsymbol{G}$, between 0.685 and 3.88 keV, and the lightest particle that delivers the 11.70 meV of the $A_1 \to T_2$ line at 0.685 keV has 130 keV, so the $1s(A_1) \to 1s(E, T_2)$ lines, absent in optics, appear at finite $q$ for MeV-scale dark matter.

A Hubbard–Anderson tight-binding model of the random donor ensemble carries the disorder. Its hopping comes from the overlaps of the multivalley envelopes (effective Bohr radius $a^*$, with the central-cell correction on each site), and it is solved for $10^4$ to $10^5$ sites by Lanczos or the kernel polynomial method and averaged over 100 realizations, for about $10^3$ core-hours. DFT on 1000- to 4000-atom silicon clusters with two to four phosphorus atoms, at the shared 60 Ry setting of Sect. 30.1, checks the central-cell part for about $10^4$ more. The ensemble has to satisfy the Thomas–Reiche–Kuhn sum per donor, reproduce the single-donor response of Sect. 16.5 in the dilute limit, and give $\langle S^D \rangle$ stable to 5 % in the number of sites and realizations. The hopping oscillates with distance because of valley interference, so a single-valley envelope is wrong; compensation changes the occupation and must be stated. The Bloch functions at the conduction minima, on the same grid, also give the coefficients $\mathcal{U}_{j'j}(\boldsymbol{G})$ that set the strengths of the intervalley lines, and with the dielectric matrix of the host from Sect. 26.1 they give the local-field ratio $\mathfrak{f}_{\mathrm{D}}^{\mathrm{loc}}$ of Eq. (16.39), which Sect. 16.5 takes as one.

### 30.4.2 Molecular crystals

In anthracene and trans-stilbene the lowest excitons are Frenkel-like, localized on one molecule, but they mix with charge-transfer states in which the electron sits on a neighbor. A two-state model with transfer integral $t_{\mathrm{CT}}$ and energy splitting $E_{\mathrm{CT}} - E_{\mathrm{Fr}}$ gives a mixing angle $\tan 2\theta_{\mathrm{CT}} = 2t_{\mathrm{CT}}/(E_{\mathrm{CT}} - E_{\mathrm{Fr}})$ (Sect. 16.4), and the charge-transfer form factor carries the intermolecular phase, which no Frenkel model has; for orthogonal orbitals it vanishes at $q = 0$ and enters at order $q^2 d^2$, with $d$ the intermolecular distance. The periodic Bethe–Salpeter response measures how much this matters: we tabulate its difference $\Delta S(\boldsymbol{q}, \omega)$ from the zero-overlap Frenkel response, the product $\Omega S(\boldsymbol{q}, \omega)$ on $|\boldsymbol{q}| \le 30$ keV times 590 directions, the charge-transfer weight $\theta_{\mathrm{CT}}^2$, and, as the summary, the largest relative change of $|f|^2$ over $q$ for each state. The periodic calculation sits at $\mathcal{X} =$ BSE with intermolecular overlap, and the Frenkel model is its zero-overlap truncation.

It is $G_0W_0$ plus BSE (Yambo or BerkeleyGW) on the measured crystal structures from the Cambridge Structural Database, with 50 to 100 atoms per cell and a $4^3$ grid at finite $q$. Anthracene has 66 and trans-stilbene 136 occupied bands, so $G_0W_0$ needs about 1000 bands, or a Sternheimer formulation that needs no empty states; the BSE needs about 200. The molecular inputs of the Frenkel model come from TDDFT or molecular $GW$–BSE (PySCF or ORCA), and the Pariser–Parr–Pople model with standard parameters gives the semi-empirical comparison. The Frenkel code takes the transition densities on a real-space grid with $\Delta r \le \pi/q_{\max} \approx 0.4\, a_0$ at 30 keV, Fourier transforms them, builds the intermolecular coupling by Ewald summation with the $\boldsymbol{G} = 0$ term removed (it belongs to $\varepsilon$), and diagonalizes the Frenkel Hamiltonian for about 30 states up to 10 eV. The exciton energies must be stable to 50 meV under doubling of bands and grid, the partial $f$-sum per molecule must hold to 2 %, and $|f|^2$ at $q = \alpha m_e$ must be stable to 5 % in the basis. Relaxed van der Waals geometries are not good enough; the BSE exchange term must carry local fields consistently; and the $\pi$-electron model fails at $q \gtrsim 10$ keV, where the $\sigma$ electrons contribute. Each crystal costs $10^3$ to $10^4$ core-hours, and its tables about 10.

### 30.4.3 Low-dimensional response tables

Nanotubes, graphene sheets, layered stacks and quantum dots each screen with their own Coulomb kernel (Sect. 16.2). For sheets and stacks we tabulate $S(q_\parallel, q_z, \omega)$, for tubes of radius $r_\mathrm{t}$ $S(q_\parallel, q_\perp, \omega)$, directional and per unit area of sheet or length of tube in the normalization of Sect. 4.2 (with $2\pi\delta$ in $S$), and for dots $S^\mathrm{QD}(\boldsymbol{q}, \omega)$ in the dilute embedding of Sect. 16.3. Two inputs are reported with them: the proper density response $\tilde{R}^\mathrm{2D}$ of the monolayer per unit area, from Yambo with a two-dimensional Coulomb cutoff or from a Wannier bubble, and the Keldysh length $r_0$ defined by its static remote-band part, $-\tilde{R}^\mathrm{2D}_\mathrm{bg} = 2r_0 q_\parallel^2/e^2$ (Sect. 16.2). The tables sit at $\mathcal{X} = \mathrm{RPA}$ on the projected low-energy bands, with the remote bands entering only through $r_0$ as a static background. The RPA is solved in one and two dimensions with an arbitrary kernel, the inter-unit lattice sums and the transverse form factor, on $q_\parallel$ from $10^{-4}$ to 10 keV (200 logarithmic points) and $\omega$ up to $v_D/r_\mathrm{t}$ for tubes or 2 eV for sheets with at least 2000 points; stacks of period $a_s$ keep $q_z$ unreduced, with at least 64 points per $2\pi/a_s$. The interlayer spacing and the substrate $\varepsilon_b(\omega)$ with its polar phonons complete the input.

A semiconducting tube screens its electron–hole pairs weakly, so its lowest excitations are excitons, which the RPA does not contain. For such a tube the RPA step is followed by a ladder (Bethe–Salpeter) step on the same projected bands, with the interaction of the tube statically screened by the projected response, which gives the exciton energies and their form factors along $q_\parallel$. The lowest exciton energy must be stable to 10 meV when the bands and the $q_\parallel$ grid are doubled and must reproduce the measured optical transition of the same tube, and the ladder adds $10^2$ to $10^3$ core-hours per chirality.

The band-projected $f$-sum of the linear-band model, $\int \omega S \,\mathrm{d}\omega = N_\mathrm{ch} v_D q^2$ per unit length of tube in that normalization, holds to $10^{-3}$. The undoped Dirac sheet reproduces its closed-form bubble to $10^{-4}$, a finite stack of about 50 layers converges to the infinite-stack form to 1 %, and the Yambo slab calculation with a Coulomb cutoff cross-checks one sheet. A metallic nanotube couples to dark matter through its charge mode only, with all four spin and valley channels; a one-channel model is low by a factor $N_\mathrm{ch} = 4$ at fixed Luttinger parameter. For dots, the envelope-function solver (spherical-well levels up to $\ell = 20$ and $n = 10$) and the dielectric-confinement factors are checked against tight-binding or DFT clusters of up to 2000 atoms. The model module runs in less than one core-hour; the DFT bubbles of real sheets cost $10^2$ to $10^3$ core-hours and the dot clusters $10^4$.

### 30.4.4 Ejection factors of graphene and nanotubes

A plane-wave final state is not orthogonal to the occupied orbital it leaves, and the overlap adds a spurious monopole term to the ejection amplitude. Orthogonalizing the plane wave to the occupied states removes it. For a carbon $2p_z$ orbital of Slater form with the exponent $1.568/a_0$ of Clementi and Raimondi, the ratio of the orthogonalized to the plain squared amplitude ranges from 0.4 to 4.3 at kinetic energies of 50 to 100 eV and reaches 740 at 10 eV (Sect. 16.3). The large values occur where $q$ far exceeds the final momentum and the plain amplitude has already fallen as $|\boldsymbol{p} - \boldsymbol{q}|^{-5}$, so a table of such ratios is not a rate correction. A rate needs the factor weighted by the halo,

$$R_\mathrm{OPW}(E_\mathrm{kin}) = \frac{\int \mathrm{d}q\, q\, \eta\, F_\mathrm{DM}^2 \sum_\mathrm{occ} |M_\mathrm{OPW}|^2}{\int \mathrm{d}q\, q\, \eta\, F_\mathrm{DM}^2 \sum_\mathrm{occ} |M_\mathrm{PW}|^2}, \tag{30.4}$$

dimensionless, per occupied orbital and in total, for graphene and nanotubes, with $\eta$ the mean inverse speed of the standard halo. The factor is computed at $\mathcal{X} = \mathrm{IP}$ with frozen orbitals and a single continuum orthogonalized to all occupied states, the carbon 1s and the $\sigma$ and $\pi$ bands.

The occupied states come from a DFT ground state (SG15) as maximally localized Wannier functions, with the 1s orbital from an all-electron atomic calculation, since the pseudopotential has none. Their plane-wave transforms and mutual overlaps come by FFT, and the integrals of Eq. (30.4) use $2 \times 10^4$ final directions on a Fibonacci set and a logarithmic $q$ grid from 1 to 60 keV, which covers heavy-mediator kinematics to $m_\chi \approx 10$ MeV. Orthogonality holds to $10^{-6}$, and $R_{\mathrm{OPW}}$ must change by less than 1 % when the directions or the $q$ grid are doubled and the Wannier spread is reminimized; the whole calculation takes CPU minutes. The final-state interaction with the hole is a separate factor, bounded above by the Fermi factor of an unscreened point charge, Eq. (2.25) (7.3, 3.4 and 2.6 at 10, 50 and 100 eV), and computed in CPU hours with scattering states of the sheet's Hartree potential plus the screened hole. An orthogonalized plane wave is not a scattering state of the hole, so multiplying it by a Fermi factor double counts. Both factors replace the bounds of Sect. 16.3.

### 30.4.5 Ionic color centers

In LiF, NaF, MgO and LiH an electronic deposit can be stored promptly as a Frenkel pair: a self-trapped exciton converts within picoseconds into an F center and an H center. The yield factorizes as in Sect. 16.6, and two of its factors have not been computed: the probability $p_{\mathrm{FH}}(T)$ that a self-trapped exciton becomes a pair at millikelvin, and the survival $p_{\mathrm{surv}}$ of the close pair against tunneling recombination. The calculation gives these two probabilities, the direction-resolved displacement threshold $E_d(\hat{\boldsymbol{r}})$ of the anion sublattice, the charge levels that set $p_q$ and the F-band cross-sections, tabulated as yields per recoil energy. On $\mathfrak{T}$ the nuclear-recoil seed is in the impulse regime of $\mathcal{K}$ and the electronic seed at $\mathcal{X} = \mathrm{RPA}$ with local fields; the yield factors themselves lie outside $\mathfrak{T}$, as properties of the final state, and come from the dynamics below.

The adiabatic surfaces of the self-trapped exciton and the pair come from constrained-occupation hybrid DFT (ΔSCF with HSE or PBE0, the hybrid built on the functional of Perdew, Burke and Ernzerhof; exact-exchange fraction 0.25 and 0.40, both reported) in 64- to 216-atom cells, large enough for pair separations up to the third neighbor (Sect. 28.2). Direct surface hopping on hybrid surfaces would cost $10^6$ to $10^7$ core-hours, so a machine-learned surface is fitted to the ΔSCF points and validated on held-out ones to 50 meV, and decoherence-corrected surface hopping runs on it with at least $10^3$ trajectories, which fixes $p_{\mathrm{FH}}$ to 10 %. The self-trapped-exciton and F–H pair energies are converged to 0.1 eV between the 64- and 216-atom cells. The threshold $E_d(\hat{\boldsymbol{r}})$ comes from recoil molecular dynamics (Sect. 29.1) over $10^4$ zero-point-sampled directions, converged to 1 eV when the directions are doubled, and $p_{\mathrm{surv}}$ from the tunneling-path calculation of Sect. 30.5 applied to the close pair. The F-band cross-section comes from a BSE on the defect supercell. The charged components need the finite-size corrections of Sect. 28.2, and at millikelvin only the athermal channel acts. The ΔSCF points dominate a cost of about $10^5$ core-hours per material. The calculation has a direct control: tagged neutron scattering on LiF at 10 mK followed by single-center optical readout measures the product of the recoil factors, and 5.9 keV X-rays measure $p_{\mathrm{FH}}$ through the quenching of the exciton luminescence.

### 30.4.6 Bond breaking in molecular solids

A nuclear recoil on one atom of a molecule splits, for an internal coordinate independent of the intermolecular potential, into a center-of-mass recoil of the molecule and an internal excitation at the reduced momentum $(\mu_{12}/M_1)\boldsymbol{q}$, with $\mu_{12}$ the reduced mass of the bond and $M_1$ the mass of the struck atom. The response is their convolution (Sect. 16.6), dissociation is the internal response above the bond energy, and in the crystal the fragments also have

Table 30.4: Dopant, color-center, molecular and low-dimensional calculations: targets, main parameters, acceptance tests and cost in core-hours. CNT, carbon nanotube; ML, machine-learned.

| quantity | targets | parameters | acceptance | cost |
|---|---|---|---|---|
| $\langle S^D(\boldsymbol{q},\omega)\rangle$, $\mathcal{U}_{j'j}(\boldsymbol{G})$ | Si:P | $10^4$–$10^5$ sites; 100 realizations | sum per donor; dilute limit | $10^3$–$10^4$ |
| $\Delta S$, $\theta^2_{\mathrm{CT}}$, tables | anthracene, t-stilbene | $4^3$, 200 bands; 590 directions | partial $f$-sum 2 % | $10^3$–$10^4$ |
| $S$ of tubes, sheets, stacks, dots; $\tilde{R}^{\mathrm{2D}}$, $r_0$ | graphene, CNT, stacks, dots | 200 $q_{\parallel}$; ≥2000 $\omega$ | projected $f$-sum $10^{-3}$ | <1 to $10^4$ |
| tube excitons | semiconducting CNT | ladder on projected bands | lowest exciton 10 meV | $10^2$–$10^3$ |
| $R_{\mathrm{OPW}}$ | graphene, CNT | $2\times10^4$ directions; $q$ 1–60 keV | orthogonality $10^{-6}$ | minutes |
| $p_{\mathrm{FH}}$, $p_{\mathrm{surv}}$, $E_d(\hat{\boldsymbol{r}})$ | LiF, NaF, MgO, LiH | 64–216 atoms; ML surface; $\geq10^3$ trajectories | tagged neutrons, X-ray quenching | $10^5$ |
| $S_{\mathrm{int}}$, $S_{\mathrm{CM}}$, $p_{\mathrm{esc}}$ | anthracene | $10^3$ trajectories per point | free-molecule limit | $10^3$ |

to escape the cage. The first target is anthracene, the molecular crystal above, with a C–H bond as the internal coordinate, and the calculation gives the internal response $S_{\mathrm{int}}$, the center-of-mass response $S_{\mathrm{CM}}$ and the escape probability $p_{\mathrm{esc}}$ per internal energy and direction. The internal motion is in the impulse regime of $\mathcal{K}$ with a bound center of mass; at the relevant momenta, about $10^2$ keV, the recoil-free fraction is essentially zero, so the center-of-mass recoil is impulsive with a Doppler width from its zero-point momentum.

The internal response comes from Morse or ab initio potential curves, the center-of-mass response from phonopy or path-integral moments (Sect. 25.3), and $p_{\mathrm{esc}}$ from molecular dynamics with a machine-learned potential whose training set contains the dissociative region (Sect. 29.1). With $10^3$ trajectories per internal energy and direction, $p_{\mathrm{esc}}$ is known to about 3 %, for about $10^3$ core-hours in all. In the free-molecule limit, $S_{\mathrm{CM}}$ replaced by free recoil and $p_{\mathrm{esc}} = 1$, the calculation must reproduce the dissociation rate of Essig et al. [24], and $S_{\mathrm{CM}}$ must satisfy the $f$-sum for the whole molecular mass. The photodissociation quantum yield of the same molecular crystal at millikelvin, with fragments detected by infrared absorption, measures $p_{\mathrm{esc}}$.

## 30.5 Defects and two-level systems

A group of atoms in fused silica that can sit in either of two configurations, a fraction of an ångström apart and equal in energy to within a meV, absorbs and scatters at micro-electronvolt energies. A close Frenkel pair in silicon stores a few eV and may give them back long after the event that made it. Sects. 16.7, 16.8 and 19.5 describe both kinds of defect with a few numbers. For defects in crystals they are the ohmic coupling $\alpha_{\mathrm{ohm}}$ to the electrons of a metal, the tunneling exponents and Huang–Rhys factors of release paths, and the barriers and masses of crowdions and dumbbells. For glasses they are the density vertex of a two-level system and the distribution of two-level parameters. None of them has been computed for a real material at the level the text needs. Each calculation below is built from the methods of Sects. 28.2 and 29.1: relaxed supercells, nudged elastic bands, density-functional perturbation theory and molecular dynamics on machine-learned potentials. Table 30.5 collects the targets, the tests and the costs.

### 30.5.1 Ohmic coupling of tunneling defects

A defect that hops between two configurations L and R in a normal metal changes the potential seen by the conduction electrons, and the Fermi sea has to rearrange at every hop. Sect. 16.8 turns this into a relaxation rate through the ohmic coupling $\alpha_{\mathrm{ohm}}$. We compute the dimensionless $\alpha_{\mathrm{ohm}}$ of Eq. (28.15) from the phase-shift differences $\tilde{\delta}_\nu$, the eigenphases of $S_R S_L^\dagger$ at the Fermi energy. The targets are a crowdion in tungsten, hydrogen in niobium, and two-level systems in aluminum, its oxide and amorphous WSi. On the truncation label $\mathfrak{T}$ (Sect. 8.5) the calculation sits at the base point for the electrons, with Kohn–Sham scattering states at clamped nuclei, $n_B = 0$; the defect enters only through its two relaxed geometries.

Two routes give the phases. The first computes $S_L$ and $S_R$ directly, from the impurity Green's function of a Korringa–Kohn–Rostoker code (JuKKR or AKAIKKR, full potential, angular momenta up to $l = 3$ or 4) with each relaxed configuration embedded in the host. The two matrices are expressed in one channel basis centered at the midpoint of the hop, and Eq. (28.15) is applied to the eigenphases. A basis centered on the defect is not enough: a hop displaces the scatterer, so for crowdions and hydrogen the eigenphase form holds and a single-center phase-shift sum does not.

The second route uses plane-wave supercells and the orthogonality catastrophe itself. The many-electron ground states of the two configurations in a cell of size $L$ overlap as $|\langle \Psi_L | \Psi_R \rangle| \propto (k_{\mathrm{F}} L)^{-\frac{1}{2}\sum_{\sigma\nu} (\tilde{\delta}_\nu/\pi)^2}$ [25], and by Eq. (28.15) the exponent is $\frac{1}{2}\sum_{\sigma\nu} \tilde{\delta}_\nu^2/\pi^2 = \alpha_{\mathrm{ohm}}$. The slope of $\ln |\langle \Psi_L | \Psi_R \rangle|$ against $\ln L$ is therefore $-\alpha_{\mathrm{ohm}}$. The overlap is the determinant of the matrix of overlaps of the occupied Kohn–Sham orbitals of the two cells. For tungsten we take cubic cells of 54, 128 and 250 atoms ($3a$, $4a$ and $5a$) with the PBE functional, an optimized norm-conserving potential with the $5s$ and $5p$ semicore states and a cutoff of 60 Ry, all at $\Gamma$ only. The length $L$ of the slope is then the cell side itself; a $\boldsymbol{k}$ grid that shrinks as the cell grows would hold the Born–von Kármán length fixed and leave the slope undefined. A sparser fixed-density grid, the same in every cell, is the alternative, with $L$ its Born–von Kármán length. The 250-atom cell holds about $2.1 \times 10^5$ plane waves and 2100 bands, about 7 GB per $\boldsymbol{k}$ point. Each defect costs about $10^3$ hours on graphics processors.

A contact coupling $\pm V_{\mathrm{d}}$ in a single channel returns $\alpha_{\mathrm{ohm}} = 4K_{\mathrm{b}}$ (Sect. 28.2); the same code also reproduces the exponent of a one-dimensional tight-binding chain with one impurity site, whose phase shift is known in closed form. The measurement is the damping of aluminum-covered nanomechanical resonators [26]. The computed $\alpha_{\mathrm{ohm}}$, which contains every order in the phase shifts, is compared through the rate of Sect. 16.8 with the measured damping, not with the Born reading of the fitted couplings. $\boldsymbol{k}$ sampling and smearing blur the Fermi surface on which the logarithm rests, so the overlaps must be taken with fixed occupations at a common Fermi level and the smearing extrapolated to zero. The host must be relaxed identically in both cells except near the defect, or the overlap measures the relaxation and not the hop. The computed couplings set the electron-assisted relaxation rates of Sect. 16.8 and, for tungsten, the coherence of crowdion motion in Sect. 19.5.

### 30.5.2 Release paths of eV-scale defects

Whether a close Frenkel pair in silicon or germanium, or a split interstitial in aluminum, releases its stored energy at millikelvin depends on the tunneling exponent through its barrier and on how many local vibrations accept the eV released in the hop (Sects. 16.8 and 19.5). Let $\Delta \boldsymbol{R}_I$ be the change of the position of atom $I$ between the two quenched minima, and $\boldsymbol{e}_{k,I}$ the mass-weighted normal modes, of energies $\omega_k$, at the final minimum. The displacement of mode $k$ is $\Delta Q_k = \sum_I M_I^{1/2} \boldsymbol{e}_{k,I} \cdot \Delta \boldsymbol{R}_I$. A harmonic oscillator displaced by $\Delta Q_k$ stores

$\frac{1}{2}\omega_k^2 \Delta Q_k^2$, which in units of its quantum is the Huang–Rhys factor [27]

$$S_{\mathrm{HR},k} = \tfrac{1}{2}\,\omega_k\,\Delta Q_k^2, \qquad E_{\mathrm{reorg}} = \sum_k S_{\mathrm{HR},k}\,\omega_k. \tag{30.5}$$

These quantities lie on the Born–Oppenheimer surface, $n_B = 0$, harmonic about each minimum, with the hop treated either as a two-state transition or, when the reduction fails, by an instanton on the full surface. The reorganization energy $E_{\mathrm{reorg}}$ decides the regime. If it is small compared with the energy $\varepsilon_{\mathrm{hop}}$ released in the hop, the hop must put that energy into many quanta of weakly coupled modes, and the rate is suppressed by the energy-gap law. If $E_{\mathrm{reorg}} \approx \varepsilon_{\mathrm{hop}}$, the reaction coordinate itself takes the energy and there is no such suppression. The tunneling exponent is $\lambda_{\mathrm{tun}} = \int \mathrm{d}s\,[2(V(s) - E)]^{1/2}$ along the mass-weighted minimum-energy path $V(s)$, and the zero-temperature ring-polymer instanton action equals $2\lambda_{\mathrm{tun}}$ when the path does not cut corners [28].

A release within a time $t_{\mathrm{obs}}$ needs the tunneling factor $\mathrm{e}^{-2\lambda_{\mathrm{tun}}}$ to beat $1/(\Gamma_0 t_{\mathrm{obs}})$, with $\Gamma_0$ the attempt rate of Sect. 29.1. For a silicon close pair in one year this is $\lambda_{\mathrm{tun}} \leq 23.6$ in the strong-coupling regime, against a two-parabola estimate of 27.5 (Sect. 19.5), so $\lambda_{\mathrm{tun}}$ and $E_{\mathrm{reorg}}/\varepsilon_{\mathrm{hop}}$ are needed to about 5 %, for real paths.

A first calculation uses a 64-atom silicon cell with a $\langle 110 \rangle$ split interstitial 2 to 4 Å from a vacancy. It uses Quantum ESPRESSO with PBE and optimized norm-conserving potentials, a cutoff of 40 Ry checked at 30 and 50 Ry, a $2^3$ Monkhorst–Pack grid, and the cell fixed at the PBE lattice constant. The climbing-image nudged elastic band of Sect. 29.1 runs with nine images to a force tolerance of 5 meV/Å. The Hessians at both minima come from finite displacements of 0.01 Å, $64 \times 3 \times 2 = 384$ calculations each. The path then gives $S_{\mathrm{HR},k}$, $E_{\mathrm{reorg}}$, $\lambda_{\mathrm{tun}}$ and the line shape of the hop, and one ring-polymer instanton with 64 to 128 beads, run with i-PI on the machine-learned potential, checks the action against $2\lambda_{\mathrm{tun}}$. Its cost is set by the roughly 900 self-consistent calculations, about 120 core-hours, at under 0.1 GB per $\boldsymbol{k}$ point. The full catalogue needs 50 paths per material, mapped on the machine-learned potential and checked against density-functional theory. A kinetic Monte Carlo over the resulting rates at 10 to 100 mK, with $10^6$ seeds, gives the release spectrum. The cost, mostly the instantons, is estimated at $10^4$ to $10^5$ hours on graphics processors.

The barrier and the exponent are converged in the cutoff at 30, 40 and 50 Ry, the grid at $2^3$ and $3^3$, and one path repeated in a 216-atom cell, since the 64-atom cell compresses the strain field of the pair. The catalogue is accepted by the release-path sum rule of Sect. 19.5: the energy released by all hops of the kinetic Monte Carlo, plus the energy still stored at the end, equals the initial stored energy, with every energy taken from the table of configurational energies of Sect. 32.1. Charged close pairs need the finite-size corrections of Sect. 28.2. A charge-state variant of the same path, computed by constrained occupations, gives the Huang–Rhys factor $S_{\mathrm{HR},R}$ and frequency $\omega_R$ of the reaction coordinate when a carrier is captured. The probability that a capture drives the reaction over a barrier $E_{\mathrm{b}}$ is then the tail above $n = E_{\mathrm{b}}/\omega_R$ of the Poisson series of Eq. (31.3). This is the athermal channel by which the carriers of a later event can release stored energy (Sect. 19.5).

### 30.5.3 Crowdions and dumbbells

A crowdion in tungsten is an extra atom spread along a $\langle 111 \rangle$ row. It glides over a Peierls barrier of meV in atomistic calculations and of micro-electronvolts in the Frenkel–Kontorova model [29], and its kink mass $M_{\mathrm{cr}}$ is a fraction of an atomic mass (Appendix C.20). Sect. 19.5 asks whether it moves coherently at millikelvin. Coherent motion needs a band that survives three effects: the phonons that distinguish neighboring sites, the conduction electrons of the normal metal (bulk tungsten becomes superconducting only near 15 mK), and the strain field

of the vacancy it left behind. The phonon dressing multiplies the hopping by $\mathrm{e}^{-S_{\mathrm{pol}}(T)}$, with

$$S_{\mathrm{pol}}(T) = \frac{1}{\pi}\int \mathrm{d}\omega\, \frac{J_{\mathrm{s}}(\omega)}{\omega^2}\coth\frac{\omega}{2T}, \tag{30.6}$$

for a linear coupling with spectral density $J_{\mathrm{s}}$ of Appendix C.20. For acoustic phonons $J_{\mathrm{s}} = J_{\mathrm{s},3}\omega^3$ up to the Debye energy $\omega_{\mathrm{D}}$, which gives a finite $S_{\mathrm{pol}}(0) = J_{\mathrm{s},3}\omega_{\mathrm{D}}^2/2\pi$ and a thermal part $\pi J_{\mathrm{s},3}T^2/3$ (from $\int_0^\infty \mathrm{d}x\, x/(\mathrm{e}^x-1) = \pi^2/6$). The electrons renormalize the hopping further by a power of $\alpha_{\mathrm{ohm}}$ (Appendix C.20), with $\alpha_{\mathrm{ohm}}$ computed as above for the crowdion geometry.

The crowdion sits on the Born–Oppenheimer surface, $n_B = 0$, with phonons and electrons coupled linearly to its position. The calculation for tungsten uses PBE, the semicore potential above, a 250-atom body-centered cubic cell, a cutoff of 50 Ry, a $3^3$ grid and Marzari–Vanderbilt smearing of 0.01 Ry. A climbing-image nudged elastic band along $\langle 111\rangle$ gives the Peierls profile $E(z_0)$ of the kink center. The Hessian of the crowdion core gives the phonon coupling and $J_{\mathrm{s}}$. A fit of the profile to the Frenkel–Kontorova form gives the kink width and mass. The interaction energy with the parent vacancy as a function of separation gives the radius inside which the strain gradient across one spacing exceeds the band width, and the crowdion is bound. For aluminum the same steps give the migration of the $\langle 100\rangle$ dumbbell, in cells of 108 to 256 atoms, and the one-dimensional glide of di- and tri-interstitials comes from machine-learned molecular dynamics at 10 to 100 K, extrapolated to low temperature with instanton rates. Each material costs about $10^5$ core-hours.

The Peierls barrier is below a meV per cell, so the energies are differences at identical meshes and smearing, extrapolated to zero smearing and checked on a $4^3$ grid, with noise below 0.1 meV per cell. The fit to the Frenkel–Kontorova form is valid only if the profile is sinusoidal; its higher harmonics measure the failure, and the kink width over the row spacing controls the continuum description (Appendix C.20). In Sect. 19.5 the barrier and mass of the kink replace the model values, and the polaron factor, the electron renormalization and the confinement radius decide whether the crowdion band is real.

### 30.5.4 Density vertex of a two-level system

Dark matter couples to a two-level system in a glass through the change $f_{\mathrm{TLS}}(\boldsymbol{q})$ of the electron density between its configurations, Eq. (16.58), which enters the scattering rate of Sect. 16.7 as $|f_{\mathrm{TLS}}|^2$. The estimate there moves one oxygen ion, eight electrons, rigidly by 1 Å, and misses the redistribution of the valence electrons, the screening of the dipole by the glass and the variety of real geometries. The vertex is a ground-state density at $n_B = 0$, with the adiabatic electrons following the atoms; the screening is that of the self-consistent density, and no further dielectric factor is applied.

The preferred method takes two self-consistent calculations of the same glass cell, relaxed in configurations L and R, and Fourier transforms the difference of the two densities at the reciprocal vectors $\boldsymbol{G}_{\mathrm{sc}}$ of the cell. It is exact to first order in the dark-matter coupling and keeps every order in the atomic displacement $\boldsymbol{d}$. That matters because $qd$ reaches 0.5 to 1 at $q = 1$ to 2 keV for $d = 1$ Å, where a linearization errs at order $(qd)^2$. The alternative is density-functional perturbation theory at $\Gamma$ of the supercell, with only the core atoms displaced. Every $\boldsymbol{G}_{\mathrm{sc}}$ is equivalent to $\Gamma$, so one response gives the density change at all of them, and the vertex is the sum over core atoms of the density response to each atom times its displacement from L to R. This route costs about ten times more and serves as a check on the relaxation field of the host.

The cells hold 216 atoms (72 formula units) of amorphous $\mathrm{SiO_2}$, a cube of side 15 Å. A two-level system found in the large glasses of the ensemble below is carved out with its

environment to that size, the outer shell re-quenched with the core held fixed, and its two configurations re-relaxed by a nudged elastic band in the small cell; systems whose asymmetry moves by more than a meV in the transplant are discarded. The calculation uses PBE, optimized norm-conserving potentials, a wavefunction cutoff of 80 Ry checked at 70 and 90 Ry, a density cutoff of 320 Ry, a self-consistency threshold of $10^{-10}$ and the $\Gamma$ point only. The momentum transfers run from $2\pi/L = 0.42\,\text{Å}^{-1}$ (0.83 keV) to $5\,\text{Å}^{-1}$, and the 576 occupied bands take about 2.5 GB. Each self-consistent calculation costs about one hour on a graphics processor, so 100 two-level systems cost about 200 hours with the preferred method.

The vertex is converged when it changes by less than 2 % with the cutoff raised by 10 Ry and with the cell enlarged from 216 to 648 atoms for one system; the ensemble average $\langle |f_{\text{TLS}}|^2 \rangle$ needs at least 100 systems for 10 %. The benchmark is the rigid-ion value, which the computed vertex approaches at large $q$, where only the core electrons of the moving atoms contribute. Below $q = 2\pi/L$ the periodic cell forces the macroscopic field to vanish, so the cell gives the unscreened dipole of the core. The long-range part, the change of the electronic dipole screened by the high-frequency dielectric constant of the glass, must be added analytically at small $q$. Rebuilding the density change from Born effective charges places point charges where the vertex needs the valence density. Sect. 16.7 uses the ratio of the computed $\langle |f_{\text{TLS}}|^2 \rangle$ to the rigid-ion value, together with the interference between core and host for cores off the center of the defect, which vanishes only for symmetric cores.

### 30.5.5 Ensemble sampling of two-level systems

The standard tunneling model of Sect. 16.7 assumes a flat distribution of asymmetries and of tunneling exponents, with density $\bar{P}$, and Sect. 16.8 builds its relaxation rates on it. Testing that assumption for a specific glass needs the distribution itself, computed from the potential-energy surface of the glass. The targets are amorphous $SiO_2$, amorphous $Al_2O_3$ and amorphous WSi. For each we need the distributions of the asymmetry $\Delta_{\text{as}}$, the tunneling amplitude $\Delta_0$ of Appendix C.20, the deformation potential and the dipole moment. From them follow tables of the two-level response of Sect. 16.7 on at least 590 directions of $\boldsymbol{q}$, 60 logarithmic points in $|\boldsymbol{q}|$ from 0.1 eV to 10 keV, and 50 energies from $10^{-7}$ to $10^{-3}$ eV.

The two-level systems that matter, with asymmetries below a meV and tunneling exponents below about 20, are rare among pairs of neighboring minima. A study of the energy minima of a model glass found of order $10^3$ active two-level systems after sampling some $10^6$ minima [30]. The sampling is classical on the Born–Oppenheimer surface of the potential, $n_B = 0$, with tunneling added through the instanton path. Following that study, glasses of $10^3$ to $10^4$ atoms are quenched with the machine-learned potential, at least $10^4$ of them, with swap Monte Carlo or a comparable method to reach stable glasses. Molecular dynamics at low temperature visits the minima of each glass, pairs of adjacent minima with energy difference below 5 meV are kept, and a nudged elastic band connects each pair. The tunneling exponent comes from the instanton path; the deformation potential from the energy difference under a finite strain; the dipole moment from the Berry-phase polarization of the two configurations. To resolve at least $10^3$ systems per decade of splitting, about $10^8$ pairs of minima are screened, some $10^4$ from each glass. The sampling costs $10^5$ to $10^6$ hours on graphics processors; the response tables, under ten core-hours.

The tables contain a principal value and a Hadamard finite part, which are implemented by subtracting the first two terms of the Taylor expansion of the integrand inside a unit interval. The ensemble average is converged to 1 %, and the finite part is stable to $10^{-3}$ when the inner interval is halved. The measured control is the acoustic loss. The computed tunneling strength $\bar{P}\gamma^2/\rho_T c^2$, with $\gamma$ the deformation potential and $c$ the sound speed, must reproduce the measured internal friction of the same glass [31]. The dielectric loss tangent $\tan\delta_0$ of Sect. 16.7

Table 30.5: Defect and two-level calculations. Costs are estimates. GPU-h are hours on a graphics processing unit (GPU); KKR, Korringa–Kohn–Rostoker; NEB, nudged elastic band; SCF, self-consistent field; MD, molecular dynamics.

| quantity | method and grid | acceptance | cost |
|---|---|---|---|
| $\alpha_{\mathrm{ohm}}$ (W, Nb–H, Al, $AlO_x$, WSi) | KKR eigenphases; or supercell overlaps, 54–250 atoms at $\Gamma$ | Born limit $4K_{\mathrm{b}}$; resonator damping (Al), beyond Born | $10^3$ GPU-h per defect |
| $S_{\mathrm{HR},k}$, $E_{\mathrm{reorg}}$, $\lambda_{\mathrm{tun}}$ (Si, Ge pairs; Al) | NEB, Hessians, instantons; 64 and 216 atoms; 50 paths | release-path sum rule; $\lambda_{\mathrm{tun}}$ to 5 % | 120 core-h (pilot); $10^4$–$10^5$ GPU-h |
| Peierls profile, $M_{\mathrm{cr}}$, $S_{\mathrm{pol}}$ (W); dumbbell glide (Al) | NEB, 250 atoms; MD at 10–100 K | noise $< 0.1$ meV per cell; sinusoidal profile | $10^5$ core-h per material |
| $\langle \lvert f_{\mathrm{TLS}} \rvert^2 \rangle$ (a-$SiO_2$) | two SCFs per system, 216 atoms, 80 Ry | 2 % per system; 100 systems for 10 % | 200 GPU-h |
| Two-level ensemble (a-$SiO_2$, $AlO_x$, a-WSi) | $10^8$ pairs of minima; NEB; response tables | 1 %; acoustic loss | $10^5$–$10^6$ GPU-h |

gives a second control through the dipole moments. The density of two-level systems in the model glass fell by two orders of magnitude between its least and most stable preparations [30]. A glass quenched on the computer is far less stable than one cooled in the laboratory, so the computed $\bar{P}$ is meaningful only as a function of the stated preparation. In Sects. 16.7 and 16.8 the tables replace the flat distribution in the scattering and absorption rates, and the distributions of $\Delta_{\mathrm{as}}$, $\Delta_0$ and the couplings test the universality that the standard tunneling model assumes.

## Summary

The harmonic base of silicon, germanium and gallium arsenide holds the force constants, the eigenpairs, the per-atom densities of states, the displacement tensors and, for GaAs, $Z^*$ and $\varepsilon_\infty$. Its frequencies are accepted against neutron data at $\Gamma$, X and L with a tolerance per branch, and the deformation potentials join it by frozen strain. The multiphonon tables follow from the per-atom density of states and are validated on an Einstein site and by the $f$-sum rule of the lattice to $10^{-4}$. The vertex at large momentum needs all-electron densities, the impulse peak needs path-integral momentum distributions, absorption by isotope disorder needs the Hessian of a disordered supercell, and the radiation of a recoiling ion needs the dynamical Born charge. For strongly anharmonic and quantum crystals the frequency moments of order 0 to 4 of the self response are exact from a static path-integral simulation, and glasses need melt-quench structures before they need responses. The magnetic, superconducting and Dirac targets need the parameters of their models, exchange constants from DFT $+U$, Eliashberg functions and Dirac velocities, and the responses that go beyond them. Donor clusters, molecular excitons, electrons ejected from graphene, color centers and dissociating molecules have final states that the band picture does not contain. The parameters of tunneling defects and two-level systems, from the ohmic coupling and the release paths to the density vertex and the distribution of two-level parameters, have not been computed for a real material at the level the text needs.

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# Chapter 31

# Electronic and mixed channels

The phonon-assisted electron-recoil rate of silicon between 0.333 and 0.405 MeV is known to a factor of ten (Sect. 13.9), and below $2q_{\mathrm{BZ}}$ the coherent zero-phonon part of its Migdal rate is not computed at all (Sect. 12.4). The electronic channels, with those that mix electrons with phonons or with the motion of a nucleus, are specified here with the methods of Chaps. 23 to 27, on the lattice base of Sect. 30.1.

## 31.1 Phonon-assisted electron recoil from first principles

The factor-of-ten estimate of the phonon-assisted rate of silicon (Sect. 13.9) takes the measured indirect absorption of silicon for the optical limit and multiplies it by a computed form factor of the zero-phonon vertex for the fall-off at finite $q$. It never evaluates the four-path amplitude of Sect. 13.2. The amplitude is hard for three reasons: the band sum over intermediate states is unrestricted and cannot stop at the top of a Wannier manifold; the density vertex is needed at momentum transfers of one to a few keV, where the small-$q$ expansion in Wannier position matrix elements fails (Sect. 27.2); and the external screening carries local fields, which the macroscopic dielectric function does not. Everything else follows the standard steps of Sect. 27.2.

### 31.1.1 The phonon-assisted loss-function grid

What a rate needs is the phonon-assisted loss function $L^{\mathrm{PA}}(\boldsymbol{q},\omega)$ of Sect. 13.3, a dimensionless number per bin of $(|\boldsymbol{q}|,\hat{\boldsymbol{q}},\omega)$, built from the four-path amplitude of Sect. 13.2 and screened as in Sect. 13.4. The bins are those of the estimate in Sect. 13.9, so that the two can be compared bin by bin. They are $|\boldsymbol{q}| = 0.25$ to $1.25\,\text{Å}^{-1}$ (0.49 to 2.47 keV) in steps of $0.05\,\text{Å}^{-1}$, 30 directions in the irreducible wedge of the cubic group, and $\omega$ from $E_{\mathrm{g}} + \omega_{\mathrm{TA}}(\boldsymbol{Q}^{*})$ up to the pointwise window edge $E_{\mathrm{min}}(\boldsymbol{q}) - \delta_w$ of Sect. 13.6. From it follow the rate $\mathrm{d}\mathcal{R}/\mathrm{d}\omega$ and the reach of Sect. 13.3, for the heavy and the light mediator. On the truncation label $\mathfrak{T}$ (Sect. 8.5) the calculation sits at $n_B = 0$, $n_g = 2$ (one density vertex and one electron–phonon vertex, all four time orderings), $n_T = 0$, lattice regime $\mathcal{K}$ = one phonon, and $\mathcal{X}$ = RPA with local fields for the external screening. The free-pair amplitude has no weight below $E_{\mathrm{g}} + \omega_{\mathrm{TA}}$; the replica shell below it (0.3332 to 0.3374 MeV in mass) is the exciton envelope times the same amplitude, matched at $E_{\mathrm{g}} + \omega_{\mathrm{TA}}$.

The ground state and the phonons are the shared harmonic base of Sect. 30.1, with its $12^3$ $\boldsymbol{k}$ grid and $8^3$ wavevector grid. On a $4^3$ grid the interpolated transverse-acoustic frequency at 0.85 X comes out above its value at X, an artifact of up to 1.5 meV on exactly the branch that sets the window. The self-consistent potential changes are kept for every wavevector, since the vertices are built from them, and the long-range quadrupolar part of $g$ is split off before

interpolation. An 18-function Wannier manifold ($sp^3$ on each atom plus $d$-like functions) is disentangled from about 60 bands on the full $12^3$ grid without symmetry. The conduction states to 10 eV above the gap then sit inside the frozen window, and the size of the manifold is tested by repeating the amplitude with 26 functions.

Since the band sum multiplies matrices in band space, $g$, the density vertex $\rho_{mn}(\boldsymbol{k}, \boldsymbol{q})$ and the energies have to be rotated with the same $U(\boldsymbol{k})$ (Appendix A.5); with a mismatch every band energy is still correct but the phases of the sum are wrong. The acoustic sum rule is imposed on the dynamical matrix, never by setting $g$ to zero as $\boldsymbol{Q} \to 0$, because the part of $g$ linear in $\boldsymbol{Q}$ is physical and is exactly what the soft acoustic limit of Sect. 13.6 tests.

The density vertex is computed for each target $\boldsymbol{q}$ from the plane-wave coefficients of one non-self-consistent run at the shifted points $\boldsymbol{k} + \boldsymbol{q}$, with $\boldsymbol{q}$ kept unreduced (Appendix A.3); a vertex at a reduced $\boldsymbol{q}$ belongs to a different Umklapp class. With 21 values of $|\boldsymbol{q}|$ and 30 directions there are 630 targets in the irreducible wedge, about 1800 once equivalent pockets are unfolded, and each needs the full $12^3$ grid without symmetry, about $10^6$ diagonalizations in all. The amplitude is then the four-path sum with an unrestricted band sum: explicit inside the Wannier manifold, and above it the projected Sternheimer solution of Eq. (25.22) contracted with $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, one linear solve per external state and perturbation. That remainder matters most on the Bethe ridge, where $q^2/2m_e$ approaches the energy deposited. Below $|\boldsymbol{q}| \approx 0.5\,\alpha m_e \approx 1.9\,\mathrm{keV}$ it is a correction to the Wannier sum; on the upper bins, 1.9 to 2.5 keV, it is carried in full. The external screening is the column $(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}(\boldsymbol{q}, \omega)$ of an RPA calculation with local fields (Sect. 26.1). The pair channel of Sect. 13.4 adds $W^{\mathrm{scr}}(\boldsymbol{q} + \boldsymbol{Q}, \omega_f)$ at the pair momentum $\boldsymbol{q} + \boldsymbol{Q}$, which kinematics pins near 0.85 X, so one dielectric matrix serves every bin. The two corrections are reported as the ratios $\delta_{\mathrm{LF}}$ and $\delta_{\mathrm{pair}}$ of Sect. 13.4. The sums over pocket pairs run on fine grids of $60^3$, refined to $120^3$, with tetrahedra and no broadening.

Quantum ESPRESSO [1], Wannier90 [2], EPW [3] and BerkeleyGW [4] carry most of this, and a long-wave DFPT code the quadrupoles. The density vertex in the Wannier gauge and the second-order amplitude with its Sternheimer remainder are in none of them; they are new code, a few hundred lines on top of EPW. Table 31.1 collects parameters and costs. The plane-wave stages total about 350 to 450 core-hours, dominated by the density vertices; the amplitude, at the $10^{16}$ operations that Sect. 27.2 counts, adds a quarter of an hour to a few hours on one accelerator.

### 31.1.2 Physical controls

A bin converged to 5 % under the refinements of Table 31.1 need not be correct, and each control below tests the physics against something the calculation was not fitted to.

The optical limit tests the vertices against data. At $|\boldsymbol{q}| = 10^{-3}\,\text{Å}^{-1}$ the density vertex reduces to its dipole form (Sect. 3.2), and $L^{\mathrm{PA}}$ has to reproduce the phonon-assisted $\varepsilon_2^{\mathrm{PA}}$ that EPW computes from the same vertices. After the gap shift it also has to reproduce the measured absorption edge at the temperature of each measurement: silicon at 20 K, where the transverse-acoustic and transverse-optical replicas are resolved [6], germanium against its edge data at 77 to 80 K, and GaAs at the temperature of its data. The comparison is made branch by branch, with $\varepsilon_2^{\mathrm{PA}}$ resolved by phonon branch, since the transfer of a measured edge to millikelvin needs the branch weights and a measured spectrum gives only their sum (Sect. 13.5). Switching $g$ off tests the labeling of final states: the phonon-assisted rate then vanishes identically, and any residue is zero-phonon weight leaking into the window through a mislabeled state.

A frozen phonon tests the adiabatic form of the amplitude derived in Sect. 13.2. At $\boldsymbol{Q} = \mathrm{X}$ an eight-atom cell holds the phonon exactly; displacing the atoms by $\pm 0.005$ Å along a mode

Table 31.1: Parameters and costs of the phonon-assisted loss function of silicon. The cutoff refers to the SG15 norm-conserving pseudopotential [5] with four valence electrons. Fine grids are those of the pockets within 0.1 eV of the band extrema, quoted as their equivalent uniform density. Costs are in core-hours unless stated; "base" is the shared DFPT run of Sect. 30.1.

| stage | grids and parameters | acceptance test | cost |
|---|---|---|---|
| ground state, phonons | PBE, $a = 5.431$ Å, 60 Ry, $12^3$ $\boldsymbol{k}$, $8^3$ $\boldsymbol{Q}$ (29 irreducible) | neutron data per branch (Sect. 30.1) | base, $\approx 100$ |
| long-range $g$ | dipoles (zero in Si), quadrupoles | small-$\boldsymbol{Q}$ angular dependence | minutes |
| Wannier gauge | 18 functions ($sp^3$, $d$); window to 40 eV | bands $\leq 10$ meV to 10 eV above the gap | 3–5 |
| density vertex | 630 targets, $12^3$ without symmetry | direct vs Wannier at 5 off-grid $\boldsymbol{k}$, $\leq 2\,\%$ | 200–300 |
| vertex $g$ | interpolated | direct DFPT at 3 off-grid $\boldsymbol{Q}$, $\leq 5\,\%$ | 10 |
| amplitude | $60^3 \rightarrow 120^3$, tetrahedra; Wannier sum plus Sternheimer remainder ($10^3$ solves per target) | 5 % per bin in $L^{\mathrm{PA}}$ | accelerator hours (Sect. 27.2) |
| screening | RPA, $(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}(\boldsymbol{q},\omega)$; $W^{\mathrm{scr}}$ at $\boldsymbol{q}+\boldsymbol{Q}$ | $\delta_{\mathrm{LF}}$, $\delta_{\mathrm{pair}}$ reported | tens |

and differencing the zero-phonon transition densities gives the adiabatic amplitude directly, which agrees with the second-order sum up to the expected $\omega_{\boldsymbol{Q}\nu}/\delta_{\mathrm{off}}$. The soft acoustic limit of Sect. 13.6, checked at $|\boldsymbol{Q}| = 10^{-2}$, $10^{-3}$ and $10^{-4}$ of $2\pi/a$ in three directions, fails at once for a vertex whose small-$\boldsymbol{Q}$ part was zeroed. Symmetry gives two exact zeros: on the valley axis the longitudinal-acoustic path of the class without Umklapp vanishes, and so does the longitudinal-optical path of the $(\bar{2}\,0\,0)$ class (Sect. 13.9); both come out at the level of numerical noise.

Between 0.333 and about 0.35 MeV the rate is a power of the distance from the onset (Sect. 13.3), so there the exponent is the meaningful output.

### 31.1.3 Zero-phonon rates on common bins

Sect. 13.9 places the end of the phonon-assisted lead at about 0.42 to 0.43 MeV, with a two-channel band to about 0.51 MeV. That is a ratio of two rates, settled only when both are computed on the same bins, so the output grid extends past the window into three regions. The first zero-phonon edge, 0.405 to 0.408 MeV with a best value of 0.407 MeV, is a joint minimum over the momentum transfer and the hole's crystal momentum and lies off the valley axis. Its state belongs to the symmetry-suppressed (glide) class, so its vertex is small and it does not end the lead. Around this mirror-plane optimum the bins are cones: $|\boldsymbol{q}| = 1.2$ to 1.9 keV, tilts $\vartheta \leq 0.4$ rad from the axis within the {001} and {011} planes and up to 0.15 rad out of plane, and $\omega$ within 0.3 eV of $E_{\mathrm{min}}(\boldsymbol{q})$. The second region is the cone of the second heavy hole on the axis, the mirror-even band that opens from 0.415 to 0.420 MeV. The third is the X-face edge of the allowed $(\bar{2}\,0\,0)$ class, where the zero-phonon line of allowed size opens near 0.50 MeV. Above $E_{\mathrm{min}}(\boldsymbol{q})$ the phonon-assisted value follows the convention of Sect. 13.6; the shell just below the edge is reported separately, and the bins contain the joint optimum at 0.407 MeV, so that the edge is the one the rate sees.

The zero-phonon side costs under one core-hour. It is a Wannier–Mott continuum (Sect. 4.9) with the anisotropic reduced-mass tensor of the pair at each bin, from the Hessians of the

interpolated bands at the pair's two crystal momenta. It uses a static $\varepsilon_\infty$ and density vertices computed directly from the plane-wave states of both top heavy-hole bands. The hole is placed where the pair is lowest at that $\boldsymbol{q}$; at the off-axis optimum it sits about 0.06 eV below the valence maximum, and placing it at $\Gamma$ misses the optimum. At the class extremum the allowed $(\bar{2}\,0\,0)$ vertex is $\sum_v |\rho_{cv}|^2 = 0.150$, the size of an ordinary allowed vertex (Sect. 13.9); the comparison needs it on every bin, with spin–orbit coupling in the bands, which shifts the first-hole bins by about 15 meV. The vertices come from matching Miller indices on a three-point stencil around each bin. They are converged when halving the stencil step from 0.02 to $0.01 \times 2\pi/a$ changes $|\rho|^2$ by less than 2 %. Halving the angular step from 0.01 to 0.005 rad has to change the rate by less than 1 %, and the ratio of the two rates per bin has to be stable to 5 % when the number of bins grows by half. Off the axis the band index no longer orders the states by mirror parity, and each state is assigned by its parity; on the axis the degenerate pair is summed.

### 31.1.4 Germanium and gallium arsenide

Germanium cannot follow silicon, because PBE gets its band order wrong. It puts the conduction state at $\Gamma$ 0.047 eV above the valence maximum and the L valley above it, the reverse of the measured order, in which the indirect gap at L (0.744 eV) lies below the direct gap at $\Gamma$ (0.89 eV). A scissor applied to that order gives a window that is wrong by the inversion of $\Gamma$ and L. Any germanium band structure is therefore checked before it is used: L below $\Gamma$, both gaps within 0.1 eV of the measured ones, and the band masses at both valleys against measurement, the transverse mass at L within $0.010\,m_e$ of $m_\perp^* = 0.0815\,m_e$ (Table 3.1) and the mass at $\Gamma$ within $0.005\,m_e$ of $0.038\,m_e$.

HSE06 with spin–orbit coupling at the standard exchange fraction of one quarter fails this check. At the measured lattice constant it puts $\Gamma$ at 0.64 to 0.69 eV and L at 0.79 to 0.83 eV above the valence maximum, so the gap at L passes while the order and the direct gap fail, with an inversion of 0.09 to 0.15 eV. Above a fraction of 0.28 the order is restored, but at 0.3 the direct and indirect gaps are still 34 % and 17 % off [7, Fig. 1(a)]. A correction that shifts only the 4s-like orbitals [7], with its single parameter fitted to both gaps, passes the energies, L at 0.749 eV below $\Gamma$ at 0.918 eV. It fails the masses: the transverse mass at L comes out $0.101\,m_e$ and the mass at $\Gamma$ $0.053\,m_e$, 24 % and 39 % heavy, so the valley energies are right and the curvatures are not. The next route is a quasiparticle *GW* correction (Sect. 23.4) on top of the hybrid or PBE bands, subject to the same check. Until bands pass, the zero-phonon edge of germanium stays the estimate of Sect. 13.1, 0.32 MeV on the L axis inside a bracket of 0.212 to 0.420 MeV.

On bands that pass, sixteen spinor $sp^3$ Wannier functions are extracted with the twenty 3d bands excluded and a frozen window to 3 eV above the valence maximum, and the interpolated edges and masses are compared with direct calculations at off-grid points. The zero-phonon edge then follows by the joint minimization over $(\boldsymbol{q}, \boldsymbol{k})$ for the L-valley class and its Umklapp copies. Whether that optimum leaves the axis is not known: the germanium heavy hole is less warped than that of silicon, and nothing transfers from the silicon result. The mirror rule of Sect. 13.9 becomes one for the diagonal mirror planes, which carry no glide phase; whether the first state is allowed is checked at the true optimum.

The germanium phonons are the shared base, with an $8^3$ $\boldsymbol{k}$ grid and a $6^3$ wavevector grid. On a $4^3$ wavevector grid the transverse-acoustic frequencies are 13.5 % (X) and 11.6 % (L) below neutron data at 80 K [8], while the optical and longitudinal branches agree to 2.3 %. Either the nearly closed PBE gap overscreens the shear modes or the $8^3$ $\boldsymbol{k}$ grid is too coarse for a 47 meV gap. One DFPT run at X on a $12^3$ grid, about three core-hours, separates the two, and $\omega_{\mathrm{TA}}(\boldsymbol{Q}^*)$ has to be stable to 0.3 meV between the $4^3$ and $6^3$ grids. If the deficit

survives, the neutron-scaled branch is used for the window; the phonon-assisted edge moves by at most 0.0003 MeV either way. The germanium rate then follows the silicon chain with spin–orbit coupling and the 3d shell in the valence (14 electrons per atom): 36 spinor Wannier functions, the sixteen $sp^3$ ones plus twenty for the 3d shell, and 120 spinor bands. Spinors double both the length of each state and the band count, which multiplies each plane-wave stage by four to eight; for germanium's smaller DFPT grid and targets the chain comes to 400 to 800 core-hours, or hours on one accelerator node.

Gallium arsenide is the direct-gap control. There the zero-phonon pair at small $q$ is out of reach for any mass: the pair at $\Gamma$ needs $E_{\mathrm{dir}} + q^2/2M_x \le q v_{\max}$, and the right side minus the recoil is at most $M_x v_{\max}^2/2 \approx 1.04\,\mathrm{eV}$ for a pair mass $M_x \approx 0.58\,m_e$, below the direct gap of 1.52 eV. A phonon can absorb the momentum instead, and the phonon-assisted rate leads until the L-valley zero-phonon state opens, from about 0.43 MeV to between 0.52 and 0.56 MeV in a parabolic estimate with the L valley at 1.815 eV (Sects. 13.1 and 13.9). The chain, with the Fröhlich vertex (Eq. (27.16)), has to reproduce that band and give a small correction above the zero-phonon threshold; without the long-range part the vertex diverges at small $\boldsymbol{Q}$. The ground state is the shared base at $a = 5.653$ Å with the gallium 3d shell in the valence. PBE puts the direct gap at about 0.5 eV, so the bands come from HSE06 or from a $\boldsymbol{k}$-dependent scissor fitted to the low-temperature conduction minima at $\Gamma$ (1.519 eV), L (1.815 eV) and X (1.981 eV) [9], with spin–orbit coupling for the holes. The Fröhlich vertex takes $\varepsilon_\infty$ and $Z^*$ from measurement through Eq. (30.1), since the PBE $\varepsilon_\infty$ of GaAs is tens of percent too large. The window costs about 8 core-hours with the scissor and 40 to 80 with the hybrid; the full rate costs about as much as silicon's.

For the other indirect-gap candidates the window comes before any rate: $E_{\min}(\boldsymbol{q})$ on the joint $(\boldsymbol{q}, \boldsymbol{k})$ space, the band masses, $\omega_{\mathrm{TA}}(\boldsymbol{Q}^*)$ and $g$ at $\boldsymbol{Q}^*$ from one DFPT run. PBE with a scissor serves wherever PBE gets the band order right, checked against measured gaps at the high-symmetry points; narrow-gap and d-electron crystals need bands that pass the germanium check of order, gaps and masses. The kinematics are checked in every case by the limit $|\boldsymbol{q}| \to |\boldsymbol{K}_{\mathrm{pk}}|$, the crystal momentum of the conduction pocket, where $E_{\min}$ returns the gap.

### 31.1.5 Fano widths and detunings in GaAs

Near the edge of the off-shell window of Sect. 13.6, a zero-phonon final state of energy $\tilde{E}$ lies inside the phonon-assisted continuum and interferes with it, the Fano term of Sect. 8.6. The term vanishes in crystals invariant under inversion combined with time reversal with an even electron number, such as silicon and germanium, at the level of one-body self-energies, but not in GaAs. Level shifts from the electron–hole vertex can split a degenerate zero-phonon manifold in a basis that is not invariant, and then only the sum of the Fano terms over the branches vanishes. The Fano term is perturbative only where the width $\Gamma$ of the resonance is small compared with its detuning $\delta_{\mathrm{F}}$, the distance of $\tilde{E}$ to the nearest feature of the continuum. With the Fano coefficient $c_\nu^F$ of each branch $\nu$, the branch probability of the first emission is $(\Gamma^{(\nu)}/\Gamma^{\mathrm{tot}})(1 + \bar{c}^F - c_\nu^F)$, with $\bar{c}^F$ the width-weighted mean. In GaAs $|c_\nu^F| \lesssim 0.3\Gamma/\delta_{\mathrm{F}}$ to first order, which bounds the change of the pair-number distribution by $0.3\Gamma/\delta_{\mathrm{F}}$ and keeps the probabilities positive for $\Gamma/\delta_{\mathrm{F}} \le 0.5$. What Sect. 13.6 needs is therefore a map of $\Gamma/\delta_{\mathrm{F}}$ over $(q, \omega)$.

The width of the pair state is the sum of the electron and hole widths $\Gamma_n(\boldsymbol{k})$. These are on-shell Fan–Migdal widths,

$$\Gamma_n(\boldsymbol{k}) = -2\,\mathrm{Im}\,\Sigma^{\mathrm{FM}}_{n\boldsymbol{k}}(\varepsilon_{n\boldsymbol{k}}),$$

computed at 10 K (Sect. 6.3), at $n_g = 2$ with a statically screened vertex. The detuning is the smallest of the distances to the renormalized edge $E^{\mathrm{ren}}_{\min}(q)$, to the nearest intervalley threshold

of the electron (L, X) or the split-off threshold of the hole, and the local scale of the joint density of states. The coefficient $c_\nu^F$ is the non-resonant remainder of the same second-order amplitude, evaluated on the final shell. Estimates give a Fröhlich LO width of a few meV from just above the conduction minimum to the Γ–L onset (about 3 meV at the onset, 0.3 eV above the minimum, with $\alpha_F = 0.068$ and $\omega_{\mathrm{LO}} = 36\,\mathrm{meV}$), and 20 to 60 meV above the onset, so $\Gamma/\delta_{\mathrm{F}}$ is 0.01 to 0.2 away from thresholds and diverges at each onset. The map excludes shells of $\pm 2\Gamma$ around the Γ–L onset and the split-off onset (0.34 eV below the valence maximum), and the shell within $3\Gamma$ of the window edge; inside them only totals from the unexpanded Dyson form are quoted.

Because the split-off band sets the hole widths, the electrons need fully relativistic pseudopotentials, at 90 Ry because the spinor calculation with the Ga 3d shell converges more slowly than the scalar base. Phonons, Born charges and $\varepsilon_\infty$ come from the shared base on a $6^3$ grid. The run uses an $8^3$ $\boldsymbol{k}$ grid, 16 spinor $sp^3$ Wannier functions with the 3d bands excluded, the polar vertex, and self-energies on a $100^3$ fine $\boldsymbol{Q}$ grid for the 13 824 $\boldsymbol{k}$ points of the final shell. The widths converge to 5 % from $80^3$ to $100^3$ and are stable as the smearing goes from 10 to 2.5 meV; $c_\nu^F$ converges to 10 % when 20 bands are added to the intermediate sum. Silicon, with eight $sp^3$ functions and a $60^3$ grid, has to return $c_\nu^F = 0$ to numerical precision, which tests the chain against the symmetry theorem of Sect. 8.6. The self-energies cost about $10^4$ core-hours (30 to 60 accelerator-hours), silicon a tenth of this. The map fixes the $(q, \omega)$ region in which Sect. 13.6 quotes the Fano term for GaAs.

### 31.1.6 The phonon-assisted kinetic-energy row

A scalar coupled to the electron's mass is absorbed below the direct gap of silicon only with a phonon, and its rate is the structure factor $S^{\mathrm{PA}}_{\mathcal{T}\mathcal{T}}(0, m_\phi)$ of Eq. (13.49), for a point group without a polar vector (Sects. 10.7.4 and 13.8). No measurement supplies it, since its amplitude Eq. (13.48) contains no velocity vertex and is not proportional to $\varepsilon_2^{\mathrm{PA}}$. The chain above computes it with one change: the density vertex at $\boldsymbol{q}$ is replaced by the kinetic vertex $\langle m|p^2/2m_e|n\rangle$ at $\boldsymbol{q} = 0$, and the energy is $\omega = m_\phi$ from the indirect edge to the direct gap. The Wannier gauge, the vertices $g$, the Sternheimer remainder above the manifold and the fine pocket grids are those of the density row. The kinetic matrix element converges like the total kinetic energy and is weighted toward the cores, so it is taken from PAW or all-electron states, at a cutoff of at least 80 Ry for the norm-conserving comparison. Since $\langle c|p^2/2m_e|v\rangle = -\langle c|V|v\rangle$ between eigenstates of different energy, Eq. (13.48) can be evaluated twice, once with the kinetic vertex and once with the matrix elements of the crystal potential, and the two must agree to the 5 % per bin of Table 31.1; a disagreement points to the core region or to an incomplete band sum. The same chain with the density vertex at $\boldsymbol{q} \to 0$ reproduces $\varepsilon_2^{\mathrm{PA}}$ and the measured edge, which tests everything except the kinetic vertex. Exciton final states near the edge enter as for the density row. With one momentum in place of the 630 targets of the density row, the row costs a small fraction of that row, and it enters Sects. 10.7.4 and 13.8 and the scalar row of Sect. 21.6.

### 31.1.7 The $f$-sum check with its parts separated

The sum rule Eq. (26.13) tests whether the intermediate-state sum is complete. With norm-conserving pseudopotentials the band sum $\sum_m (\varepsilon_{m,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}})|\rho_{mv}(\boldsymbol{k}, \boldsymbol{q})|^2$ at $\boldsymbol{k} = \Gamma$ reaches 0.81 of $q^2/2m_e$ with 16 bands and 0.88 with 100 bands at $|\boldsymbol{q}| = 0.1\,\text{Å}^{-1}$, and 0.78 and 0.89 at $0.5\,\text{Å}^{-1}$, still rising slowly (Sect. 13.9). That shortfall mixes the bands left out with the nonlocal term $\delta_{\mathrm{NL}}$, which a comparison with $q^2/2m_e$ alone ignores, and the two can be separated exactly.

The truncation part needs no more bands. Let $P_N^{\perp}$ project out the lowest $N$ eigenstates of $H(\boldsymbol{k}+\boldsymbol{q})$ and let $|\phi\rangle = P_N^{\perp}|u_{v\boldsymbol{k}}\rangle$. Since $\rho_{mv}(\boldsymbol{k},\boldsymbol{q}) = \langle u_{m,\boldsymbol{k}+\boldsymbol{q}}|u_{v\boldsymbol{k}}\rangle$ and $P_N^{\perp}$ commutes with $H(\boldsymbol{k}+\boldsymbol{q})$,

$$\sum_{m>N} (\varepsilon_{m,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}})\,|\rho_{mv}(\boldsymbol{k},\boldsymbol{q})|^2 = \langle\phi|H(\boldsymbol{k}+\boldsymbol{q}) - \varepsilon_{v\boldsymbol{k}}|\phi\rangle, \tag{31.1}$$

one application of the Hamiltonian to a projected vector, the cost of a single eigensolver iteration. The nonlocal part is a ground-state expectation value, $\delta_{\mathrm{NL}} = \langle u_{v\boldsymbol{k}}|V_{\mathrm{NL}}(\boldsymbol{k}+\boldsymbol{q}) - V_{\mathrm{NL}}(\boldsymbol{k})|u_{v\boldsymbol{k}}\rangle$ minus its part linear in $\boldsymbol{q}$, from the same projectors as the Hamiltonian. The truncated sum plus Eq. (31.1) then equals $\boldsymbol{q}\cdot\boldsymbol{v}_v^{\mathrm{g}} + q^2/2m_e + \delta_{\mathrm{NL}}$ to the precision of the eigenvectors, a test of the code rather than of the physics. The ratio $\delta_{\mathrm{NL}}/(q^2/2m_e)$ against $q$ says how much of the shortfall left at 100 bands, 0.12 at 0.1 $\text{Å}^{-1}$ and 0.11 at 0.5 $\text{Å}^{-1}$, is the nonlocal term rather than missing bands. The remainder also measures the intermediate-state weight above the Wannier manifold that the Sternheimer remainder supplies: with the nonlocal term, 60 bands should exhaust at least 95 % of the sum. The check runs at Γ and five off-grid points, for the three cubic directions, in minutes, and Worked example 31.1 works it through at one momentum.

**Worked example 31.1: The $f$-sum check of a phonon-assisted calculation of silicon**

Take $|\boldsymbol{q}| = 1\,\mathrm{keV} = 0.507\,\text{Å}^{-1}$ along [100], where the off-shell window of Sect. 13.6 is open and $q^2/2m_e = 0.9785\,\mathrm{eV}$. The phonon-assisted amplitude of Sect. 13.2 sums over intermediate bands $m$ products of a density vertex $\rho_{mv}(\boldsymbol{k},\boldsymbol{q})$ and an electron–phonon vertex, each pair over an energy denominator. At 0.5 $\text{Å}^{-1}$ the norm-conserving sum at Γ reaches 0.89 of $q^2/2m_e$ with 100 bands. The check has three parts: the remainder, the nonlocal term and the transfer rule.

*The remainder.* The sum over the bands above $N$ is one expectation value, Eq. (31.1), so the truncated sum plus the remainder must equal the right side of Eq. (26.13) to the precision of the eigenvectors. A local model shows what passing looks like. Take the empirical pseudopotential of Sect. 3.3, which has no nonlocal term ($\delta_{\mathrm{NL}} = 0$), with 531 plane waves, and sum over the four valence bands at Γ. The lowest 8, 16, 30, 60 and 100 bands then give 0.1514, 0.9265, 0.9939, 0.9992 and 0.9996 of $4q^2/2m_e = 3.9139\,\mathrm{eV}$, and the truncated sum plus the remainder equals it to $9\times10^{-14}$. At $\boldsymbol{k} = (0.10, 0.20, 0.30)\,2\pi/a$ the group-velocity terms lower the target to 3.1135 eV, and the identity closes to $5\times10^{-14}$. A code that misses the identity by more than its eigenvector precision has an error in its vertices or its Hamiltonian, whatever its physics. Passing it says nothing about convergence in the basis, since the identity is exact in any finite basis.

*The nonlocal term.* Once the remainder is added, what is left of the shortfall against $q^2/2m_e$ is the ground-state expectation value $\delta_{\mathrm{NL}}$. In the local model 100 bands leave $4\times10^{-4}$ of the sum, so a shortfall of 0.11 at the same $N$ is large on the scale that a smooth local potential sets. The closure says how it divides between bands left out and the nonlocal term. An all-electron calculation, or a PAW calculation whose density vertices are reconstructed inside the augmentation spheres, satisfies the local $f$-sum of the true electrons and checks the split. The remainder also bounds the part of the amplitude carried by the bands above $N$. With $\Delta E_m = \varepsilon_{m,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{v\boldsymbol{k}} > 0$ there and $b_m$ the electron–phonon vertex divided by its energy denominator, the Cauchy–Schwarz inequality gives

$$\Big|\sum_{m>N} \rho_{mv}^{*}\, b_m\Big|^2 \le \Big[\sum_{m>N} \Delta E_m |\rho_{mv}|^2\Big]\Big[\sum_{m>N} \frac{|b_m|^2}{\Delta E_m}\Big],$$

whose first factor is the remainder. A small remainder therefore guarantees a small truncation error once the second factor is bounded. The converse does not hold: a large remainder leaves the truncation error unbounded, but small vertices or cancellations can still keep it small.
*The transfer rule.* The total $f$-sum cannot see the sideband. The first moment of $S^{\mathrm{PA}}$ is of relative order $\theta_{\mathrm{ep}} = 0.010$ to 0.016 of the $f$-sum (Sects. 8.5 and 13.3), while a converged electronic table holds the $f$-sum to 2 % (Sect. 31.3), 1.25 to 2 times $\theta_{\mathrm{ep}}$. A calculation can pass the total test at that tolerance with its sideband missing or doubled. What sees the sideband is Eq. (13.36). The first moment of $S^{\mathrm{PA}}$, integrated over all $\omega$ at the test momentum and not only over the window, must equal the first moment that the zero-phonon response loses through its self-energies and vertex corrections, computed from the same vertices. The rule holds up to terms of order $g^4$ and the share of the electrons that follow the nuclei, $Z_{\mathrm{val}} m_e / M_{\mathrm{cell}} = 7.8 \times 10^{-5}$ of the $f$-sum with $Z_{\mathrm{val}} = 8$ valence electrons per cell. If the order-$g^4$ term is estimated as $\theta_{\mathrm{ep}}^2$ of the $f$-sum, the two neglected pieces add to $\theta_{\mathrm{ep}}^2 + Z_{\mathrm{val}} m_e / M_{\mathrm{cell}}$, and relative to the sideband's own weight $\theta_{\mathrm{ep}}$ the two sides must agree to about 2 %: 1.8 % at $\theta_{\mathrm{ep}} = 0.010$ and 2.1 % at 0.016. A calculation that passes all three parts has a complete band sum, a known nonlocal term and a sideband whose weight is consistent with the renormalization of the lines it accompanies.

The loss function and the rate replace the factor-ten estimate in the table of the window of Sect. 13.9, and the reach replaces its curve.

## 31.2 Excitons and core states

In silicon an electron and a hole attract each other strongly enough to bind 14.7 meV below the gap [10], and the same attraction enhances the continuum just above it. Every near-threshold rate that a random-phase calculation gives misses both effects. The finite-momentum Bethe–Salpeter equation (BSE) of Sects. 4.9 and 26.1 contains them, and Sect. 10.5 shows how its output enters a rate. We need the BSE with its coupling block, and with it the exciton series, the phonon replicas of Sect. 13.8, the exciton lines of the Migdal effect (Sect. 12.3) and the self-trapped exciton of an alkali halide; the semicore response and the spin–orbit mixing at the band edges (Sect. 10.6) complete the set.

### 31.2.1 Finite-momentum BSE with coupling

What a rate near the gap needs is the loss function $L(\boldsymbol{q}, \omega)$ of Sect. 10.2 on bins of $(|\boldsymbol{q}|, \hat{\boldsymbol{q}}, \omega)$ for $|\boldsymbol{q}| \le 10\,\mathrm{keV}$ and $\omega \le E_{\mathrm{g}} + 10\,\mathrm{eV}$, with the energies and weights of the bound lines listed separately. Outside that box the RPA with local fields serves, and the two are matched on a strip where both are computed; their difference is the many-body theory uncertainty of Sect. 8.8. On the truncation label $\mathfrak{T}$ (Sect. 8.5) the calculation sits at $n_B = 0$ and $n_g = 0$, with $\mathcal{X} =$ BSE on quasiparticle bands.

At finite $q$ the coupling block between pairs and antipairs carries the long-range exchange $v_{\mathrm{C}}(\boldsymbol{q} + \boldsymbol{G})$, which turns the proper response into the screened loss function (Sect. 4.9). The largest error comes from a resonant-only response, which keeps one time ordering: in silicon at $|\boldsymbol{q}| = 1.39\,\mathrm{keV}$ and $\omega = 1.3\,\mathrm{eV}$, in the momentum range of the phonon-assisted window, it overstates the loss function by a factor of 2.1 (Eq. (10.22)). A retarded Tamm–Dancoff calculation has no such error, since it keeps both poles $\pm E_\lambda$ and takes $\varepsilon_1$ by Kramers–Kronig; what it drops is the coupling block itself, an error set by the ratio of the local-field exchange to the pair energy, about 10 % on the proper $\varepsilon_2$ (Sect. 10.5). The error is avoided by the full

coupling, or by keeping the Tamm–Dancoff BSE for the proper $\varepsilon_2$ (line shapes and bound lines) and taking the external screening from the RPA with local fields, with the head removed from the BSE exchange so that it is not counted twice.

The bands come from Quantum ESPRESSO with PBE and $G_0W_0$ quasiparticle energies (Sect. 23.4), and a code with a finite-momentum BSE, such as BerkeleyGW [4], solves the pair problem on them. For germanium the PBE starting point has Γ below L and a nearly closed gap, so the quasiparticle bands must first pass the check of Sect. 31.1; the germanium BSE, and the exciton dispersion along Γ–L below, wait for bands that pass. The pair basis has 4 valence bands (for germanium plus the 3d shell) and 12 conduction bands. On a coarse $8^3$ grid the kernel holds $8^3 \times 4 \times 12 = 24{,}576$ pairs, about 10 GB per reduced momentum, and it is interpolated to a fine $24^3$ grid of $6.6 \times 10^5$ pairs, whose dense kernel would hold $4.4 \times 10^{11}$ complex numbers, 7 TB per momentum. That kernel is never stored: a Haydock (Lanczos) solver of about 200 iterations applies it to a vector on the fly. The momenta are the reduced points of the $24^3$ set with reciprocal-lattice shells up to 10 keV, about $10^3$ points after the cubic symmetry, each of order $10^{13}$ to $10^{14}$ floating-point operations, so the whole run takes about $10^3$ node-hours, a few times $10^4$ core-hours.

Even the fine grid cannot resolve the silicon exciton. Its radius is $a_x = 4.2\,\mathrm{nm}$ (Sect. 10.5), and a hydrogenic envelope needs a $\boldsymbol{k}$ spacing below about $1/(3a_x) \approx 0.008\,\text{Å}^{-1}$, against $0.08\,\text{Å}^{-1}$ on the $24^3$ grid. The bound lines are therefore added from the effective-mass model of Sect. 10.5, with ab initio masses, $\varepsilon_\infty = 11.7$ and the vertices and their gradients at the band extrema, and the BSE supplies the continuum and its Sommerfeld enhancement. The pair basis is evaluated around the off-axis pair optimum of Sect. 13.9, where the hole sits off Γ, not only at the band extrema.

The coupling block restores the $f$-sum rule of Sect. 4.3, which the scissor and the band cutoff also strain, so with the coupling on, the sum rule of the full loss function holds to better than 1 %. At $q < 3\,\alpha m_e$ the loss function is compared with inelastic x-ray and electron energy-loss data for silicon, and the bound-line model has to reproduce the free-exciton binding of silicon, 14.7 meV. The structure factor is reported with the full coupling and in the retarded Tamm–Dancoff form at $|\boldsymbol{q}| = 0.5$, 0.7, 1.0 and 1.4 keV; the Tamm–Dancoff route passes if it agrees with the full one to 15 %.

### 31.2.2 Exciton dispersion and the hydrogenic series

A bound exciton of total momentum $\boldsymbol{Q}$ sits below the lowest pair at that momentum by its binding energy,

$$E_x(\boldsymbol{Q}) = E_{\min}(\boldsymbol{Q}) - E_x^b[\mu(\boldsymbol{Q}), \varepsilon_\infty], \tag{31.2}$$

where the reduced-mass tensor $\mu(\boldsymbol{Q})^{-1} = \nabla^2\varepsilon_c(\boldsymbol{k}_v + \boldsymbol{Q}) - \nabla^2\varepsilon_v(\boldsymbol{k}_v)$ is evaluated at the crystal momenta $\boldsymbol{k}_v(\boldsymbol{Q})$ and $\boldsymbol{k}_v(\boldsymbol{Q}) + \boldsymbol{Q}$ of the lowest pair. Since the hole is not at Γ near the zero-phonon optimum (Sect. 13.9), the mass changes along the dispersion. The Hessians of the Wannier-interpolated bands are taken by finite differences with a step of $0.005 \times 2\pi/a$, on the momentum grid of the zero-phonon edge along Γ–X on both sides of X and across the {001} planes (along Γ–L and near Γ for germanium, on bands that pass the check). A variational anisotropic hydrogen problem in a basis of 30 anisotropic Gaussians, with the measured $\varepsilon_\infty = 11.7$, then gives $E_x(\boldsymbol{Q})$ with the weights of the 1s and 2p states, converged when $E_x^b$ is stable to 0.1 meV in the basis size. Near Γ the heavy and light holes are degenerate, so scalar masses fail and the hole needs the six-band $\boldsymbol{k} \cdot \boldsymbol{p}$ form with spin–orbit coupling (split-off energy 44 meV in silicon). The calculation takes minutes; it enters Sect. 10.5, and its bound lines feed the BSE above.

### 31.2.3 Exciton–phonon replicas

The replicas of Sect. 13.8, lines at $E_x + \omega_\nu(\boldsymbol{Q}^*)$ below the free-pair edge, need the exciton–phonon vertex: the electron and hole vertices $g$ weighted by the envelopes of the initial and final excitons. Their strength per phonon branch and their position are wanted at pair momenta near 0.85 X and, following Sect. 13.9, at the off-axis momenta where the hole sits off Γ. A finite-momentum BSE at that pocket set gives them, with the vertices $g$ from EPW in the same Wannier gauge $U(\boldsymbol{k})$ as the exciton eigenvectors; about $10^3$ $\boldsymbol{k}$ points per pocket resolve an exciton of radius 4.2 nm, at about $10^4$ core-hours. A cheaper route, a Wannier–Mott envelope multiplying the free-pair amplitude of Sect. 31.1, is the cross-check, and both must reproduce the replica edge at 0.3332 MeV (Sect. 13.9).

Whether the replica formula of Sect. 13.8 is complete cannot be tested in a crystal. The band sum of Sect. 13.2 has one pair in every intermediate state; states with two pairs enter only through the interaction beyond the Tamm–Dancoff space of one pair (Sect. 13.8), and whether the two-pair sector reduces to the one-pair terms when the electron and hole interact is proven only without interaction. A model that can be solved exactly decides it. Take a one-dimensional two-band chain of 8 to 12 cells, with spinless electrons in two cosine bands of hopping $t$ separated by $E_\mathrm{g} = 2t$. Add a contact attraction $W$ between electron and hole, and one phonon mode of frequency $0.05\,E_\mathrm{g}$ coupled linearly with strength $g$, truncated at three phonons (checked at four). The Lanczos method diagonalizes it in the sectors with zero, one and two pairs. A linear fit in $g$ between $10^{-3}$ and $10^{-2}$ extracts the exact first-order phonon-assisted amplitude, which is compared with the replica formula built from the model's exact one-pair excitons. At $W = 0$ the two must agree to $10^{-6}$, as the theorem requires. For $W/E_\mathrm{g} = 0.05$ to 0.3 the deviation is the two-pair correction, to be compared with its estimated size $\sim E_x^b/E_\mathrm{g}$, about 1 % in silicon; it is extrapolated in the chain length (8, 10, 12) with a Lanczos residual below $10^{-10}$. The two-pair sector must be in the basis, whose dimension grows as $N^4$, since without it the test is circular. It runs in minutes to an hour on one core.

### 31.2.4 Exciton lines in the Migdal response

A nuclear kick can leave the crystal in a bound exciton rather than an ionized pair (Sect. 12.3). The Migdal convolution integrates the electronic momentum $\boldsymbol{p}$ against the lattice response at $\boldsymbol{q} - \boldsymbol{p}$ over the whole zone, so the exciton lines are needed everywhere, not only near $\boldsymbol{p} \to 0$: the line strengths $\rho_{\lambda 0}(\boldsymbol{p} + \boldsymbol{G})$, including the cross terms between different $\boldsymbol{G}$ and $\boldsymbol{G}'$, and the energies $E_\lambda(\boldsymbol{p})$. The boost operator carries the unreduced $\boldsymbol{p}$, so the Umklapp copies $\boldsymbol{p} + \boldsymbol{G}$ carry the same exciton band with different strengths, and a reduced $\boldsymbol{p}$ would merge them. The finite-momentum BSE above, in the resonant form for the proper lines with external screening from the RPA with local fields, supplies them on the 29 irreducible points of an $8^3$ grid of $\boldsymbol{p}$ and their Umklapp copies to $|\boldsymbol{p}| \le 5$ keV, about 60 points, with the three lowest exciton bands per point, up to 20 lines and 50 reciprocal-lattice vectors. The line strengths are interpolated in $\boldsymbol{p}$ and convolved with the lattice response. They must reproduce the optical exciton strength as $\boldsymbol{p} \to 0$, close the $f$-sum rule with the continuum to 1 % and approach the Wannier–Mott line weights in that limit, and change by less than 5 % when the $\boldsymbol{k}$ points per pocket go from $10^3$ to $8 \times 10^3$ and when the bands are doubled. At about one node-hour per momentum the lines cost 60 to 100 node-hours, or 2 to $3 \times 10^3$ core-hours.

### 31.2.5 Self-trapped excitons

In an alkali halide such as NaI the hole localizes on a pair of anions, the lattice relaxes around it, and the energy of the self-trapped state differs from that of the free pair. Under linear

coupling, a harmonic lattice and the Condon approximation, dark matter creates the exciton $\lambda$ vertically, and its line is dressed by a Poisson series of phonons [11],

$$S_\lambda(\boldsymbol{q},\omega) = |\rho_{\lambda 0}(\boldsymbol{q})|^2 \sum_{n\geq 0} \mathrm{e}^{-S_{\mathrm{HR}}} \frac{S_{\mathrm{HR}}^n}{n!}\, 2\pi\delta(\omega - E_\lambda - n\bar{\omega}), \tag{31.3}$$

with the Huang–Rhys factor $S_{\mathrm{HR}}$ and a mean phonon energy $\bar{\omega}$; for $S_{\mathrm{HR}} \gg 1$ it is a Gaussian of mean $E_\lambda + S_{\mathrm{HR}}\bar{\omega}$ and variance $S_{\mathrm{HR}}\bar{\omega}^2$. The relaxation into the self-trapped state comes after the creation and belongs to the cascade of Sect. 18.5, not to the vertex.

A finite-$q$ BSE for NaI, with the Na 2p and I 4d semicore frozen and a $6^3$ grid interpolated to $18^3$, gives $\rho_{\lambda 0}(\boldsymbol{q})$ and $E_\lambda$. The relaxation energy comes from a constrained-occupation hybrid calculation of the self-trapped exciton (a hole on an anion pair plus an electron) in 64- to 216-atom supercells at the measured lattice constant, with exact-exchange fractions 0.25 and 0.40, $\Gamma$-point sampling checked on a $2^3$ grid in the 64-atom cell, and the cutoff of the SG15 sodium and iodine potentials converged to 1 meV per atom; $S_{\mathrm{HR}}$ is that energy divided by $\bar{\omega}$, taken from the projected phonon density of states. A semilocal functional delocalizes the hole and must not be used, and the dependence on the exchange fraction is reported as the theory uncertainty. Iodine's spin–orbit coupling, about 0.9 eV, splits the exciton and is included. The result is trusted when $S_{\mathrm{HR}}$ and $\bar{\omega}$ change by less than 10 % from 64 to 216 atoms (otherwise 512 are added), when constrained occupations and time-dependent DFT give relaxation energies within 20 % of each other, and when the computed luminescence energy of the self-trapped exciton matches the measured one to 0.2 eV. The hybrid relaxations of the 216-atom cell with iodine spin–orbit coupling dominate the cost, about $10^4$ core-hours, and the time-dependent check doubles it; the factors enter Sect. 10.5.

### 31.2.6 Core levels with spin–orbit coupling

The semicore shells of germanium and gallium, 3d at about 30 eV below the valence maximum, and the silicon 2p shell contribute to $L(\boldsymbol{q},\omega)$ between 30 and 200 eV, and relativistic operators enter at order $\alpha^2$ (Sect. 10.6). The core part of the loss function is wanted for $q \leq 30$ keV and $\omega \leq 200$ eV, resolved into the $3d_{5/2}$ and $3d_{3/2}$ components. An all-electron calculation with spin–orbit coupling in the zeroth-order regular approximation, in a full-potential code such as WIEN2k or exciting, or PAW with core reconstruction (Sect. 24.2), followed by a core-level BSE at finite $q$ on a $12^3$ grid, provides it. No crystal code computes the four-component response with projection onto positive-energy states; the two-component calculation misses only terms of order $(Z_{\mathrm{eff}}\alpha)^4 \approx 10^{-6}$ for the germanium 3d shell and a projection error that is at most 2 % there, with a central estimate near $10^{-3}$ (Sect. 10.6). The binding energy of the 3d shell is computed with PBE, HSE06, $G_0W_0$ and total-energy differences, and their spread about the measured doublet, $3d_{5/2}$ at 29.2 eV and $3d_{3/2}$ at 29.8 eV with the centroid at 29.4 eV (Sect. 10.6), is the theory uncertainty. The decisive check is the mass–velocity shift of the $f$-sum rule of germanium, $-0.58$ %, the mass–velocity part of the total relativistic shift of $-0.76$ % in Sect. 10.6: this crystal calculation and the atomic one of Sect. 31.3 both compute it, and they have to agree to 0.1 %. The step from $12^3$ to $16^3$ must change the shift by less than 0.05 % and $S(q,\omega)$ at 10 and 30 keV by less than 2 %. By analogy with the valence BSE, each material costs about $10^3$ core-hours.

### 31.2.7 Spin–orbit mixing at the band edges

Spin–orbit coupling mixes spin into the band edges and can lift the symmetry zeros of the density vertex. In silicon the on-axis zero is lifted to $|\rho|^2 \approx 3.6 \times 10^{-6}$ (summed over the four $\Gamma_8$ states) and the mirror-plane zero to $1.3 \times 10^{-4}$ (Sect. 13.9), too small to move the window.

For germanium the mixing near the L-valley edge is estimated at up to 10 % in amplitude, and it needs bands in the right order, which neither PBE with spin–orbit coupling nor HSE06 with it gives. The step therefore runs on germanium bands with spin–orbit coupling that pass that check. On those bands a non-self-consistent run at the crystal momenta $(\boldsymbol{k}_v, \boldsymbol{k}_v + \boldsymbol{q})$ of the zero-phonon optimum gives the vertices from spinor wavefunctions matched by Miller index. Single spinor bands are gauge dependent, so only sums over Kramers pairs, or over degenerate quartets, are compared. Those sums are converged when they change by less than 2 % from 60 to 80 Ry and when the L and Γ energies of the underlying bands are stable to 10 meV in the number of bands and the screening cutoff. The silicon result, which must vanish as the spin–orbit strength is scaled to zero, is the control. Once bands that pass exist, the step costs less than a core-hour, and it replaces the estimate of Sect. 10.6.

## 31.3 Electronic response libraries

A spin-dependent coupling probes the spin density of the crystal, a velocity-dependent one its current, a scalar coupled to the electron's mass its kinetic-energy density, and each of these is screened differently from the charge density (Sects. 10.4 and 10.7). A rate for any of them needs a response function that most published calculations do not export: the full matrix of the independent-particle response, from which the screened responses to every operator follow by one Dyson equation. A library of these matrices, and the calculations that use it, follow the methods of Sects. 26.1 and 26.2.

### 31.3.1 The response library

For a crystal symmetric under the combination of inversion and time reversal, such as silicon or germanium, the response to the density $n_e(\boldsymbol{q})$ and the two transverse components of the current $j_{\mathrm{T}}(\boldsymbol{q})$ is fixed at each $(\boldsymbol{q}, \omega)$ by 12 real functions (Sect. 10.4). Six belong to the screened singlet block: the density structure factor $S_{nn}$, two density–current cross terms, and three symmetric current–current terms. Six more give the same block for independent particles, which also fixes the independent-particle triplet block, a quarter of the singlet one. The library holds these 12 functions on a grid of $(|\boldsymbol{q}|, \hat{\boldsymbol{q}}, \omega)$ and, in addition, the full independent-particle matrix $R^0_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q}, \omega)$, so that other kernels can be applied later without repeating the band calculation. For operator $\mathcal{O}_2$, whose velocity squared brings in the kinetic-energy density $\mathcal{T}_e$, the library adds four functions in each version, the kinetic rows below, so that it holds 20 functions in all (Sect. 33.1). On the truncation label $\mathfrak{T}$ (Sect. 8.5) it sits at $n_B = 0$, $n_g = 0$, with $\mathcal{X} = \mathrm{IP}$ and $\mathcal{X} = \mathrm{RPA}$ with local fields.

For silicon the ground state is the shared setting of Sect. 30.1 with a scissor to the measured gap. A scissor keeps the band order of PBE, which in germanium is wrong (Sect. 23.3), so the germanium library is built only on germanium bands that pass the check of Sect. 31.1; until such bands exist, no germanium version is released. Density vertices $\rho_{cv}(\boldsymbol{k}, \boldsymbol{q} + \boldsymbol{G})$ come from matching Miller indices. The current vertex $\langle c|\frac{1}{2}\{\boldsymbol{p}, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\}|v\rangle/m_e$ needs the correction for the nonlocal pseudopotential of Eq. (26.9); without it the transverse current of silicon is off by about 10 %. The Dyson equation of Eq. (26.2) is solved on the joint index set of the density at $\boldsymbol{q} + \boldsymbol{G}$, with $|\boldsymbol{q} + \boldsymbol{G}| \le 8\,\alpha m_e$, and the current, which enters only as an external row and column. The grids are $10^3$ $\boldsymbol{k}$ points without symmetry, $|\boldsymbol{q}| \le 25\,\alpha m_e$ in steps of $0.1\,\alpha m_e$, 30 directions in the irreducible wedge, $\omega \le 50$ eV in steps of 0.05 eV, and 60 bands. The tables are converged when $S_{nn}$ at 1 keV changes by less than 3 % from $8^3$ to $12^3$ $\boldsymbol{k}$ points and by less than 1 % when the local-field cutoff goes from 6 to $8\,\alpha m_e$. With the nonlocal term included, the $f$-sum rule of Eq. (26.13) has to hold to 2 % below the band cutoff, the tolerance at which Sect. 33.1 releases a version and on which the transfer rule of Worked example 31.1 relies;

where the spectrum ends early, the fraction it holds is stored as a value per $|\boldsymbol{q}|$. Symmetry gives a check that costs nothing: in such a crystal the imaginary part of the density–current cross term and the antisymmetric part of the current–current block vanish, and numerically they stay below $10^{-3}$ of the real parts.

Storing the full matrix is impossible. At $|\boldsymbol{q}+\boldsymbol{G}| \le 8\,\alpha m_e$ ($15\,\text{Å}^{-1}$) it has about 2300 reciprocal-lattice vectors, 85 MB per $(\boldsymbol{q},\omega)$, and on the full grid of $7.5\times10^6$ points it would fill some 600 TB. The 12 functions need only the screened column at $\boldsymbol{G}=0$ and the current rows, so the Dyson equation is solved for those few right-hand sides by an iterative solver, about $10^9$ operations per point, and never by inverting the matrix; the full $R^0_{\boldsymbol{G}\boldsymbol{G}'}$ is stored only on a coarse subset (25 values of $|\boldsymbol{q}|$, 3 directions, 100 frequencies) at a cutoff of $4\,\alpha m_e$, about 10 GB. Building the bubble then dominates, about $10^{16}$ operations, some $10^3$ core-hours for silicon and four times that for germanium with spin–orbit coupling. The output is a set of versioned tables, which Sect. 33.1 uses as the reference electronic response.

The library also gives the screening of the current by local fields and the spin channel beyond independent particles. With the Dyson equation solved on the joint index set, the screened current–current response differs from its independent-particle value by a term built from the mixed blocks $R^0_{j_\mathrm{T},n(\boldsymbol{q}+\boldsymbol{G})}$, the density response to a transverse current. At $\boldsymbol{G}=0$ this block vanishes when $\hat{\boldsymbol{q}}$ lies on a three- or fourfold axis, where a transverse current cannot drive a longitudinal density, so the screening of the current is purely a local-field effect, estimated at 10 to 30 %. Its value per $(|\boldsymbol{q}|,\omega)$ is what settles the in-medium factor for the velocity-dependent operator in Sect. 10.4. The spin channel has a Dyson equation that uses the matrix $R^0_{\boldsymbol{G}\boldsymbol{G}'}$ with a kernel made of the triplet exchange–correlation kernel and the screened electron–hole ladder, but without the exchange term that the charge channel carries. Since the exchange matrix element of a Wannier–Mott exciton is of relative order $(a/a_x)^3 \approx 10^{-3}$ in silicon, apart from its long-range part, which is the external screening and is absent from the spin channel by construction, the triplet ladder equals the singlet one to that accuracy. The ladder is that of the BSE run of Sect. 31.2 with the exchange switched off, so it costs little on top of that run, and the limit without interaction, which must return a quarter of the independent-particle density response, controls it.

### 31.3.2 Anisotropy of the electronic response

A cubic crystal's response depends on the direction of $\boldsymbol{q}$, and that dependence is what turns the Earth's rotation into a daily modulation (Sect. 11.7). It is carried by the cubic-harmonic components $S_\ell(|\boldsymbol{q}|,\omega) = \int \mathrm{d}\hat{\boldsymbol{q}}\, Y^c_\ell(\hat{\boldsymbol{q}})\, S(\boldsymbol{q},\omega)$ on the cubic harmonics $Y^c_\ell$ of Appendix C.3, for $\ell = 4$, 6 and 8, reported as $S_\ell/S_0$. The loss function with local fields is computed at full $\boldsymbol{q}$ (GPAW or BerkeleyGW) on a Lebedev grid of directions and projected. Near the gap the response has structure on a scale of 0.1 rad: the minima of the zero-phonon edge of silicon lie 0.13 to 0.22 rad from the valley axis (Sect. 13.9). A 110-point Lebedev rule integrates polynomials exactly only to degree 17, which is not enough once the response has components with $\ell \ge 10$. The grid is therefore refined from 110 to 230 to 434 points, exact to degrees 17, 25 and 35, and the test is that $S_4$, $S_6$ and $S_8$ change by less than 1 % between the last two. Near the edge the adaptive cone grid of Sect. 31.1 is added. The components with $\ell = 1$ and 3 vanish by time reversal and $\ell = 2$ by cubic symmetry, which gives checks that cost nothing: those components stay below $10^{-3}$ of the isotropic part. Projecting a line that is sharp in $\omega$ before binning gives meaningless coefficients, so $\omega$ is binned first. The run shares its band calculation with the library and adds little cost.

### 31.3.3 Local fields in the phonon-assisted window

The rate estimate of silicon in Sect. 13.9 is normalized to the measured optical edge and carries a screening correction relative to that calibration: 0.3 to 2.6 for the external factor and the local-field wings, an estimate, and the pair channel of Sect. 13.4 beyond the calibration, which no estimate sizes. A supercell calculation and a bound from zero-phonon vertices size the correction.

For silicon, a supercell displaced along the special (zero-gradient) configuration of Sect. 25.3 samples the phonon average deterministically, and the loss function of the displaced supercell, unfolded to the primitive momentum, contains the phonon-assisted response at the momenta the supercell resolves. Three runs of it separate the corrections. The supercell with RPA local fields in its own reciprocal-lattice basis gives the full result; the same supercell with independent particles screened by the head alone, $1/\varepsilon_{\boldsymbol{00}}(\boldsymbol{q},\omega)$, gives the reference of Sect. 13.4, and the ratio of the two is $(1+\delta_{\mathrm{LF}})(1+\delta_{\mathrm{pair}})$; a third run with local fields restricted to the primitive reciprocal lattice gives $\delta_{\mathrm{LF}}$ alone. On $\mathfrak{T}$ this is $n_B=0$, $n_g=2$, $\mathcal{X}=$ RPA with local fields; the special displacements are the adiabatic quadrature, and the correction of order $\omega_Q$ of Sect. 13.2 is applied afterwards. Cubic supercells of $n\times n\times n$ conventional cells resolve only certain momenta: $n=3$ (216 atoms) reaches $|\boldsymbol{q}|=0$, 0.76, 1.08, 1.32 and 1.52 keV, and $n=4$ (512 atoms) reaches 0, 0.57, 0.81, 0.99 and 1.14 keV. The two meshes share no nonzero momentum below 2.28 keV, so a convergence test between them is impossible at fixed $|\boldsymbol{q}|$. The tests are instead the change from $n=2$ (64 atoms) to $n=4$ at their common 1.14 keV, below 5 %, and the agreement of $n=3$ and $n=4$ after interpolation in $|\boldsymbol{q}|$ between 0.76 and 1.52 keV, also within 5 %. At $n=2$ the special configuration samples only a $2^3$ phonon mesh, so the first test also measures phonon sampling; both antithetic configurations are reported. The ground state is PBE with a scissor; the screening uses Yambo or a Liouville–Lanczos solver, which scales better, with a local-field cutoff of 3 Ry and bands to 30 eV above the Fermi level, and Γ-only sampling checked against $2^3$. The output is the screening ratio $\mathfrak{r}_{\mathrm{scr}}(\boldsymbol{q})=(1+\delta_{\mathrm{LF}})(1+\delta_{\mathrm{pair}})$, the full result over the reference, at the resolved momenta, and the test $\mathfrak{r}_{\mathrm{scr}}(\boldsymbol{q})\approx\mathfrak{r}_{\mathrm{scr}}(0)$. The three cell sizes cost about $10^2$, $2\times10^3$ and $2\times10^4$ core-hours, hours to a day on accelerators. A second pass with the adiabatic local-density kernel isolates its extra source term, estimated at a few percent of $L^{\mathrm{PA}}$ (Sect. 13.4).

The bound needs no amplitude. The vertex part of the external local-field correction, the wing terms $\boldsymbol{G}\neq0$ of Sect. 13.4, can be bounded from the zero-phonon vertices at the same $(\boldsymbol{q},\omega)$,

$$1+\delta^{(0)}_{\mathrm{wing}}(\boldsymbol{q},\omega)=\frac{\sum_{cv}\left|\sum_{\boldsymbol{G}}(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}(\boldsymbol{q},\omega)\,\rho_{cv}(\boldsymbol{q}+\boldsymbol{G})\right|^2}{\sum_{cv}|\rho_{cv}(\boldsymbol{q})/\varepsilon_{\mathrm{mac}}|^2},\qquad \varepsilon_{\mathrm{mac}}=1/(\tilde{\varepsilon}^{-1})_{00},\tag{31.4}$$

a ratio that is one without local fields. Numerator and denominator carry the same macroscopic factor $1/\varepsilon^2_{\mathrm{mac}}$, so the ratio measures the wings alone. With zero-phonon vertices the local-field ratio of Sect. 13.4 is $1+\delta_{\mathrm{LF}}=(\varepsilon_{1,\boldsymbol{00}}/\varepsilon_{\mathrm{mac}})^2\,(1+\delta^{(0)}_{\mathrm{wing}})$, whose first factor is the increase of 25 to 31 % at 1 keV. It needs the column $(\tilde{\varepsilon}^{-1})_{\boldsymbol{G}0}(\boldsymbol{q},\omega)$ in the primitive cell for $\boldsymbol{q}$ on the library grid and $\omega$ from 1.2 to 1.6 eV, with 40 bands, a $12^3$ grid and a cutoff of 4 Ry (about 37 reciprocal-lattice vectors), converged to 1 % between 3 and 5 Ry. It costs about 10 core-hours.

### 31.3.4 Loss functions of the candidate targets

The ranking of targets in Sect. 21.6 needs the loss function $L(\boldsymbol{k},\omega)=\mathrm{Im}[-(\varepsilon^{-1})_{00}(\boldsymbol{k},\omega)]$ of each candidate for $|\boldsymbol{k}|$ up to about 5 keV and $\omega$ from the gap to about five valence plasma energies, 75 to 150 eV. Its inverse fourth moment in $\omega$ sets the Migdal yield, and its weight near the gap sets the electron-recoil rate at low mass. The level is RPA with the full matrix

$\varepsilon_{\boldsymbol{G}\boldsymbol{G}'}$, since $L$ is not $\operatorname{Im}\varepsilon_{00}/|\varepsilon_{00}|^2$ once local fields matter, on a ground state with a scissor or quasiparticle gap matched to the measured one, with the adiabatic local-density kernel as the cross-check. A scissor serves only where PBE gets the band order right, checked against measured gaps at the high-symmetry points; elsewhere the bands must pass the check of order, gaps and masses, as for germanium. Semicore shells within five plasma energies of the gap are included: Ga 3d, Ge 3d, W 5p and 4f, Pb 5d, Te 4d, Cs 5p, I 4d, Mo 4p. A cubic crystal needs three directions, a uniaxial one the directions parallel and perpendicular to the axis and one oblique. Every directional average is over $\hat{\boldsymbol{k}}$, never one direction.

Since the inverse fourth moment weights the gap as $E_\mathrm{g}^{-5}$, every grid test is run on that moment, not on $L$ itself. The $f$-sum rule $\int \omega L \,\mathrm{d}\omega = (\pi/2)\omega_p^2$ holds to 2 % with the electrons counted (valence plus the semicore included), which also fixes the split between valence and core. The number of bands, the $\boldsymbol{k}$ grid and the local-field cutoff are each increased until $\int L/\omega^4 \,\mathrm{d}\omega$ changes by less than 1 %. The broadening is at most $E_\mathrm{g}/20$ and is extrapolated to zero. Missing semicore weight biases that moment low at a fixed $f$-sum, because the weight then sits at high $\omega$. The scissor moves the onset, and the computed moment is compared with the bound $(\pi/2)\omega_p^2/E_\mathrm{g}^5$ that the $f$-sum rule gives. Excitonic weight below the gap, absent in the RPA, adds to the moment; the BSE bounds it. Beyond the first zone, the element at $\boldsymbol{G} = 0$ must be taken at the unreduced $\boldsymbol{k}$ (Sect. 26.1). We estimate the cost from Worked example 26.1 at tens to a hundred core-hours for a two-atom cell and several hundred for a ten-atom oxide.

### 31.3.5 Relativistic shift of the $f$-sum rule

The $f$-sum rule holds exactly only for a nonrelativistic Hamiltonian. At order $\alpha^2$, the mass–velocity term reduces it, by $1.8 \times 10^{-3}$ in silicon (Sect. 4.3), and so does the orbit–orbit (Breit) interaction between electron pairs, which Sect. 10.6 writes as an integral of the all-electron static structure factor over the Coulomb kernel with a transverse projector. The all-electron atoms of Sect. 10.6 put the total relativistic shift at small $q$ at $-0.24$ % for silicon and $-0.76$ % for germanium, of which the mass–velocity part is $-0.18$ % and $-0.58$ %; a Thomas–Fermi estimate gives $-0.23$ and $-0.73$ %. What is missing is the orbit–orbit shift as a function of $q$ from 1 to 100 keV, and the total with the mass–velocity term. The cores dominate it, so the calculation is atomic: Hartree–Fock with the exact two-component (X2C) scalar-relativistic Hamiltonian in pyscf, in an uncontracted large basis (the atomic natural orbital (ANO) set for relativistic core correlation (RCC), ANO-RCC, or cc-pwCV5Z) with tight $s$ and $p$ exponents up to $10^7$, gives the two-electron energy and, from the one-particle density matrix in momentum space, the static structure factor. A crystal correction for the valence comes from a PAW all-electron density with the electron-gas exchange–correlation hole. The two-electron energy is converged to $10^{-4}$ when the basis goes from quadruple to quintuple zeta, and the $k$ integral to $10^{-3}$ with a radial grid out to $50/a_0$ for the germanium 1s shell; contracted valence basis sets would underestimate the core cusp by several percent. The orbit–orbit term is about 0.31 of the mass–velocity one in both atoms, so including only one misstates the total by about 24 %. The formula subtracts the uniform Hartree term itself, so it is never subtracted by hand, and for the valence, where $q$ times the pair distance exceeds one, the full angular dependence is kept rather than its average. The atoms cost minutes, the crystal correction about 10 core-hours, and the result replaces the estimate in Sect. 10.6.

### 31.3.6 Static screening with lattice relaxation

A probe that couples only to electrons still moves the nuclei in a static measurement, because the electrons drag them. The static response of the electron channel with the lattice allowed

to relax is the clamped-nuclei step plus a lattice term (Sect. 10.2),

$$1-\varepsilon_e^{-1}(\boldsymbol{q},0)=[1-\varepsilon_{\mathrm{cl}}^{-1}(\boldsymbol{q})]+\mathcal{G}_{\mathrm{rel}}(\boldsymbol{q}),$$
$$\mathcal{G}_{\mathrm{rel}}(\boldsymbol{q})=\frac{e^2}{q^2\Omega}\sum_{\nu}\frac{\left|\sum_\kappa \boldsymbol{q}\cdot Z_\kappa^{(e)}(\boldsymbol{q})\cdot \boldsymbol{e}_{\kappa\nu}(\boldsymbol{q}_r)\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{\tau}_\kappa}/\sqrt{M_\kappa}\right|^2}{\omega_\nu(\boldsymbol{q}_r)^2}, \tag{31.5}$$

at adiabatic, harmonic, one-phonon order. Here $Z_\kappa^{(e)}(\boldsymbol{q})$ is the electron charge that follows atom $\kappa$ when it moves, which at $q\to 0$ in an insulator is $Z_\kappa - Z_\kappa^*$ with all electrons counted, 14 for silicon. As $q\to 0$ the acoustic branch dominates and $\mathcal{G}_{\mathrm{rel}}\to \Omega_{P,e}^2/(c_{\mathrm{LA}}q)^2$, with $\Omega_{P,e}^2=e^2\boldsymbol{n}_e^2/\rho_T$ and $c_{\mathrm{LA}}$ the longitudinal sound speed: the step diverges as $1/q^2$ in every solid. For silicon $\Omega_{P,e}=0.51\,\mathrm{eV}$, and the limiting form reaches one at 18 keV, above the zone boundary; it gives $3\times10^2$ at 1 keV, where neither the linear dispersion nor $Z^{(e)}\approx 14$ holds, so the true value there is only estimated at $\gtrsim 10^2$.

The calculation needs $Z_\kappa^{(e)}(\boldsymbol{q})$ at finite $q$. Frozen-phonon supercells commensurate with $\boldsymbol{q}$, for silicon $1\times1\times N$ conventional cubic cells along [100] with $N=2$ to 8, displace one sublattice by $u\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}}$; the Fourier component of the induced electron density at $\boldsymbol{q}$, at two amplitudes to check linearity, gives the charge. The DFPT induced density is equivalent. A pseudopotential calculation gives only the valence part; the frozen cores are added back as rigid charges, valid for $q$ well below the inverse core radius, about 30 keV for the silicon 1s shell. The phonons come from the shared base, the clamped step from the loss functions above. $Z^{(e)}$ has to be stable to 1 % in cutoff and $\boldsymbol{k}$ grid and $\mathcal{G}_{\mathrm{rel}}$ to 2 % under refinement in $q$, and the small-$q$ limit has to match $c_{\mathrm{LA}}$ from the computed elastic constants. A violated acoustic sum rule sends $\omega_\nu\to 0$ with a finite numerator and fakes a divergence, so the sum rule is enforced on the force constants. The momentum is kept unreduced, $\boldsymbol{q}=\boldsymbol{q}_r+\boldsymbol{G}$, so that the step follows the density to several zone momenta, with its Umklapp terms; above a Debye–Waller exponent $2W\approx 0.1$ multiphonon terms are added; and in GaAs the direction dependence of the sound speed and of the nonanalytic optical term must be kept. Silicon and GaAs together cost about $10^2$ core-hours.

### 31.3.7 Kinetic-energy response

A scalar coupled to the electron's mass is absorbed through the kinetic-energy density $\mathcal{T}_e$ (Sect. 10.7). For every crystal whose point group has no polar vector, the absorption rate is proportional to $-\operatorname{Im} R_{\mathcal{T}\mathcal{T}}(0,m_\phi)$ (Eq. (10.36)), whose independent-particle form is a joint density of states weighted by second-order momentum matrix elements; with $\mathcal{T}_e=\sum_j p_j^2/2m_e$ as in Sect. 8.1, the squared element carries $1/4m_e^2$,

$$\operatorname{Im} R^0_{\mathcal{T}\mathcal{T}}(0,\omega)=-\frac{2\pi}{\mathcal{V}}\sum_{cv\boldsymbol{k}}\frac{|\langle c\boldsymbol{k}|p^2|v\boldsymbol{k}\rangle|^2}{4m_e^2}\,\delta(\omega-\varepsilon_{c\boldsymbol{k}}+\varepsilon_{v\boldsymbol{k}}), \tag{31.6}$$

with spin summed. The matrix element is computed directly from the plane-wave coefficients,

$$\langle c\boldsymbol{k}|p^2|v\boldsymbol{k}\rangle=\sum_{\boldsymbol{G}}|\boldsymbol{k}+\boldsymbol{G}|^2\,c^*_{c\boldsymbol{k}}(\boldsymbol{G})\,c_{v\boldsymbol{k}}(\boldsymbol{G}),$$

plus the PAW one-center correction. It equals $\sum_m \boldsymbol{p}_{cm}\cdot\boldsymbol{p}_{mv}$ over all bands including the core, a sum that converges slowly. It converges like the total kinetic energy, so silicon needs at least 80 Ry with SG15; the grid needs at least $24^3$ points, the delta function a broadening of at most 20 meV, and the gap a scissor. The $f$-sum rule is not a check here. Its analogue for $\mathcal{T}_e$ fixes the energy-weighted integral by $\langle|\nabla V|^2\rangle/2m_e$ per electron (Sect. 10.7), which the core region dominates, and the check uses that sum rule restricted to the valence. For the same reason,

norm-conserving pseudo-wavefunctions, which lack the core oscillations, are compared with a PAW or all-electron calculation. $\mathcal{T}_e$ is the canonical kinetic operator; velocity matrix elements $\mathrm{i}[H, \boldsymbol{r}]$ are not a substitute. Since $\mathcal{T}_e$ is even under inversion, the direct edge of silicon at $\Gamma$ is suppressed, and absorption inside the gap needs the phonon-assisted row of Sect. 31.1. Polar crystals add a mixing term through the current response. Local fields, the next many-body approximation on $\mathfrak{T}$, are added at RPA level. Silicon takes minutes; a ten-atom oxide about one accelerator-hour.

Operator $\mathcal{O}_2$ needs the same row at finite momentum. Its four functions per version, $\mathcal{T}_e(\boldsymbol{q})$ with itself and with the density and the two transverse currents, enter the Dyson solve of the library as external rows and columns, like the current, so that they are screened by the same density block (Sect. 33.1). The vertex $\langle c, \boldsymbol{k}+\boldsymbol{q}|\mathcal{T}_e(\boldsymbol{q})|v\boldsymbol{k}\rangle$ is built from the plane-wave coefficients with the matrix element $(\boldsymbol{k}+\boldsymbol{k}')^2/8m_e$ between plane waves of Eq. (8.3), plus the PAW one-center correction, on the grid of the library, and as $\boldsymbol{q} \to 0$ it must return Eq. (31.6).

## 31.4 Migdal and nonadiabatic quantities

The drag mass of silicon has two values that must not be confused. The trace value, $14\,m_e$ or $\delta M/M = 2.73 \times 10^{-4}$ for natural silicon ($M = 26.16\,\mathrm{GeV}$, Sect. 5.3), is the sum of the drag-mass tensor over all nuclei and counts every electron; it is exact, but a kick does not measure it. A kick measures the on-site element, which the tables of Sect. 12.5 put at $28.7\,m_e$, $\delta M_{II}/M = 5.60 \times 10^{-4}$: $10.1\,m_e$ from the cores and $18.6\,m_e$ from the valence, with the band 12 to $24\,m_e$ for the valence and the sign of the off-site coupling not fixed. The other nonadiabatic inputs of Sects. 12.3, 12.5 and 12.6 are likewise estimates from tables or models. The crystal Migdal rate is written there as a convolution of an electronic and a lattice response, and no single code has yet evaluated both pieces from first principles. A scalar that couples to nuclei out of proportion to their mass needs electronic matrix elements at reciprocal-lattice momenta. All the calculations below sit at the leading nonadiabatic order of the Born–Oppenheimer parameter on the truncation label $\mathfrak{T}$ (Sect. 8.5), harmonic ($n_A = 0$) and at zero temperature ($n_T = 0$). They differ in the many-body approximation $\mathcal{X}$, the lattice regime $\mathcal{K}$ and the Migdal regime $\mathfrak{M}$. Table 31.2 collects their parameters.

### 31.4.1 Drag masses and inverse moments

The electronic spectral measure of the force on nucleus $\kappa$ fixes every quantity of Sect. 12.5. With $|0\rangle$ the clamped-nuclei ground state, $|a\rangle$ its excited states and $\Delta_a = E_a - E_0$, we define

$$\mathrm{d}\sigma_\kappa(\Delta) = \sum_a |\hat{\boldsymbol{q}} \cdot \langle a|\nabla_{\boldsymbol{R}_\kappa} V|0\rangle|^2 \, \delta(\Delta - \Delta_a), \qquad \mathcal{I}_k \equiv \int \Delta^{-k} \, \mathrm{d}\sigma_\kappa(\Delta), \tag{31.7}$$

where $V$ is the electron–nucleus potential. Sect. 12.5 shows that the drag mass along $\hat{\boldsymbol{q}}$ is $\delta M_\kappa = 2\mathcal{I}_3$ and that the Migdal probability is $P_{\mathrm{Mig}} = q^2\mathcal{I}_4/M_\kappa^2$. Because $\Delta \geq E_{\mathrm{g}}$ on the support of $\mathrm{d}\sigma_\kappa$, $\mathcal{I}_4 \leq \mathcal{I}_3/E_{\mathrm{g}}$, and so $P_{\mathrm{Mig}} \leq E_R\,\delta M_\kappa/(M_\kappa E_{\mathrm{g}})$ with $E_R = q^2/2M_\kappa$. With the on-site drag mass $28.7\,m_e$ and the gap of $1.17\,\mathrm{eV}$ this bound keeps $P_{\mathrm{Mig}}$ below $10^{-3}$ only up to $E_R \approx 2.1\,\mathrm{eV}$ in silicon, and to less in germanium; the trace value $14\,m_e$ would give $4.3\,\mathrm{eV}$, but it does not bound the probability (Sect. 12.5.3). Above that the computed $\mathcal{I}_4$ is needed.

The three moments follow from two Sternheimer solves per occupied state, the static form of Eq. (25.7). The first, $|\psi_v^{(1)}\rangle = -(H_{\mathrm{KS}} - \varepsilon_v)^{-1}\hat{P}^{\mathrm{c}}\,\partial_{\hat{\boldsymbol{q}}} V|\psi_v\rangle$, is the ordinary phonon response; the second, $|\psi_v^{(2)}\rangle = (H_{\mathrm{KS}} - \varepsilon_v)^{-1}\hat{P}^{\mathrm{c}}|\psi_v^{(1)}\rangle$, uses $|\psi_v^{(1)}\rangle$ as its source. Expanding both in conduction states $|c\rangle$ gives one power of $\Delta_{cv}^{-1}$ per solve, so

$$\mathcal{I}_2 = 2\sum_v \langle\psi_v^{(1)}|\psi_v^{(1)}\rangle, \qquad \mathcal{I}_3 = 2\sum_v \langle\psi_v^{(1)}|\psi_v^{(2)}\rangle, \qquad \mathcal{I}_4 = 2\sum_v \langle\psi_v^{(2)}|\psi_v^{(2)}\rangle, \tag{31.8}$$

with the factor 2 for spin and the sums over occupied bands and $\boldsymbol{k}$. These are sums over Kohn–Sham pairs. The interacting drag mass at $\mathcal{X} = \mathrm{RPA}$ is the $\omega^2$ coefficient of the on-site electronic force constant at imaginary frequency (Sect. 12.5), and a fit in $\omega^2$ through five points between 0.1 and 0.5 eV returns it. The difference between the pair sum and the RPA slope is the many-body theory uncertainty of Sect. 8.8. We compute $\delta M_\kappa$ both for the $\boldsymbol{Q} = 0$ sublattice displacement and on site (the Brillouin-zone average over an $8^3$ grid of $\boldsymbol{Q}$), with the inputs of the shared harmonic base of Sect. 30.1. The pseudopotential freezes the core, whose electrons move rigidly with the nucleus; their contribution, $N_{\mathrm{core}} m_e$ to $\delta M_\kappa$, is added exactly, and the core part of $P_{\mathrm{Mig}}$ is added shell by shell from the atomic calculation of Sect. 12.2.

The trace rule tests the result. Displacing the whole cell rigidly drags every electron, so the sum over all sites of the $\boldsymbol{Q} = 0$ tensor equals $N_e m_e$ per cell. It holds only with the core term included, and it tests the $\boldsymbol{k}$ grid and the band completeness of Eq. (31.8) at once. $\delta M_\kappa$ must change by less than 1 % from $12^3$ to $16^3$ $\boldsymbol{k}$ points. Each perturbation needs one Sternheimer solve beyond those of the shared DFPT run, so the cost is about twice that run: some 200 core-hours for silicon, 30 for germanium and 40 for gallium arsenide, at a few gigabytes of memory, and 2.4 times as much for the $16^3$ test. The Kohn–Sham gap of gallium arsenide, 0.5 eV in PBE against 1.52 eV measured, puts spurious resonances below the real gap and inflates $\mathcal{I}_3$ and $\mathcal{I}_4$, which carry $\Delta^{-3}$ and $\Delta^{-4}$; a scissor or a hybrid functional is required. Germanium is worse: PBE at the measured lattice constant puts the conduction state at $\Gamma$ only 0.047 eV above the valence maximum, so the inflation of $\mathcal{I}_3$ and $\mathcal{I}_4$ is far larger, and a scissor keeps the wrong valley order. The germanium drag masses, like the electronic piece of the germanium Migdal rate below, are computed on germanium bands that pass the check of Sect. 31.1. In a polar crystal the $\boldsymbol{Q} = 0$ response must be taken at zero macroscopic field, without the nonanalytic term of Sect. 25.2, because the drag mass belongs to the transverse displacement. The results replace the table-based on-site values of Sect. 12.5 and fix the sign of the off-site coupling.

### 31.4.2 The crystal Migdal rate

The Migdal rate of a crystal is set by the Migdal entry $S_{\kappa\kappa}(\boldsymbol{q}; \omega_l, \omega_e)$ of the joint response, Eq. (12.6), a density in two frequencies, in $\mathrm{eV}^{-2}$ per cell. It is a convolution of two pieces: the density response $S^{ee}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{p}, \omega_e)$ at the momentum $\boldsymbol{p}$ that the boost gives the electrons, with local fields, and the multiphonon or impulse response of the struck nucleus. We want it for silicon and germanium with $p \leq 30\,\mathrm{keV}$ and $\omega_e \leq 100\,\mathrm{eV}$. On $\mathfrak{T}$ it sits at the leading nonadiabatic order with $\mathfrak{M} =$ sudden. It uses the Condon approximation with the lattice in its ground vibrational manifold, at $\mathcal{X} = \mathrm{RPA}$ with local fields, and with $\mathcal{K}$ the multiphonon or impulse regime of the lattice tables.

The electronic piece is all-electron: GPAW with PAW reconstruction, or Yambo [12] and BerkeleyGW [4] with the 2p shell of silicon and the 3d shell of germanium in the valence; the deeper shells enter one at a time from the atomic calculation of Sect. 12.2. The convolution uses the exact impulse bracket of Sect. 12.3 wherever $|\boldsymbol{q} \cdot \boldsymbol{p}|/(M_\kappa \omega_e) > 10^{-2}$, and the soft limit elsewhere. The matrix is not computed on the Cartesian $50^3$ grid of $\boldsymbol{p}$ per shell that the convolution integrates over; that would be some $10^{19}$ operations. It is computed at the 72 irreducible reduced momenta of a $12^3$ grid, whose Umklapp copies $\boldsymbol{p}_{\mathrm{red}} + \boldsymbol{G}$ cover the unreduced momenta, and interpolated in $\boldsymbol{p}$ for the convolution. Local fields are converged from $|\boldsymbol{G}| \leq 4$ to $6\,a_0^{-1}$, which is about 300 to 1000 plane waves in silicon. A $10^3 \times 10^3$ complex matrix takes 16 MB per $(\boldsymbol{p}, \omega_e)$ and the 0.05 eV grid has 2000 frequencies, so a single $\boldsymbol{p}$ holds 32 GB; the frequencies are processed in blocks, contracted with the lattice response on the fly, and only the convolved $S_{\kappa\kappa}$ is stored. Building the independent-particle matrix costs about $3 \times 10^{12}$ operations per $(\boldsymbol{p}, \omega_e)$ on the $12^3$ $\boldsymbol{k}$ grid, and the whole calculation about $10^4$

core-hours. The same code, fed the same electronic and lattice tables, reports the rate at each truncation of Fig. 12.1, so that the steps between them are differences of one calculation and not of different inputs.

The third electronic moment, summed over shells and valence, must equal the contact-density sum rule of Sect. 12.5. The $\omega_e^{-3}$ moment must reproduce the drag mass of Sect. 31.4.1, and its trace $N_e m_e$. The computed spectrum must respect the bound on $P_{\mathrm{Mig}}$ of Sect. 31.4.1. No measurement of the crystal Migdal effect exists yet; once the tagged-recoil neutron measurement proposed by Adams et al. [13] has been made on silicon, the computed rate is compared with it at the same recoil energies. The difference between the result with the full matrix and with its diagonal is reported as the local-field uncertainty.

The Migdal term must not be added to a lattice response computed with the bare mass and unit weight (Sect. 12.3): the lattice part loses the weight $P_{\mathrm{Mig}}$ and moves with the mass $M_\kappa + \delta M_\kappa$, or the same recoil is counted twice. A response diagonal in $\boldsymbol{G}$ keeps the local fields on the diagonal of $\varepsilon^{-1}$, which lower the silicon probability by a factor 1.35 against a scalar dielectric function (Limit Check 12.2). It drops the elements $\boldsymbol{G} \neq \boldsymbol{G}'$, which raise it again by a factor of about 1.2 to 1.4 in an empirical-pseudopotential model that is not converged in the reciprocal-lattice sum (Sect. 12.3.2), and this calculation is what sizes them. A valence-only electronic piece misses the core-dominated $\omega_e^3$ moment entirely. The results enter Sect. 12.3 as the crystal Migdal spectra of silicon and germanium.

### 31.4.3 Coherent zero-phonon Migdal response

The convolution above treats the lattice factor in its incoherent form, one nucleus at a time and diagonal in $\boldsymbol{G}$. A zero-phonon Migdal event is a Bragg process instead: the crystal takes a reciprocal-lattice vector $\boldsymbol{G}$ as a whole, with weight $\mathrm{e}^{-2W_\kappa(\boldsymbol{G})}$ and the structure factor of the cell, and the electrons take exactly $\boldsymbol{p} = \boldsymbol{q} - \boldsymbol{G}$ (Sect. 12.4). The amplitudes of different $\boldsymbol{G}$ interfere, so the rate is a double sum over $\boldsymbol{G}$ and $\boldsymbol{G}'$ with the cross-site terms $S_{\kappa\kappa'}$ and the full matrix $S^{ee}_{\boldsymbol{q}-\boldsymbol{G},\boldsymbol{q}-\boldsymbol{G}'}$. Above $2q_{\mathrm{BZ}}$ the direction-averaged double sum agrees with the incoherent form. Far below $q_{\mathrm{BZ}}$ the boost identity Eq. (8.32) fixes it at 1/14.9 of the diagonal form in silicon, with the median $\omega_e$ at the plasmon (Sect. 12.4); a model puts the diagonal form within 15 % of the coherent one only for $q \leq 0.75\,\mathrm{keV}$. Between 0.75 keV and $2q_{\mathrm{BZ}}$, where nearly all of the silicon Migdal rate up to 1 MeV sits, 33 % at 2 MeV and 7 % at 3 MeV (Table 12.2), nothing fixes it.

The calculation evaluates the coherent zero-phonon response of silicon on that band of $q$, at $\omega_l = 0$, with every cross-site term and local fields at $\mathcal{X} = \mathrm{RPA}$. Each $\boldsymbol{q}$ is its own reduced momentum plus a reciprocal-lattice vector, so the matrices $S^{ee}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{p}_{\mathrm{red}}, \omega_e)$ of the crystal rate serve unchanged; only the head and wings, which are nonanalytic as $\boldsymbol{p}_{\mathrm{red}} \to 0$, need reduced momenta added on three lines toward $\Gamma$, at the cost per momentum of the crystal rate. The ion vertex is the all-electron or PAW one of the crystal rate, and the Bragg weights carry $\mathrm{e}^{-2W_\kappa(\boldsymbol{G})}$ from the shared base. The result is reported as the ratio of the coherent to the diagonal rate per bin of $(q, \omega_e)$, averaged over directions on a 590-point Lebedev grid and kept per direction for the daily modulation. It has three tests: as $q$ falls toward 0.75 keV it must approach the boost identity, the factor 1/14.9 and the plasmon median; above $2q_{\mathrm{BZ}}$ it must return the diagonal form; and at every $q$ it must satisfy the partial sum rule Eq. (8.56). The ratio replaces the unsized entry of Table 12.2 and the Migdal band of Sect. 17.1.

### 31.4.4 Couplings not proportional to mass

A scalar that couples to nucleus $\kappa$ with strength $c_\kappa$ is absorbed at momentum $k \to 0$ through the acceleration form Eq. (12.50), with the electronic factor $\langle f|\nabla_{\boldsymbol{R}_\kappa} V|i\rangle$. If $c_\kappa/M_\kappa$ is the

Table 31.2: Nonadiabatic calculations of silicon (and germanium, gallium arsenide or diamond where named): the computed quantity, the main convergence parameters, the acceptance test and the cost. $N_e$ is the number of electrons per cell; the cost is in core-hours.

| quantity | grids and parameters | acceptance test | cost |
|---|---|---|---|
| $\delta M_\kappa$, $\mathcal{I}_2$–$\mathcal{I}_4$ | shared base (60 Ry); $\boldsymbol{k}$: $12^3 \to 16^3$ (1 %); $8^3$ $\boldsymbol{Q}$; $\mathrm{i}\omega = 0.1$–$0.5\,\mathrm{eV}$ | trace $N_e m_e$ per cell | 2× base: 200 (Si), 30 (Ge), 40 (GaAs) |
| $S_{\kappa\kappa}(\boldsymbol{q};\omega_l,\omega_e)$ | $p \le 30\,\mathrm{keV}$ from 72 reduced $\boldsymbol{p}$; $\omega_e \le 100\,\mathrm{eV}$ in 0.05 eV; $\lvert\boldsymbol{G}\rvert \le 4 \to 6\,a_0^{-1}$; $\boldsymbol{k}$: $8^3 \to 12^3$ | third moment, drag mass, $P_{\mathrm{Mig}}$ bound; neutron data when measured | $10^4$ |
| coherent zero-phonon ratio | 0.75 keV to $2q_{\mathrm{BZ}}$; matrices of the row above plus reduced $\boldsymbol{p}$ toward Γ; 590 directions | boost limit (1/14.9); diagonal form above $2q_{\mathrm{BZ}}$; Eq. (8.56) | per added $\boldsymbol{p}$ as above |
| $S^{ee}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{k}\to 0,\omega)$ | $\lvert\boldsymbol{G}\rvert \le 6(2\pi/a)$; $\boldsymbol{k}$: $12^3$ | (200) zero; mass-proportional limit | $10^2$ |

same for every species, Sect. 12.6 reduces the amplitude exactly to the electron density and the rate to the loss function. Write $c_\kappa/M_\kappa = \tilde{c} + \delta\tilde{c}_\kappa$. The force operator of a point nucleus is $\nabla_{\boldsymbol{R}_I} V = \mathcal{V}^{-1}\sum_{\boldsymbol{p}} \mathrm{i}\boldsymbol{p}\, Z_\kappa v_{\mathrm{C}}(\boldsymbol{p})\, \mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{R}_I} n_e(\boldsymbol{p})$, and the sum over sites of $\delta\tilde{c}_\kappa\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_I}$ forces $\boldsymbol{p} = \boldsymbol{k} + \boldsymbol{G}$. The remainder of the rate is then

$$\Gamma \propto \frac{1}{\Omega^2\omega^4} \sum_{\boldsymbol{G}\boldsymbol{G}'} [\boldsymbol{k}\cdot(\boldsymbol{k}+\boldsymbol{G})][\boldsymbol{k}\cdot(\boldsymbol{k}+\boldsymbol{G}')]\, \Xi_{\boldsymbol{G}}(\boldsymbol{k}) \Xi^*_{\boldsymbol{G}'}(\boldsymbol{k})\, S^{ee}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{k}\to 0,\omega),$$
$$\Xi_{\boldsymbol{G}}(\boldsymbol{k}) = \sum_\kappa \delta\tilde{c}_\kappa Z_\kappa v_{\mathrm{C}}(\boldsymbol{k}+\boldsymbol{G})\, \mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{\tau}_\kappa}, \tag{31.9}$$

and every Umklapp term is $O(k)$, like the $\boldsymbol{G} = 0$ term. The whole effect therefore sits in the off-diagonal elements $S^{ee}_{\boldsymbol{0}\boldsymbol{G}}$ and $S^{ee}_{\boldsymbol{G}\boldsymbol{G}'}$ at vanishing reduced momentum. For equal $\delta\tilde{c}$ on both sites of the diamond structure, $\Xi_{\boldsymbol{G}}$ is proportional to the basis structure factor, which vanishes for (200) and has $\lvert S^{\mathrm{bas}}\rvert^2 = 2$ and 4 for (111) and (220).

The matrix $S^{ee}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{k}\to 0,\omega)$ is needed for $\lvert\boldsymbol{G}\rvert \le 6(2\pi/a)$, about 230 vectors in silicon, and $\omega$ from $E_{\mathrm{g}}$ to 100 eV plus the core edges, contracted with Eq. (31.9) for each coupling model. On $\mathfrak{T}$ it is $\mathcal{X}$ = RPA with local fields (then the adiabatic local-density kernel), at clamped nuclei because $\omega \gg \omega_{\max}$. Yambo or BerkeleyGW give $\chi_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q}\to 0,\omega)$ directly. A diagonal dielectric matrix would miss every Umklapp term, so the local-field set is enlarged until $\Gamma$ changes by less than 2 %, on a $12^3$ grid; each frequency needs under 1 MB, and the cost is about $10^2$ core-hours. The $\boldsymbol{G} = 0$ term uses the full $S^{ee}$, which already contains the factor $1/\lvert\varepsilon\rvert^2$, and screening it again would double count. The calculation is accepted when the (200) channel vanishes to numerical noise for equal $\delta\tilde{c}$, and when for $c_\kappa \propto M_\kappa$, where $\delta\tilde{c}_\kappa$ is only the mass defect of order $10^{-3}$, the result reduces to the loss-function rate Eq. (12.52). The result enters Sect. 12.6.

## 31.5 Coherent and second-order absorption

An axion-like field of mass 1 μeV stays coherent for $t_{\mathrm{coh}} = 1/(m_\phi v_0^2) \approx 1.04\,\mathrm{ms}$, with $v_0$ the circular speed of the standard halo (Sect. 1.2), and over $1/(m_\phi v_0) \approx 249\,\mathrm{m}$, so a nuclear-spin sample or an acoustic resonator sees one classical drive, not a stream of quanta. A fermion of 100 keV absorbed in germanium deposits nearly its whole mass at a momentum transfer of the same size. A dark photon can be absorbed in pairs, or scattered into a photon. These processes lie outside the single-vertex golden rule of Sects. 31.1 to 31.3, and Sects. 10.8, 10.9,

11.8 and 14.5 derive them; what they need from the material is mostly a response that the methods of Sects. 25.2, 26.1 and 26.2 already compute, read on a different line of the $(q, \omega)$ plane.

### 31.5.1 Coherent-regime material inputs

The first decision is the regime, and it is made before any material calculation. The target responds on a time $1/\Gamma_T$, which is $T_2$ for spins and the mode quality over the mode frequency for a resonator. For an integration time $t_{\mathrm{int}}$ long compared with the smaller of $t_{\mathrm{coh}}$ and $1/\Gamma_T$, the mean absorbed power equals the velocity-averaged golden rule, and the rate tables of Chaps. 30 to 32 apply. A single record matches that mean only for $t_{\mathrm{int}} \gg t_{\mathrm{coh}}$. Below $t_{\mathrm{coh}}$ the drive is a single Rayleigh-distributed amplitude, the signal is the windowed linear response of Sect. 14.5, and the sensitivity scales as $t_{\mathrm{int}}^{-1/2}$ instead of $(t_{\mathrm{int}} t_{\mathrm{coh}})^{-1/4}$. The material inputs differ by platform.

For nuclear-spin precession, the chemical shielding and electric-field-gradient tensors of every nuclear site give the inhomogeneous line as a site sum, and come from the gauge-including projector augmented-wave method [14]. The effective electric field at the nucleus that converts a nucleon electric dipole moment into an energy shift comes from density-functional theory, and $T_1$ and $T_2$ come from data. Prepolarized samples are out of equilibrium, with $T_1$ of hours, so their spin noise is evaluated at the spin temperature fixed by the polarization $p$ through $\tanh(\beta_s \omega_{\mathrm{Lar}}/2) = p$, with $\omega_{\mathrm{Lar}}$ the Larmor frequency, not at the lattice temperature [15]. Bulk acoustic resonators need the inputs listed in Sect. 11.8: elastic and piezoelectric tensors from density-functional perturbation theory, mode shapes from finite elements, the shape factor and the measured mode quality. For resonant-mass detectors driven by a scalar that changes $\alpha$ or $m_e$, the inputs are $\mathrm{d}\ln a/\mathrm{d}\ln\alpha$ and $\mathrm{d}\ln a/\mathrm{d}\ln m_e$ of the lattice constant $a$ [16]. A nonrelativistic density-functional calculation with scaled constants must return $-1$ for both to numerical precision (Eq. (11.59)); that is the benchmark. A fully relativistic calculation of a new crystal is accepted once it reproduces the relativistic and finite-nuclear-mass corrections of copper and silicon given in Sect. 11.8 [17, Table II]. For magnetoelectric multiferroics, linear spin-wave theory or time-dependent density-functional theory with spin–orbit coupling on a magnetic structure computed at $T = 0$ gives the response only far below the ordering temperature. Near it, where the proposed signal peaks, a finite-temperature spin-dynamics method is needed, which Chaps. 23 to 29 do not supply.

The frequency grid resolves the one-sided dark-matter line, $\omega \in [m_\phi, (1 + 3.5 \times 10^{-6}) m_\phi]$, with at least 3000 points, and $t_{\mathrm{int}}$ runs from $10^{-2}$ to $10^{3}\, t_{\mathrm{coh}}$. The Asimov test statistic of Sect. 14.5 must change by less than 1 % when the frequency step is halved, and the resonator mode frequencies must be converged to 0.1 %. Two pitfalls are statistical rather than material: the random amplitude is either in the likelihood or in a penalty factor, never in both [18], and an analysis with fixed amplitude overestimates the reach, by a factor 1.99 for the median limit of a record shorter than $t_{\mathrm{coh}}$ and 2.7 for data at the detection threshold (up to 10 with other priors; Sect. 14.5). Once the responses are tabulated, a point in $(m_\phi, t_{\mathrm{int}})$ takes seconds. The gauge-including projector-augmented-wave (GIPAW) and DFPT inputs use the shared setting, converged to 1 % in cutoff and $\boldsymbol{k}$ grid, and cost $10^2$ to $10^4$ core-hours per target, the multiferroic response about $10^3$.

### 31.5.2 Fermionic absorption

A fermion $\chi$ absorbed by the target turns into a light neutral fermion $\tilde{\nu}$ (Sect. 10.9), and for $v \to 0$ and a massless $\tilde{\nu}$ the target is probed along the line $\omega = m_\chi - q$, at $q \approx m_\chi$. For $m_\chi$ from $E_{\mathrm{g}}$ to 1 MeV this reaches momentum transfers far above the valence scale, into the cores.

We need $\mathrm{d}\mathcal{R}/\mathrm{d}\omega$ for vector, scalar and axial operators. The scalar operator needs only the density row $S(q,\omega)$. The vector operator needs the density row and the transverse current row of $S_{ab}$: continuity turns the longitudinal current into $(\omega/q)$ times the density, so that the density enters with weight $m_\chi^2/q^2$ and the transverse current with weight one. A light mediator multiplies the scalar rate by $(m_{\mathrm{med}}^2 - 2m_\chi\omega + m_\chi^2)^{-2}$, normalized to its contact value, where $m_{\mathrm{med}}$ is the mediator mass, since the four-momentum transfer is set by the relativistic $\tilde{\nu}$ leg and not by $-q^2$; its expansion, $1 - 2(m_\chi^2 - 2m_\chi\omega)/m_{\mathrm{med}}^2 + \ldots$, is a correction of relative order $(m_\chi/m_{\mathrm{med}})^2$ with an $\omega$ dependence that tilts the spectrum.

The responses are all-electron. In crystals they come from the plane-wave calculation of Sect. 26.2 with augmentation reconstructed to high $q$ (Sect. 24.2); for $q \gtrsim m_e/3$, where the core shells dominate, from Dirac–Hartree–Fock orbitals of the atom. The grid is about 200 points in $q$ per mass and 40 masses per decade. The electron spectra must change by less than 2 % when the augmentation is enlarged, and the nuclear line, an impulse response broadened by the zero-point momentum $\langle p_\kappa^2 \rangle$ from density-functional perturbation theory, must have its width converged to 1 %. The spin convention is the commonest error: the atomic ionization form factor that the literature uses counts both spins in its sum over occupied states [19, Eqs. (A.18)–(A.21)], and a table that does not state its convention in the header can be off by a factor of two. The transverse current and the relativistic corrections to the bound states are computed with the same Dirac–Hartree–Fock states. The cost is $10^4$ to $10^5$ core-hours per crystal; the nuclear part is negligible.

### 31.5.3 Pair, Compton and Barkas responses

Pair absorption, dark Compton scattering and the second-order Barkas term need responses at two insertions of the field (Sect. 10.8). Contact pair absorption reduces to single absorption at $\omega \approx 2m_\phi$ and $k \approx 0$, so it uses the absorption tables of Sect. 31.3 (charge-like rows) or, for the pseudoscalar coupling, the axioelectric (spin-current) row of Eq. (10.33) at $\omega = 2m_\phi$ (Eq. (10.44)), contracted with the pair tensor of Sect. 10.8 [20, Eqs. (5)–(8)]. The pseudoscalar route inherits the caveats of that row: the measured $\sigma_1$ belongs to singlet pairs while the row creates triplets, spin–orbit coupling spoils the factor $\frac{3}{4}$ (Sect. 10.8), and in a superconductor the spin-current conductivity replaces $\sigma_1$ only under the conditions of Sect. 15.4.3. The tests below accept the pair tensor and its velocity moments, not the row. For the scalar and pseudoscalar couplings the tensor is constant; for the vector and axial couplings it carries the relative momentum of the two absorbed particles, and its velocity moments are computed over the product of the two velocity distributions on a $20^3$ quadrature. Two tests accept it: for the scalar coupling the pair rate reduces to the single-absorption rate at $\omega = 2m_\phi$, $k \to 0$, times the constant tensor, and the velocity moments reproduce their analytic values for the standard halo to $10^{-3}$. Dark Compton scattering of a vector of mass $m_\phi$ on electrons bound with energies small compared with $m_\phi$ needs the shell Compton profiles, or the high-$q$ density response. It also needs the refractive index $n_r(\omega')$ of the medium at the outgoing photon energy $\omega' \approx m_\phi$; the grid is 100 points in $\omega' \in m_\phi(1 \pm 0.1)$ and 32 angles. For a free electron at rest the rate must reproduce the Thomson-limit form $n_e \zeta^2 \sigma_{\mathrm{Th}}\, \omega'/[m_\phi(1 + \omega'/m_e)]$, with $\sigma_{\mathrm{Th}} = e^4/(6\pi m_e^2)$ the Thomson cross-section and $\omega' + \omega'^2/2m_e = m_\phi$, which agrees with the tree-level result [21, Eq. (2)] through second order in $m_\phi/m_e$; this is the acceptance test. The Barkas term needs the quadratic density response of the electron gas, tabulated for 50 densities with $r_s$ from 1 to 6 and mapped onto the crystal by the local-density approximation on the $(q,\omega)$ grid of the loss functions of Sect. 31.3. It is accepted when the electron gas returns the known $Z_1^3$ (Barkas) correction to the stopping of slow charges and the linear response when the quadratic kernel is switched off. The Barkas table is new code of about $10^2$ core-hours; the other two calculations take minutes.

## Summary

The phonon-assisted loss function of silicon is computed from the four-path amplitude, with the sum over intermediate states completed above the Wannier manifold by the Sternheimer remainder. It is computed on bins shared with the zero-phonon rate, so that the end of the phonon-assisted lead is settled as a ratio of two rates on the same bins. Its plane-wave stages cost some 350 to 450 core-hours. A bin converged to 5 % is then tested against the measured indirect absorption in the optical limit and against the $f$-sum rule with the truncation and the nonlocal term separated. A germanium calculation first needs bands with the measured order of its valleys. Near the gap the finite-momentum Bethe–Salpeter equation with its coupling block adds the bound excitons, the exciton–phonon replicas and the exciton lines of the Migdal response, and the core levels and the spin–orbit mixing at the band edges complete the electronic response. A response library holds 12 real functions per $(\boldsymbol{q}, \omega)$ for silicon, and the full independent-particle matrix on a coarse subset, from which the screened response to every operator follows by one Dyson equation; building the bubble dominates its cost, some $10^3$ core-hours. The nonadiabatic inputs, the drag masses and the crystal Migdal rate, are specified at the leading nonadiabatic order, the rate as a convolution of an electronic and a lattice response that no single code has yet evaluated. Coherent and second-order absorption read the same responses on other lines of the $(q, \omega)$ plane.

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# Chapter 32

# Detector response and inputs

Silicon recoils of 100 eV, below the binary threshold for ionization, are measured to produce charge, and no calculation yet predicts the yield (Sect. 19.2). No first-principles cascade reaches the precision of the measured pair energy and Fano factor of silicon (Sect. 18.4). No collection kernel has been computed for a real detector, and the readout conclusions of Chap. 20 stay conditional until the rates, kernels and efficiencies of a stated detector are computed. These calculations use the methods of Chap. 29, and their costs run from seconds for the pair-number library to about $10^6$ core-hours for the close-collision kernel.

## 32.1 Nuclear-recoil partition and charge yield

In silicon a binary collision with a valence electron creates an electron–hole pair only above the recoil energy $E_{R,\mathrm{th}} \approx 352\,\mathrm{eV}$ of Sect. 19.2. Yet the measured ionization yield at 100 eV is $0.102^{+0.034}_{-0.030}$, and the data show no sign of an ionization threshold above 100 eV [1, Table II]. In germanium a recoil of 254 eV, below its threshold of 399 eV, gives $(64 \pm 8)\,\mathrm{eV}$ of electron-equivalent energy, a quenching factor of $25 \pm 3\,\%$ [2]. The charge at these energies comes from processes that the Lindhard model of Sect. 19.3 does not contain, and the share of the recoil energy that stays in the lattice as stored energy is bounded by calibration data but not computed (Sect. 19.4.4). Single recoils need the electronic excitation in a close collision of two atoms, the partition of the recoil energy into heat, stored energy and charge, and the charge yield below the binary threshold; dense tracks need the interaction of their carriers, the recombination of the track plasma and the amorphous pockets left by recoils of keV. Molecular dynamics on machine-learned potentials (Sect. 29.1) carries all of them except the excitation kernel and the track recombination. Table 32.1 collects their grids, tests and costs.

### 32.1.1 Close-collision excitation kernel

The kernel is $\mathrm{d}P/\mathrm{d}\omega_e(b_{\mathrm{imp}}, E_{\mathrm{coll}})$ of Eq. (19.10). The grid has 10 logarithmic values of $E_{\mathrm{coll}}$ from 50 eV to 5 keV and 8 values of $b_{\mathrm{imp}}$ from 0 to 1.5 Å, 80 points in all.

The Born–Oppenheimer order is not small here, since the Massey parameter of Sect. 19.1 is of order one near 1 keV, so the calculation sits off the perturbative axes of $\mathfrak{T}$ (Sect. 8.5): the nuclei are classical, the electrons are propagated in time.

Real-time time-dependent density-functional theory with Ehrenfest nuclei propagates the collision, in the real-space codes Octopus [3] and SALMON (Scalable Ab initio Light–Matter simulator for Optics and Nanoscience) or the plane-wave code Qb@ll. Cells hold 216 atoms (16.3 Å across) or more; the projectile is launched along random directions; and the excitation is counted by projecting the time-evolved orbitals on the conduction manifold of the ground state about 20 fs after the collision, 20 to 50 trajectories per point. The time step is at most

2 as near closest approach, the real-space grid at most 0.2 Å (or the equivalent plane-wave cutoff), and the pseudopotentials include semicore states and are hard enough for the close approach at $E_{\mathrm{coll}} \geq 1\,\mathrm{keV}$. The cell has to be at least 15 Å across, so that the wake does not re-enter; the 216-atom cell on a 0.2 Å grid has $5.4 \times 10^5$ points, and its 432 occupied orbitals take about 3.7 GB. A trajectory of about 30 fs costs some 250 core-hours, and the 1600 to 4000 trajectories of the grid about $10^6$.

The adiabatic exchange–correlation functional misplaces level crossings that involve charge transfer, so the gap of the distorted configuration comes out too small. The theory uncertainty is therefore the spread between three treatments of a 10 % subset of trajectories, which costs a few times $10^6$ core-hours more because the hybrid functional is some thirty times dearer per step: Ehrenfest with a hybrid functional (HSE), decoherence-corrected surface hopping [4], and Landau–Zener probabilities from diabatic couplings of the $Si_2$ or $Ge_2$ quasi-molecule computed by multireference configuration interaction. The hybrid arm serves silicon only: in germanium HSE06 fails the band check of Sect. 31.1, and the germanium spread rests on the other two treatments. Ehrenfest dynamics gives the mean excitation, not its distribution, so the kernel is built from the projection statistics, never from the mean electronic energy. Above $E_{R,\mathrm{th}}$ the mean energy lost per unit path must approach the electronic stopping that the loss function gives (Sect. 19.3).

### 32.1.2 Partition tables

The calculation tabulates the partition of Sect. 19.1 into heat $E_{\mathrm{heat}}$, all of the phonon energy, stored energy $E_{\mathrm{st}}$ and charge, as the joint distribution $P(E_{\mathrm{heat}}, E_{\mathrm{st}}, N_{\mathrm{eh}} | E_R, \hat{\boldsymbol{r}}_R, T)$. The grid, for silicon and germanium, is $E_R$ from 1 eV to 10 keV (40 logarithmic points), 100 recoil directions, and 100 initial conditions per direction, drawn from the zero-point and thermal distribution of the lattice. It also gives the threshold surface $E_{\mathrm{d}}(\hat{\boldsymbol{r}}_R)$ and the Frenkel-pair probability $p_{\mathrm{FP}}(E_R, \hat{\boldsymbol{r}}_R)$, and the spectrum of the phonons the recoil creates. The dynamics sits on $\mathfrak{T}$ where Sect. 29.1 places it, at $n_B = 0$ with classical nuclei, and is controlled between the quantum–classical boundary at 0.41 eV and $E_{\mathrm{d}}$, 12.5 to 36 eV by direction, where $\xi_{\mathrm{M}} \gg 1$; above, the close-collision kernel is attached as stochastic events.

The potential is a machine-learned interatomic potential (MLIP) of the fast linear atomic cluster expansion (ACE) family, spliced to the screened Coulomb repulsion as in Sect. 29.1. It is trained on about $10^4$ configurations computed with PBE and SCAN: dimers down to 0.5 Å, point defects, close pairs, amorphous and liquid structures, and snapshots of recoils. It is accepted at force errors below 50 meV/Å and energy errors below 3 meV per atom. $E_{\mathrm{d}}$ and $p_{\mathrm{FP}}$ follow its protocol, with its directions, energy steps, cell, border and run length, with this potential. What differs is the quench at $t_{\mathrm{pr}}$ below and a check of twenty directions by Born–Oppenheimer molecular dynamics in density-functional theory, 512 atoms, about $10^5$ core-hours. The partition tables use cells from $10^3$ atoms below 100 eV to $10^5$ atoms at 10 keV, run for 1 to 10 ps, and cost $10^4$ to $10^5$ hours on graphics processors per material.

The stored energy is defined by the quench of Sect. 29.1, Eq. (29.6), at a time $t_{\mathrm{pr}}$ of 10 to 100 ps after the recoil, by steepest descent or the fast inertial relaxation engine (FIRE) [5] at fixed cell, without annealing. Its reference is the relaxed perfect cell, computed once for each potential, cell and boundary condition, and never the initial configuration of the run. The initial atoms are drawn from the zero-point distribution, and in a 4096-atom cell of silicon they lie 126 eV above the minimum (Sect. 19.1), 17 to 19 times the 6.8 to 7.5 eV that a close Frenkel pair stores. The change of zero-point energy between the end basin and the perfect crystal comes from the Hessians at both minima; it is booked in the heat, not in $E_{\mathrm{st}}$. One table of configurational energies, with one potential, cell and quench, serves the stored energy here, the end states of the release paths of Sect. 30.5 and the pockets below; otherwise the

end states and the stored energies disagree by the potential's error on each defect. The spread of its entries between the potential and density-functional theory is the potential's uncertainty on $E_{\mathrm{st}}$.

The created-phonon spectrum is recorded at $t_{\mathrm{pr}}$, before the quench. The excess lattice state is projected on the harmonic modes $s$ of the quenched end state, $E_s = \frac{1}{2}(p_s^2 + \omega_s^2 u_s^2)$, and the same projection of the initial state is subtracted to remove the zero-point spectrum. The result is binned at 0.5 meV, with branch labels from the overlap of eigenvectors with those of the perfect crystal. The energy that the projection misses, the anharmonic residual, must stay below 10 %. The energy absorbed by the boundary thermostat is recorded with its spectrum, from the flux into the thermostat layer, or the spectrum is taken before the front reaches it. The acceptance test for the spectrum is the hand-over of Sect. 19.1: for recoils small enough to stay harmonic, the created minus absorbed spectrum must reproduce the phonon spectrum of the nuclear response of Sect. 11.3.

### 32.1.3 Charge yield below the binary threshold

The measured silicon yields at 100 to 460 eV straddle $E_{R,\mathrm{th}}$. The calculation gives the full $P(N_{\mathrm{eh}}|E_R, \hat{\boldsymbol{r}}_R, T)$ for $N_{\mathrm{eh}} \leq 30$ on $E_R$ from 20 eV to 1 keV, with at least 25 logarithmic points, dense near 352 and 399 eV. From it follow the yield $Y_q(E_R) = \langle N_{\mathrm{eh}} \rangle \varepsilon_{\mathrm{eh}} / E_R$, the probability $P(N_{\mathrm{eh}} = 0|E_R)$ of a recoil without charge, the Fano factor $F_{\mathrm{NR}}(E_R)$ of the nuclear-recoil charge, and the correlations of $N_{\mathrm{eh}}$ with the heat and with $E_{\mathrm{st}}$ at fixed $E_R$, which Sect. 19.4 needs.

The trajectories are those of the partition tables, with excitation events drawn from the close-collision kernel at every close approach and the electronic friction of the potential switched off where the kernel acts. Three channels are flagged separately: binary excitation above $E_{R,\mathrm{th}}$, Landau–Zener promotion, and linear-response excitation through the high-momentum tail of the valence density, which makes the binary threshold soft (Sect. 19.2). Each excitation $\omega_e$ is converted into pairs with the cascade distribution $P_{\mathrm{casc}}$ at that seed (Sects. 29.3 and 32.2). Slow-ion excitations lie near the gap, where the cascade is far from its asymptote, so dividing $\omega_e$ by the high-energy $\varepsilon_{\mathrm{eh}}$ is wrong. The Fano factor follows from the law of total variance, $\mathrm{Var}\, N_{\mathrm{eh}} = \mathbb{E}[\mathrm{Var}(N_{\mathrm{eh}}|\mathrm{seeds})] + \mathrm{Var}(\mathbb{E}[N_{\mathrm{eh}}|\mathrm{seeds}])$, which separates the fluctuations of the cascade from those of the excitation events.

The number of trajectories per point is set by the variance. For a sample of $n$ values with excess kurtosis $\kappa$, the relative error of the sample variance is $[2/(n-1) + \kappa/n]^{1/2}$; for 6 % at $\kappa = 2$ this needs $n \geq 1112$. The cost lies within the molecular-dynamics budget of the partition tables, about $10^3$ to $5 \times 10^3$ hours on graphics processors. The result is validated, not fitted, against the silicon data: $Y_q = 0.102^{+0.034}_{-0.030}$ and $F_{\mathrm{NR}} = 0.9^{+0.7}_{-0.4}$ at 0.10 keV, $0.108^{+0.009}_{-0.010}$ and $0.5^{+0.2}_{-0.1}$ at 0.22 keV, $0.136^{+0.009}_{-0.008}$ and $1.8^{+0.6}_{-0.5}$ at 0.46 keV [1, Table II], and against the germanium point at 254 eV [2]. The silicon yields assume 3.8 eV per pair and a total phonon energy equal to the recoil energy plus the work of the pairs in the bias field, so that no energy is stored [1, Eq. (1)]. The computed yield is converted to the same definition before the comparison. The stored energy, at most a few percent of $E_R$ (Sect. 19.4.4), lies inside their errors.

### 32.1.4 Dense tracks

The cascade of Sect. 18.1 treats the pairs of one event as independent, each subtree evolving alone. At keV, a nuclear-recoil track packs its carriers so densely that they interact: the independence is the zeroth order of an expansion in the pair density times the exciton volume, $n_{\mathrm{pair}} a_x^3$, and the first correction is a carrier–carrier collision term. In germanium

the lattice itself can melt locally in a thermal spike, beyond the description of the ion as a sequence of binary collisions. The calculation gives the ratio of $P(N_{\mathrm{eh}}|E_R)$ with and without interactions and the change of $F_{\mathrm{NR}}$, per recoil energy and species. The ratio is the error of the independence hypothesis at keV, which Sect. 18.1 leaves unestimated; the change of $F_{\mathrm{NR}}$ is the excess variance that Sect. 18.5.2 expects when excitons form from carriers of different pairs.

The carriers are followed by ensemble Monte Carlo with molecular-dynamics Coulomb forces among all of them, computed by the particle–particle particle–mesh method, seeded with the end-of-cascade distribution and the geometry of the track, with screened impact ionization and Auger recombination as extra channels. The lattice is followed by machine-learned molecular dynamics of the recoil coupled to a two-temperature model of the electrons [6], with the electron–phonon coupling from the vertex of Sect. 27.2. The time step is 0.1 fs, the Coulomb mesh at most 1 nm, the electron-temperature grid at most 1 nm, and at least $10^3$ tracks are run per energy and species. The dynamics costs about $10^4$ hours on graphics processors and the carriers $10^3$ core-hours. In the dilute limit, with the carrier density scaled down, the calculation must recover the independent cascade. Three errors must be avoided: the electronic stopping friction of the potential and the close-collision kernel must not both act; carriers that leave the molecular-dynamics box must be continued in the cascade, not dropped; and the thermostat must not absorb the heat of the spike before it has spread.

### 32.1.5 Track recombination

A dense track is a plasma of electrons and holes that shields an applied field from its interior and loses carriers by Auger recombination, at a rate $C_{\mathrm{Aug}} n^3$ per volume. Take a cylindrical Gaussian track of $\lambda_{\mathrm{tr}}$ pairs per length and initial radius $r_{\mathrm{tr}}$, spreading by ambipolar diffusion with constant $D_{\mathrm{amb}}$ at zero field. Its density is $n = \lambda_{\mathrm{tr}} \mathrm{e}^{-\rho^2/s}/\pi s$ with $s = r_{\mathrm{tr}}^2 + 4 D_{\mathrm{amb}} t$, and $\int \mathrm{d}^2\rho\, n^3 = \lambda_{\mathrm{tr}}^3/3\pi^2 s^2$. To first order in $C_{\mathrm{Aug}}$ the fraction lost is

$$f_{\mathrm{A}} = \frac{C_{\mathrm{Aug}} \lambda_{\mathrm{tr}}^2}{3\pi^2} \int_0^\infty \frac{\mathrm{d}t}{(r_{\mathrm{tr}}^2 + 4 D_{\mathrm{amb}} t)^2} = \frac{C_{\mathrm{Aug}} \lambda_{\mathrm{tr}}^2}{12 \pi^2 D_{\mathrm{amb}} r_{\mathrm{tr}}^2} . \tag{32.1}$$

The loss is set at early times, as $1/r_{\mathrm{tr}}^2$, so the initial radius from the dense-track and pocket calculations matters more than the late diffusion. In a biased detector the plasma screens the field until diffusion erodes it; the time this takes, the plasma time, delays the induced current.

The full calculation solves drift–diffusion and Poisson equations with rate terms for Auger recombination, exciton formation and Shockley–Read–Hall capture on the geometry of the track, two-dimensional with axial symmetry, with a mesh of at most 1 nm near the core and fields from 0 to 250 V/cm. It gives $f_{\mathrm{A}}(E_R, \mathcal{E}_{\mathrm{b}})$, with $\mathcal{E}_{\mathrm{b}}$ the bias field, the plasma time and the delay of the induced current, for under $10^3$ core-hours, and it must reproduce Eq. (32.1) for $f_{\mathrm{A}} \ll 0.1$; above $f_{\mathrm{A}} \approx 0.1$ the loss is nonlinear and only the full solution holds. The weak input is $C_{\mathrm{Aug}}$, which has not been measured at millikelvin in silicon or germanium; it is carried as a stated parameter, scanned from a tenth to ten times its room-temperature value. Exciton formation in the track does not by itself remove gap energy, since a bound pair still holds $E_{\mathrm{g}} - E_x^b$; only loss channels, Auger recombination and exciton–exciton annihilation, do. The computed loss and its field dependence feed the dense-track charge of Sects. 18.5 and 19.4 and the kinetics of Sect. 32.3.

### 32.1.6 Amorphous pockets

Recoils of keV in silicon and germanium leave small amorphous pockets as well as point defects. A pocket that recrystallizes later releases its stored energy at once, and Sect. 19.5 counts such

Table 32.1: Nuclear-recoil calculations for silicon and germanium. Costs are estimates. GPU-h are hours on graphics processors; TDDFT, time-dependent density-functional theory; MLIP-MD, molecular dynamics with a machine-learned interatomic potential; MC, Monte Carlo.

| quantity | method and grid | acceptance | cost |
|---|---|---|---|
| $\mathrm{d}P/\mathrm{d}\omega_e(b_{\mathrm{imp}}, E_{\mathrm{coll}})$ | real-time TDDFT on 216 atoms, step 2 as, grid 0.2 Å; $10 \times 8$ points from 50 eV to 5 keV | stopping above $E_{R,\mathrm{th}}$; HSE (Si only), surface hopping, Landau–Zener spread | $10^6$ core-h; hybrid subset a few $10^6$ |
| $P(E_{\mathrm{heat}}, E_{\mathrm{st}}, N_{\mathrm{eh}})$, $E_{\mathrm{d}}$, $p_{\mathrm{FP}}$, phonon spectra | MLIP-MD, 40 energies $\times$ 100 directions $\times$ 100 seeds | DFT on 20 directions; harmonic hand-over | $10^4$–$10^5$ GPU-h |
| $Y_q$, $F_{\mathrm{NR}}$, $P(N_{\mathrm{eh}} = 0)$ | MD with excitation events, 20 eV–1 keV, $\geq 1112$ runs per point | silicon data at 0.10, 0.22, 0.46 keV | $10^3$–$5 \times 10^3$ GPU-h |
| Dense-track ratio, $\Delta F_{\mathrm{NR}}$ | carrier MC with Coulomb forces; MD with two-temperature model | dilute limit | $10^4$ GPU-h |
| $f_{\mathrm{A}}$, plasma time | drift–diffusion–Poisson, mesh $\leq 1$ nm | Eq. (32.1) for $f_{\mathrm{A}} \ll 0.1$ | $10^3$ core-h |
| $P(N_a\|E_R)$, barrier spectrum | MLIP-MD, 0.2–20 keV, $10^5$–$10^6$ atoms | two cluster cuts; energy closure | $10^4$–$10^5$ GPU-h |

releases among the candidate sources of the low-energy excess. A release is visible only if its energy, after collection, exceeds the threshold; with a recrystallization energy of order 0.1 eV per atom, an estimate, a threshold of tens of eV needs pockets of hundreds of atoms. The calculation gives the size distribution $P(N_a|E_R, \hat{\boldsymbol{r}}_R)$ of the number $N_a$ of atoms in pockets at $t_{\mathrm{pr}}$, and the spectrum of recrystallization barriers.

The calculation sits at the same place on $\mathfrak{T}$ as the partition tables. The cascades are run with the spliced machine-learned potential at $E_R$ from 0.2 to 20 keV, at least $10^3$ per energy and material, in cells of $10^5$ to $10^6$ atoms for 10 to 50 ps with a boundary thermostat, and quenched as above. A pocket is a connected set of atoms whose local order parameter, a bond-angle distribution or a Steinhardt parameter, falls below a crystalline cut. The barrier spectrum comes from nudged elastic bands and accelerated molecular dynamics on about 100 pockets. The cost is $10^4$ to $10^5$ hours on graphics processors. The definition of a pocket changes $N_a$ by a factor of about two, so the distribution is reported for two cuts. Thermal spikes in germanium need the two-temperature coupling of the dense-track calculation. The pocket energies come from the configurational table above, so that pockets and point defects add up to the stored energy of the partition tables. The size distribution and barriers give the pocket term of the delayed-release spectrum of Sect. 19.5.

## 32.2 Electron cascades

For most of the materials ranked in Chap. 17 no first-principles cascade exists. The method is the cascade of Sect. 29.3; Table 32.2 collects the grids, tests and costs of the calculations that apply it.

### 32.2.1 Silicon, germanium and GaAs

The output is the distribution $P_n(\boldsymbol{q},\omega,T) = P(N_{\mathrm{eh}} = n|\boldsymbol{q},\omega,T)$ of the pair number for a deposit of energy $\omega$ at momentum transfer $\boldsymbol{q}$, the distribution that Sect. 18.1 builds from the branching process and Sect. 18.4 compares with measurements. Its moments are $\varepsilon_{\mathrm{eh}}(\boldsymbol{q},\omega,T)$ and $F_{\mathrm{eh}}(\boldsymbol{q},\omega,T)$ of Eq. (29.10), and the variance is also reported split into the terms $\mathcal{V}_<$, $\mathcal{V}_>$ and $\mathcal{V}_{<>}$ of Sect. 18.2, from below and above the impact-ionization threshold and their covariance. For silicon the run also records the energies of the carriers at the end of the cascade and the separations of the pairs they form, which Sect. 18.5 compares with the field-separation radius $r_{\mathcal{E}}$ and over which the capture probability of the last subsection is averaged. The grid is $\omega$ from $E_{\mathrm{g}}$ to 50 eV in steps of 50 meV, $|\boldsymbol{q}|$ from 0 to $25\alpha m_e$ in steps of $\alpha m_e$ along three directions plus the angular average, and five temperatures, 0, 50, 100, 150 and 300 K. Above 50 eV the moments are matched to their asymptotic forms, a mean $\omega/\varepsilon_{\mathrm{eh}}$ plus a constant offset and a variance $F_{\mathrm{eh}}\omega/\varepsilon_{\mathrm{eh}}$. On $\mathfrak{T}$ (Sect. 8.5) the bands are $G_0W_0$ quasiparticles, the screening is RPA with local fields ($\mathcal{X}$ = RPA), the phonon kernel is at $n_g = 2$ and $n_B = 0$, and the events are incoherent (Sect. 29.3). Temperature enters three times: through the occupation factors of the phonon kernel, through the Bose factor $1 + n_{\mathrm{B}}(\omega)$ of the pair or plasmon excited at energy transfer $\omega$ in the impact-ionization kernel, and through the temperature-renormalized gap (Sect. 25.3).

The bands come from Quantum ESPRESSO and $G_0W_0$ in BerkeleyGW [7] or Yambo [8], Wannier-interpolated up to about 15 eV above the conduction minimum. Silicon needs 16 to 20 disentangled $sp^3d$ Wannier functions; germanium and GaAs need about as many, with spin–orbit coupling. A germanium band structure enters only if it passes the check of Sect. 31.1: L below Γ, both gaps within 0.1 eV of the measured ones, and the transverse mass at L and the mass at Γ against measurement. The germanium bands are $G_0W_0$ with spin–orbit coupling on a PBE or hybrid ground state. Above 15 eV the states are taken as free-electron-like with $GW$ lifetimes, which is where the free-particle form of the single-parameter model is accurate, and the matching point serves as a check. The screened interaction is the full-frequency $\varepsilon^{-1}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{p},\omega)$ for $\omega$ from 0 to 60 eV, on about 200 frequencies that are denser below 20 eV; carriers above 60 eV, up to the energy cap, use the free-electron kernel of Sect. 29.3, so the grid of $W^{\mathrm{scr}}$ need not reach the cap. It uses $10^3$ to $12^3$ momenta $\boldsymbol{p}$, $N_G \approx 200$ to 300 and at least 200 empty bands, and $\mathrm{Im}\,\varepsilon^{-1}$ at the plasmon, whose loss peak lies at 16.4 eV in the optical loss function of silicon (Sect. 4.4), is converged to 2 %. For GaAs the total dielectric function includes the ionic part built from the measured $\varepsilon_\infty$ and the measured LO–TO splitting, which gives $Z^*/\varepsilon_\infty = 0.194$ (Sects. 5.4 and 11.9); a semilocal $\varepsilon_\infty$ of GaAs is not a safe input. The impact-ionization kernel of Eq. (29.11) is tabulated with local fields on iso-energy surfaces, about $10^3$ $\boldsymbol{k}$ points per 50 meV bin. The electron–phonon kernel comes from EPW [9], with $8^3$ coarse grids, fine $\boldsymbol{k}$ grids of $80^3$ to $100^3$ and $50^3$ phonon wavevectors sampled by Monte Carlo. GaAs needs the long-range dipole and quadrupole terms, and every material needs the same vertex and phonons as the response code (Sect. 27.2).

The pair number is computed in two stages. The first generation, the primary carriers and the products of their first impact ionization, keeps its full $\boldsymbol{k}$ resolution, because the reduction to energy alone is least accurate there; it is seeded with the final states of the dark-matter response (Sect. 18.3). Later generations use the energy-only reduction of Sect. 18.1, solved deterministically by the moment equations on an energy grid of bin $h$, with kernels averaged over iso-energy surfaces. For $N_s$ sampled first steps with subtree means $M_i$ and variances $V_i$, the estimator is $\bar{n} = 1 + \mathrm{mean}_i\, M_i$ and $\mathrm{Var}\, N = \mathrm{mean}_i\, V_i + \mathrm{var}_i\, M_i$, the law of total variance, with errors from 20 batch means. The relative error of $\mathrm{var}_i\, M_i$ is $(2/N_s)^{1/2}$ for a near-Gaussian $M$, so a Fano factor to 0.5 % needs at most $2/0.005^2 = 8 \times 10^4$ samples; the run starts at $10^4$ and doubles until the batch errors fall below 0.5 % on $F_{\mathrm{eh}}$ and 0.1 % on $\varepsilon_{\mathrm{eh}}$. The plasmon

part of $\mathrm{Im}\, W^{\mathrm{scr}}$ is tagged and its share of the variance reported (Sect. 29.3).

The grid is converged when the bin, the energy cap, the matching point, the fit window and the $\boldsymbol{k}$ mesh each pass. Halving $h$ from at most 10 meV changes $\varepsilon_{\mathrm{eh}}$ by less than $10^{-3}$ and $F_{\mathrm{eh}}$ by less than $3 \times 10^{-3}$, relative. Raising the energy cap from 100 to 150 eV, and the free-electron matching point from 15 to 20 eV, change both by less than $10^{-3}$. The asymptotic slopes fitted over $[X/2, X]$ and over $[3X/4, X]$, for a top energy $X$, agree to $10^{-4}$. A second $\boldsymbol{k}$ mesh for the iso-energy averages changes both by less than $10^{-3}$. Energy is conserved along every path to $10^{-12}$, and $F_{\mathrm{eh}}$ computed from the pair number and from the heat agree to $10^{-9}$. Before the full kernels are switched on, the code must reproduce, with free-particle kernels in their place, the single-parameter model of Sect. 29.3 within its Monte Carlo scatter of about 2 %. The validation, not used in any fit, is the silicon measurement $\varepsilon_{\mathrm{eh}} = (3.749 \pm 0.001)$ eV and $F_{\mathrm{eh}} = 0.119 \pm 0.002$ at 123 K [10] and the silicon quantum yield of photons between 3 and 8 eV and between 50 and 1500 eV [11], with the 50 to 1500 eV data run at 300 K, their measurement temperature. For germanium no measurement at 77 K uses the same definition of the pair energy; until one does, the germanium value is an input, not a test. A photon seed at $q \approx 0$ and a dark-matter seed at $q$ of 1 to 10 keV, run in one code, give the $q$ dependence of the yield directly.

The cost per material is $10^4$ to $10^5$ core-hours for the screened interaction, as much for the impact-ionization kernel, and about $10^4$ for the phonon kernel; the three materials together need $10^5$ to $6 \times 10^5$ core-hours. Memory is set by $W^{\mathrm{scr}}$: at $N_G = 300$, 200 frequencies and the 72 irreducible points of a $12^3$ grid, $72 \times 300^2 \times 200 \times 16$ byte $\approx 21$ GB. Beyond the pitfalls of Sect. 29.3, each phonon step must be an integer number of energy bins to better than 1 %, either by a bin commensurate with the dominant phonon or by interpolating the other branches, since an off-grid step biases every phonon loss the same way. The quasiparticle residue enters once, in the rate or in the weight, never in both. Kohn–Sham and quasiparticle energies must not be mixed between kernels. $F_{\mathrm{eh}}$ is a small difference of large terms (Sect. 18.2), so the three variance terms are reported with their errors. The results replace the model values of Sect. 18.4 for the three materials and fix the $q$ dependence of the yield used in Sect. 18.1.

### 32.2.2 Wide-gap and polar crystals

Sapphire, $CaWO_4$, LiF and NaF, among the materials ranked in Chap. 17, have no measured Fano factor and no validated pair energy, and Sect. 18.4 shows that they lie outside the range in which the single-parameter model was validated. These four come first; 4H-SiC, diamond, 3C-SiC, GaN, AlN and $TeO_2$ follow. The quantities and grids are those of the silicon cascade. Polar crystals add the long-range Fröhlich vertex, built with the same Born charges and dielectric constants as the response code (Sects. 25.2 and 27.2). LiF and NaF have excitons bound by an eV or more, so the Bethe–Salpeter equation of Sect. 26.1 is solved as well. The bookkeeping must then state whether a bound exciton counts as a pair. A carrier between $E_{\mathrm{g}} - E_x^b$ and the ionization threshold can also create a bound pair directly, and the cascade must include that channel. Tungsten in $CaWO_4$ needs spin–orbit coupling.

The cells are small, two atoms for LiF and NaF, 10 for sapphire and 12 for the primitive cells of $CaWO_4$ and $TeO_2$, but the oxides have many bands. The cost is estimated at about $10^5$ core-hours for each oxide, $10^4$ to $10^5$ for each halide plus the Bethe–Salpeter step, and twice the silicon cost for diamond, the silicon carbides, GaN and AlN; the set needs $6 \times 10^5$ to $2.5 \times 10^6$ core-hours. The silicon validation comes first, because silicon alone has a measurement at the precision that tests a Fano factor, and the rest are run after it passes, reusing its kernels. Their benchmark is the full-band Monte Carlo with density-functional inputs, which gives 7.3 eV against the measured 7.8 eV for 4H-SiC, 6.6 eV against 6.9 eV for 3C-SiC and 16.7 eV

against 18.0 eV for $SiO_2$ [12, Table II], a few to 15 % below measurement where measurements exist. The computed pair energies and the first ab initio Fano factors of these materials replace the estimates of Sect. 18.4.

### 32.2.3 Impact ionization beyond first order

Sect. 29.3 gives two definitions of the impact-ionization rate, from the imaginary part of the $GW$ self-energy with its residue $z_{\mathrm{qp}}$ and from the golden rule with exchange, which differ by a factor between 0.5 and 1. A constant factor does not matter once a model is calibrated on silicon. In the single-parameter model a factor $z_{\mathrm{qp}} = 0.8$ on the rate raises the fitted phonon-to-ionization constant $A_{\mathrm{AB}} = 5.703\,\mathrm{eV}^3$ by 25 %, $\Delta A_{\mathrm{AB}} = 1.43\,\mathrm{eV}^3$, and with the secant slope $\partial\varepsilon_{\mathrm{eh}}/\partial A_{\mathrm{AB}} = 0.076\,\mathrm{eV}^{-2}$ of Sect. 18.4 it raises $\varepsilon_{\mathrm{eh}}$ by 0.11 eV, 2.9 %, which recalibration removes. A factor that varies with energy cannot be removed, and it changes $F_{\mathrm{eh}}$, which depends on how the rate rises above threshold (Sect. 18.5). We compute the ratio $r_{\mathrm{ii}}(E)$ of the impact-ionization rate beyond first order in $W^{\mathrm{scr}}$ to the first-order rate, on 50 meV bins from threshold to about 10 eV, for silicon.

The calculation evaluates $\mathrm{Im}\,\Sigma_{n\boldsymbol{k}}$ in $G_0W_0$ with and without $z_{\mathrm{qp}}$, and adds the second-order screened exchange as the first vertex correction beyond $GW$. Vertex corrections of this kind shift the absolute band energies of solids by tenths of an eV [13], and the question here is how much they change the rate. The grids are those of the cascade: $N_G \approx 250$, 200 empty bands, the full-frequency $\varepsilon^{-1}$ and $10^3$ $\boldsymbol{k}$ points per bin. The ratio is converged when it changes by less than 2 % per bin with $N_G$ raised to 350, the bands to 400 and the $\boldsymbol{k}$ points to $2 \times 10^3$; the second-order exchange is reported separately to the same precision. A variation below 2 % across the window counts as constant. The cascade is then rerun with $\Gamma_{\mathrm{ii}} \to r_{\mathrm{ii}}(E)\Gamma_{\mathrm{ii}}$, and the change of $F_{\mathrm{eh}}$, for about $3 \times 10^4$ core-hours in all, is the uncertainty that the order of the rate adds to $F_{\mathrm{eh}}$ in Sect. 18.5.

### 32.2.4 Momentum memory between events

The energy-only reduction of Sect. 18.1 assumes that a carrier forgets its momentum on its iso-energy surface between two events that change its energy. Suppose momentum-randomizing scattering, by intervalley and acoustic phonons, is a Poisson process of rate $\Gamma_k$ that leaves the carrier uniform on the surface, while energy-changing events occur at rate $\Gamma_E = \Gamma_{\mathrm{ii}} + \Gamma_{\mathrm{ph,loss}}$. The probability that the next event changes the energy before any randomization is

$$w_{\mathrm{mem}}(E) = \frac{\Gamma_E}{\Gamma_E + \Gamma_k}, \tag{32.2}$$

the ratio of competing Poisson rates. At the next energy-changing event the momentum distribution is then $1 - w_{\mathrm{mem}}$ times the uniform one plus $w_{\mathrm{mem}}$ times the remembered one. For any cascade observable $m_{\boldsymbol{k}}(E)$ that depends on the starting $\boldsymbol{k}$, such as the mean pair number or its square, the error of the reduction per generation is therefore at most $w_{\mathrm{mem}}(E)$ times the spread of $m_{\boldsymbol{k}}$ over the surface, $\max_{\boldsymbol{k}} m_{\boldsymbol{k}} - \min_{\boldsymbol{k}} m_{\boldsymbol{k}}$. Within about 1 eV of threshold, intervalley scattering at $10^{13}$ to $10^{14}\,\mathrm{s}^{-1}$ dominates and $w_{\mathrm{mem}}$ is small. Above 3 to 5 eV in silicon, impact ionization outpaces the phonons and $w_{\mathrm{mem}}$ can reach one half, so the bound is useful only if the spread is small.

The calculation gives the spread of the mean pair number and of its second moment, and $w_{\mathrm{mem}}(E)$, for silicon between 3 and 8 eV on 50 meV bins. They come from the $\boldsymbol{k}$-resolved second generation of the cascade, with at least 200 $\boldsymbol{k}$ samples per bin, weighted by area on the surface, and at least $10^3$ cascades per sample. $\Gamma_k$ comes from EPW, $\Gamma_E$ from the impact-ionization and phonon rates of the same run. The spread is converged to 10 % when

the $\boldsymbol{k}$ samples are doubled to 400 and the cascades quadrupled, for $10^3$ to $10^4$ core-hours. The maximum minus the minimum of $n_k$ noisy estimates is biased upward by about $2\sigma(2\ln n_k)^{1/2}$, with $\sigma$ the noise of one estimate, so the range between the 5th and 95th percentiles is used, or the noise is subtracted. The bound rests on the assumption that randomization leaves the carrier uniform on the surface, itself a form of the energy-only reduction; it bounds the error under that assumption and is not an independent test of it.

### 32.2.5 A library of pair-number distributions

The response libraries of Sect. 33.1 need, for every final-state class $\Lambda$ of every material, not just the moments but the full distribution $P_{\mathrm{casc}}(N_{\mathrm{eh}}|\Lambda)$ (Sect. 18.3). The reduced cascade gives it cheaply. With energy-only branching, random final momenta for impact ionization, one phonon energy $\omega_{\mathrm{ph}}$ and one phonon-to-ionization constant, the generating function $\mathcal{G}_\Lambda(z) = \sum_N z^N P_{\mathrm{casc}}(N|\Lambda)$ obeys the recursion of Sect. 18.1 in the energies $x_e$ and $x_h$ of the seed electron and hole above their band edges. We evaluate it at the $N_z$ roots of unity $z_j = \mathrm{e}^{2\pi \mathrm{i} j/N_z}$. The discrete Fourier sum

$$P_{\mathrm{casc}}(N|\Lambda) = \frac{1}{N_z} \sum_{j=0}^{N_z-1} z_j^{-N}\, \mathcal{G}_\Lambda(z_j), \tag{32.3}$$

returns $\sum_{N'} P_{\mathrm{casc}}(N + N'N_z|\Lambda)$. That is the exact probability as long as $N_z$ exceeds every pair number with appreciable weight, so $N_z$ is taken as the smallest power of two above $\bar{N} + 8\sigma_N$, with $\sigma_N$ the standard deviation of the pair number.

The energy grid has a step $h$ of about $E_{\mathrm{g}}/50$, chosen so that it divides $E_{\mathrm{g}}$, $\theta_{\mathrm{ii}}E_{\mathrm{g}}$ and $\omega_{\mathrm{ph}}$. With phonons the discretization error falls as $h^{3/2}$, so the values at $h$ and $h/2$ are extrapolated to zero step as $f(h/2) + [f(h/2) - f(h)]/(2^{3/2} - 1)$ (Appendix E.13); the $h^2$ formula would leave a bias. The energies run to $20E_{\mathrm{g}}$, and beyond the table the distribution is the asymptotic lattice Gaussian of mean $\bar{N}$ and variance $\sigma_N^2$, sampled at the integers, since integrating it over unit bins adds $1/12$ to the variance, with the Edgeworth terms added. The Gaussian is approached slowly: with $x$ the carrier energy in units of $E_{\mathrm{g}}$, the absolute pointwise error falls as $C/x$ and the relative skewness as $x^{-1/2}$, with $C \approx 0.24$ for silicon, so the match at $20E_{\mathrm{g}}$ carries an error of about $1.2 \times 10^{-2}$ in each probability. The tests are internal: the probabilities sum to one to machine precision and are non-negative; the derivatives of $\mathcal{G}_\Lambda$ at $z = 1$ reproduce the moment recursion to $10^{-9}$; and below the first impact-ionization threshold $P_{\mathrm{casc}}(N|\Lambda) = \delta_{N1}$. The reader that interpolates the tables must return that exact value below $\theta_{\mathrm{ii}}E_{\mathrm{g}}$ and interpolate only from the threshold upward, with a table row at the threshold itself: for $\theta_{\mathrm{ii}} > 1$ the distribution jumps there, and an interpolation across the jump puts weight on extra pairs below threshold.

The same tables carry the marks $\mathfrak{m}$ of Eq. (18.19) that a readout needs (Sect. 33.1): the phonon spectrum and the separate electron and hole counts, which follow from the same recursion with one further generating variable for each, the carrier positions, taken from the ranges between events in the $\boldsymbol{k}$-resolved cascade of silicon, and the stored energy, taken from the table of configurational energies of Sect. 32.1. A table takes seconds per material and parameter set, so five parameter sets for every ranked material take minutes on one core. The phonon step must be commensurate with the energy grid. Rounding $\omega_{\mathrm{ph}}$ to the grid changes it by up to $h/2$; with $h = 0.02E_{\mathrm{g}}$ in silicon that is up to 12 meV of a 63 meV phonon, and every phonon loss of the cascade inherits the error. Two smaller ones are aliasing in $z$, which Eq. (32.3) avoids only if $N_z$ is large enough, and the seed dependence of the constant offset of $\bar{N}$. A seed that gives all the energy to one carrier and a seed that shares it have different offsets. Temperature beyond its effect on the gap, and excitons, are outside the reduced model.

When the full-band kernels of the silicon cascade exist, they replace the free-particle forms in the same recursion.

### 32.2.6 Exciton and capture parameters

At millikelvin the pairs at the end of the cascade bind into free excitons, bind to impurities, are trapped, or recombine, and Sect. 18.5 writes these processes as rate equations whose rates are unknown in silicon and germanium. The exciton itself needs its binding energy $E_x^b$, radius $a_x$ and oscillator strengths, and its binding to neutral donors and acceptors. Capture needs the cross-section $\sigma_{\mathrm{cap}}(E, T, \mathcal{E}_{\mathrm{b}})$ of a carrier of energy $E$ on shallow centers in a bias field $\mathcal{E}_{\mathrm{b}}$, and multiphonon capture on deep levels. The kinetics needs the rates of exciton formation, trapping of free carriers, binding of excitons to each impurity species and decay of bound excitons, the luminescence fractions, the bimolecular formation coefficient, exciton–exciton annihilation and Auger recombination, all at 10, 20 and 50 mK. The targets are silicon and germanium, then LiF, NaF, sapphire and $CaWO_4$.

On $\mathfrak{T}$ (Sect. 8.5) the exciton parameters sit at $\mathcal{X}$ = BSE on quasiparticle bands, the bound excitons at effective-mass level with a density-functional central cell, and the capture rates at $n_g = 2$. The exciton parameters come from $G_0W_0$ and the Bethe–Salpeter equation (Sect. 26.1) in BerkeleyGW or Yambo, which in silicon and germanium (for germanium on quasiparticle bands that pass the band check) checks the effective-mass binding of Sect. 31.2, with 4 to 8 valence and 4 to 8 conduction bands. In silicon and germanium the exciton radius spans many cells, and the $\boldsymbol{k}$ grid must resolve it: a spacing below $1/(3a_x)$ means $10^4$ to $10^5$ points, reached by double-grid interpolation, against $8^3$ to $12^3$ for the ionic crystals. When $E_x^b$ approaches the longitudinal optical phonon energy, the lattice part of the screening enters through a polaronic correction. Donor- and acceptor-bound excitons come from effective-mass theory with central-cell corrections from density-functional calculations in 216- to 512-atom cells. Capture on shallow centers follows cascade-capture theory [14], a carrier captured if it emits phonons inside the Coulomb radius, with acoustic and optical emission rates from EPW and the vertex of the cascade. Deep levels use charge-transition levels and Huang–Rhys factors (Sects. 28.2 and 30.5) [15]. The Bethe–Salpeter step costs $10^3$ to $10^4$ core-hours for the semiconductors and about $10^4$ for the oxides; capture about $10^5$ core-hours per material; the six materials about $6.5 \times 10^5$.

The exciton levels of the ionic crystals are converged to 1 meV when the $\boldsymbol{k}$ mesh goes from $8^3$ to $12^3$, and those of silicon and germanium when the double-grid density near the band edges is doubled, with the bands from four to six on each side; central-cell binding energies to 10 % between 216 and 512 atoms. A rate that changes by more than 10 % between 10 and 50 mK is reported as a temperature dependence. The measured control is the free-exciton binding of silicon, 14.7 meV [16]. Formation of an exciton from two free carriers moves gap energy between pairs and excitons and releases only $E_x^b$, while exciton–exciton annihilation and Auger recombination remove gap energy, so the bimolecular channel must be named. The excitons cool by acoustic phonons slowly at millikelvin, so their temperature is not the lattice temperature. Capture at the end of the cascade changes the pair statistics. A capture probability shared by every carrier changes $\varepsilon_{\mathrm{eh}}$ and $F_{\mathrm{eh}}$ by Eq. (18.27). A capture probability $f(x)$ that depends on the final energy $x$ of the carrier is instead averaged, with the law of total variance, over the end-of-cascade energy distribution that the silicon cascade records. The parameters close the rate equations and the end-of-cascade populations of Sect. 18.5.

Table 32.2: Electron-cascade calculations, with estimated costs in core-hours. GW+BSE, quasiparticle and Bethe–Salpeter calculation.

| quantity | method and grid | acceptance | cost |
|---|---|---|---|
| $P_n(\boldsymbol{q},\omega,T)$, $\varepsilon_{\mathrm{eh}}$, $F_{\mathrm{eh}}$, end-of-cascade energies and pair separations (Si, Ge, GaAs) | $G_0W_0$ bands (Ge after the band check), full-frequency $W^{\mathrm{scr}}$ to 60 eV, EPW; $k$-resolved first generation | Si 3.749 eV, 0.119 at 123 K; quantum yield at 300 K | $10^5$–$6\times10^5$ |
| $\varepsilon_{\mathrm{eh}}$, $F_{\mathrm{eh}}$ (wide-gap, polar) | as above, Fröhlich vertex, BSE for halides | silicon validation first; DFT full-band benchmarks | $6\times10^5$–$2.5\times10^6$ |
| $r_{\mathrm{ii}}(E)$, Si | $G_0W_0$ with and without $z_{\mathrm{qp}}$, plus second-order exchange | 2 % per 50 meV bin | $3\times10^4$ |
| $w_{\mathrm{mem}}(E)$ and spread, Si, 3–8 eV | $k$-resolved second generation, 200 $\boldsymbol{k}$ per bin | 10 %; percentile range | $10^3$–$10^4$ |
| $P_{\mathrm{casc}}(N\|\Lambda)$ library, marks | reduced recursion, $h=E_{\mathrm{g}}/50$, Eq. (32.3), $h^{3/2}$ extrapolation | sum to one; moments; $\delta_{N1}$ | $\leq 1$ |
| Exciton and capture set | GW+BSE, central-cell DFT, cascade capture | Si $E_x^b=14.7$ meV | $6.5\times10^5$ |

## 32.3 Event-level transport

A 100 eV electron recoil in silicon ends its cascade with about 27 electron–hole pairs, at 3.75 eV per pair [10], and with the rest of its energy in phonons. What the detector reports is decided afterwards, in a readout window of a microsecond to a second. In that window the pairs drift, bind into excitons or get trapped, the gap energy they carry is released somewhere as phonons, and each sensor absorbs some share of each phonon population. Sects. 18.5 and 19.5 and Chap. 20 derive the physics of every step, and Sect. 29.2 sets out the transport Monte Carlo that chains the steps. Every calculation below sits at the place on the truncation label $\mathfrak{T}$ (Sect. 8.5) that the transport Monte Carlo occupies: lowest-order golden-rule rates, with $n_g=2$ for phonon emission by carriers, $n_A=2$ for three-phonon decay, $n_B=0$, and incoherent propagation between events. Table 32.3 collects grids, acceptance tests and costs.

### 32.3.1 Charge and exciton kinetics

The measured charge and the measured heat of an event both depend on what the pairs do during the window $t_{\mathrm{win}}$, and both follow from the joint distribution of the measured pair number $N_{\mathrm{meas}}$ and of the energy $E_{\mathrm{st}}(t_{\mathrm{win}})$ still stored at the end of the window. It is conditional on the recoil energy $E_R$, the recoil direction $\hat{\boldsymbol{r}}_R$, the applied field $\mathcal{E}$, the window and the temperature $T$. From it follow two tables that Sects. 19.4 and 19.5 use: the phonon-yield ratio $Y_{\mathrm{ph}}(t_{\mathrm{win}})$ of nuclear to electron recoils evaluated at a finite window, for $t_{\mathrm{win}}\in\{1\,\mu\mathrm{s},100\,\mu\mathrm{s},10\,\mathrm{ms},1\,\mathrm{s}\}$, and the same ratio at detector thresholds of 0.3, 1, 3 and 16 eV. The grid is 20 logarithmic recoil energies from 10 eV to 10 keV, fields of 0, 1, 4 and 40 V/cm, and the impurity census of the detector modeled (boron, phosphorus and the deep centers its growth certificate lists, at their stated concentrations).

That distribution comes from the rate equations of Sect. 18.5 for free pairs, free and bound excitons and, at high density, electron–hole droplets, coupled to drift; the luminescent share leaves the crystal as escaping energy $E_{\mathrm{esc}}$ (Sect. 19.4). The field enters in two ways: pairs separated by drift faster than the binding time do not form excitons, and above $E_x^b/(ea_x)=3.5\times10^4$ V/cm excitons are field-ionized (Sect. 18.5). Drift, Luke emission, trapping and Ramo induction are those of the carrier Monte Carlo of Sect. 29.2, with four

additions: the trap occupations $n_j(t)$ carry over from event to event, trap energies are referred to the Fermi level, the electrostatic energy of the trapped charge is a separate share, and the drift during the prompt phase is kept. The exciton module is new code. The exciton and capture parameters come from Sect. 32.2.

Two measured numbers serve as acceptance tests. The silicon phonon-yield ratio of the CDMS II detectors, $Y_{\mathrm{ph}} = 0.952^{+0.009}_{-0.007}$ (95 % confidence) at 4 V/cm [17, Sect. 4], and the defect energy loss of $^{206}$Pb recoils in germanium, $(6.08 \pm 0.18)$ % [18], must both be reproduced by one parameter set. The Hecht limit of Sect. 29.2 is the numerical check. The calculation is meaningful only together with the collection efficiencies and the athermal emission rates specified below, since the measured yield mixes all three. A parameter set costs up to $10^3$ core-hours; the full table over windows, thresholds and the two materials $10^3$ to $10^4$.

### 32.3.2 Athermal emission and space charge

At millikelvin temperatures thermal emission from traps is frozen out, and two effects remain that a constant trapping length cannot describe. Trapped charge is released athermally, by phonons and by hot carriers, and the charge that stays trapped distorts the drift field. The calculation gives the emission rates $\Gamma_j^{\mathrm{em}}$ of each trap class $j$, in $\mathrm{s}^{-1}$, over a stated exposure history, and the field distortion $\xi_{\mathrm{SC}} = \max_z |\Delta\mathcal{E}|/\mathcal{E}$ of Sect. 20.1.

The field distortion follows from the one-dimensional Poisson equation for a sheet of trapped charge (Eq. (20.9)). The grid satisfies $\mathrm{d}z \le d/200$ for a crystal of thickness $d$, and the event sequence runs until the distortion saturates or is reset by flashing with infrared light.

The emission rates need cross-sections that are not known at millikelvin. A neutral phosphorus donor in silicon is bound by 45 meV, and phonons above 10.9 THz live below a nanosecond and travel about a micrometer (Sect. 20.2), so phonon-driven emission from neutral shallow centers happens only next to the track of a later event. The Luke phonons, a few meV, cannot ionize neutral centers, but can ionize the over-charged centers $\mathrm{D}^-$ and $\mathrm{A}^+$, which bind their second carrier by a few meV; that is the dominant athermal route once flashing has populated them. The phonon-ionization cross-sections follow from the golden rule for one-phonon absorption with the deformation-potential coupling of Sect. 6.2, between effective-mass envelopes that include the valley–orbit splitting (six valleys in silicon, four in germanium), since the splitting sets the threshold. For $\mathrm{D}^-$ and $\mathrm{A}^+$ the bound state is a correlated two-electron envelope of Chandrasekhar type. Impact ionization by drifting carriers uses Born cross-sections on the same envelopes away from threshold and a distorted-wave calculation near it, where the hot tail of a Luke-limited distribution sits. The envelope basis is converged to 1 % in binding energies and cross-sections, on a threshold grid no coarser than a fiftieth of the binding energy. A 216-atom supercell (Sect. 28.2), 1.6 nm across, cannot hold a donor envelope of radius 2 to 3 nm and supplies only the central-cell correction, at about $10^3$ core-hours with the shared 60 Ry setting; the envelopes and the event-sequence driver take minutes.

Ramo's theorem stays exact with space charge, but the heat dissipated by drifting carriers differs from the induced-charge work by the change of electrostatic energy, which the energy balance of Sect. 19.4 carries; and flashing creates the $\mathrm{D}^-$ and $\mathrm{A}^+$ populations whose emission it was meant to remove.

### 32.3.3 The mark-sufficiency test

The two-stage factorization of Sects. 18.3 and 19.5 lets the detector stage see the cascade only through a few marks (the pair number, the centroid and spread of the charge cloud, the phonon spectrum), and it is exact if the marks are a sufficient statistic for the seed class. The

test compares the detector stage run on full cascade microstates with the same stage run on microstates resampled to match only the marks, for a dark-matter seed class and a photon seed class. The test statistic is the difference $\Delta P$ between the two arms in the distributions of the measured charge and of the phonon signal. A difference resolved at two standard deviations needs $2p(1-p)(z/\Delta P)^2$ trajectories per arm, with $z=2$, which for $\Delta P=0.01$ at $p=0.5$ is $2\times10^4$ per seed class. Common random numbers between the arms, as in Sect. 29.2, lower this, and the test costs $10^2$ to $10^3$ core-hours. The test can fail but cannot prove: a null difference can hold while the omitted features differ, if the detector stage happens to ignore them.

### 32.3.4 Low-energy-excess inputs

The collective-release model of Sect. 19.5 lacks two inputs: the relaxable elastic energy $U_{\mathrm{rel}}$ stored at cooldown, with its spread of release rates $[\Gamma_{\min},\Gamma_{\max}]$ for a given holder and glue, and the branching ratio $b_{\mathrm{br}}$, the mean number of further releases triggered by the phonons of one release.

$U_{\mathrm{rel}}$ comes from a thermo-elastic finite-element calculation of the cooldown from 300 K to 10 mK for the crystal, the glue and the holder, with measured elastic tensors $C_{ij}(T)$ and expansion coefficients, or with the integrated thermal contraction of Sect. 30.1 where no measurement exists. The elastic energy of each component is converged to 5 % under a doubling of the mesh. The relaxable fraction and its spread in $\ln\Gamma$ come from low-temperature dynamic-mechanical data of the glue, not from the simulation. The energy law of single release events comes from molecular dynamics of a crystal–epoxy interface with $10^6$ atoms at 1 to 10 K, which needs a machine-learned potential for cured epoxy (Sect. 29.1). None exists; one has to be trained on dispersion-corrected density-functional energies of the cured network and the interface, validated on elastic constants and thermal expansion, and extended to bond breaking by active learning. Event energies are read from energy drops, and the law is stable when the box grows from $10^6$ to $4\times10^6$ atoms. Molecular dynamics reaches nanoseconds, so it gives the energy law and never the rates, which come from the finite-element step.

The branching ratio needs the release-path catalogue of Sect. 30.5. The phonon pulse of one release is propagated by the transport Monte Carlo of Sect. 29.2, and a catalogued site with barrier $E_b$ triggers with probability $1-\exp[-\int \mathrm{d}t\,\Gamma(t)]$, where $\Gamma(t)$ is its phonon-assisted rate in the absorbed flux above $E_b$. The triggering volume is the integral of this probability over space, and $b_{\mathrm{br}}$ is the density of sites with barriers below the phonon energy times that volume, averaged by kinetic Monte Carlo over $10^6$ seeds. It is first-generation and linear in the flux, and it has to change by less than 10 % when the catalogue grows from 50 to 100 paths, the phonon packets from $10^5$ to $4\times10^5$, and the box from $L$ to $2L$. An incomplete catalogue, missing the low barriers, biases $b_{\mathrm{br}}$ low; a box smaller than the phonon mean free path truncates the triggering volume; and the thermal bath and the pulse must not both drive the same rate. The finite-element part costs $10^2$ to $10^3$ core-hours, the molecular dynamics $10^3$ to $10^4$ GPU-hours. A fit of the model to the binned rates of a glued and an unglued crystal [19] then tests the predicted scaling with $U_{\mathrm{rel}}$ without free parameters.

### 32.3.5 Collection kernels of a stated detector

The collection contrasts $\varpi_O$ and $\varpi_{\mathrm{gap}}$, the residual position spread and the collection ratio $\mathfrak{r}_{\mathrm{coll}}$, defined in Sects. 19.4 and 20.5, are linear functionals of the collection kernels $\mathfrak{C}_s(E_{\mathrm{ath}};\boldsymbol{r}_{\mathrm{ev}})$ of Eq. (20.41). Electron and nuclear recoils enter only through their birth spectra, never through the kernels.

Two geometries are specified: the $1\times1\times0.4\,\mathrm{cm}^3$ silicon crystal of Ref. [20], whose top face

carries two sensor arrays, and a germanium cylinder 1 cm thick of the CDMS type. Table 32.3 gives the grids: 40 logarithmic points in $E_{\mathrm{ath}}$ between 0.41 and 62 meV (0.1 to 15 THz), $8^3$ event positions in the symmetry-reduced octant, and $10^4$ phonons launched per cell as monochromatic sources, which is what makes the tally a kernel. Luke phonons are generated by the carrier transport at biases of 2, 4 and 10 V. The approximation level, including the lumped films, is that of Sect. 29.2. Each geometry uses its actual sensor films; the mean-field estimate of Sect. 20.5, by contrast, takes one film with a niobium-like gap ($2\Delta_{\mathrm{BCS}} = 3.0$ meV) and one of aluminum ($2\Delta_{\mathrm{BCS}} = 0.36$ meV).

The kernels are accepted on numerical and physical grounds. $\varpi_O$ must change by less than 0.01 when the energy grid is refined, and the statistical error per cell must fall below 1 % when the launch count grows from $10^4$ to $4 \times 10^4$. In the limit of diffuse, grey surfaces and diffusion faster than absorption, the position-averaged kernels must reproduce the mean-field solution of Sect. 20.5, with a bare-surface loss of about $10^{-3}$ per bounce [21, Sect. III G]; there $\varpi_O$ lies between 0.02 and 0.39 for mean Luke energies from 4.7 down to 1.5 meV and the position spread is zero by construction. Three systematic scans make up the uncertainty of the contrasts: the surface coefficients varied between the acoustic- and diffuse-mismatch values, the probability of specular reflection, which no code default fixes, and, for germanium, the anharmonic constant over its factor-of-ten range.

Luke-phonon birth spectra depend on the field, so they must come from the carrier transport at each bias; a sampler that assigns one phonon energy per emission fixes $\mathfrak{r}_{\mathrm{coll}}$ by construction. Phonons below $2\Delta_{\mathrm{BCS}}$ of aluminum are lost energy, not absorbed energy, unless a tungsten sensor or another film takes them, and that assignment sets the sign of $\mathfrak{r}_{\mathrm{coll}} - 1$ (Sect. 20.5); it must be stated with every result. Accelerated propagation modes that skip scattering events are checked against full tracking before use; and the transport code has to pass the film validation of Sect. 32.3.9 first. The cost is about $2 \times 10^{12}$ propagation steps, some 140 core-hours per geometry; a first pass at $10^3$ phonons per cell takes under an hour at about 3 % per cell.

### 32.3.6 Gap-energy release and created-phonon spectra

Each pair created at zero field carries the gap energy $E_{\mathrm{g}}$, which comes back as phonons when the pair recombines, and where and as what spectrum it comes back decides $\varpi_{\mathrm{gap}}$. Its phonon-borne part has a birth density $N_{\mathrm{eh}} E_{\mathrm{g}} \sum_\ell \pi_\ell(\boldsymbol{r}, t)\, s_\ell(E_{\mathrm{ath}})$ summed over release channels $\ell$: bulk recombination of excitons by Auger or multiphonon processes, recombination at a surface, collection at an electrode, and quasiparticle injection into a film. Here $\pi_\ell$ is the channel flux in space and time and $s_\ell$ the spectrum emitted per channel. Then $\varpi_{\mathrm{gap}}$ is the sum over channels of the time-integrated channel fraction times the contrast of $s_\ell$ born at the release site, read from the kernels above. Bulk release of a hard spectrum, above the memory-loss scale of Sect. 20.2, contributes nothing.

Besides the kinetic rates of Sect. 32.2, the release needs the exciton diffusion constant $D_X$, the surface recombination velocity $S_{\mathrm{surf}}$ and the spectra $s_\ell$. The channel fluxes come from a one-dimensional implicit finite-difference solution of exciton diffusion with Robin faces, with $\mathrm{d}z \le L_X/20$ for the diffusion length $L_X = (D_X/\Gamma_X)^{1/2}$ and exciton decay rate $\Gamma_X$. An exciton born at depth $z_0$ recombines at a face with probability $[S_{\mathrm{surf}}/(S_{\mathrm{surf}} + D_X/L_X)]\, \mathrm{e}^{-z_0/L_X}$, which reduces to $\mathrm{e}^{-z_0/L_X}$ only for $S_{\mathrm{surf}} \to \infty$. The fluxes enter the transport as time-tagged sources; the released fraction is converged to $10^{-3}$ under grid and time step, with $S_{\mathrm{surf}}$ scanned from 10 to $10^4$ cm/s. The surface and electrode spectra cannot be assumed: taking them as a hard line at the optical-phonon energy decides $\varpi_{\mathrm{gap}} = 0$ in advance. Until they are computed, they are scanned as mixtures $(1 - \kappa_s) s_{\mathrm{hard}} + \kappa_s s_{\mathrm{soft}}$, with $\kappa_s \in \{0, 0.25, 0.5, 0.75, 1\}$ and $s_{\mathrm{soft}}$ a Debye spectrum cut at 2, 4 or 8 meV. Soft surface spectra are measured: in germanium at $\pm 3$ V,

recombination phonons from the surfaces carry 13 % of the phonon signal in a low-frequency, ballistic regime [21, Sect. V A]. The scan is replaced by the multiphonon line shape of the recombination center (a dangling bond at the $Si/SiO_2$ interface, or a state of a metal contact), with the Huang–Rhys factor and local modes from defect supercells (Sect. 28.2) converged to 10 %, at about $10^3$ core-hours per center. At millikelvin $D_X$ and the binding rates depend on the impurity census of the actual crystal, droplets lie outside linear kinetics, and luminescence belongs to $E_{\mathrm{esc}}$, not to $\varpi_{\mathrm{gap}}$.

The primary phonons are treated the same way. The created-phonon spectra of electron and nuclear recoils at $E_R = 50\,\mathrm{eV}$, 100 eV, 300 eV and 1000 eV, branch-resolved on a 0.5 meV grid with the zero-point energy of the final state subtracted (Sect. 32.1), are convolved with the kernels to give the arrival spectra and the observable difference between the two recoil types. Sect. 20.2 bounds that difference by the $L^1$ distance of the normalized birth spectra below the memory-loss scale plus a term $\Delta_{\mathrm{h}}$ from the hard part; the bound reduces to the soft-fraction difference only when the spectral difference has one sign below that scale. The test is that refining the grid to 0.25 meV, or moving the memory-loss scale by a factor of two, changes the bound by no more than $\Delta_{\mathrm{h}}$. Thermal phonons absorbed during the recoil enter as negative sources, never clipped, and the energy the molecular-dynamics thermostat removed enters with its own spectrum. The convolution costs nothing; a direct transport check samples births from the tables at about $10^8$ steps per seed set, minutes.

### 32.3.7 Collection efficiencies of the calibration detectors

The silicon phonon-yield ratio $Y_{\mathrm{ph}} = 0.952$ is degenerate between stored energy and collection (Sect. 19.4). Breaking the degeneracy needs the collection efficiencies $f_{\mathrm{coll}}^{(p)}$ of each population $p$, primary phonons of nuclear and of electron recoils, Luke phonons and gap-energy phonons, for the detectors that measured the data. These are a CDMS II silicon detector, 76 mm across and 10 mm thick at 4 V/cm [17, Sect. 2], and a SuperCDMS germanium detector at $\pm 2\,\mathrm{V}$ with a $^{206}$Pb surface source [18]. The gap part of the collection offset $\delta_{\mathrm{coll}}$ is fixed by $\rho_g$, the efficiency for gap-energy phonons over that for hard primaries; $\rho_g - 1 = 0.21$ alone would account for the silicon deficit $1 - Y_{\mathrm{ph}} = 0.048$ beyond the 0.006 that the pair energy of the reconstruction produces, so $\rho_g$ must be known to $\pm 0.02$. The two-bias inversion, at 4 and 40 V, assumes that $\mathfrak{r}_{\mathrm{coll}}$, the ionized fraction $f_{\mathrm{ion}}$ and the collection offset do not change with bias and that the reconstruction keeps its pair energy; the run computes the residual from the bias dependence of $\mathfrak{r}_{\mathrm{coll}}$, and a third bias tests all three.

The calculation adds charge transport to the kernels: Herring–Vogt valleys, Luke emission, intervalley scattering inside the field range of its fitted parameters, and trapping at constant cross-section (Sect. 29.2), with event positions from source maps of a Geant4 simulation [22] of the neutron, gamma and lead sources. A carrier collected at an electrode releases its kinetic energy plus its energy relative to the Fermi level of that contact, supplied per contact; booking $E_{\mathrm{g}}/2$ per carrier with no phonons would set $\rho_g$ by construction. Each population needs $10^5$ events, and the inversion must be consistent between the two biases. The surface source of the germanium data makes the escape term of $\delta_{\mathrm{coll}}$ the leading one there. The cost is about $10^{11}$ steps, some 8 core-hours, per geometry.

### 32.3.8 Anisotropic film interfaces

The escape time $t_l = 4d_{\mathrm{f}}/(v_s \bar{t}_{\mathrm{esc}})$ of a phonon from a sensor film (Sect. 20.3) needs the escape probability $\bar{t}_{\mathrm{esc}}$ of the actual interface. The isotropic acoustic- and diffuse-mismatch values differ by a factor 1.31 in $t_l$ for aluminum on silicon. The welded interface between two anisotropic elastic media, with Christoffel-matched plane waves and the six-by-six boundary

problem for displacement and traction, gives the transmissions $t_\nu(\theta,\phi)$ per mode and incidence direction for Al(111) on Si(001) and for tungsten on silicon, in both directions. Thin-film interference enters through a transfer matrix when the wavelength is comparable to the thickness, and $t_l(E_{\mathrm{ath}})$ is tabulated for $d_{\mathrm{f}}$ from 40 to 600 nm. The aluminum constants are the measured ones extrapolated to zero temperature, the set of that section; finite-strain density-functional constants (Sect. 25.2), half an hour for fcc aluminum at a 60 Ry cutoff, a $16^3$ grid and 0.02 Ry smearing, serve only where no measurement exists, as for tungsten films.

The transmitted and reflected fluxes sum to one to $10^{-14}$; in the isotropic limit the code returns the escape probabilities of Sect. 20.3 for aluminum on silicon, $\bar{t}_{\mathrm{esc}} = 0.368$ (acoustic mismatch) and 0.282 (diffuse mismatch); the angle averages satisfy the reciprocity $\mathfrak{n}_1\bar{t}_{12} = \mathfrak{n}_2\bar{t}_{21}$; and $\bar{t}$ is stable to $10^{-3}$ under refinement of the angular grid. The anisotropic result for Al(111) on Si(001) is compared with the sensitivity found there, where giving isotropic silicon its [100] or [111] speeds moves $\bar{t}_{\mathrm{esc}}$ to 0.327 or 0.405; a value outside that range points to the treatment of the aluminum side. The film formula carries its own hypotheses. Its flux average of the per-angle absorption is $1 - 2E_3(2d_{\mathrm{f}}/\mathfrak{l}_{\mathrm{f}})$, with $E_3$ the exponential integral, which the lumped form $1 - \mathrm{e}^{-4d_{\mathrm{f}}/\mathfrak{l}_{\mathrm{f}}}$ of Sect. 29.2 matches only to first order: with the escape probability set to one in both, the lumped form lies 3 % to 13 % above it for $d_{\mathrm{f}}/\mathfrak{l}_{\mathrm{f}}$ from 0.01 to 0.2. And $t_l$ holds for a film population that is isotropized faster than it escapes: without mixing in the film, transverse modes beyond 37.7° are totally reflected at Al/Si, some 56 % of the film's phonon flux, and never escape. Real interfaces carry oxide or amorphous interlayers, and the fitted silicon-to-aluminum value is an absorption probability $\bar{t}\,P_{\mathrm{abs}}$, not an escape probability. The whole calculation costs 2 to 10 core-hours.

### 32.3.9 Validation of the film cascade

Every number that Chap. 20 takes from a lumped film rests on the quasiparticle yield $Y_{\mathrm{qp}}(E_{\mathrm{ath}})$ of Sect. 20.4, and the code that produces it is tested before any kernel is trusted. Without film loss, $Y_{\mathrm{qp}}$ must equal $2\Delta_{\mathrm{BCS}}/E_{\mathrm{ath}}$ to round-off below $4\Delta_{\mathrm{BCS}}$, reach the maximum of 0.600 at $5.46\,\Delta_{\mathrm{BCS}}$ and settle at the 0.590 that the Kaplan cascade gives; the test points are $4.5\Delta_{\mathrm{BCS}}$, $6\Delta_{\mathrm{BCS}}$ and 10 to $100\Delta_{\mathrm{BCS}}$. With the escape law of a 600 nm aluminum film, $Y_{\mathrm{qp}}$ falls with energy at 3, 5, 10 and $40\Delta_{\mathrm{BCS}}$, because the fastest phonons escape before they break pairs. Energy closes per event to $10^{-14}$ relative for incident phonons from 50 to $350\Delta_{\mathrm{BCS}}$. The last range matters because germanium prompt phonons near 36 meV arrive at an aluminum film with some $200\Delta_{\mathrm{BCS}}$. The closure counts escaped energy as phonons actually returned to the substrate, so that an escape bin cannot absorb lost energy. The mean Luke-phonon energy against field is compared with the carrier model of Sect. 29.2. Each test runs in seconds and a benchmark of a few thousand events in minutes. Below $100\Delta_{\mathrm{BCS}}$ the yields agree with the loss-free forms within statistical error; above it the closure test decides.

Table 32.3: Event-level transport: grids, acceptance tests and cost of each calculation. Energies of phonons are $E_{\mathrm{ath}}$; core-hours are CPU core-hours unless marked GPU. Costs are per geometry or per parameter set as stated.

| quantity | grid | acceptance | cost |
|---|---|---|---|
| Charge and exciton kinetics | $t_{\mathrm{win}}$: 1 µs to 1 s; thresholds 0.3–16 eV | $Y_{\mathrm{ph}}$ = 0.952 (Si), 6.08 % (Ge), Hecht limit | $\leq 10^3$ core-h per set |
| Athermal emission, space charge | envelope basis to 1 %; $\mathrm{d}z \leq d/200$ | saturation of the field distortion | $\sim 10^3$ core-h |
| Mark sufficiency | $2 \times 10^4$ trajectories per arm | $\Delta P$ resolved to 0.01 | $10^2$–$10^3$ core-h |
| Low-energy-excess inputs | mesh ×2; $10^6 \rightarrow 4 \times 10^6$ atoms; catalogue 50 → 100 | elastic energy per component to 5 %; $b_{\mathrm{br}}$ to 10 % | $10^2$–$10^3$ core-h; $10^3$–$10^4$ GPU-h |
| Collection kernels | 40 energies, 0.41–62 meV; $8^3$ positions; $10^4$ phonons per cell | $\varpi_O$ to 0.01; mean-field limit | $\sim 140$ core-h |
| Gap-energy release | $\mathrm{d}z \leq L_X/20$; $S_{\mathrm{surf}}$ 10–$10^4$ cm/s | released fraction to $10^{-3}$ | seconds; $10^3$ per center |
| Created-phonon spectra | 0.5 meV → 0.25 meV | bound stable to $\Delta_{\mathrm{h}}$ | minutes |
| Collection efficiencies | $10^5$ events per population | $\rho_g$ to ±0.02; two biases | $\sim 8$ core-h |
| Film interfaces | angular grid; $d_{\mathrm{f}}$ 40–600 nm | flux sum to $10^{-14}$; isotropic 0.368, 0.282; reciprocity | 2–10 core-h |
| Film cascade validation | 3–350$\Delta_{\mathrm{BCS}}$ | $2\Delta_{\mathrm{BCS}}/E_{\mathrm{ath}}$; 0.590; closure $10^{-14}$ | < 1 core-h |

## 32.4 Halo, background and nuclear-spin inputs

A limit from a silicon crystal that is valid for every velocity distribution of the halo, not only the standard one, is the solution of a linear program whose columns are the counts that single streams of dark matter would produce in each bin. A crystal kept at sea level for a hundred days before it goes underground carries tritium whose $\beta$ spectrum puts a fraction of an event per kilogram, day and keV below 1 keV. And a spin-dependent coupling to unpolarized nuclei produces no coherent phonon at all. These three facts belong to Sects. 11.6, 21.2 and 21.6, and each needs one computation, specified below.

### 32.4.1 The directional halo-independent bound

Sect. 21.2 writes the expected count in bin $i$ as $\int_{\mathcal{B}} \mathsf{k}_i(\boldsymbol{w})\,\mathrm{d}\Xi(\boldsymbol{w})$, linear in the nonnegative measure $\Xi$ of Eq. (21.9) on the ball $\mathcal{B} = \{|\boldsymbol{w}| \leq v_{\mathrm{esc}}\}$ of galactic-frame velocities ($v_{\mathrm{esc}}$, $v_0$ and $\rho_\chi$ as in Sect. 1.2), with the stream kernel $\mathsf{k}_i(\boldsymbol{w})$ the count that a single stream of unit $\Xi$-mass at $\boldsymbol{w}$ would leave in bin $i$. A crystal fixed in the laboratory sees the stream at velocity $\boldsymbol{w} - \boldsymbol{v}_E(t)$ and turns with the sidereal rotation. The kernel is then the time average over the bin of the master rate of Sect. 8.2 with the crystal's direction-dependent response $S(\boldsymbol{q}, \omega; \Lambda)$ and the final-state weights of Sect. 8.7. The largest value of a linear functional $\int \mathsf{f}\,\mathrm{d}\Xi$ allowed by the measured upper counts $N_i^{\mathrm{up}}$ is the optimum of the semi-infinite program of Eq. (C.109), with nonnegative dual weights $\lambda_i$; it is attained by at most as many streams as there are bins. The functional $\mathsf{f}$ can be a halo integral above a speed in a direction, the rate of a second crystal or of a second orientation of the same crystal, or a modulation amplitude; a signed

$\mathsf{f}$ is allowed provided some bin sees every stream on which $\mathsf{f} > 0$. The method extends the stream approach of nuclear-recoil analyses [23, 24] to electronic and directional responses.

The computation is a sweep followed by a small linear program. The response must be resolved in the direction of $\boldsymbol{q}$, not averaged over angles. The non-self-consistent band calculation behind it (Sect. 23.3) therefore runs on the full zone without symmetry reduction, and the library sits at the truncation label $\mathfrak{T}$ of the target's response tables (Sects. 31.1 and 31.3). The stream grid has 768 directions on an equal-area pixelization of the sphere [25], 55 galactic-frame speeds in steps of 10 km/s up to $v_{\mathrm{esc}}$, and 24 sidereal phases, about $10^6$ kernel columns. For each stream and phase the energy delta function of the master rate fixes $\omega$ on every vector of the $\boldsymbol{q}$ grid. Each column is then one pass over about $10^6$ $\boldsymbol{q}$ vectors, some $10^{12}$ multiply-adds per mass, 5 to 50 GPU-hours for about 50 masses. The linear program has one constraint per bin and about $4 \times 10^4$ variables and is solved by a dual simplex code [26] in seconds.

The grid is refined around the few optimal streams by column generation, and the dual vector $\boldsymbol{\lambda}$ must satisfy $\sum_i \lambda_i \mathsf{k}_i \geq \mathsf{f}$ on the refined grid; that inequality certifies the bound, and the result is reported at two grid levels. In the laboratory frame, a kernel that depends only on the speed must reproduce the single-bin result $N_i^{\mathrm{up}} / \inf_{v \geq u_0} [v\, \mathsf{k}_i(v)]$ for the halo integral above $u_0$; the same statement in galactic speeds does not hold, because there the kernel depends on $|\boldsymbol{w} - \boldsymbol{v}_E(t)|$. And the kernel must be nonnegative for every stream. A response truncated at some order can have negative pieces. A dual vector that satisfies the dual constraint of Eq. (C.109) still bounds the objective by weak duality, but the mass bound behind the grid bracket fails for the computed kernel, and the leakage-corrected bound of Sect. 21.2.4 applies instead. The support constraint is imposed in the galactic frame, a per-bin confidence level needs the Bonferroni correction across bins, and slow streams alias on a coarse $\boldsymbol{q}$ grid, where the energy delta function falls between grid points.

### 32.4.2 Cosmogenic production

A nuclide produced at a rate $R_k$ during exposure $k$ of length $t_k$ and decaying with constant $\lambda_{\mathrm{dec}}$ has, after the whole history, the activity $\mathcal{A}_{\mathrm{act}} = \sum_k R_k (1 - \mathrm{e}^{-\lambda_{\mathrm{dec}} t_k})\, \mathrm{e}^{-\lambda_{\mathrm{dec}} t'_k}$, where $t'_k$ is the time from the end of exposure $k$ to the measurement; at saturation $\mathcal{A}_{\mathrm{act}} = R$. The sub-keV rate density is $\mathcal{A}_{\mathrm{act}}$ times the fraction of the $\beta$ spectrum below 1 keV, per keV, plus the lines of electron capture. L- and M-shell captures produce lines near 1 and 0.1 keV [27, Sect. 1], so capture nuclides land in the sub-keV region through their L and M fractions. Tritium in silicon is produced at sea level at $124 \pm 25$ atoms per kilogram and day from all cosmic-ray components, $112 \pm 24$ of them from neutrons [28], has a half-life of 12.3 years, and puts 8.9 % of its $\beta$ spectrum below 1 keV; a hundred days of exposure then give 0.17 events $\mathrm{kg}^{-1}\,\mathrm{d}^{-1}\,\mathrm{keV}^{-1}$ below 1 keV, and saturation gives 11.

What is missing is the sea-level production rate of each relevant nuclide, in atoms per kilogram and day, for every target retained in Sect. 21.6 (the survivors) that has no measured or published rate. The semi-empirical cross-sections of ACTIVIA [29], with measured cross-sections where they exist, are folded with the measured sea-level neutron spectrum, scaled to New York City at the midpoint of solar modulation, and with the measured sea-level proton spectra, the inputs of the silicon measurement [28]. For the light nuclides tritium and $^7$Be, where the semi-empirical systematics fail, cross-sections from a Geant4 hadronic list [22] or from TALYS are folded with the same spectra. The inputs are the composition and isotopic abundances, the spectrum parameterization and the exposure history. The calculation must reproduce the measured rates of silicon [28] and germanium [27] within their errors before any other material is computed. The models disagree most for tritium: in germanium they span 27.7 to 210 nuclei per kilogram and day [27, Sect. 6.3], so the light nuclides carry the largest

error. Tritium also diffuses out of silicon, which makes its activity in the crystal depend on the thermal history, and the K:L:M capture fractions are needed per nuclide. ACTIVIA takes seconds per material and Geant4 with $10^8$ primaries 1 to 3 CPU-hours, a few tens to a hundred core-hours for the survivor pool.

### 32.4.3 Spin-dependent nucleon inputs

For unpolarized nuclear spins only the self response $S_i^{\mathrm{self}}$ of Eq. (11.38) survives, for the isotopes $i$ that carry a spin $J_{\mathrm{nuc},i}$. For $\mathcal{O}_4$ it is weighted by the nuclear coefficient $c_4^{(i)}$ of Eq. (8.8), which needs the expectation values $\langle S_p \rangle$ and $\langle S_n \rangle$ of the proton and neutron spins in the nuclear ground state, and its one-phonon term contains the local density of states $g_i$ of Eq. (11.40). A coherent piece proportional to the squared polarization returns only in a polarizing field.

Two inputs are missing for the survivors. The spin expectation values come from large-scale shell-model calculations: $\langle S_n \rangle = 0.156$ and $\langle S_p \rangle = 0.016$ for $^{29}$Si, and $\langle S_n \rangle = 0.439$ and $\langle S_p \rangle = 0.031$ for $^{73}$Ge, where a second interaction gives 0.450 and 0.006 [30, Table III]. Each odd-$A$ isotope of the survivors needs values from at least two interactions, and their spread is the theory uncertainty. The host density of states $g_{\mathrm{ph},\kappa}$ comes from density-functional perturbation theory (Sects. 25.2 and 30.1), interpolated to a fine grid, with the $\boldsymbol{Q}$ grid refined until $\langle \omega^{-1} \rangle$ over $g_{\mathrm{ph},\kappa}$ changes by less than 1 % and an energy bin no wider than a quarter of the threshold; the local $g_i$ of each isotope follows from it by Eq. (11.40). Reusing the coherent one-phonon response with spin weights in place of the scalar couplings gives zero for the wrong reason, because the coherent amplitude averages out. Even–even isotopes have no spin and contribute nothing, so an element's weight is set by its odd-$A$ abundance alone. The density-of-states runs cost 10 to 100 core-hours over the survivors.

## Summary

The nuclear-recoil calculations give the close-collision excitation kernel on 80 points of collision energy and impact parameter and the partition of the recoil energy into heat, stored energy and charge as a joint distribution. They also give the charge yield on both sides of the binary threshold, 352 eV in silicon and 399 eV in germanium. Dense tracks add the interaction of their carriers, the recombination of the track plasma and the amorphous pockets of keV recoils. Molecular dynamics on machine-learned potentials carries all of them except the excitation kernel and the track recombination. The electron cascades give the pair-number distribution $P_n(\boldsymbol{q}, \omega, T)$ of silicon, germanium and GaAs, and then of the wide-gap and polar crystals that have no measured Fano factor, sapphire, $CaWO_4$, LiF and NaF first. A reduced cascade with energy-only branching gives the library of full distributions cheaply. In the readout window the pairs bind into excitons, are trapped or recombine at rates that are unknown in silicon and germanium, and the gap energy returns as phonons. The transport Monte Carlo then gives the collection kernels of a stated detector, the collection efficiencies that break the degeneracy of the silicon phonon-yield ratio 0.952, and the escape probability of a real film interface. Three inputs complete a limit: the stream counts of the directional halo-independent bound, the cosmogenic activity of a crystal from its exposure history, and the spin expectation values of the nuclei that carry a spin.

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# Chapter 33

# Libraries and open calculations

The specifications of Chaps. 30 to 32 end in tables: 12 real functions per $(\boldsymbol{q}, \omega)$ for the electronic response of silicon, $1.2 \times 10^7$ values per quantity for its lattice response, generating functions on a grid of carrier energies for the cascade. A limit can be recomputed or recast from them only if their axes, conventions, versions and checks are declared, and Sect. 33.1 gives the schema. Most of the specified calculations have not been run. Sect. 33.2 lists those that would change a printed result, limit or ranking, each with the check it must pass, and sorts the channels by what limits them: theory, calculation or data.

## 33.1 Shared response libraries

At $|\boldsymbol{q}| = \alpha m_e$ and $\omega = 10\,\mathrm{eV}$, the screened structure factor of silicon with local fields is $4.2 \times 10^7\,\mathrm{eV}^2$ in the published random-phase tables of Knapen, Kozaczuk and Lin [1] and $4.9 \times 10^7\,\mathrm{eV}^2$ in the empirical-pseudopotential model of Worked example 8.9. Both calculations sit at the same point of the truncation label, $\mathcal{X} = \mathrm{RPA} + \mathrm{LF}$. They agree on the independent-particle response to 3 percent and differ in the screening, which leaves 0.24 of the unscreened value in the tables and 0.29 in the model (Worked example 33.1). At 3 MeV two tabulations of the same independent-particle response give rates that differ by a third (Sect. 10.10). A limit can therefore be reproduced only from the table it was computed with, and recast for another coupling, halo or readout only if that table holds the rows, directions and final states the new calculation needs. The schema of Table 33.1 lists what a response library must tabulate for this, which conventions each table must state, and which section of Chaps. 23 to 32 specifies the calculation that fills it.

### 33.1.1 Tables of a response library

The axes of a library follow from the rate kernel Eq. (21.2). The velocity of the dark matter enters it in two places: in the coupling matrix $\mathfrak{D}_{ab}(\boldsymbol{q}, \boldsymbol{v})$ and in the energy $\omega_{\boldsymbol{q}}$ fixed by energy conservation. The response $S_{ab}(\boldsymbol{q}, \omega; \Lambda)$ depends on $\boldsymbol{q}$, $\omega$ and the final-state class, and $P_{\mathrm{casc}}(N_{\mathrm{eh}}|\Lambda)$ on the class alone. A library therefore has no velocity axis and holds nothing about the halo. It tabulates functions of $|\boldsymbol{q}|$, $\hat{\boldsymbol{q}}$ and $\omega$, labeled by a pair of coupling rows $(a, b)$, a class $\Lambda$ and a pair number. The coupling model enters through $\mathfrak{D}$, built from the operator coefficients of Table 8.1 and contracted by the user as $\sum_{ab} \mathfrak{D}_{ab} S_{ab}$, so the response is stored uncontracted, one function per row pair. Velocity-dependent couplings need no velocity axis either: their kernels are polynomials in the transverse velocity, and after the orientation average every power is fixed by $\eta(v_{\min})$ (Eq. (21.6)).

The electronic response is the largest part. For silicon and germanium the minimal table of Table 10.1 reduces every nonrelativistic operator to the 12 real functions per $(\boldsymbol{q}, \omega)$ of

Sect. 31.3, with four more in each version for operator $\mathcal{O}_2$. A crystal without the combined symmetry of inversion and time reversal, such as GaAs, keeps the imaginary parts of the cross terms as well. On the grid of Sect. 31.3, 250 values of $|\boldsymbol{q}|$ up to $25\,\alpha m_e$, 30 directions and 1000 energies up to 50 eV, the 12 functions fill $7.5 \times 10^6$ nodes, 720 MB in double precision. A screened response at a higher level, a time-dependent density-functional kernel or the Bethe–Salpeter equation, needs the full independent-particle matrix $R^0_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q},\omega)$, which is too large to store on the same grid and is kept on a coarse subset of the nodes.

The direction of $\boldsymbol{q}$ is kept as far as the analyses that use it require. A speed-only kernel, Eq. (21.4), needs the direction average $\bar{S}_{ab}$. The daily modulation (Sect. 11.7) and the directional halo-independent bound of Eq. (21.13) need the dependence on $\hat{\boldsymbol{q}}$ in crystal axes. In a cubic crystal it is carried, away from the edges, by the cubic-harmonic components $S_\ell/S_0$ for $\ell = 4$, 6 and 8. Near the zero-phonon edge the response changes on a scale of 0.1 rad in direction (Sect. 13.9), and there the library keeps the full grid of directions. The lattice tables of Sect. 30.1 hold the response per cell, the probability $P_{\mathrm{ph}}(n|\boldsymbol{q},\omega)$ that $n$ phonons were created and the coherence correction, on a grid of $1.2 \times 10^7$ values per quantity.

The energy grid has to resolve the edges. The power law of the rate near an edge, Eq. (21.19), rests on the onset of the response, and the computed phonon-assisted loss function of silicon follows its $x^3$ onset only for the first 1.3 meV above the onset, the zero-phonon continuum for 49 meV (Sect. 21.3.3). A step of 0.05 eV, adequate for the bulk of the spectrum, has no node inside the phonon-assisted range, which is 38 times narrower. The library therefore adds a sub-grid with steps of a fraction of a meV within some 50 meV of each onset, and lists for each channel the derived edge entries of Table 21.1: the edge mass $m_{\mathrm{edge}}$, the onset energy and the onset exponent $p_{\mathrm{on}}$.

The final-state tables carry the per-pair weights of Eq. (8.68). A full list is out of reach: the library grid has $10^3$ points $\boldsymbol{k}$ and $4 \times 56$ band pairs in silicon, $2.24 \times 10^5$ pairs per node and $1.7 \times 10^{12}$ numbers, 13 TB for each coupling row. The cascade does not need the list. In the energy-only reduction of Sect. 18.3 it needs the seed distribution $P_{\mathrm{seed}}(E_{\mathrm{e}}, E_{\mathrm{h}}|\boldsymbol{q},\omega,\boldsymbol{v})$ of Eq. (18.14), which for one coupling row depends on neither the velocity nor the model. In the zero-phonon sector $E_{\mathrm{e}} + E_{\mathrm{h}} = \omega - E_{\mathrm{g}}$, so one variable, the electron share $f_e$, describes the seed, and 20 bins of it take 1.2 GB per row pair on the full grid. A seed with a phonon adds the phonon energy, since then $E_{\mathrm{e}} + E_{\mathrm{h}} = \omega - \omega_{\boldsymbol{Q}} - E_{\mathrm{g}}$, and a valley label for the electron keeps the momentum memory of Sect. 18.1. The weights are the retarded ones of Sect. 8.7, which label the late-time state the cascade receives; in silicon and germanium they equal the advanced weights pair by pair, and in GaAs the two differ by 15 percent of the total. The split between the zero-phonon and the phonon-dressed classes follows either the off-shell window or the Dyson convention of Sect. 8.6, and the totals are independent of the choice only when one convention is declared for every table of the library.

The cascade tables hold $P_{\mathrm{casc}}(N_{\mathrm{eh}}|\Lambda)$ for each cascade model and parameter set, through the subtree generating functions $\mathfrak{G}_{\mathrm{e}}(x;s)$ and $\mathfrak{G}_{\mathrm{h}}(x;s)$ of Eq. (18.15) on a grid of carrier energies $x$. Sect. 32.2 specifies that grid, a step of about $E_{\mathrm{g}}/50$ up to $20\,E_{\mathrm{g}}$ and the asymptotic Gaussian beyond, with its acceptance tests. The same tables carry the marks $\mathfrak{m}$ of Eq. (18.19) that a readout needs: the phonon spectrum, the separate electron and hole counts, the carrier positions and the stored energy. The library ends there. The detector response $P_{\mathrm{det}}(\mathsf{R}|N_{\mathrm{eh}},\mathfrak{m})$ of Eq. (18.19) belongs to the experiment, which measures and publishes it with its collection efficiencies (Chap. 20), and the velocity distribution belongs to the analysis, which contracts it with the kernel in Eq. (21.1).

The uncertainty tables make the theory band of a limit reproducible. For each truncation $k$ active at a node, the library stores the next-order table $S^{\mathfrak{T}^+_k}$ itself rather than the ratio $\delta_k$ of Eq. (8.71). The matrix form Eq. (8.73) is built from $S^{\mathfrak{T}^+_k}$, and a ratio is undefined wherever $S^{\mathfrak{T}}$ vanishes, which is where a new-support template acts (Eq. (8.75)). Beside it stand the

type of the template in the classes of Sect. 8.8 (normalization, tilt, threshold or new support), its basis (computed, model, estimate) and a key naming its parameter $\vartheta_k$ in the likelihood of Sect. 21.4. The key is what enforces coherence: one $\vartheta_k$ acts on every node, class, target and experiment that shares the approximation (Sect. 8.8), and two tables carrying the same key are varied together. A template known only as an inequality, such as the tabulation spread of Sect. 10.10, a bound on the width of a spread that enters a likelihood with that width (Sect. 21.4.2), is flagged as a bound. A node with no template carries a flag that names the missing calculation. Energies where no central value can be defended, those below $3\Delta_{\mathrm{BCS}}$ at the superconducting edge of an aluminum film (Sect. 15.2), are flagged as excluded. The band of Sect. 21.4 is then built with these omissions named rather than with invented widths. A second calculation at the same truncation label is stored as an alternative table; the spread between the two is the uncertainty of the approximations that the label leaves out, as for the two silicon tabulations of Sect. 10.10.

The derived tables are those that analyses would otherwise recompute for every search. The step column $\mathcal{A}_i(v)$ of Eq. (21.11), per bin and cascade model, turns the halo-independent bound of Eq. (21.12) into a minimum over a few numbers, and it follows from the response by cumulative sums over $v_{\min}$. It is also a check: for a nonnegative kernel it never decreases with $v$ (Eq. (E.51)), and a decrease flags either a sign error in a template or the final-state leakage of Eq. (21.15), at most $0.035\,\varrho_i^{\mathrm{leak}}$ per silicon pair bin. The direction-resolved stream kernels of Sect. 32.4 complete the set for directional searches.

### 33.1.2 Conventions declared with each table

A number in a table is usable only with the conventions that turn it into a rate, and each convention that a user has to guess is a possible error of a factor of order one. The momentum $\boldsymbol{q}$ points from the dark matter to the target and $\omega$ is the energy given to the target, negative at finite temperature for a de-excitation; detailed balance, $S(\boldsymbol{q},-\omega) = \mathrm{e}^{-\beta\omega} S(\boldsymbol{q},\omega)$, relates the two signs, and the table states the temperature it holds. The temperature is metadata, distinct from the thermal order $n_T$ of the label, which counts powers of the phonon occupations (Sect. 8.5).

Next comes the normalization, the most common source of factors of $2\pi$ and $4\pi$. The structure factor of Eq. (4.13) is per volume, in $\mathrm{eV}^2$ in natural units; a table per cell holds $\Omega S$, in $\mathrm{eV}^{-1}$, with $\Omega = 40.05\,\text{Å}^3$ for silicon; a rate per target mass divides by $\rho_T$ in the halo prefactor. Spin is summed. A table of loss functions is turned into structure factors by Eq. (4.45), $S = (2q^2/e^2)\,L$ at zero temperature with $e^2 = 4\pi\alpha$; in Gaussian units the same symbol $e^2$ means $\alpha$ and the factor changes by $4\pi$. A dielectric table must say which function it lists: the head $\varepsilon_{00}$ of the dielectric matrix, whose loss function $\varepsilon_2/|\varepsilon_{00}|^2$ is the random-phase value without local fields, or the macroscopic function $\varepsilon_{\mathrm{mac}} = 1/(\varepsilon^{-1})_{00}$, whose loss function includes them. The two differ by the local-field step, 9 percent at the node of Worked example 33.1. Momenta are unreduced (Appendix A.3); a table that lists $|\boldsymbol{q}+\boldsymbol{G}|$ for the points of its calculation mixes directions at equal length, and an average over bins of $|\boldsymbol{q}|$ is then an average over directions, which the table states.

The row basis and its index order are declared with the matrix. The contraction is $\sum_{ab} \mathfrak{D}_{ab} S_{ab}$ with the rows of Table 10.1, and the order of the indices of $R^0_{\boldsymbol{G}\boldsymbol{G}'}$ fixes which factor is conjugated. Transposing it conjugates the cross terms, which is immaterial in silicon, where every element is real, but not in GaAs (Sect. 12.3). The final-state tables add the labeling of pairs (retarded) and the convention of the sector split, and a table of Migdal responses states its energy axis, the total electronic energy transfer $\omega_e$ (Sect. 12.1).

The truncation label is written out in all eight components of Eq. (8.44), not in the deviation notation that suits running text, together with the reference state (normal, superconducting,

Table 33.1: Schema of a response library for one material. Every table carries the truncation label $\mathfrak{T}$ of Eq. (8.44) with the metadata and conventions of Sect. 33.1.2, a content hash and the list of its parents (Sect. 33.1.3). Energies $\omega$ are given to the target, momenta $\boldsymbol{q}$ point from the dark matter to the target and are not reduced to the first zone. The last column names the sections that specify the calculation filling the table.

| Table | Axes and labels | Content | Specified in |
|---|---|---|---|
| Material inputs | — | structure and measured lattice constant; the gap of the scissor and the gap used by the cascade, separately; functional, pseudopotentials, cutoffs, grids, number of bands; force constants of the shared harmonic base | Sects. 23.3 and 30.1 |
| Electronic response | $\lvert\boldsymbol{q}\rvert$, $\hat{\boldsymbol{q}}$, $\omega$; rows $(a, b)$ | minimal table of Table 10.1 (12 real functions in Si and Ge, 20 with $\mathcal{O}_2$), screened and independent-particle, per cell; $R^0_{\boldsymbol{G}\boldsymbol{G}'}$ on a coarse subset of nodes | Sects. 26.1, 26.2 and 31.3 |
| Directional components | $\lvert\boldsymbol{q}\rvert$, $\omega$; $\ell$ | $S_\ell/S_0$ for $\ell = 4$, 6, 8; full direction grid within the edge sub-grid | Sect. 31.3 |
| Lattice response | $\lvert\boldsymbol{q}\rvert$, $\hat{\boldsymbol{q}}$, $\omega$; $T$ | response per cell; $P_{\mathrm{ph}}(n\lvert\boldsymbol{q}, \omega)$; branch content; coherence correction | Sect. 30.1 |
| Further channels | as the parent table | phonon-assisted loss function; exciton lines and core shells; joint response of the Migdal effect; spin-wave, superconducting and Dirac responses | Sects. 30.3, 31.1, 31.2 and 31.4 |
| Final-state classes | $\lvert\boldsymbol{q}\rvert$, $\omega$, rows; $f_e$, $\omega_{\boldsymbol{Q}}$, valley | seed distribution $P_{\mathrm{seed}}$ per class from retarded pair weights; convention of the zero-phonon/phonon-dressed split | Sects. 26.1 and 26.2 |
| Cascade | carrier energy $x$; model and parameters | $\mathfrak{G}_{\mathrm{e}}$, $\mathfrak{G}_{\mathrm{h}}$ and $P_{\mathrm{casc}}(N_{\mathrm{eh}}\lvert\Lambda)$; marks $\mathfrak{m}$ for the readout | Sects. 29.3 and 32.2 |
| Theory uncertainties | as the parent table; key of $\vartheta_k$ | next-order table $S^{\mathfrak{T}^+_k}$; type and basis of the template; flags for bounds, missing templates and excluded energies; alternative tables at the same $\mathfrak{T}$ | as the parent table |
| Checks | as the parent table | $f$-sum fraction; smallest eigenvalue of $S_{ab}$; Kramers–Kronig residual; detailed balance; symmetry-forbidden parts; cascade normalization, moments and the single-pair limit | as the parent table |
| Derived | bin $i$, speed $v$; channel | step columns $\mathcal{A}_i(v)$; edge mass, onset energy, $p_{\mathrm{on}}$; direction-resolved stream kernels | Sect. 32.4 |

spin-wave), the screening set and the split convention. The approximations that the label leaves out, the exchange–correlation functional, the pseudopotentials, the scissor and the grids, are measured by tests and convergence (Chaps. 23 to 25 and 27), and they go into the material inputs with the results of those tests. The scissor gap and the pair-conversion gap need separate entries. The gap to which a scissor shifts the bands and the gap with which a cascade or a step model converts energy into pairs are different numbers in general, and a rate that uses both inherits two thresholds. Near 1 eV the energy nodes of the tables of Worked example 33.1 are 0.22 eV apart, more than the 0.06 eV between the room-temperature and low-temperature gaps of silicon (Sect. 10.10), so the gap of such a table cannot be read off its nodes and has to be stated. The tables of Worked example 33.1 state a scissor to the measured gap at zero temperature, 1.17 eV; the 1.11 eV that accompanies them in their package is the gap of the step model of pair creation (Sect. 10.10). The interpolation rule between nodes is part of the table too: near 10 eV the same loss function changes by 10 percent from one energy node to the next.

### 33.1.3 Versions and provenance

Every array of a library is identified by a hash of its content. It carries the list of its parents: the structure, the pseudopotentials, the code and its version, the input files, and for a derived table the tables it was computed from. Two arrays with the same name and different content can then not be confused, and a limit can name the exact arrays it used. Versions follow one rule for each kind of change. A change of a truncation label or of a convention changes rates and makes a new major version. Added tables, such as a new row pair, a new material or a new uncertainty template, change no existing rate and make a minor version. A numerical refinement, a finer grid or a larger cutoff, is a patch only if no tabulated value moves by more than the convergence tolerance the table declares. A larger move means the old values were outside their stated accuracy, and it makes a major version as well. A published version is never changed and carries a persistent identifier, so a limit cites the library the way it cites a paper.

A version is released with its checks, each stored as a value rather than a verdict. The $f$-sum rule Eq. (4.30) holds to 2 % below the cutoff of the bands, and a spectrum that ends early stores the fraction it holds at each $|\boldsymbol{q}|$. Positivity, Eq. (4.20), is checked through the smallest eigenvalue of every stored matrix $S_{ab}$, with a tolerance stated by the library, for instance $-10^{-6}$ of its trace; the Kramers–Kronig residual is stored likewise, for instance below 1 %, and the two signs of $\omega$ satisfy detailed balance. In silicon and germanium the symmetry-forbidden parts of the density–current and current blocks stay below $10^{-3}$ of the allowed parts. The cascade tables sum to one and are nonnegative, their derivatives reproduce the moment recursion to $10^{-9}$, and below the first impact ionization, at $(1+\theta_{\mathrm{ii}})E_{\mathrm{g}} = 2.32\,\mathrm{eV}$ in silicon (Sect. 18.3), they give exactly one pair. The step columns never decrease. Where optical, energy-loss or inelastic x-ray measurements exist, the comparison with them is stored in the same way. A table that misses a check by a few percent may still serve an analysis that needs less, and only the stored value lets its user decide.

A limit published with a library names the library and its version, the truncation label of each table it used and the cascade model, and gives per bin the count, the exposure and the detector response. Three things then become recomputations. The limit itself is reproduced from the tables. Its theory band follows from the uncertainty tables and their keys (Sect. 21.4). A recast for another coupling, another halo or a corrected response evaluates the kernel Eq. (21.2) again instead of rescaling a published curve, which Sect. 21.5 shows to be exact only under a condition on the bins. The guide to targets of Sect. 21.6 reads its inputs from the same records: the loss function $L(\boldsymbol{k},\omega)$ of each candidate and its elastic constants.

The provenance it reports for each figure of merit, measured, computed or estimated, is the provenance stored with those tables.

**Worked example 33.1: The silicon loss function at one table node**

Knapen, Kozaczuk and Lin computed the random-phase dielectric function of silicon from Kohn–Sham states [1, Sect. III B]. They used the TB09 functional, the meta-generalized-gradient functional of Tran and Blaha (2009) chosen for band gaps, a plane-wave cutoff of 500 eV, an $8 \times 8 \times 8$ grid of $\boldsymbol{k}$ points, 70 bands per spin and a scissor to the measured gap. The tables distributed with their package [2] list $\varepsilon_1$ and $\varepsilon_2$ against $\omega$ and $|\boldsymbol{q}+\boldsymbol{G}|$, in one file for the head $\varepsilon_{00}$ and in one for $\varepsilon_{\mathrm{mac}}$. We write the entry at the node nearest to $(q_0, 10\,\mathrm{eV})$, with $q_0 = \alpha m_e = 3728.9\,\mathrm{eV}$.

*Node.* $|\boldsymbol{q}| = 3732.0\,\mathrm{eV} = 1.0008\,q_0$ and $\omega = 9.8585\,\mathrm{eV}$. There the head is $\varepsilon_{00} = 1.838 + 0.591\,\mathrm{i}$ and the macroscopic function $\varepsilon_{\mathrm{mac}} = 1.763 + 0.482\,\mathrm{i}$.

*Value.* With local fields, $L = \mathrm{Im}(-1/\varepsilon_{\mathrm{mac}}) = 0.1442$. By Eq. (4.45), $2q^2/e^2 = 3.038 \times 10^8\,\mathrm{eV}^2$, so $S = 4.380 \times 10^7\,\mathrm{eV}^2$ per volume and $\Omega S = 0.2283\,\mathrm{eV}^{-1}$ per cell. The same node gives $L = \varepsilon_2/|\varepsilon_{00}|^2 = 0.1585$ without local fields and, from the head's $\varepsilon_2 = 0.591$, the independent-particle $S^{\mathrm{IP}} = 1.796 \times 10^8\,\mathrm{eV}^2$.

*Label.* $\mathfrak{T} = (n_B = 0,\ n_g = 0,\ n_A = 0,\ \mathcal{X} = \mathrm{RPA} + \mathrm{LF},\ n_T = 0;\ \mathcal{K}, \mathfrak{M}, n_D$ not applicable), with $T = 0$, a normal reference state, external and local-field screening, and no sector split, since a table at $n_g = 0$ has no phonon-dressed part. Off the label: TB09, the cutoff, the grid, the bands and the scissor, as above. The package keeps a separate file of material constants, with $E_{\mathrm{g}} = 1.11\,\mathrm{eV}$ and $\varepsilon_{\mathrm{eh}} = 3.6\,\mathrm{eV}$ for the conversion of energy into pairs by the step model, and an entry of the library would carry that gap beside the gap of the scissor.

*Uncertainty.* The local-field step is computed, so it moves the central value: $\delta_{\mathcal{X}} = 0.1442/0.1585 - 1 = -0.090$ at this node, a tilt by Eq. (8.76). The theory uncertainty that remains is the excitonic correction. No calculation at finite momentum gives it for silicon, so the node carries the flag of a missing template. Sect. 10.10 estimates the step at $+0.28$ over the energies a 3 MeV particle reaches, from a model at 0.91 keV, and a band takes it from there with that basis named.

*Checks.* At this $|\boldsymbol{q}|$ the table holds 0.916 of the valence $f$-sum $(\pi/2)\omega_p^2$, $\omega_p = 16.6\,\mathrm{eV}$, below its last energy, 89.2 eV; the missing weight lies above about 70 eV, which the 70 bands do not reach [1, Fig. 1]. The three other nodes within 50 eV of $|\boldsymbol{q}|$, at 3689, 3722 and 3776 eV, give $L = 0.1431$, 0.1402 and 0.1445. The four nodes span 3.0 percent, partly from direction and partly from $|\boldsymbol{q}|$, and the entry states that its direction average is a bin average over such nodes. The next energy node, 10.26 eV, gives $L = 0.1295$, so at 10 eV linear interpolation gives $L = 0.1390$ and $S = 4.22 \times 10^7\,\mathrm{eV}^2$.

*Comparison.* The empirical-pseudopotential model of Worked example 8.9 gives, at $q_0$ along [100] and 10 eV, $S^{\mathrm{IP}} = 1.7 \times 10^8\,\mathrm{eV}^2$ against $1.74 \times 10^8\,\mathrm{eV}^2$ here, and $S = 4.89 \times 10^7\,\mathrm{eV}^2$ with local fields against $4.22 \times 10^7\,\mathrm{eV}^2$, a ratio of 0.864. The screening factor $S/S^{\mathrm{IP}}$ is 0.29 in the model and 0.242 in the tables, and the local-field step $-0.06$ against $-0.09$. The two calculations carry the same label. What separates them is off the label: a model potential with a gap of 0.82 eV against TB09 with a scissor, and one direction against a mixture. Only the provenance of the table tells a reader which of the two a limit used.

## 33.2 Open calculations

The phonon-assisted loss function of silicon is specified down to its grids, its Wannier manifold and its tolerances, and its plane-wave stages cost some 350 to 450 core-hours (Sect. 31.1). Done, it would turn the estimate of the silicon window into a number with an error bar. Most open calculations of the book are of this kind: specified in Chaps. 30 to 32, and each tied to a definite result. They differ in what they would move, from the reach of the running example to a correction of a few percent on a special target, and in the test that separates a correct result from a converged one.

### 33.2.1 Sum rules and controls

An exact relation holds for every Hamiltonian of a stated class. A result that violates it has an error in its sum over states, in its labeling of final states or in its normalization, whatever it approximates in the interactions. A control compares the result with a measurement that the calculation was not fitted to. The first kind tests completeness and bookkeeping, the second tests the physics, and an open calculation needs both. Exact relations have a limit of their own: an approximation that does not conserve particle number at its vertices violates them by itself, as the bare BCS bubble does (Sect. 7.2), and its error then has to be read from a conserving version of the same response.

The exact relations of the book are the $f$-sum rule and its shares among the sectors of Sect. 8.6 (Eqs. (4.30), (8.54), (8.56) and (8.57)) and its transfer between zero-phonon lines and sideband (Eq. (13.36)); positivity of $S_{ab}$ and Kramers–Kronig (Eqs. (4.20) and (4.24)); the translation Ward identity with its soft and boost limits (Eqs. (8.29), (8.31) and (8.32)); the zeros that symmetry imposes; and the conservation laws of the cascade and of detector transport (Eqs. (18.10), (20.3) and (20.4)). Three of them carry an instruction for a calculation. A table that stores $S_{ab}$ with a negative eigenvalue, or a real part that fails the transform, fails before any comparison with data. The acoustic sum rule Eq. (5.18) is imposed on the dynamical matrix, never by setting the electron–phonon vertex to zero at small phonon momentum (Sect. 31.1). And a calculation that returns the symmetry zeros at the level of its numerical noise has its phases and labels right: the interference (Fano) term between a zero-phonon state and the phonon-assisted continuum, which vanishes at the level of one-body self-energies in crystals invariant under inversion combined with time reversal (Sect. 8.6), the paths that the glide rule Eq. (13.50) removes on the valley axis of silicon, and the $\ell = 2$ component of any scalar response of a cubic crystal. The controls are measured absorption for a finite-$q$ response in its optical limit, neutron data for computed phonons, the measured pair energy and Fano factor of silicon for a cascade, and the neutron-calibrated charge yields of silicon and germanium. Table 33.2 gives the relation and the control that each calculation must pass.

### 33.2.2 Priority list of calculations

In Table 33.2 a calculation that serves several sections appears once, in the row of its largest effect, with the others named. The order starts with the silicon electron recoil, the running example.

Table 33.2: Open calculations that would change a printed result, limit or ranking of the book, ordered by the result each would change; the remaining calculation steps are stated in the sections indexed by Table 22.1. Calculation: what is to be computed, with the section that specifies it. Result: the number or statement of the book that the calculation replaces, with its present size; masses are dark-matter masses. Check: the exact relation (sum rule, positivity, symmetry zero, conservation law) and the control against data that the result must pass. Section: where the item is stated. The guide is the ranking of crystals by figure of merit in Sect. 21.6, its survivors are the eleven crystals with a documented detector record, and $P$(first) is the probability that a crystal ranks first in a row of the guide.

| Calculation | Result it changes | Check it must pass | Section |
|---|---|---|---|
| *Silicon electron recoil* | | | |
| Phonon-assisted loss function from the four-path amplitude, with the zero-phonon rate on common bins (Sect. 31.1) | rate and reach from the onset at 0.333 MeV to the first zero-phonon edge at 0.405 MeV, now an estimate normalized to the measured cold edge; mass up to which the phonon-assisted rate leads (about 0.43 MeV, at least 0.42 MeV); band of comparable rates up to 0.51 MeV | optical limit against the measured 20 K edge; zero rate at $g = 0$; frozen phonon at X; soft acoustic limit Eq. (8.31); symmetry zeros on the valley axis; band-sum $f$-sum with the nonlocal part separated (Worked example 31.1); transfer rule Eq. (13.36) | Sect. 13.9 |
| Pair channel at finite $q$ in a supercell with local fields (Sect. 31.3) | screening correction 0.3 to 2.6 relative to the optical calibration; pair-channel residual unsized | reduces at $q \to 0$ to the optical calibration, which contains the pair channel | Sect. 13.4 |
| Finite-momentum Bethe–Salpeter response of silicon and germanium with the coupling block, bound lines added from computed masses and vertices (Sect. 31.2) | attraction, ×1.28 on the rate from 3.6 to 10 MeV (heavy mediator), 39 % of the variance of the one-standard-deviation factor 1.48 at 3 MeV, 93 % together with the converged table of the next row; triplet channel; spin–orbit mixing at the germanium edge | $f$-sum Eq. (4.30) to a percent; measured optical $\varepsilon_2$ at $q \to 0$; coupling block against its estimate of 10 % | Sects. 10.5 and 10.10 |
| One converged independent-particle table, and all-electron random-phase response with local fields to 10 keV with the current and kinetic rows as external rows (Sects. 26.1 and 31.3) | tabulation spread, ×3.2 at 1 MeV and ×1.34 at 3 MeV; local fields above $2q_0$, ×1.24 at 10 MeV; velocity couplings, now at least 16 times the static-electron formula | $f$-sum at small $q$, where the published tables give 1.1–1.2 against 0.93 at 0.3–0.7 $q_0$, and at 10 keV; positivity of $S_{ab}$; Kramers–Kronig | Sects. 10.1, 10.2 and 10.4 |
| Crystal final states of the germanium 3$d$ shell; four-component response (Sect. 31.2) | 3$d$ share near 30 MeV: 3$d$/valence between 0.07 and 0.69 | relativistic $f$-sum shift, −0.76 % | Sect. 10.6 |
| *Germanium window* | | | |
| Quasiparticle bands of germanium with spin–orbit coupling; transverse-acoustic phonons on a $6^3$ grid (Sect. 31.1) | zero-phonon edge, not known: 0.32 MeV in an on-axis parabolic estimate that orients but does not bound it (Sect. 13.1), above the phonon-assisted onset 0.2119–0.2141 MeV; germanium electronic window and crossovers of Sect. 17.1 | band check of Sect. 31.1: valley order, both gaps, masses at L and Γ; transverse-acoustic branch at X and L against neutron data | Sects. 13.9 and 17.1 |
| *Migdal rate and ranking* | | | |
| Coherent zero-phonon Migdal response of silicon from 0.75 keV to $2q_{\mathrm{BZ}}$, all cross-site terms and local fields (Sect. 31.4) | nearly all of the silicon Migdal rate up to 1 MeV, 33 % at 2 and 7 % at 3 MeV, now unsized | boost limit Eq. (8.32) as $q \to 0$ (the diagonal form is 14.9 times too high there); diagonal form above $2q_{\mathrm{BZ}}$; partial sum rule Eq. (8.56) | Sects. 12.4 and 17.1 |
| Off-diagonal $\varepsilon^{-1}_{\boldsymbol{GG'}}$ in the Migdal rate, with an all-electron or PAW ion vertex and a converged $\boldsymbol{G}$ sum (Sect. 31.4) | factor 1.2–1.4 from a model whose lower end is not bounded | diagonal local fields, ×1.35 (Limit Check 12.2); plain $1/\varepsilon_{00}$ | Sect. 12.3 |
| $\boldsymbol{Q}$-resolved drag mass $\delta M_{\kappa\kappa}(\boldsymbol{Q})$ of crystalline silicon (Sect. 31.4) | on-site drag mass $\delta M_{II}$, valence part 18.6 $m_e$ bounded to 12–24 $m_e$ (the trace value 14 $m_e$ is exact but is not the on-site element) | trace rule Eq. (12.39), 14 $m_e$ per atom at $\boldsymbol{Q} = 0$; acoustic sum rule; partial sum rule Eq. (8.56) | Sects. 12.3 and 12.5 |
| Loss function $L(\boldsymbol{k}, \omega)$ near the gap, $\lvert\boldsymbol{k}\rvert$ up to a few keV, for every survivor (Sect. 31.3) | Migdal row of the guide, without a reading (largest $P$(first) 0.315 among the survivors, 0.498 in the full pool) | inverse fourth moment against the partial $f$-sum bound from Eq. (17.16); gap of the ground state equal to the measured one | Sects. 17.5 and 21.6 |

| Calculation | Result it changes | Check it must pass | Section |
|---|---|---|---|
| *Charge of a nuclear recoil and the cascade* | | | |
| Close-collision excitation kernel Eq. (19.10) for Si–Si and Ge–Ge collisions, with its exchange–correlation spread (Sect. 32.1) | charge yield below the binary thresholds, 352 and 399 eV; measured silicon yield 0.10 at 100 eV with no threshold, germanium $(25 \pm 3)$ % at 254 eV; probability of a recoil without charge | the measured yields and Fano factors of silicon and germanium | Sect. 19.2 |
| Cascade without a fitted constant: impact ionization from the *GW* self-energy, computed phonon couplings of both carriers, excitonic final states, converged photon seed; then germanium and GaAs with spin–orbit coupling (Sects. 29.3 and 32.2) | pair energy without a fitted constant, 4.15 eV [3.95, 4.73] with a class systematic of at least $\pm 0.4$ eV; two-pair fraction at 5 eV, 0.097 in $[0.030, 0.173]$; $F_{\mathrm{eh}} = 0.114$; photon against dark-matter seed | measured $\varepsilon_{\mathrm{eh}} = 3.749$ eV and $F_{\mathrm{eh}} = 0.119$ of silicon, 2.96 eV of germanium at 77 K; normalization, moments Eq. (18.10), one pair below $(1 + \theta_{\mathrm{ii}})E_{\mathrm{g}}$ | Sects. 18.3 and 18.4 |
| Pair energies of the crystals outside the validated range, and cascade tables $P_{\mathrm{casc}}$ for every survivor (Sects. 29.3 and 32.2) | threshold-in-quanta row of the guide: GaSb ranks ahead of germanium with probability 0.83, and the lead salts are not separated, on estimates with the conventional width 0.21 | measured pair energies of the validated range; one pair below $(1 + \theta_{\mathrm{ii}})E_{\mathrm{g}}$ in every table | Sects. 18.4 and 21.6 |
| Binding, trapping and capture rates of pairs at millikelvin in silicon and germanium (Sect. 32.2) | capture fraction (0.54 %, 0.32 to 0.77 %, closes the silicon Fano gap); fraction of excitons never split; release law of the gap energy | measured bound-exciton Auger lifetime; the measured silicon Fano factor | Sects. 18.5 and 20.5 |
| *Readout and delayed signals* | | | |
| Collection kernels of a stated detector from validated phonon and charge transport (Sect. 32.3) | collection ratio, 0.81–1.02 in the mean-field estimate against the 1.128 that collection alone needs; stored-energy bound | Hecht relation, charge sum, bias work, Fano factor of the drifted charge; energy conservation of film phonons | Sects. 19.4 and 20.5 |
| Emission cross-sections of shallow and over-charged traps; Luke emission into both acoustic branches (Sects. 29.2 and 32.3) | delayed charge and space-charge field over an exposure; spectrum of Luke phonons | measured drift velocity against field; phonon signal at fixed field | Sect. 20.1 |
| Release paths of close Frenkel pairs: tunneling exponent and $E_{\mathrm{reorg}}/\varepsilon^{\mathrm{hop}}$ to about 5 % (Sect. 30.5) | delayed release of stored energy at millikelvin; bound on the low-energy excess | stored-energy sum rule Eq. (19.45) | Sect. 19.5 |
| *Lattice, spin, superconducting and novel targets* | | | |
| Anharmonic template of silicon from cubic and quartic force constants, anchored to a path-integral kinetic energy (Sect. 30.2) | few-phonon weights between $q_{\mathrm{BZ}}$ and about 20 keV | exact moments with Hankel positivity Eq. (8.23); kinetic energy within $\pm 0.2$ % | Sects. 11.5 and 11.9 |
| All-electron density response of displaced atoms above the zone (Sect. 30.1) | multiphonon rate of a charge coupling far above $q_{\mathrm{BZ}}$ | translation Ward identity Eq. (8.29) | Sects. 11.3 and 11.9 |
| Zero-temperature two-phonon absorption of silicon below 65 meV, from the derivatives of the Born charges (Sect. 30.1) | electromagnetic channel of vector absorption in the one-phonon band; the isotope channel now exceeds it at least $5.0 \times 10^3$ times at 50 meV | the measured 293 K absorption, reproduced at that temperature; the bound Eq. (11.44) | Sect. 11.6 |
| Cubic-harmonic components of the silicon loss function with local fields (Sect. 31.3) | electronic part of the daily modulation | $\ell = 2$ component zero; no first moment in the components with $\ell \geq 4$, since Eq. (4.30) is isotropic | Sect. 11.7 |
| GaAs dynamical quadrupoles (Sect. 27.2); kinetic energy of the core electrons of free gallium and arsenic atoms | acoustic charge of the polar rate, at most 3 %; GaAs scalar rate, $-19$ % in a parametric estimate | small-$\boldsymbol{Q}$ angular dependence of the vertex; virial theorem of the free atom | Sects. 11.2 and 11.8 |
| Magnetic response of NiO beyond linear spin waves, with its anisotropy near the lightest masses (Sect. 30.3) | one-magnon weight, high by 7.8 %; rate near the edge | total-spin sum rule (Sect. 14.1) | Sect. 14.1 |

| Calculation | Result it changes | Check it must pass | Section |
|---|---|---|---|
| Two-magnon ladder Eq. (14.12) for NiO and YIG on $30^3$ to $40^3$ wavevectors (Sect. 30.3) | two-magnon continuum, about 0.04 of the NiO spin response moved in energy; existence of a bound pair below the continuum | sum rule of the ladder (weight moved, not added); total-spin sum rule | Sect. 14.2 |
| Current response $\boldsymbol{\mathcal{G}}_{\mathrm{OA}}(\boldsymbol{q},\omega)$ of tellurium or $\alpha$-quartz at $\pm\boldsymbol{q}$, with spin–orbit coupling and the nonlocal current vertex (Sect. 30.3) | chiral response at keV momenta, now only the scale $q\,g_{33} = 0.034$ at 1 keV in quartz | density-functional optical activity at small $q$; zero in silicon | Sect. 14.6 |
| Disorder-averaged loss function of amorphous WSi at keV momenta and the distribution of its local gap (Sect. 30.3) | rate of an amorphous film, unsized; it differs from the rate of the averaged $\varepsilon$ at the order of the variance of $\varepsilon$ | computed conductivity against the measured resistivity | Sect. 15.3 |
| Transport function $\alpha^2 F_{\mathrm{tr}}$ of aluminum below a few meV (Sect. 30.3) | phonon-assisted absorption within a few meV of $2\Delta_{\mathrm{BCS}}$; the free-electron model holds $1.25 \times 10^4$ times too much weight there | acoustic sum rule of the vertex; $\lambda_{\mathrm{tr}}$ against the resistivity value 0.39 | Sect. 15.4 |
| Superconducting phonon-assisted absorption with Eliashberg spectral functions, with and without the strong-coupling factor (Sect. 30.3) | rate near the edge; the normal-metal factor $(1+\lambda_{\mathrm{tr}})^{-2} = 0.52$ in aluminum | normal-metal limit at $\omega \ll \omega_{\mathrm{D}}$ | Sect. 15.4 |
| Electron–hole ladder of $ZrTe_5$ near its edge (Sect. 30.3) | rate below 1.33 times the edge (Sommerfeld factor above 1.5) | $f$-sum of the loss function Eq. (4.46) | Sect. 16.1 |
| Ejected-electron final states; periodic finite-$q$ | rates of these targets beyond their model forms | Thomas–Reiche–Kuhn sum per donor; partial $f$-sum per molecule to 2 %; closed-form bubble of an undoped Dirac sheet; Born limit of $\alpha_{\mathrm{ohm}}$; the measured damping of aluminum-covered resonators | Sect. 16.9 |
| Bethe–Salpeter response of a molecular crystal; excitons of a semiconducting tube; valley-orbit coefficients of a donor; layered stack beyond its effective medium; ohmic coupling $\alpha_{\mathrm{ohm}}$ of a tunneling defect (Sects. 28.2, 29.1, 30.4 and 30.5) | | | |
| *Fermionic absorption* | | | |
| Transverse-current response of bound shells with Dirac–Hartree–Fock states along $\omega = m_\chi - q$ (Sect. 31.5) | vector-operator rate on bound electrons, bounded from below by $1/r$ times the free-electron-vertex rate (1.7 for xenon $1s$ at 56.5 keV) | continuity Eq. (4.29) of the shell current | Sect. 10.9 |

No single calculation changes as much as the phonon-assisted one at the head of the table. Up to the first zero-phonon edge it is the whole electron-recoil rate of silicon, and on common bins with the zero-phonon rate it fixes where the zero-phonon line takes over, which Sect. 13.9 places only by estimate. It cannot settle the window alone, however. The rate estimate rests on measured absorption, and the two cold datasets of Sect. 13.9.4 differ by a factor of twenty at 1.20 eV. In Table 21.2 the choice of data moves the rate by a factor of nine at 0.36 MeV, more than the factor three of the finite-$q$ form factor that the calculation replaces. The calculation and a millikelvin measurement of the absorption edge within 25 meV of threshold test each other through the optical limit, and the window needs both.

The sum-rule check of the phonon-assisted calculation is the subtle one, since what the calculation adds is a small part of a sum rule that a complete calculation already satisfies, and Worked example 31.1 works it out.

Above 1 MeV the zero-phonon rate of silicon is computed, and two calculations would narrow its band. Its spread is dominated by the attraction between electron and hole and by the disagreement between published independent-particle tables (Sect. 10.10). A finite-momentum Bethe–Salpeter calculation removes the first, and one converged table that passes the $f$-sum at small $q$ removes the second. The all-electron calculation with local fields to

10 keV serves three sections at once: the local fields at large $q$, the screening of core transitions and the current rows that a velocity coupling needs. Until it exists, a limit on a velocity coupling can use only the bound Eq. (10.15).

The germanium window needs bands before it needs a rate. The phonon-assisted onset follows from the measured transverse-acoustic branch, but the zero-phonon edge depends on the order, the energies and the curvatures of the conduction valleys at L and Γ, which PBE gets wrong and which neither a hybrid functional nor a shift of the 4s-like orbitals gets right together (Sect. 31.1). The check in Table 33.2 therefore includes the band masses beside the two gaps, since the edge enters the window through the pair energy at each momentum transfer, and the germanium window stays an estimate until bands pass both.

Of the Migdal entries, the coherent response acts below $2q_{\mathrm{BZ}}$ and the off-diagonal elements of $\varepsilon^{-1}$ change the rate at all masses (Sect. 12.3.2), while the drag mass enters the energy bookkeeping of Eq. (8.56) at a few parts in $10^4$. The loss function of each survivor near its gap decides whether the Migdal figure of merit ranks crystals at all; between the plasmon-pole and the gap-weighted limits it spans a factor $5.7 \times 10^5$ in silicon (Sect. 17.5).

One kernel decides the charge of a low-energy nuclear recoil: the close-collision kernel Eq. (19.10). Since the Born parameter of the collision, $Z_\kappa \alpha / v_{\mathrm{F}}$ with $v_{\mathrm{F}}$ the Fermi velocity of the target electrons, is 15 in silicon and 35 in germanium, no linear response describes the close collisions that promote electrons below the binary threshold (Sect. 19.2). The exchange–correlation spread of the non-perturbative methods is then part of the result rather than an error to be converged away. The full-band cascade of Sect. 18.3.3 already has no fitted constant, but its envelope over screening, phonon couplings and exchange factor spans a factor of almost six in the two-pair fraction at 5 eV. Computed impact-ionization rates and phonon couplings remove most of that envelope. The test is the pair energy, which the cascade without a fitted constant misses and which is set by the impact-ionization rate below about 4.5 eV (Sect. 18.4); the Fano factor cannot separate the models, since the full-band and the calibrated cascade give the same value. For crystals outside the validated range of Sect. 18.4.4 the same calculation replaces a conventional width by a computed pair energy, and with it the order of the threshold-in-quanta row of the guide.

The readout entries change the interpretation of measured phonon energies rather than any rate. The collection kernels of a stated detector decide whether collection alone can explain the silicon phonon datum (Sects. 19.4 and 20.5); the two-bias measurement of Sect. 19.4 decides the same question from data.

For the running example the lattice calculations hardly matter: for a nucleon coupling the silicon lattice rate is already known to a quarter of a percent above 100 meV at 30 MeV (Sect. 11.9). They matter for charge couplings, quantum crystals and polar targets, and the spin, superconducting and novel entries each move a special target.

### 33.2.3 Theory, calculation and data limits

A better calculation sharpens a channel only if the channel is limited by calculation. Fig. 33.1 sorts every channel of the book by theory, calculation and data, the three classes by which Sect. 16.9.2 sorts the novel targets. It assigns each channel the class of the item that dominates its band at its reference point, read from the open problems and uncertainty budgets of its section and from Table 21.2, and marks the classes that contribute.

Of the 22 channels in the figure, 12 are limited first by calculation, 9 by data and 2 by theory; the phonon-assisted window of silicon counts under both calculation and data. The first theory-limited row is that of strongly anharmonic and quantum crystals at detector temperatures, whose response is bounded only by the band of its exact moments (Sect. 11.5). The second joins Dirac crystals at strong coupling and silicon doped near the Mott density,

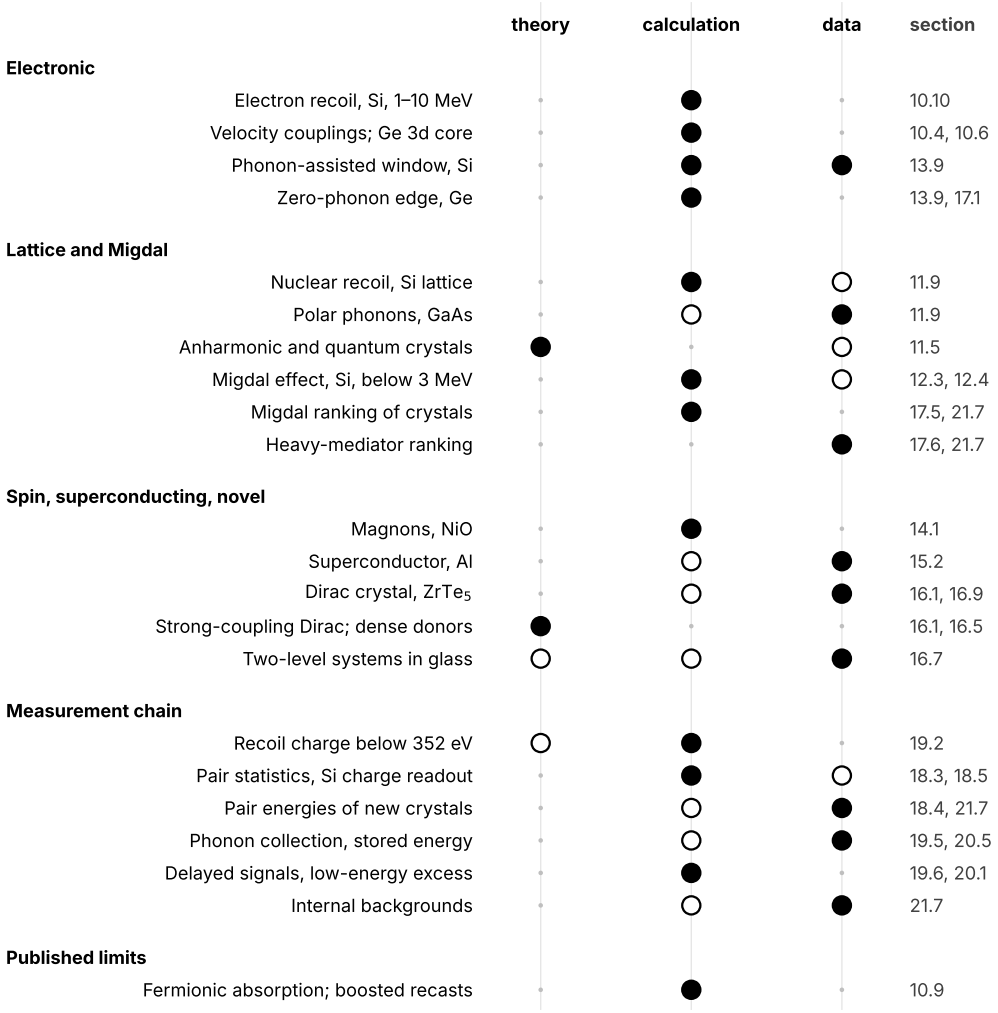



Figure 33.1: Channels of the book by the factor that limits them: theory, calculation or data, as defined in Sect. 16.9.2. Filled circles mark the class of the item that dominates the uncertainty of the channel at its reference point, open circles a class that contributes. The right column gives the section that states the items.

where no small parameter remains (Sects. 16.1 and 16.5). In both, the next step is a method, not a run. Theory also contributes to the recoil charge, where the exchange–correlation spread of the close-collision kernel has a known sign and no known size, and to glass targets, whose two-level systems interact through their strain fields, outside the standard tunneling model of independent systems (Sect. 16.7).

The calculation-limited rows hold the leading electron-recoil and Migdal results of the running example: the silicon rate above 1 MeV, the phonon-assisted window, the germanium edge, the Migdal rate below 3 MeV and the Migdal ranking. The silicon lattice rate belongs here too, since its small band is the difference between computed and measured phonon frequencies. In these channels the framework is complete and the numbers are missing. The data-limited rows are of two kinds. Some are rankings in which computed inputs stand in for measured ones: the elastic tensor of $TeO_2$ for the heavy-mediator row and the pair energies of crystals outside the validated range for the threshold in quanta (Sect. 21.6). Others are properties of a given sample or detector. The polar rate of GaAs rests on the measured ratio of Born charge to dielectric constant, and the background tensor of $ZrTe_5$ moves its reach by a factor 2.8 (Worked example 16.9). The gap and mean free path of an aluminum film, the parameters of the two-level systems of one film, the collection of a stated detector and the $^{210}$Pb and cosmogenic activities of a crystal are of the same kind. For these no calculation in this book substitutes for the measurement.

Four calculations each replace a factor of order ten or an unsized rate in a leading result: the phonon-assisted loss function of silicon with the zero-phonon rate on common bins, the coherent Migdal response of silicon below $2q_{\mathrm{BZ}}$, the loss functions of the survivors near their gaps, and the close-collision kernel. Germanium bands that pass the valley-order and mass checks would open the zero-phonon window of germanium, the survivor with the lowest electron-recoil threshold mass. On the data side, five measurements move a leading result: the cold absorption edge of silicon near its threshold (Sect. 13.9.8), the pair energy of germanium at millikelvin with its phonon collection calibrated separately, the elastic constants of $TeO_2$, the $^{210}$Pb activity of lead that can be grown into a crystal (Table 21.4), and the two-bias measurement that separates collection from stored energy (Sect. 19.4). A tagged-recoil Migdal measurement in silicon would test the chain from the kick to the count (Sect. 12.6.5).

## Summary

Since a limit can be reproduced and recast only from the response tables it was computed with, a response library stores the response uncontracted, one function per pair of coupling rows and without a velocity axis, and declares with every table its truncation label, conventions, version and checks (Sect. 33.1). Its next-order and alternative tables make the theory band of a limit reproducible. Of the 22 channels of Fig. 33.1, 12 are limited first by calculation, 9 by data and 2 by theory (Sect. 33.2). The first group holds the silicon electron recoil above 1 MeV, the phonon-assisted window, the germanium edge and the Migdal rate below 3 MeV. Their calculations are specified, and Table 33.2 orders them by the result each would change; the phonon-assisted loss function of silicon, on common bins with the zero-phonon rate, changes the most. In the data-limited channels no calculation substitutes for a measurement of the sample or the detector. In the two theory-limited ones, strongly anharmonic and quantum crystals at detector temperatures, and Dirac crystals at strong coupling with silicon doped near the Mott density, the next step is a new method.

## References

[1] S. Knapen, J. Kozaczuk, and T. Lin. "Dark matter-electron scattering in dielectrics". In: *Phys. Rev. D* 104 (2021), p. 015031. DOI: 10.1103/PhysRevD.104.015031. arXiv: 2101.08275.

[2] S. Knapen, J. Kozaczuk, and T. Lin. "DarkELF: A python package for dark matter scattering in dielectric targets". In: *Phys. Rev. D* 105 (2022), p. 015014. DOI: 10.1103/PhysRevD.105.015014. arXiv: 2104.12786.

# Appendix A

# Conventions and units

## A.1 Natural units

We set $\hbar = c = k_\mathrm{B} = 1$. Energy, mass, momentum, frequency and temperature are then all measured in electronvolts, length and time in $\mathrm{eV}^{-1}$, and speeds are pure numbers, fractions of the speed of light. A dark-matter speed of $250.5\,\mathrm{km\,s^{-1}}$ is $8.356 \times 10^{-4}$, and the energy and momentum it carries at mass $m_\chi$ are $\frac{1}{2}m_\chi v^2$ and $m_\chi v$ in the same unit. The choice suits a subject in which a particle-physics mass, a crystal momentum, a phonon frequency and a temperature enter one energy-conservation condition. Particle physics uses it routinely, and Ref. [1, Appendix A, Table III] tabulates the astrophysical units it is mixed with; condensed-matter texts often keep $\hbar$ and $k_\mathrm{B}$, and their formulas are converted by restoring the factors below.

### A.1.1 Restoring the constants

A formula in natural units is converted to conventional units by inserting the powers of $\hbar$, $c$ and $k_\mathrm{B}$ that make every term dimensionally consistent. Four constants suffice; their values follow from the 2022 adjustment of CODATA (the Committee on Data of the International Science Council) [2], in which $c$, $h$, $e$ and $k_\mathrm{B}$ are exact:

$$\begin{aligned} \hbar c &= 1.973\,270 \times 10^{-5}\,\mathrm{eV\,cm} = 197.3270\,\mathrm{MeV\,fm}, & \hbar &= 6.582\,120 \times 10^{-16}\,\mathrm{eV\,s}, \\ k_\mathrm{B} &= 8.617\,333 \times 10^{-5}\,\mathrm{eV\,K^{-1}}, & c &= 2.997\,925 \times 10^{5}\,\mathrm{km\,s^{-1}}. \end{aligned} \tag{A.1}$$

A length $L$ becomes $L/(\hbar c)$ in $\mathrm{eV}^{-1}$, a time $t$ becomes $t/\hbar$, a temperature $T$ becomes $k_\mathrm{B}T$ and a speed $u$ becomes $u/c$. A density per unit volume picks up $(\hbar c)^3$, a cross-section $(\hbar c)^{-2}$, and a rate per unit time $\hbar$.

Table A.1 lists the factors. Each entry is a power of one of the products in Eq. (A.1). For example, $1\,\mathrm{cm^{-3}} = (\hbar c)^3/\mathrm{cm^3} = (1.973\,270 \times 10^{-5}\,\mathrm{eV})^3 = 7.6835 \times 10^{-15}\,\mathrm{eV^3}$, and $1\,\mathrm{b} = 10^{-24}\,\mathrm{cm^2}$ divided by $(\hbar c)^2 = 3.8938 \times 10^{-10}\,\mathrm{eV^2\,cm^2}$ gives $2.5682 \times 10^{-15}\,\mathrm{eV^{-2}}$. A mass density in $\mathrm{g\,cm^{-3}}$ first becomes an energy density through $1\,\mathrm{g} = 5.610 \times 10^{23}\,\mathrm{GeV}$ and then picks up $(\hbar c)^3$. Spectroscopic frequencies need care: a frequency $\nu$ quoted in THz corresponds to the energy $h\nu = 2\pi\hbar\nu$, while an angular frequency $\omega$ in $\mathrm{rad\,s^{-1}}$ corresponds to $\hbar\omega$. Phonon tables usually quote $\nu$, so the optical phonon of silicon at the zone center, at 15.5 THz [3], has $\omega_{Q\nu} = 64.1\,\mathrm{meV}$.

### A.1.2 Checking dimensions

In natural units every quantity has a single dimension, a power of energy, and a formula is checked by counting powers of eV. The rate per unit target mass of Eq. (1.46) is the

Table A.1: Conversion factors between conventional units and natural units ($\hbar = c = k_\mathrm{B} = 1$), computed from CODATA 2022 values [2]. Examples in the last column are values used in the book.

| Quantity | Conversion | Example |
|---|---|---|
| length | $1\,\mathrm{cm} = 5.0677 \times 10^4\,\mathrm{eV}^{-1}$, $1\,\text{Å} = 5.0677 \times 10^{-4}\,\mathrm{eV}^{-1}$ | Bohr radius $1/(\alpha m_e) = 0.5292\,\text{Å}$ |
| time | $1\,\mathrm{s} = 1.5193 \times 10^{15}\,\mathrm{eV}^{-1}$ | $1\,\mathrm{yr} = 3.156 \times 10^7\,\mathrm{s} = 4.795 \times 10^{22}\,\mathrm{eV}^{-1}$ |
| temperature | $1\,\mathrm{K} = 8.6173 \times 10^{-5}\,\mathrm{eV}$, $1\,\mathrm{eV} = 11\,605\,\mathrm{K}$ | $300\,\mathrm{K} = 25.85\,\mathrm{meV}$ |
| speed | $1\,\mathrm{km\,s^{-1}} = 3.3356 \times 10^{-6}$ | $v_\mathrm{max} = 794.5\,\mathrm{km\,s^{-1}} = 2.650 \times 10^{-3}$ |
| number density | $1\,\mathrm{cm^{-3}} = 7.6835 \times 10^{-15}\,\mathrm{eV}^3$ | $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}} = 2.31 \times 10^{-6}\,\mathrm{eV}^4$ |
| mass density | $1\,\mathrm{g\,cm^{-3}} = 4.310 \times 10^{18}\,\mathrm{eV}^4$ | silicon, $2.33\,\mathrm{g\,cm^{-3}} = 1.004 \times 10^{19}\,\mathrm{eV}^4$ |
| mass | $1\,\mathrm{kg} = 5.610 \times 10^{26}\,\mathrm{GeV}$ | – |
| cross-section | $1\,\mathrm{cm}^2 = 2.5682 \times 10^9\,\mathrm{eV}^{-2}$, $1\,\mathrm{b} = 2.5682 \times 10^{-15}\,\mathrm{eV}^{-2}$ | $1\,\mathrm{GeV}^{-2} = 0.3894\,\mathrm{mb} = 3.894 \times 10^{-28}\,\mathrm{cm}^2$ |
| spectroscopy | $1\,\mathrm{THz} = 4.1357\,\mathrm{meV}$ $(h\nu)$, $1\,\mathrm{cm^{-1}} = 0.123\,98\,\mathrm{meV}$ | – |

standard example: it is dimensionless, as a rate (eV) per unit mass (eV) must be (Sect. 1.5). The dimensions of the main objects are fixed by their definitions: the dynamic structure factor $S(q,\omega)$ has dimension $\mathrm{eV}^2$ (Sect. 4.2), the transform $\tilde{V}(\boldsymbol{q})$ of a potential has $\mathrm{eV}^{-2}$, the density vertex $\rho_{mn}(\boldsymbol{k},\boldsymbol{q})$ and the dielectric function are dimensionless, and $\bar{\sigma}_e$ has $\mathrm{eV}^{-2}$. Final numbers are always converted back: rates to $\mathrm{kg^{-1}\,yr^{-1}}$, cross-sections to $\mathrm{cm}^2$, momenta to keV, and energies to eV or meV. A rate per unit mass in natural units is converted with $1\,\mathrm{kg^{-1}\,yr^{-1}} = 3.718 \times 10^{-59}$, the inverse of the product of 1 kg in eV and 1 yr in $\mathrm{eV}^{-1}$.

The metric signature is $(+,-,-,-)$, with $x^\mu = (t,\boldsymbol{x})$ and $\partial_\mu\partial^\mu = \partial_t^2 - \nabla^2$, so that a free particle obeys $p^\mu p_\mu = E^2 - |\boldsymbol{p}|^2 = m^2$; it is the signature of most particle-physics texts, and Appendix C.9 uses it. Physical values in the text are quoted in conventional units with a thin space, 1.17 eV or $0.3\,\mathrm{GeV\,cm^{-3}}$, and natural-unit values are written without units only for speeds.

## A.2 Charge conventions

The elementary charge is $e > 0$ with

$$e^2 = 4\pi\alpha, \qquad \alpha = 1/137.035999, \tag{A.2}$$

the rationalized Heaviside–Lorentz convention of particle physics [2]. The electron carries charge $-e$ and a nucleus of species $\kappa$ carries $+Z_\kappa e$, and the charge density operator is $e$ times the sum of the nuclear densities weighted by $Z_\kappa$ minus the electron density. Two electrons at distance $r$ have the interaction energy $e^2/(4\pi r) = \alpha/r$, and Gauss's law states that $\nabla\cdot\boldsymbol{E}$ equals the charge density, without a factor $4\pi$ (Appendix C.9). The Coulomb interaction in momentum space, with the Fourier convention of Appendix A.3, is the kernel

$$v_\mathrm{C}(k) \equiv \frac{e^2}{k^2} = \frac{4\pi\alpha}{k^2}, \tag{A.3}$$

the transform of $e^2/(4\pi r)$ by Eq. (C.79) at $m_V = 0$. In a crystal it is a diagonal matrix in reciprocal-lattice vectors, $(v_\mathrm{C})_{\boldsymbol{G}\boldsymbol{G}'} = v_\mathrm{C}(|\boldsymbol{q}+\boldsymbol{G}|)\,\delta_{\boldsymbol{G}\boldsymbol{G}'}$ (Sect. 4.4).

### A.2.1 Heaviside–Lorentz units and the loss function

Dark-matter couplings are written in field theory, where Heaviside–Lorentz units are standard: the dark-photon coupling to the electron, $y_e = -\zeta e$, is the electromagnetic one scaled by $\zeta$ (Sect. 1.4), and the Yukawa potential of Eq. (1.30) has the same $1/(4\pi r)$ as Coulomb's law. The dielectric response of a solid is expressed in the same units, so the relation Eq. (A.36) between the structure factor and the loss function carries no factors of $4\pi$ [4, Eq. (6), 5, Eq. (S.7)]. In a dark-photon model the factor $1/e^2$ cancels against $y_e^2 = \zeta^2 e^2$, and the rate depends only on $\zeta^2$ and the loss function [5, below Eq. (S.8)].

### A.2.2 Converting from Gaussian and SI units

Condensed-matter sources usually use Gaussian units, in which the electron charge $e_\mathrm{G}$ obeys $e_\mathrm{G}^2 = \alpha$ (with $\hbar = c = 1$), the Coulomb energy is $e_\mathrm{G}^2/r$ and the Coulomb kernel is $4\pi e_\mathrm{G}^2/k^2$. SI sources write $e^2/(4\pi\varepsilon_0^\mathrm{SI} r)$, with $\varepsilon_0^\mathrm{SI}$ the vacuum permittivity; with $\hbar = c = 1$, $e_\mathrm{SI}^2/\varepsilon_0^\mathrm{SI} = 4\pi\alpha$. The three charges are therefore related by

$$e^2 = 4\pi e_\mathrm{G}^2 = \frac{e_\mathrm{SI}^2}{\varepsilon_0^\mathrm{SI}} = 4\pi\alpha. \tag{A.4}$$

A formula is converted by replacing every $e^2$ of this book with $4\pi e_\mathrm{G}^2$ (to Gaussian) or $e_\mathrm{SI}^2/\varepsilon_0^\mathrm{SI}$ (to SI), and the reverse for the other direction [6, Appendix]. Dimensionless response functions, $\varepsilon$, $\varepsilon^{-1}$ and $L$, have the same value in every system; only the formulas that express them through charges change.

The plasma frequency is

$$\omega_p^2 = \frac{e^2 n_e}{m_e} = \frac{4\pi e_\mathrm{G}^2 n_e}{m_e} = \frac{e_\mathrm{SI}^2 n_e}{\varepsilon_0^\mathrm{SI} m_e},$$

in the three systems; all three give 16.6 eV for the four valence electrons per atom of silicon, $n_e = 2.00 \times 10^{23}\,\mathrm{cm}^{-3}$ at $a = 5.431\,\text{Å}$ (Sect. 4.3). The dielectric matrix is $\varepsilon_{\boldsymbol{GG'}} = \delta_{\boldsymbol{GG'}} - v_\mathrm{C}(|\boldsymbol{q}+\boldsymbol{G}|)\tilde{R}_{\boldsymbol{GG'}}$ (Appendix A.7); in the Gaussian sources of the local-field literature [7, 8] the same expression appears with $4\pi e_\mathrm{G}^2/|\boldsymbol{q}+\boldsymbol{G}|^2$, and the polarizability there is our $\tilde{R}$ up to the sign and the factor $1/\mathcal{V}$ that each source fixes in its own definition. For the optical conductivity of Appendix A.7, $\varepsilon = 1 + \mathrm{i}\sigma_\mathrm{cond}/\omega$, the Gaussian form is $\varepsilon = 1 + 4\pi\mathrm{i}\sigma_\mathrm{cond}/\omega$ and the SI form $\varepsilon = 1 + \mathrm{i}\sigma_\mathrm{cond}/(\varepsilon_0^\mathrm{SI}\omega)$, so a measured conductivity $\sigma_1^\mathrm{SI}$ in $\mathrm{S\,m^{-1}}$ gives $\varepsilon_2 = \sigma_1^\mathrm{SI}/(\varepsilon_0^\mathrm{SI}\omega)$. The prefactor $8\pi^2\alpha/(\Omega\omega^2)$ of the independent-particle $\varepsilon_2$ in the optical limit (Sect. 4.4) reads $8\pi^2 e_\mathrm{G}^2/(\Omega\omega^2)$ in the Gaussian sources, and the two agree under Eq. (A.4).

Fields convert as $\boldsymbol{E} = \boldsymbol{E}_\mathrm{G}/\sqrt{4\pi} = (\varepsilon_0^\mathrm{SI})^{1/2}\boldsymbol{E}_\mathrm{SI}$, so that $e\boldsymbol{E} = e_\mathrm{G}\boldsymbol{E}_\mathrm{G} = e_\mathrm{SI}\boldsymbol{E}_\mathrm{SI}$ and the force on a charge is the same in every system. Magnetic fields convert in the same way, $\boldsymbol{B} = \boldsymbol{B}_\mathrm{G}/\sqrt{4\pi} = (\varepsilon_0^\mathrm{SI})^{1/2}\boldsymbol{B}_\mathrm{SI}$ with $c = 1$. The Bohr magneton is then $\mu_\mathrm{B} = e/(2m_e)$, and the spin coupling of an electron is $g_e\mu_\mathrm{B}\,\boldsymbol{s}\cdot\boldsymbol{B}$, with $\boldsymbol{s}$ its spin and $g_e \simeq 2$; the product of $\mu_\mathrm{B}$ and the field strength is an energy with the same value in every system, $5.788 \times 10^{-5}\,\mathrm{eV}$ per tesla.

The symbol $\varepsilon_0$ alone is the static dielectric constant of a crystal, lattice included (Sect. 5.4); the vacuum permittivity appears only as $\varepsilon_0^\mathrm{SI}$, inside quoted SI formulas.

## A.3 Fourier transforms and normalization

A function of position is transformed with a plus sign in the exponent,

$$O(\boldsymbol{q}) = \int \mathrm{d}^3 r\; \mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\, O(\boldsymbol{r}), \qquad O(\boldsymbol{r}) = \int \frac{\mathrm{d}^3 q}{(2\pi)^3}\; \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\, O(\boldsymbol{q}), \tag{A.5}$$

and a function of time with the same sign,

$$f(\omega) = \int \mathrm{d}t\, \mathrm{e}^{+\mathrm{i}\omega t} f(t), \qquad f(t) = \int \frac{\mathrm{d}\omega}{2\pi}\, \mathrm{e}^{-\mathrm{i}\omega t} f(\omega). \tag{A.6}$$

The factor $(2\pi)^{-1}$ per dimension sits with the inverse transform. Applied to the density of point electrons, $n_e(\boldsymbol{r}) = \sum_i \delta^3(\boldsymbol{r} - \boldsymbol{r}_i)$, Eq. (A.5) gives the density operator

$$n_e(\boldsymbol{q}) = \sum_i \mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}, \tag{A.7}$$

and applied to the dark-matter–electron potential it gives $\tilde{V}(\boldsymbol{q}) = \int \mathrm{d}^3x\, \mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}} V(\boldsymbol{x})$, the form used in Eq. (1.31).

### A.3.1 Sign of the spatial transform

The sign is fixed by the direction of $\boldsymbol{q}$, which is the momentum the dark matter gives to the target (Appendix A.4). Take dark-matter plane waves $\langle \boldsymbol{r}_\chi | \boldsymbol{p} \rangle = \mathscr{V}^{-1/2} \mathrm{e}^{\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{r}_\chi}$ in a box of volume $\mathscr{V}$, and a potential $V(\boldsymbol{r}_\chi - \boldsymbol{r}_e)$ between the dark matter and one electron. Shifting the integration variable to $\boldsymbol{x} = \boldsymbol{r}_\chi - \boldsymbol{r}_e$,

$$\langle \boldsymbol{p} - \boldsymbol{q} | V(\boldsymbol{r}_\chi - \boldsymbol{r}_e) | \boldsymbol{p} \rangle = \frac{1}{\mathscr{V}} \int \mathrm{d}^3 r_\chi\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_\chi} V(\boldsymbol{r}_\chi - \boldsymbol{r}_e) = \frac{1}{\mathscr{V}}\, \tilde{V}(\boldsymbol{q})\, \mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_e}. \tag{A.8}$$

With Eq. (A.5), the transform of the potential and the operator left on the target both carry $+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}$, and summed over electrons the target operator is $n_e(\boldsymbol{q})$ of Eq. (A.7). With the opposite spatial convention, $\tilde{V}$ would be evaluated at $-\boldsymbol{q}$ and the target operator would be $n_e(-\boldsymbol{q})$. Single-particle form factors follow the same rule: $f_{i\to f}(\boldsymbol{q}) = \langle f | \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} | i \rangle$ [9, Eq. (3.10)]. Ref. [5, Eq. (S.4)] uses the other spatial convention with the same direction of $\boldsymbol{q}$, and writes the density operator as $\sum_i \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}_i}$; Ref. [10, Eq. (4)] keeps $\mathrm{e}^{+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{x}}$ on the target by writing the potential's transform at $-\boldsymbol{q}$. Appendix A.4 gives the condition under which these differences matter.

The time convention fixes where retarded functions are analytic. A response that starts at $t = 0$ and oscillates at a transition energy $\Omega$ transforms as $-\mathrm{i}\int_0^\infty \mathrm{d}t\, \mathrm{e}^{\mathrm{i}(\omega-\Omega)t - 0^+ t} = 1/(\omega - \Omega + \mathrm{i}0)$. Its pole lies in the lower half of the complex $\omega$ plane, so retarded functions are analytic for $\mathrm{Im}\,\omega > 0$, and their denominators carry $\omega - (E_f - E_i) + \mathrm{i}0$ (Sect. 4.1). A time-dependent perturbation of frequency $\omega > 0$ is $\propto \mathrm{e}^{-\mathrm{i}\omega t}$, the convention of quantum mechanics; engineering texts use $\mathrm{e}^{+\mathrm{i}\omega t}$ (written $\mathrm{e}^{j\omega t}$), and their dielectric functions are the complex conjugates of ours, with $\mathrm{Im}\,\varepsilon < 0$ for absorption. Here $\varepsilon = \varepsilon_1 + \mathrm{i}\varepsilon_2$ with $\varepsilon_2 > 0$ for an absorbing medium at $\omega > 0$.

### A.3.2 Box normalization

States are normalized in a box of volume $\mathscr{V}$ with periodic boundary conditions, and the limit $\mathscr{V} \to \infty$ is taken at the end. Allowed wavevectors form a grid of density $\mathscr{V}/(2\pi)^3$, so sums and Kronecker deltas become integrals and delta functions by

$$\sum_{\boldsymbol{q}} \longleftrightarrow \mathscr{V} \int \frac{\mathrm{d}^3 q}{(2\pi)^3}, \qquad \delta_{\boldsymbol{q}\boldsymbol{q}'} \longleftrightarrow \frac{(2\pi)^3}{\mathscr{V}}\, \delta^3(\boldsymbol{q} - \boldsymbol{q}'). \tag{A.9}$$

The second follows from the first, since $\sum_{\boldsymbol{q}} \delta_{\boldsymbol{q}\boldsymbol{q}'} = 1$. The box contains $N$ unit cells of volume $\Omega$, so $\mathscr{V} = N\Omega$, and a uniform grid of crystal momenta in the first Brillouin zone has $N$ points (Sect. 3.1). Sums over the zone become

$$\frac{1}{N} \sum_{\boldsymbol{k}\in\mathrm{BZ}} \longleftrightarrow \Omega \int_{\mathrm{BZ}} \frac{\mathrm{d}^3 k}{(2\pi)^3}, \qquad \langle\!\langle \cdots \rangle\!\rangle \equiv \frac{1}{N^2} \sum_{\boldsymbol{k},\boldsymbol{Q}} (\cdots) = \Omega^2 \int_{\mathrm{BZ}} \frac{\mathrm{d}^3 k\, \mathrm{d}^3 Q}{(2\pi)^6}\, (\cdots), \tag{A.10}$$

where the double bracket is the zone average over an electron momentum $\boldsymbol{k}$ and a phonon momentum $\boldsymbol{Q}$, used for phonon-assisted processes (Sect. 6.5).

The reason for this bookkeeping is that every intensive quantity must come out independent of $N$, and the powers of $N$ and $\mathcal{V}$ make that check mechanical. The structure factor of Sect. 4.2 has a prefactor $2\pi/\mathcal{V}$, an independent-particle sum $\sum_{\boldsymbol{k}}$ contributes a factor $N$, and the density vertex $\rho_{mn}(\boldsymbol{k},\boldsymbol{q})$ is of order one (Appendix A.5). Their product is $2\pi N/\mathcal{V} = 2\pi/\Omega$, independent of the size of the crystal. A factor of $N$ or $\sqrt{N}$ that survives this count signals an error in a normalization. In the same way the rate per unit target mass contains $1/\rho_T$ and no $N$ (Appendix A.8).

### A.3.3 Lattice-periodic functions

A function with the periodicity of the lattice, such as the crystal potential or the periodic part of a Bloch state, has components only at reciprocal-lattice vectors. With Eq. (A.5) restricted to one cell,

$$f_{\boldsymbol{G}} = \frac{1}{\Omega}\int_\Omega \mathrm{d}^3 r\; \mathrm{e}^{+\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}} f(\boldsymbol{r}), \qquad f(\boldsymbol{r}) = \sum_{\boldsymbol{G}} f_{\boldsymbol{G}}\, \mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}, \tag{A.11}$$

and the transform over the whole crystal is

$$f(\boldsymbol{q}) = \mathcal{V}\sum_{\boldsymbol{G}} f_{\boldsymbol{G}}\,\delta_{\boldsymbol{q},\boldsymbol{G}} \;\longrightarrow\; (2\pi)^3 \sum_{\boldsymbol{G}} f_{\boldsymbol{G}}\,\delta^3(\boldsymbol{q}-\boldsymbol{G}),$$

by Eq. (A.9). Plane-wave codes expand in the opposite direction, with coefficients $c_{n\boldsymbol{k}}(\boldsymbol{G})$ of $\mathrm{e}^{+\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}$ in $u_{n\boldsymbol{k}}(\boldsymbol{r})$ [9, Eq. (3.14)]. A code coefficient $c(\boldsymbol{G})$ is therefore our component at $-\boldsymbol{G}$, and a formula that combines code coefficients with the components of Eq. (A.11) needs $\boldsymbol{G}\to-\boldsymbol{G}$ in one of them. For a real function, $f_{-\boldsymbol{G}} = f^*_{\boldsymbol{G}}$, and the distinction only conjugates; for a Bloch function it does not. Ref. [11, Appendix A] states the transforms between the direct and the reciprocal lattice that the electron–phonon literature uses.

### A.3.4 Crystal momenta and reciprocal-lattice matrices

In a crystal, a momentum transfer $\boldsymbol{q}$ is split into a first-zone part and a reciprocal-lattice vector, $\boldsymbol{q} = \boldsymbol{q}_r + \boldsymbol{G}_q$ with $\boldsymbol{q}_r$ in the first Brillouin zone (Sect. 3.1). Response functions are matrices in reciprocal-lattice vectors, such as $(\varepsilon^{-1})_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q}_r,\omega)$: the transform of a function of two positions $\boldsymbol{r}$ and $\boldsymbol{r}'$ with $\mathrm{e}^{+\mathrm{i}(\boldsymbol{q}_r+\boldsymbol{G})\cdot\boldsymbol{r}}$ and $\mathrm{e}^{-\mathrm{i}(\boldsymbol{q}_r+\boldsymbol{G}')\cdot\boldsymbol{r}'}$. The book indexes such a matrix by the unreduced $\boldsymbol{q}$ at $\boldsymbol{G}=0$:

$$(\varepsilon^{-1})_{00}(\boldsymbol{q},\omega) \equiv (\varepsilon^{-1})_{\boldsymbol{G}_q\boldsymbol{G}_q}(\boldsymbol{q}_r,\omega), \tag{A.12}$$

and likewise for every reciprocal-lattice matrix. Both forms appear in the literature, often for the same object; Ref. [4, Eq. (24)] writes the second with an explicit selection $\delta_{\boldsymbol{k},|\boldsymbol{q}+\boldsymbol{G}|}$. The momentum transfer $\boldsymbol{q}$ itself is never folded: it is the momentum the dark matter loses, and the dark-matter kinematics of Sect. 1.3 depend on $|\boldsymbol{q}|$, not on $|\boldsymbol{q}_r|$.

## A.4 Kinematic signs

The momentum $\boldsymbol{q}$ is transferred from the dark matter to the target: a dark-matter particle of momentum $\boldsymbol{p}$ leaves with $\boldsymbol{p}-\boldsymbol{q}$. The energy $\omega$ is the energy given to the target, so a final target state has $E_f = E_i + \omega$. For a dark-matter particle of laboratory velocity $\boldsymbol{v}$ that changes its mass by $\Delta_\chi = m_{\chi 2} - m_{\chi 1}$ in the collision, energy conservation gives

$$\omega = \omega_{\boldsymbol{q}} \equiv \boldsymbol{q}\cdot\boldsymbol{v} - \frac{q^2}{2m_\chi} - \Delta_\chi, \tag{A.13}$$

as in Eq. (1.21). The minimum velocity needed for a momentum transfer $\boldsymbol{q}$ and an energy deposit $\omega$ is $v_{\min}(q,\omega) = |(\omega + \Delta_\chi)/q + q/(2m_\chi)|$ (Eq. (1.22)). The scattering is endothermic for $\Delta_\chi > 0$ and exothermic for $\Delta_\chi < 0$; elastic scattering has $\Delta_\chi = 0$. The component of the dark-matter velocity along $\hat{\boldsymbol{q}}$ that the delta function selects is $(\omega + \Delta_\chi)/q + q/(2m_\chi)$, and the speed integral of every rate depends on $(\boldsymbol{q},\omega)$ only through it and through $\hat{\boldsymbol{q}}$ (Sect. 1.3).

### A.4.1 Negative energy transfer

At zero temperature the target starts in its ground state and $\omega \geq 0$. At $T > 0$ it starts in a thermal mixture and can give energy to the dark matter, so $\omega < 0$ is allowed. Such transfers are part of the structure factor, which at negative $\omega$ is fixed by its value at positive $\omega$ through detailed balance, $S(-\boldsymbol{q},-\omega) = \mathrm{e}^{-\beta\omega} S(\boldsymbol{q},\omega)$ (Appendix A.7). A negative $\omega$ is therefore suppressed by the Boltzmann factor of the target excitation it de-excites. For electronic excitations across a gap of 1 eV at the 0.1 K or less of a cryogenic detector, $\beta\omega$ exceeds $10^5$, and $\omega < 0$ is dropped; for phonons, magnons and other excitations of a few meV in a target at a few kelvin, it is not. The book does not restrict $\omega$ to positive values; an expression that holds only for $\omega > 0$ says so.

### A.4.2 Direction of the momentum transfer

The direction convention for $\boldsymbol{q}$ differs between papers. Some take $\boldsymbol{q}$ as the momentum gained by the dark matter; Ref. [10, below Eq. (12)] lists papers on each side. Others keep our direction but transform with the opposite sign and write the target vertex as $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ [5, Eq. (S.4)]. Each choice replaces $S(\boldsymbol{q},\omega)$ by $S(-\boldsymbol{q},\omega)$.

The two agree whenever the target responds in the same way at $\boldsymbol{q}$ and $-\boldsymbol{q}$. Inversion symmetry gives this directly, since it maps $n_e(\boldsymbol{q})$ to $n_e(-\boldsymbol{q})$ and leaves the Hamiltonian unchanged. Time-reversal symmetry gives it too: the antiunitary time-reversal operator maps $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ to $\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$, and the thermal ensemble of a time-reversal-invariant target is mapped to itself. Each term $|\langle f|n_e(\boldsymbol{q})|i\rangle|^2$ of the structure factor then equals $|\langle \tilde{f}|n_e(-\boldsymbol{q})|\tilde{i}\rangle|^2$, with $|\tilde{i}\rangle$ and $|\tilde{f}\rangle$ the time-reversed states, which have the same energies, and the sum over the mapped states is the same sum, so $S(\boldsymbol{q},\omega) = S(-\boldsymbol{q},\omega)$. Silicon, germanium and GaAs are time-reversal invariant, and silicon and germanium also have inversion symmetry. A crystal with neither, for example a magnetically ordered crystal without an inversion center, can have $S(\boldsymbol{q},\omega) \neq S(-\boldsymbol{q},\omega)$, and a daily modulation computed with the wrong sign of $\boldsymbol{q}$ is then wrong. A formula written in the other convention becomes ours under $\boldsymbol{q} \to -\boldsymbol{q}$ in every target quantity.

The first-zone part $\boldsymbol{q}_r$ of the momentum transfer labels the crystal-momentum change of the target, and its reciprocal-lattice part $\boldsymbol{G}_q$ labels an Umklapp component (Appendix A.3 and Sect. 3.1); both are needed, because $\omega_{\boldsymbol{q}}$ and $v_{\min}$ depend on the full $\boldsymbol{q}$. The phonon wavevector is a different object, written $\boldsymbol{Q}$ and always in the first zone (Appendix A.6).

## A.5 Bloch and Wannier conventions

A Bloch state is normalized to one over the whole crystal, and its periodic part to one over a unit cell:

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = \frac{1}{\sqrt{N}}\,\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\,u_{n\boldsymbol{k}}(\boldsymbol{r}), \qquad \int_{\mathscr{V}} \mathrm{d}^3r\,|\psi_{n\boldsymbol{k}}|^2 = 1, \qquad \int_{\Omega} \mathrm{d}^3r\,|u_{n\boldsymbol{k}}|^2 = 1. \tag{A.14}$$

The two normalizations are consistent: $|\psi_{n\boldsymbol{k}}|^2 = |u_{n\boldsymbol{k}}|^2/N$ is periodic, and $\mathscr{V}$ holds $N$ cells. This is the convention of Ref. [11, Eq. (29)]. Ref. [9, Eqs. (3.14) and (3.15)] writes the same

states as $\mathcal{V}^{-1/2}\sum_{\boldsymbol{G}} u_n(\boldsymbol{k}+\boldsymbol{G})\mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{G})\cdot\boldsymbol{r}}$ with $\sum_{\boldsymbol{G}}|u_n(\boldsymbol{k}+\boldsymbol{G})|^2=1$; its plane-wave coefficients are the Fourier components of our $u_{n\boldsymbol{k}}$ times $\Omega^{1/2}$.

### A.5.1 Periodic gauge

A Bloch state is fixed only up to a phase, $\psi_{n\boldsymbol{k}}\to\mathrm{e}^{\mathrm{i}\theta_n(\boldsymbol{k})}\psi_{n\boldsymbol{k}}$, and for degenerate or entangled bands up to a unitary mixing [12, Eqs. (6), (7) and (10)]. We fix one part of this freedom everywhere: states at $\boldsymbol{k}$ and at $\boldsymbol{k}+\boldsymbol{G}$ are the same state,

$$\psi_{n,\boldsymbol{k}+\boldsymbol{G}}(\boldsymbol{r})=\psi_{n\boldsymbol{k}}(\boldsymbol{r}),\qquad\text{equivalently}\qquad u_{n,\boldsymbol{k}+\boldsymbol{G}}(\boldsymbol{r})=\mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}\,u_{n\boldsymbol{k}}(\boldsymbol{r}).\tag{A.15}$$

This *periodic gauge* lets a crystal momentum outside the first zone be used directly, without bookkeeping of the reciprocal-lattice vector it differs by. It is the gauge in which Wannier functions are built [12, Sect. II A, footnote 1, and Eq. (10)].

### A.5.2 The density vertex

The matrix element of $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}$ between Bloch states, the density vertex, is

$$\rho_{mn}(\boldsymbol{k},\boldsymbol{q})\equiv\langle\psi_{m,\boldsymbol{k}+\boldsymbol{q}}|\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}|\psi_{n\boldsymbol{k}}\rangle=\int_\Omega \mathrm{d}^3r\;u^*_{m,\boldsymbol{k}+\boldsymbol{q}}(\boldsymbol{r})\,u_{n\boldsymbol{k}}(\boldsymbol{r}),\qquad \rho_{nn}(\boldsymbol{k},0)=1.\tag{A.16}$$

The second form follows from Eq. (A.14): the phases $\mathrm{e}^{-\mathrm{i}(\boldsymbol{k}+\boldsymbol{q})\cdot\boldsymbol{r}}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$ cancel, the integrand is periodic, and the factor $1/N$ cancels the $N$ cells. When $\boldsymbol{k}+\boldsymbol{q}$ lies outside the first zone it is folded back, $\boldsymbol{k}+\boldsymbol{q}=\boldsymbol{k}'+\boldsymbol{G}$ with $\boldsymbol{k}'$ in the zone, and by Eq. (A.15) the vertex becomes $\int_\Omega u^*_{m\boldsymbol{k}'}\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}u_{n\boldsymbol{k}}$: the Umklapp vector stays in the plane wave. The vertex is of order one and carries no factor of $N$ (Appendix A.3). In the notation of Ref. [9, Eq. (3.16)], $\rho_{mn}(\boldsymbol{k},\boldsymbol{q})=f_{[n\boldsymbol{k},m\boldsymbol{k}',\boldsymbol{G}]}$ with $\boldsymbol{q}=\boldsymbol{k}'-\boldsymbol{k}+\boldsymbol{G}$; Ref. [4, Eq. (B2)] defines the same integral over the unit cell, divided by $\Omega$, for $u$ normalized to $\Omega$ rather than to one.

At small $q$ and $m\neq n$, the vertex reduces to a velocity matrix element. Expanding $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\simeq 1+\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}$ and using $\langle m|\boldsymbol{r}|n\rangle=-\mathrm{i}\,\boldsymbol{v}_{mn}/(\varepsilon_m-\varepsilon_n)$, which follows from $\mathrm{i}[H,\boldsymbol{r}]=\boldsymbol{v}$ for $m\neq n$,

$$\rho_{mn}(\boldsymbol{k},\boldsymbol{q})\xrightarrow{q\to0}\frac{\boldsymbol{q}\cdot\boldsymbol{v}_{mn}(\boldsymbol{k})}{\varepsilon_{m\boldsymbol{k}}-\varepsilon_{n\boldsymbol{k}}},\qquad m\neq n,\tag{A.17}$$

with $\boldsymbol{v}_{mn}(\boldsymbol{k})=\langle\psi_{m\boldsymbol{k}}|\boldsymbol{v}|\psi_{n\boldsymbol{k}}\rangle$ (Sect. 3.2). The velocity is $\boldsymbol{p}/m_e$ for a local potential; a nonlocal pseudopotential adds $\mathrm{i}[V_{\mathrm{NL}},\boldsymbol{r}]$, where $V_{\mathrm{NL}}$ is its nonlocal part, and the momentum matrix elements of a code must then be corrected before they are used as velocities (Sect. 26.2).

### A.5.3 Gauge consistency

Under a change of gauge $\psi_{n\boldsymbol{k}}\to\mathrm{e}^{\mathrm{i}\theta_n(\boldsymbol{k})}\psi_{n\boldsymbol{k}}$, the vertex and the velocity transform as

$$\rho_{mn}(\boldsymbol{k},\boldsymbol{q})\to\mathrm{e}^{\mathrm{i}[\theta_n(\boldsymbol{k})-\theta_m(\boldsymbol{k}+\boldsymbol{q})]}\rho_{mn}(\boldsymbol{k},\boldsymbol{q}),\qquad \boldsymbol{v}_{mn}(\boldsymbol{k})\to\mathrm{e}^{\mathrm{i}[\theta_n(\boldsymbol{k})-\theta_m(\boldsymbol{k})]}\boldsymbol{v}_{mn}(\boldsymbol{k}),\tag{A.18}$$

consistently with Eq. (A.17). A first-order rate contains $|\rho_{mn}|^2$ and does not depend on the gauge. A second-order amplitude does not have this protection: it sums products such as $g_{cm\nu}\rho_{mv}$ over intermediate bands $m$ (Sect. 6.5), and each product is gauge-covariant only if both factors were computed from the same set of states. Every quantity that enters one amplitude, the vertex $\rho$, the electron–phonon vertex $g$ of Appendix A.6 and any tensor interpolated with Wannier functions, is therefore computed in one gauge, with one set of gauge matrices $U(\boldsymbol{k})$. Codes that diagonalize at each $\boldsymbol{k}$ independently assign random phases, and independent runs for $\rho$ and $g$ do not share them. Projector-augmented-wave and ultrasoft inputs need the augmentation term in $\rho$ at every $\boldsymbol{q}$, or $\rho_{nn}(\boldsymbol{k},0)=1$ fails (Sect. 24.1).

### A.5.4 Wannier functions and band labels

Wannier functions are built from the Bloch states of one gauge [12, Eqs. (10) and (14)],

$$w_n(\boldsymbol{r}-\boldsymbol{R}) = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{k}\in \mathrm{BZ}} \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} \sum_m U_{mn}(\boldsymbol{k})\, \psi_{m\boldsymbol{k}}(\boldsymbol{r}). \tag{A.19}$$

They are orthonormal over the crystal, and $w_n$ has dimension $\Omega^{-1/2}$ when it is localized within a few cells. The real-space Hamiltonian $H^W(\boldsymbol{R})$ and the interpolated vertices are the matrix elements between them (Sect. 27.1).

With spin–orbit coupling, Bloch states are two-component spinors, $u_{n\boldsymbol{k}}$ carries a spin index, and Eq. (A.16) includes the sum over it; each band then holds one electron per $\boldsymbol{k}$, and the spin factor 2 of the independent-particle formulas is dropped (Appendix A.7). Without spin–orbit coupling the states are spin-degenerate, the band index runs over spatial bands only, and the factor 2 is written explicitly. A degenerate subspace has no preferred basis, so only quantities summed over it, such as $\sum_m |\rho_{mn}|^2$ over the degenerate partners $m$, are compared between codes.

Band indices are $n$ for a generic band and $m$ for an intermediate one. The letters $v$ and $c$ label occupied (valence) and empty (conduction) bands and appear only as subscripts, $\varepsilon_{v\boldsymbol{k}}$ or $\rho_{cv}$; bare $v$ is the dark-matter velocity, and $c$ is the speed of light, set to one.

## A.6 Phonon conventions

The displacement of nucleus $\kappa$ in the cell at $\boldsymbol{R}$ is expanded in normal modes as

$$\boldsymbol{u}_{\boldsymbol{R}\kappa} = \sum_{\boldsymbol{Q}\nu} \frac{1}{\sqrt{2NM_\kappa\omega_{\boldsymbol{Q}\nu}}}\, \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\, B_{\boldsymbol{Q}\nu}, \qquad B_{\boldsymbol{Q}\nu} = b_{\boldsymbol{Q}\nu} + b^\dagger_{-\boldsymbol{Q}\nu}, \tag{A.20}$$

with $\boldsymbol{Q}$ in the first Brillouin zone, $\nu$ the branch, $M_\kappa$ the nuclear mass and $[b_{\boldsymbol{Q}\nu}, b^\dagger_{\boldsymbol{Q}'\nu'}] = \delta_{\boldsymbol{Q}\boldsymbol{Q}'}\delta_{\nu\nu'}$. The polarization vectors are normalized over the basis and obey

$$\sum_\kappa |\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})|^2 = 1, \qquad \boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q}) = \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*, \qquad \omega_{-\boldsymbol{Q}\nu} = \omega_{\boldsymbol{Q}\nu}. \tag{A.21}$$

These are the conventions of Refs. [11, Eqs. (16)–(18)] and [10, Eq. (29)]. The second relation, a choice of phase, makes $\boldsymbol{u}_{\boldsymbol{R}\kappa}$ Hermitian: the terms at $\boldsymbol{Q}$ and $-\boldsymbol{Q}$ are conjugates of each other. Ref. [11] and most of the phonon literature write the wavevector as $\boldsymbol{q}$ and the operators as $a_{\boldsymbol{q}\nu}$. In this book $\boldsymbol{q}$ is the dark-matter momentum transfer and $a_{\boldsymbol{k}\nu}$ is reserved for magnons (Sect. 7.1), so sources are converted by $\boldsymbol{q} \to \boldsymbol{Q}$ and $a \to b$.

The frequencies and polarization vectors are the eigenvalues and eigenvectors of the dynamical matrix,

$$\sum_{\kappa'\beta} D^{\mathrm{dyn},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{Q})\, e^\beta_{\kappa'\nu}(\boldsymbol{Q}) = \omega^2_{\boldsymbol{Q}\nu}\, e^\alpha_{\kappa\nu}(\boldsymbol{Q}), \qquad D^{\mathrm{dyn},\alpha\beta}_{\kappa\kappa'}(\boldsymbol{Q}) = \frac{1}{\sqrt{M_\kappa M_{\kappa'}}} \sum_{\boldsymbol{R}} \Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R})\, \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}, \tag{A.22}$$

where $\Phi^{(2),\alpha\beta}_{\kappa\kappa'}(\boldsymbol{0},\boldsymbol{R})$ is the second derivative of the Born–Oppenheimer energy with respect to the displacements $u^\alpha_{\boldsymbol{0}\kappa}$ and $u^\beta_{\boldsymbol{R}\kappa'}$, and $\alpha, \beta$ are Cartesian indices (Sect. 5.2) [11, Eq. (14)]. The sign of the exponent matches Eq. (A.20): a displacement pattern $\propto \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}$ is an eigenvector of $D^{\mathrm{dyn}}(\boldsymbol{Q})$, not of $D^{\mathrm{dyn}}(-\boldsymbol{Q})$. A code that uses the opposite sign exchanges $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})$ and $\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*$; by Eq. (A.21) this maps $\boldsymbol{Q}$ to $-\boldsymbol{Q}$ and must be undone before polarization vectors from such a code are combined with vertices from another.

### A.6.1 Zero-point length

The scale of the displacement is the zero-point length of a mode,

$$\ell_{\boldsymbol{Q}\nu} \equiv \frac{1}{\sqrt{2M_{\mathrm{cell}}\,\omega_{\boldsymbol{Q}\nu}}}, \tag{A.23}$$

with $M_{\mathrm{cell}} = \sum_\kappa M_\kappa$ and the factor $1/\sqrt{N}$ kept outside. For an acoustic mode at small $Q$ all nuclei move together, $\boldsymbol{e}_{\kappa\nu} = (M_\kappa/M_{\mathrm{cell}})^{1/2}\hat{\boldsymbol{e}}_\nu$ with a unit vector $\hat{\boldsymbol{e}}_\nu$, which satisfies Eq. (A.21). The amplitude in Eq. (A.20) is then $(2NM_\kappa\omega_{\boldsymbol{Q}\nu})^{-1/2}(M_\kappa/M_{\mathrm{cell}})^{1/2} = \ell_{\boldsymbol{Q}\nu}/\sqrt{N}$ for every $\kappa$, so one mode displaces each nucleus by $\ell_{\boldsymbol{Q}\nu}\hat{\boldsymbol{e}}_\nu/\sqrt{N}$ per unit of $B_{\boldsymbol{Q}\nu}$. Keeping $1/\sqrt{N}$ outside $\ell_{\boldsymbol{Q}\nu}$ matches the vertex normalization below, in which all such factors are explicit. Ref. [11, Eq. (34)] writes the same scale with an arbitrary reference mass $M_0$ and the ratio $(M_0/M_\kappa)^{1/2}$.

### A.6.2 The electron–phonon vertex

The first-order coupling of electrons to phonons has the form

$$H_{\mathrm{ep}} = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{k}\boldsymbol{Q}} \sum_{mn\nu} g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})\, c^\dagger_{m,\boldsymbol{k}+\boldsymbol{Q}} c_{n\boldsymbol{k}}\, B_{\boldsymbol{Q}\nu}. \tag{A.24}$$

The vertex is

$$g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q}) = \frac{1}{\sqrt{2\omega_{\boldsymbol{Q}\nu}}} \sum_\kappa \frac{\boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})}{\sqrt{M_\kappa}} \cdot \langle \psi_{m,\boldsymbol{k}+\boldsymbol{Q}} | \partial_{\kappa,\boldsymbol{Q}} V_{\mathrm{KS}} | \psi_{n\boldsymbol{k}} \rangle. \tag{A.25}$$

Here $\partial_{\kappa,\boldsymbol{Q}} V_{\mathrm{KS}} = \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{Q}\cdot\boldsymbol{R}}\, \partial V_{\mathrm{KS}}/\partial \boldsymbol{u}_{\boldsymbol{R}\kappa}$ is the change of the self-consistent Kohn–Sham potential for a displacement wave of nucleus $\kappa$. It is computed with density-functional perturbation theory and therefore screened by the electrons (Sects. 6.2 and 25.2). The matrix element, normalized as in Eq. (A.14), is of order one in $N$, and so is $g$; the single factor $1/\sqrt{N}$ of Eq. (A.24) carries the size dependence. This is the normalization of Refs. [11, Eqs. (37) and (38)] and [13, Eq. (2)], and it is what the EPW code outputs, so computed vertices enter the formulas of this book without rescaling. The vertex has dimension of energy.

Hermiticity of $H_{\mathrm{ep}}$ fixes the conjugate vertex. The Hermitian conjugate of the term $(\boldsymbol{k},\boldsymbol{Q},m,n)$ is $g^*_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})\, c^\dagger_{n\boldsymbol{k}} c_{m,\boldsymbol{k}+\boldsymbol{Q}} B^\dagger_{\boldsymbol{Q}\nu}$, and $B^\dagger_{\boldsymbol{Q}\nu} = B_{-\boldsymbol{Q}\nu}$. It must equal the term $(\boldsymbol{k}+\boldsymbol{Q},-\boldsymbol{Q},n,m)$ of the sum, so

$$g_{mn\nu}(\boldsymbol{k},\boldsymbol{Q})^* = g_{nm\nu}(\boldsymbol{k}+\boldsymbol{Q},-\boldsymbol{Q}), \tag{A.26}$$

which also follows from Eq. (A.25) with $\boldsymbol{e}_{\kappa\nu}(-\boldsymbol{Q}) = \boldsymbol{e}_{\kappa\nu}(\boldsymbol{Q})^*$ and $(\partial_{\kappa,\boldsymbol{Q}} V_{\mathrm{KS}})^\dagger = \partial_{\kappa,-\boldsymbol{Q}} V_{\mathrm{KS}}$. The vertex is split into a short-range and a long-range part, $g = g^{\mathrm{S}} + g^{\mathrm{L}}$, in polar crystals, where $g^{\mathrm{L}}$ is the Fröhlich term (Sect. 6.2). Both carry the gauge of the electronic states, and the rule of Appendix A.5 applies: $g$ and $\rho$ in one amplitude come from one set of states.

### A.6.3 Emission and absorption

In Eq. (A.24) the electron gains crystal momentum $\boldsymbol{Q}$. It does so either by absorbing a phonon $(\boldsymbol{Q},\nu)$, through $b_{\boldsymbol{Q}\nu}$, or by emitting a phonon $(-\boldsymbol{Q},\nu)$, through $b^\dagger_{-\boldsymbol{Q}\nu}$; both use the same vertex $g(\cdot,\boldsymbol{Q})$. With $n_{\boldsymbol{Q}\nu} = n_{\mathrm{B}}(\omega_{\boldsymbol{Q}\nu})$ phonons in the mode, $n_{\mathrm{B}}(x) = (\mathrm{e}^{\beta x}-1)^{-1}$, the squared matrix elements $|\langle n+1|b^\dagger|n\rangle|^2 = n+1$ and $|\langle n-1|b|n\rangle|^2 = n$ give the thermal weights

$$n_{\boldsymbol{Q}\nu} + \tfrac{1}{2} \pm \tfrac{1}{2}, \tag{A.27}$$

with the upper sign for emission of $-\boldsymbol{Q}$ and the lower for absorption of $+\boldsymbol{Q}$. The energy given to the lattice is $\pm\omega_{\boldsymbol{Q}\nu}$ with the same sign convention. At $T = 0$ only emission survives.

### A.6.4 Second order and the Debye–Waller factor

The coupling at second order in the displacements has a vertex quadratic in $B$, which we call the Allen–Heine vertex $g^{\mathrm{AH}}$; its self-energy is $\Sigma^{\mathrm{AH}}$ (Sect. 6.3). Ref. [11, Eqs. (39) and (40)] calls the same vertex $g^{\mathrm{DW}}$, a "Debye–Waller" vertex. This book reserves "Debye–Waller" for one object only, the reduction of the zero-phonon weight of a momentum transfer $\boldsymbol{q}$ to nucleus $\kappa$ by the factor $\mathrm{e}^{-2W_\kappa(\boldsymbol{q})}$, with

$$2W_\kappa(\boldsymbol{q}) = \boldsymbol{q}\cdot B_\kappa\cdot\boldsymbol{q}, \qquad B_\kappa = \langle \boldsymbol{u}_\kappa \boldsymbol{u}_\kappa\rangle, \tag{A.28}$$

where $B_\kappa$ is the thermal displacement tensor, isotropic only at a site of cubic symmetry (Sect. 5.5). The two objects share $B_\kappa$ and appear next to the same double commutator, that of the first-moment sum rule Eq. (4.28), in different roles. In Sect. 5.5 the factor $\mathrm{e}^{-2W_\kappa(\boldsymbol{q})}$ is the elastic share of the scattering weight, whose mean energy the sum rule fixes at the free recoil energy. In Sect. 6.3 the commutator, taken for a rigid translation, makes $\Sigma^{\mathrm{AH}}$ cancel the Fan–Migdal term for long-wavelength acoustic phonons.

## A.7 Response conventions

Response functions are written $R$, never $\chi$, because $\chi$ is the dark-matter particle. For two target operators $O_a(\boldsymbol{q})$ and $O_b(\boldsymbol{q})$, the retarded response and the response matrix are

$$R_{ab}(\boldsymbol{q},\omega) = -\frac{\mathrm{i}}{\mathscr{V}}\int_0^\infty \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t}\,\langle [O_a(\boldsymbol{q},t)^\dagger, O_b(\boldsymbol{q},0)]\rangle, \tag{A.29}$$

$$S_{ab}(\boldsymbol{q},\omega) = \frac{2\pi}{\mathscr{V}}\sum_{i,f} p_i\,\langle i|O_a(\boldsymbol{q})^\dagger|f\rangle\langle f|O_b(\boldsymbol{q})|i\rangle\,\delta(\omega - E_f + E_i), \tag{A.30}$$

with Heisenberg operators $O(t) = \mathrm{e}^{\mathrm{i}Ht}O\mathrm{e}^{-\mathrm{i}Ht}$, thermal averages with Gibbs weights $p_i = \mathrm{e}^{-\beta E_i}/\mathscr{Z}$, and the time convention of Appendix A.3. The dynamic structure factor of the electron density is the diagonal element $S \equiv S_{ee}$ with $O_e = n_e(\boldsymbol{q})$ (Sects. 4.1 and 4.2). Both carry the factor $1/\mathscr{V}$, so both are intensive, and for density operators both have dimension $\mathrm{eV}^2$. Ref. [4, Eqs. (2) and (4)] defines the same objects for the density, with $\chi$ for $R$, $n_{-\boldsymbol{k}}$ for the operator inside $S$ and $\boldsymbol{k}$ for the momentum transfer. The literature's $\chi$, $\Pi$ and $W$ for response, polarization and loss are listed in Appendix B.2.

Inserting eigenstates of the target into Eq. (A.29) gives its spectral (Lehmann) form,

$$R_{ab}(\boldsymbol{q},\omega) = \frac{1}{\mathscr{V}}\sum_{i,f} p_i\left[\frac{\langle i|O_a^\dagger|f\rangle\langle f|O_b|i\rangle}{\omega - (E_f - E_i) + \mathrm{i}0} - \frac{\langle i|O_b|f\rangle\langle f|O_a^\dagger|i\rangle}{\omega + (E_f - E_i) + \mathrm{i}0}\right], \tag{A.31}$$

with every pole below the real axis, as Appendix A.3 requires. For a diagonal element at $T = 0$ and $\omega > 0$ only the resonant term has an imaginary part, $-\pi\delta(\omega - E_f + E_i)$ times a positive weight, so $\operatorname{Im} R_{aa} < 0$ for $\omega > 0$. The sign is a consequence of the time convention: with $\mathrm{e}^{+\mathrm{i}\omega t}$ in the inverse transform it would be reversed. Analyticity in the upper half plane gives the Kramers–Kronig relations, and they hold for $R$, for $\varepsilon^{-1} - 1$ and, in a crystal with a gap, for $\varepsilon - 1$ (Sects. 4.1 and 4.4).

### A.7.1 Fluctuation–dissipation theorem and detailed balance

Inserting energy eigenstates into Eqs. (A.29) and (A.30) and using $\int_{-\infty}^{\infty}\mathrm{d}t\,\mathrm{e}^{\mathrm{i}\omega t}\langle O_a^\dagger(t)O_b\rangle = \mathscr{V}\,S_{ab}(\boldsymbol{q},\omega)$ and $\int \mathrm{d}t\,\mathrm{e}^{\mathrm{i}\omega t}\langle O_b O_a^\dagger(t)\rangle = \mathrm{e}^{-\beta\omega}\mathscr{V}\,S_{ab}(\boldsymbol{q},\omega)$ gives

$$(1 - \mathrm{e}^{-\beta\omega})\,S_{ab}(\boldsymbol{q},\omega) = \mathrm{i}\big[R_{ab}(\boldsymbol{q},\omega) - R_{ba}(\boldsymbol{q},\omega)^*\big], \qquad S_{aa} = -\frac{2\operatorname{Im} R_{aa}}{1 - \mathrm{e}^{-\beta\omega}}, \tag{A.32}$$

the fluctuation–dissipation theorem (Sect. 4.2). The diagonal form is Ref. [4, Eq. (3)]. For operators with $O(-\boldsymbol{q}) = O(\boldsymbol{q})^\dagger$, such as densities, exchanging $i$ and $f$ in Eq. (A.30) gives detailed balance,

$$S_{ab}(-\boldsymbol{q}, -\omega) = \mathrm{e}^{-\beta\omega}\, S_{ba}(\boldsymbol{q}, \omega). \tag{A.33}$$

At $T \ll \omega$ the factor $1 - \mathrm{e}^{-\beta\omega}$ is one.

### A.7.2 Dielectric function and loss

The longitudinal dielectric matrix of a crystal, with local fields, is built from the Coulomb-proper response $\tilde{R}$. This is the part of the response that is irreducible with respect to one Coulomb line (Sect. 4.4), and

$$\varepsilon_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q}, \omega) = \delta_{\boldsymbol{G}\boldsymbol{G}'} - v_{\mathrm{C}}(|\boldsymbol{q} + \boldsymbol{G}|)\, \tilde{R}_{\boldsymbol{G}\boldsymbol{G}'}(\boldsymbol{q}, \omega), \tag{A.34}$$

with $\tilde{R}$ the response of the total charge in units of $e$ and $v_{\mathrm{C}}$ from Eq. (A.3). This is the unsymmetrized form; the symmetrized form uses $v_{\mathrm{C}}^{1/2} \tilde{R}\, v_{\mathrm{C}}^{1/2}$, has the same diagonal and differs off the diagonal [4, footnote 3]. The inverse matrix, the macroscopic dielectric function and the loss function are

$$\varepsilon_{\mathrm{mac}}(\boldsymbol{q}, \omega) = \frac{1}{(\varepsilon^{-1})_{00}(\boldsymbol{q}, \omega)}, \qquad L(\boldsymbol{q}, \omega) = \mathrm{Im}\big[-(\varepsilon^{-1})_{00}(\boldsymbol{q}, \omega)\big], \tag{A.35}$$

with the unreduced-$\boldsymbol{q}$ indexing of Eq. (A.12), $\varepsilon = \varepsilon_1 + \mathrm{i}\varepsilon_2$, and $L \geq 0$ for $\omega > 0$. For the electron channel alone, $\varepsilon_e^{-1} = 1 + v_{\mathrm{C}} R_{ee}$; it equals $\varepsilon^{-1}$ when the nuclei are clamped. The proper part gives the absorptive part directly, $\varepsilon_2 = -v_{\mathrm{C}}\, \mathrm{Im}\, \tilde{R}$.

Combining $\varepsilon_e^{-1} = 1 + v_{\mathrm{C}} R_{ee}$ with Eq. (A.32) gives the relation between the structure factor and the loss function,

$$S(\boldsymbol{q}, \omega) = \frac{2q^2}{e^2}\, \frac{L_e(\boldsymbol{q}, \omega)}{1 - \mathrm{e}^{-\beta\omega}}, \qquad \varepsilon_2(\boldsymbol{q}, \omega) = \frac{e^2}{2q^2}\, (1 - \mathrm{e}^{-\beta\omega})\, S^{\mathrm{prop}}(\boldsymbol{q}, \omega), \tag{A.36}$$

where $L_e$ is the loss of the electron channel and $S^{\mathrm{prop}}$ the structure factor built from $\tilde{R}_{ee}$. The first relation is Ref. [4, Eq. (6)], with $e^2 = 4\pi\alpha$. It holds with the total loss $L$ at clamped nuclei, and for a dark photon, which couples to the total charge, it holds exactly with the total $\varepsilon$ (Sect. 4.4). The optical conductivity is defined by $\varepsilon = 1 + \mathrm{i}\sigma_{\mathrm{cond}}/\omega$ in the transverse, long-wavelength limit, so that $\sigma_1 = \omega\varepsilon_2$ (Appendix A.2 gives the Gaussian and SI forms).

### A.7.3 Spin factor

The structure factor and $\varepsilon_2$ are summed over spin. For a spin-independent vertex without spin–orbit coupling, each band is doubly occupied, and the independent-particle form carries an explicit factor 2:

$$S(\boldsymbol{q}, \omega) = \frac{4\pi}{\mathscr{V}} \sum_{v,c,\boldsymbol{k}} |\rho_{cv}(\boldsymbol{k}, \boldsymbol{q})|^2\, \delta(\omega - \varepsilon_{c,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}}) \qquad (T = 0), \tag{A.37}$$

where $4\pi/\mathscr{V}$ is $2 \times 2\pi/\mathscr{V}$ and the sums run over spatial bands. With spin–orbit coupling the factor 2 is dropped and the sums run over spinor bands. Refs. [9, Eq. (A.29)] and [14, Eq. (6)] write the same factor outside their band sums, and Ref. [4, Appendix B, above Eq. (B7)] states that it is implicit in its main-text formulas. A factor of 2 between two published rates for the same material is often this factor.

### A.7.4 The first frequency moment

The first frequency moment of the density structure factor is fixed by the f-sum rule. For a target with $S(\boldsymbol{q},\omega) = S(-\boldsymbol{q},\omega)$,

$$\int_{-\infty}^{\infty} \mathrm{d}\omega\, \omega\, S(\boldsymbol{q},\omega) = \frac{\pi\, \boldsymbol{n}_e\, q^2}{m_e}, \tag{A.38}$$

at any temperature, with $\boldsymbol{n}_e$ the mean electron density (Sect. 4.3 and Appendix D.4). Splitting the integral at $\omega = 0$ and using Eq. (A.33) turns the left side into $\int_0^\infty \mathrm{d}\omega\, \omega(1 - \mathrm{e}^{-\beta\omega})S$, and Eq. (A.36) then gives [5, Eq. (S.10)]

$$\int_0^{\infty} \mathrm{d}\omega\, \omega\, L_e(\boldsymbol{q},\omega) = \frac{\pi}{2}\,\omega_p^2, \qquad \omega_p^2 = \frac{e^2 \boldsymbol{n}_e}{m_e}. \tag{A.39}$$

At $T = 0$, where $S$ vanishes for $\omega < 0$, dividing Eq. (A.38) by $2\pi\boldsymbol{n}_e$ gives the per-particle form of many-body texts, $\int_0^\infty \mathrm{d}\omega\, \omega\, S(\boldsymbol{q},\omega)/(2\pi\boldsymbol{n}_e) = q^2/(2m_e)$.

## A.8 Halo conventions and rate normalization

Every rate in this book is a rate per unit target mass, $\mathcal{R}$ in $\mathrm{kg}^{-1}\,\mathrm{yr}^{-1}$. Every halo average uses the standard halo model with the parameters of Ref. [15, Table 1]: $\rho_\chi = 0.3\,\mathrm{GeV\,cm}^{-3}$, $v_0 = 238\,\mathrm{km\,s}^{-1}$, $v_{\mathrm{esc}} = 544\,\mathrm{km\,s}^{-1}$, and the solar velocity $\boldsymbol{v}_\odot = (11.1, 250.2, 7.3)\,\mathrm{km\,s}^{-1}$ in galactic coordinates (Eq. (1.5), Table 1.1). The laboratory distribution is $f_\chi(\boldsymbol{v},t) = N_{\mathrm{esc}}^{-1}\mathrm{e}^{-|\boldsymbol{v}+\boldsymbol{v}_{\mathrm{E}}(t)|^2/v_0^2}\,\Theta(v_{\mathrm{esc}} - |\boldsymbol{v} + \boldsymbol{v}_{\mathrm{E}}(t)|)$ with $\boldsymbol{v}_{\mathrm{E}}(t) = \boldsymbol{v}_\odot + \boldsymbol{v}_\oplus(t)$ (Eq. (1.8)). These are the standard values of the field; the measured ranges are given in Sect. 1.2.

### A.8.1 Two Earth speeds

The Earth's speed enters in two roles. Halo integrals, such as $\eta(v_{\min})$ and the daily and annual modulation, use the full time-dependent $\boldsymbol{v}_{\mathrm{E}}(t)$, averaged over the exposure when a time-averaged rate is quoted. Reference scalars, such as kinematic edges, threshold masses and the windows in $(q,\omega)$ that a mass can reach, use one fixed speed,

$$v_{\mathrm{E}}^{\mathrm{ref}} \equiv |\boldsymbol{v}_\odot| = 250.5\,\mathrm{km\,s}^{-1}, \qquad v_{\max} \equiv v_{\mathrm{esc}} + v_{\mathrm{E}}^{\mathrm{ref}} = 794.5\,\mathrm{km\,s}^{-1} = 2.650\times10^{-3}, \tag{A.40}$$

as in Eq. (1.9). The fixed value keeps every threshold time-independent. Over a year, $v_{\mathrm{esc}} + |\boldsymbol{v}_{\mathrm{E}}(t)|$ ranges from 781 to 810 $\mathrm{km\,s}^{-1}$ (Sect. 1.2); that spread belongs to the uncertainty of the halo, not to the definition of the edge. The annual mean of $|\boldsymbol{v}_{\mathrm{E}}(t)|$ differs from $|\boldsymbol{v}_\odot|$ at second order in $v_\oplus/v_\odot$, and the difference is absorbed in the same way. Papers that quote "$v_{\mathrm{E}}$" mean one or the other.

### A.8.2 Speed integrals

The halo enters through the velocity kernel $\int \mathrm{d}^3v\, f_\chi(\boldsymbol{v})\,\delta(\omega - \omega_{\boldsymbol{q}})$, which depends on $\hat{\boldsymbol{q}}$ and on $v_{\min}(q,\omega)$. For a structure factor that depends only on $|\boldsymbol{q}|$, or after averaging $S$ over the directions of $\boldsymbol{q}$, the angular integral over $\hat{\boldsymbol{q}}$ can be done first, and

$$\int \frac{\mathrm{d}^3q}{(2\pi)^3}\, S(q,\omega) \int \mathrm{d}^3v\, f_\chi(\boldsymbol{v})\,\delta(\omega - \omega_{\boldsymbol{q}}) = \int \frac{q\,\mathrm{d}q}{4\pi^2}\,\eta(v_{\min}(q,\omega))\, S(q,\omega), \tag{A.41}$$

with $\eta(v_{\min}) = \langle\Theta(v - v_{\min})/v\rangle$ (Eq. (1.27)). Per unit $\mathrm{d}^3q/(2\pi)^3$ the kernel is $\eta(v_{\min})/(2q)$; the factor $1/2$ comes from the angular integral $\frac{1}{2}\int_{-1}^{1}\mathrm{d}\cos\theta\,\delta(\cdots - qv\cos\theta) = \Theta(v - v_{\min})/(2qv)$

(Sect. 1.3). The step requires only that the target response be independent of the direction of $\boldsymbol{q}$; the halo itself need not be isotropic, since the integral over $\hat{\boldsymbol{q}}$ at fixed $\boldsymbol{v}$ is the same for every direction of $\boldsymbol{v}$. A second speed moment, $\eta_2(v_{\min}) = 2\int_{v_{\min}}^{\infty} \mathrm{d}v'\, v'\eta(v')$, appears for interactions whose strength grows with the dark-matter velocity. Halo-independent analyses absorb the particle physics into $\tilde{\eta}(v_{\min}) = \rho_\chi \bar{\sigma}_e \eta(v_{\min})/m_\chi$ [16, Eq. (2.8)].

For a crystal the structure factor depends on the direction of $\boldsymbol{q}$, and Eq. (A.41) does not apply. The full kernel is then kept. The halo distribution is written in the laboratory frame at the time of the event, the crystal axes are rotated into that frame with the Earth's rotation (Eq. (1.14)), and the integral over $\hat{\boldsymbol{q}}$ is done with $S(\boldsymbol{q},\omega)$ evaluated in crystal axes. The daily modulation of Sect. 1.2 is the result. A time-averaged rate averages this kernel over the exposure; it is not the rate computed with the angle-averaged $\bar{S}$ unless the exposure samples all orientations of $\boldsymbol{v}_{\mathrm{E}}$ relative to the crystal equally, which a fixed detector does not.

### A.8.3 Rate per unit target mass

For an interaction through the electron density with transform $\tilde{V}(\boldsymbol{q})$, a dark-matter particle in the box of Appendix A.3 scatters at the rate $\int \mathrm{d}^3q\,(2\pi)^{-3}|\tilde{V}(\boldsymbol{q})|^2 S(\boldsymbol{q},\omega_{\boldsymbol{q}})$. By Eq. (A.8) the matrix element is $\mathcal{V}^{-1}\tilde{V}\langle f|n_e(\boldsymbol{q})|i\rangle$, the golden rule squares it, the sum over final dark-matter momenta gives $\mathcal{V}\int \mathrm{d}^3q/(2\pi)^3$, and Eq. (A.30) absorbs the remaining $2\pi/\mathcal{V}$ (Eq. (2.67)). The box holds $\rho_\chi \mathcal{V}/m_\chi$ dark-matter particles and a target mass $\rho_T\mathcal{V}$, so the spectrum per unit target mass is

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega} = \frac{\rho_\chi}{m_\chi\,\rho_T}\int \mathrm{d}^3v\, f_\chi(\boldsymbol{v}) \int \frac{\mathrm{d}^3q}{(2\pi)^3}\,|\tilde{V}(\boldsymbol{q})|^2\, S(\boldsymbol{q},\omega)\,\delta(\omega-\omega_{\boldsymbol{q}}), \tag{A.42}$$

in which the volume has cancelled. With $|\tilde{V}|^2 = \pi\bar{\sigma}_e F_{\mathrm{DM}}^2/\mu_{\chi e}^2$ (Eq. (1.33)) and Eq. (A.41),

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\omega} = \frac{\rho_\chi\,\bar{\sigma}_e}{4\pi\, m_\chi\,\rho_T\,\mu_{\chi e}^2}\int q\,\mathrm{d}q\; F_{\mathrm{DM}}(q)^2\,\eta\big(v_{\min}(q,\omega)\big)\,\bar{S}(q,\omega), \tag{A.43}$$

with $\bar{S}$ the average of $S$ over the directions of $\boldsymbol{q}$. This is Eq. (1.46), and it agrees with Refs. [4, Eq. (10)] and [5, Eq. (S.9)], the latter after Eq. (A.36). For operators other than the density, $|\tilde{V}|^2 S$ is replaced by a contraction of a dark-matter-side tensor with the response matrix $S_{ab}$ (Sect. 4.2).

Other papers normalize per target or per unit cell. Ref. [9, Eq. (3.13)] gives the rate of a whole target of $N_{\mathrm{cell}}$ cells, and Ref. [14, Eq. (36)] the same with $n_\chi = \rho_\chi/m_\chi$. Dividing by the target mass $N_{\mathrm{cell}}M_{\mathrm{cell}}$, with $\rho_T = M_{\mathrm{cell}}/\Omega$, and comparing with Eq. (A.43) identifies the crystal form factor of Ref. [9] with the angle-averaged structure factor,

$$|f_{\mathrm{crys}}(q,\omega)|^2 = \frac{q^3\,\Omega}{4\pi\alpha\, m_e^2}\,\bar{S}(q,\omega) = \frac{q^5\,\Omega}{8\pi^2\alpha^2 m_e^2}\,\varepsilon_2(q,\omega), \tag{A.44}$$

where the second form uses Eq. (A.36) at $T=0$ without screening and agrees with Ref. [4, Eq. (16)]. The product $\Omega\bar{S}$, of dimension $\mathrm{eV}^{-1}$, is the natural per-cell quantity to tabulate.

## A.9 Cross-section conventions

The strength of a dark-matter–electron interaction is quoted as the reference cross-section $\bar{\sigma}_e$ and its momentum dependence as the form factor $F_{\mathrm{DM}}(q)$ of Eq. (1.32), normalized at $q_0 = \alpha m_e = 3.73\,\mathrm{keV}$. For a mediator of mass $m_V$, $F_{\mathrm{DM}}(q) = (m_V^2+\alpha^2m_e^2)/(m_V^2+q^2)$, which is 1 for a heavy and $(q_0/q)^2$ for a light mediator [4, Eqs. (11) and (12)]. The form factor

multiplies the amplitude, so it enters every rate as $F_{\rm DM}^2$. Papers differ in the reference point, the medium and the amplitude normalization of this definition.

The first is the reference point. The momentum $q_0 = \alpha m_e$ is the inverse Bohr radius, typical of bound electrons. A paper that normalizes at another momentum $q_{\rm ref}$ quotes $\sigma_{\rm ref} = \bar\sigma_e\, F_{\rm DM}(q_{\rm ref})^2$; for a light mediator this is $\bar\sigma_e\,(q_0/q_{\rm ref})^4$, so a limit quoted at $q_{\rm ref} = 2q_0$ is smaller by a factor 16 for the same physics.

The second is the medium. Here $\bar\sigma_e$ is defined in vacuum: $\tilde V$ is the bare potential between a dark-matter particle and a free electron, and screening, local fields and every other effect of the target are in the response $S$ (Sect. 4.4). The same convention is used in Refs. [4, Eq. (8) and the paragraph after it] and [5, below Eq. (S.8)]. Some papers put the in-medium dielectric factor into the coupling, which is common for dark photons and for Dirac materials with a background dielectric constant. They quote a cross-section that depends on the target; it is converted by removing that factor from the coupling and putting the corresponding screening into the response.

The third is the amplitude normalization. Ref. [9, Eq. (3.8)] defines

$$\bar\sigma_e = \frac{\mu_{\chi e}^2\,|\mathcal{M}_{\rm free}(q_0)|^2}{16\pi m_\chi^2 m_e^2},$$

with the relativistically normalized free amplitude $\mathcal{M}_{\rm free}$, averaged over initial and summed over final spins. Comparison with Eq. (1.32) gives the dictionary

$$|\mathcal{M}_{\rm free}(q)|^2 = 16\, m_\chi^2 m_e^2\,|\tilde V(q)|^2, \tag{A.45}$$

the factor $4m_\chi m_e$ per amplitude being the ratio of relativistic to nonrelativistic state normalizations. A nonrelativistic effective operator with coefficient $c_1$ in the relativistic normalization therefore has $\tilde V = c_1/(4m_\chi m_e)$ and $\bar\sigma_e = \mu_{\chi e}^2|c_1|^2/(16\pi m_\chi^2 m_e^2)$.

The spin average is part of this definition. The amplitude squared in Eq. (A.45) is averaged over the initial spins of the dark matter and the electron and summed over the final spins [9, below Eq. (3.8)]. For a spin-independent interaction this average equals the squared amplitude of a single spin state, so $\bar\sigma_e$ is the same for scalar and fermionic dark matter with the same $\tilde V$ [9, below Eq. (2.2)]. The electron spin sum is then not repeated in the target response: there the factor 2 of Appendix A.7 counts occupied spin states, not final spins of a free electron [9, below Eqs. (A.28) and (A.29)].

### A.9.1 Differential cross-section

The differential cross-section of free elastic scattering in the first Born approximation is Eq. (2.15),

$$\frac{\mathrm{d}\sigma_T}{\mathrm{d}q} = \frac{q\,\bar\sigma_e}{2\mu_{\chi e}^2 v^2}\,F_{\rm DM}(q)^2, \qquad 0 \le q \le 2\mu_{\chi e} v.$$

For $F_{\rm DM} = 1$ it integrates to $\bar\sigma_e$, which is the meaning of the reference cross-section for a heavy mediator. The same normalization holds for nuclei, $\mathrm{d}\sigma_N/\mathrm{d}q = q\sigma_N F_N(q)^2/(2\mu_{\chi N}^2 v^2)$ (Eq. (2.27)). A differential cross-section defined per unit $q^2$ is $\mathrm{d}\sigma_T/\mathrm{d}q^2 = (2q)^{-1}\mathrm{d}\sigma_T/\mathrm{d}q$; a factor of 2 between two papers often traces to this choice.

### A.9.2 Nucleons and nuclei

For a mediator coupled to nucleons, the dark-matter–proton cross-section $\sigma_p$ is the analogue of $\bar\sigma_e$ with $\mu_{\chi p}$, and the spin-independent nuclear cross-section at small $q$ follows from it. It is $\sigma_N = \sigma_p(\mu_{\chi N}/\mu_{\chi p})^2 A^2$ (Eq. (1.36)), with $A^2 \to [Z + (A-Z)y_n/y_p]^2$ for unequal proton and

neutron couplings; Ref. [1, Eq. (104)] writes the equivalent rescaling relative to the neutron coupling. For a dark photon, which couples to protons only, $A^2 \to Z^2$. At larger $q$ the nuclear form factor $F_{\mathrm{Helm}}(q)$ multiplies the amplitude (Sect. 2.2). For a dark photon the electron and proton couplings have equal magnitude $\zeta e$, so for a heavy dark photon, $m_V \gg q_0$, $\sigma_p/\bar{\sigma}_e = (\mu_{\chi p}/\mu_{\chi e})^2$; this ratio is how electron- and nuclear-scattering limits on the same model are compared.

### A.9.3 Kinetic mixing

The dark photon mixes with the photon through the term $+(\zeta/2)F_{\mu\nu}F'^{\mu\nu}$ in the Lagrangian density, with $\mathcal{L} \supset -eA_\mu J^\mu_{\mathrm{em}}$ for the photon (Eq. (C.83)). The sign of the mixing term follows Ref. [1, Eq. (70)], which writes $\kappa$ for $\zeta$ and $+eA_\mu J^\mu_{\mathrm{em}}$ for the photon coupling; Ref. [9, Eq. (2.1)] writes $\varepsilon$ and mixes with hypercharge, $(\varepsilon/2\cos\theta_W)F_{Y\,\mu\nu}F'^{\mu\nu}$, where $\theta_W$ is the weak mixing angle. With our signs, the dark photon couples to electrons with $y_e = -\zeta e$ and to protons with $+\zeta e$ (Sect. 1.4). A change of either sign flips the sign of $y_e$ and of $\tilde{V}$, and no rate changes, because every rate contains $\zeta^2$. The symbols used for $\zeta$ in the literature are listed in Appendix B.2.

### A.9.4 Relativistic dark matter

The form factor of Eq. (1.32) is the nonrelativistic limit, $\omega \ll q$, of the mediator propagator $1/(q^2 - \omega^2 + m_V^2)$ (Appendix C.9). For a light mediator in scattering, $\omega \leq q v_{\max}$ and the correction is below $10^{-5}$. Dark matter boosted to relativistic speeds is outside this limit: its rates need the full propagator and the coupling to the transverse current, and $F_{\mathrm{DM}}(q)$ alone does not describe them.

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# Appendix B

# Notation

Every symbol first used in this volume is listed in Appendix B.1 with its meaning, its units and the section of its first use, and every acronym in Appendix B.3 with its expansion. Appendix B.2 gives the symbols that common papers and textbooks use for the same quantities; the sign and normalization conventions in which they differ from this book are in Appendix A.

## B.1 List of symbols

The list holds the symbols first used in Volume III; a symbol first used in another volume is listed there. Each symbol is listed once, under the chapter or appendix in which it first appears; the last column gives that section. The Units column gives the dimension in natural units ($\hbar = c = k_\mathrm{B} = 1$, Appendix A.1) and, after a semicolon, the conventional unit in which values are quoted. The entry 1 marks a dimensionless quantity, and a dash an object without a dimension of its own, such as an operator, a state or a label.

| Symbol | Meaning | Units | Sect. |
|---|---|---|---|
| **Chapter 23: Density-functional theory** | | | |
| $h$, $h_\mathrm{HF}$ | One-body Hamiltonian $-\nabla^2/2m_e + U_1$ of a determinant; Hartree–Fock (Fock) operator | eV | 23.1 |
| $\varrho(x, x')$ | One-body density matrix $\sum_\mathrm{occ} \psi_\gamma(x)\psi^*_\gamma(x')$, the kernel of $\hat{\varrho}(x = (r, s))$ | $\mathrm{Å}^{-3}$ | 23.1 |
| $\lambda_{\gamma\gamma'}$ | Lagrange multipliers of orthonormality | eV | 23.1 |
| $\mathcal{M}_{\gamma\gamma'}$ | Unitary matrix of occupied-orbital rotations | 1 | 23.1 |
| $E_\mathrm{HF}$, $E^\mathrm{HF}_{\boldsymbol{k}}$, $E_x$ | Hartree–Fock total energy; Hartree–Fock band $k^2/2m_e + \Sigma_\mathrm{x}(k)$ of the gas; exchange energy ($E_x/N_e$ per electron) | eV | 23.1 |
| $E_\mathrm{HOMO}$ | Highest occupied Kohn–Sham eigenvalue | eV | 23.2 |
| $N_\mathrm{PW}$ | Number of plane waves per state, $\Omega k^3_\mathrm{cut}/6\pi^2$ | 1 | 23.3 |
| $N_\mathrm{b}$, $N_\mathrm{occ}$, $N_\mathrm{proj}$ | Numbers of bands, of occupied bands and of nonlocal projectors of a calculation | 1 | 23.3 |
| $\sigma_\mathrm{sm}$, $\tilde{f}(x)$ | Occupation-smearing width; smearing function, $f_{n\boldsymbol{k}} = \tilde{f}((\mu_\mathrm{ch} - \varepsilon_{n\boldsymbol{k}})/\sigma_\mathrm{sm})$ | eV; 1 | 23.3 |
| $F_\mathrm{sm}$, $S_\mathrm{sm}$ | Smeared free energy $E - \sigma_\mathrm{sm}S_\mathrm{sm}$; occupation entropy | eV; 1 | 23.3 |
| $E(\sigma_\mathrm{sm} = 0)$ | Energy per cell of the unsmeared metal | eV | 23.3 |
| $\beta_\mathrm{mix}$, $\varepsilon_\lambda$ | Density-mixing parameter; eigenvalues of the static dielectric matrix $\varepsilon^\mathrm{T}$ | 1 | 23.3 |
| $\boldsymbol{n}_\mathrm{in}$, $\boldsymbol{n}_\mathrm{out}$ | Input and output densities of a self-consistency step | $\mathrm{eV}^3$ | 23.3 |
| $E_\mathrm{qp}(nk)$ | Quasiparticle energy | eV | 23.4 |
| $E^\varepsilon_\mathrm{cut}$ | Dielectric-matrix (screening) cutoff | eV | 23.4 |

| Symbol | Meaning | Units | Sect. |
|---|---|---|---|
| $\tilde{\omega}_{\boldsymbol{GG'}}(q)$ | Plasmon-pole frequency of the element $GG'$ | eV | 23.4 |
| $v^Q P_{mn}$, $v^{\mathrm{KS}}_{mn}$ | Velocity matrix elements between quasiparticle and between Kohn–Sham states | 1 | 23.4 |
| **Chapter 24: Pseudopotentials and core states** | | | |
| $E_{\mathrm{cut}}$ | Plane-wave kinetic-energy cutoff | eV | 24.1 |
| $r_c$ | Pseudopotential core radius | Å; $a_0$ | 24.1 |
| $\tilde{p}_j$, $\phi_j^{\mathrm{AE}}$, $\phi_j^{\mathrm{PS}}$ | PAW projectors; all-electron and pseudo partial waves | $\text{Å}^{-3/2}$ | 24.1 |
| $Ry$ | Rydberg unit $\alpha^2 m_e/2 = 13.61$ eV | eV | 24.1 |
| $p_f$ | Momentum scale of a low-lying final state, $[2m_e(\omega + E_{\mathrm{F}})]^{1/2}$ | eV | 24.1 |
| $u_l(r,\varepsilon)$, $V_l(r)$, $P_l$ | Radial function $rR_l$ at energy $\varepsilon$; pseudopotential of channel $l$; projector on angular momentum $l$ | —; eV; 1 | 24.1 |
| $\psi^1$, $\tilde{\psi}^1$ | One-center all-electron and pseudo partial-wave expansions of a state inside an augmentation sphere | as $\psi_{n\boldsymbol{k}}$ | 24.1 |
| $\Phi$, $\epsilon$ | Normalized two-component state $(1 + p^2/8m_e^2)\psi_{\mathrm{A}}$ (in Sect. 24.1); energy $E - m_e$ (in Sect. 24.1) | —; eV | 24.1 |
| $N_{nl}$, $\phi_{nl}(p)$ | Occupation of the shell nl; its radial momentum-space orbital | 1; $\text{eV}^{-3/2}$ | 24.1 |
| $\psi_{\mathrm{A}}$, $\psi_{\mathrm{B}}$ | Large and small two-component parts of a Dirac spinor | $\text{eV}^{3/2}$ | 24.1 |
| $\boldsymbol{\alpha}_{\mathrm{D}}$, $\beta_{\mathrm{D}}$, $\gamma_{\mathrm{D}}$ | Dirac matrices; $\gamma_{\mathrm{D}} = [1 - (Z\alpha)^2]^{1/2}$ | 1 | 24.1 |
| $H_{\mathrm{mv}}$, $H_{\mathrm{Dar}}$, $H_{\mathrm{so}}$ | Mass–velocity, Darwin and spin–orbit one-body terms | eV | 24.1 |
| $k_{\mathrm{cut}}$ | Plane-wave cutoff wavevector $(2m_e E_{\mathrm{cut}})^{1/2}$ | eV (quoted in $\text{bohr}^{-1}$) | 24.1 |
| $\tilde{\psi}$, $\tilde{\psi}_{n\boldsymbol{k}}$ | Pseudo (smooth) Bloch state of a pseudopotential or PAW calculation | as $\psi_{n\boldsymbol{k}}$ | 24.1 |
| $\Delta_{\kappa,ij}(q)$ | One-center (PAW) correction to the density vertex | 1 | 24.1 |
| $E_{\mathrm{core}}$ | Depth of a core shell below the valence maximum | eV | 24.1 |
| $p_{1/2}$, $p_{3/2}$, $d_{3/2}$, $d_{5/2}$ | Spin–orbit-split $p$ and $d$ shells in spectroscopic labels | — | 24.1 |
| $V_{\mathrm{loc}}$, $D_{\kappa,ij}$ | Local radial part and projector coefficients of a separable (Kleinman–Bylander) pseudopotential | eV | 24.2 |
| $q_{\mathrm{opt}}$, $E^{\mathrm{res}}(q_{\mathrm{opt}})$ | Optimization wavevector of an optimized norm-conserving pseudopotential; residual kinetic energy above it | eV (quoted in $\text{bohr}^{-1}$); eV | 24.2 |
| $R_l(r)$, $j_l$, $j_{\mathrm{L}}$ | Radial solution of angular momentum $l$ (logarithmic-derivative test); spherical Bessel functions ($j_{\mathrm{L}}$ in Sects. 24.2 and 26.2) | —; 1 | 24.2 |
| $\Delta_{\mathrm{gauge}}$, $\mathcal{V}_0$, $E_1(\mathcal{V})$, $E_2(\mathcal{V})$ | Equation-of-state $\Delta$ gauge; equilibrium volume and the two equations of state it compares | meV/atom; $\text{Å}^3$; eV | 24.2 |
| $\delta_{\mathrm{NL}}(q)$ | Nonlocal-potential correction to the $f$-sum rule | eV | 24.2 |
| **Chapter 25: Phonons from first principles** | | | |
| $\partial_{\kappa\alpha,\boldsymbol{Q}}$ | Derivative with respect to the amplitude of the displacement wave $u^\alpha_{\boldsymbol{R}\kappa} \propto \mathrm{e}^{\mathrm{i}Q\cdot R}$, $\sum_R \mathrm{e}^{\mathrm{i}Q\cdot R}\partial/\partial u^\alpha_{\boldsymbol{R}\kappa}$ | $\text{Å}^{-1}$ | 25.1 |
| $\Phi^{\mathrm{SC}}_{\kappa\kappa'}(R_s)$, $T$, $R_s$ | Aliased (supercell) force constants $\sum_T \Phi^{(2)}(R_s + T, 0)$ of an $N_Q^3$ supercell; supercell translation; cell vector within the supercell | $\text{eV Å}^{-2}$; Å | 25.1 |
| $\hat{P}^{\mathrm{c}}$ | Projector $1 - \hat{P}^{\mathrm{occ}}$ on the empty Kohn–Sham states | 1 | 25.1 |
| $E_{NN}$, $E_{\mathrm{el}}$, $v_\kappa$ | Nucleus–nucleus (Ewald) energy; electronic energy; potential of one nucleus or pseudo-ion of species $\kappa$, $v_\kappa(r)$ at the origin, its Fourier component $v_\kappa(G)$, $v_\kappa(\boldsymbol{p})$ at the momentum given to the electrons | eV | 25.1 |
| $\Delta^{\mathrm{ASR}}_\kappa$, $\omega_{\mathrm{ASR}}$ | Acoustic-sum-rule violation $\sum_{\boldsymbol{R}\kappa'} \Phi^{(2)}_{\kappa\kappa'}(0, R)$ of computed force constants; the $\Gamma$ acoustic frequency it leaves before the repair | $\text{eV Å}^{-2}$; meV | 25.1 |
| $D^{\mathrm{dyn},el}$ | Electronic part of the dynamical matrix | $\text{eV}^2$ | 25.1 |

| Symbol | Meaning | Units | Sect. |
|---|---|---|---|
| $\psi^{(l)}$, $E^{(l)}$ | Order-$l$ coefficients of an orbital and of the energy in a perturbation of strength $\lambda$ | —; eV | 25.1 |
| $\partial_{\kappa\alpha,Q} V_{\mathrm{KS}}$ | First-order change of the Kohn–Sham potential for a displacement wave of sublattice $\kappa$ along $\alpha$ | $\mathrm{eV\,Å^{-1}}$ | 25.1 |
| $T_{\mathrm{d}}$, $O_h$ | Tetrahedral and octahedral point groups (Schoenflies labels) | — | 25.1 |
| $H_{\mathrm{KS}}$, $V_{\mathrm{ext}}$ | Kohn–Sham Hamiltonian; external (ionic) potential of DFPT | eV | 25.1 |
| $a_{\mathrm{sh}}$, $\hat{P}^{\mathrm{occ}}_{\boldsymbol{k}+\boldsymbol{Q}}$ | Sternheimer shift (larger than the valence bandwidth); projector on the occupied states at $k+Q$ | eV; 1 | 25.1 |
| $N_Q$, $Q_c$ | Phonon-grid points per direction; radius of the $\Gamma$-cell sphere | 1; eV (quoted in $\mathrm{Å^{-1}}$) | 25.1 |
| $\hat{P}^{\mathrm{deg}}_c$, $\mathcal{W}$, $\hat{P}^{\perp}_{\mathcal{W}}$ | Projector on the states degenerate with $c$; band window of the empty-state Sternheimer equation; projector on its complement | 1; —; 1 | 25.2 |
| $\partial\psi^{\pm}_v$ | First-order state changes of the frequency-dependent Sternheimer equation at $\pm\omega$ | as $\psi_{n\boldsymbol{k}}$ | 25.2 |
| $\boldsymbol{P}_{\mathrm{el}}$, $u_{v'\boldsymbol{k}+\Delta\boldsymbol{k}}$ | Electronic (Berry-phase) polarization; periodic part at the next point of a Berry-phase string of spacing $\Delta k$ | $\mathrm{e\,Å^{-2}}$; as $u_{n\boldsymbol{k}}$ | 25.2 |
| $\Delta B^{\alpha\alpha}$ | $\Gamma$-cell correction to the displacement tensor $B_\kappa$ | $\mathrm{Å^2}$ | 25.2 |
| $R_{\mathrm{IFC}}$ | Range of the interatomic force constants in units of the cell size (cost estimate) | 1 | 25.2 |
| $C_{ij}$, $C_{ijkl}$ | Elastic constants in Voigt and Cartesian form | GPa | 25.2 |
| $c_{\mathrm{TA}}$, $c_{\mathrm{T}}$ | Transverse-acoustic sound speed; transverse sound speed of an isotropic model | $\mathrm{km\,s^{-1}}$ | 25.2 |
| $\mathcal{F}(\Omega,T)$, $E^{\mathrm{ad}}_0(\Omega)$ | Free energy per cell; static-lattice energy at cell volume $\Omega$ | eV | 25.3 |
| $\alpha_{\mathrm{th}}$ | Volume thermal-expansion coefficient | $\mathrm{K^{-1}}$ | 25.3 |
| $N_{\mathrm{conf}}$, $N_{\mathrm{eff}}$, $w_j$ | Configurations of an SCHA population; effective sample number $(\sum_j w_j)^2/\sum_j w_j^2$; reweighting weights | 1 | 25.3 |
| $N_t$, $\Delta_{\mathrm{fit}}$ | MD time steps of a TDEP fit; the force mismatch it minimizes | 1; $\mathrm{eV^2\,Å^{-2}}$ | 25.3 |
| $\Delta\varepsilon^{\mathrm{FM}}_{n\boldsymbol{k}}(T)$ | Fan–Migdal part of the temperature-dependent band-energy shift | eV | 25.3 |
| $x_{\boldsymbol{Q}\nu}$, $\sigma_{\boldsymbol{Q}\nu}$ | Mass-weighted normal coordinate of mode $Q\nu$; width of its thermal Gaussian | $\mathrm{eV^{-1/2}}$ | 25.3 |
| $\boldsymbol{u}^{\mathrm{ZG}}_{R\kappa}$, $\varsigma_{\boldsymbol{Q}\nu}$, $\mathcal{B}$ | ZG (special) displacement of nucleus $\kappa$ in cell $R$; its signs $\pm1$; the set of wavevectors it uses | Å; 1; — | 25.3 |
| **Chapter 26: Dielectric response and form factors** | | | |
| $\alpha_{\mathrm{xc}}$ | Strength of the long-range xc kernel $-\alpha_{\mathrm{xc}}/\lvert q+G\rvert^2$ | 1 | 26.1 |
| $N_G$, $N_v$, $N_c$, $N_k$ | Numbers of reciprocal-lattice vectors, valence bands, conduction bands and $k$ points of a response calculation | 1 | 26.1 |
| $N_{\mathrm{orb}}$ | Number of orbitals of a molecular GW-BSE calculation | 1 | 26.1 |
| $t_{\mathrm{f}}$, $\boldsymbol{k}_{\mathrm{rot}}$ | Fractional translation of a symmetry operation; rotated wavevector $k+G$ | Å; $\mathrm{Å^{-1}}$ | 26.2 |
| $\psi^{\mathrm{at}}_{\kappa nlm}$, $R^{\mathrm{at}}_{nl}(r)$ | Atomic orbital of species $\kappa$; its radial function | $\mathrm{Å^{-3/2}}$ | 26.2 |
| $C_{jln}$, $Z_{jl}$, $\tilde{n}_{jl}$ | Slater-type expansion coefficients, exponents and integer powers of $R^{\mathrm{at}}_{nl}$ | 1 | 26.2 |
| $l'_{\max}$ | Partial-wave cutoff of the continuum state | 1 | 26.2 |
| **Chapter 27: Wannier interpolation** | | | |
| $w_n(r-R)$ | Wannier function | $\Omega^{-1/2}$ | 27.1 |
| $U(k)$ | Bloch-to-Wannier gauge matrix | 1 | 27.1 |
| $H^W(R)$ | Real-space (Wannier) Hamiltonian | eV | 27.1 |
| $\Omega_W$ | Wannier spread functional | $\mathrm{Å^2}$ | 27.1 |

| Symbol | Meaning | Units | Sect. |
|---|---|---|---|
| $\psi^W_{n\boldsymbol{k}}$, $u^W_{n\boldsymbol{k}}$ | Bloch state and periodic part in the Wannier gauge, $\sum_m U_{mn}(k)\psi_{m\boldsymbol{k}}$ | as $\psi_{n\boldsymbol{k}}, u_{n\boldsymbol{k}}$ | 27.1 |
| $\theta_n(k)$ | Gauge phase of a Bloch state | 1 | 27.1 |
| $P(k)$, $Q(k)$ | Projector $\sum_n \lvert u^W_{n\boldsymbol{k}}\rangle\langle u^W_{n\boldsymbol{k}}\rvert$ on the $N_W$-dimensional Wannier manifold at $k$, and its complement | 1 | 27.1 |
| $P$, $Q$ | Projector $\sum_{m\boldsymbol{R}} \lvert w_{m\boldsymbol{R}}\rangle\langle w_{m\boldsymbol{R}}\rvert$ on a group of Wannier functions and its complement | 1 | 27.1 |
| $\phi^{\mathrm{tr}}_n(r)$, $A^{\mathrm{pr}}_{mn}(k)$ | Trial orbital of the projection gauge; projection matrix $\langle\psi_{m\boldsymbol{k}}\rvert\phi^{\mathrm{tr}}_n\rangle$ | $\Omega^{-1/2}$; 1 | 27.1 |
| $\bar{r}_n$, $\langle r^2\rangle_n$ | Wannier centre $\langle w_{n\boldsymbol{0}}\rvert\boldsymbol{r}\rvert w_{n\boldsymbol{0}}\rangle$; second moment $\langle w_{n\boldsymbol{0}}\rvert r^2\rvert w_{n\boldsymbol{0}}\rangle$ | Å; Å$^2$ | 27.1 |
| $\Delta k$, $\lambda_{\Delta\boldsymbol{k}}$ | Vector from a grid point to a neighbour in the finite-difference spread; its weight, $\sum \lambda_{\Delta\boldsymbol{k}}\Delta k_\alpha \Delta k_\beta = \delta_{\alpha\beta}$ | eV; eV$^{-2}$ | 27.1 |
| $M_{mn}(k,\Delta k)$ | Overlap $\langle u^W_{m\boldsymbol{k}}\rvert u^W_{n,k}+\Delta k\rangle_\Omega$ | 1 | 27.1 |
| $\Omega^{\mathrm{I}}_W$, $\tilde{\Omega}_W$ | Gauge-invariant and gauge-dependent parts of the spread, $\Omega_W = \Omega^{\mathrm{I}}_W + \tilde{\Omega}_W$ | Å$^2$ | 27.1 |
| $H^W(k)$, $\tilde{H}^W(R)$ | Wannier-gauge Bloch Hamiltonian $\sum_R \mathrm{e}^{\mathrm{i}k\cdot R}H^W(R) = U^\dagger E U$; coarse-grid (aliased) real-space Hamiltonian | eV | 27.1 |
| $E(k)$ | Diagonal matrix $\mathrm{diag}(\varepsilon_{n\boldsymbol{k}})$ of band energies | eV | 27.1 |
| $R_{\max}$ | Range of $H^W(R)$ | Å | 27.1 |
| $\xi_W$ | Decay length of a Wannier function, $\lvert w\rvert \sim \mathrm{e}^{-\lvert R\rvert/\xi_W}$ | Å | 27.1 |
| $n_k$, $n_1$, $n_2$, $n_3$, $n^{\mathrm{irr}}_k$ | Number of points of a $k$ grid, $n_k = n_1 n_2 n_3$ (points along the three reciprocal primitive vectors); the inequivalent points among them | 1 | 27.1 |
| $N_W$ | Number of Wannier functions | 1 | 27.1 |
| $L_{\mathrm{sc}}$, $\alpha_{\mathrm{M}}$ | Supercell side; Madelung constant of the simple-cubic lattice, 2.8373 (in Sect. 28.2) | Å; 1 | 27.1 |
| $N_{\mathrm{fine}}$ | Fine-grid points per direction | 1 | 27.2 |
| $k_\delta$, $V_{\mathrm{BZ}}$, $\Omega_{\mathrm{BZ}}$ | Radius $(2m^*\delta)^{1/2}$ of the sphere of states within $\delta$ of a band edge; Brillouin-zone volume $(2\pi)^3/\Omega$ | eV; Å$^{-3}$ | 27.2 |
| **Chapter 28: Magnets, superconductors and defects** | | | |
| $E_U$, $n^{I\sigma}_{mm'}$, $P^I_{mm'}$ | Hubbard energy correction; occupation matrix of the correlated shell of ion I and spin $\sigma$; projector on its orbitals | eV; 1; 1 | 28.1 |
| $R^U_{IJ}$, $R^{U,0}_{IJ}$, $\delta V^U_I$ | Interacting and bare responses of the shell occupations to the potential shift $\delta V^U_I$ of linear-response $U$ | eV$^{-1}$; eV$^{-1}$; eV | 28.1 |
| $E_{\mathrm{FM}}$, $E_{\mathrm{AFI}}$, $E_{\mathrm{AFII}}$ | Energies per ion of the ferromagnetic and the two antiferromagnetic orders of NiO (energy mapping) | eV | 28.1 |
| $E(Q,\theta)$ | Energy of a spin spiral of wavevector $Q$ and cone angle $\theta$ | eV | 28.1 |
| $\Delta_i$ | Exchange splitting of site $i$ in the LKAG formula | eV | 28.1 |
| $J^{\mathrm{LKAG}}_{ij}$ | Exchange constant of the LKAG formula in the convention of its source (pairs twice, unit vectors), $J^{\mathrm{ex}}_{ij} = -2J^{\mathrm{LKAG}}_{ij}/\lvert\boldsymbol{S}\rvert^2$ | meV | 28.1 |
| $Z^{\mathrm{El}}(\mathrm{i}\omega^{\mathrm{F}}_j)$, $\Delta^{\mathrm{El}}(\mathrm{i}\omega^{\mathrm{F}}_j)$ | Eliashberg renormalization function and gap on the fermionic Matsubara axis | 1; eV | 28.1 |
| $\mu^*_{\mathrm{C}}$ | Coulomb pseudopotential | 1 | 28.1 |
| $\omega_{\log}$ | Logarithmic average phonon frequency of the Allen–Dynes formula | eV (quoted in K) | 28.1 |
| $z_{\mathrm{D}}$, $z'_{\mathrm{D}}$ | Defect charge number in units of $e$; a second charge state | 1 | 28.2 |
| $E^{\mathrm{f}}_{\mathrm{D}}(z_{\mathrm{D}})$, $E_{\mathrm{D}}$, $E_{\mathrm{host}}$ | Formation energy of a defect $D$ in charge state $z_{\mathrm{D}}$; total energies of the defective and the host supercells | eV | 28.2 |
| $n_s$, $\mu^{\mathrm{at}}_s$ | Number of atoms of species $s$ added; atomic chemical potentials | 1; eV | 28.2 |

| Symbol | Meaning | Units | Sect. |
|---|---|---|---|
| $\varepsilon_{\mathrm{VBM}}$, $E_{\mathrm{corr}}$ | Kohn–Sham energy of the valence-band maximum; finite-size (image-charge) correction | eV | 28.2 |
| $\Delta Q_k$, $\omega_k$ | Mass-weighted displacement projected on vibronic mode $k$; its frequency (in Sect. 28.2) | $\mathrm{eV}^{-1/2}$; eV | 28.2 |
| $V_{\mathrm{d}}$ | Half the difference of the conduction-electron contact potentials of the two configurations, which sit at $\pm V_{\mathrm{d}}$ | eV | 28.2 |
| $S_L$, $S_{\mathrm{R}}$, $\tilde{\delta}_\nu$, $\tilde{\delta}$ | Scattering matrices of the conduction electrons at the Fermi energy in the two configurations; phase-shift differences, the eigenvalues $\mathrm{e}^{2\mathrm{i}\tilde{\delta}_\nu}$ of $S_{\mathrm{R}} S_L^\dagger$ | 1 | 28.2 |
| **Chapter 29: Recoils, transport and cascades** | | | |
| $E_\kappa^{\mathrm{at}}(\mathcal{N}_I)$, $\mathcal{N}_I$, $\mathcal{N}_s$ | Local atomic energy of an interatomic potential; neighborhood of atom I within the cutoff $r_c$; representative environments of a GAP fit | eV; —; — | 29.1 |
| $V^{(2)}$, $\alpha_s$, $M_{\mathrm{rep}}$, $K_{\mathrm{GAP}}$ | GAP pair term; kernel weights; number of representative environments; similarity kernel | eV; eV; 1; 1 | 29.1 |
| $V_{\mathrm{ZBL}}(r)$, $a_{\mathrm{U}}$, $Z_1$, $Z_2$ | ZBL screened pair potential; screening length $0.8854 a_0/(Z_1^{0.23}+Z_2^{0.23})$; atomic numbers of the pair | eV; Å; 1 | 29.1 |
| $r_{\min}$ | Head-on distance of closest approach | Å | 29.1 |
| $f_{\mathrm{sw}}$, $r_{\mathrm{sw}}$, $b_{\mathrm{sw}}$, $E_{\mathrm{ML}}^{\mathrm{ad}}$ | ZBL switch function, its center and steepness; machine-learned energy surface | 1; Å; $\mathrm{Å}^{-1}$; eV | 29.1 |
| $\hat{r}_R$ | Recoil direction | 1 | 29.1 |
| $p_{\mathrm{FP}}(E_R, \hat{r}_R)$, $N_{\mathrm{dir}}$ | Probability that a recoil leaves a Frenkel pair once the collision has died out; number of recoil directions | 1 | 29.1 |
| $N_{\mathrm{im}}$ | Number of NEB images | 1 | 29.1 |
| $E_{\min}^{\mathrm{ad}}$ | Energy surface at a quenched (inherent) structure | eV | 29.1 |
| $B_{\mathrm{iso}}$, $g_{\mathrm{md}}$, $\bar{M}$ | Isotope rate constant, $\Gamma_{\mathrm{iso}} = B_{\mathrm{iso}}(E_{\mathrm{ath}}/h)^4$; mass-variance coefficient $\langle (M-\bar{M})^2\rangle/\bar{M}^2$; mean isotope mass | $\mathrm{s}^3$; 1; amu | 29.2 |
| $\Gamma_{\mathrm{anh}}$, $A_{\mathrm{anh}}$ | Three-phonon (anharmonic) decay rate $A_{\mathrm{anh}}(E_{\mathrm{ath}}/h)^5$ and its constant | $\mathrm{s}^{-1}$; $\mathrm{s}^4$ | 29.2 |
| $E_{\mathrm{L}}$, $E_{\mathrm{L}'}$, $\theta_{\mathrm{L}'}$ | Energies of the parent and of the longitudinal daughter of $L \to L' + T'$ decay; angle of the daughter | meV; 1 | 29.2 |
| $\Gamma_{\mathrm{L}}(k)$, $k_{\mathrm{s}}$, $l_0$ | Luke emission rate; sound-speed momentum; scattering length | $\mathrm{s}^{-1}$; eV; m | 29.2 |
| $m_{\mathrm{c}}$ | Conductivity mass, $3/m_{\mathrm{c}} = 1/m_\parallel^* + 2/m_\perp^*$ | $m_e$ | 29.2 |
| $\mathfrak{l}_{\mathrm{f}}(E_{\mathrm{ath}})$ | Phonon mean free path in a superconducting film | $\mu$m | 29.2 |
| $\boldsymbol{r}_{\mathrm{ev}}$, $\phi_s$, $N_{\mathrm{l}}$ | Event position; Ramo weighting potential of electrode $s$; number of launched phonons | Å; 1; 1 | 29.2 |
| $\Gamma_{\mathrm{ii}}(nk)$, $\Gamma_{\mathrm{ph}}(nk)$, $\Gamma_{\mathrm{tot}}$, $\Gamma_{\max}$ | Impact-ionization and phonon rates of a carrier state; their sum; self-scattering ceiling | $\mathrm{s}^{-1}$ | 29.3 |
| $E_0$ | Seed energy of a cascade, $\varepsilon_{\mathrm{eh}} = E_0/\langle N_{\mathrm{eh}}\rangle$ | eV | 29.3 |
| $A_{\mathrm{AB}}$ | Alig–Bloom constant, ratio of the squared phonon and Coulomb matrix elements times phase-space factors | $\mathrm{eV}^3$ | 29.3 |
| $\omega_{\mathrm{AB}}$ | Energy of the single optical phonon of the Alig–Bloom model, 63 meV | eV | 29.3 |
| $N_{\mathrm{casc}}$, $k_E$ | Number of simulated cascades (in Sect. 29.3); radius $(2m_e E)^{1/2}$ of the free-electron sphere below energy $E$ | 1; eV | 29.3 |
| $M_{\mathrm{D}}$, $M_{\mathrm{E}}$ | Direct and exchange matrix elements of the screened interaction of impact ionization | eV | 29.3 |
| **Chapter 30: Lattice and special targets** | | | |
| $t_{\max}$, $S_{\mathrm{coh}}^{(2)}(\boldsymbol{q},\omega)$ | Time window of the time-domain multiphonon sum; coherent (cross-site) part of the two-phonon structure factor at fixed $\boldsymbol{q}$ and $\omega$ | $\mathrm{eV}^{-1}$; $\mathrm{eV}^{-1}$ | 30.1 |
| $P_{\mathrm{wall}}(\hat{\boldsymbol{n}})$ | Pressure of the reflected dark matter on a face of outward normal $\hat{\boldsymbol{n}}$ of a test mass | Pa | 30.2 |

| Symbol | Meaning | Units | Sect. |
|---|---|---|---|
| $L_{\mathrm{th}}$, $\mathsf{M}_{\lambda\lambda'}$, $b_\lambda$, $\mathcal{N}_\lambda^{\mathrm{ch}}$ | Threshold on $\lvert L^{\mathrm{ph}}\rvert$ of the long-lived chiral band; mean number of daughters in mode $\lambda'$ per decay of mode $\lambda$; indicator of the terminal chiral set, 1 inside and 0 outside; expected number of detectable chiral phonons from one phonon created in mode $\lambda$ | 1 | 30.2 |
| $P_{\mathrm{K}}(\boldsymbol{u})$, $E_{\mathrm{K}}$, $H_{\mathrm{CF}}(\boldsymbol{u})$ | Projector on the ground Kramers doublet of a lanthanide ion at displacement $\boldsymbol{u}$; energy of that doublet; crystal-field Hamiltonian of the 4f shell | 1; eV; eV | 30.2 |
| $R_{n\ell}(r)$, $j_K$, $\langle j_K(q)\rangle$ | Radial function of the open $n\ell$ shell of a magnetic ion; spherical Bessel function of even order $K$; its radial average $\int r^2 \mathrm{d}r R_{n\ell}(r)^2 j_K(qr)$ | $\text{Å}^{-3/2}$; 1; 1 | 30.3 |
| $a_{\mathrm{D}}$, $v_{\mathrm{D},x}$, $v_{\mathrm{D},y}$, $v_{\mathrm{D},z}$, $\varepsilon_{\mathrm{bg},zz}$ | Bohr radius of the Dirac exciton; principal Dirac velocities along the crystal axes; zz component of the background dielectric tensor | Å; 1; 1 | 30.3 |
| $T_N$, $T_{\mathrm{C}}$ | Néel and Curie temperatures of a magnetic target | K | 30.3 |
| $\boldsymbol{k}_\iota$ | Wavevector of the conduction minimum of one valley of silicon, $\lvert\boldsymbol{k}_\iota\rvert = k_0$ | keV | 30.4 |
| $R_{\mathrm{OPW}}(E_{\mathrm{kin}})$ | Ratio of the ejection factor with the orthogonalized plane wave to that with the plain plane wave, at final kinetic energy $E_{\mathrm{kin}}$ | 1 | 30.4 |
| $\Psi_L$, $\Psi_R$ | Many-electron ground states of the two configurations $L$ and $R$ of a defect in a plane-wave supercell | — | 30.5 |
| $\varepsilon_{\mathrm{hop}}$ | Energy released in a defect hop | eV | 30.5 |
| **Chapter 31: Electronic and mixed channels** | | | |
| $\omega_{\mathrm{LO}}$ | Longitudinal optical phonon energy at the zone center in the Fröhlich width estimate of GaAs | meV | 31.1 |
| $S_\lambda(\boldsymbol{q},\omega)$ | Dynamic structure factor of the self-trapped exciton line $\lambda$, its vertical creation dressed by a Poisson series of phonons | $\text{eV}^{-1}$ | 31.2 |
| $\mathfrak{r}_{\mathrm{scr}}(\boldsymbol{q})$ | Screening ratio of the phonon-assisted amplitude, screened over reference, $\lvert\tilde{M}\rvert^2/\lvert M\rvert^2$ | 1 | 31.3 |
| $\delta^{(0)}_{\mathrm{wing}}$ | Vertex (wing, $\boldsymbol{G}\neq 0$) part of the local-field ratio from zero-phonon vertices; $1+\delta_{\mathrm{LF}} = (\varepsilon_{1,\mathbf{00}}/\varepsilon_{\mathrm{mac}})^2(1+\delta^{(0)}_{\mathrm{wing}})$ | 1 | 31.3 |
| $N_{\mathrm{core}}$, $\boldsymbol{p}_{\mathrm{red}}$ | Number of core electrons per atom frozen in the pseudopotential, which move rigidly with the nucleus; reduced (first-zone) momentum whose Umklapp copies $\boldsymbol{p}_{\mathrm{red}}+\boldsymbol{G}$ cover the unreduced momenta | 1; keV | 31.4 |
| **Chapter 32: Detector response and inputs** | | | |
| $n_{\mathrm{pair}}$, $C_{\mathrm{Aug}}$, $r_{\mathrm{tr}}$, $\lambda_{\mathrm{tr}}$ | Electron–hole pair density of a recoil track; Auger recombination coefficient, rate $C_{\mathrm{Aug}}n^3$ per volume; initial radius of a Gaussian track; its pairs per unit length | $\text{cm}^{-3}$; $\text{cm}^6\,\text{s}^{-1}$; nm; $\text{nm}^{-1}$ | 32.1 |
| $N_z$, $z_j$ | Number of roots of unity at which a pair-number generating function is evaluated in the discrete Fourier inversion; the roots $\mathrm{e}^{2\pi ij/N_z}$ | 1 | 32.2 |
| $\Gamma_{\mathrm{ph,loss}}$, $w_{\mathrm{mem}}$ | Rate of the phonon emissions that change the carrier energy; weight of the remembered momentum distribution at the next energy-changing event | $\text{s}^{-1}$; 1 | 32.2 |
| $\mathcal{G}_\Lambda(z)$, $x_e$, $x_h$ | Generating function $\sum_N z^N P_{\mathrm{casc}}(N\lvert\Lambda)$ of the pair number for seed class $\Lambda$; energies of the seed electron and hole above their band edges | 1; eV | 32.2 |
| $N_{\mathrm{meas}}$ | Measured pair number of an event in the read window | 1 | 32.3 |
| $s_\ell(E_{\mathrm{ath}})$, $D_X$, $S_{\mathrm{surf}}$, $L_X$, $s_{\mathrm{hard}}$, $s_{\mathrm{soft}}$ | Phonon spectrum of release channel $\ell$, energy fraction per unit $E_{\mathrm{ath}}$; exciton diffusion constant; surface recombination velocity; exciton diffusion length $(D_X/\Gamma_X)^{1/2}$; hard-line and soft (Debye) trial spectra of the surface and electrode channels | $\text{meV}^{-1}$; $\text{cm}^2\,\text{s}^{-1}$; $\text{cm}\,\text{s}^{-1}$; $\mu$m; $\text{meV}^{-1}$ | 32.3 |

| Symbol | Meaning | Units | Sect. |
|---|---|---|---|
| $f_{\mathrm{coll}}^{(p)}$ | Collection efficiency of the phonon population $p$: primary phonons of nuclear and of electron recoils, Luke phonons, gap-energy phonons | 1 | 32.3 |
| $\lambda_{\mathrm{dec}}$, $R_k$, $t_k$, $t'_k$ | Decay constant of a cosmogenic nuclide; production rate, length of exposure $k$, time from its end to the measurement | $\mathrm{s}^{-1}$; $\mathrm{kg}^{-1}\,\mathrm{d}^{-1}$; s; s | 32.4 |
| **Chapter 33: Libraries and open calculations** | | | |
| $P_{\mathrm{ph}}$ | Phonon-energy distribution of a phonon-dressed seed in the library | $\mathrm{eV}^{-1}$ | 33.1 |
| **Appendix A: Conventions and units** | | | |
| $\mu_{\mathrm{B}}$, $g_e$ | Bohr magneton $e/2m_e$; electron $g$-factor | $\mathrm{eV}^{-1}$; 1 | A.2 |
| $\boldsymbol{E}_{\mathrm{G}}$, $\boldsymbol{B}_{\mathrm{G}}$; $\boldsymbol{E}_{\mathrm{SI}}$, $\boldsymbol{B}_{\mathrm{SI}}$ | Electric and magnetic fields in Gaussian and in SI units | as quoted | A.2 |
| $f_{\boldsymbol{G}}$ | Fourier component of a lattice-periodic function, $\Omega^{-1}\int_\Omega \mathrm{e}^{+\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}} f$ | as $f$ | A.3 |
| $\langle\langle\cdot\rangle\rangle$ | Zone average $N^{-2}\sum_{kQ}$ | — | A.3 |
| $\boldsymbol{r}_\chi$, $\boldsymbol{r}_e$ | Positions of the dark matter and of one electron | $\mathrm{eV}^{-1}$ | A.3 |
| $\tilde{i}$, $\tilde{f}$ | Time-reversed initial and final states, as kets $\lvert\tilde{i}\rangle, \lvert\tilde{f}\rangle$ | — | A.4 |
| $V_{\mathrm{NL}}$ | Nonlocal part of a pseudopotential | eV | A.5 |
| $D^{\mathrm{dyn},\alpha\beta}_{\kappa\kappa'}$, $\Phi^{(2),\alpha\beta}_{\kappa\kappa'}$ | Cartesian components of the dynamical matrix and of the force constants, written as superscripts | as $D^{\mathrm{dyn}}, \Phi^{(2)}$ | A.6 |
| $\hat{e}_\nu$ | Unit polarization vector of an acoustic branch $\nu$ at small $\boldsymbol{Q}$ | 1 | A.6 |
| $b$, $b^\dagger$ | Annihilation and creation operators of a single phonon mode, written without labels | — | A.6 |
| $L_e$ | Electron-channel loss function $\mathrm{Im}[-(\varepsilon_e^{-1})_{\mathbf{00}}]$ | 1 | A.7 |
| $S^{\mathrm{prop}}$ | Structure factor of $\tilde{R}_{ee}$ | $\mathrm{eV}^2$ | A.7 |
| $O_e$, $\varepsilon_e(q,\omega)$ | Electron density operator $n_e(\boldsymbol{q})$ as an operator $O_a$; electron-channel dielectric function, $\varepsilon_e^{-1} = 1 + v_{\mathrm{C}} R_{ee}$ (equals $\varepsilon$ at clamped nuclei) | —; 1 | A.7 |
| $R_{ee}(q,\omega)$, $\tilde{R}_{ee}(q,\omega)$, $S_{ee}(q,\omega)$ | Density–density response function, its proper part, and the structure factor of the electron channel ($S \equiv S_{ee}$) | $\mathrm{eV}^2$; $\mathrm{eV}^2$; $\mathrm{eV}^2$ | A.7 |
| $\bar{S}(q,\omega)$ | $S$ averaged over directions of $q$ | $\mathrm{eV}^2$ | A.8 |
| $\eta_2(v_{\mathrm{min}})$ | Second speed moment $2\int_{v_{\mathrm{min}}}^\infty \mathrm{d}v' v' \eta(v')$ | 1 | A.8 |
| $\tilde{\eta}(v_{\mathrm{min}})$ | $\rho_\chi \bar{\sigma}_e \eta / m_\chi$ | — | A.8 |
| $N_{\mathrm{cell}}$ | Number of unit cells of a target of mass $M_{\mathrm{det}}$, $M_{\mathrm{det}}/M_{\mathrm{cell}}$ | 1 | A.8 |
| $\mathcal{M}_{\mathrm{free}}$ | Relativistic free dark-matter–electron amplitude | 1 | A.9 |
| $q_{\mathrm{ref}}$, $\sigma_{\mathrm{ref}}$ | Reference momentum transfer of a cross-section conversion and the cross-section $\bar{\sigma}_e F_{\mathrm{DM}}(q_{\mathrm{ref}})^2$ quoted there | eV; $\mathrm{cm}^2$ | A.9 |

## B.2 Literature notation

The dark-matter literature and the condensed-matter literature use the same letters for different objects. The letter $\chi$ is the dark-matter particle in one and a susceptibility in the other, $q$ is the momentum transfer in one and the phonon wavevector in the other, and $W$ is the loss function, the screened interaction or the Debye–Waller exponent, depending on the paper. This book gives each letter one meaning, so several common symbols are renamed. Table B.2 lists the renamings that a reader moving between the book and the papers will meet.

Table B.2: Symbols of the literature and of this book. Where a paper's symbol differs from the book's only by a factor or a sign, the factor is given in the Meaning column.

| Literature | Here | Meaning | Section |
|---|---|---|---|
| *Response and screening* | | | |
| $\chi_{AB}$, $\chi$, $C^R_{AB}$ | $R_{ab}$ | retarded response function; the literature usually omits the factor $1/\mathscr{V}$ | 4.1 |
| $\Pi$ | $\tilde{R}$ | polarization, the Coulomb-proper (irreducible) response | 4.4 |
| $W$, $\mathrm{Im}(-1/\varepsilon)$ | $L$ | loss function $\mathrm{Im}[-(\varepsilon^{-1})_{00}]$ | 4.4 |
| $W$ (as in $GW$) | $W^{\mathrm{scr}}$ | screened Coulomb interaction | 4.8 |
| $-2\,\mathrm{Im}\,G^{\mathrm{R}}$ | $\mathsf{A}$ | spectral function; the book's $\mathsf{A}$ integrates to one, the literature's to $2\pi$ | 4.7 |
| $G(\boldsymbol{r},t)$ | $\mathcal{G}_{\mathrm{VH}}(\boldsymbol{r},t)$ | Van Hove correlation function | 4.2 |
| $\lambda_{\mathrm{TF}}$ | $1/k_{\mathrm{TF}}$ | Thomas–Fermi screening length | 4.4 |
| $\kappa$ | $\varepsilon_{\mathrm{bg}}$ | background (remote-band) dielectric constant | 7.3 |
| $\alpha(\omega)$ | $\alpha_{\mathrm{abs}}(\omega)$ | optical absorption coefficient | 6.5 |
| *Dark matter and kinematics* | | | |
| $\varepsilon$, $\kappa$, $\chi$ | $\zeta$ | kinetic mixing of the dark photon | 1.4 |
| $g_\chi$, $g_e$ | $y_\chi$, $y_e$ | mediator couplings to dark matter and electrons | 1.4 |
| $n_\chi$ | $n_\chi$ | dark-matter number density $\rho_\chi/m_\chi$ | 1.1 |
| $\boldsymbol{k}$ | $\boldsymbol{q}$ | momentum transferred from the dark matter to the target | 1.3 |
| $\mathrm{d}\Omega$ | $\mathrm{d}\Omega_{\boldsymbol{q}}$ | solid-angle element of the scattered momentum | 2.2 |
| $\sigma_e$ (fermionic absorption) | $\sigma_F$ | absorption cross-section of fermionic dark matter | 10.9 |
| $Q$ | $N_{\mathrm{eh}}$ | number of electron–hole pairs (ionization yield) | 1.5 |
| *Nuclei and phonons* | | | |
| $\boldsymbol{q}$, $\boldsymbol{k}$ | $\boldsymbol{Q}$ | phonon wavevector in the first zone | 5.2 |
| $r_n$, $R_1$; $s$ | $R_{\mathrm{nuc}}$; $s_{\mathrm{skin}}$ | Helm radius; skin thickness | 2.2 |
| $c$, $a$ | $c_{\mathrm{nuc}}$, $a_{\mathrm{nuc}}$ | Woods–Saxon half-density radius and diffuseness | 2.2 |
| $\lambda$, $\mu$ (Lamé) | $C_{12}$, $C_{44}$ | elastic constants of an isotropic medium | C.18 |
| $\lambda$ | $\lambda_{\mathrm{ep}}$ | electron–phonon coupling constant | 6.4 |
| $\mu^*$ | $\mu^*_{\mathrm{C}}$ | Coulomb pseudopotential | 28.1 |
| $Z$, $\Delta$ | $Z^{\mathrm{El}}$, $\Delta^{\mathrm{El}}$ | Eliashberg renormalization function and gap | 28.1 |
| *Electrons in crystals* | | | |
| $\mu$ | $\mu_{\mathrm{ch}}$ | chemical potential | 2.5 |
| $N(0)$, $N(E_{\mathrm{F}})$ | $\mathcal{N}_{\mathrm{F}}$ | density of states per spin and volume at the Fermi level | 2.5 |
| $v_{\mathrm{F}}$ (Dirac material) | $v_{\mathrm{D}}$ | Dirac velocity; $v_{\mathrm{F}}$ is the Fermi velocity of the electron gas | 7.3 |
| $\chi_{nl}$ | $\tilde{\psi}_{nl}$ | radial momentum-space wavefunction of an atomic shell | 2.2 |
| $T_{\mathrm{s}}$ | $K_{\mathrm{s}}$ | Kohn–Sham (independent-electron) kinetic energy | 3.4 |
| $V_2$ | $V_{\mathrm{cov}}$ | covalent energy between two bond hybrids | 3.2 |

| Literature | Here | Meaning | Section |
|---|---|---|---|
| $\Omega$; $\Omega_{\mathrm{I}}$; $\Omega_{\mathrm{D}} + \Omega_{\mathrm{OD}}$ | $\Omega_W$; $\Omega_W^{\mathrm{I}}$; $\tilde{\Omega}_W$ | Wannier spread and its gauge-invariant and gauge-dependent parts | 27.1 |
| $V_{\mathrm{cell}}$ | $\Omega$ | unit-cell volume | 3.1 |
| $\chi$, $\chi^0$, $\alpha$ | $R^U$, $R^{U,0}$, $\delta V^U$ | interacting and bare occupation responses, and the potential shift, of linear-response Hubbard $U$ | 28.1 |
| $\alpha$, $K$ | $\alpha_{\mathrm{ohm}}$ | ohmic coupling of a tunneling defect to conduction electrons | 16.8 |
| $e$ (Gaussian) | $e_{\mathrm{G}}$ | Gaussian elementary charge, $e_{\mathrm{G}}^2 = \alpha$; the book's $e$ has $e^2 = 4\pi\alpha$ | A.2 |
| $k$ (elliptic modulus) | $s$ | modulus of the complete elliptic integrals | C.8 |

The Fourier sign (Appendix A.3), the sign of the mixing term (Appendix A.9), the units of charge (Appendix A.2) and the normalization of the structure factor (Appendix A.7) differ from parts of the literature by a sign or a factor rather than by a letter, and a formula taken from a paper must be converted before it is combined with the book's.

## B.3 Acronyms

The last column gives the section or appendix in which the acronym first appears.

| Acronym | Expansion | First use |
|---|---|---|
| ACE | atomic cluster expansion | 32.1 |
| AHC | Allen–Heine–Cardona | 6.3 |
| ALDA | adiabatic local-density approximation | 4.6 |
| ALP | axion-like particle | 1.4 |
| ANO | atomic natural orbital | 30.2 |
| BCS | Bardeen–Cooper–Schrieffer | 1.5 |
| BSE | Bethe–Salpeter equation | 8.5 |
| CASSCF | complete-active-space self-consistent field | 30.2 |
| CBM | conduction-band minimum | 3.3 |
| CCD | charge-coupled device | 1.5 |
| CDMS | Cryogenic Dark Matter Search | 18.5 |
| CMB | cosmic microwave background | 1.1 |
| CNT | carbon nanotube | 30.4 |
| CODATA | Committee on Data of the International Science Council | A.1 |
| COHSEX | Coulomb-hole plus screened-exchange | 23.4 |
| CPU | central processing unit | 29.2 |
| DAMIC | Dark Matter in CCDs | 10.9 |
| DFPT | density-functional perturbation theory | 5.2 |
| DFT | density-functional theory | 7.1 |
| DLMF | Digital Library of Mathematical Functions | C.8 |
| EPW | Electron–Phonon Wannier | 6.2 |
| FFT | fast Fourier transform | 23.3 |
| FIRE | fast inertial relaxation engine | 32.1 |
| FP | Frenkel pair | 19.1 |
| FT | fast transverse | 29.2 |
| G4CMP | Geant4 Condensed Matter Physics | 20.1 |
| GGA | generalized-gradient approximation | 3.4 |
| GIPAW | gauge-including projector-augmented-wave | 31.5 |
| GPAW | grid-based projector-augmented wave | 12.5 |
| GPU | graphics processing unit | 30.5 |

| Acronym | Expansion | First use |
|---|---|---|
| GRASP | General-purpose Relativistic Atomic Structure Package | 30.3 |
| HOMO | highest occupied molecular orbital | 23.2 |
| HSE | hybrid functional of Heyd, Scuseria and Ernzerhof | 3.4 |
| HSE06 | hybrid functional of Heyd, Scuseria and Ernzerhof, 2006 parametrization | 10.6 |
| IP | independent particles | 4.6 |
| JARVIS | Joint Automated Repository for Various Integrated Simulations | 17.6 |
| KK | Kramers–Kronig | 30.3 |
| KKR | Korringa–Kohn–Rostoker | 30.5 |
| LA | longitudinal acoustic | 5.2 |
| LCAO | linear combination of atomic orbitals | 3.2 |
| LDA | local-density approximation | 3.4 |
| LF | local fields | 8.5 |
| LO | longitudinal optical | 4.9 |
| MC | Monte Carlo | 32.1 |
| MD | molecular dynamics | 30.5 |
| ML | machine-learned | 30.4 |
| MLIP | machine-learned interatomic potential | 32.1 |
| NEB | nudged elastic band | 30.5 |
| OPW | orthogonalized plane wave | 16.3 |
| PAW | projector-augmented-wave | 10.6 |
| PBE | generalized-gradient functional of Perdew, Burke and Ernzerhof | 3.3 |
| PBE0 | hybrid functional built on the functional of Perdew, Burke and Ernzerhof | 30.4 |
| QED | quantum electrodynamics | 6.3 |
| RAVE | Radial Velocity Experiment | 1.2 |
| RCC | relativistic core correlation | 31.3 |
| RPA | random-phase approximation | 8.5 |
| SALMON | Scalable Ab initio Light–Matter simulator for Optics and Nanoscience (code) | 32.1 |
| SCAN | strongly constrained and appropriately normed | 30.1 |
| SCF | self-consistent-field | 16.6 |
| SCHA | self-consistent harmonic approximation | 5.6 |
| SH | transverse polarization perpendicular to the plane of incidence (shear horizontal) | 20.3 |
| SHM | standard halo model | 1.2 |
| SOC | spin–orbit coupling | 8.4 |
| ST | slow transverse | 29.2 |
| SV | transverse polarization in the plane of incidence (shear vertical) | 20.3 |
| TA | transverse acoustic | 5.2 |
| TB09 | meta-generalized-gradient functional of Tran and Blaha (2009) | 33.1 |
| TDDFT | time-dependent density-functional theory | 8.5 |
| TDEP | temperature-dependent effective potential | 25 |
| TO | transverse optical | 4.9 |
| TRIM | Transport of Ions in Matter | 19.4.4 |
| VBM | valence-band maximum | 3.3 |
| X2C | exact two-component (scalar-relativistic Hamiltonian) | 31.3 |
| XCOM | photon cross-section database of the National Institute of Standards and Technology | 10.8 |
| YIG | yttrium iron garnet | 14.1 |
| ZBL | Ziegler, Biersack and Littmark | 29.1 |
| ZG | Zacharias–Giustino (special displacement) | 25.3 |
| ZPR | zero-point renormalization | 6.3 |